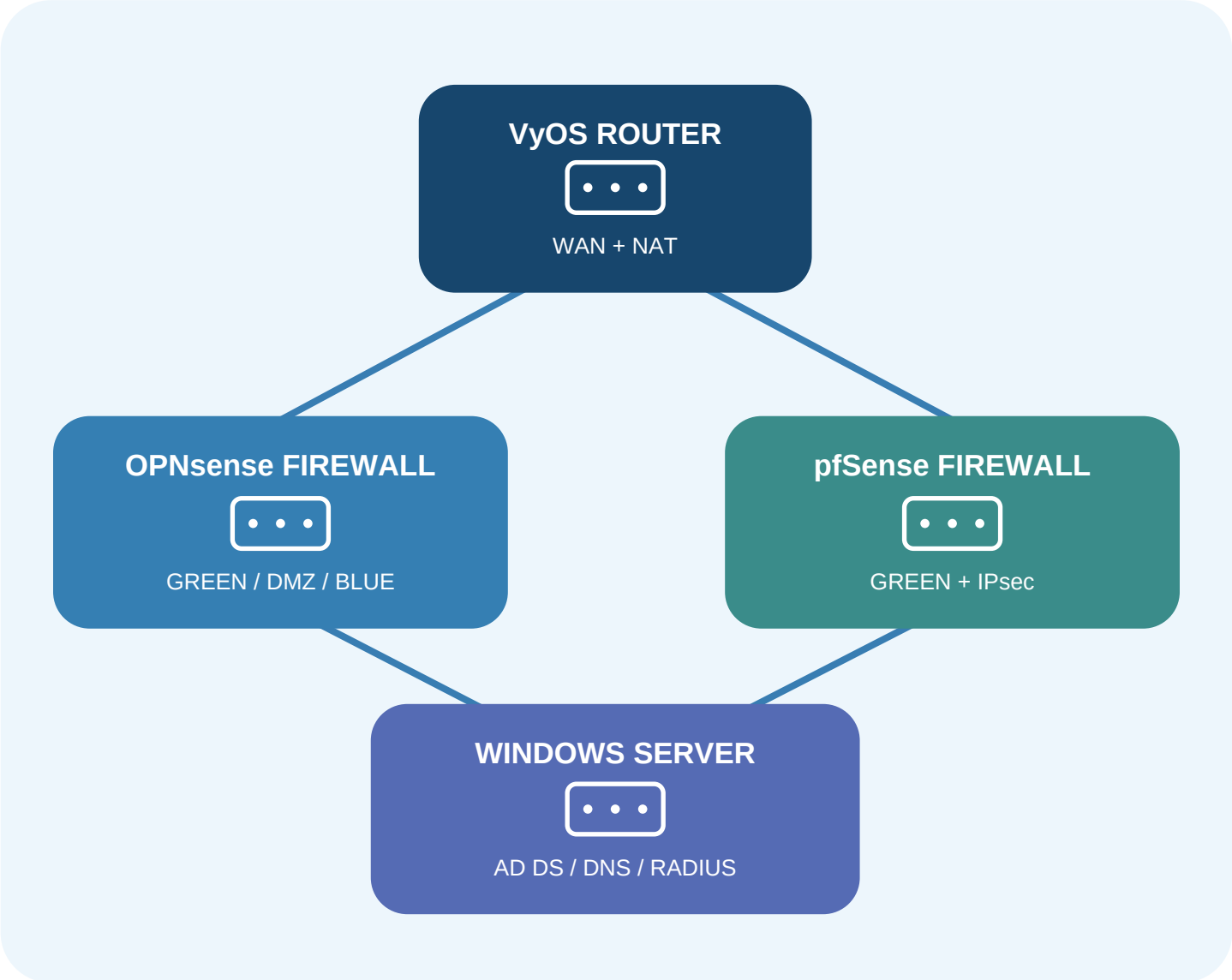
ACTIVE CYBERSECURITY

Network Security Architecture

OPNsense | pfSense | VyOS | Secure VPNs

Technical Project - Pedro Goncalves



Network Segmentation - IPsec - OpenVPN - WireGuard

Firewall Rules | DNS Security | Proxy | Captive Portal

Introdução

O presente trabalho de avaliação insere-se no âmbito da unidade de formação UF9196-Cibersegurança Ativa, tendo como principal objetivo a implementação prática de uma arquitetura de rede segura, recorrendo a diferentes tecnologias de firewall, VPN, autenticação e controlo de acesso.
Ao longo do trabalho foi construída uma infraestrutura segmentada em várias zonas de segurança (Green, F2Green, WAN, DMZ e Blue), recorrendo às plataformas OPNsense, pfSense, Windows Server 2022 e VyOS, permitindo simular um ambiente empresarial real.
Foram implementados mecanismos de:

- Resolução e controlo de DNS
- Autenticação centralizada via RADIUS
- VPNs seguras (IPsec, OpenVPN e WireGuard)
- Proxy com filtragem de conteúdos
- Captive Portal com controlo de largura de banda
- Regras de firewall e NAT
- Simulação de cenários como DNS Poisoning e bloqueio de conteúdos

O trabalho teve como foco não só a configuração técnica, mas também a validação funcional através de testes práticos (ping, tracert, nslookup, acesso remoto e navegação controlada).

Trabalho Proposto

O trabalho proposto consistiu na implementação completa de uma arquitetura de segurança ativa, contemplando os seguintes objetivos técnicos:

Configuração de DNS Interno e Controlo de Resolução

- Instalação e configuração do serviço DNS no Windows Server.
- Criação de zonas diretas.
- Implementação de bloqueio de DNS externo na firewall.
- Simulação de ataque de DNS Poisoning.
- Testes de resolução via nslookup, ping e browser.

Objetivo:
Garantir que todos os clientes utilizam exclusivamente o DNS interno e impedir bypass para DNS públicos.

Implementação de Autenticação Centralizada (RADIUS)

- Instalação do serviço NPAS no Windows Server.
- Criação de clientes RADIUS.
- Configuração de políticas de acesso.
- Integração do servidor RADIUS no OPNsense.

Objetivo:
Centralizar a autenticação de utilizadores para serviços VPN.

Configuração de VPN OpenVPN com RADIUS

- Criação de instância OpenVPN (modo servidor).
- Integração com autenticação RADIUS.
- Configuração de certificados (CA, Server e Cliente).
- Exportação de perfil .ovpn.
- Instalação no cliente Windows.
- Testes de ligação e conectividade interna.

Objetivo:
Permitir acesso remoto seguro com autenticação centralizada.

Implementação de VPN IPsec (OPNsense ↔ pfSense)

- Criação de Phase 1 (IKE).
- Configuração de Phase 2 (Child SA).
- Definição de algoritmos de encriptação.
- Criação de regras WAN:
  - ESP
  - ISAKMP
  - NAT-T
- Validação em Status → IPsec Overview.
- Testes de comunicação entre redes distintas.
- Acesso remoto ao Windows Server via RDP através do túnel.

Objetivo:
Estabelecer comunicação site-to-site segura entre firewalls distintas.

Implementação de VPN WireGuard

- Criação de túnel no OPNsense.
- Geração de chaves públicas e privadas.
- Configuração de peer.
- Configuração manual no cliente Windows.
- Criação de interface WG no firewall.
- Ajuste de regras WAN.
- Testes de ping e resolução DNS através do túnel.

Objetivo:
Comparar desempenho e simplicidade de configuração face ao IPsec e OpenVPN.

Implementação de Proxy (Squid) com Filtragem

- Instalação do plugin Squid.
- Configuração de Forward Proxy.
- Ativação de SSL Inspection.
- Criação de certificados self-signed.
- Importação manual no Windows (MMC).
- Bloqueio manual de domínios.
- Download e aplicação de blacklists externas.
- Testes de bloqueio (gmail, sapo, , iefp, playboy, bmw.de).

Objetivo:
Controlar e monitorizar o tráfego HTTP/HTTPS e aplicar políticas de navegação.

Implementação de Captive Portal

- Criação de zona.
- Criação de vouchers.
- Autenticação por senha.
- Aplicação de regras específicas na interface Blue.
- Monitorização de sessões ativas.
- Implementação de Traffic Shaper (10MB e 5MB).

Objetivo:
Controlar acesso à rede convidada e limitar largura de banda.

Implementação de Traffic Shaping

- Criação de pipes (10MB e 5MB).
- Associação de regras por interface.
- Testes com Speedtest.
- Validação de limitação efetiva.

Objetivo:
Gerir qualidade de serviço e largura de banda por zona.

Validação Final

Foram realizados testes práticos em:

- Conectividade interna
- Acesso remoto via VPN
- Resolução DNS
- Bloqueio de conteúdos
- Estabilidade dos túneis
- Monitorização no Dashboard

Projeto de avaliação UFCD 9196, (Pedro Gonçalves)

- Router VyOS (R1)

- OPNsense (yinPG)

- pfSense (yangPG)

- Windows Server 2022 (AD DS + DNS + NPS/RADIUS + RDP + SSH)

- Windows 10 (cliente de testes + OpenVPN + WireGuard + certificados + testes)

- CentOS 7 (web + auth + SSL/TLS + FTP)

Notas :

- Passwords a usar: **Passw0rd.**
  - Hostnames das firewalls: **yinPG** e **yangPG** (P=Pedro, G=Gonçalves).
  - Os testes/validação devem partir do **Windows 10/11 cliente**
  - VMs a usar e interfaces (yin com 4 NICs, yang com 2 NICs, etc.).

1) Redes atribuídas a Pedro Gonçalves e plano de IPs

- Yin Green: 172.19.130.0/23

- Yin Orange (DMZ): 172.29.130.64/29

- Yin Blue (Captive): 10.55.130.0/24

- Yang Green: 10.120.130.0/24

- WAN pessoal (rede WAN): 10.103.130.0/27

Endereçamento

Router VyOS (R1)
- eth0 = NAT (DHCP) (internet)

- eth1 = WAN pessoal: 10.103.130.1/27

OPNsense yinPG
- WAN = 10.103.130.10/27 gateway 10.103.130.1

- GREEN = 172.19.130.1/23

- ORANGE = 172.29.130.65/29

- BLUE = 10.55.130.1/24

pfSense yangPG

- WAN = 10.103.130.11/27 gateway 10.103.130.1

- F2 green = 10.120.130.1/24

Windows Server 2022 (DC/DNS/NPS)

- NIC (yin green) = 172.19.130.10/23

- GW = 172.19.130.1

- DNS = 172.19.130.10 (ele próprio) /127.0.0.1

CentOS 7 (Web/FTP)

- NIC (yin orange) = 172.29.130.66/29

- GW = 172.29.130.65

- DNS = 172.19.130.10

Windows 10 (Cliente de testes)

- NIC WAN: 10.103.130.20/27 GW 10.103.130.1

- NIC green : 172.19.130.20/23 GW 172.19.130.1

- NIC f2 green: 10.120.130.20/24 GW 10.120.130.1

- NIC yin blue : 10.55.130.20/24 GW 10.55.130.1

Windows 10 (Cliente de testes - cenário WAN)

- NIC “WAN” do Windows 10 → WAN_PESSOAL_10.103.130
- IP: 10.103.130.20/27
- Gateway: 10.103.130.1

Nota :O enunciado indica que **todas as NICs** (exceto eth0 do router em NAT) devem estar em **Internal Network**.

Instalação das máquinas virtuais :

Opnsense :

OPNsense yinPG — Configuração base (interfaces + gateways + NAT)
O enunciado define a firewall yin como OPNsense com 4 interfaces.

Atribuição de interfaces

Na consola do OPNsense:
1. Assign interfaces (WAN, LAN, OPT1, OPT2)

- WAN → NIC ligada à WAN_PESSOAL_10.103.130

- LAN → GREEN

- OPT1 -> ORANGE

- OPT2 → BLUE

IPs das interfaces

No OPNsense (Interfaces →[WAN/GREEN/ORANGE/BLUE]):

- WAN: 10.103.130.10/27, Gateway 10.103.130.1

- GREEN: 172.19.130.1/23

- ORANGE: 172.29.130.65/29

- BLUE: 10.55.130.1/24

DNS do firewall

System → Settings → General
- DNS Servers: colocar temporariamente 1.1.1.1 e 8.8.8.8

- Gateway: WAN

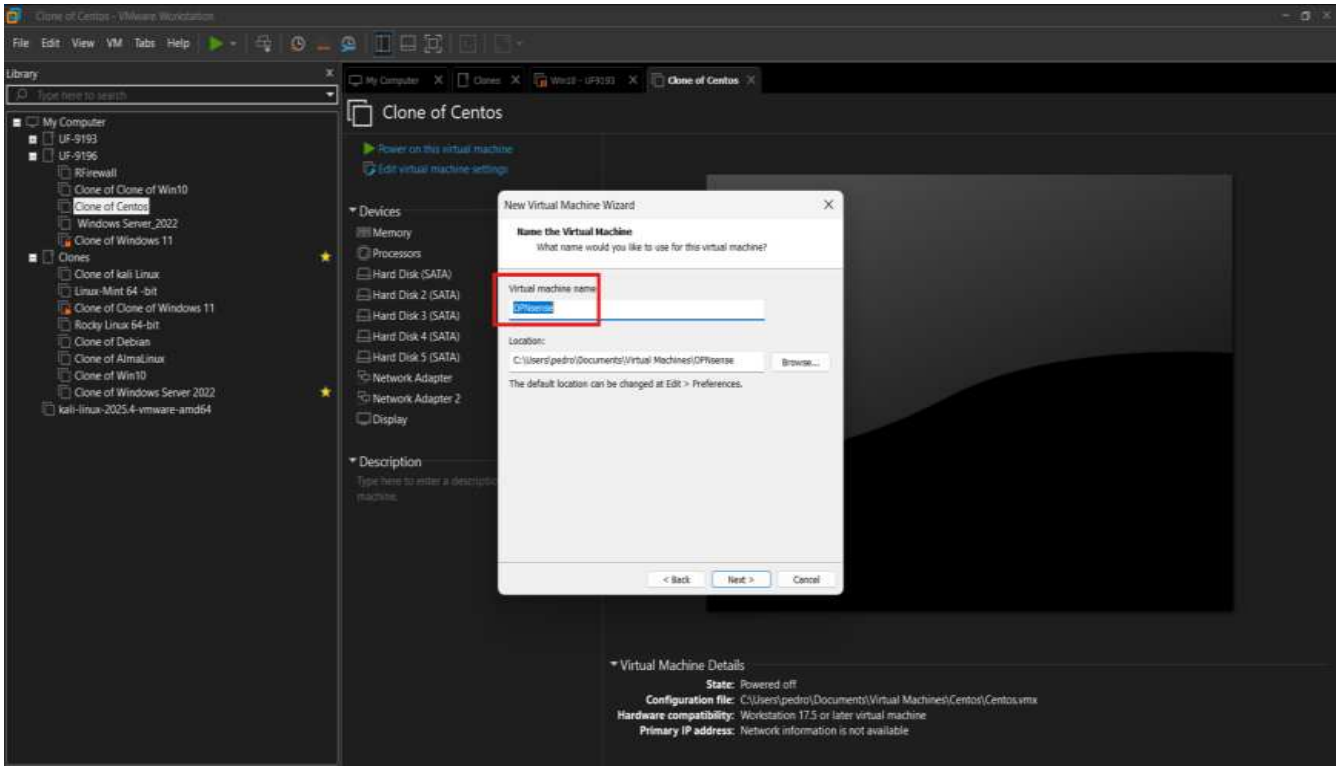
(Mais tarde, para o Captive Portal na BLUE, o enunciado é mencionado o DNS tem de ser o do próprio firewall nessa zona.)

Outbound NAT

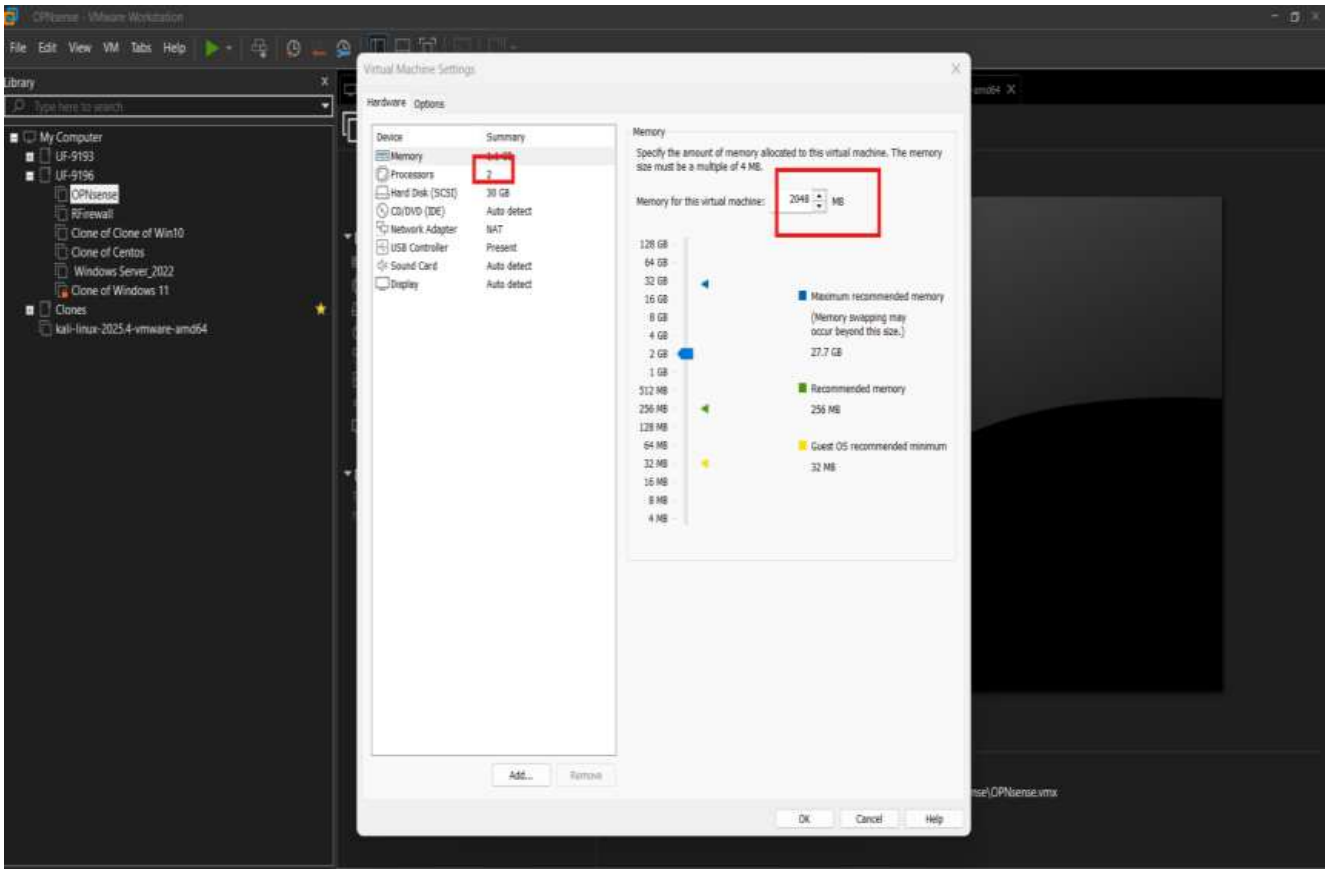
Firewall →NAT >Outbound
- Modo: **Automatic**

.

1- Confirmação se as redes GREEN/ORANGE/BLUE (e depois VPNs) estão a sair pela WAN

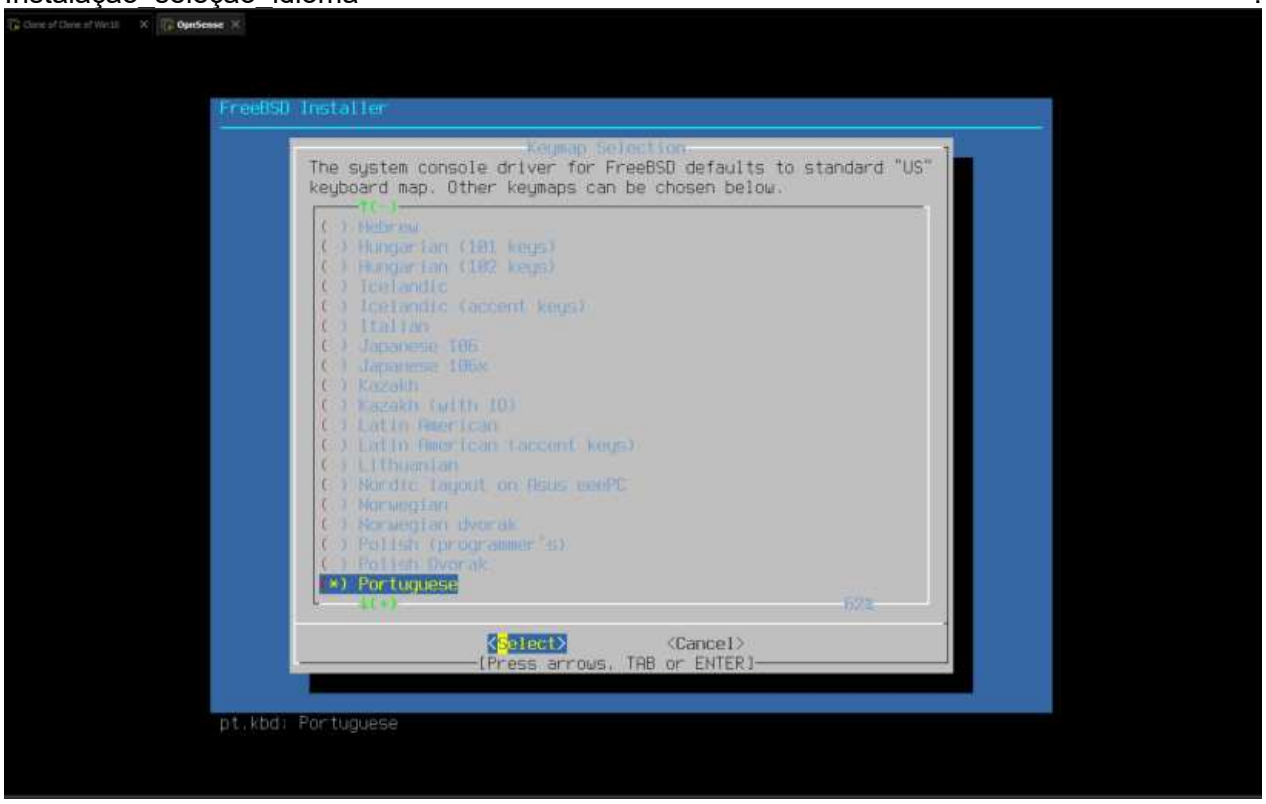


1.1 memoria ; 2 cores -Processador

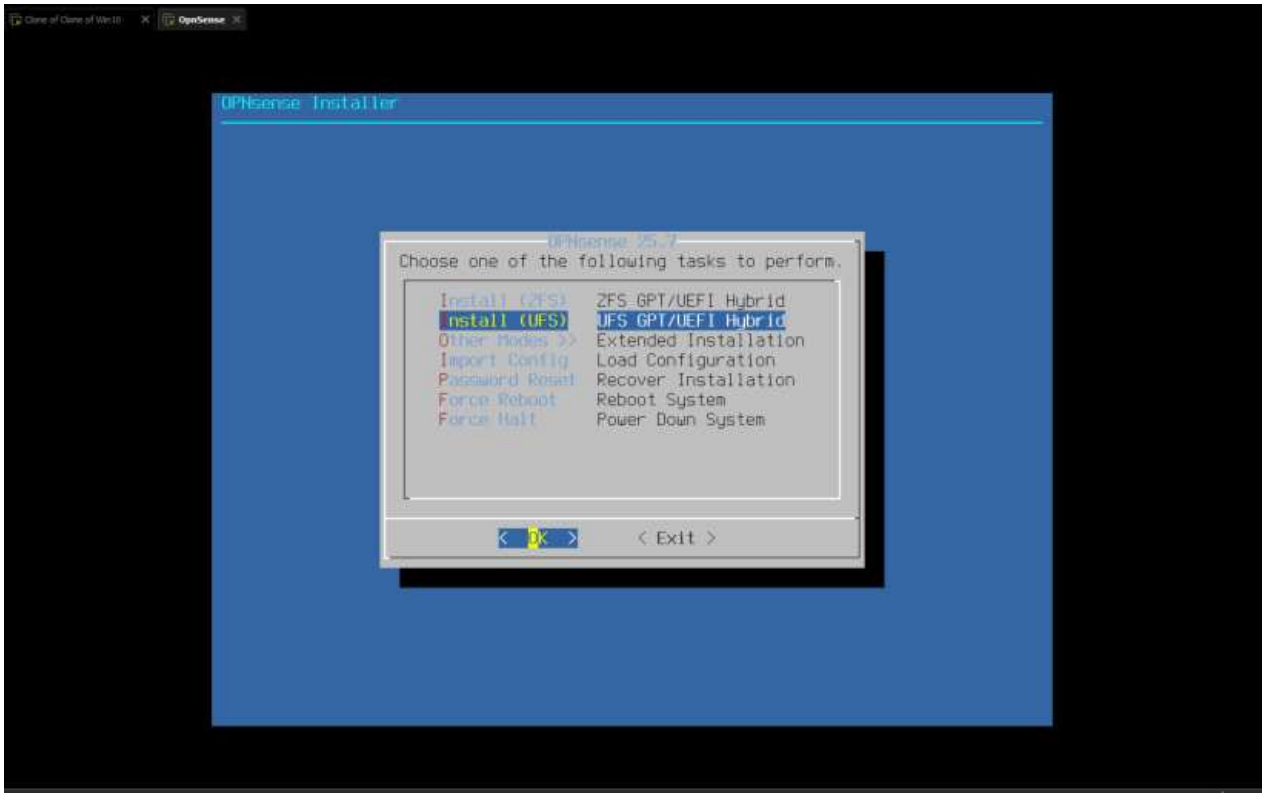


1.3 login como:**installer** ;password:**opnsense**

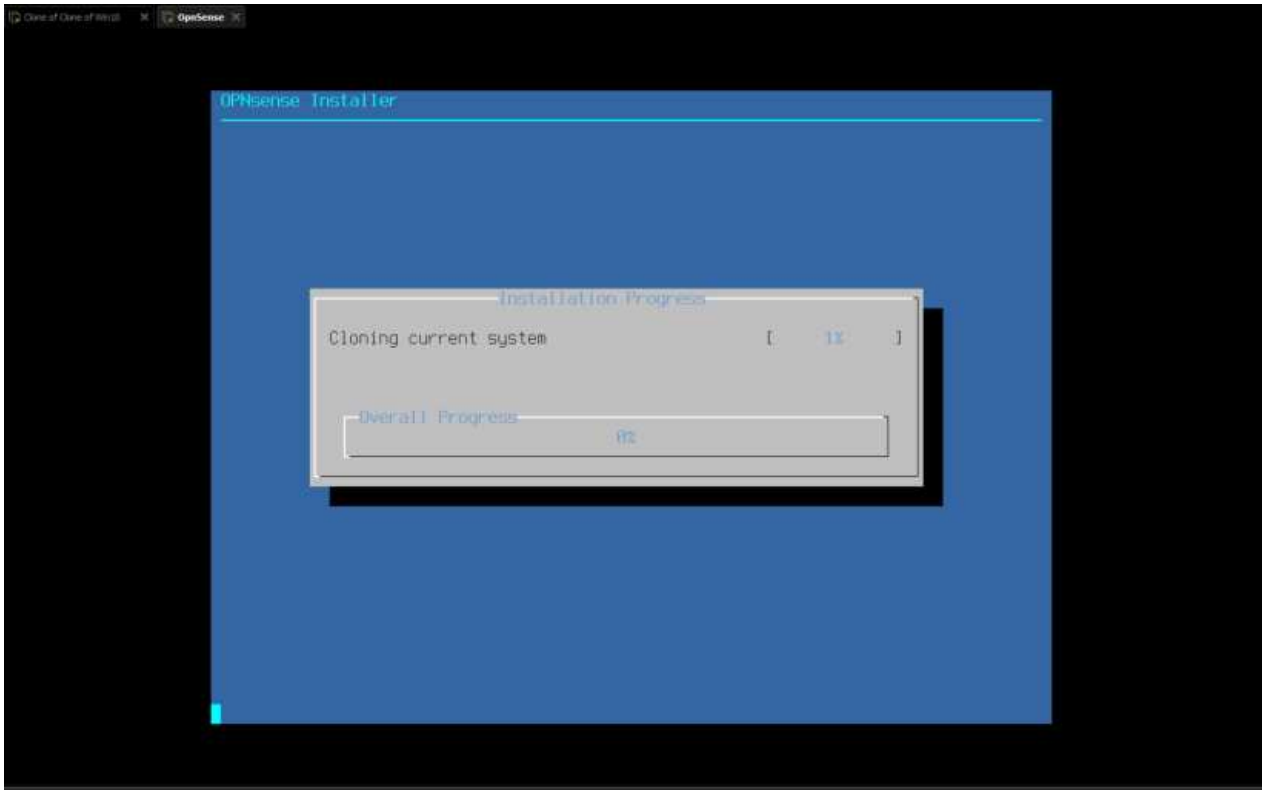
Instalação_seleção_idioma :



1.4 Install UFS

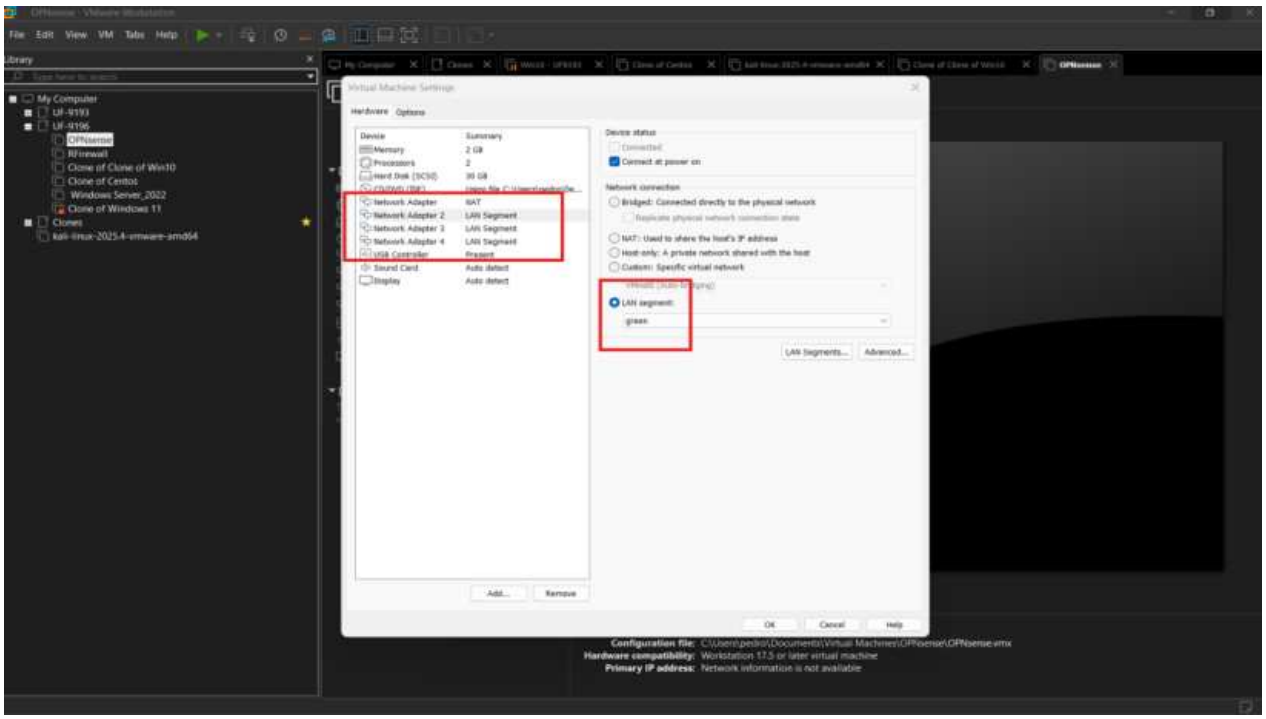


Continução …

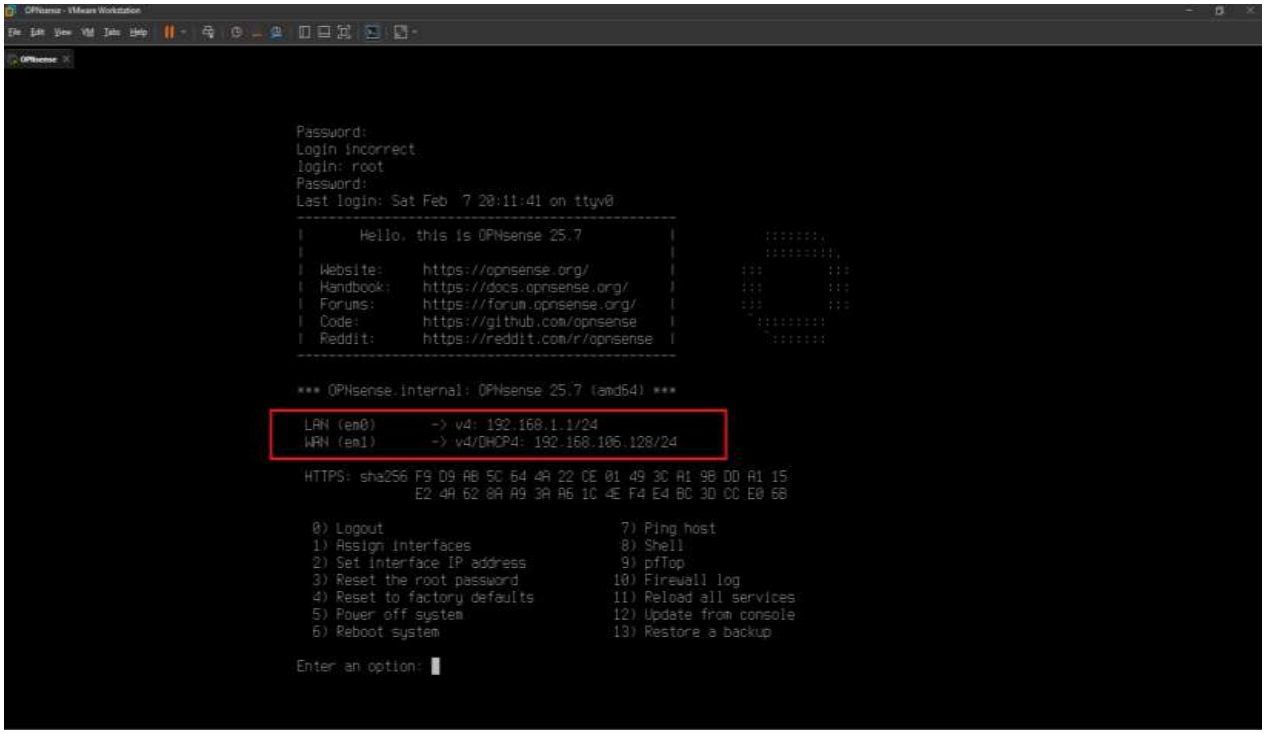


1.5 Atribuição de interfaces :

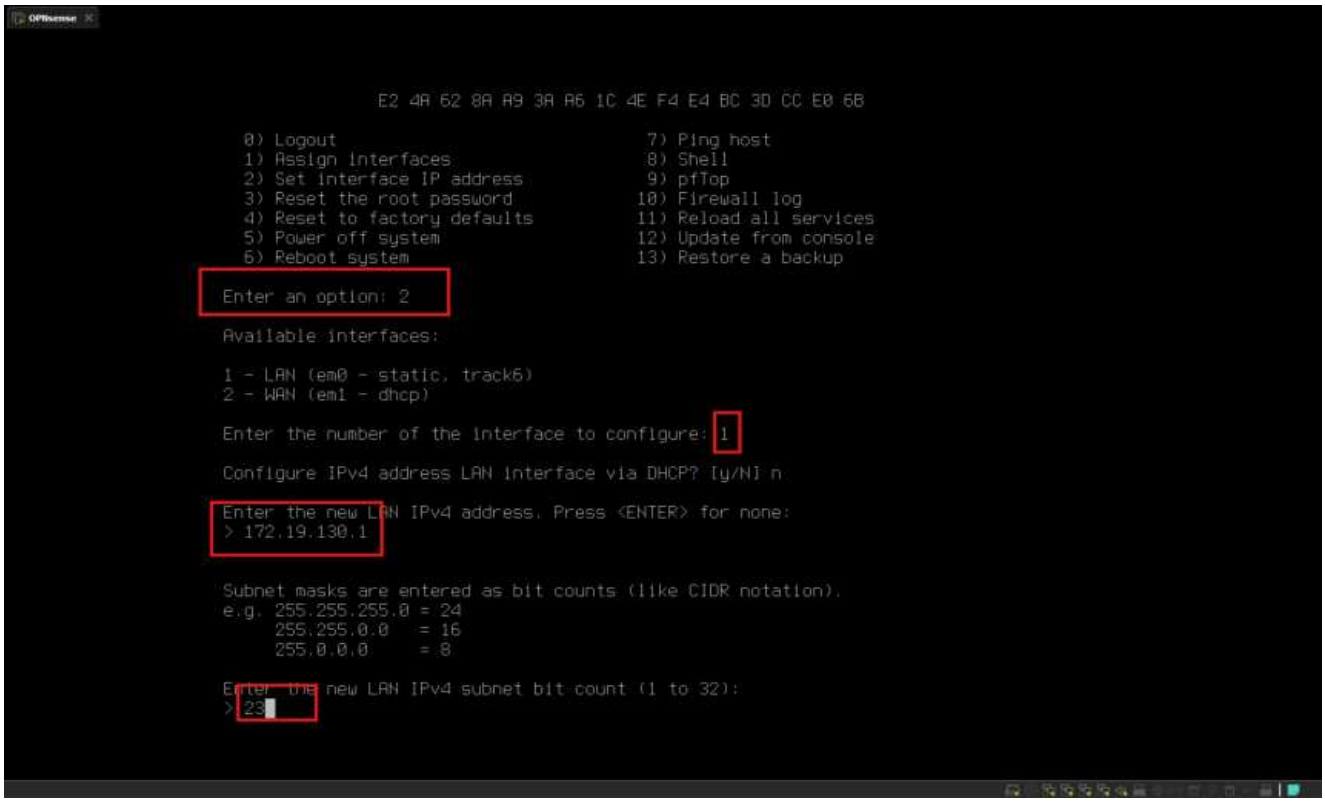
Green -LAN Segment;-exemplo
NAT- Network Adapater 2
Green - Lan Segment 3 (network Adapter 3)
Orange- Lan Segment 4 (network Adapater 4)



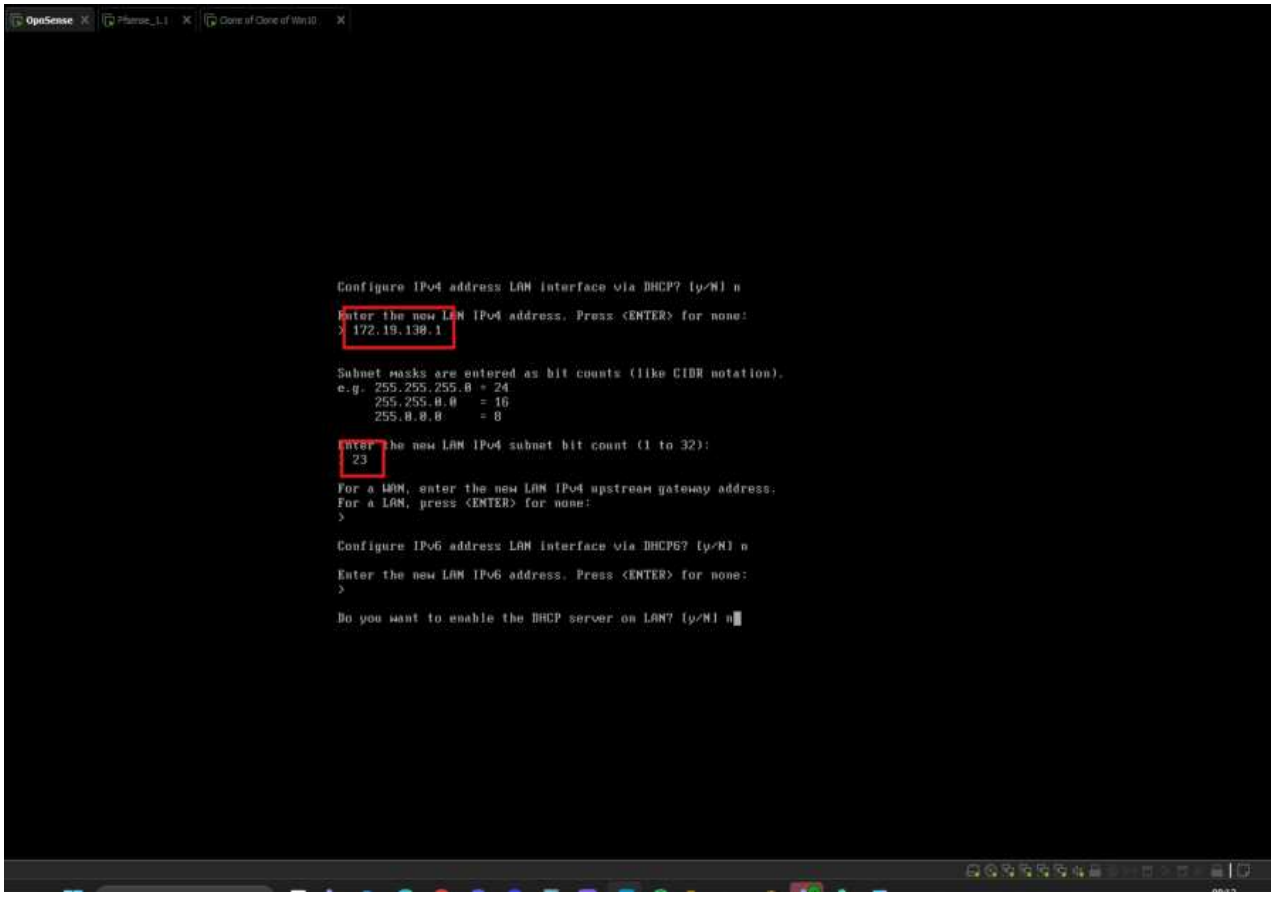
1.6 Instalação …
User : **root**
Password :**opnsense**



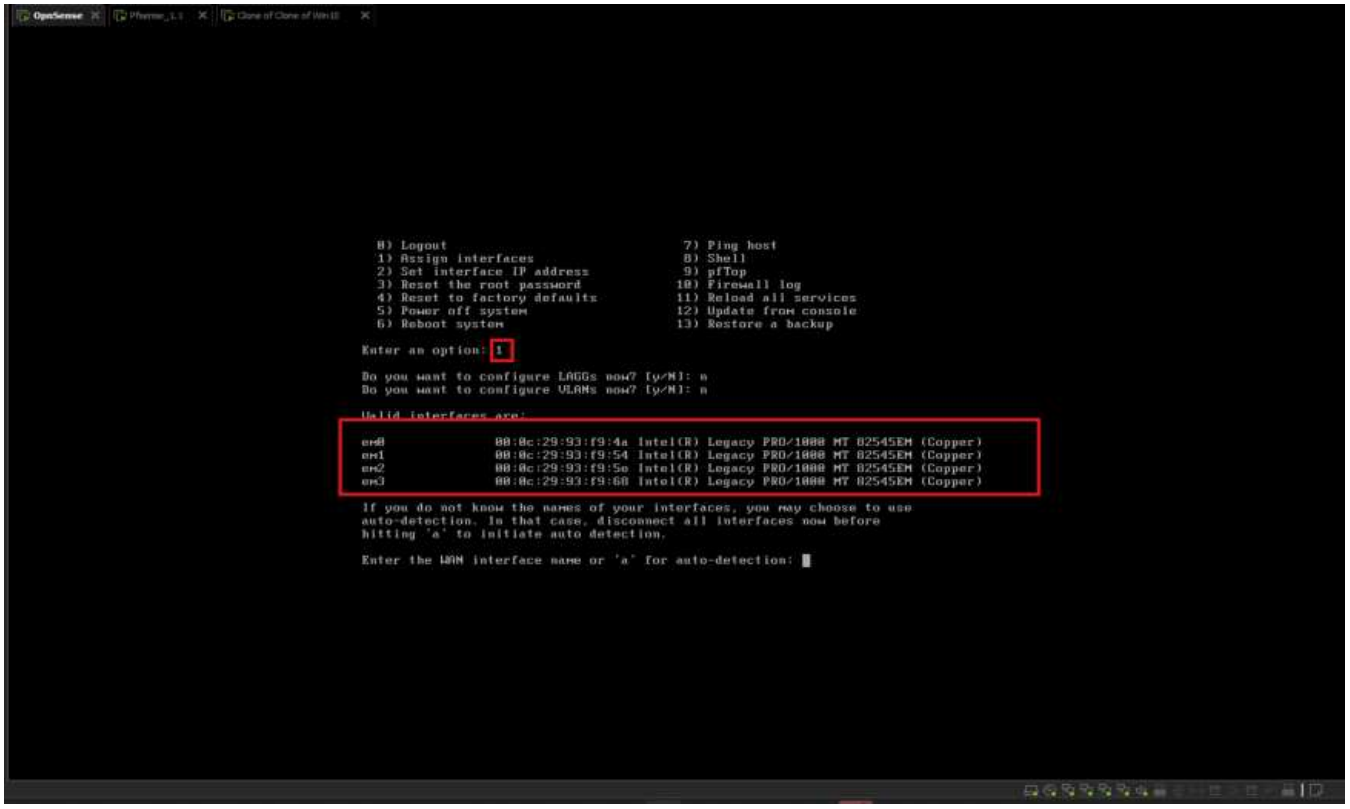
1.7 Configurações -LAN- green ; IPv4 LAN -172.19.130.1- (green)- em0



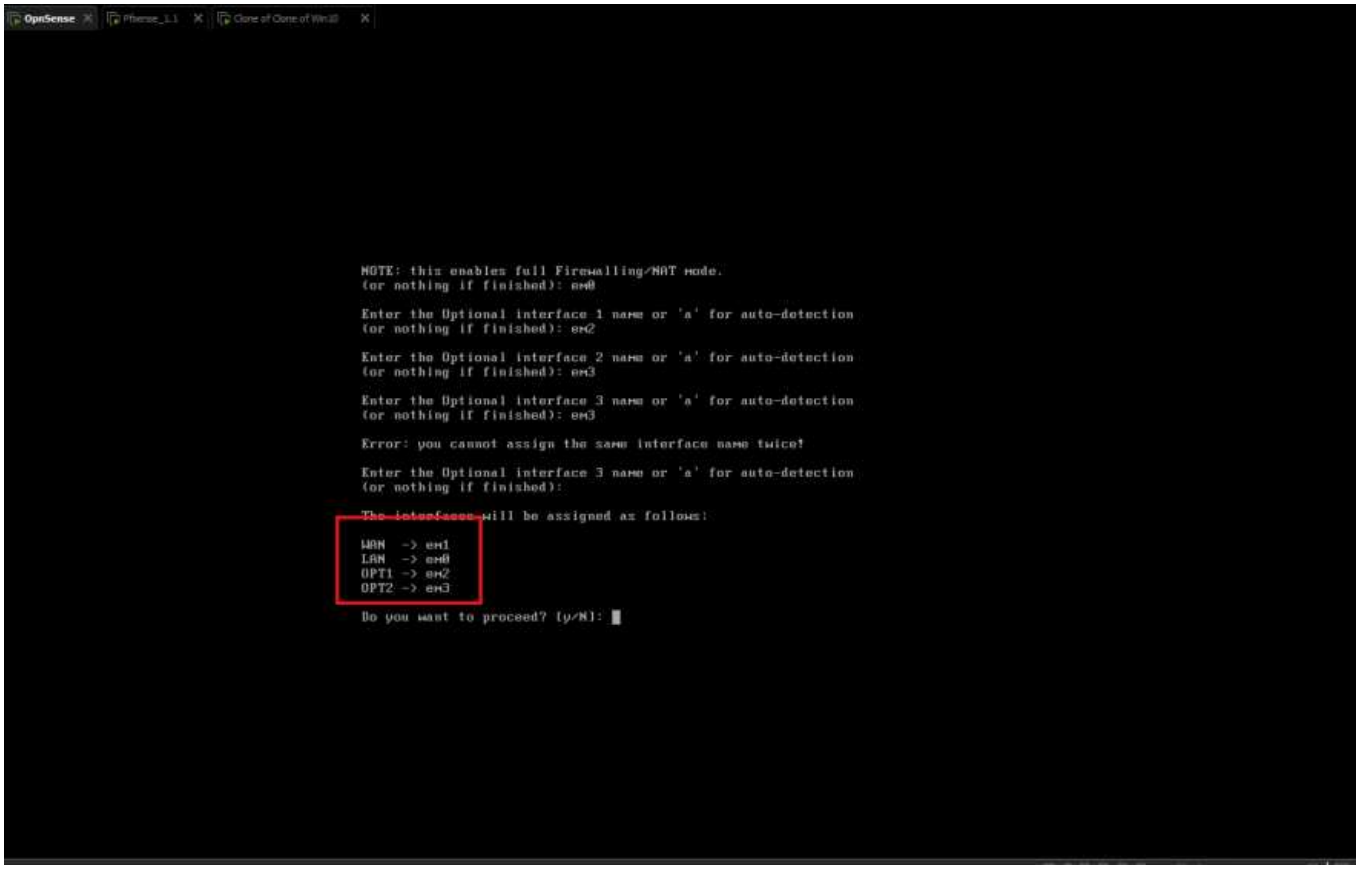
Continuação…configuração da rede – Green-LAn



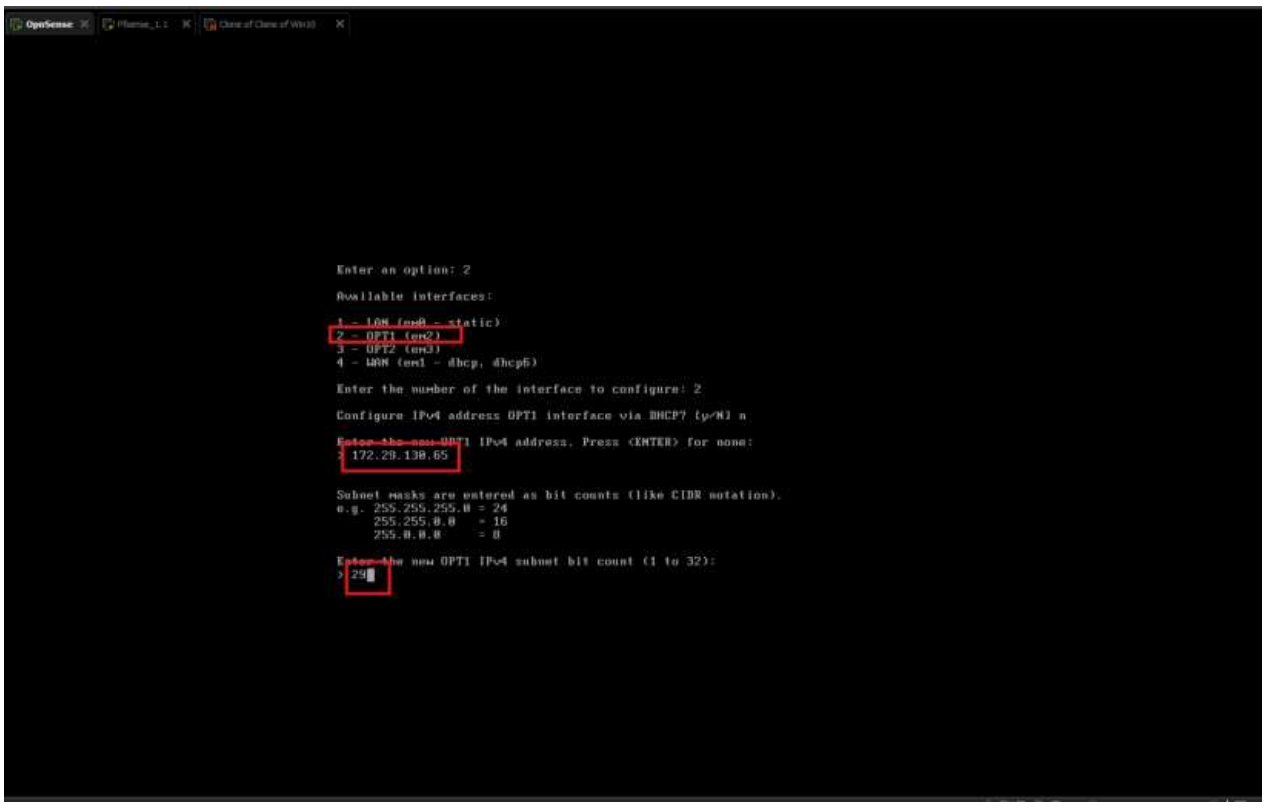
1.8 Opção 1 Assign interfaces :



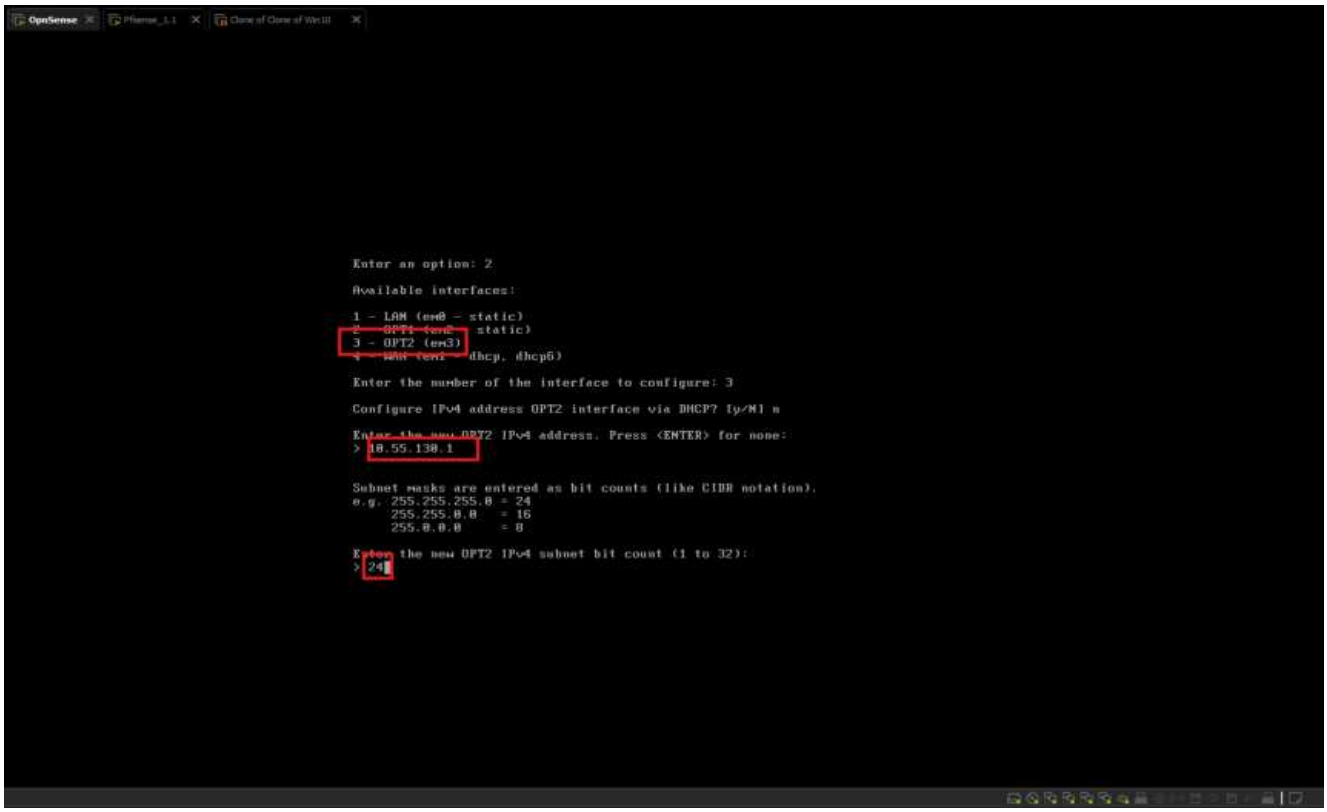
1.9 Atribuição de nome ás interfaces :



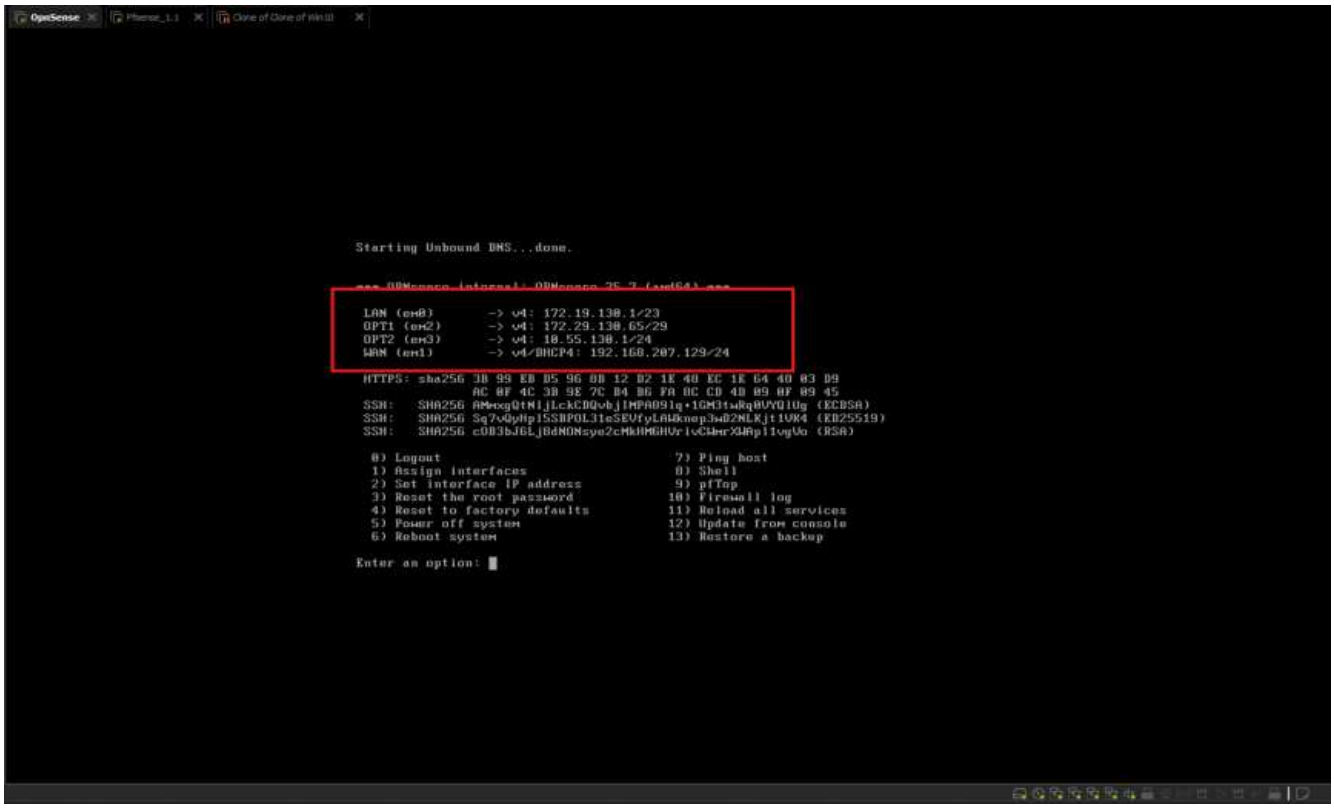
1.10 Configuração da Lan- Orange -OPT1



1.11 Configurações da OPT2 – em3 -blue



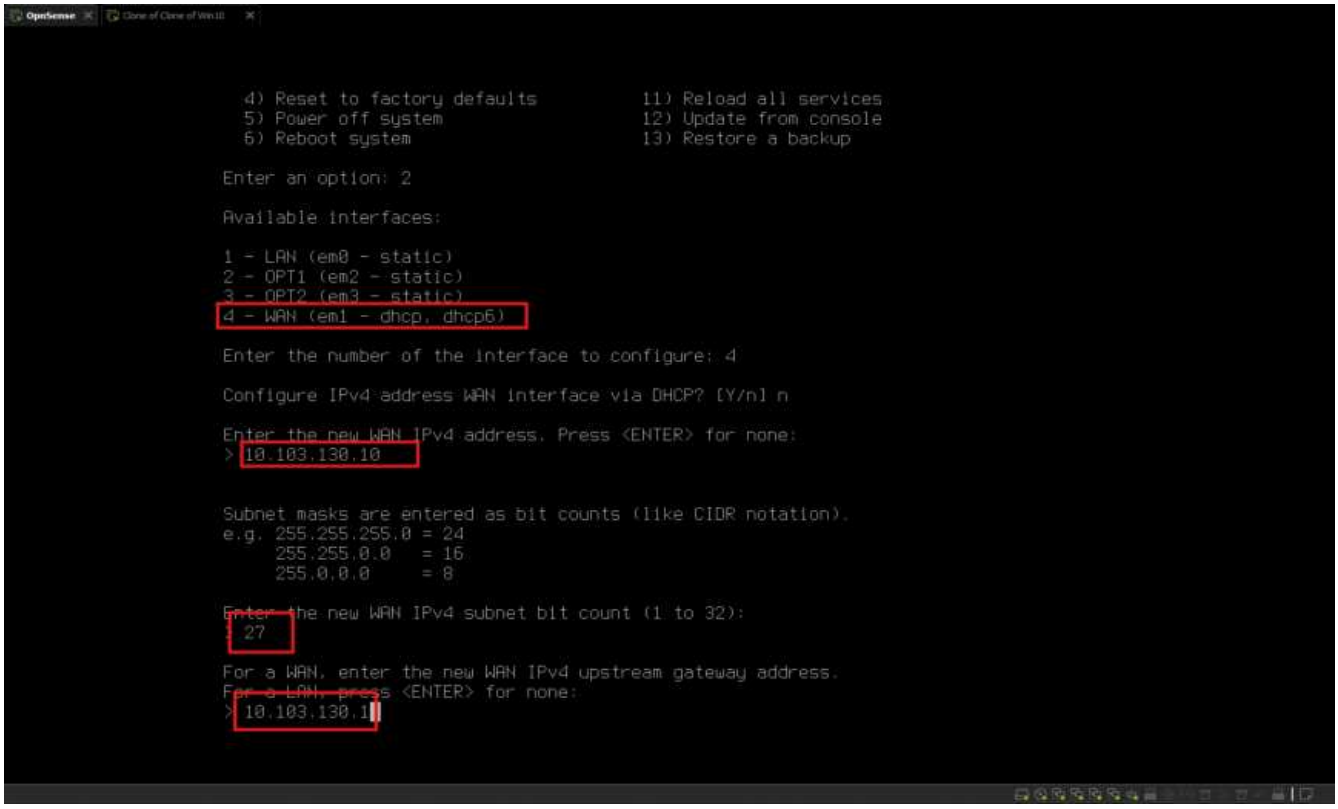
Nota : a rede Wan fica temporariamente em dhcp , para que se possa posteriormente proceder ás instalações dos plugins , já que o modo bridge do vmWare não funciona – firewall foi desligada



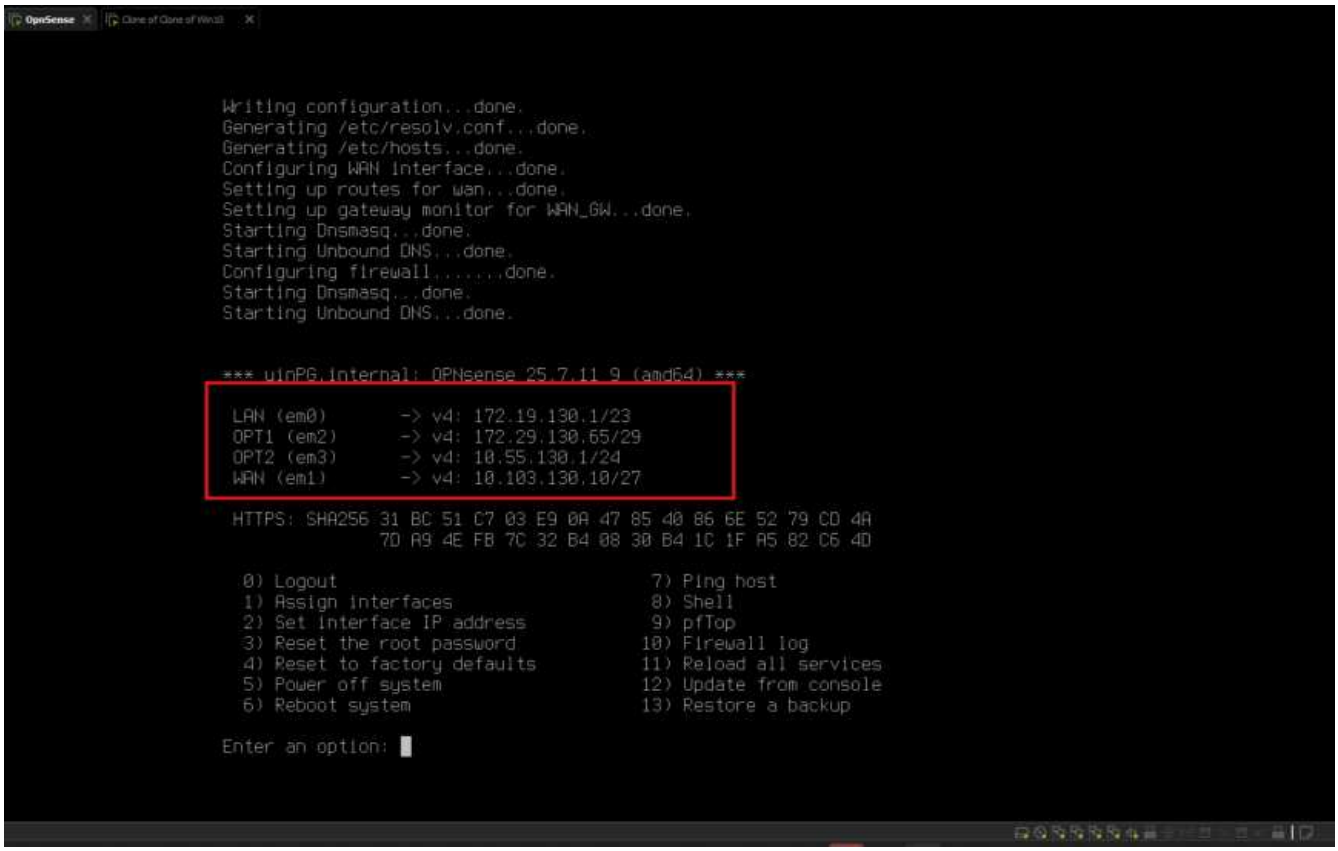
Configurações : já com ip da em Wan configurado

Nota Importante:

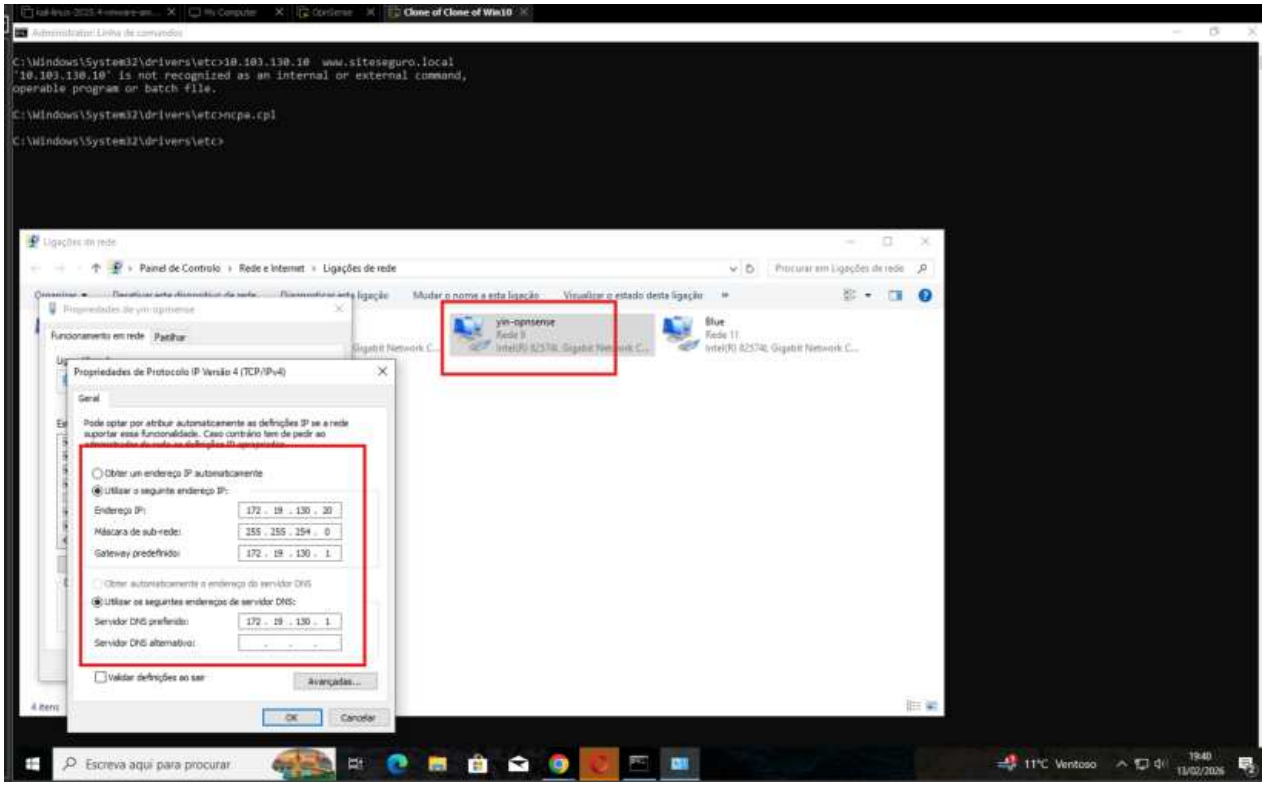
1.12 Atribuição de ips --- WAN (que se enconta atualmente em dhcp) = 10.103.130.10/27 gateway 10.103.130.1- após updates plugins



1.13 Resumo configurações finais :

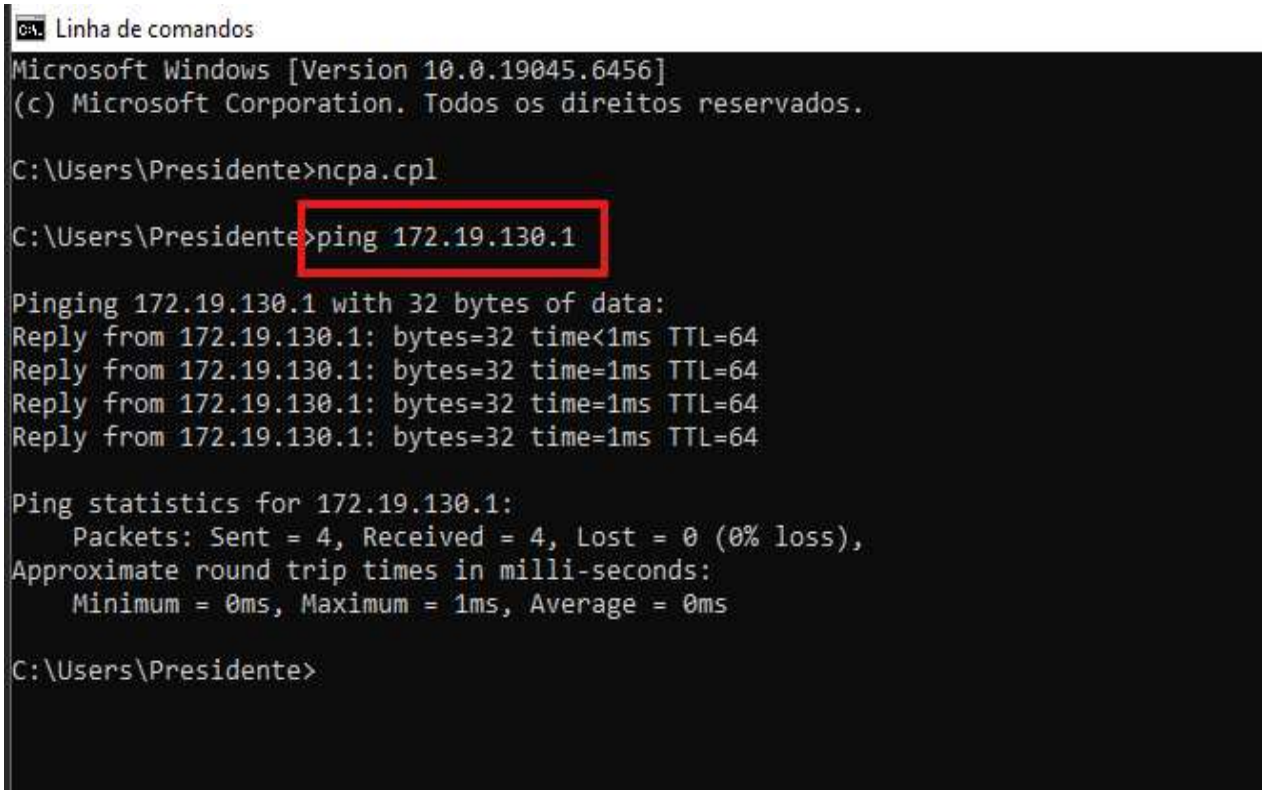


1.14 Configurações para a rede Yin (OPNsense yinPG)

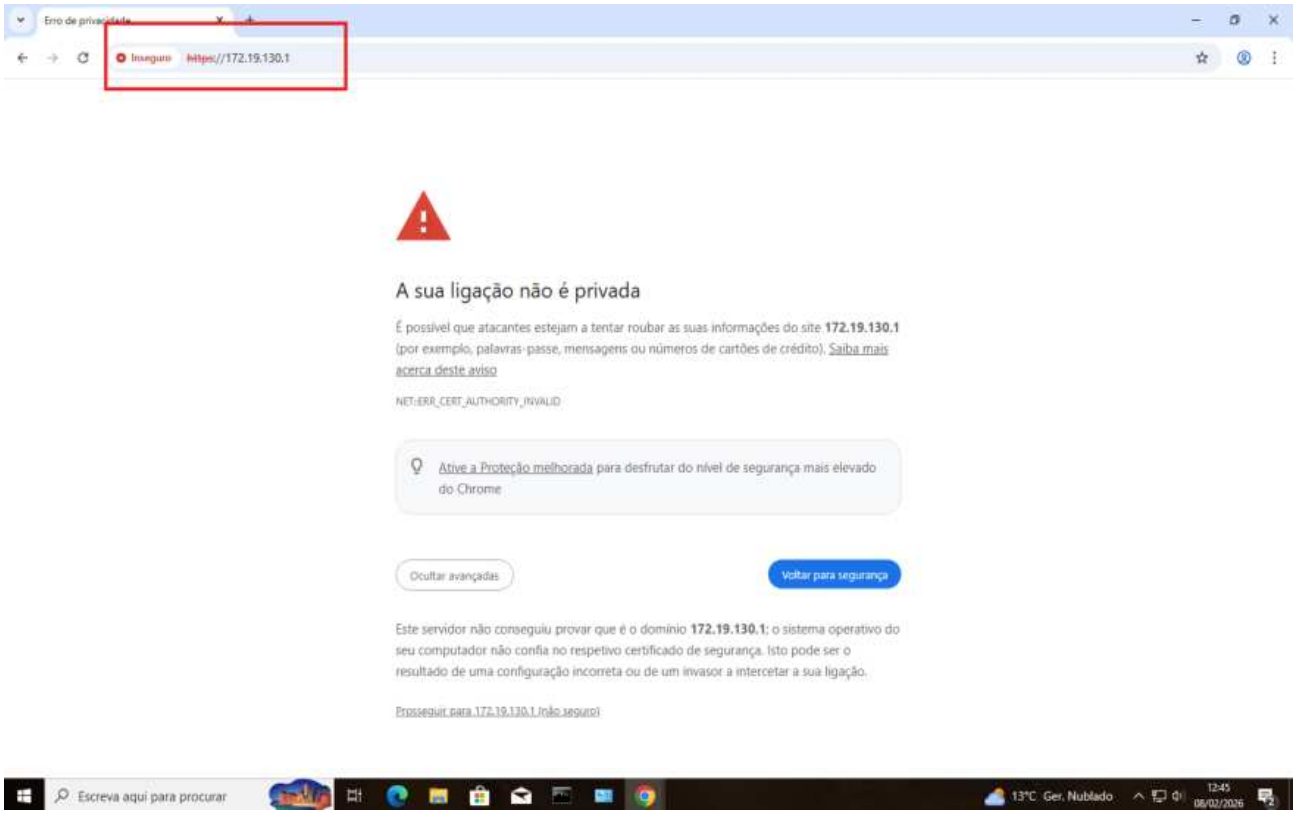


Nota :Após configurações , desativar e ativar interfaces

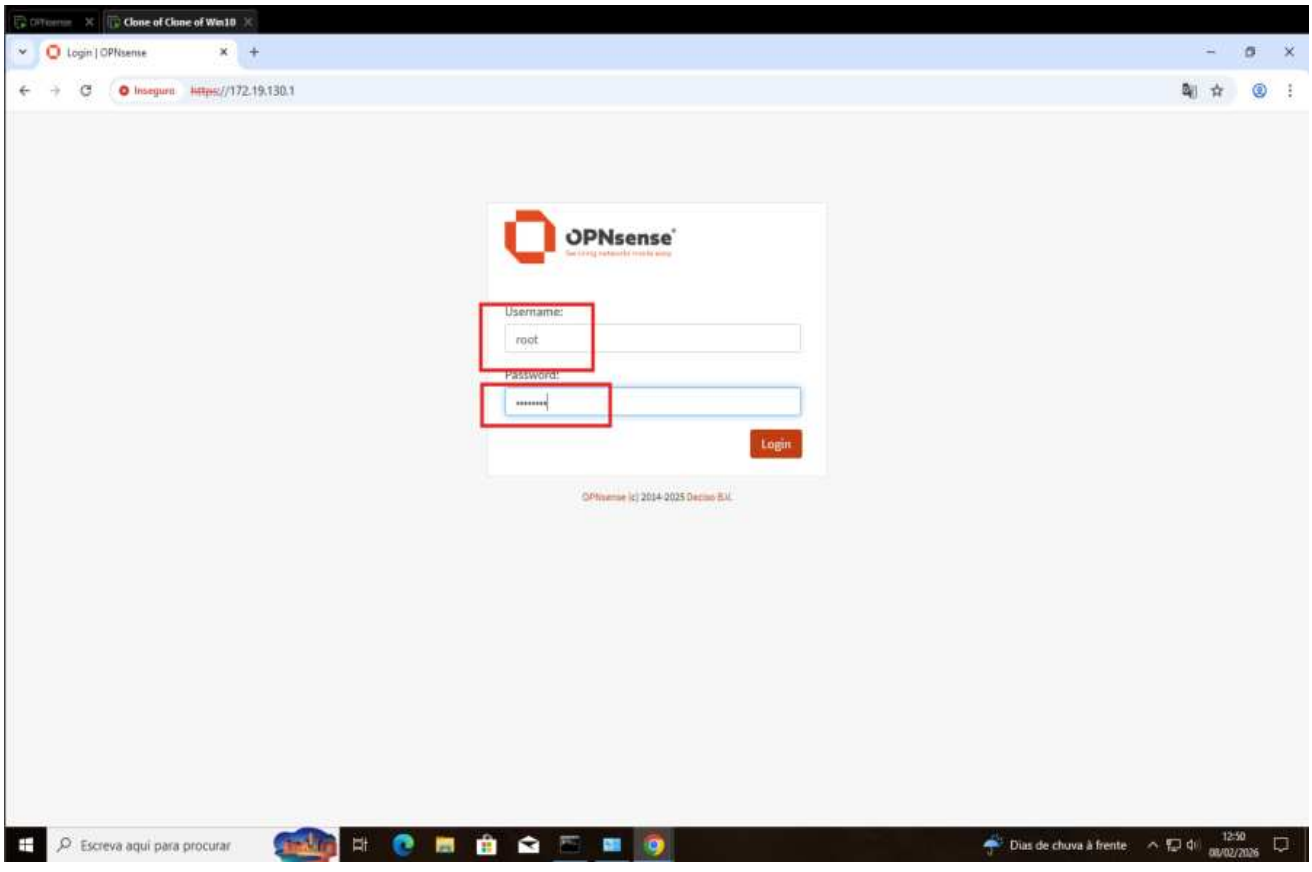
1.15 Teste “ping” no Windows -10



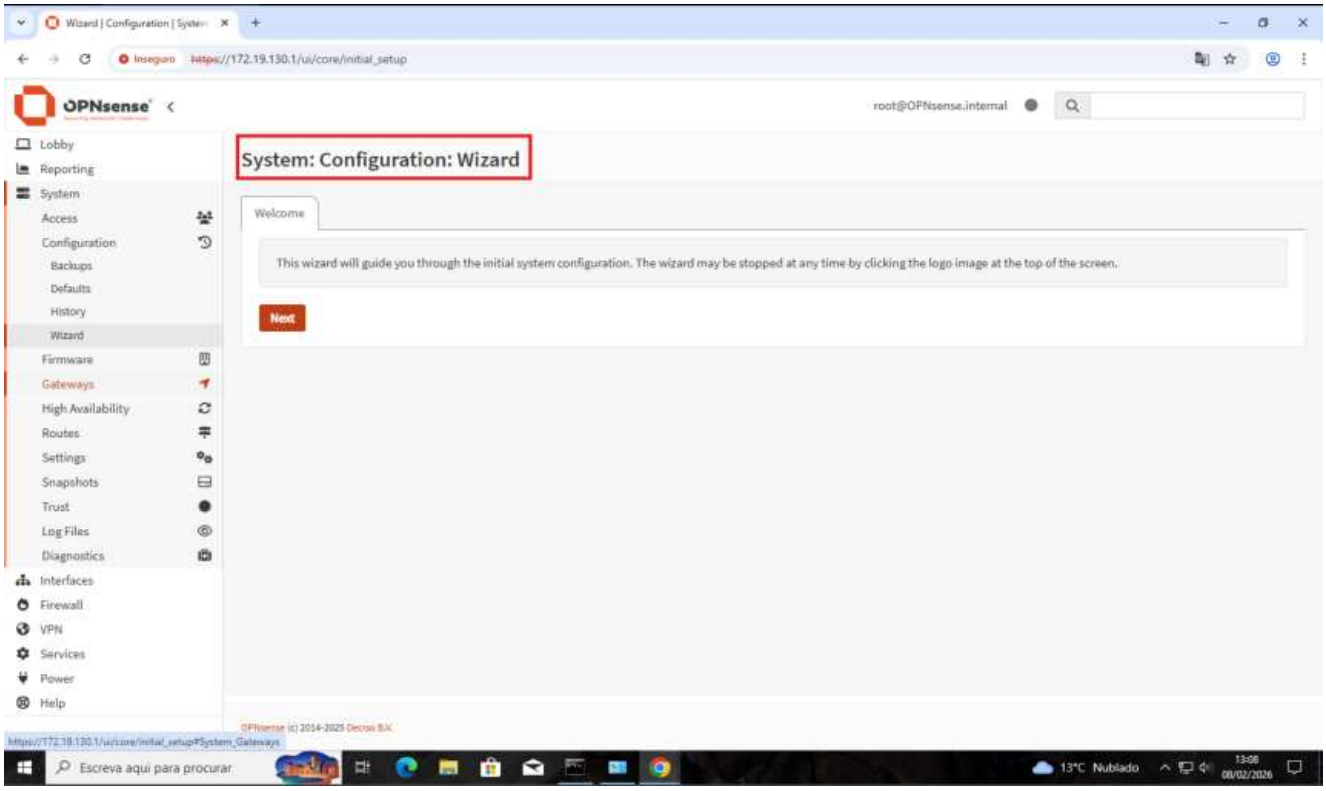
1.16 No google chrome : aceder ao opnsense



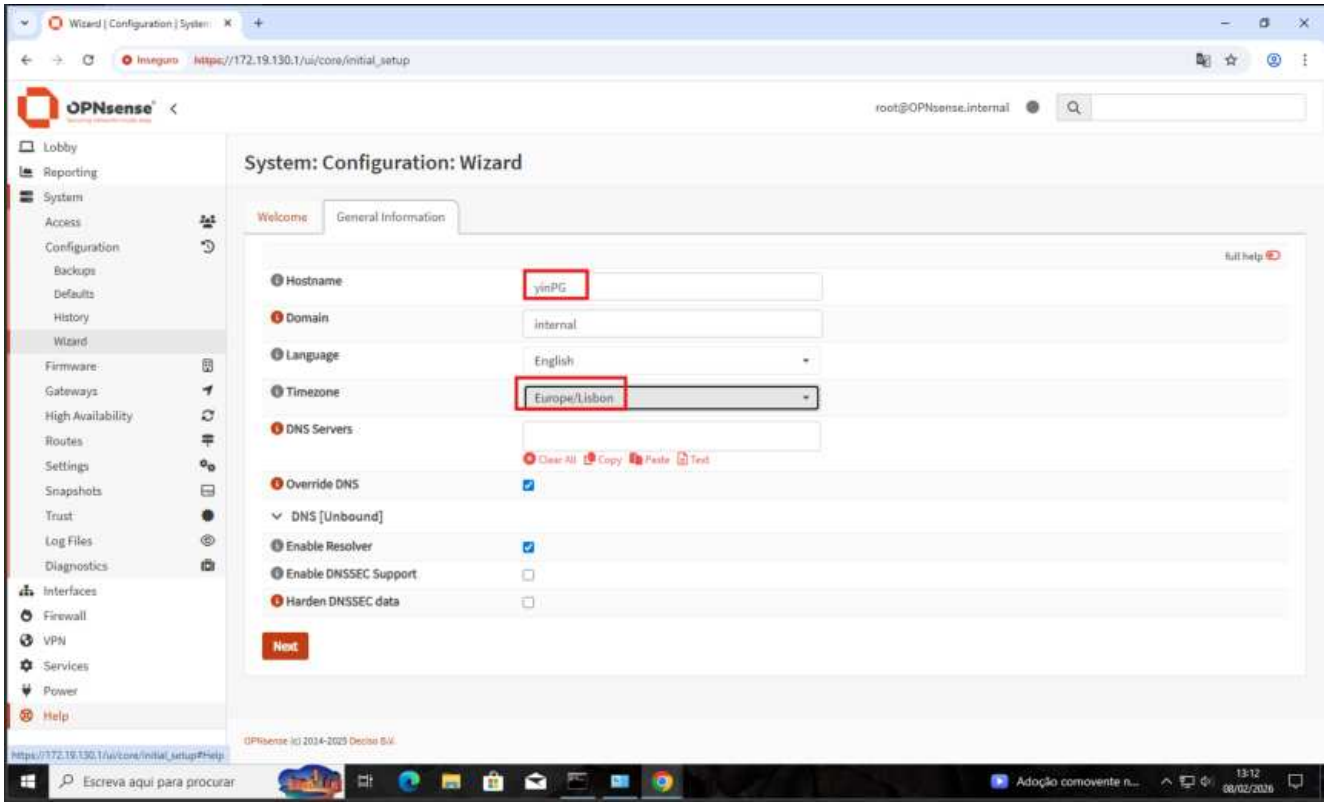
1.17 Acesso :Usename: **root**; Password : **opnsense**



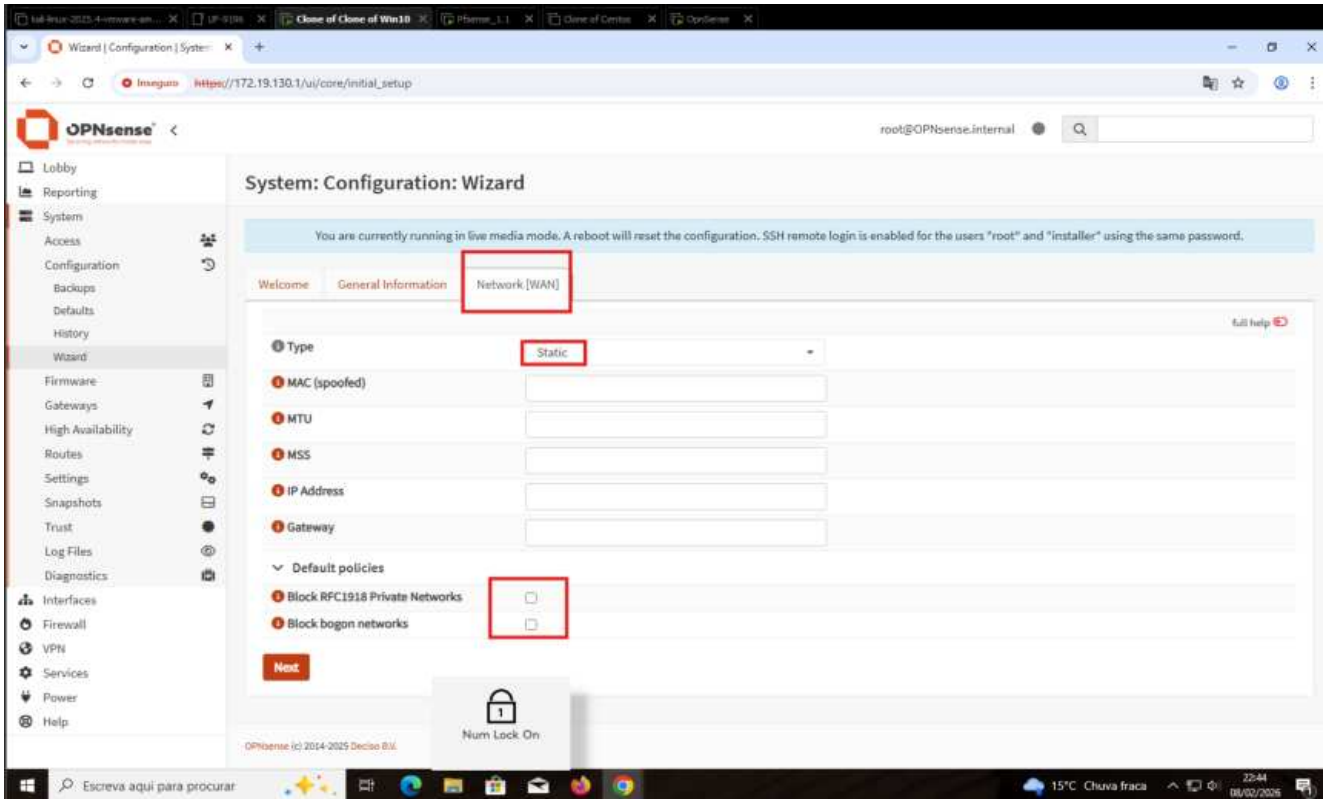
1.18 Configurações e updates :"wizard"



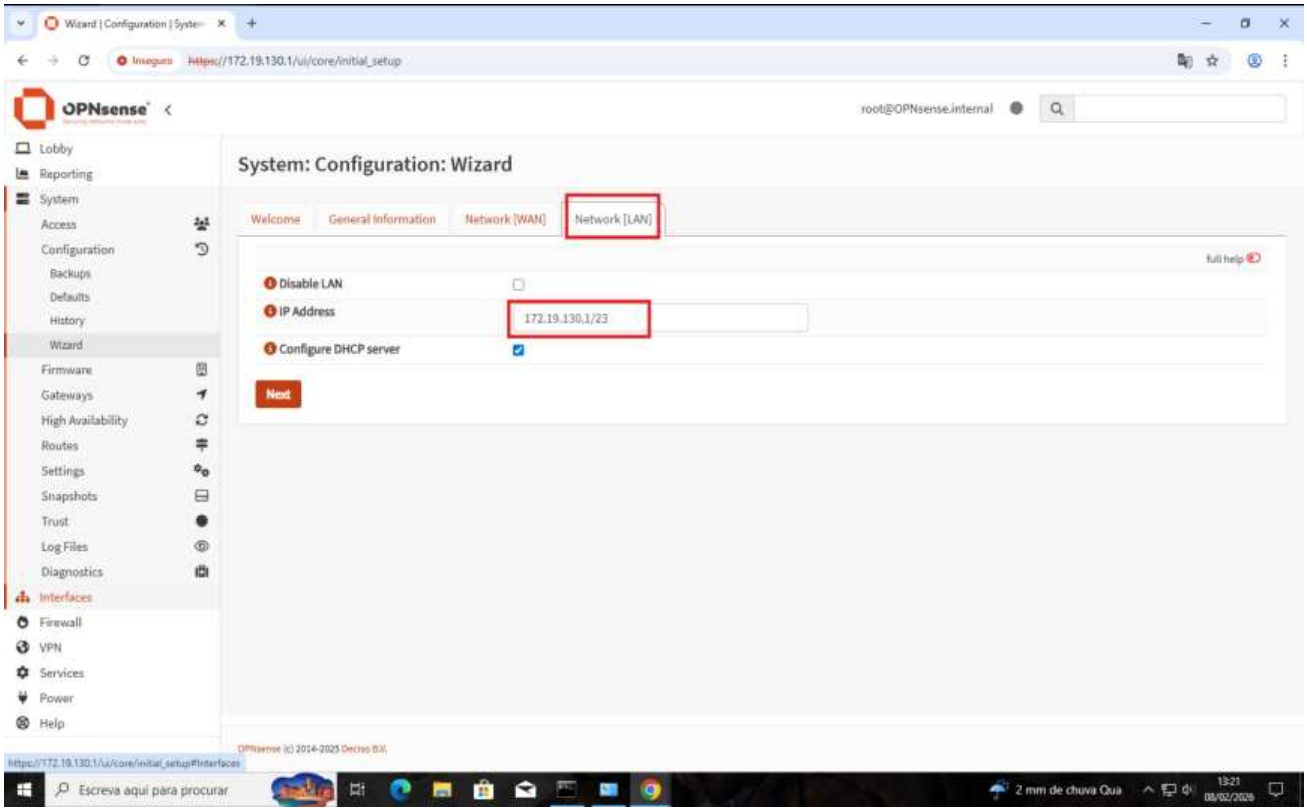
"General Information":



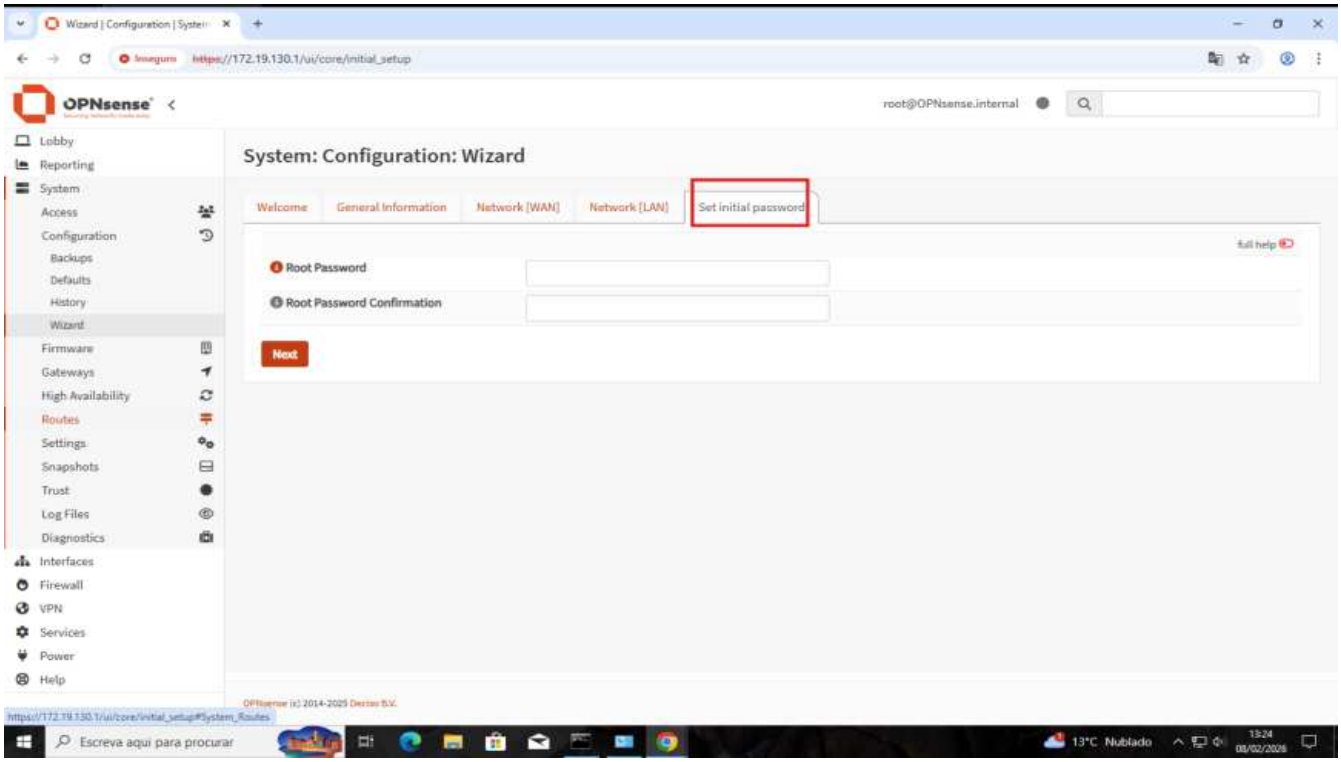
1.19 Network[wan] – nota a wan inicialmente encontra-se em "DHCP" – de forma a proceder a atualizações- "ver 25.7.11"e plug ins



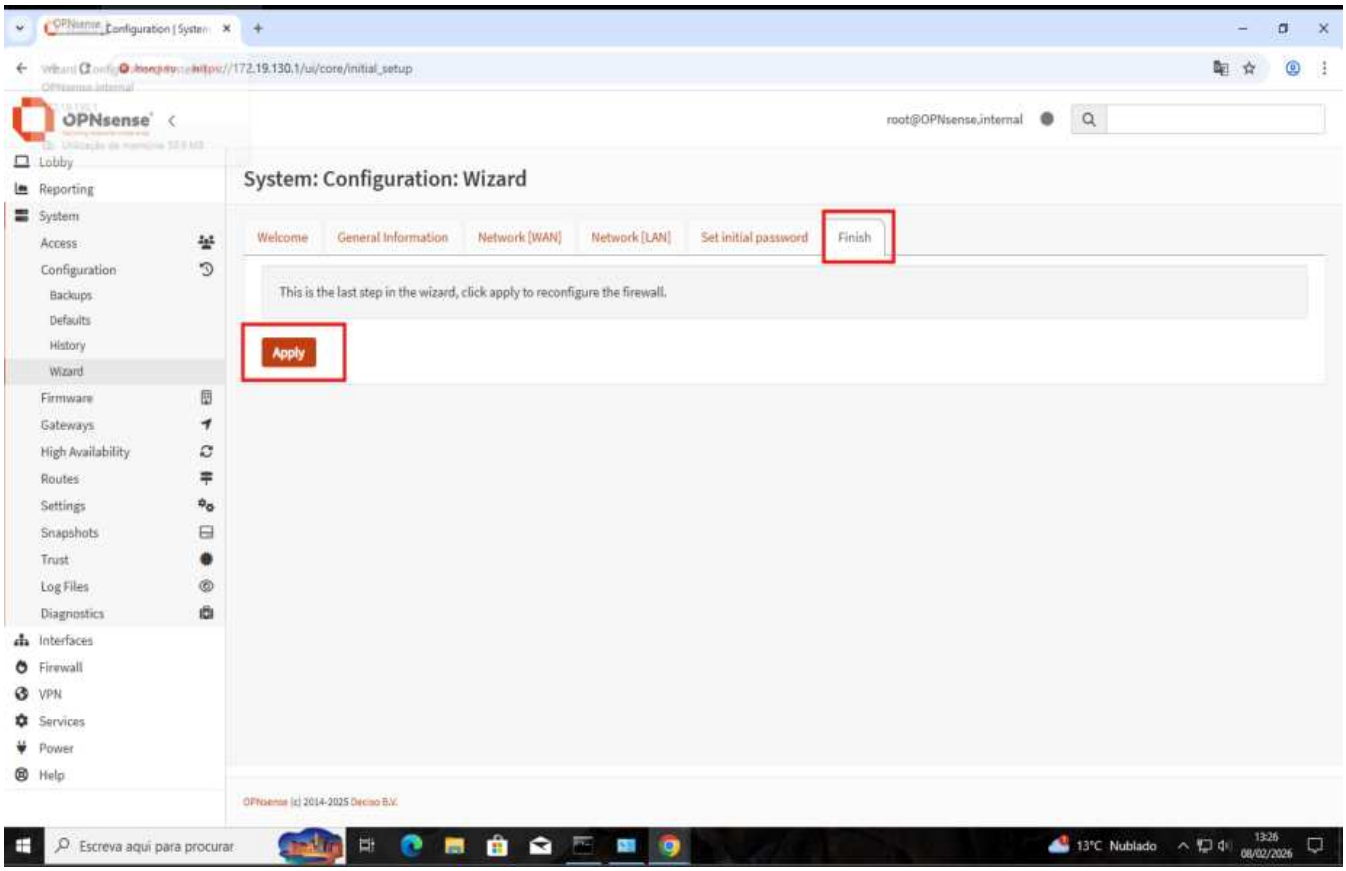
1.20 Network[LAN]



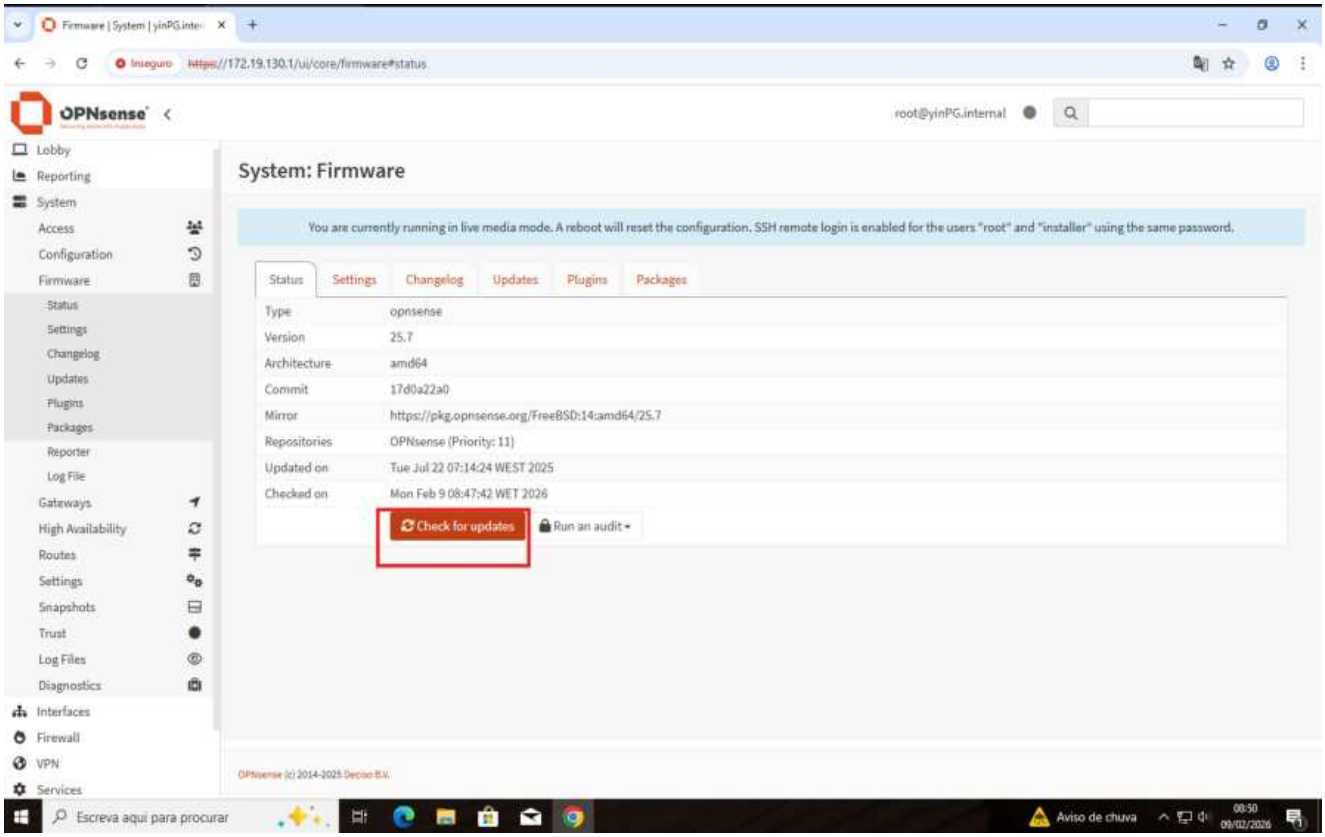
1.21 “Set Initial password”

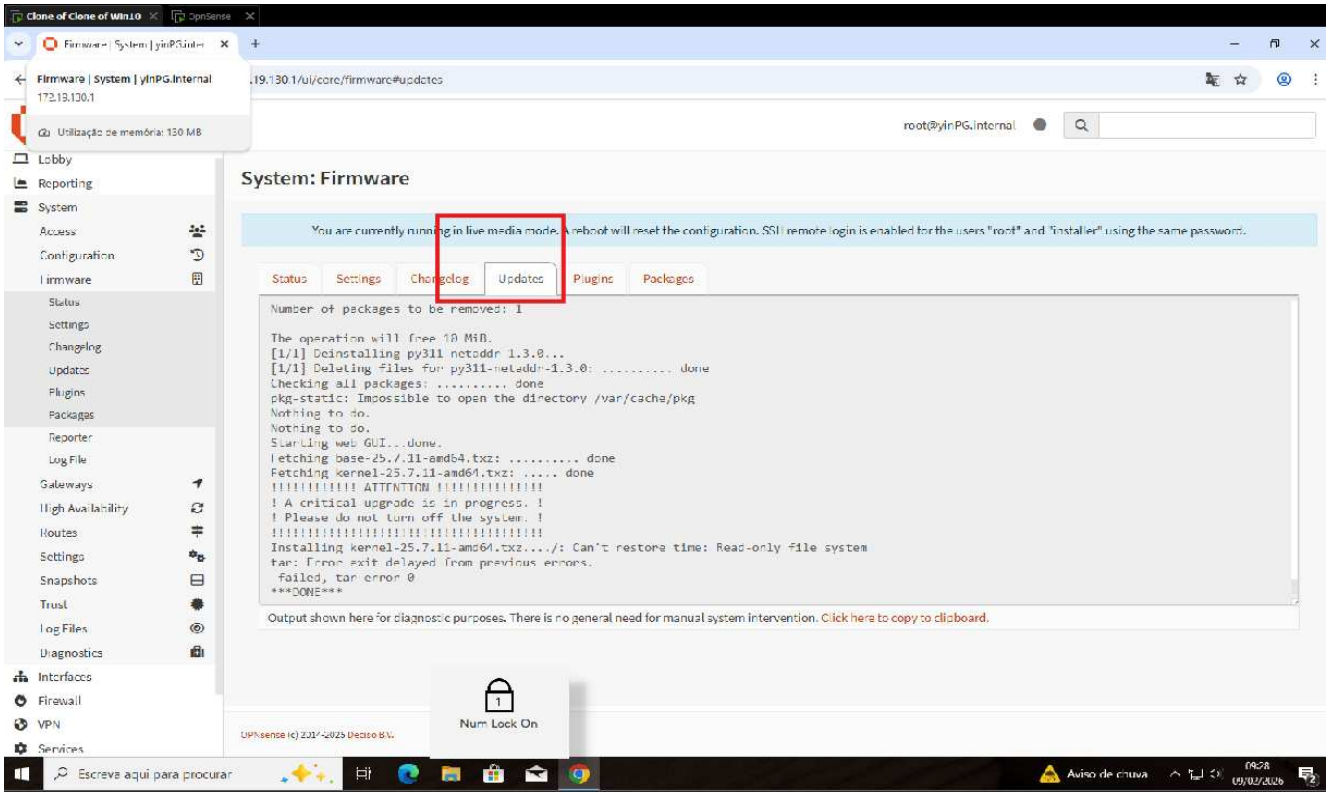


1.22 “Finish”

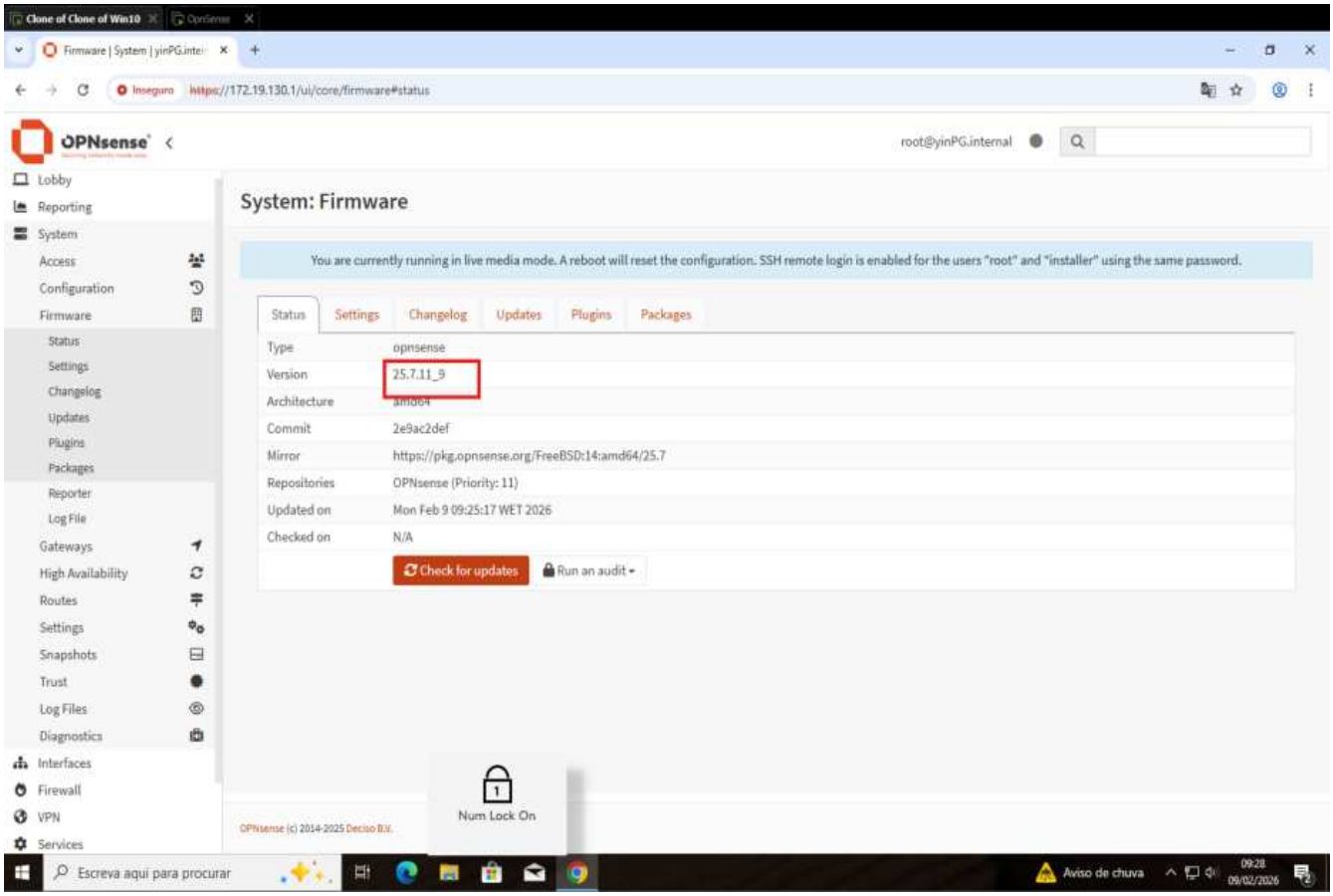


1.23 Proceder a atualizações :

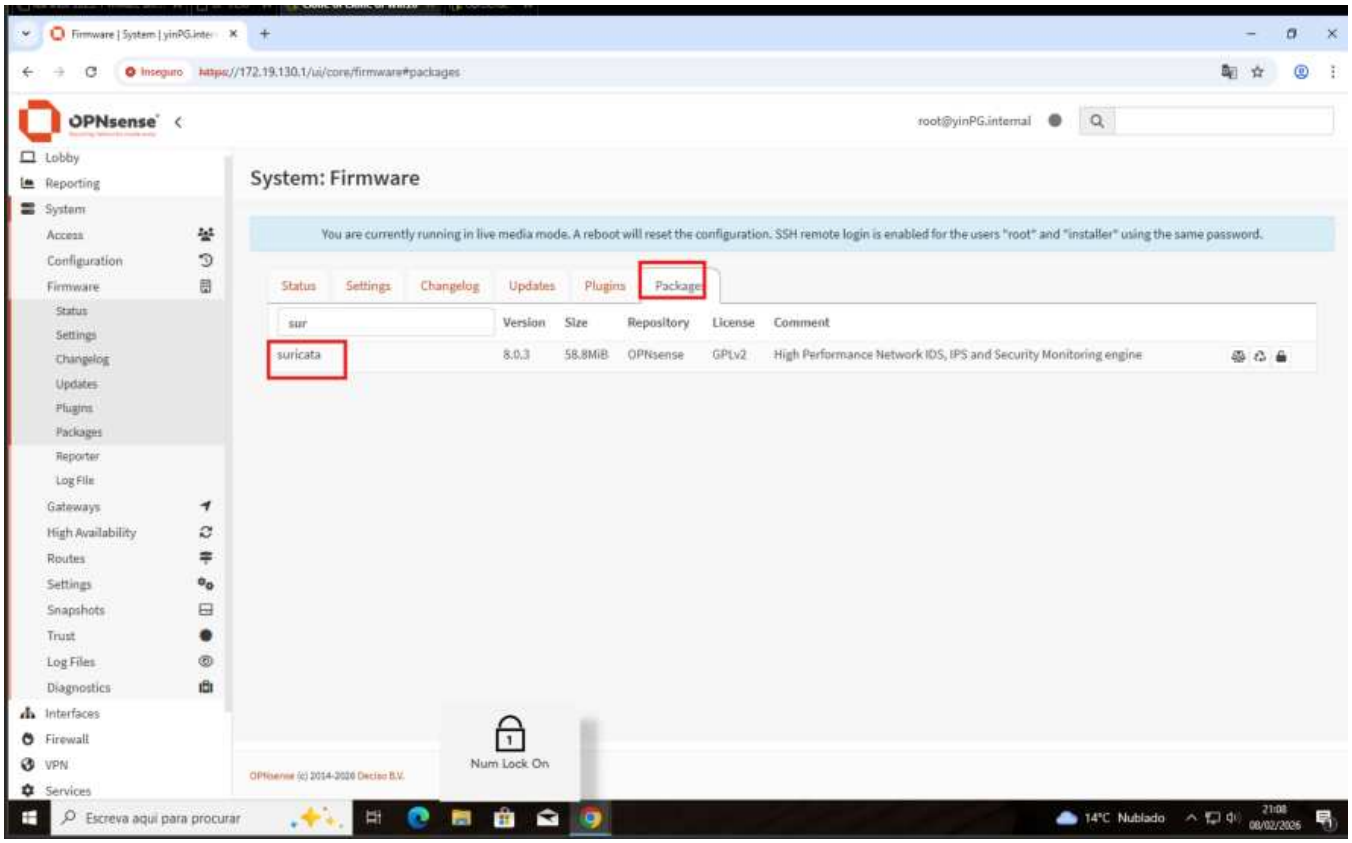




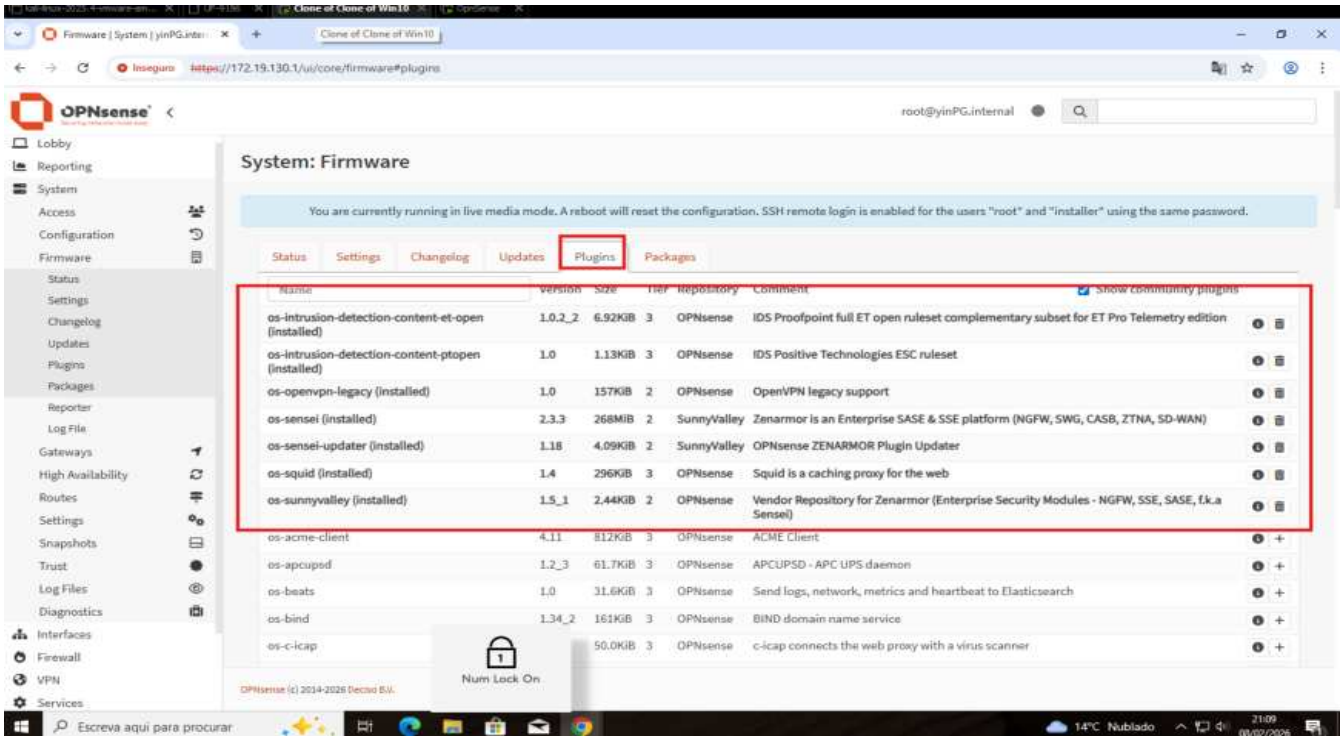
Nota : Versão -25.7.11 – versão essencial para instalação de plugins



1.24 “Package” e “plugins” – instalados



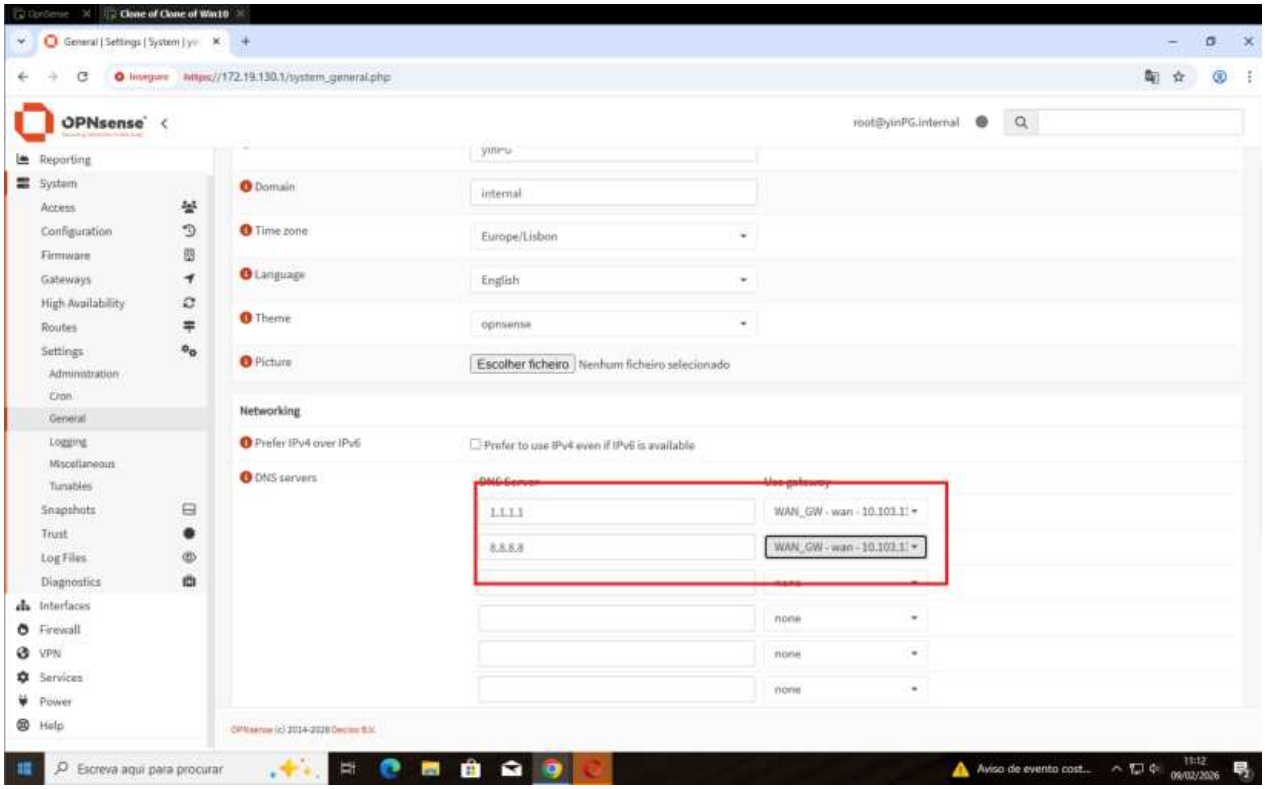
Plugins:



1.25 DNS do firewall (recomendado)

System →Settings → General
- DNS Servers: colocar temporariamente 1.1.1.1 e 8.8.8.8

- Gateway: WAN

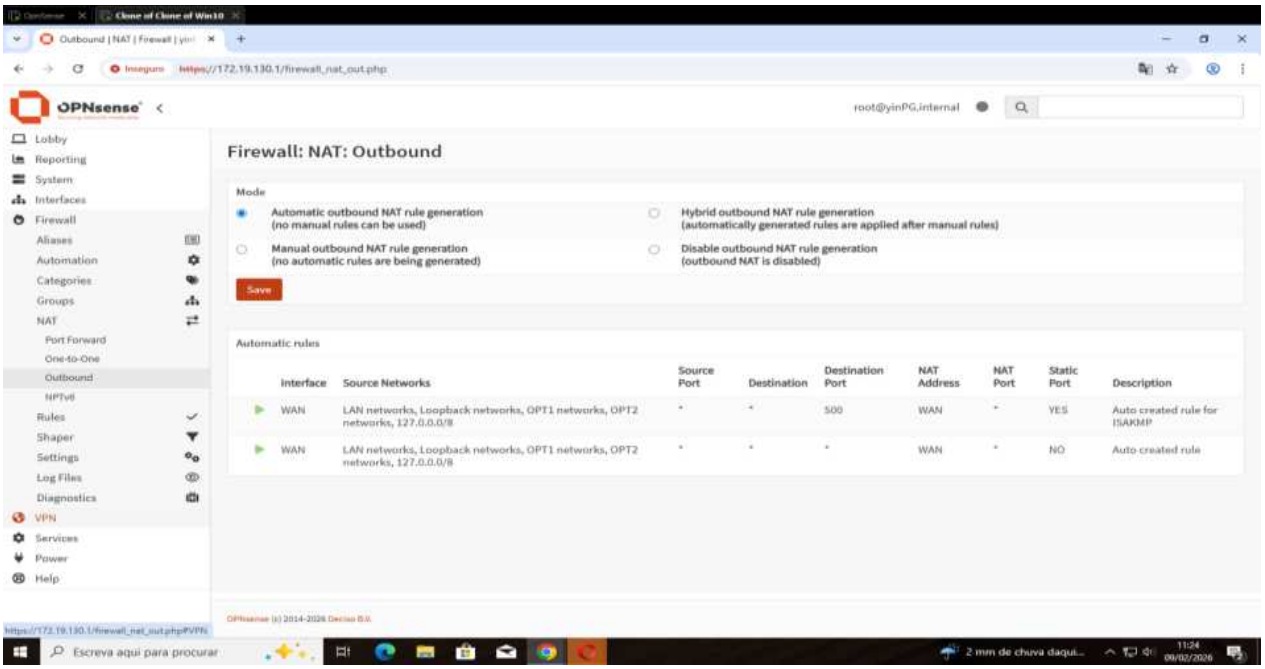


1.27 Outbound NAT

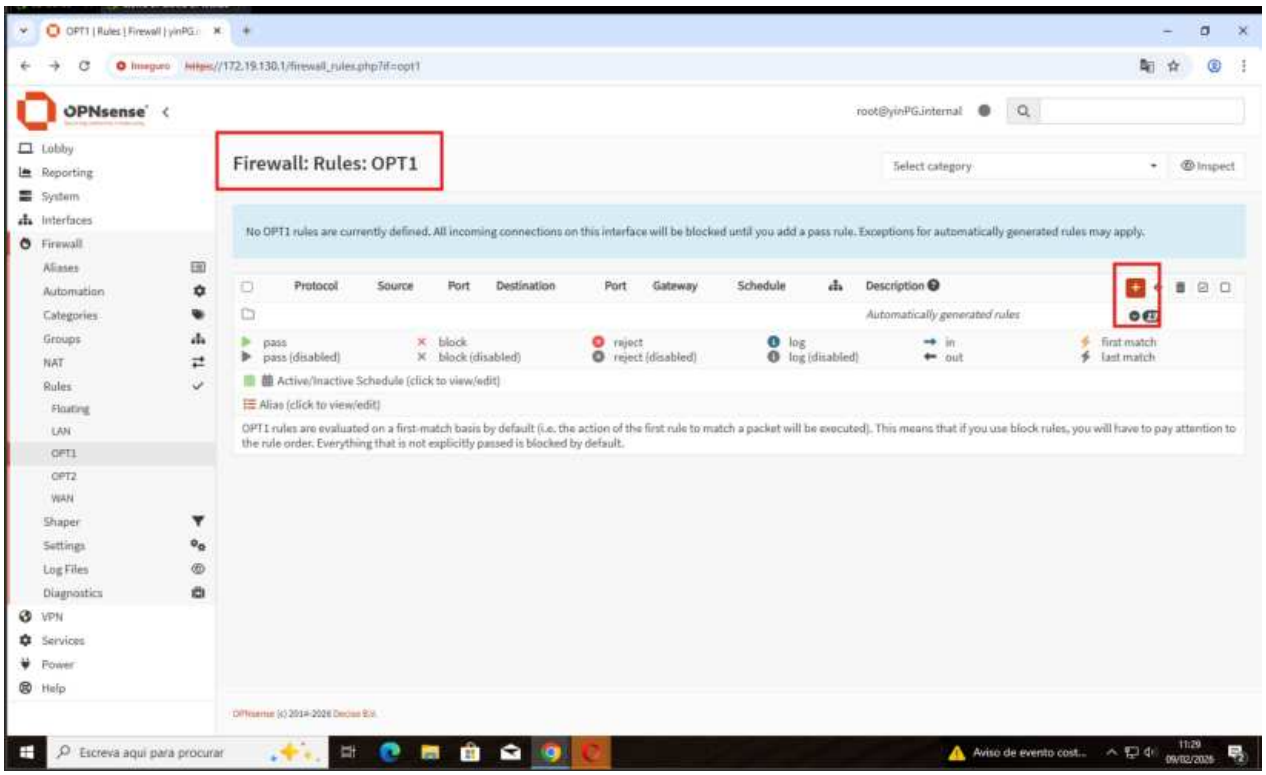
Firewall –> NAT → Outbound

- Modo: **Automatic** (normalmente chega)

Confirmar se as redes GREEN/ORANGE/BLUE (e depois VPNs) estão a sair pela WAN.



1.28 Criação de rules OPT1 e OPT2



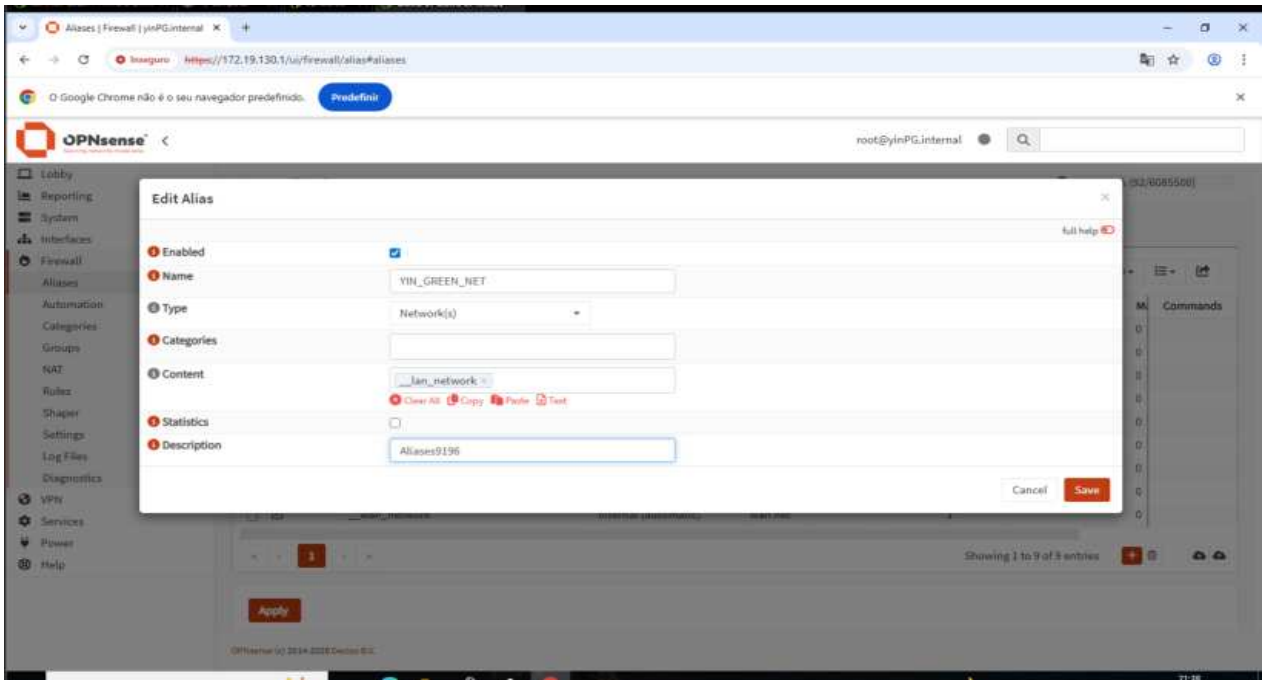
2 Configurações de aliases e regras (ORANGE/DMZ e BLUE)

Aliases

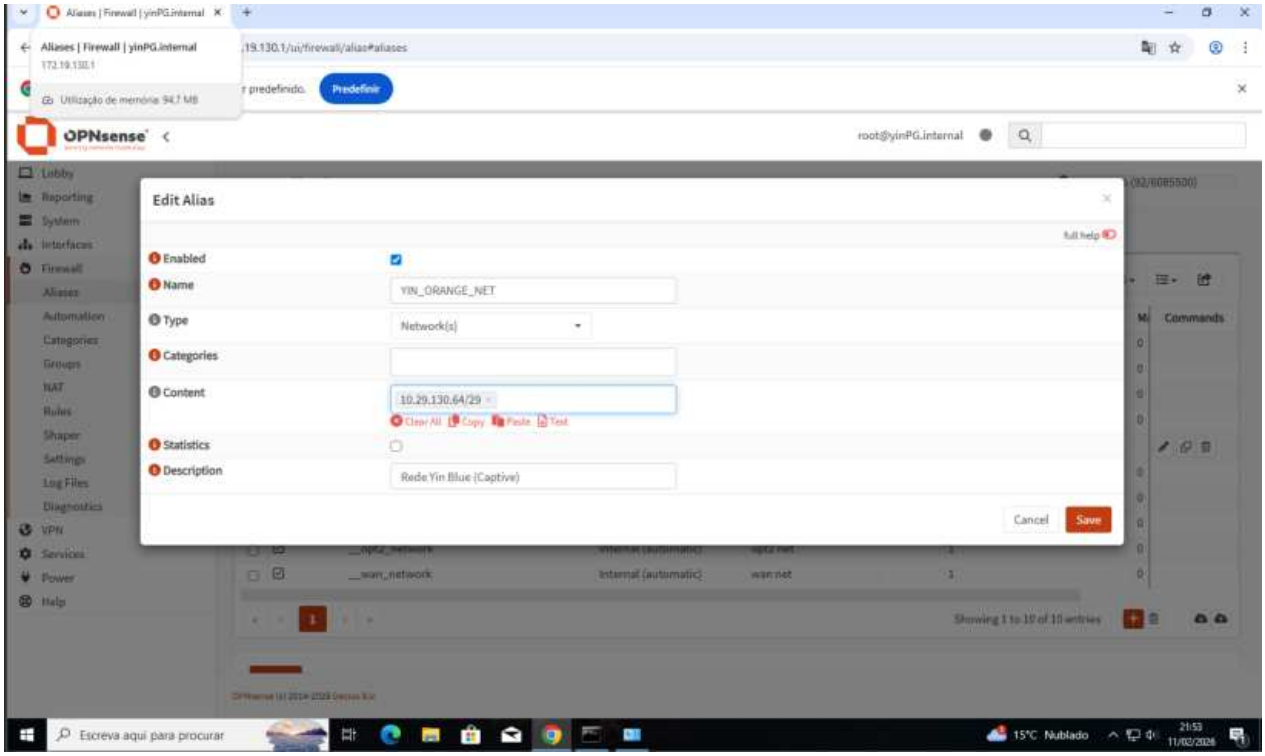
Ir a **Firewall** → **Aliases** e criar

A) Networks

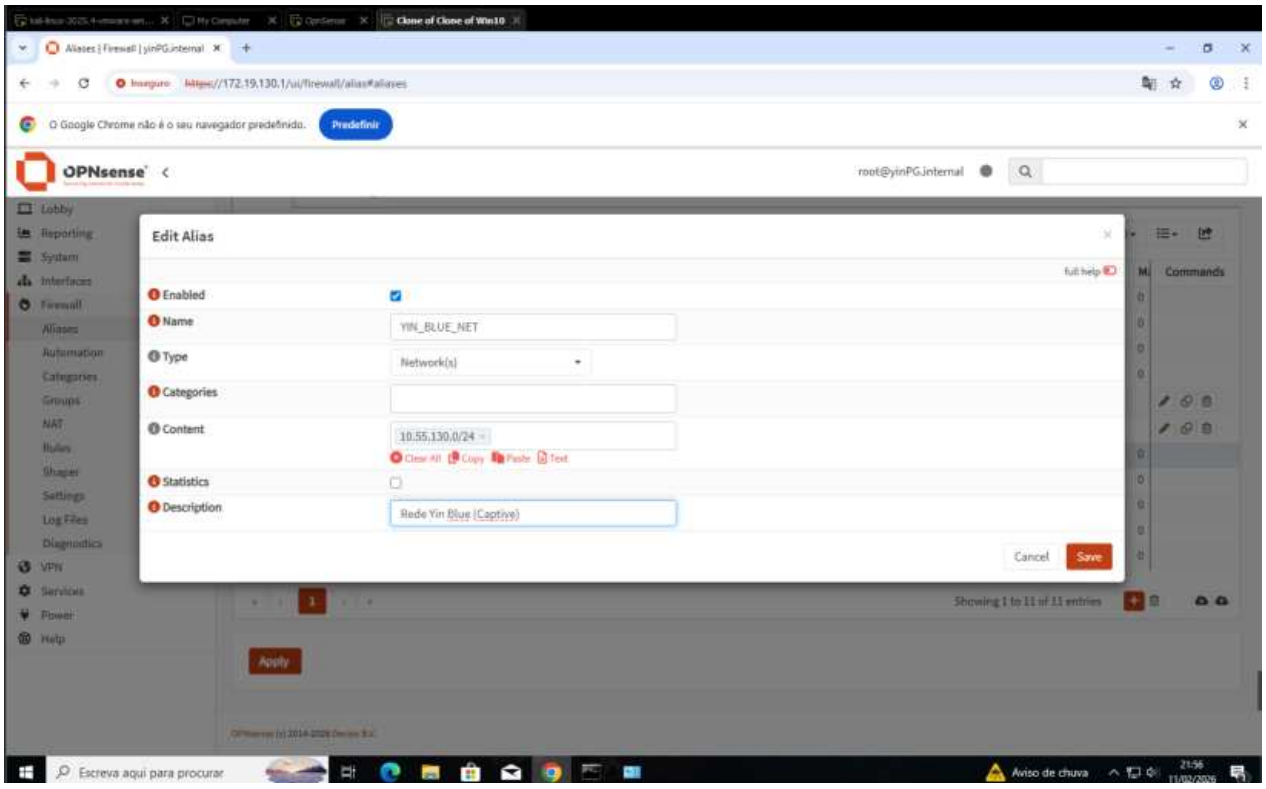
2.1 YIN_GREEN_NET = 172.19.130.0/23



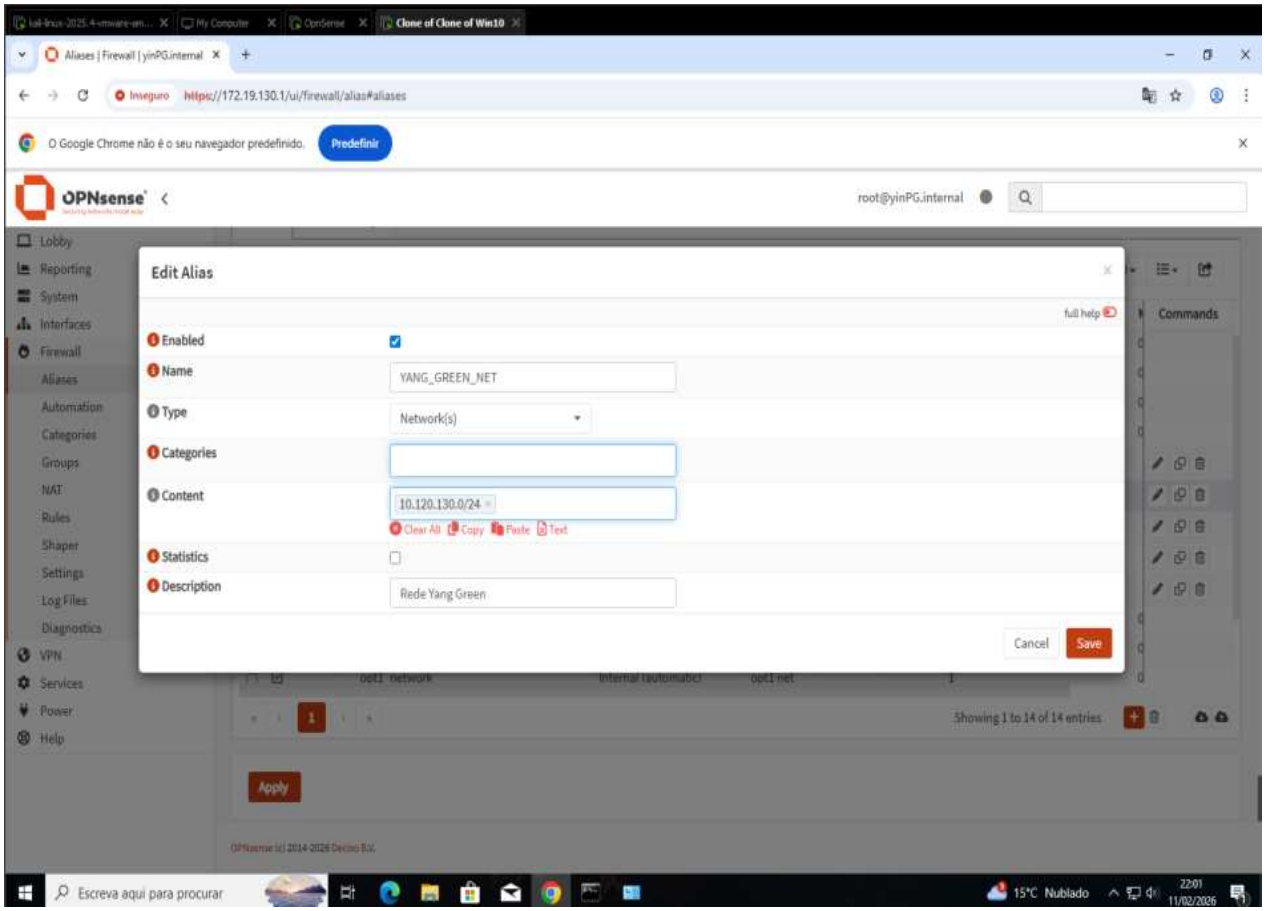
2.2 YIN_ORANGE_NET = 172.29.130.64/29



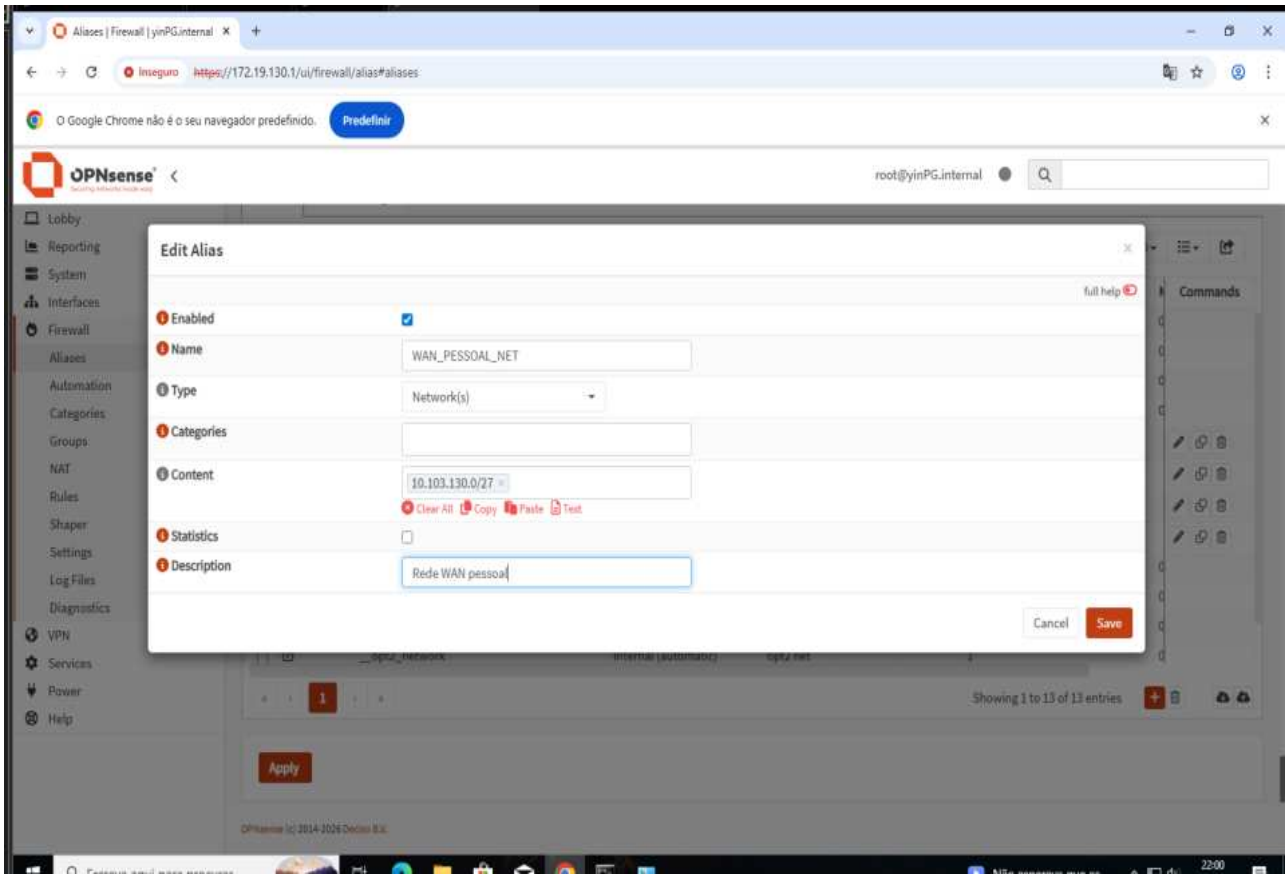
2.3 YIN_BLUE_NET = 10.55.130.0/24



2.4 YANG_GREEN_NET = 10.120.130.0/24



2.4 WAN_PESSOAL_NET = 10.103.130.0/27



Aliases "grupo" (para bloquear redes internas)

2.5 INTERNAL_NETS_NO_ORANGE

Usado para a regra: **ORANGE bloquear acesso a redes internas**

- Name: INTERNAL_NETS_NO_ORANGE

-Type: Network(s)

- Content: (1 por linha)

-172.19.130.0/23

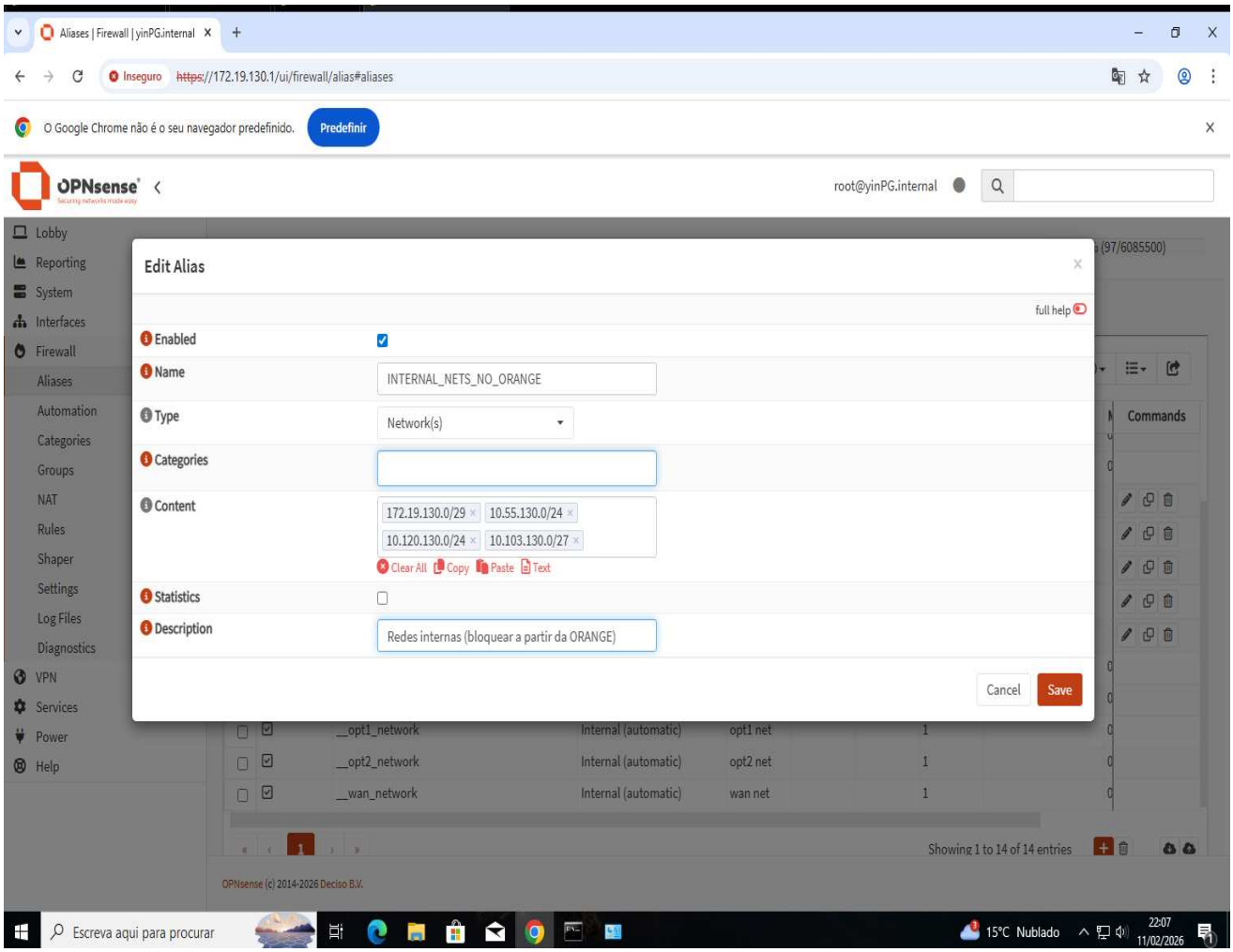
- 10.55.130.0/24

- 10.120.130.0/24

- 10.103.130.0/27

Description: Redes internas (bloquear a partir da ORANGE)

Continuação...



2.6 INTERNAL_NETS_NO_BLUE

Usado para a regra: **BLUE bloquear acesso a redes internas**

-m Name: INTERNAL_NETS_NO_BLUE

- Type: Network(s)

- Content: (1 por linha)

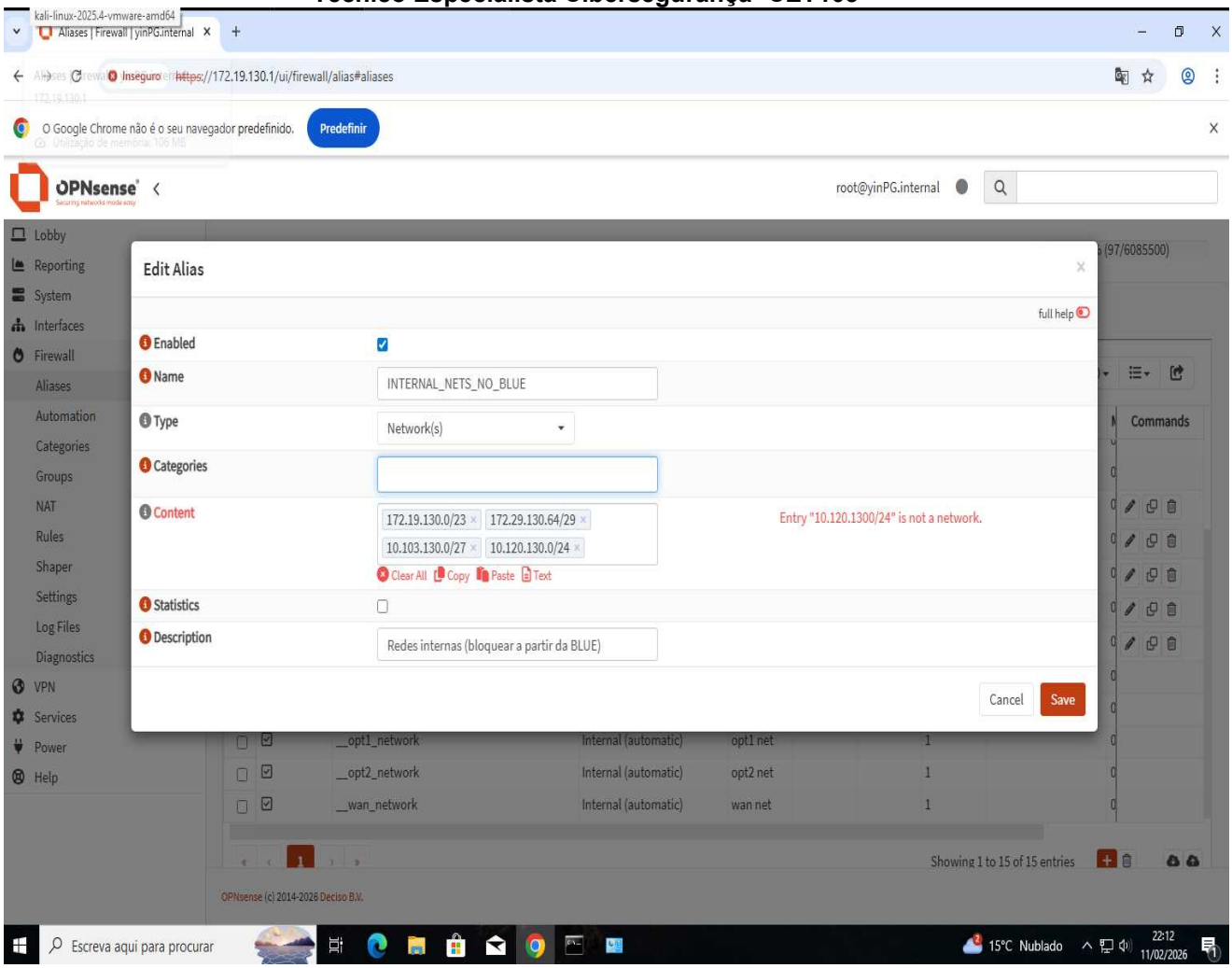
- 172.19.130.0/23

- 172.29.130.64/29

- 10.120.130.0/24

- 10.103.130.0/27

Description: Redes internas (bloquear a partir da BLUE)



Alias de host (Windows Server DNS)

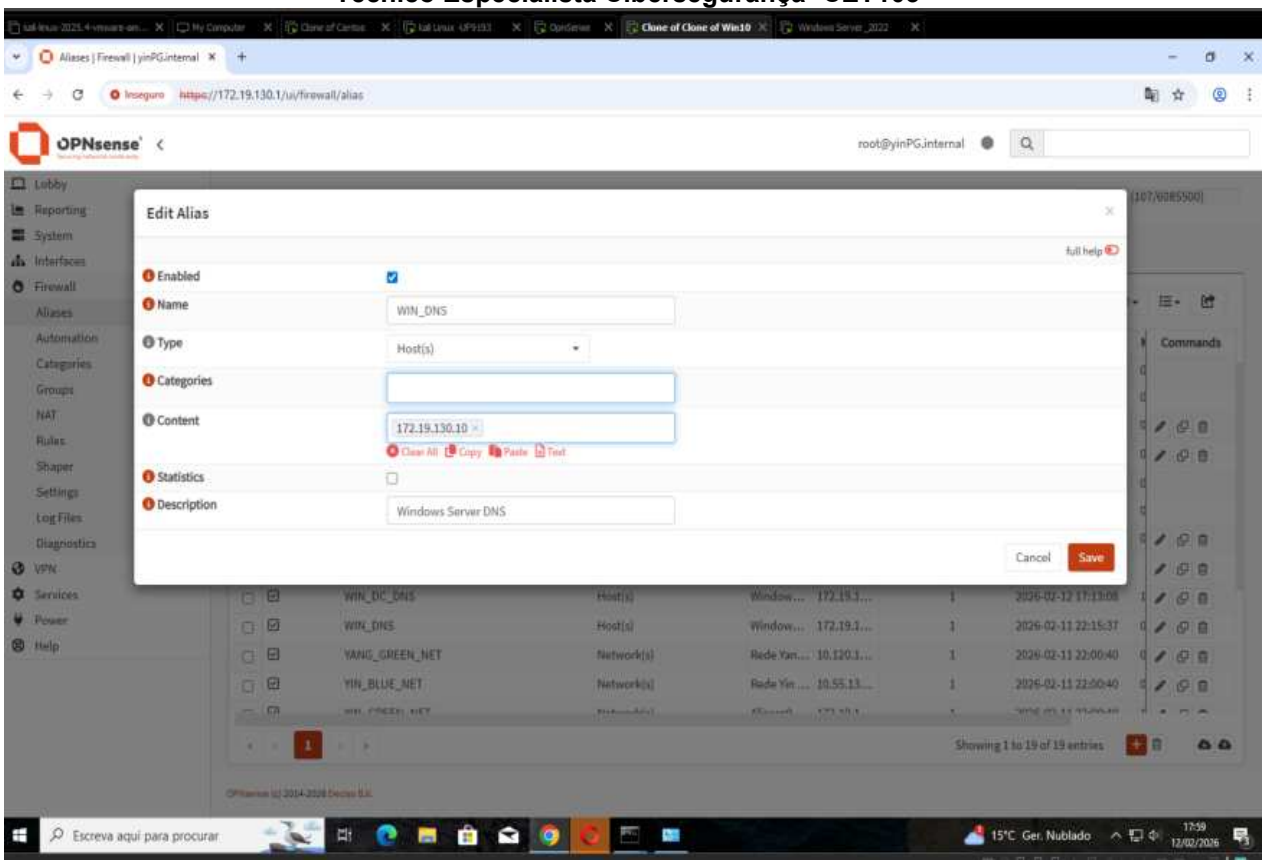
2.7 WIN_DNS

- Name: WIN_DNS

- Type: Host(s)

- Content:172.19.130.10

- Description: Windows Server DNS



2.8 Alias de portas (Web)

WEB_PORTS

- Name: WEB_PORTS

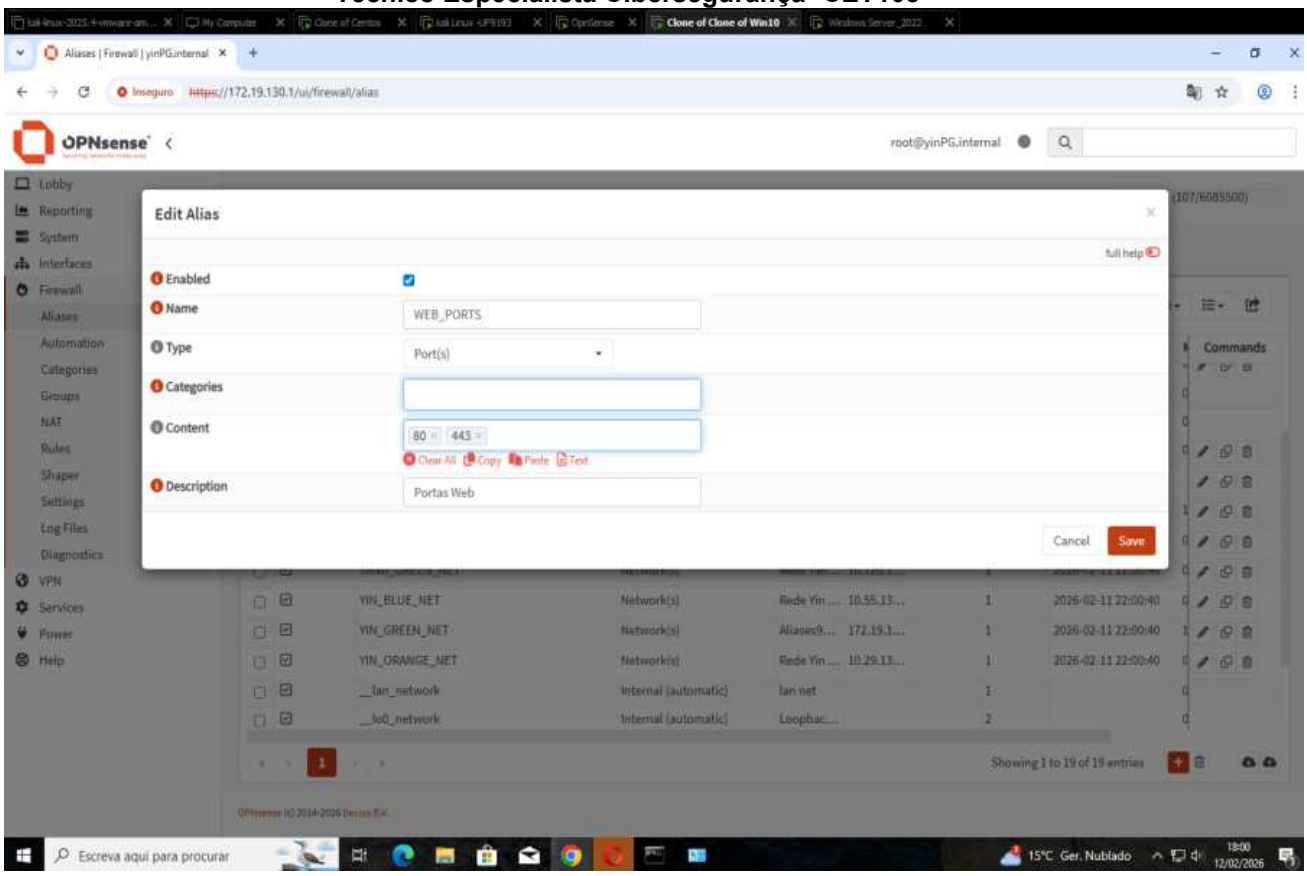
- Type: Port(s)

- **Content:** ()

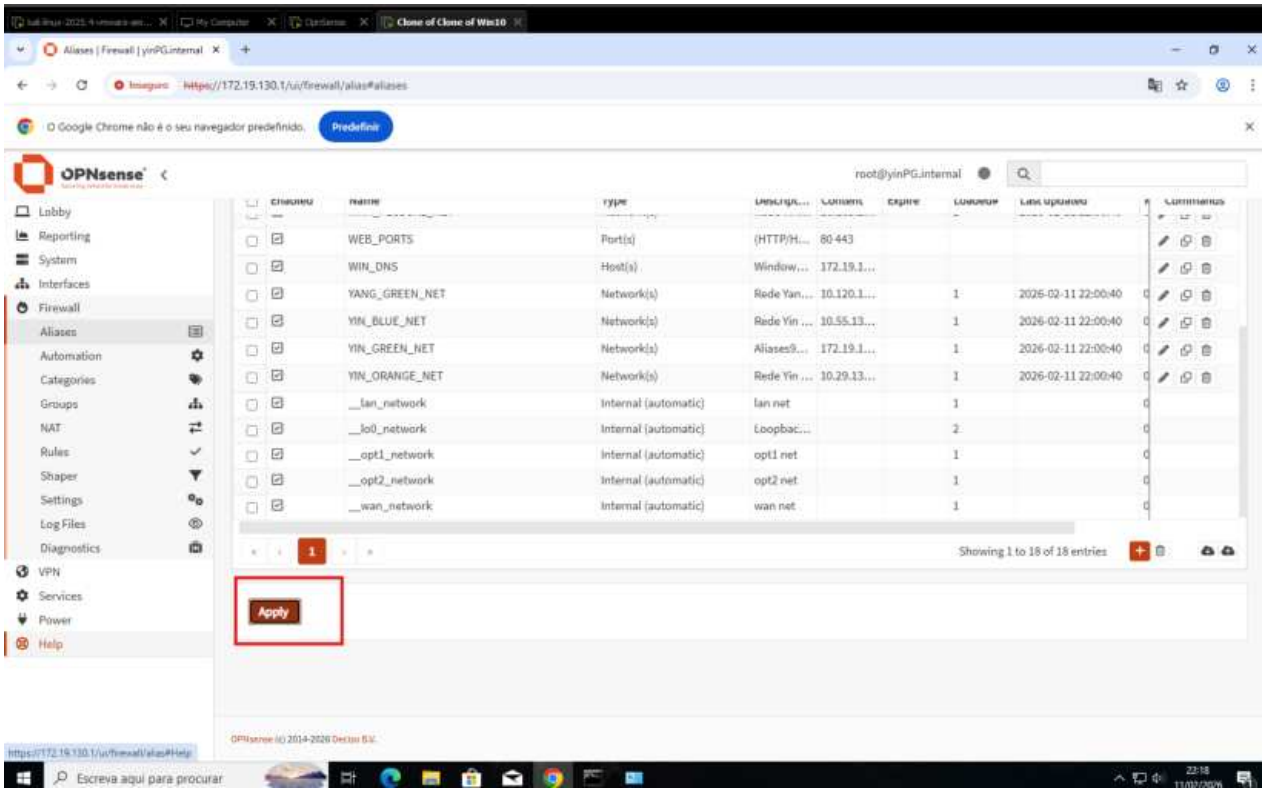
- 80

- 443

- **Description:** Portas Web



2.9 Carregar Apply (para aplicar as alterações)



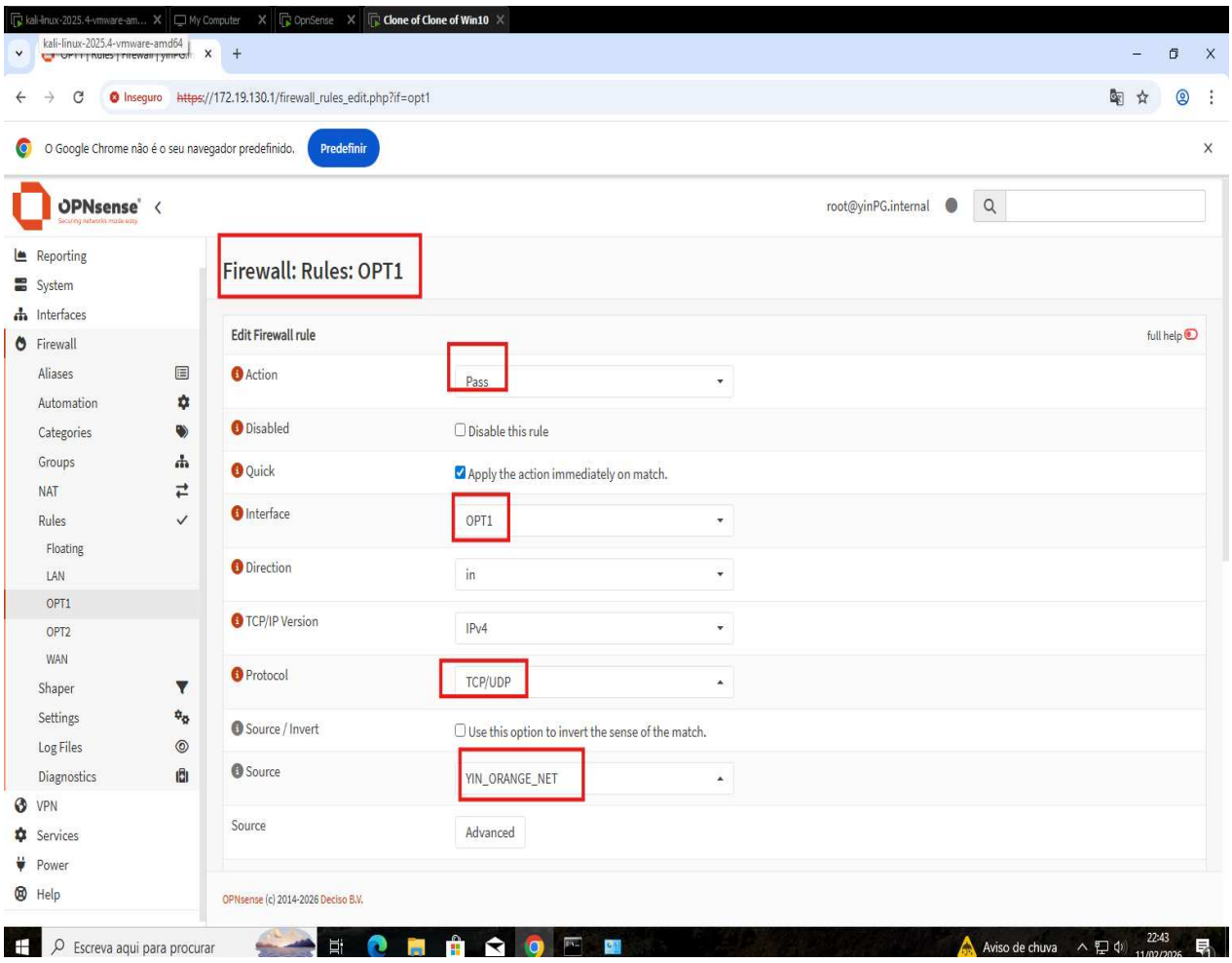
2.10 ORANGE (OPT1/DMZ) — criar as regras

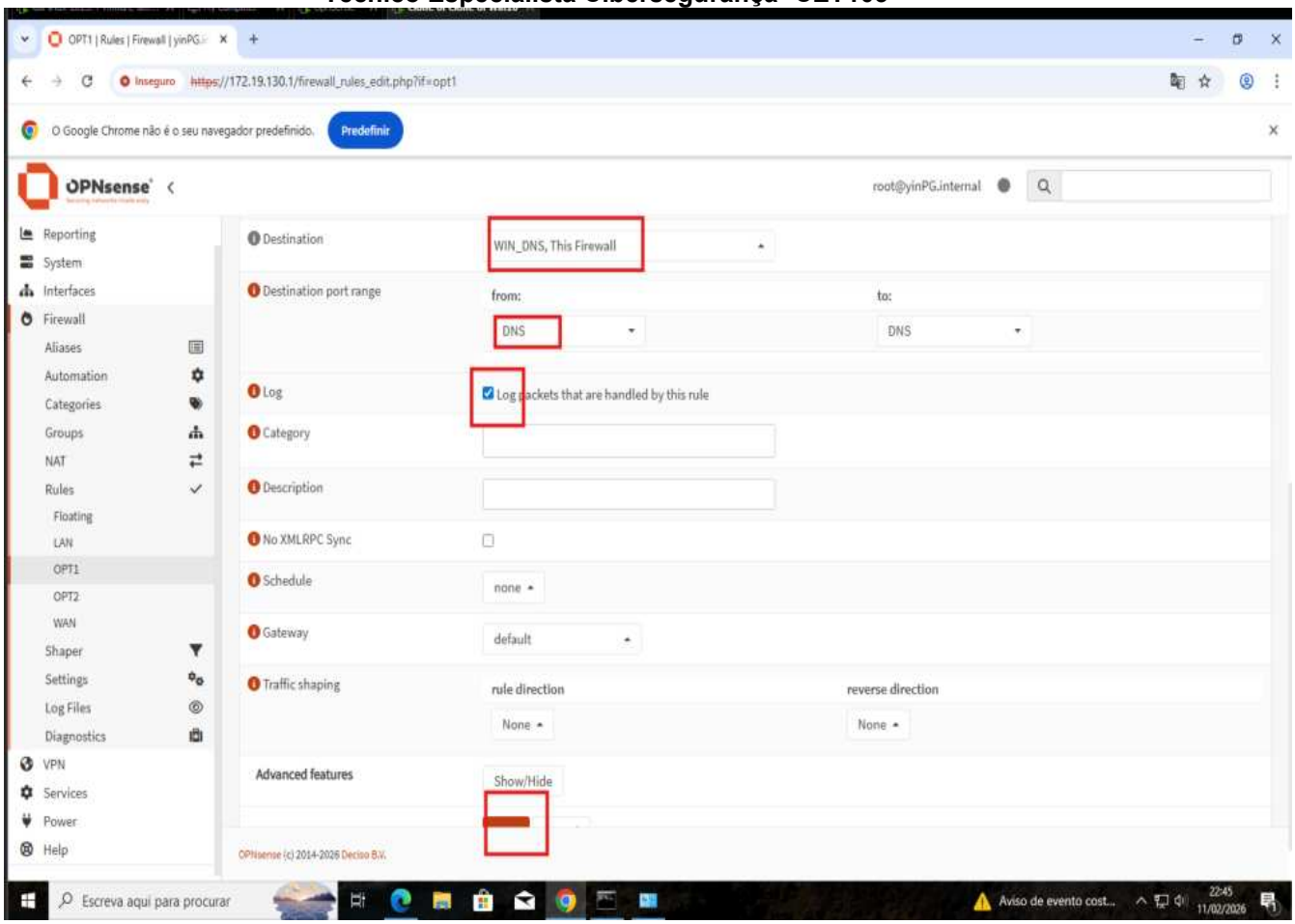
Ir a: **Firewall → Rules → ORANGE (OPT1)**

Criar estas regras **por esta ordem** (ordem é crítica!):

Regra 1 — DNS da DMZ para o Windows Server (DC/DNS)

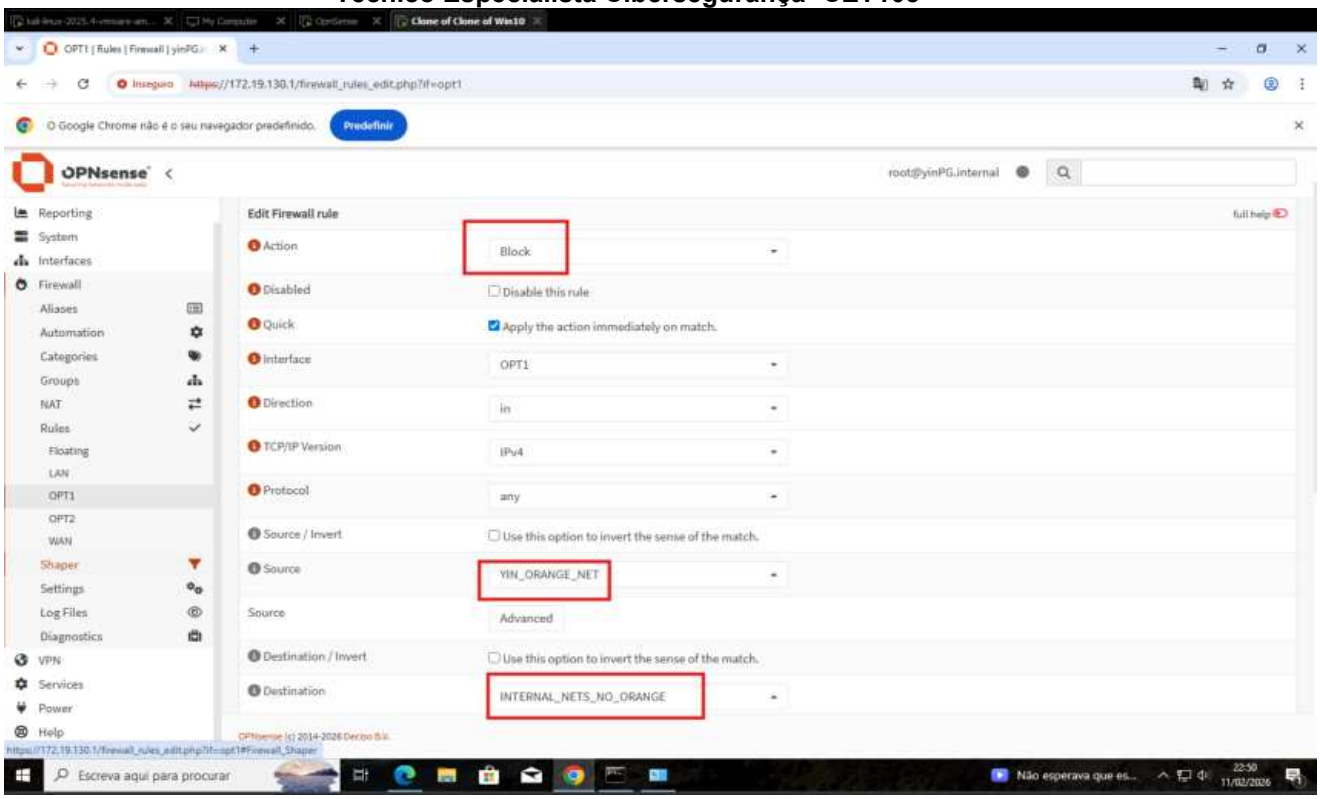
- **Action:** Pass
- **Protocol:** TCP/UDP
- **Source:** YIN_ORANGE_NET
- **Destination:** alias WIN_DNS (172.19.130.10)
- **Destination port:** 53 (DNS)
- **Log:** ON
- **Save**





2.11 Regra 2 — Bloquear DMZ para redes internas

- **Action:** Block (ou Reject)
- **Protocol:** any
- **Source:** ORANGE net
- Destination: INTERNAL_NETS_NO_ORANGE
- **Log:** ON
- **Save**



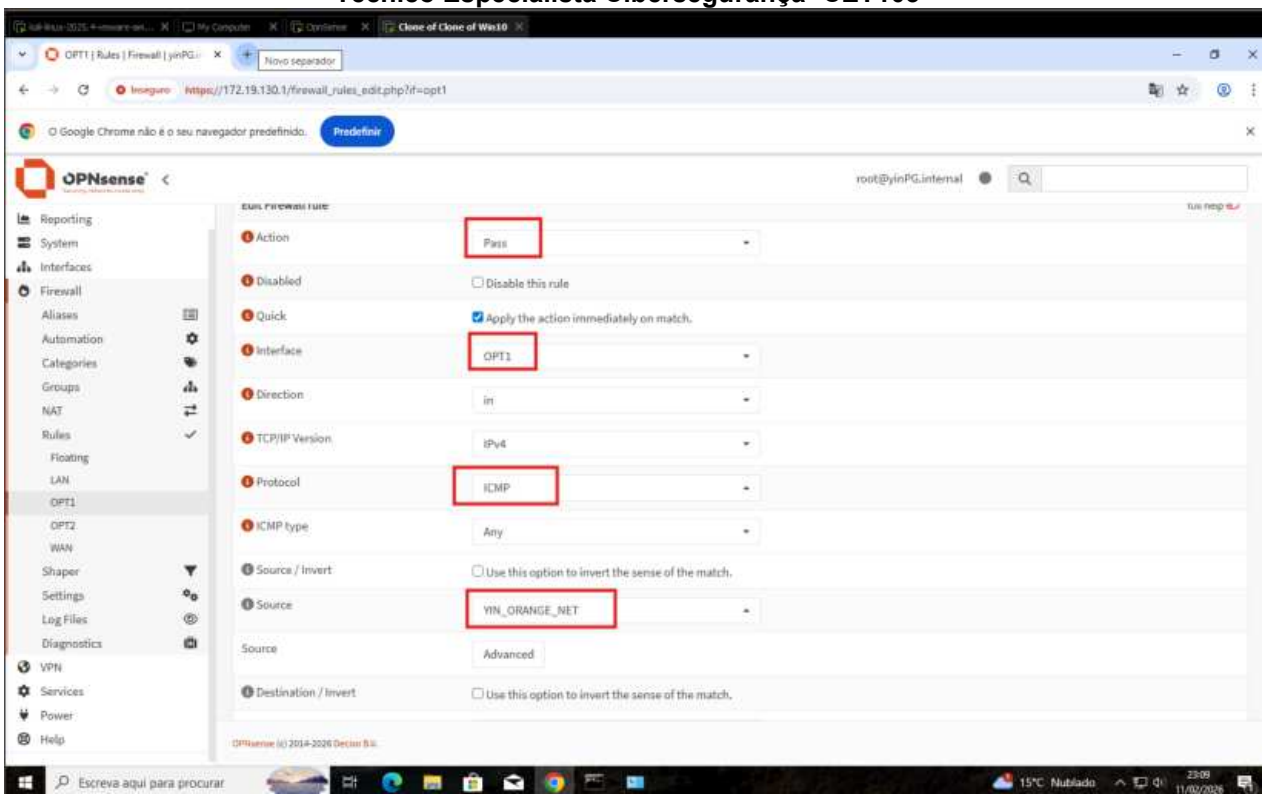
2.13 — (Opcional) Permitir internet para updates (80/443)

Action: Pass

- **Protocol:** TCP
- **Source:** ORANGE net
- **Destination:** any
- **Destination port:** alias WEB_PORTS
- **Save**

2.14 — (Opcional) ICMP para testes

- **Action:** Pass
- **Protocol:** ICMP
- **Source:** ORANGE net
- **Destination:** any
- **Save**



3) BLUE (OPT2/Captive) — DHCP + Shaper + regras

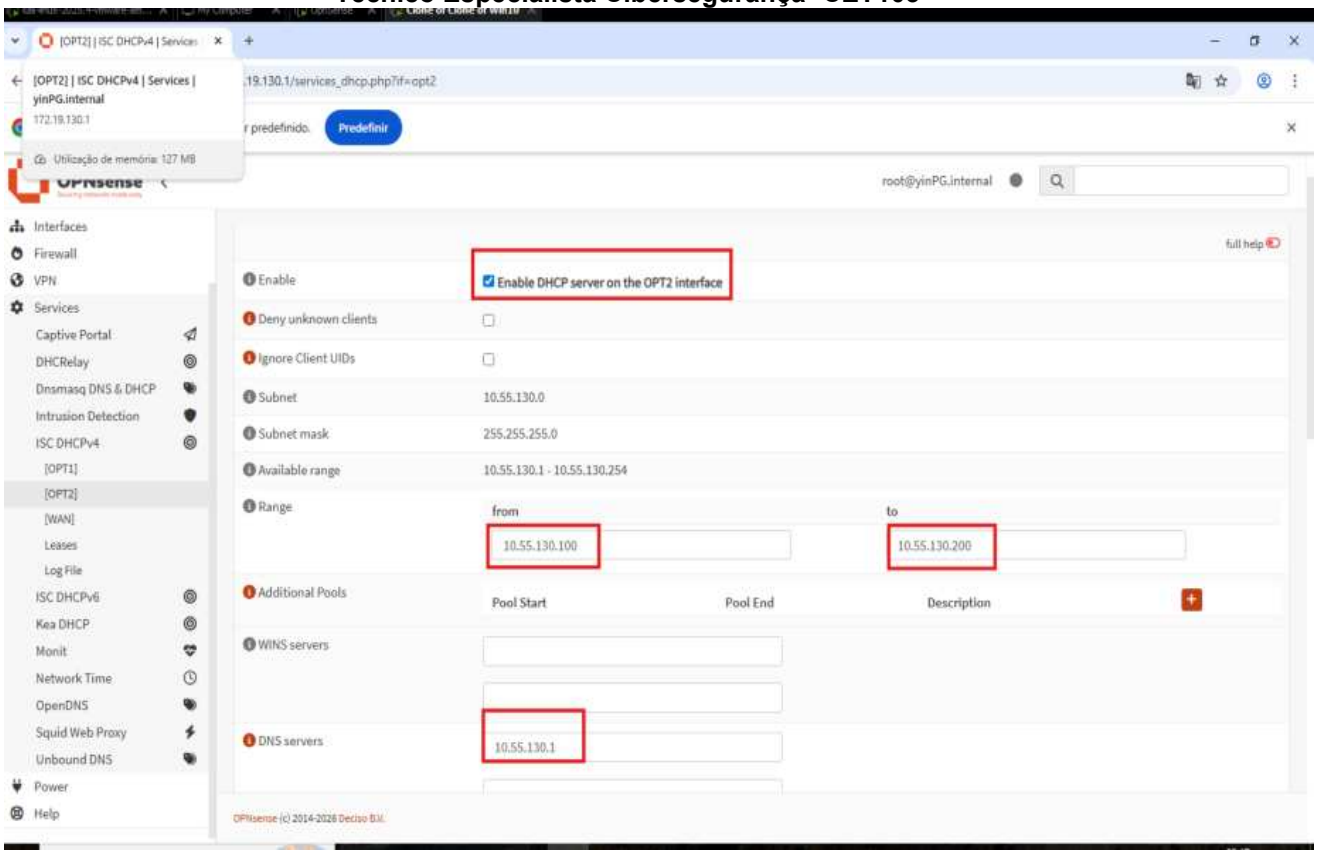
A BLUE tem requisitos explícitos no enunciado:

- DHCP no firewall
- DNS **só pode ser o DNS do próprio firewall**
- limite por utilizador: **10Mb download / 5Mb upload**

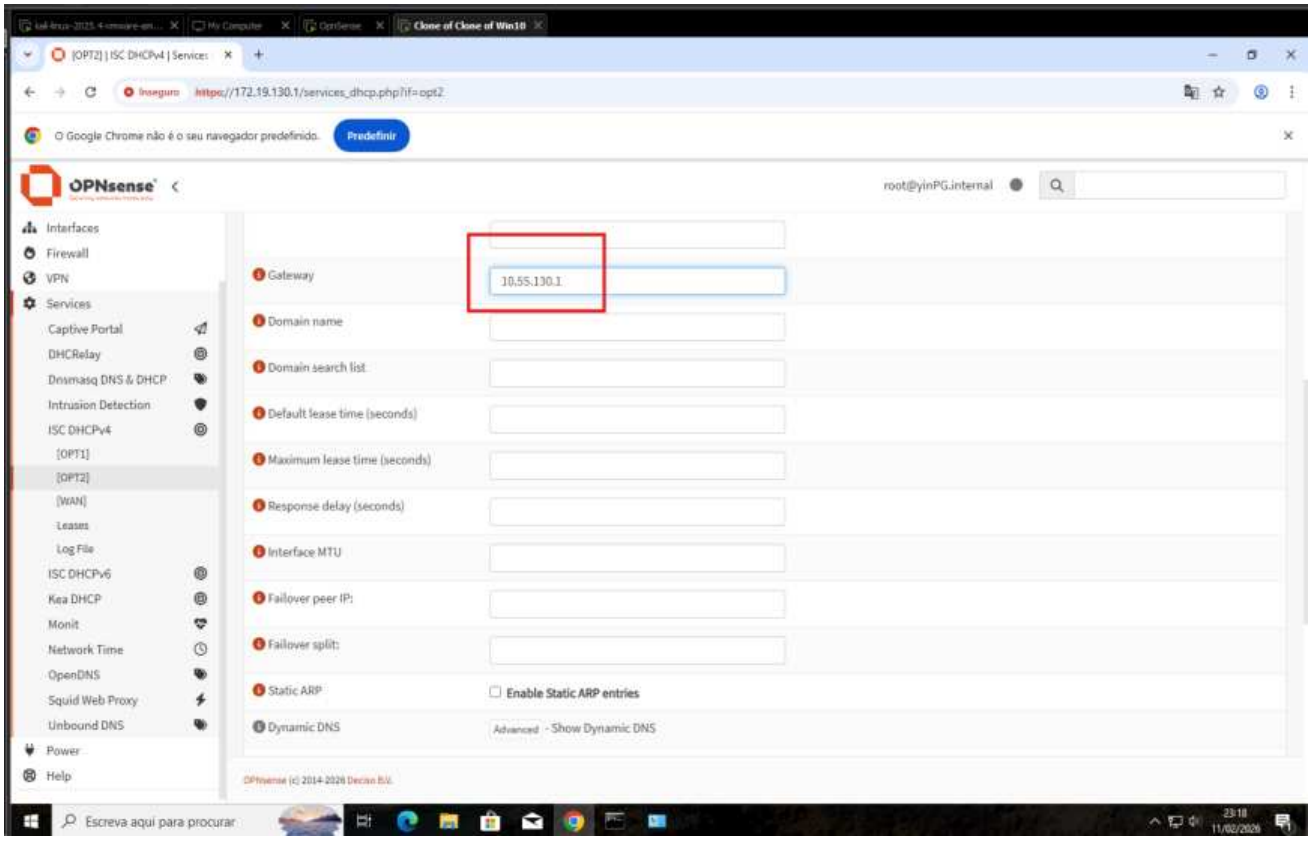
3.1 Ativar DHCP na BLUE

Ir a: **Services → DHCPv4 → [BLUE]**

- Enable DHCP server
- Range: 10.55.130.100 – 10.55.130.200 (exemplo)
- DNS servers: 10.55.130.1 (o IP da BLUE do firewall)
- Gateway: 10.55.130.1
- **Save / Apply**



Continuação ..

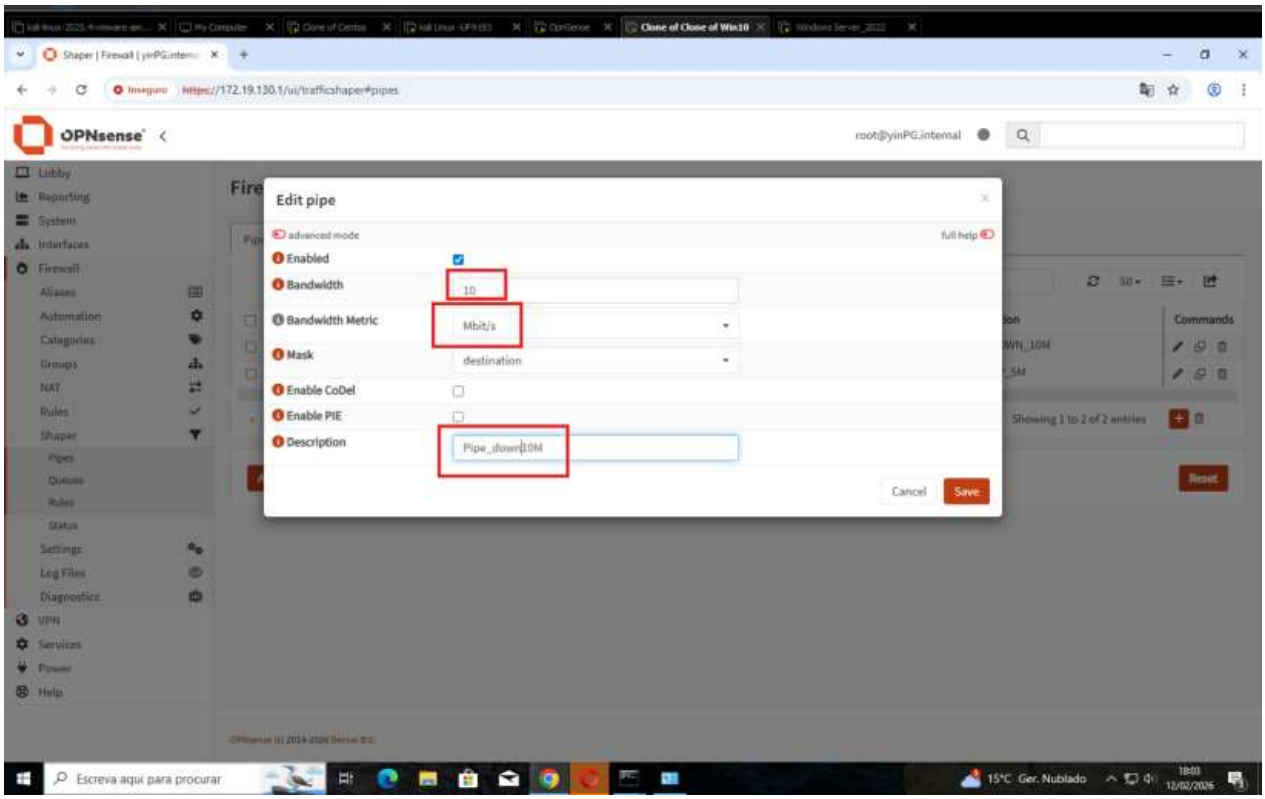


3.2 Criar Shaper Pipes (para o limite 10/5)

Vai a: **Firewall → Shaper → Pipes → Add**

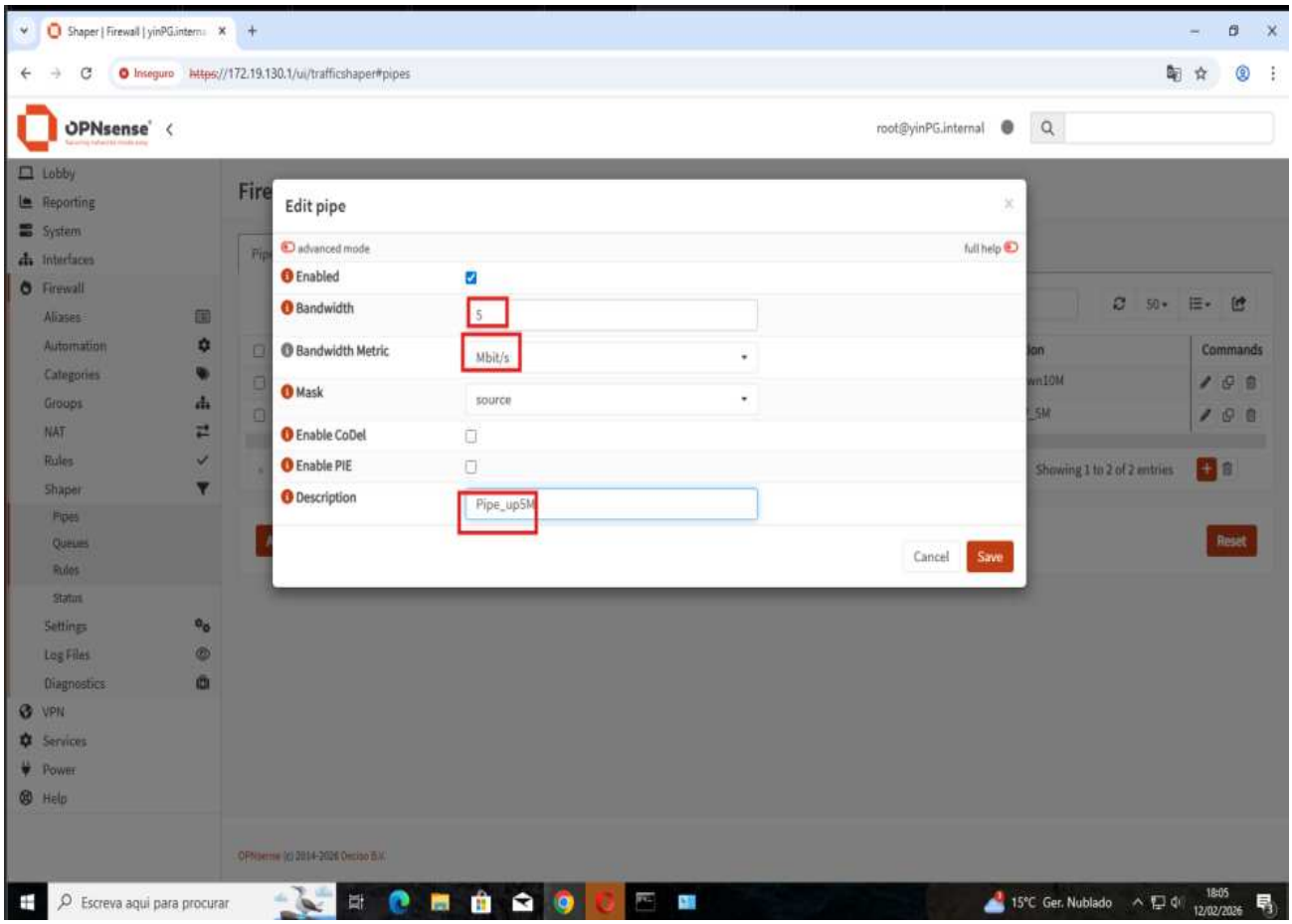
Pipe DOWN (10Mb)

- Bandwidth: 10
- Mask: **destination addresses**
- Name/Description: Pipe_Down10M
- Save



Pipe UP (5Mb)

- Bandwidth: 5 Mbit/s
-
- +Mask: **source addresses**
- Name/Description: Pipe_up5M
- Save

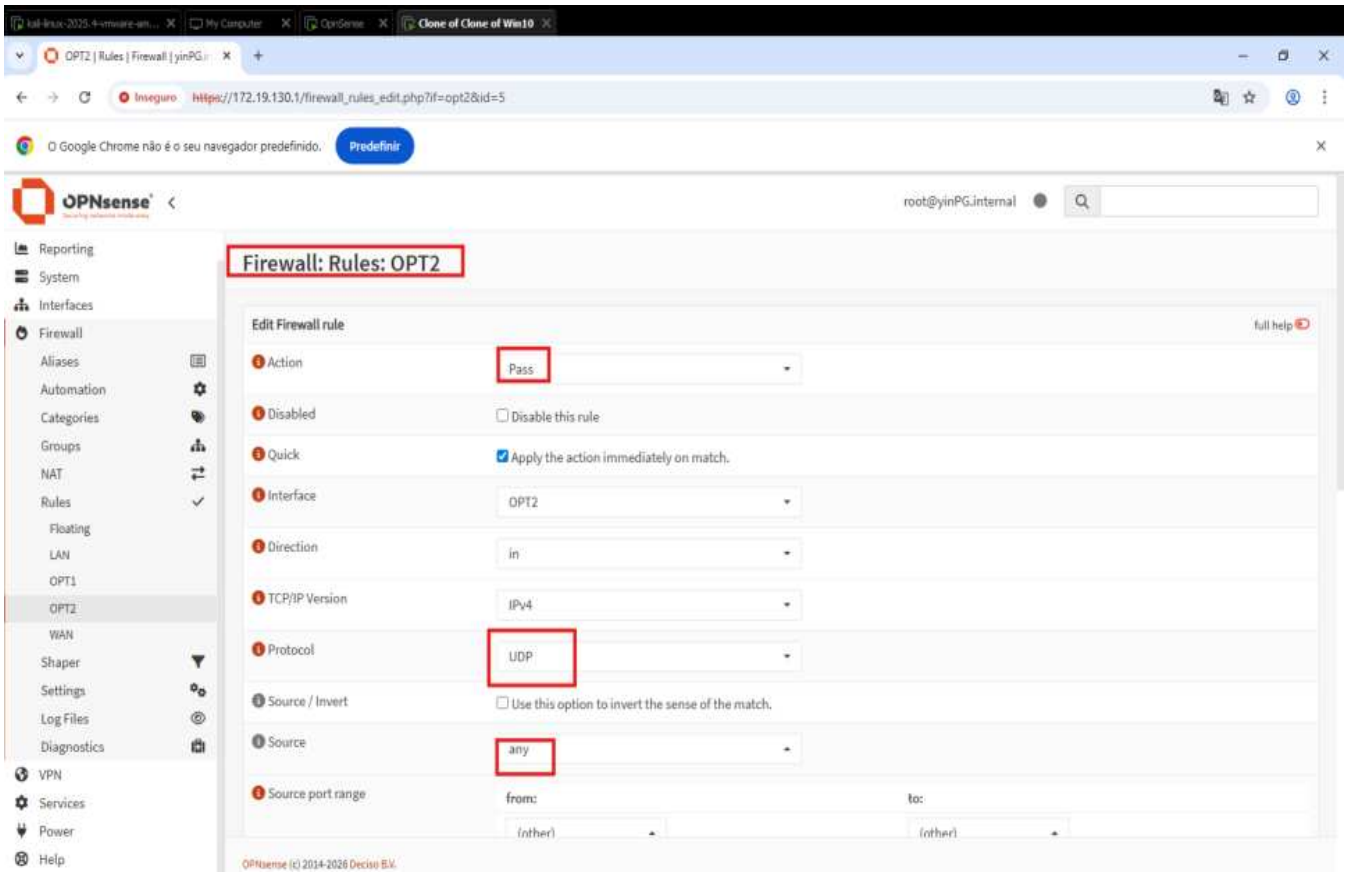


3.3 Criar regras na BLUE (por ordem)

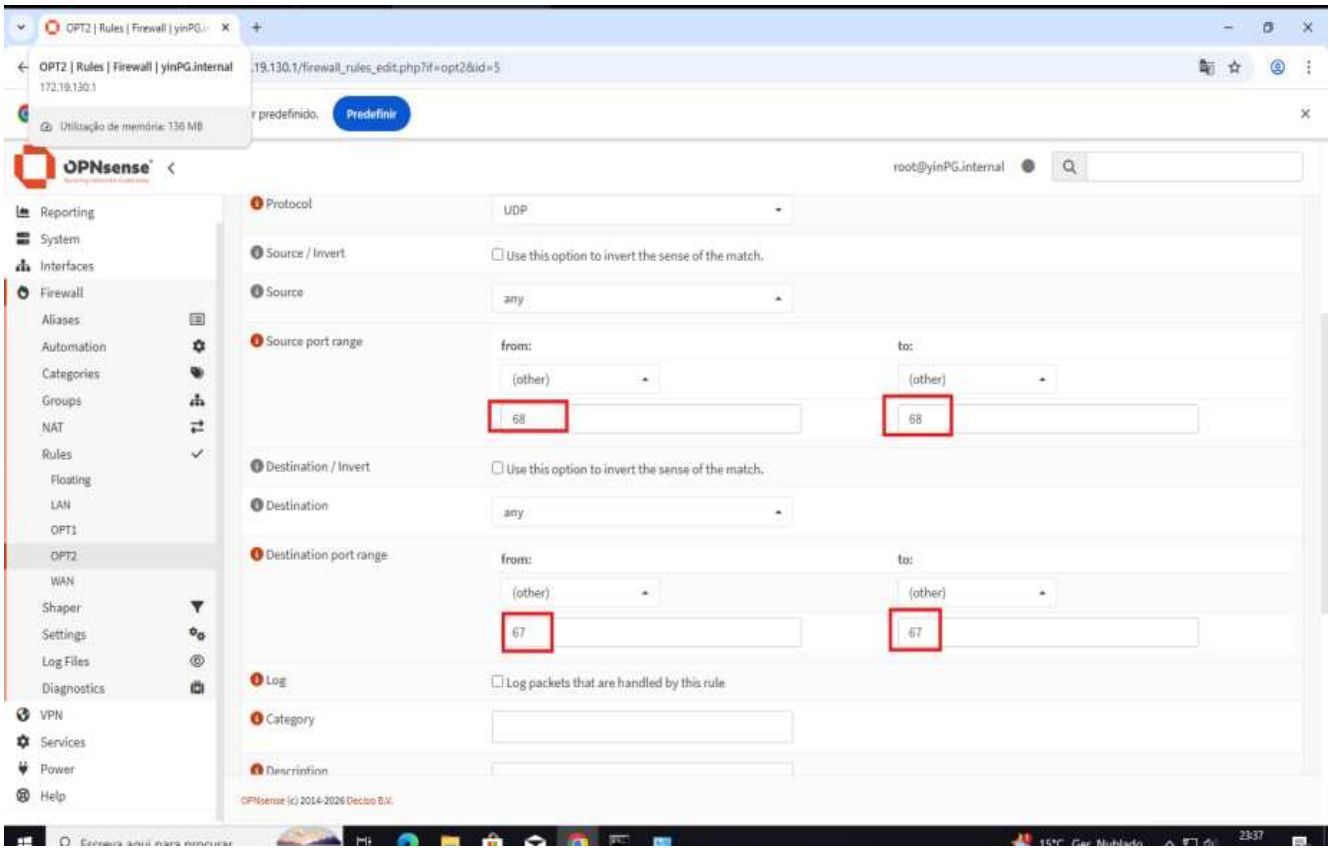
Ir a: **Firewall → Rules → BLUE (OPT2)**

Regra 1 — Permitir DHCP (obrigatório)

- **Action:** Pass
- **Protocol:** UDP
- **Source:** any
- **Source port:** 68
- **Destination:** any
- **Destination port:** 67
- **Save**

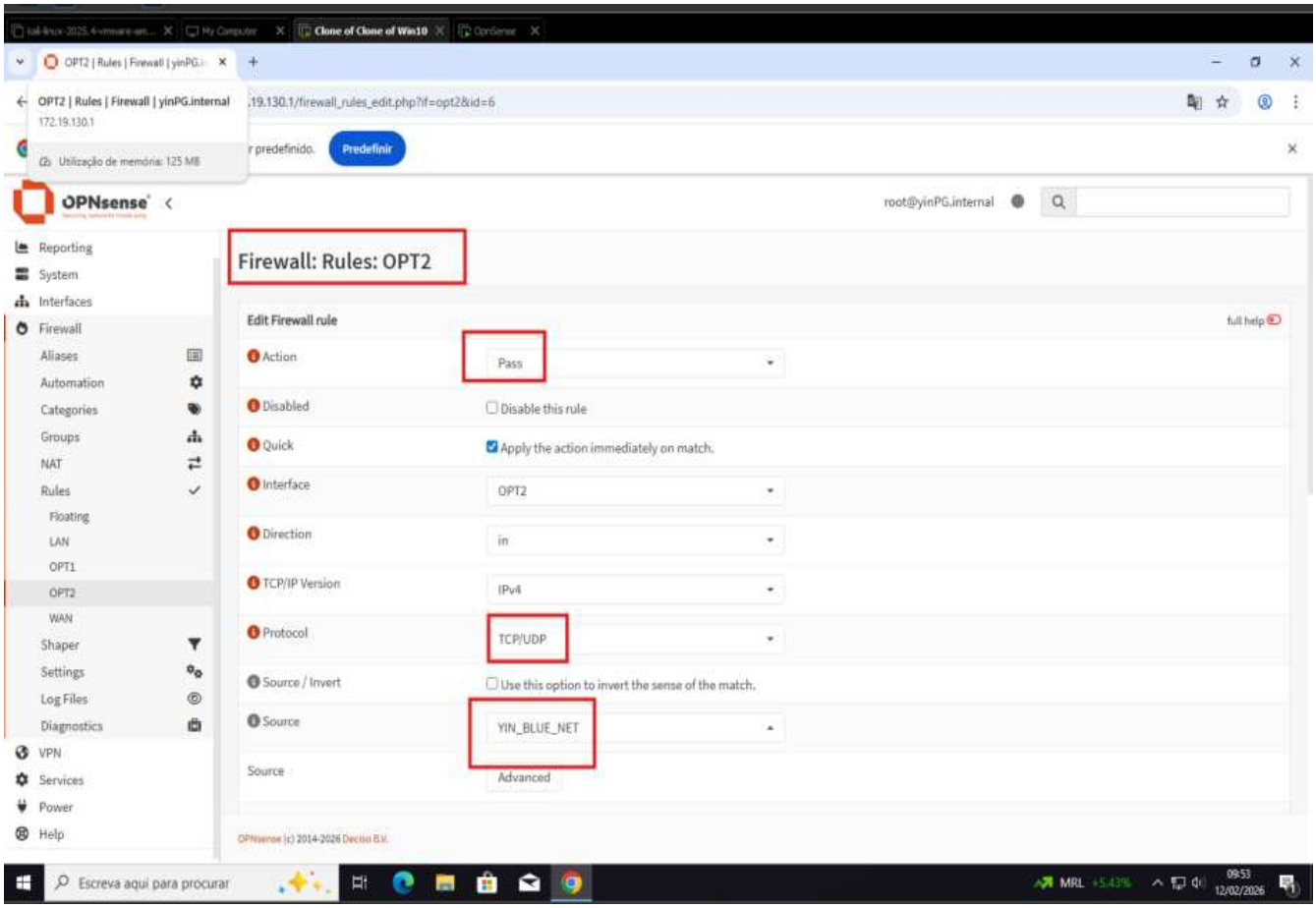


Continuação …

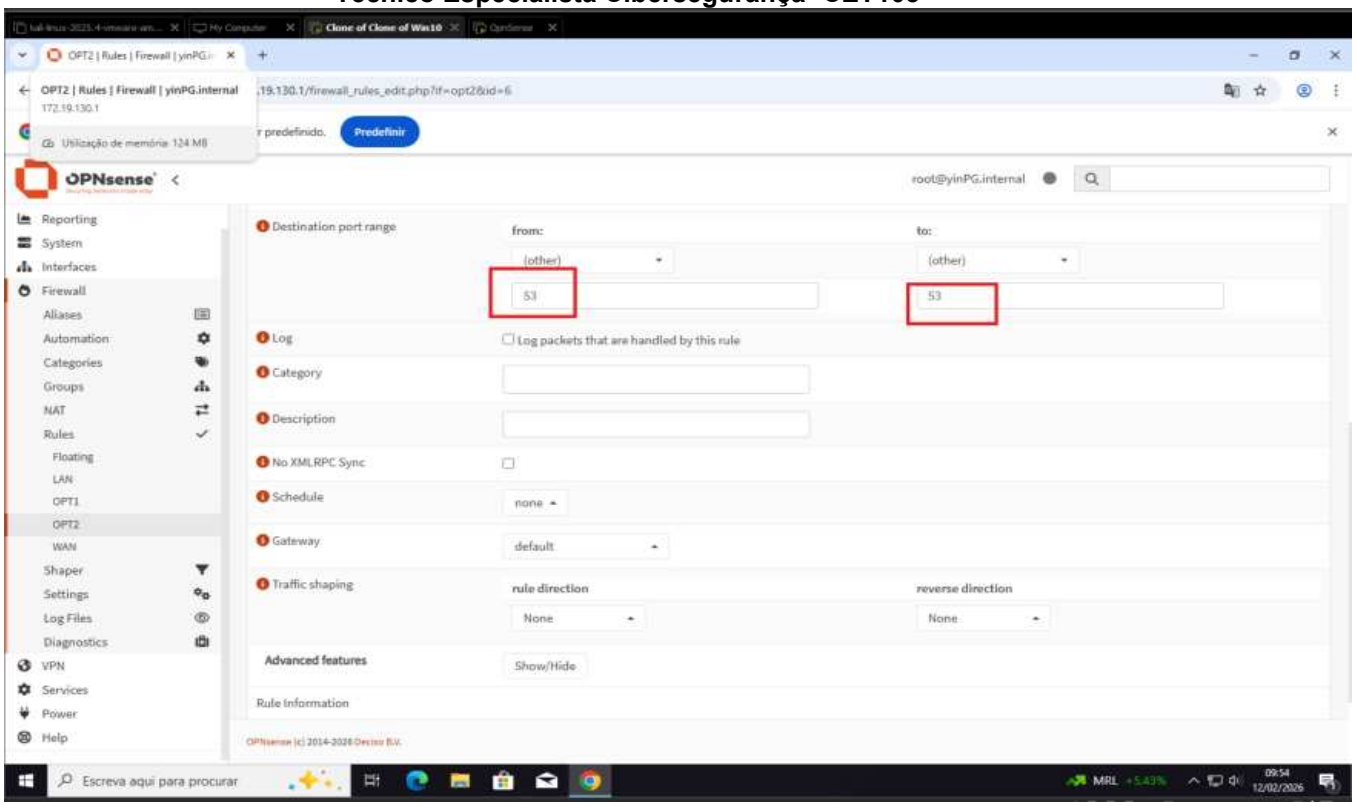


3.4 Regra 2 — Permitir DNS só para o firewall (obrigatório)

- **Action:** Pass
- **Protocol:** TCP/UDP
- Source: YIN_BLUE_NET
- **Destination: This Firewall**
- **Destination port:** 53
- **Save**



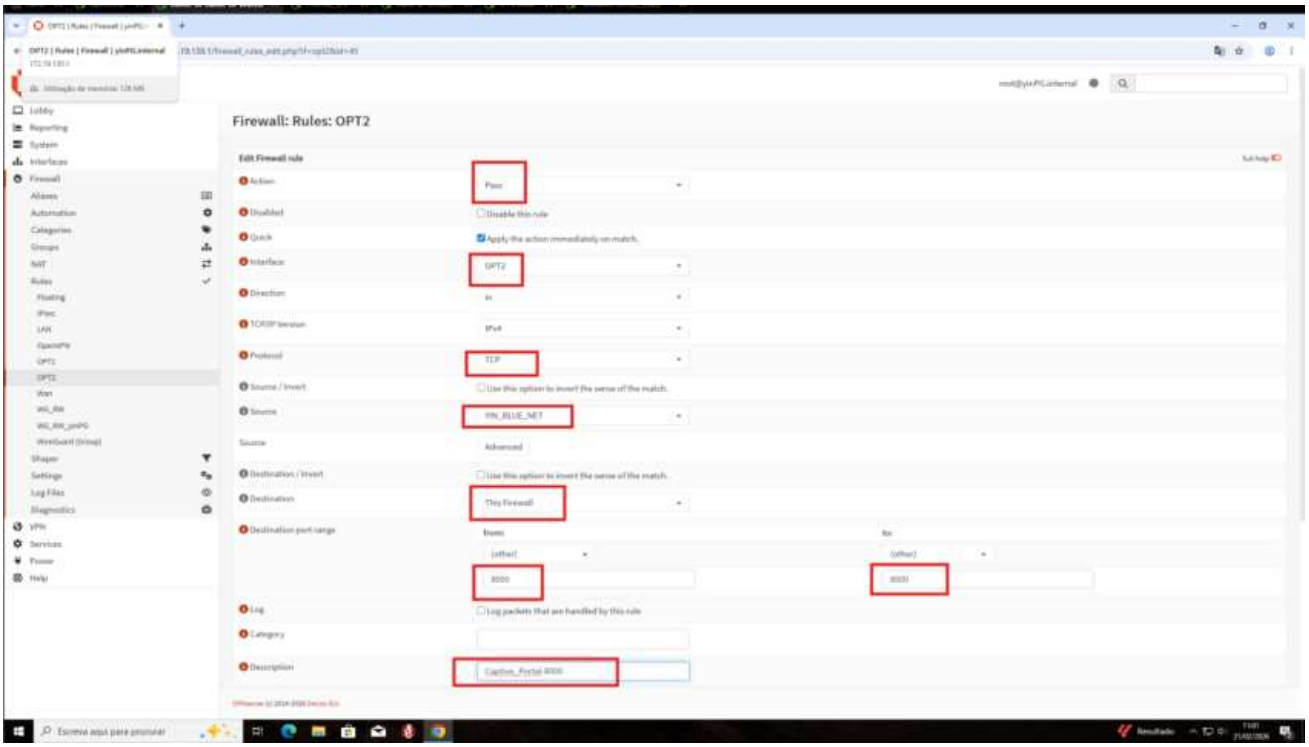
Continuação…



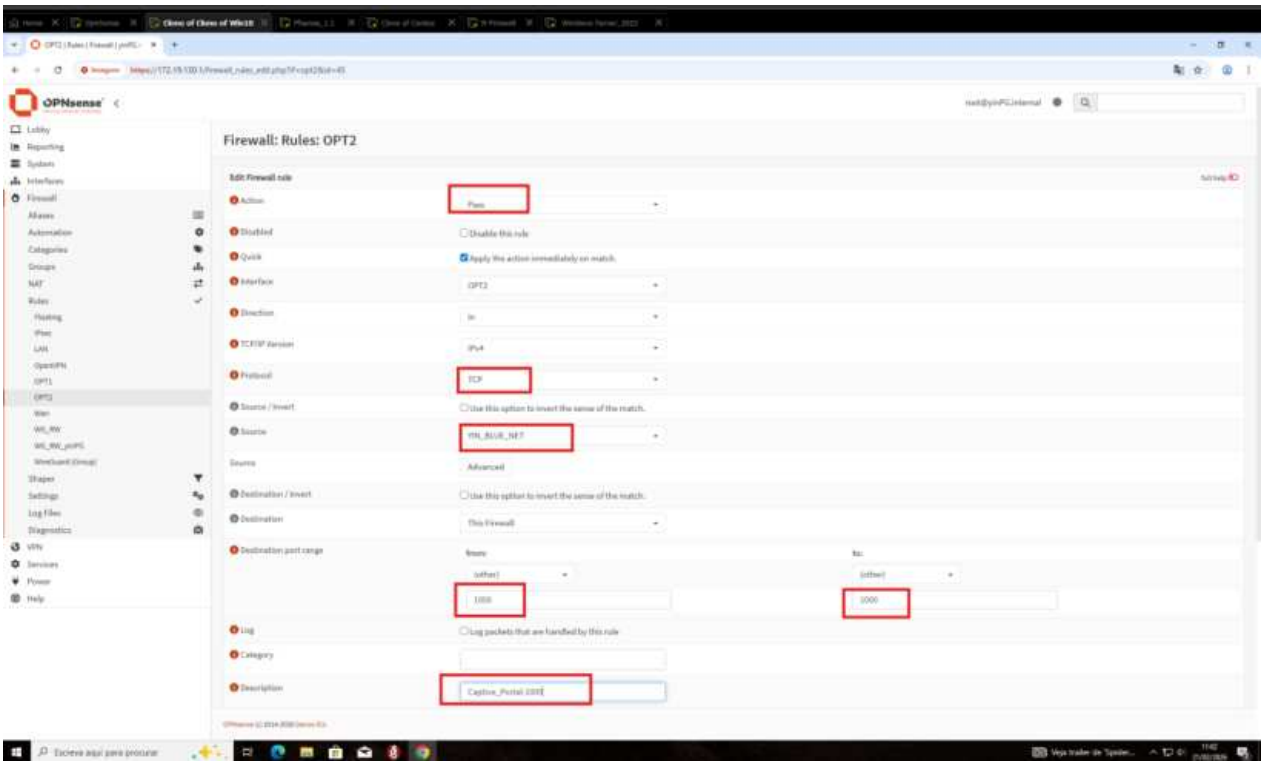
3.5 Regra 3 — Permitir acesso ao Captive Portal na firewall (portas 8000-10000)

- **Action:** Pass
- **Protocol:** TCP
- **Source:** YIN_BLUE_NET
- **Destination: This Firewall**

PORT -8000

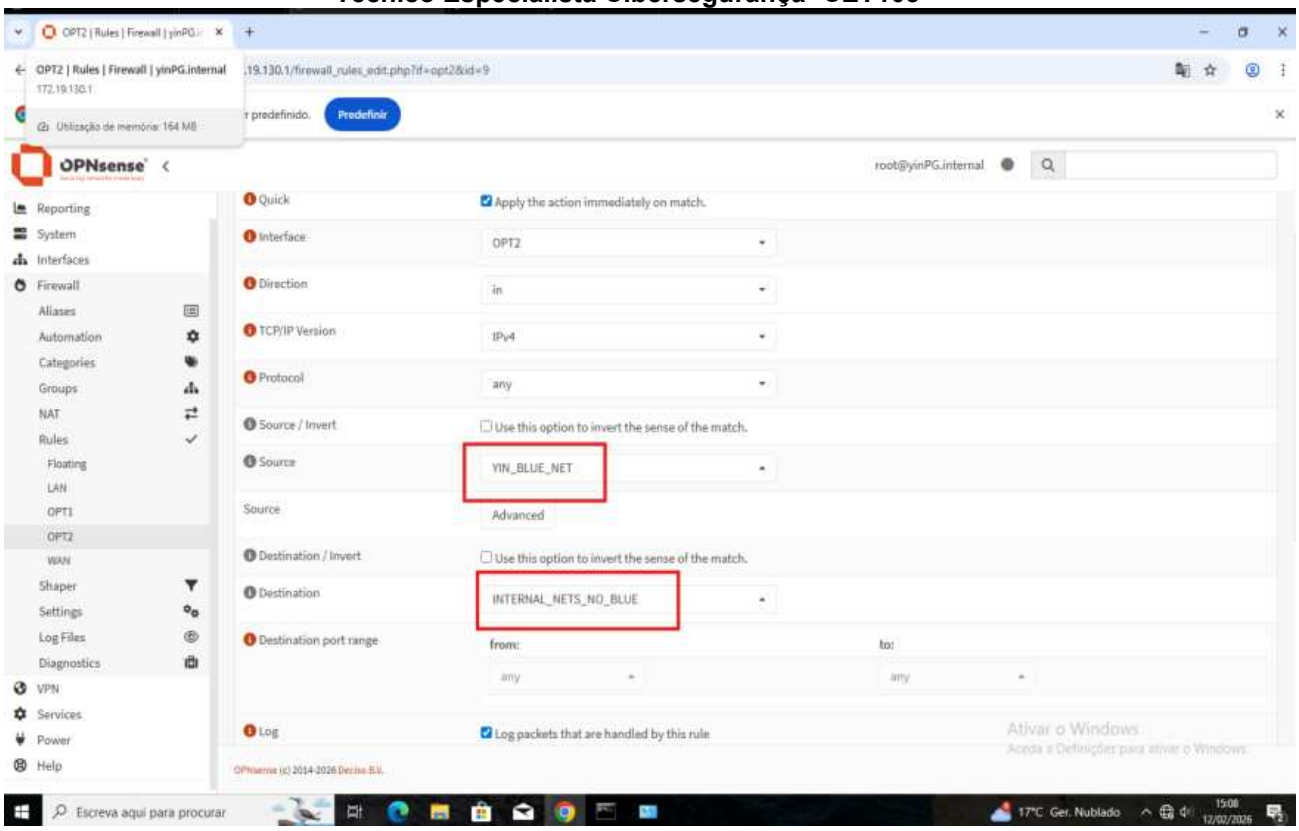


PORT -1000



3.6 Regra 4 — Bloquear BLUE → redes internas

- **Action:** Block
- **Protocol:** any
- **Source:** BLUE net
- Destination: INTERNAL_NETS_NO_BLUE
- **Log:** ON
- **Save**



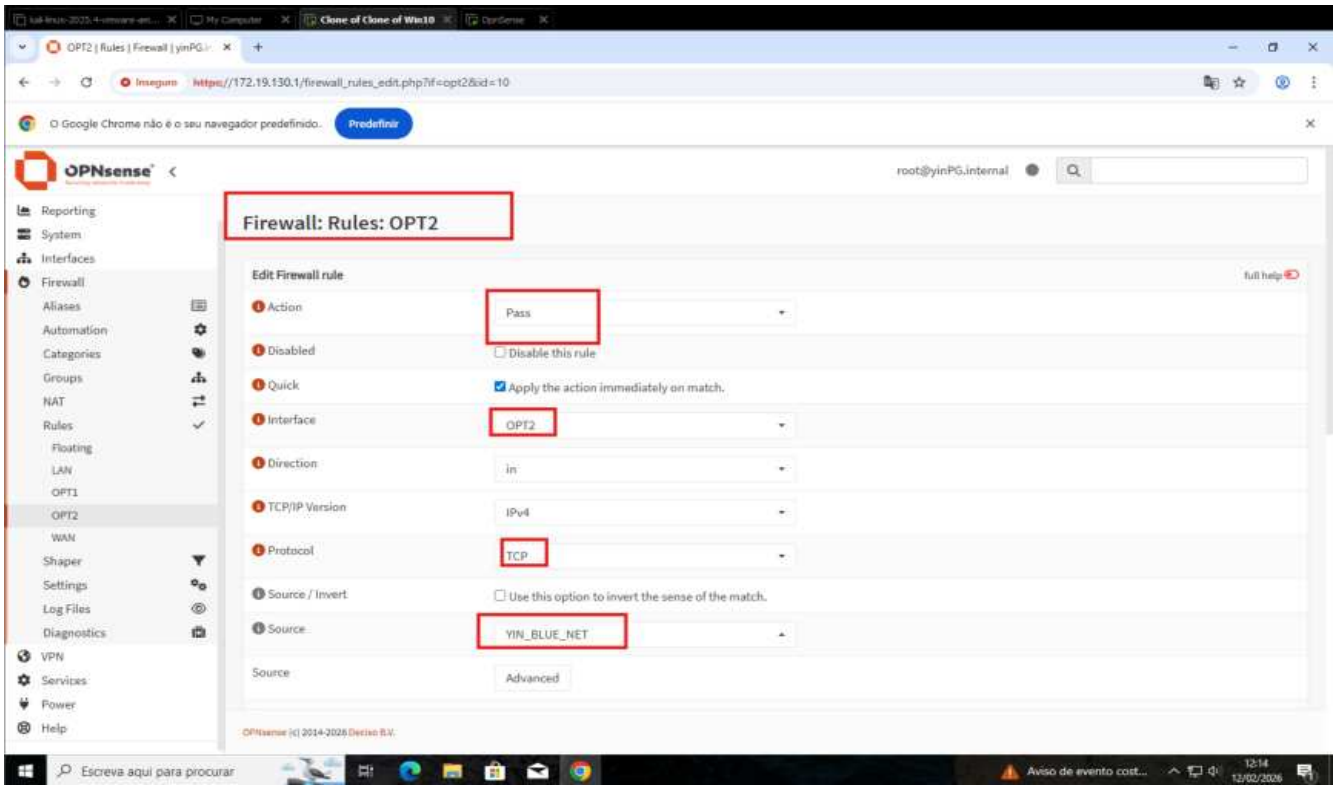
3.7 Regra 5 — Permitir internet (80/443) com Shaper 10/5

- **Action:** Pass
- **Protocol:** TCP
- **Source:** BLUE net
- **Destination:** any
- **Destination port:** alias WEB_PORTS
- **Traffic Shaper / Advanced options:**

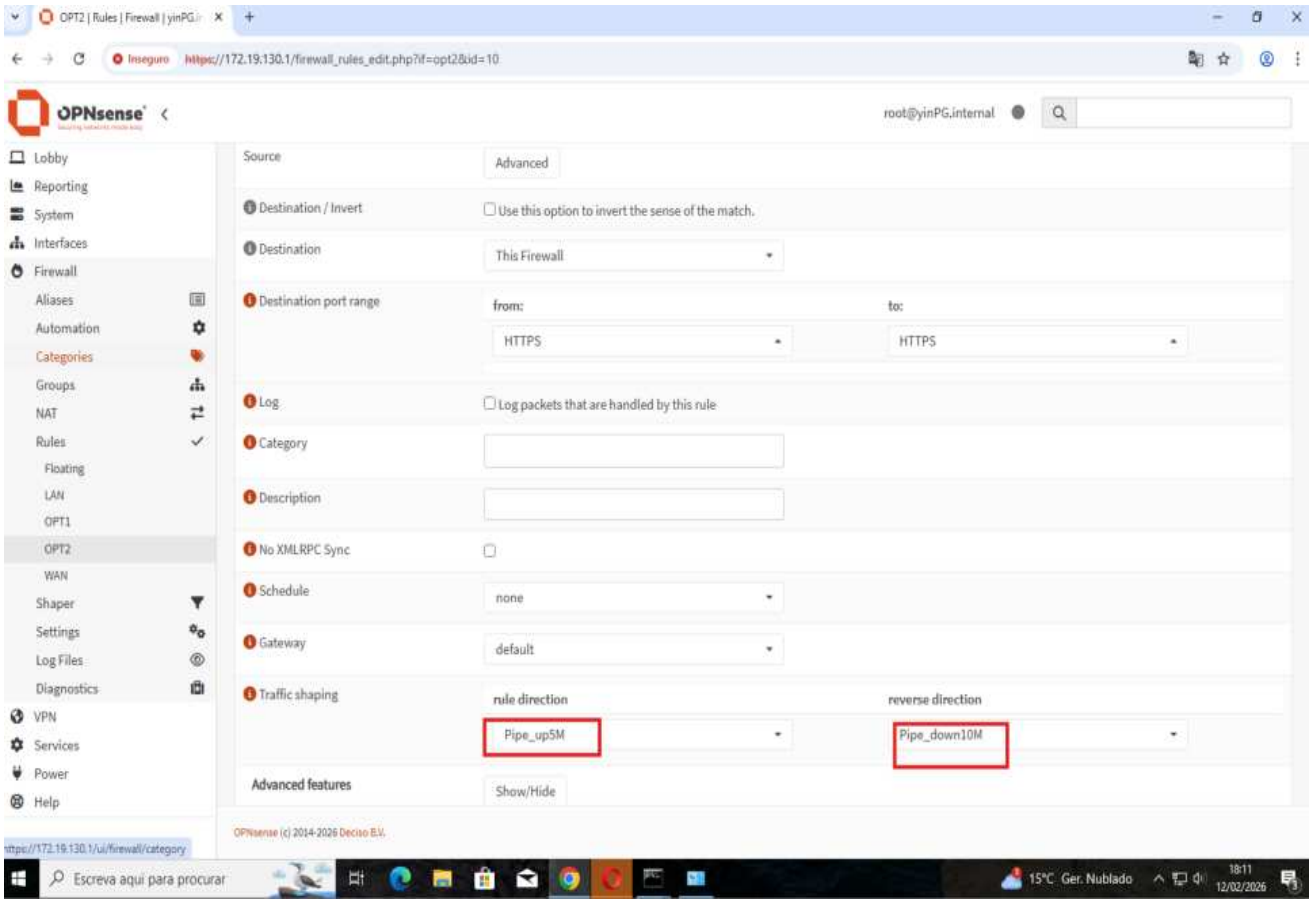
- In pipe: Pipe_up5M

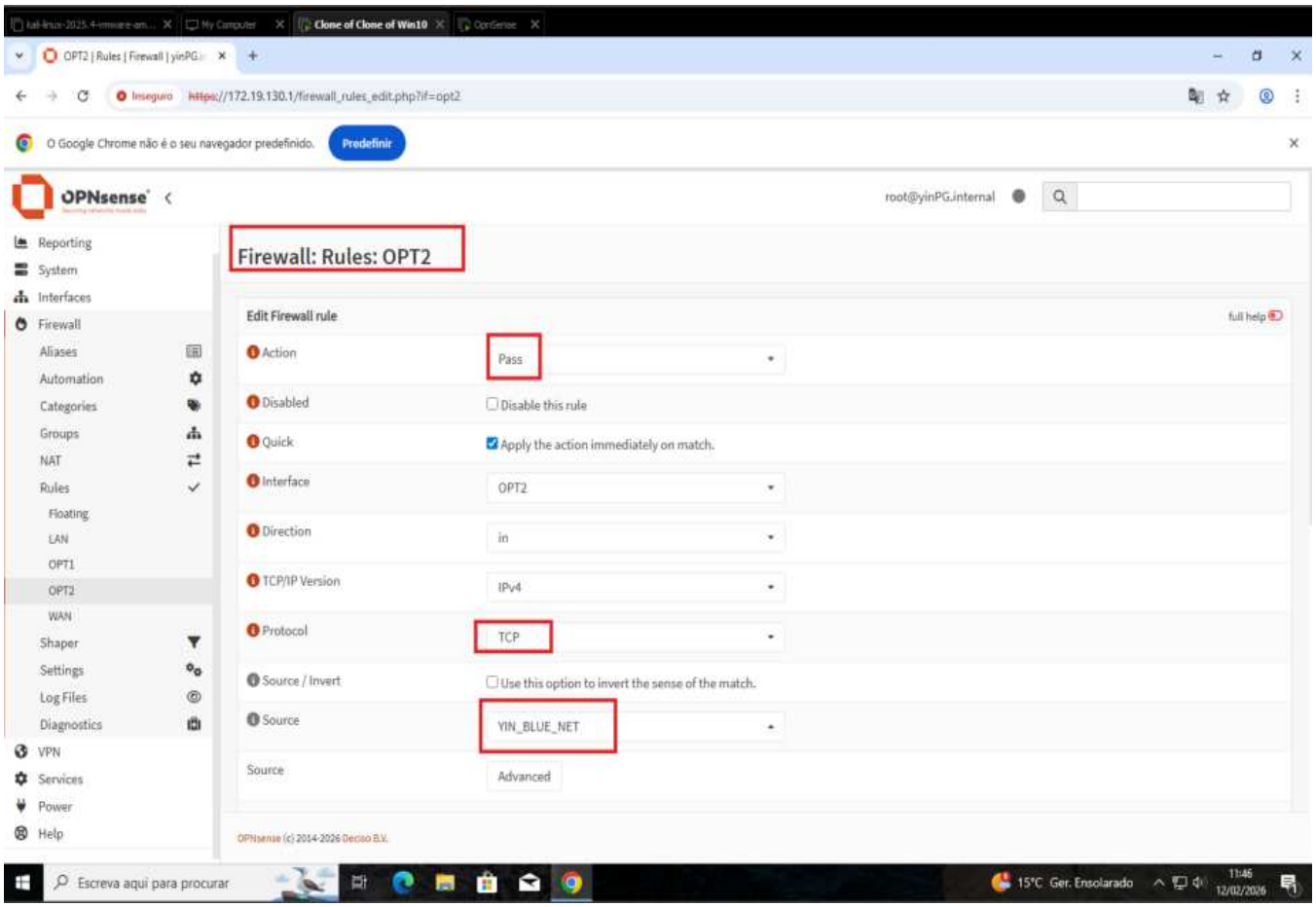
- Out pipe: Pipe_down10M

- **Save**



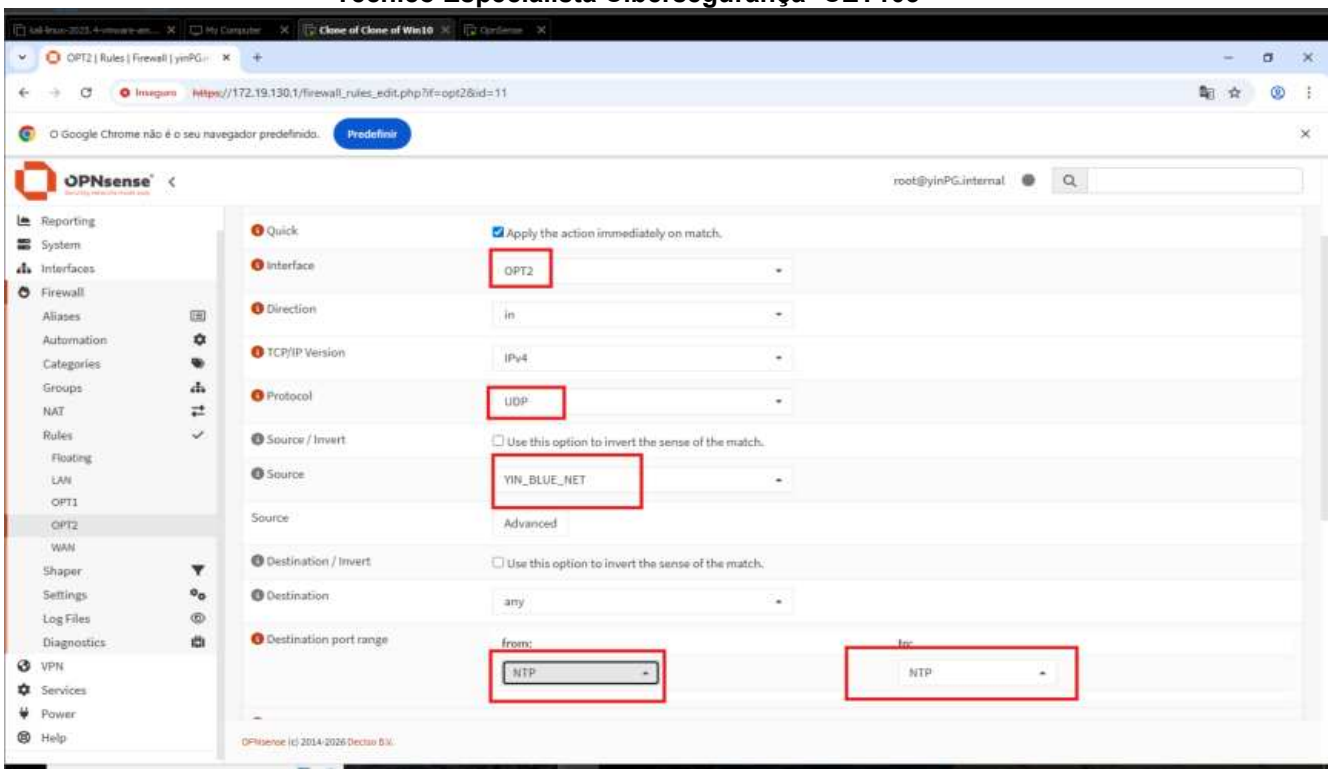
Continuação…



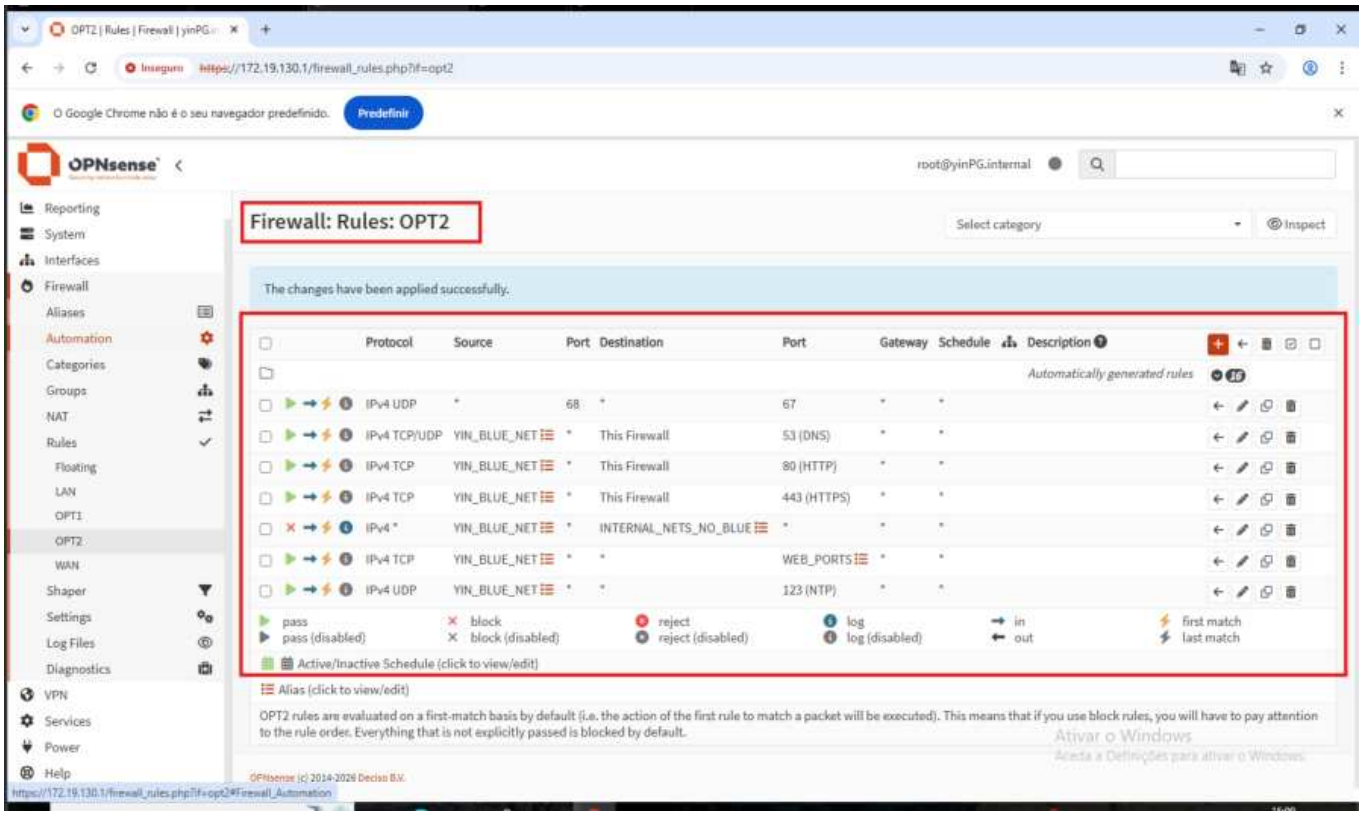


3.8 Regra 6 — (Opcional) NTP

- **Action:** Pass
- **Protocol:** UDP
- **Source:** BLUE net
- **Destination:** any
- **Destination port:** NTP
- **Save**



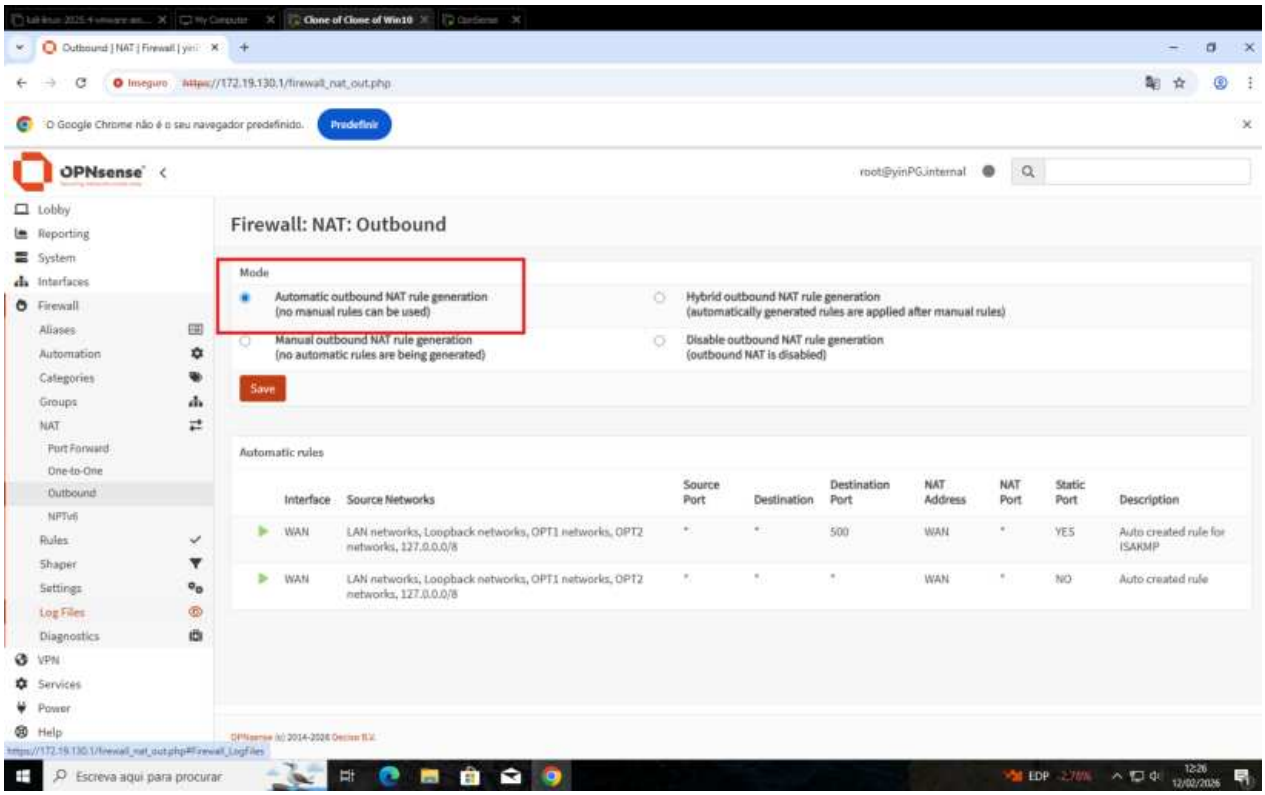
Resumo :



3) Verificação do Outbound NAT (para haver internet)

BLUE (e ORANGE, quando ativar a regra 3) tenham internet, confirmar:
Firewall → NAT → Outbound

- Modo **Automatic**



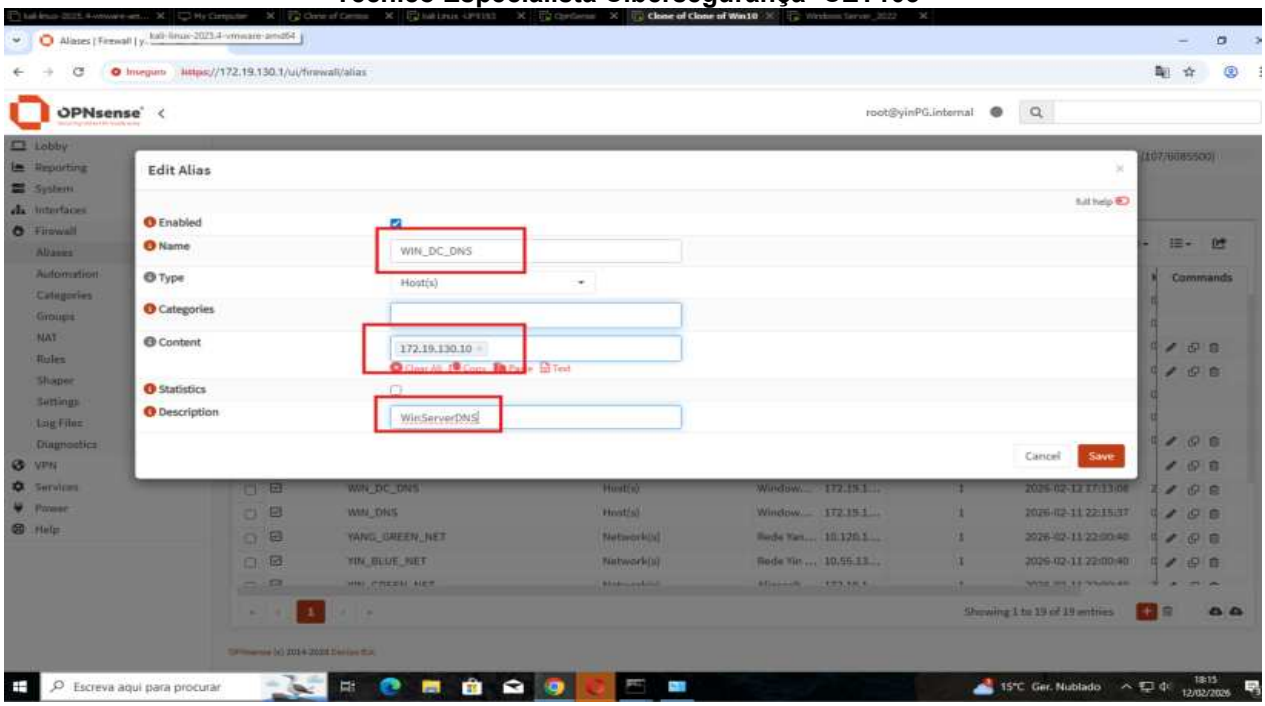
4) Regras do yinPG (OPNsense) — (continuação)

2- a) Os pedidos de DNS, originários da zona Green deste firewall, apenas podem ter sucesso, se iniciados pelo servidor Windows DC. Ainda que os clientes coloquem nas interfaces dos seus hosts, IPs de DNS públicos, a resolução de nomes, apenas pode ocorrer da seguinte forma: cliente◊ windows Server ◊ yinNA ◊ WAN, cliente recebe resolução do pedido no sentido inverso ao indicado
7.A)

4.1 Criar alias do DC

Firewall → Aliases → +Add

- Name: WIN_DC_DNS
- Type: Host(s)
- Content: 172.19.130.10
- Description: WinServerDNS

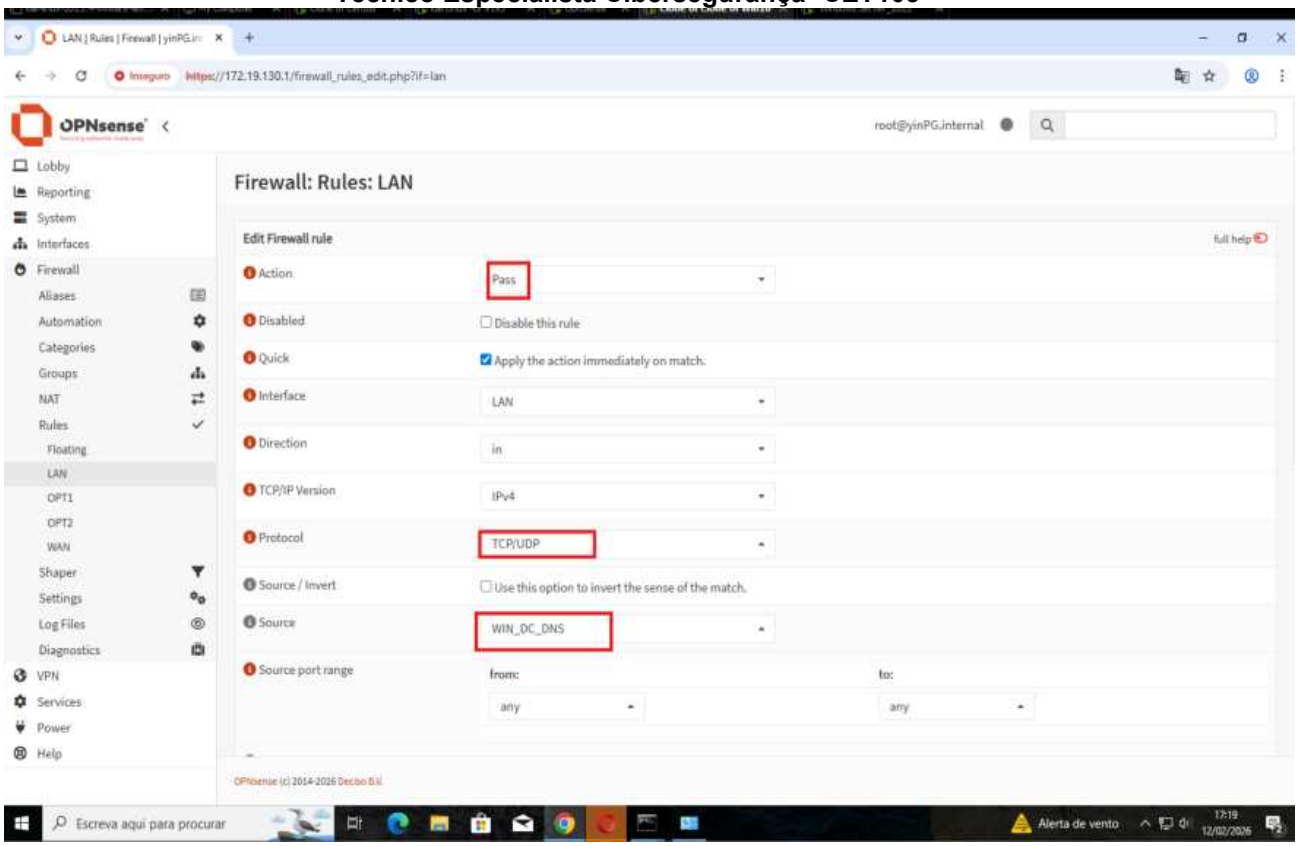


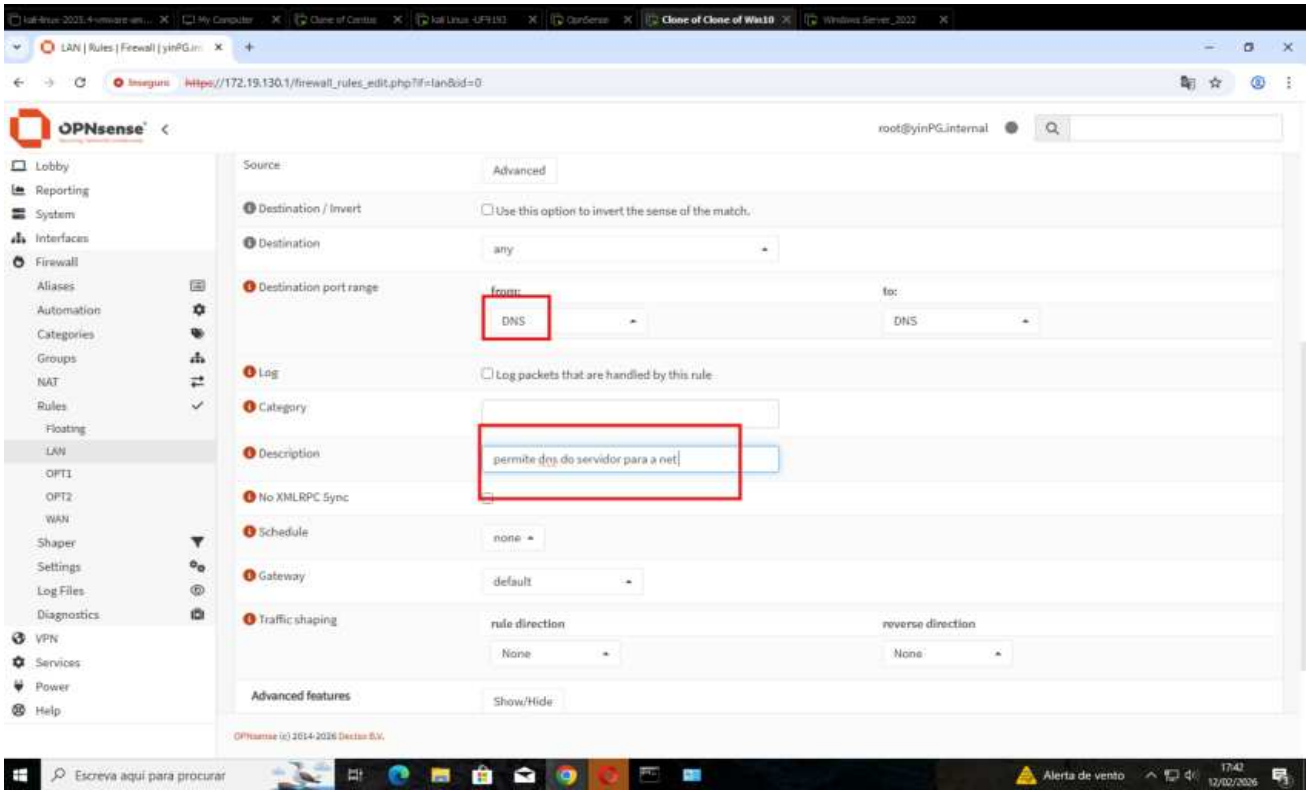
4.2 Regras na interface GREEN

Firewall → Rules → GREEN

Regra 1 — permitir DNS do Windows Server para a internet

- Action: Pass
- Interface: GREEN
- Protocol: TCP/UDP
- Source: WIN_DC_DNS
- Destination: any
- Destination port: 53 (DNS)
- Description: permite dns do servidor para a net



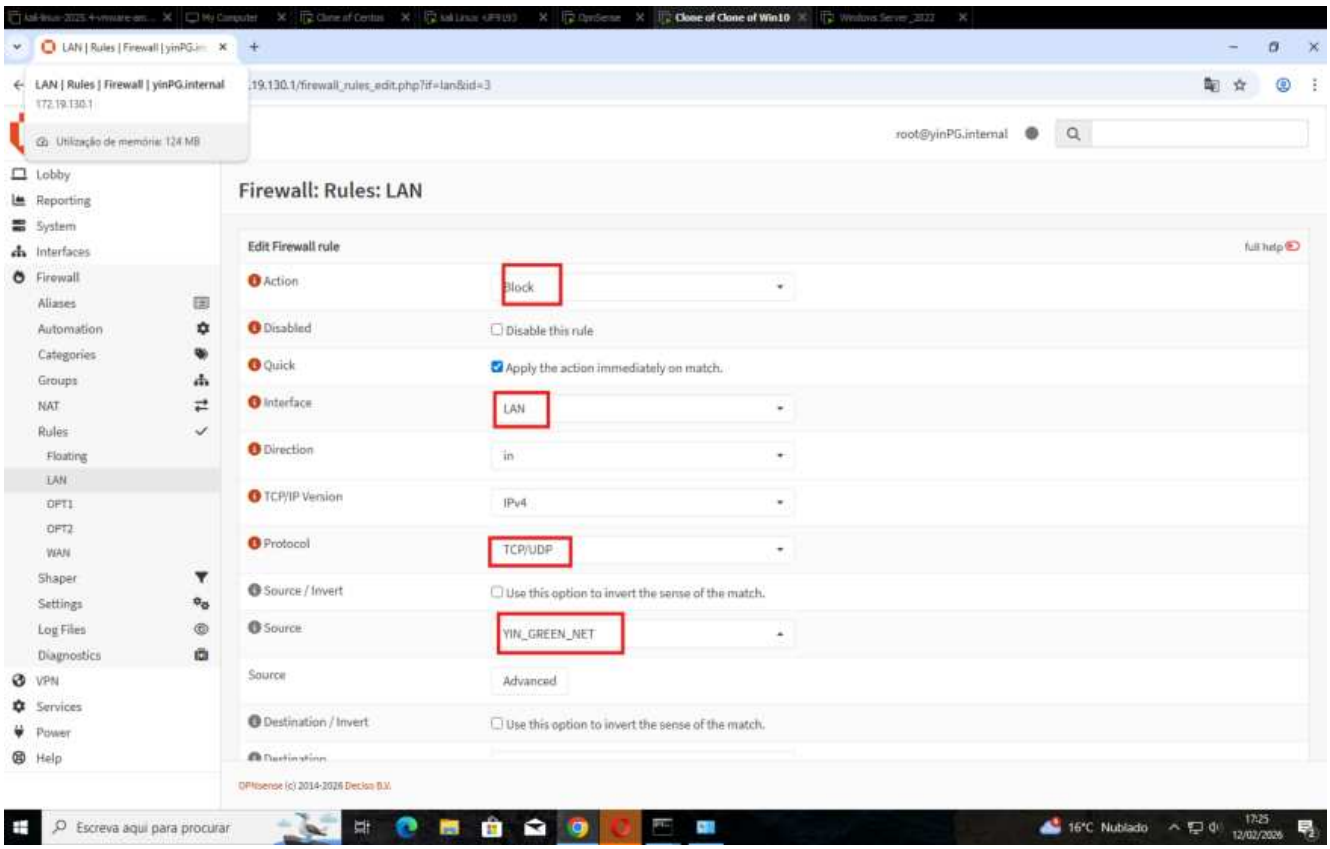


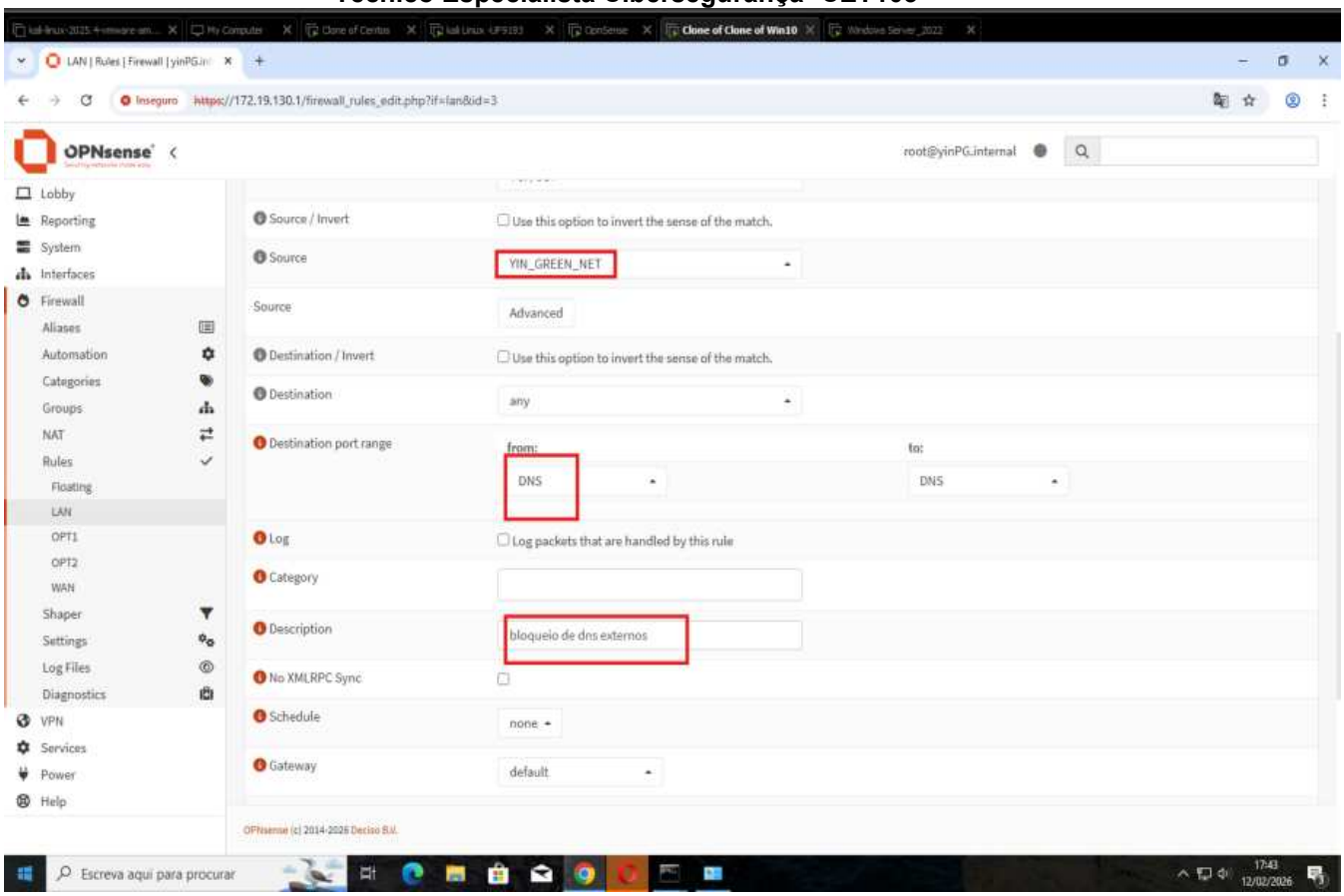
Nota : colocar esta regra no topo

4.3 Regra 2 — bloquear DNS de todos os outros

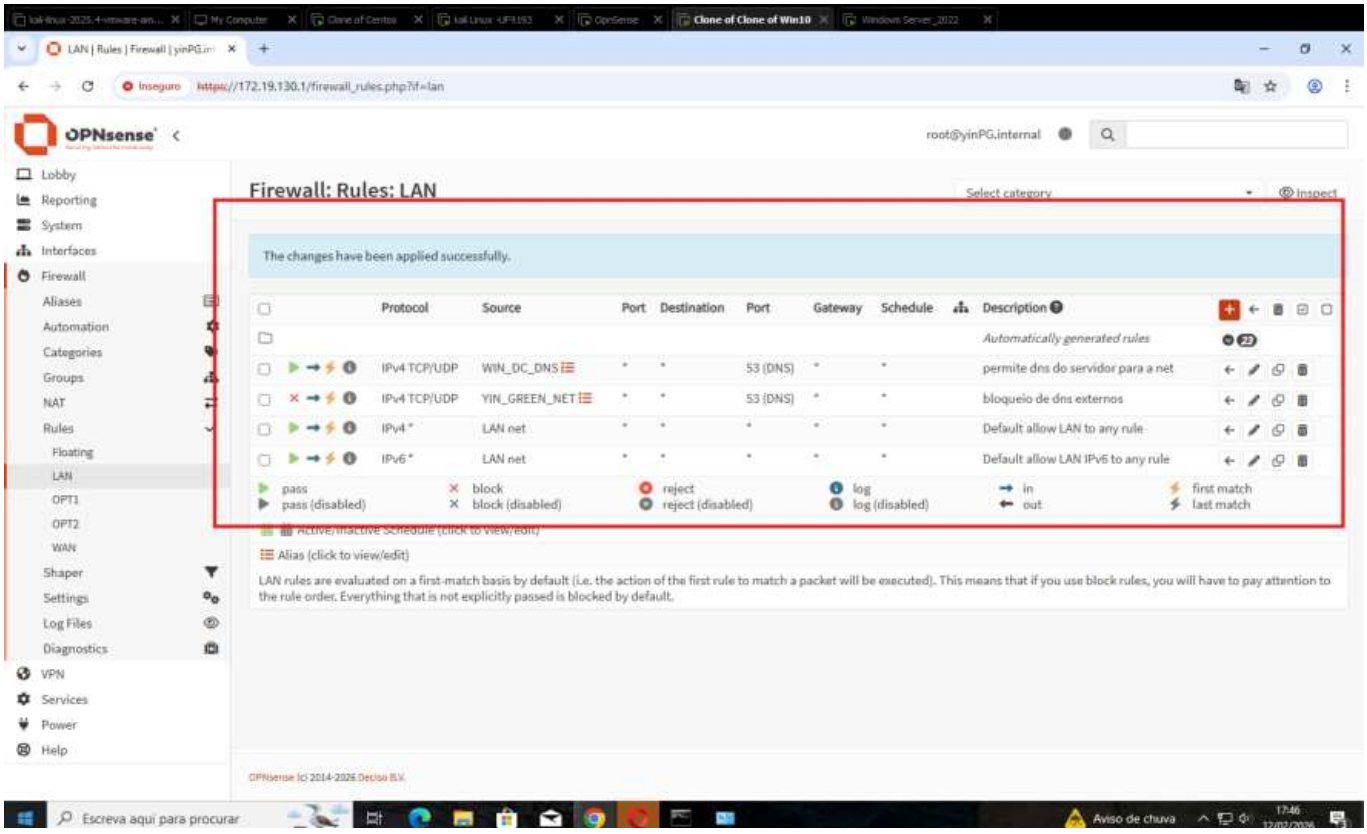
- Action: Block

- Interface: GREEN
- Protocol: TCP/UDP
- Source: GREEN net
- Destination: any
- Destination port: 53
- Description: bloqueio de dns externos




LAN | Rules | Firewall | yinPG.in
Inseguro https://172.19.130.1/firewall_rules_edit.php?if=lan&id=3
OPNsense
root@yinPG.internal
Lobby
Reporting
System
Interfaces
Firewall
Aliases
Automation
Categories
Groups
NAT
Rules
Floating
LAN
OPT1
OPT2
WAN
Shaper
Settings
Log Files
Diagnostics
VPN
Services
Power
Help
Source / Invert
Use this option to invert the sense of the match.
Source
YIN_GREEN_NET
Source
Advanced
Destination / Invert
Use this option to invert the sense of the match.
Destination
any
Destination port range
from:
to:
DNS
DNS
Log
Log packets that are handled by this rule
Category
Description
bloqueio de dns externos
No XMLRPC Sync
Schedule
none
Gateway
default
OPNsense (c) 2014-2026 Deciso B.V.
Escreva aqui para procurar
17:43
12/02/2026

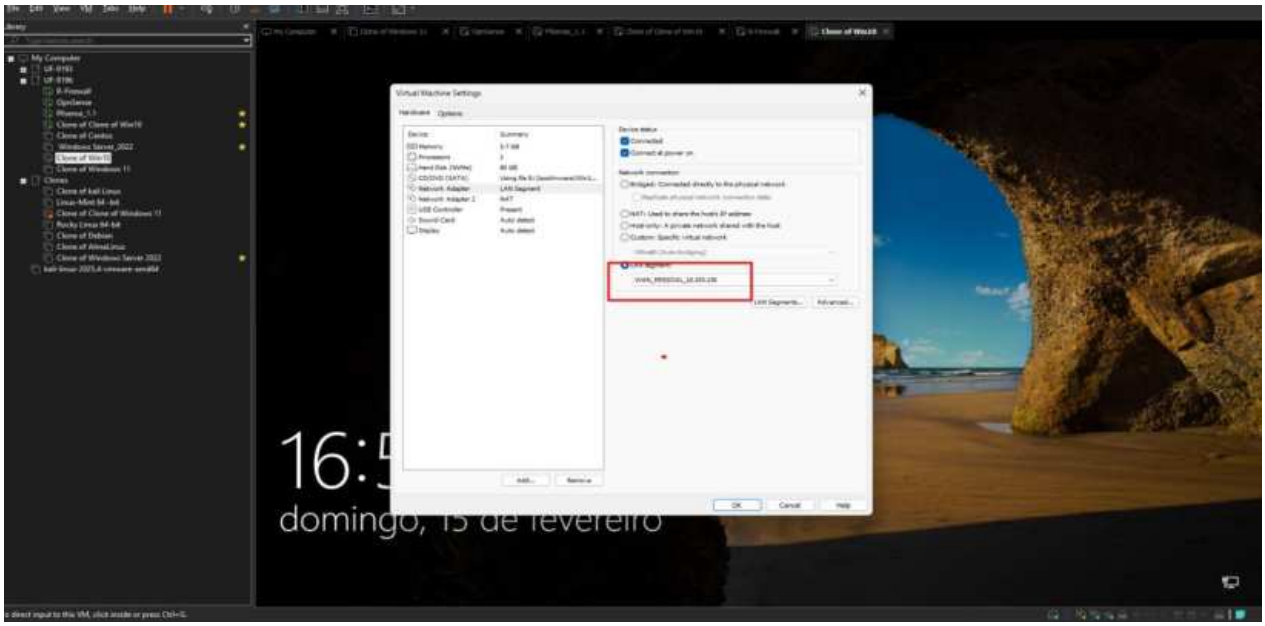
Resumo :


LAN | Rules | Firewall | yinPG.in
Inseguro https://172.19.130.1/firewall_rules.php?if=lan
OPNsense
root@yinPG.internal
Firewall: Rules: LAN
Select category
Inspect
The changes have been applied successfully.
Protocol
Source
Port
Destination
Port
Gateway
Schedule
Description
Automatically generated rules
IPv4 TCP/UDP
WIN_DC_DNS
53 (DNS)
permite dns do servidor para a net
IPv4 TCP/UDP
YIN_GREEN_NET
53 (DNS)
bloqueio de dns externos
IPv4 *
LAN net
Default allow LAN to any rule
IPv6 *
LAN net
Default allow LAN IPv6 to any rule
pass
pass (disabled)
block
block (disabled)
reject
reject (disabled)
log
log (disabled)
in
out
first match
last match
Alias (click to view/edit)
LAN rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.
OPNsense (c) 2014-2026 Deciso B.V.
Escreva aqui para procurar
Aviso de chuva
17:46

4.4 Windows 10 (PC na WAN)-Externo

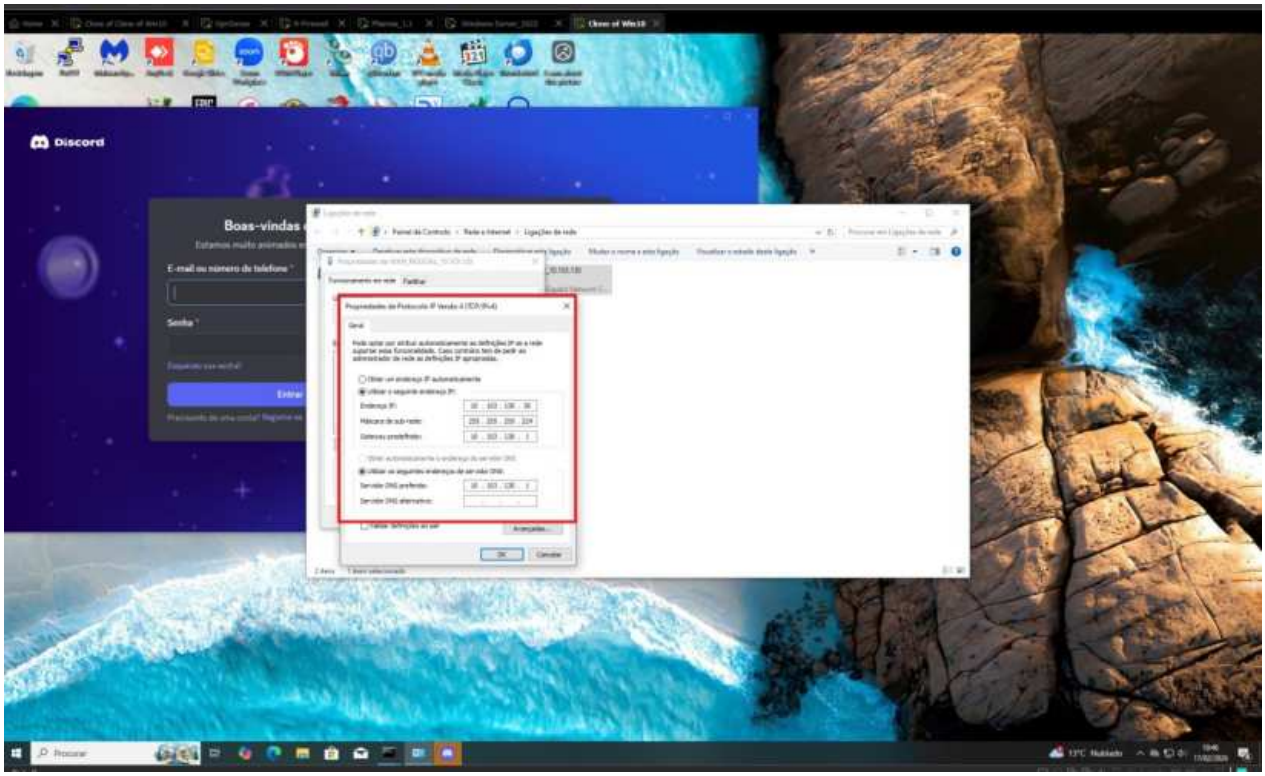
VMware Workstation

- **VM VyOS → Settings → Network Adapter –" WAN_PESSOAL_10.103.130"**



4.5 Configurações :

- **NIC "WAN" do Windows 10 → WAN_PESSOAL_10.103.130**
-
- **IP: 10.103.130.30/27**
- **Gateway: 10.103.130.1**

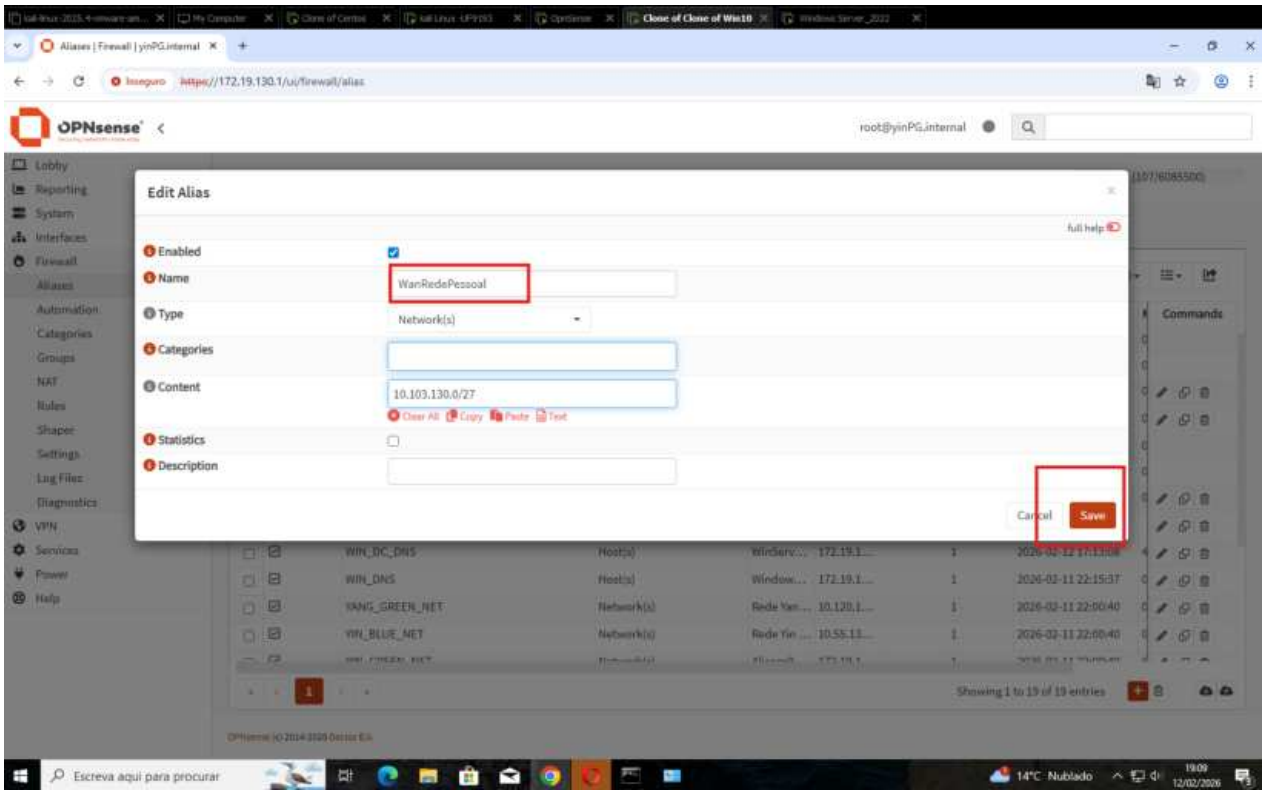


2- b) Pedidos de ICMP, não podem pingar a rede que foi definida para o router (“WAN pessoal”), apenas podem ter sucesso para IPs que não estejam abrangidos nessa rede.

5.1 Alias da rede WAN pessoal

Firewall → Aliases → +Add

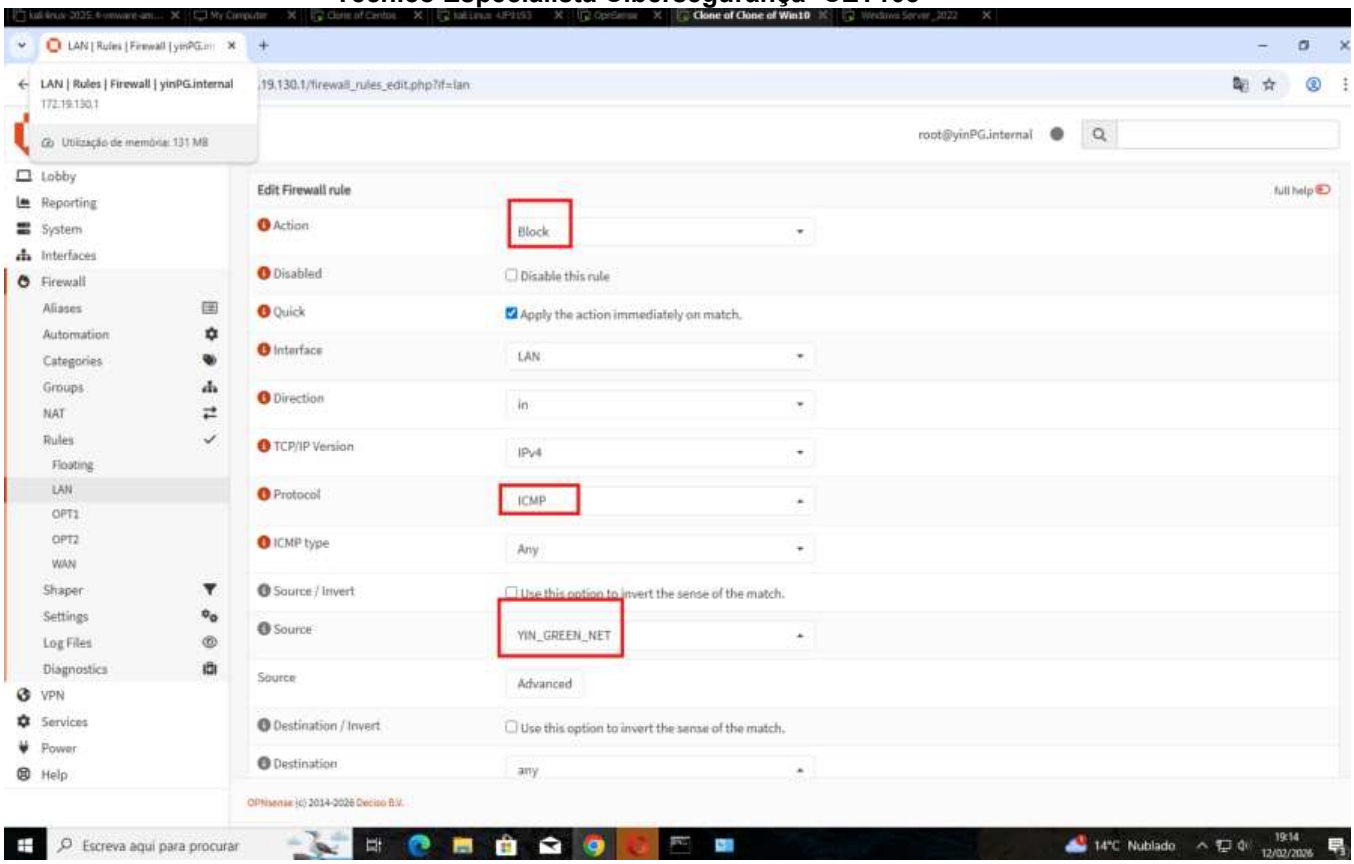
- Name: WanRedePessoal
- Type: Network(s)
- Content: 10.103.130.0/27



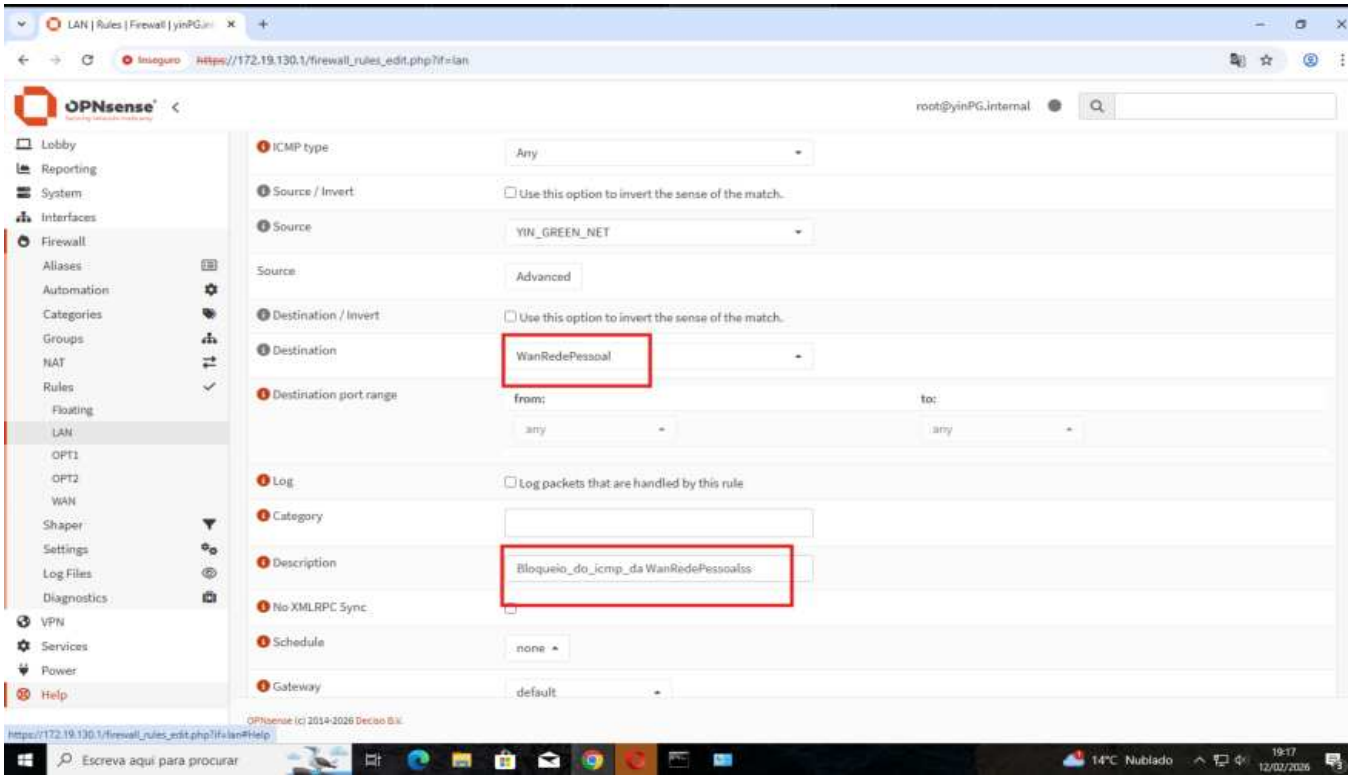
5.2 Regra na GREEN para bloquear ICMP para essa rede

Firewall → Rules → GREEN (colocar acima das regras)

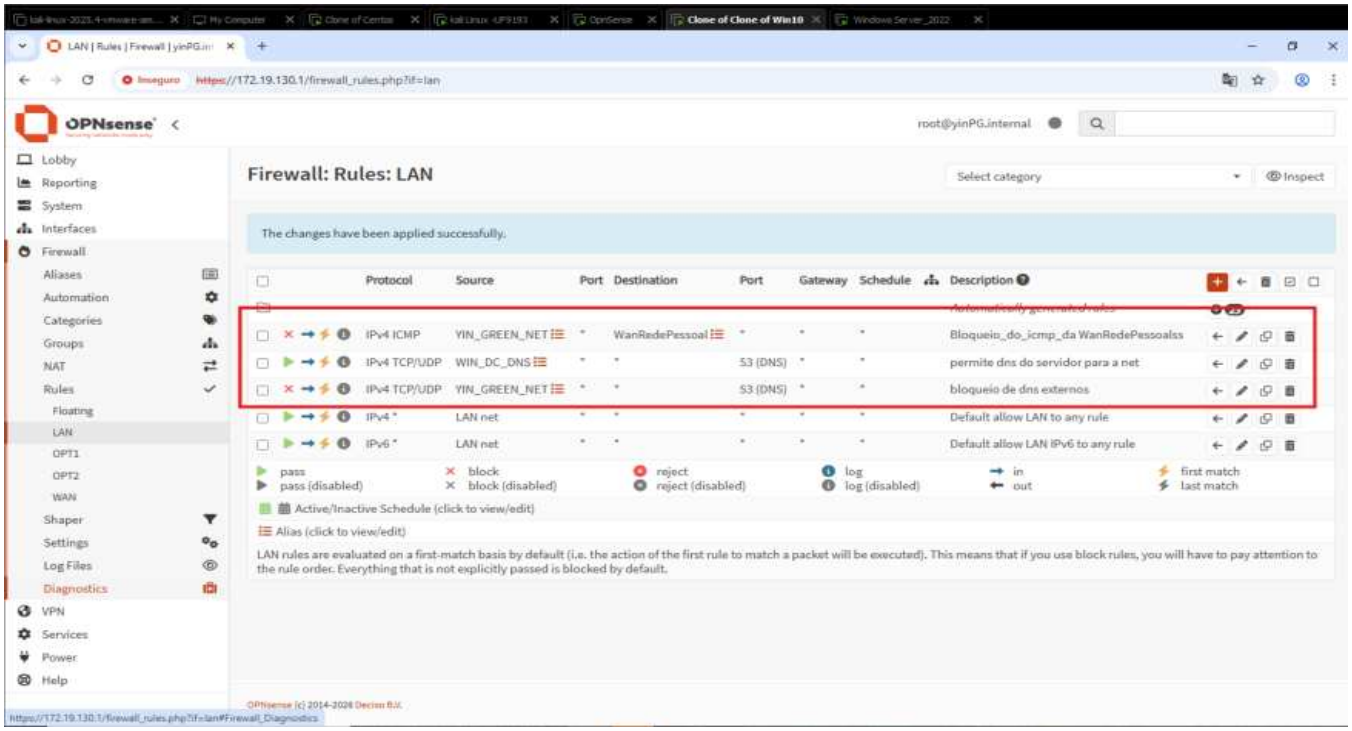
- Action: Block
- Protocol: ICMP
- Source: YIN-GREEN_NET
- Destination: WanRedePessoal
- Description: Bloqueio_do_icmp_daWanRedePessoal



Continuação…



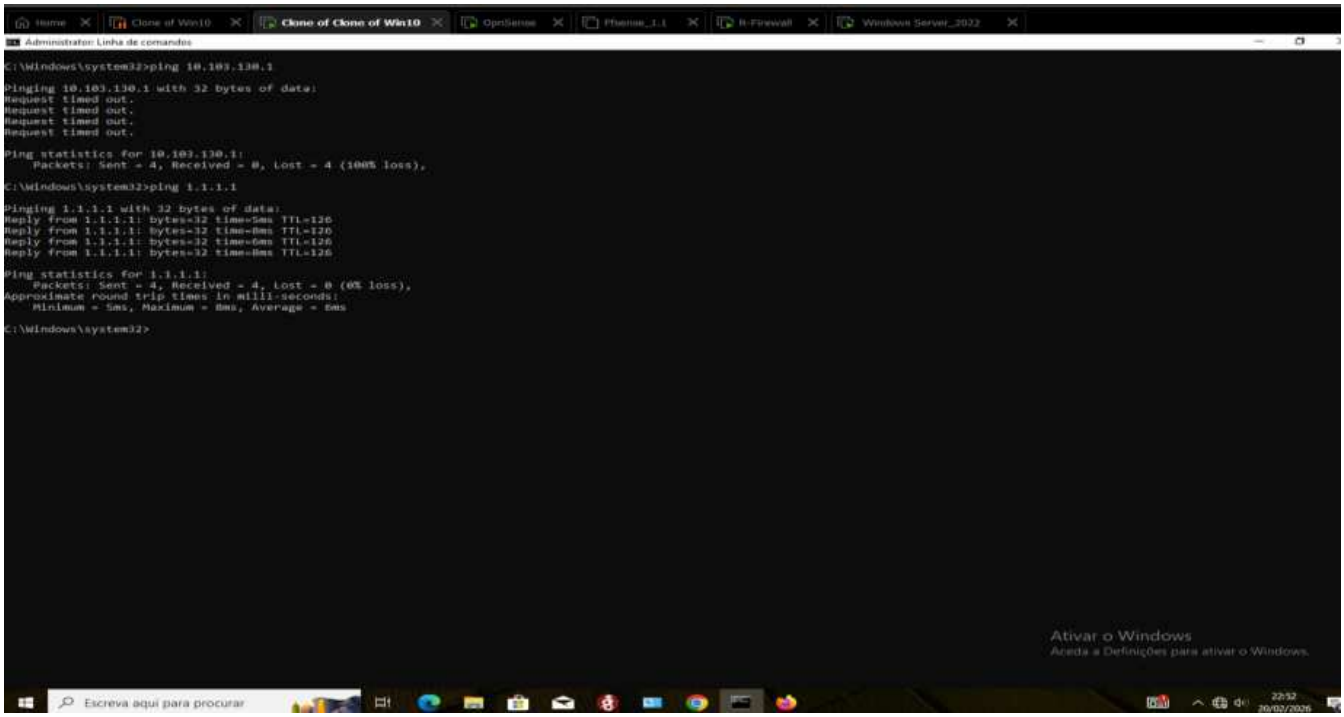
Resumo:



5.3 Teste

Windows 10 (GREEN):

- ping 10.103.130.1 → **falha**
- ping 1.1.1.1 → **ok**



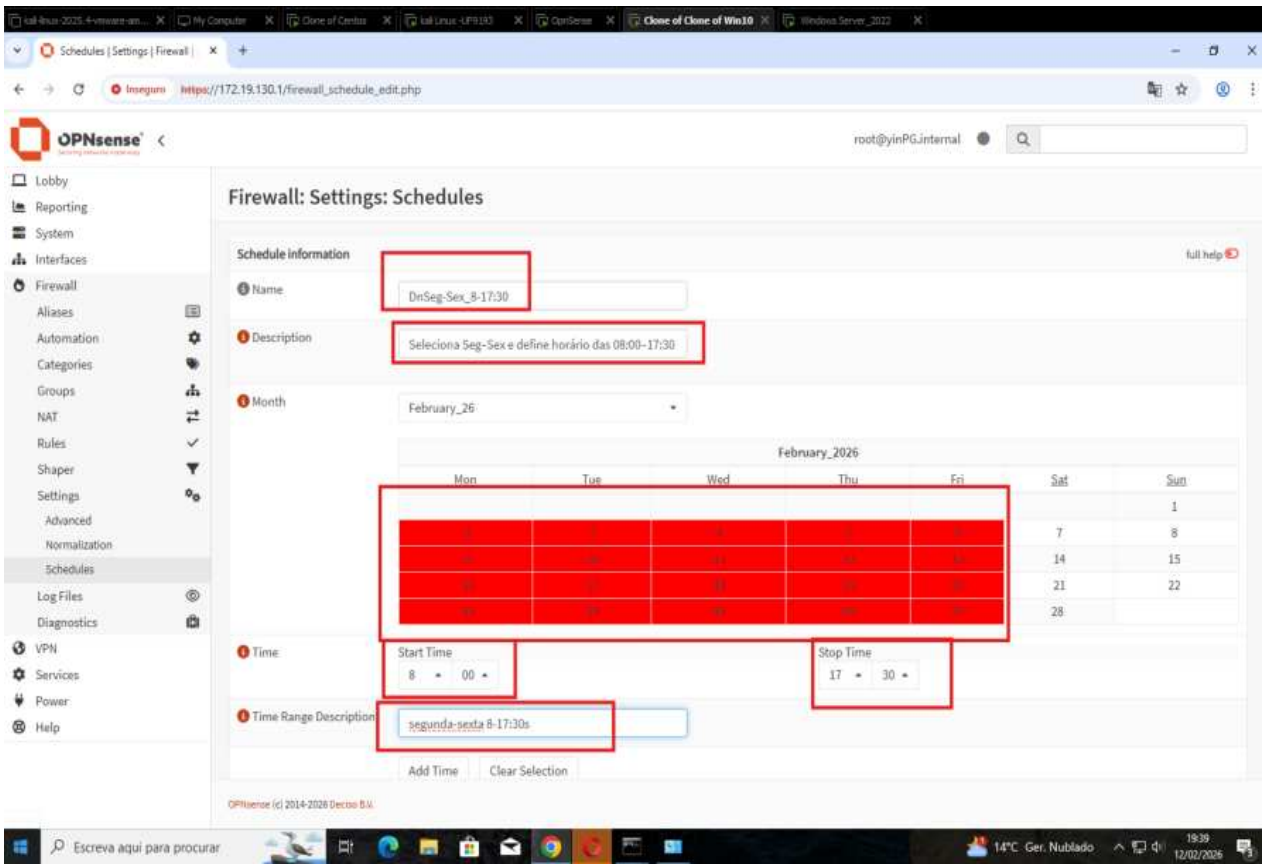
2- c) Bloquear o acesso ao site do DN (http e https), dias da semana (seg-sexta)das 8h-17h:30.

6 -Requisito: bloquear DN por horário/dias.

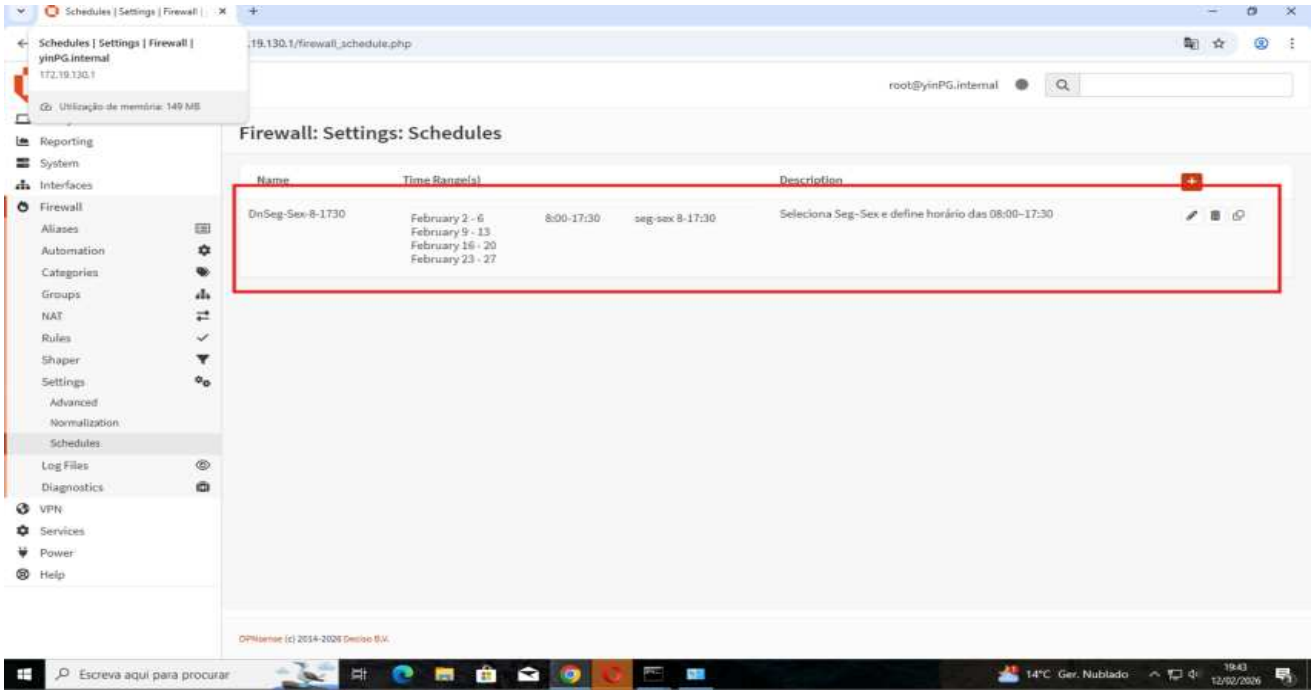
6.1 Criar Schedule

Firewall → Settings → Schedules → +Add

- Name:DnSegunda-Sexta_8-17:30
- Seleciona Seg–Sex e define horários das 08:00–17:30
- Save



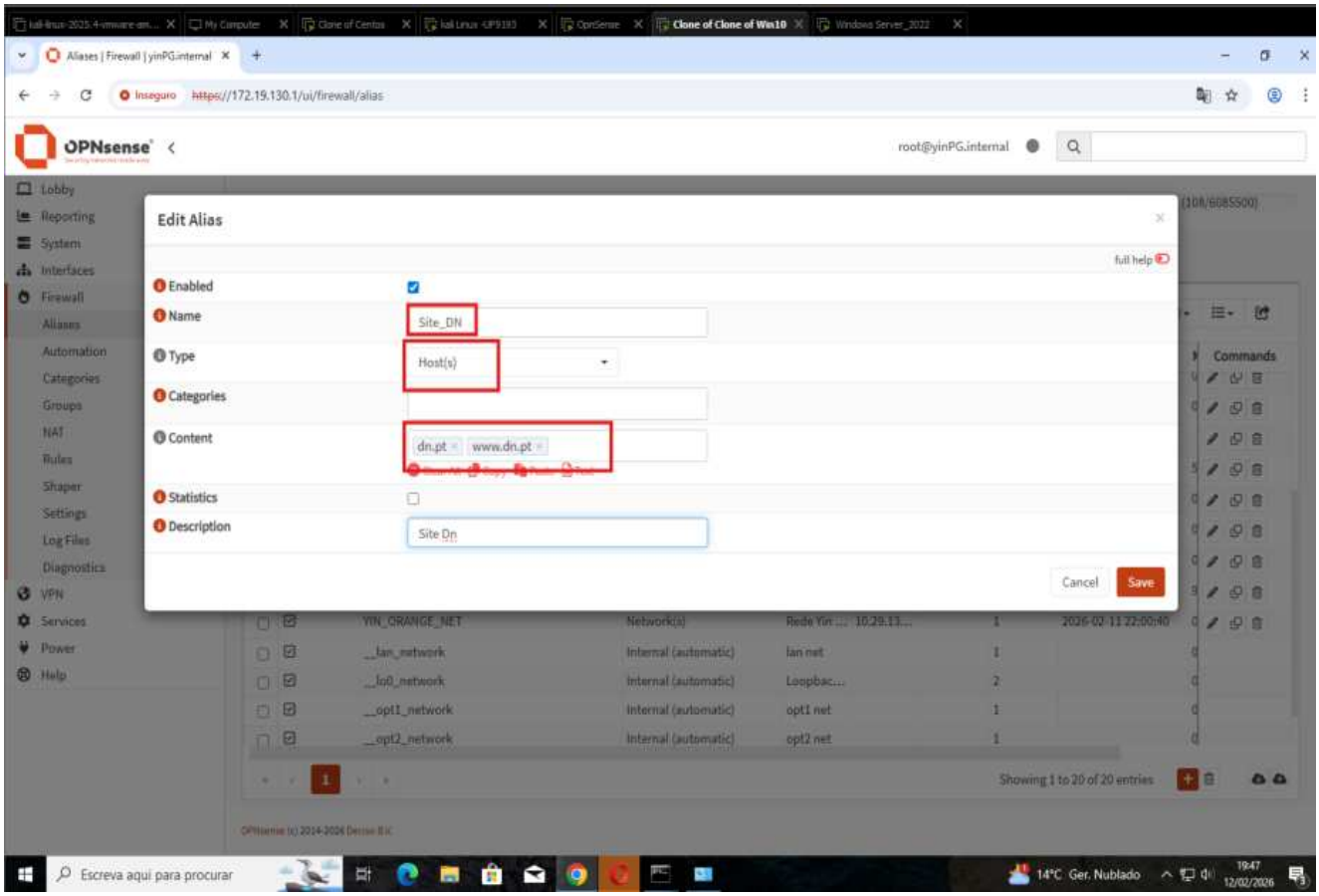
Resumo:



6.2 Criar alias FQDN do DN

Firewall → Aliases → +Add

- Name: Site_DN
- Type: Host(s)
- Adicionar: dn.pt ; www.dn.pt

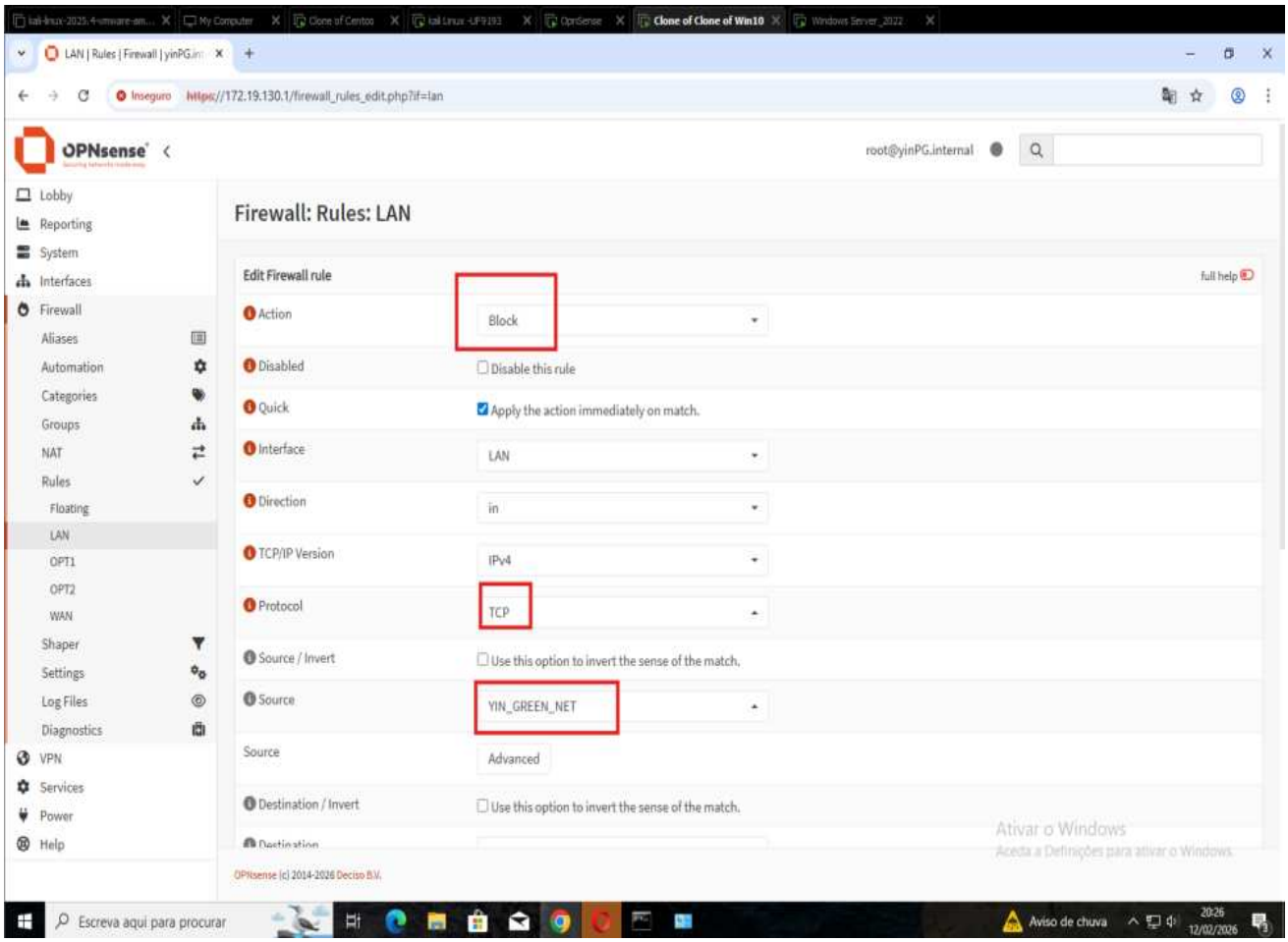


6.3 Regra a bloquear http/https com schedule

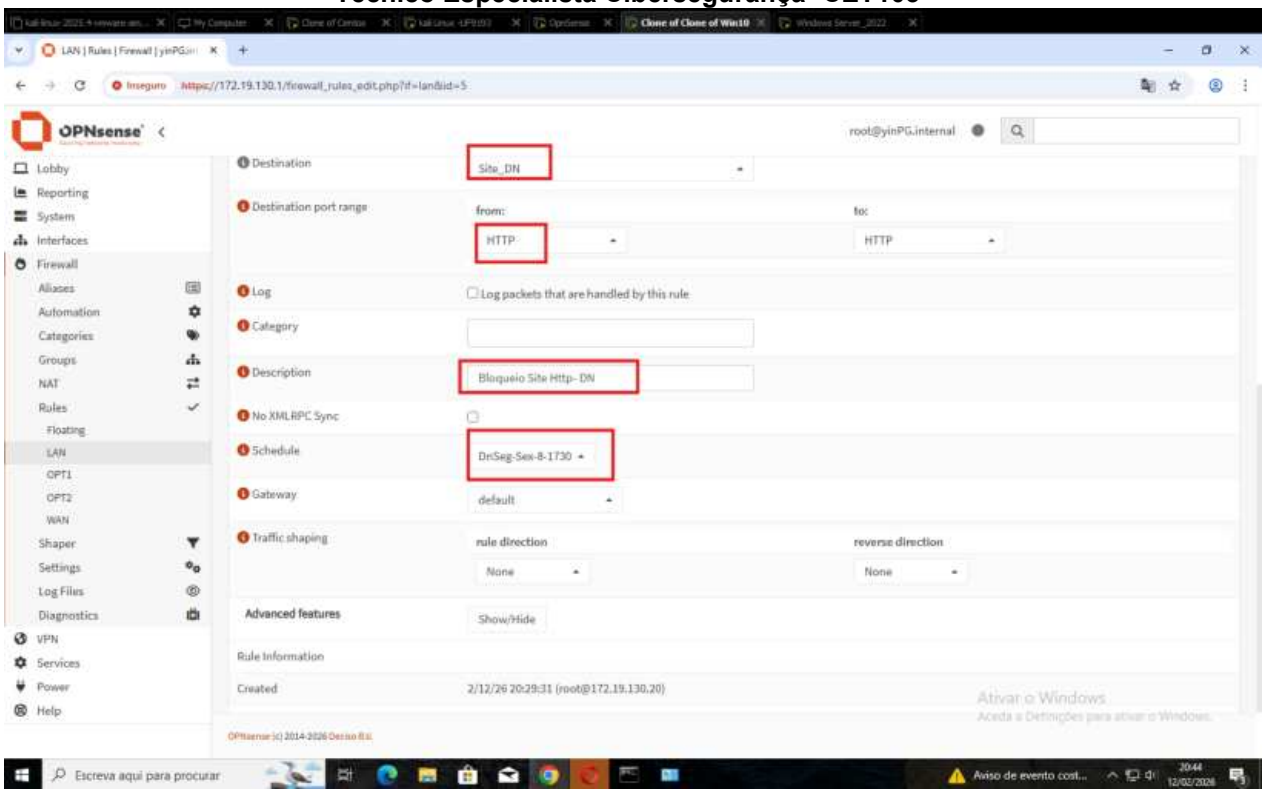
Firewall → Rules → GREEN

- Action: Block
- Protocol: TCP
- Source: YIN_GREEN_NET
- Destination: Site_DN
- Destination port: 80 (HTTP)
- Schedule: DNSeg-Sex-8-1730
- Description: Bloqueio Site http DN

A lógica “clone da regra http para https” e schedules aparece nos teus resumos.
Resumo_detalhado_01

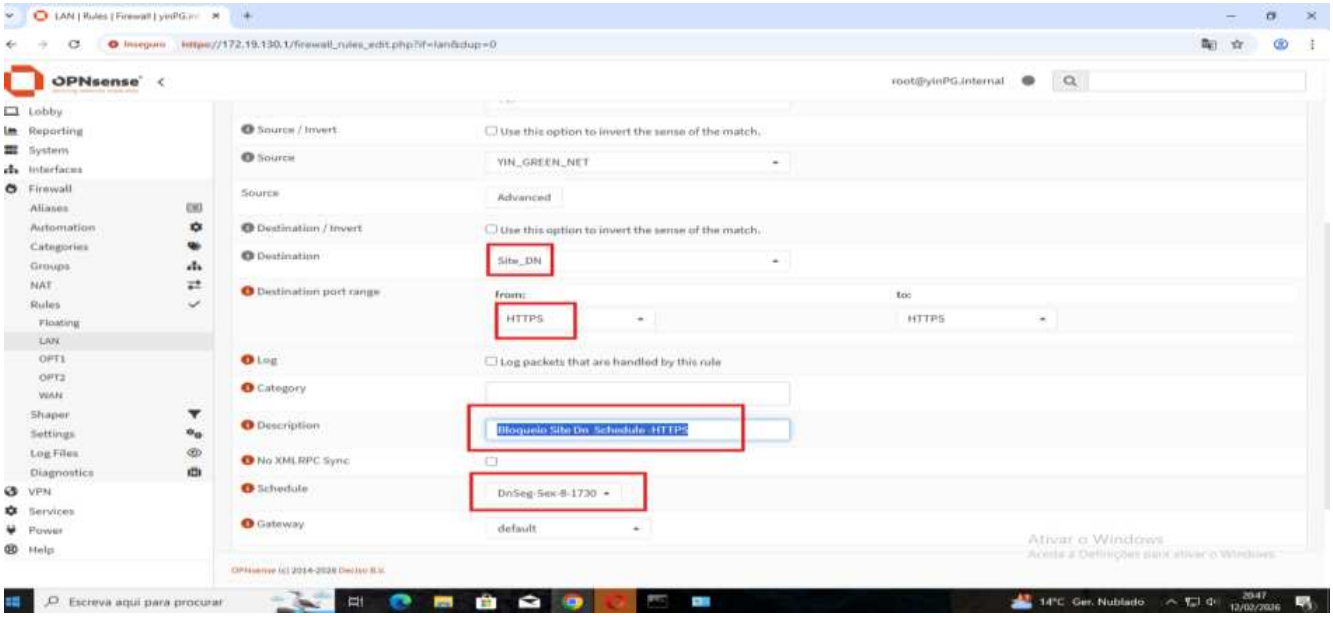


Continuação …

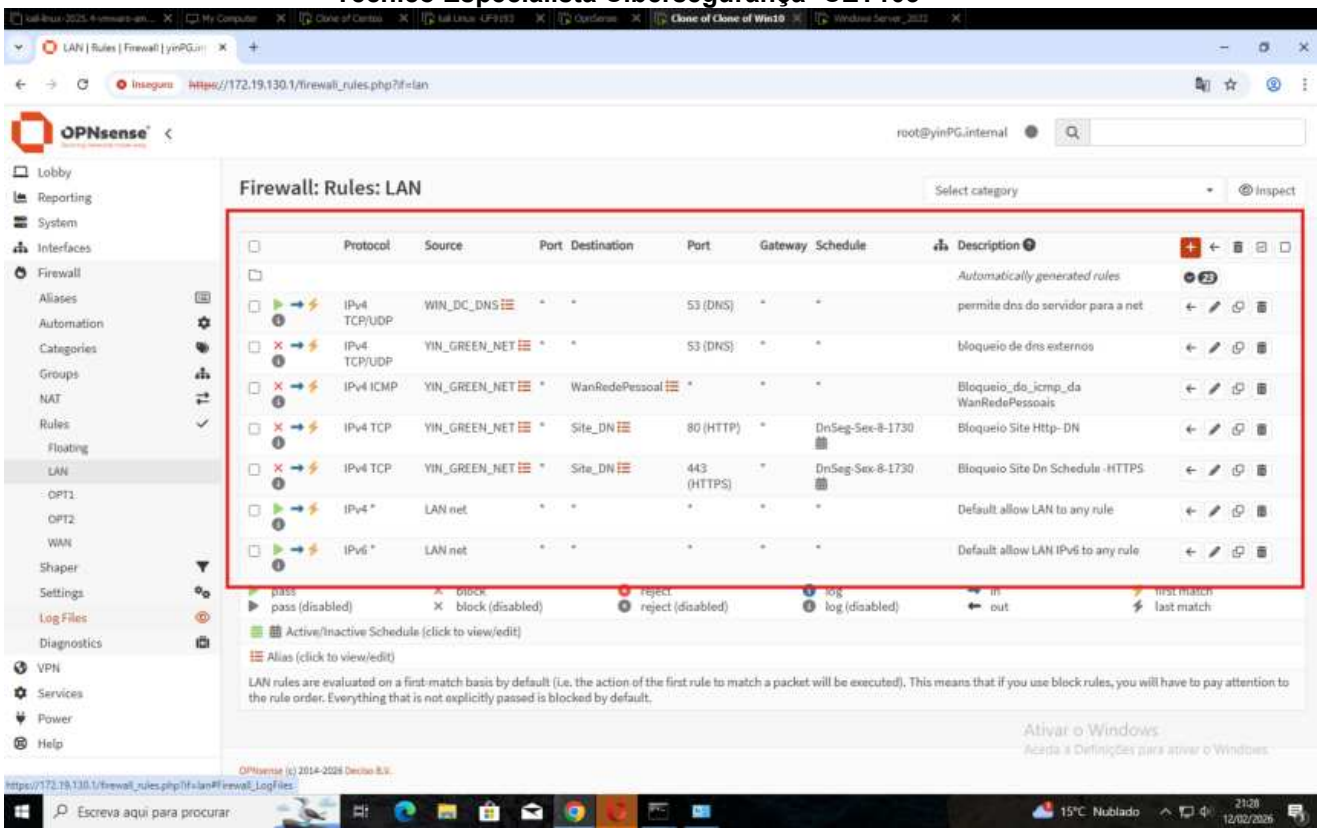


6.4 Duplicar (clone) para HTTPS:

- Destination port: 443
- Description: Bloqueio Site Dn Schedule -HTTPS



Resumo:



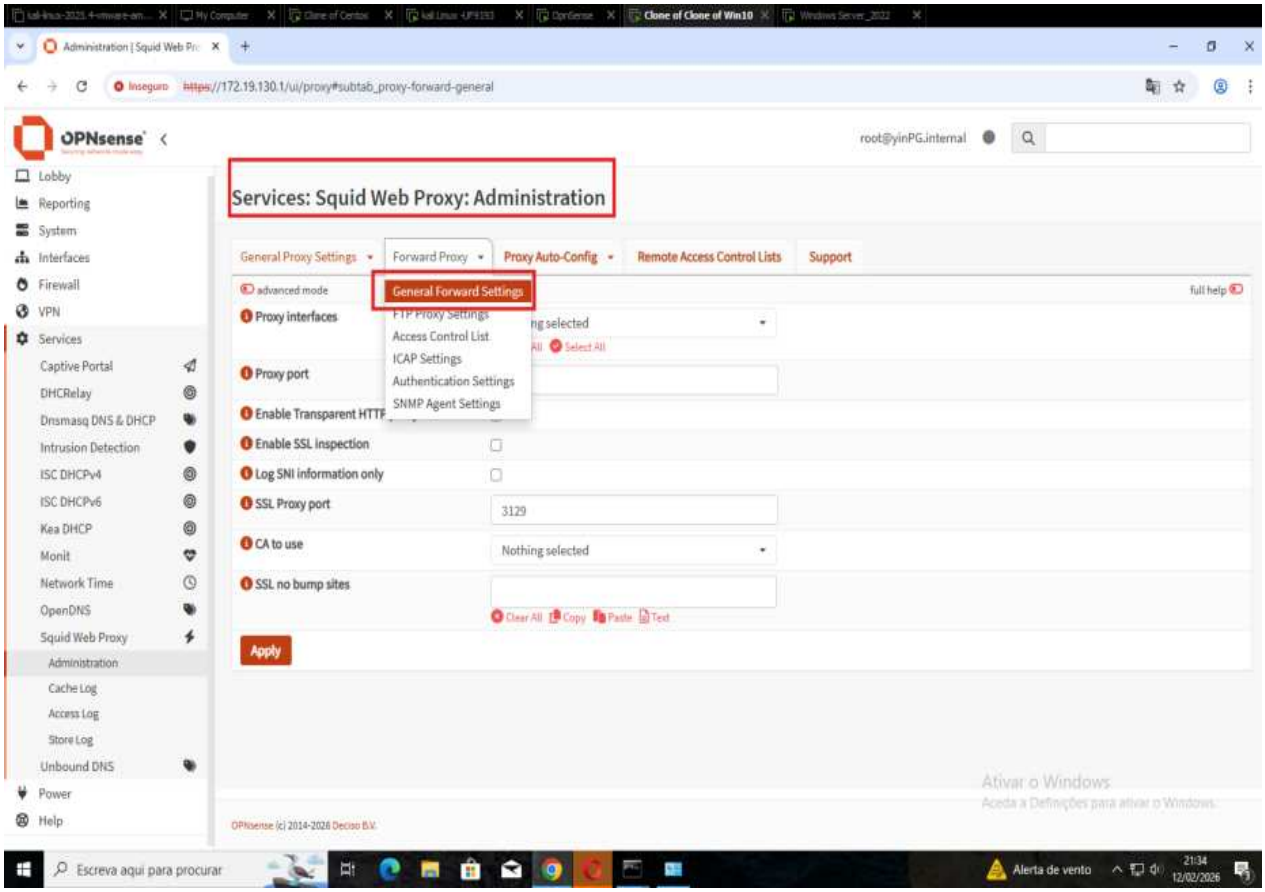
6.5 Teste

- Fora do horário: DN abre



6.6 OPNsense yinPG — Instalar e configurar Proxy Squid (com SSL inspection + blacklists)

- Instalação e configuração do Squid (Services → Squid Web Proxy, forward proxy, regras)

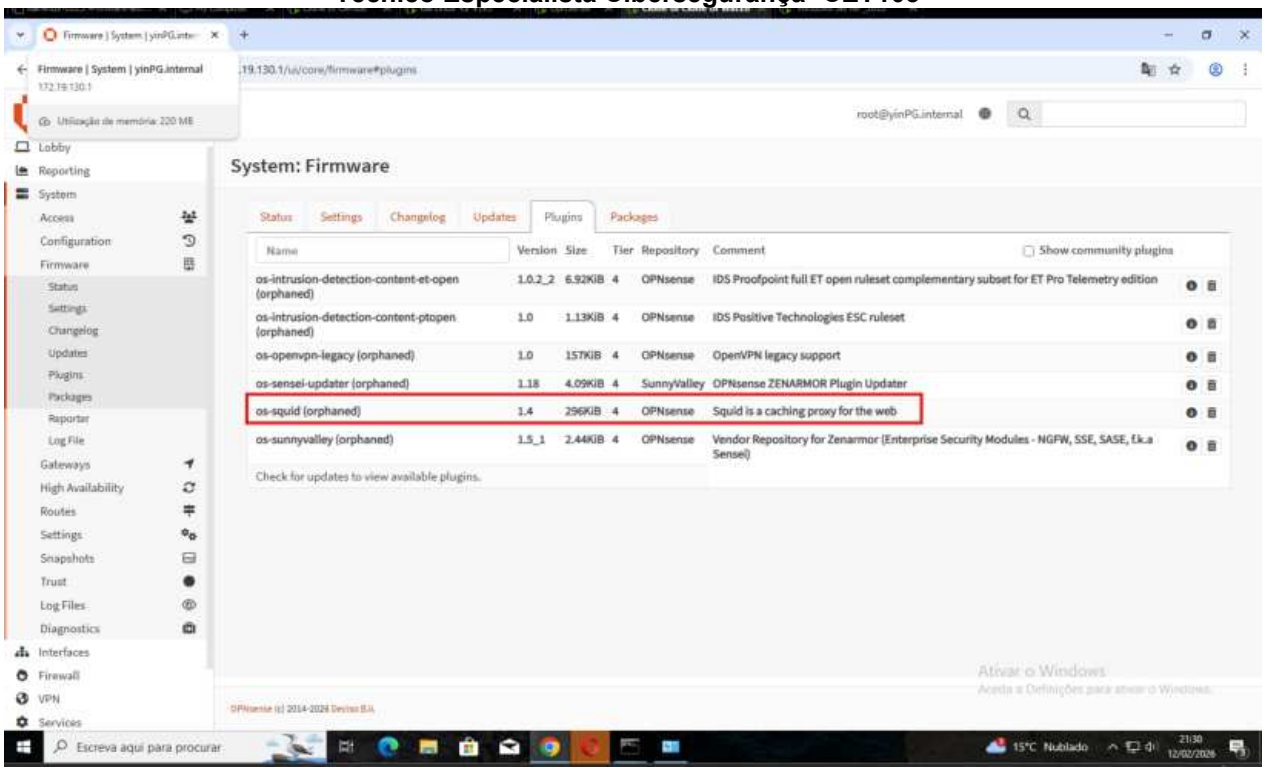


- Criação de CA “self sign” e importação no Windows via mmc
- Download e aplicação de blacklists UT Capitole e seleção de categorias (adult/bitcoin, etc.)

6.7 Instalar plugin

System → Firmware → Plugins

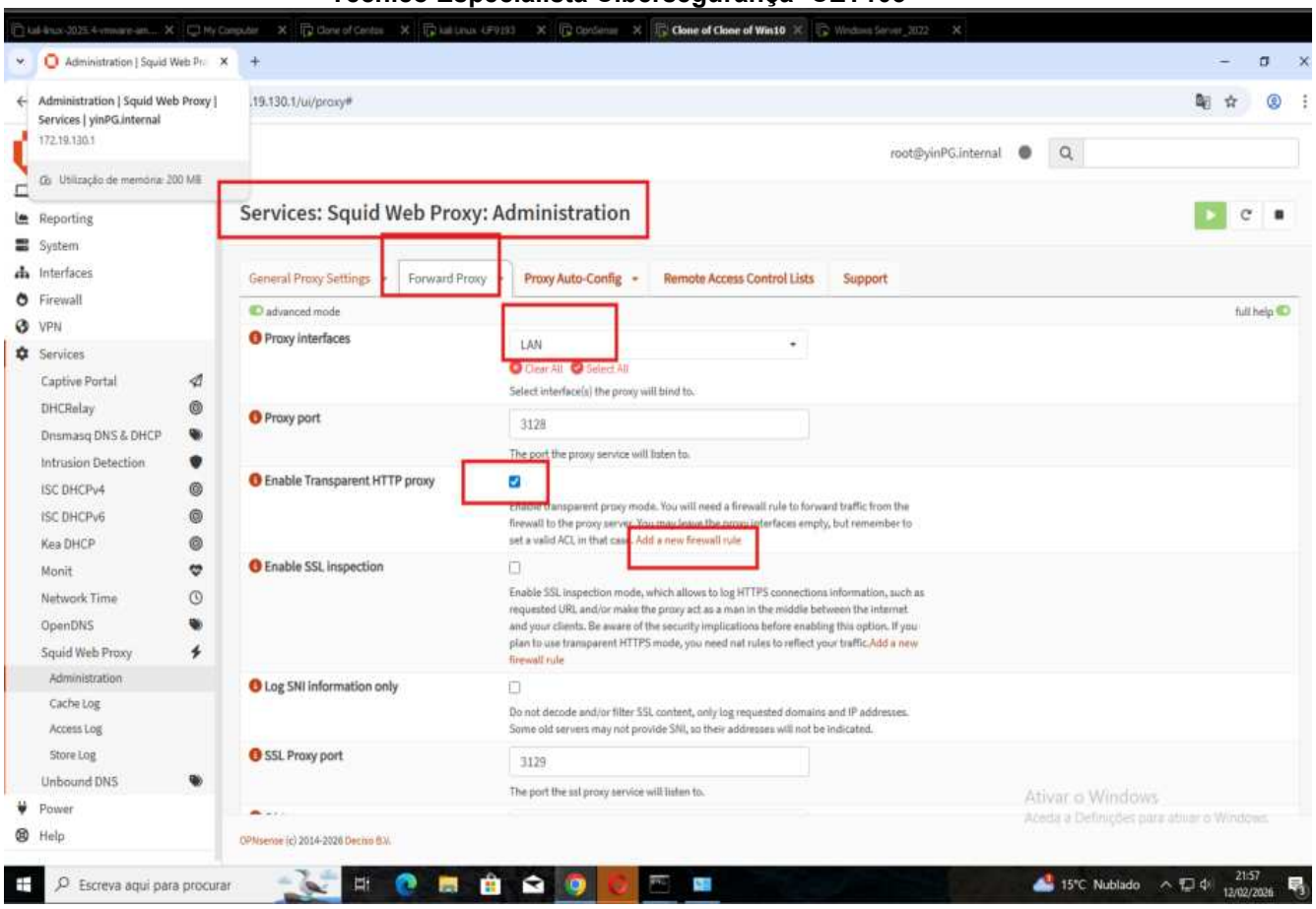
- Instalar: **os-squid** (Squid Web Proxy)



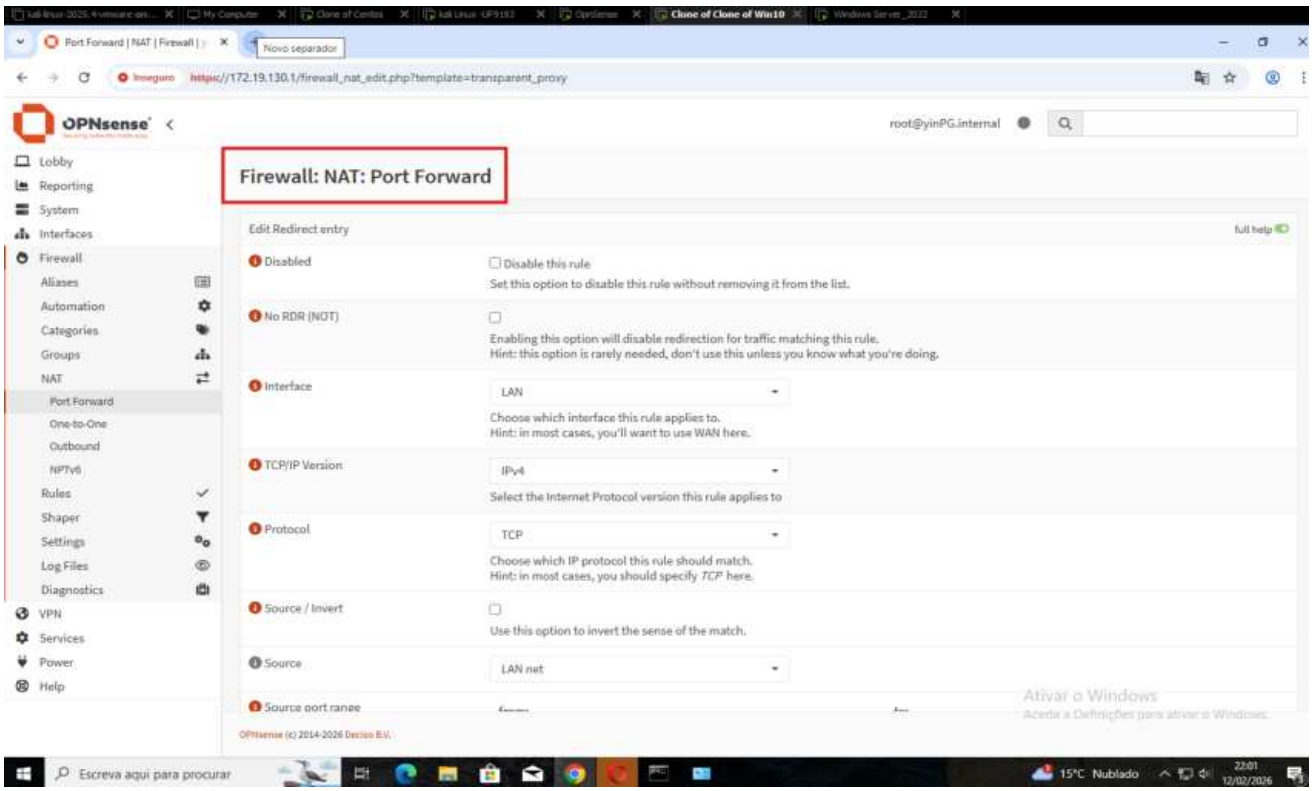
6.8 Ativar Forward Proxy (modo transparente)

Services → Web Proxy → Administration (ou “Squid Web Proxy”)

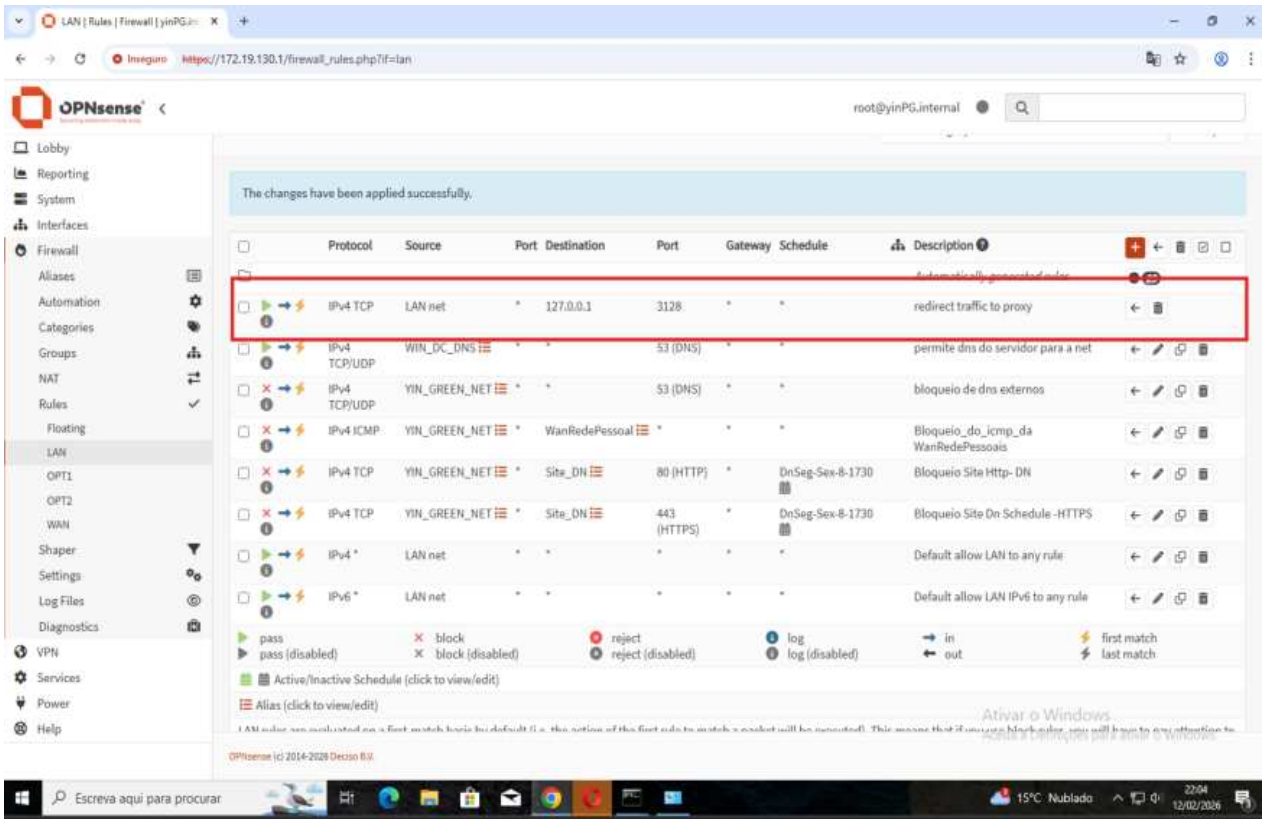
- Enable proxy: sim
- Transparent mode / Intercept (HTTP): sim
- Proxy interface(s): selecionar:
- **GREEN- “Lan “**



Continuação …

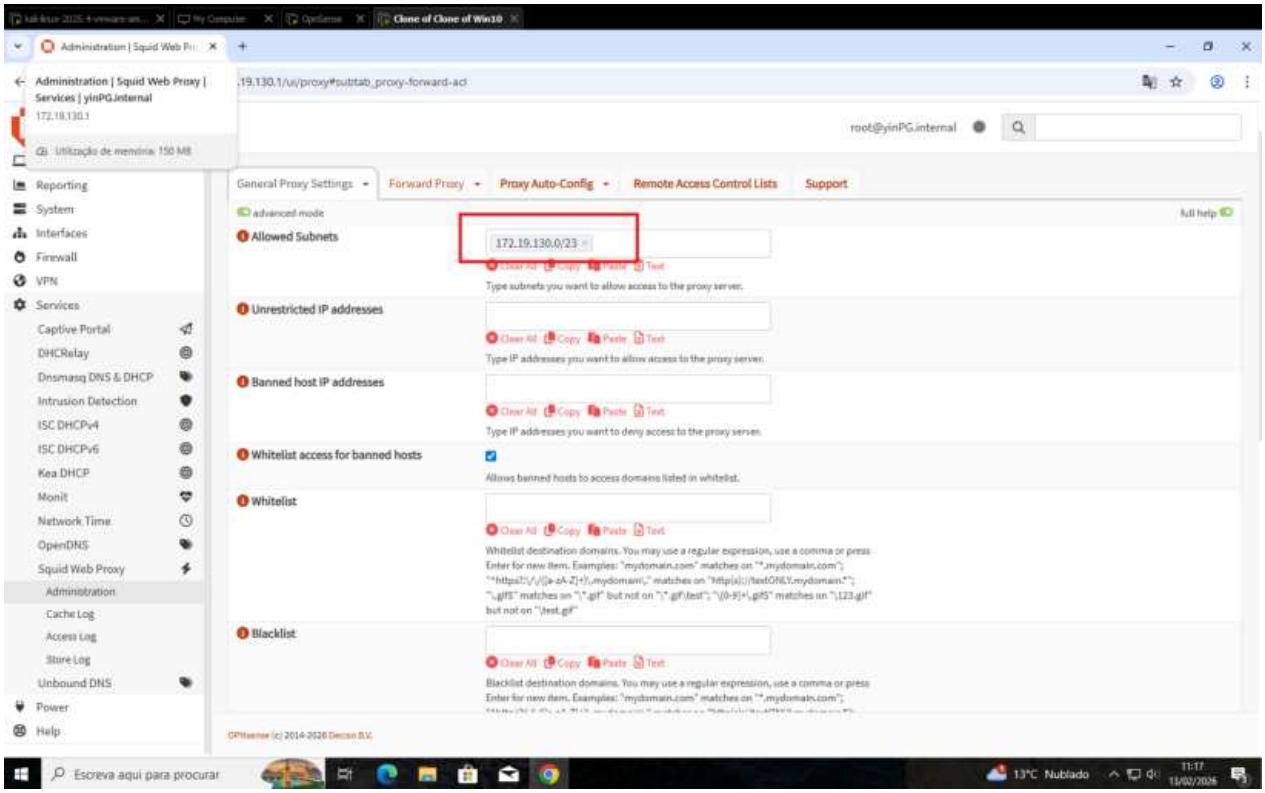


6.9 Colocar a regra na primeira posição – Firewall LAN



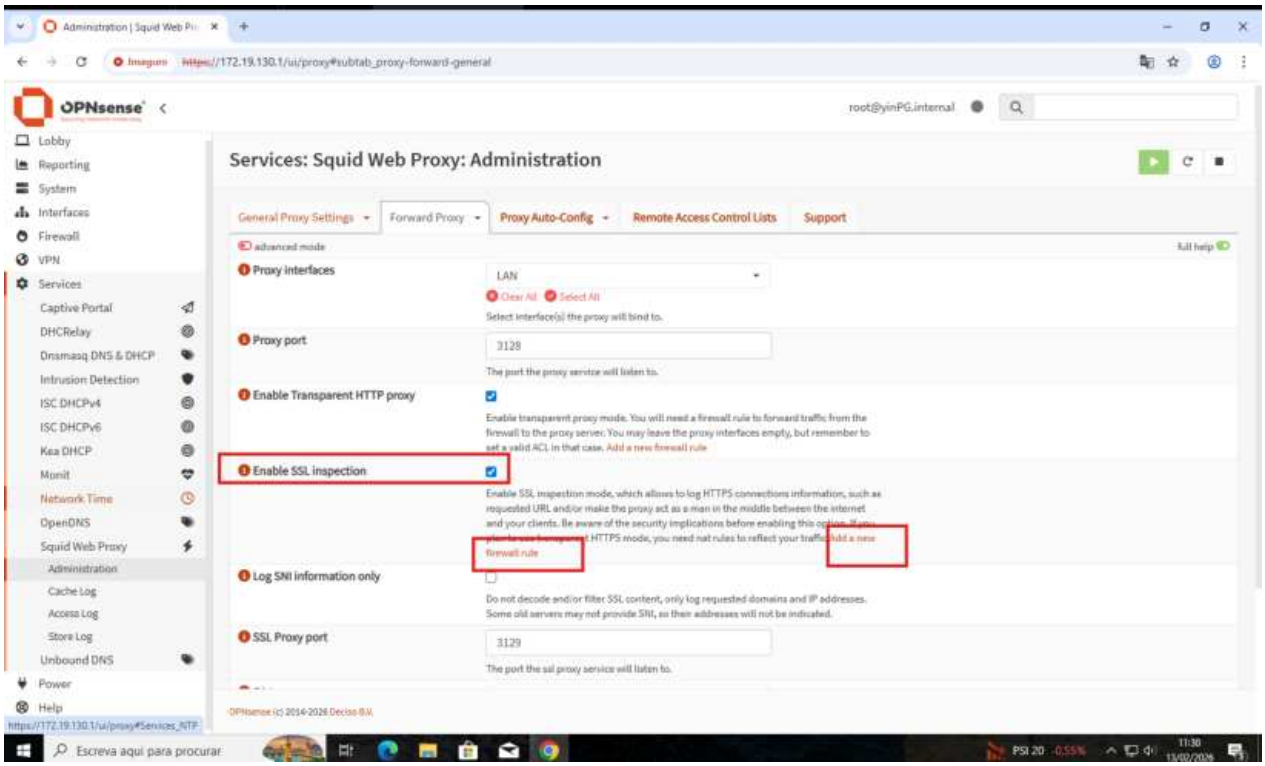
Nota seguidamente : **OpenVPN** e **WireGuard**, para aplicar as mesmas regras aos clientes VPN)

6.10 Allowed subnets: 172.19.130.0/23 (GREEN)



6.11 SSL inspection (para filtrar HTTPS por domínio)

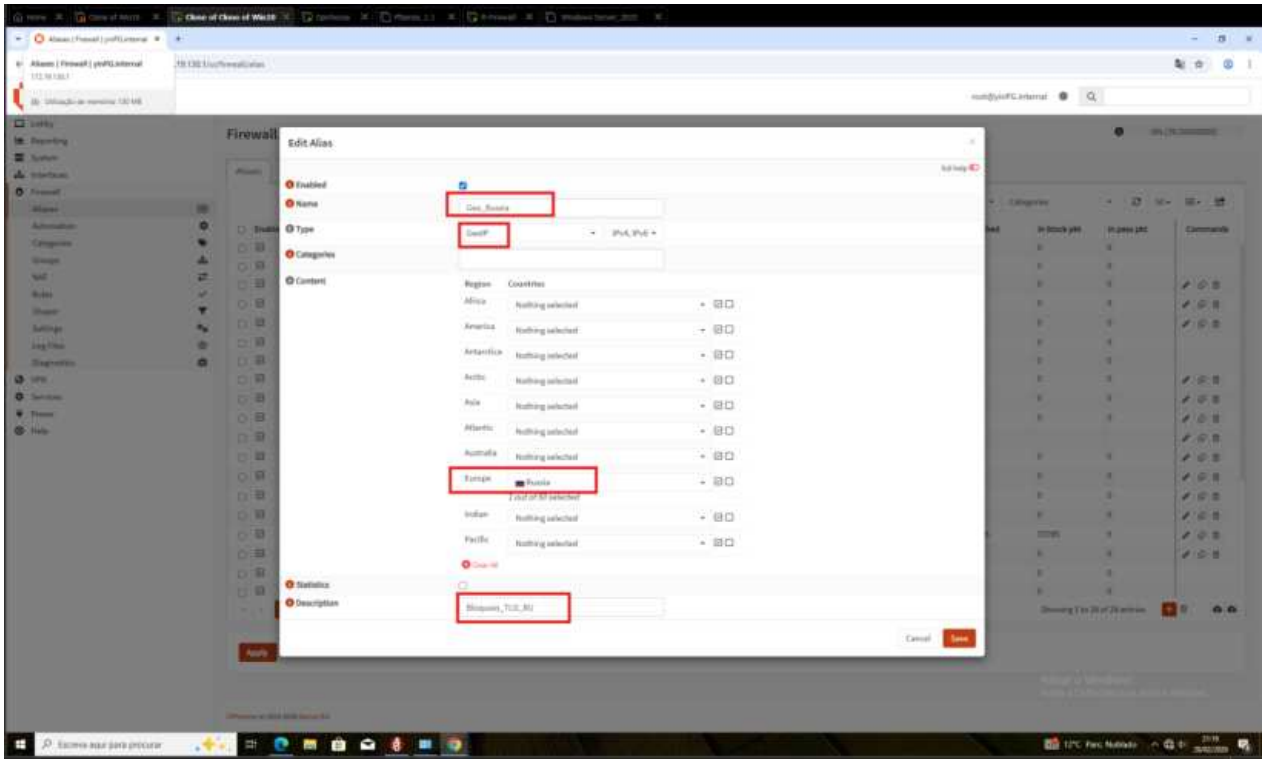
Bloqueio Webmail (HTTPS) e .ru



2 - h) Bloquear acesso a sites TLD de domínio da Russia a partir da zona green (.ru ◊Ex: www.nomesite.ru)

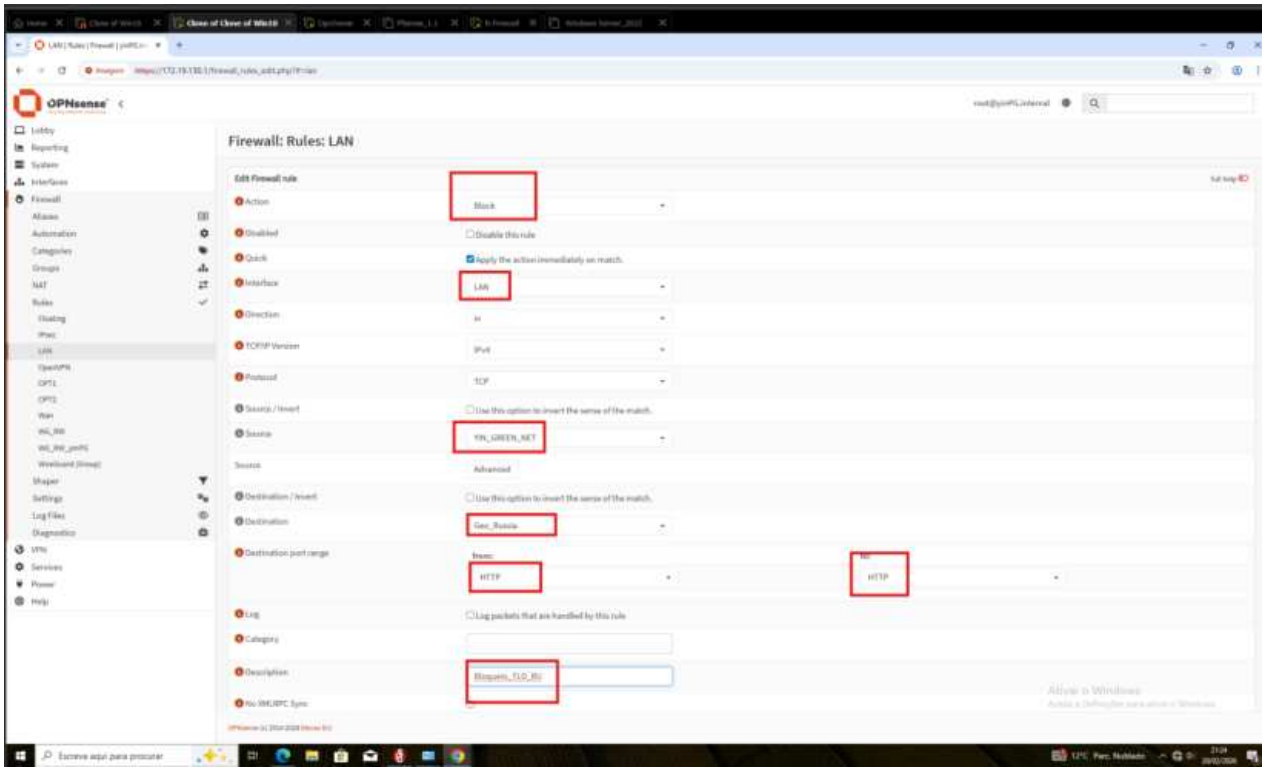
7- Bloqueio de Site Russo :"**www.yandex.ru" - 95.213.56.1**

Criar Alias GeoIP

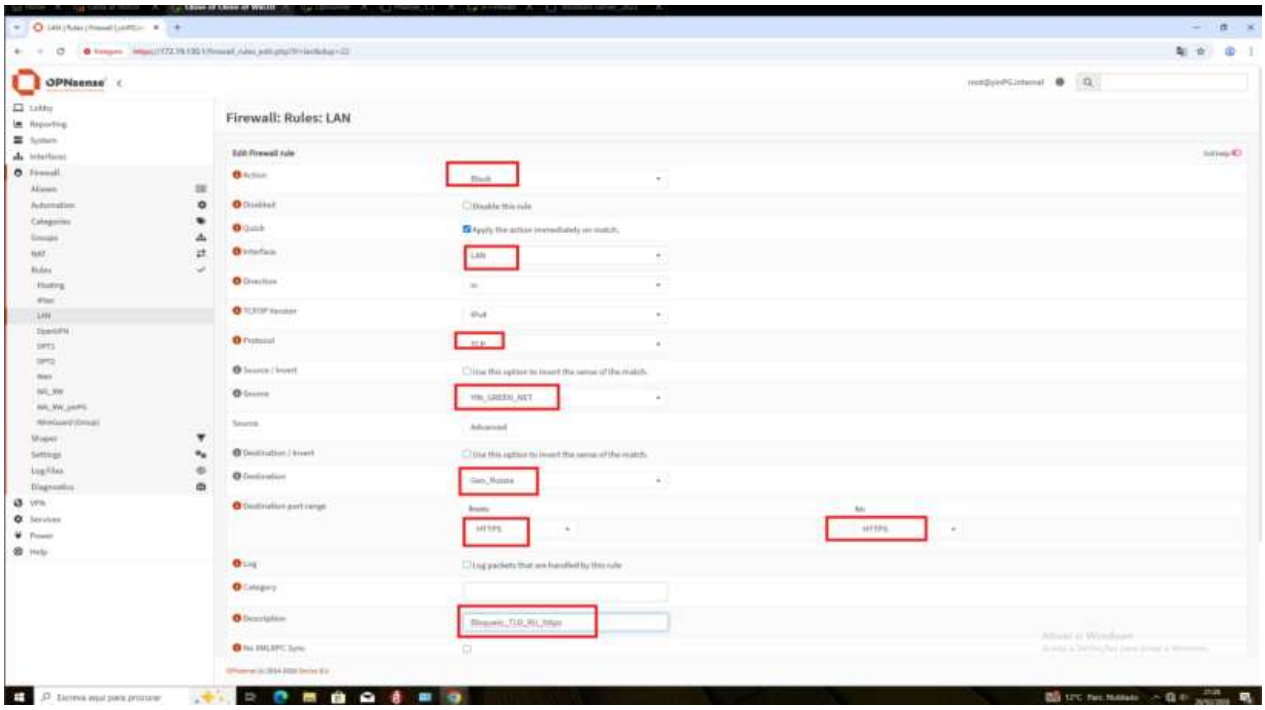


7.1 Firewall → Rules → - Lan

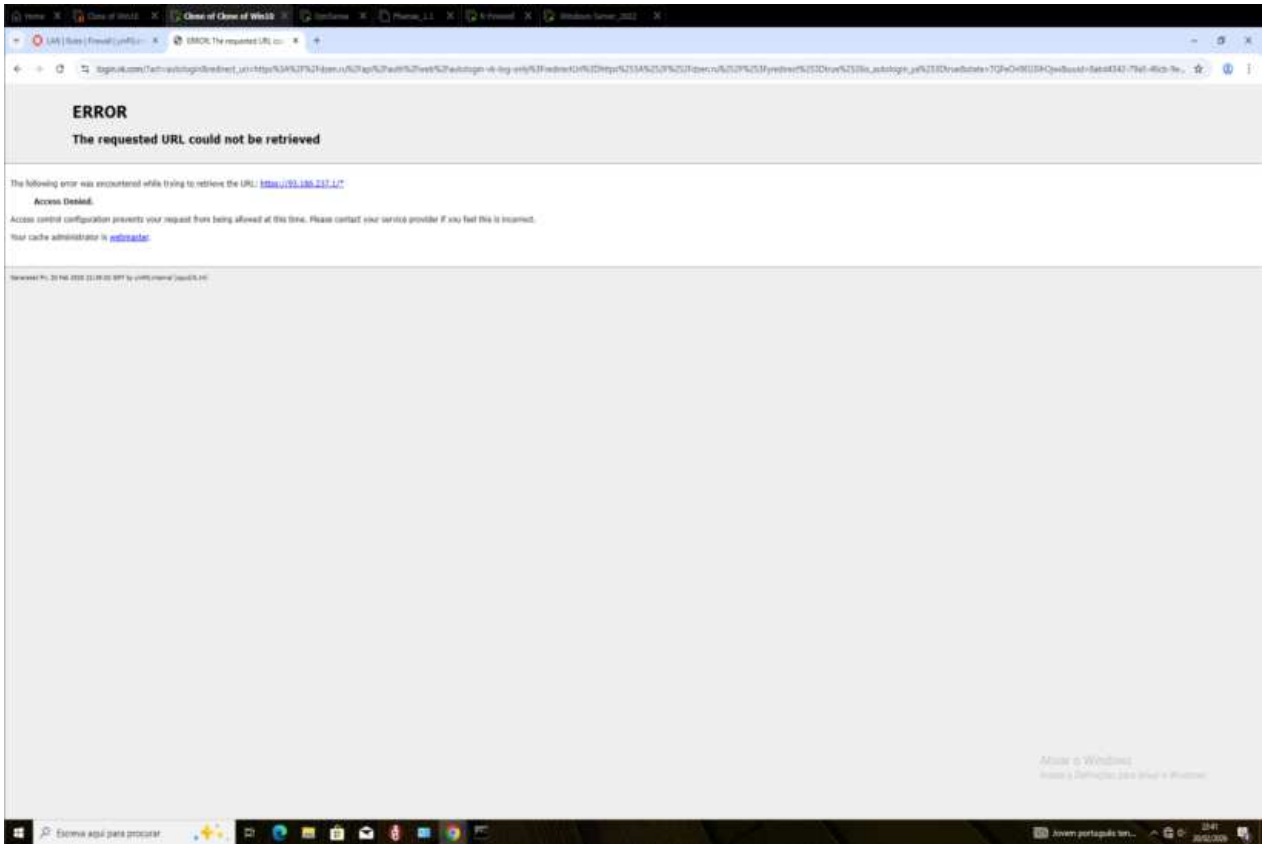
Bloqueio http



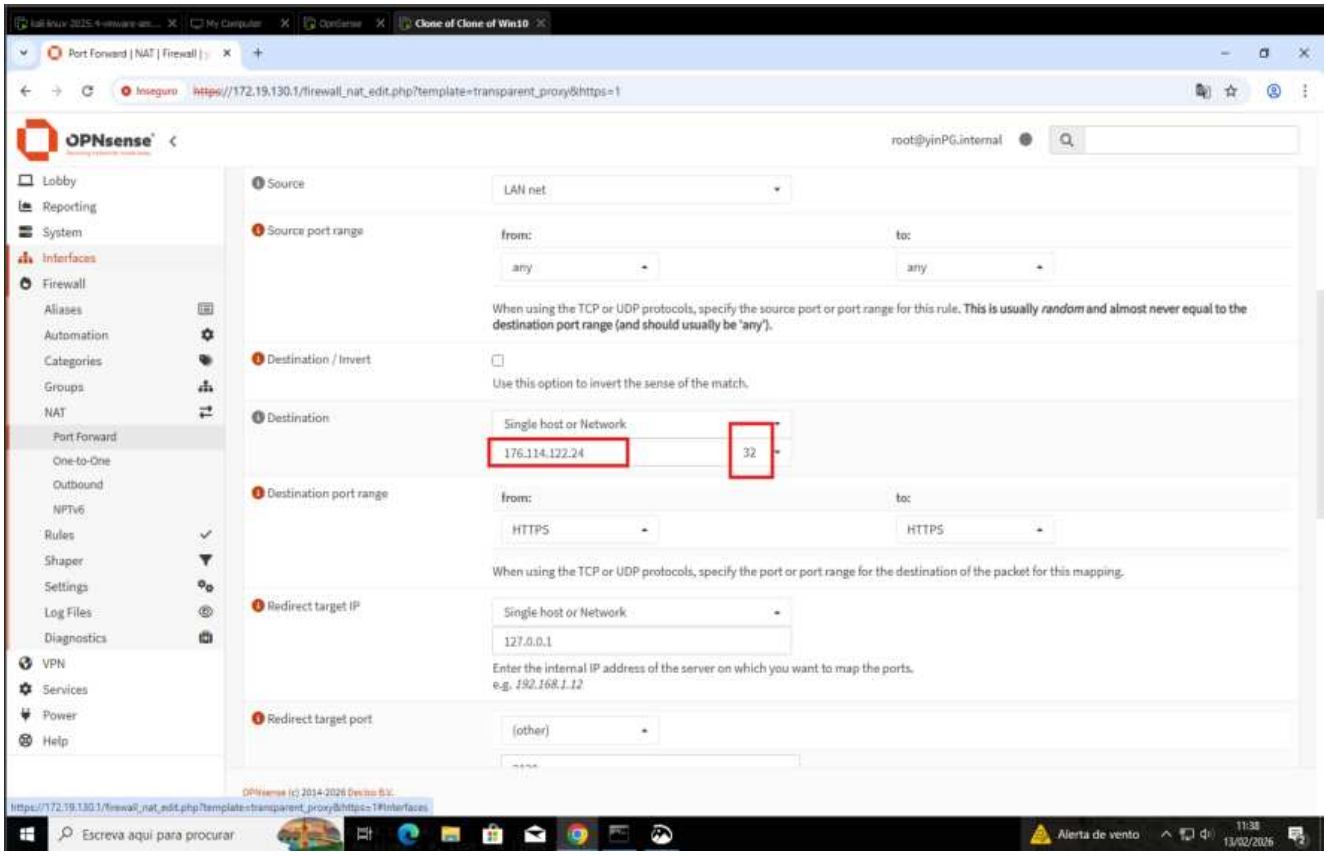
7.2 Bloqueio https



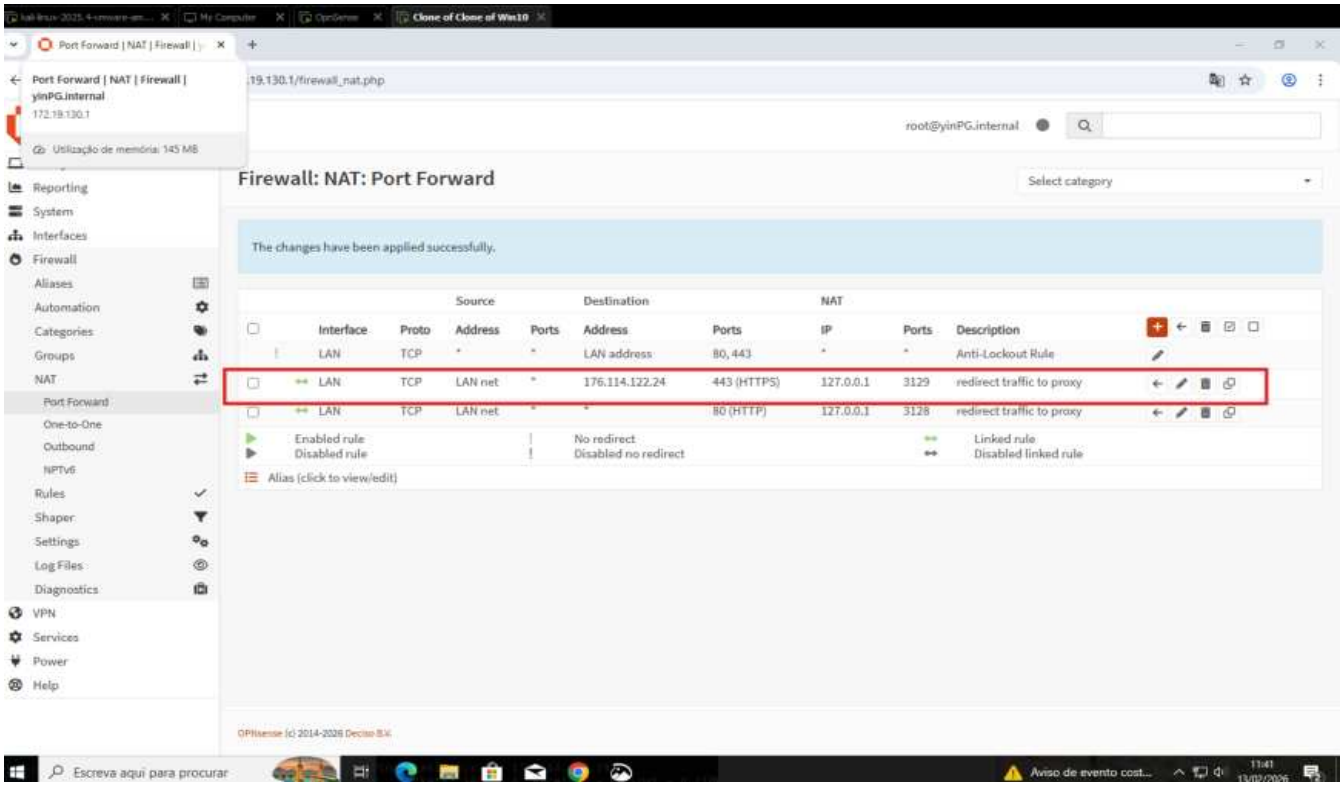
7.3 Bloqueio



Continuação..



Resumo :

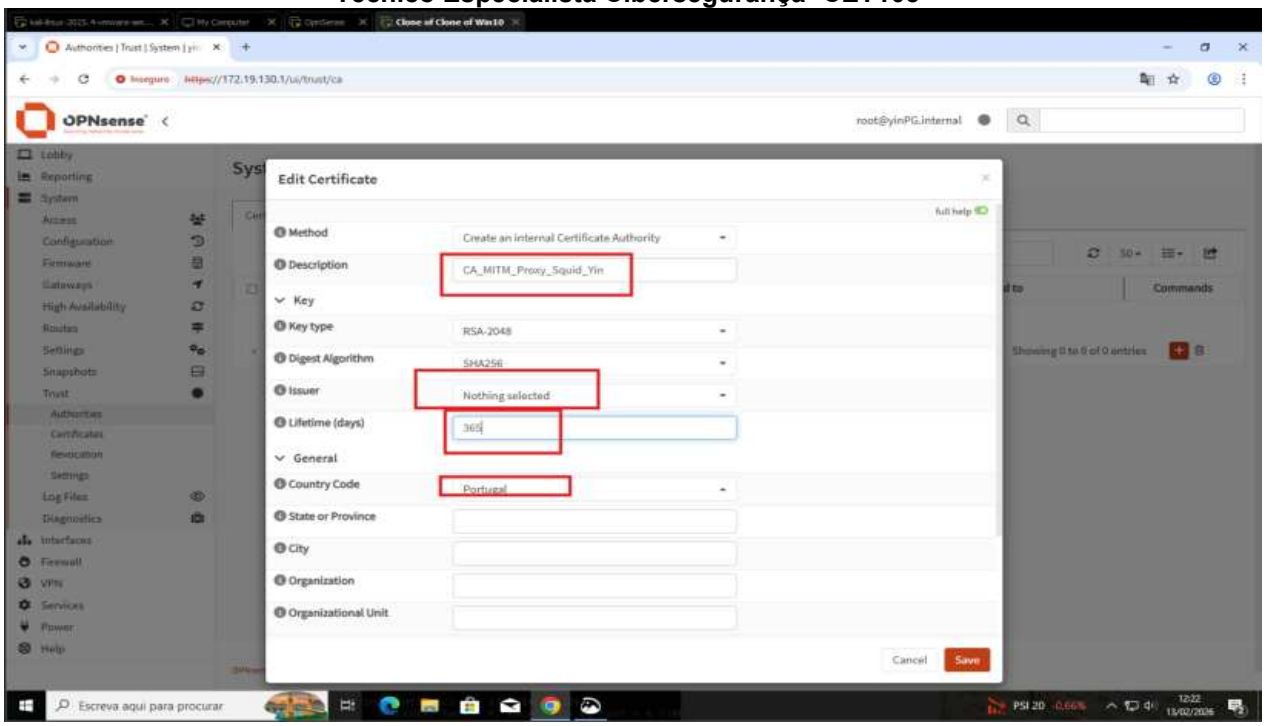


7.3 Criar CA (Authority) no OPNsense

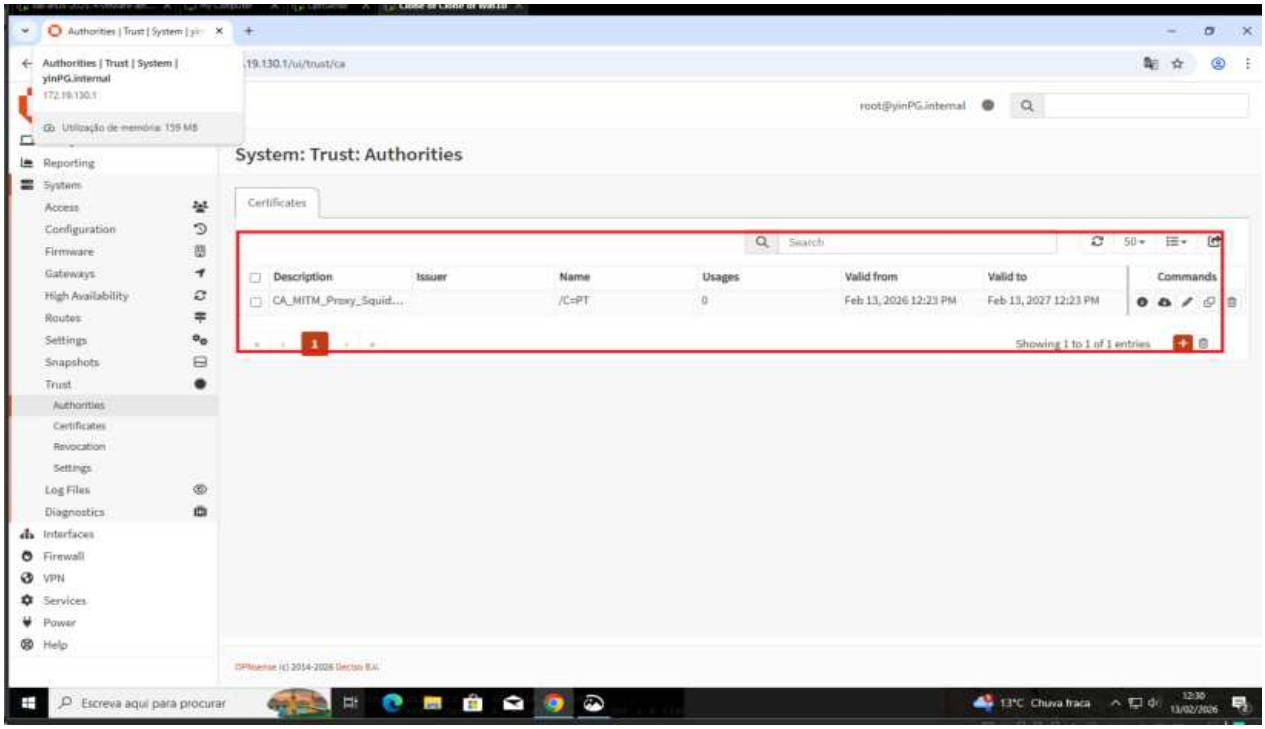
System → Trust → Authorities → +Add

Method: Create internal Certificate Authority

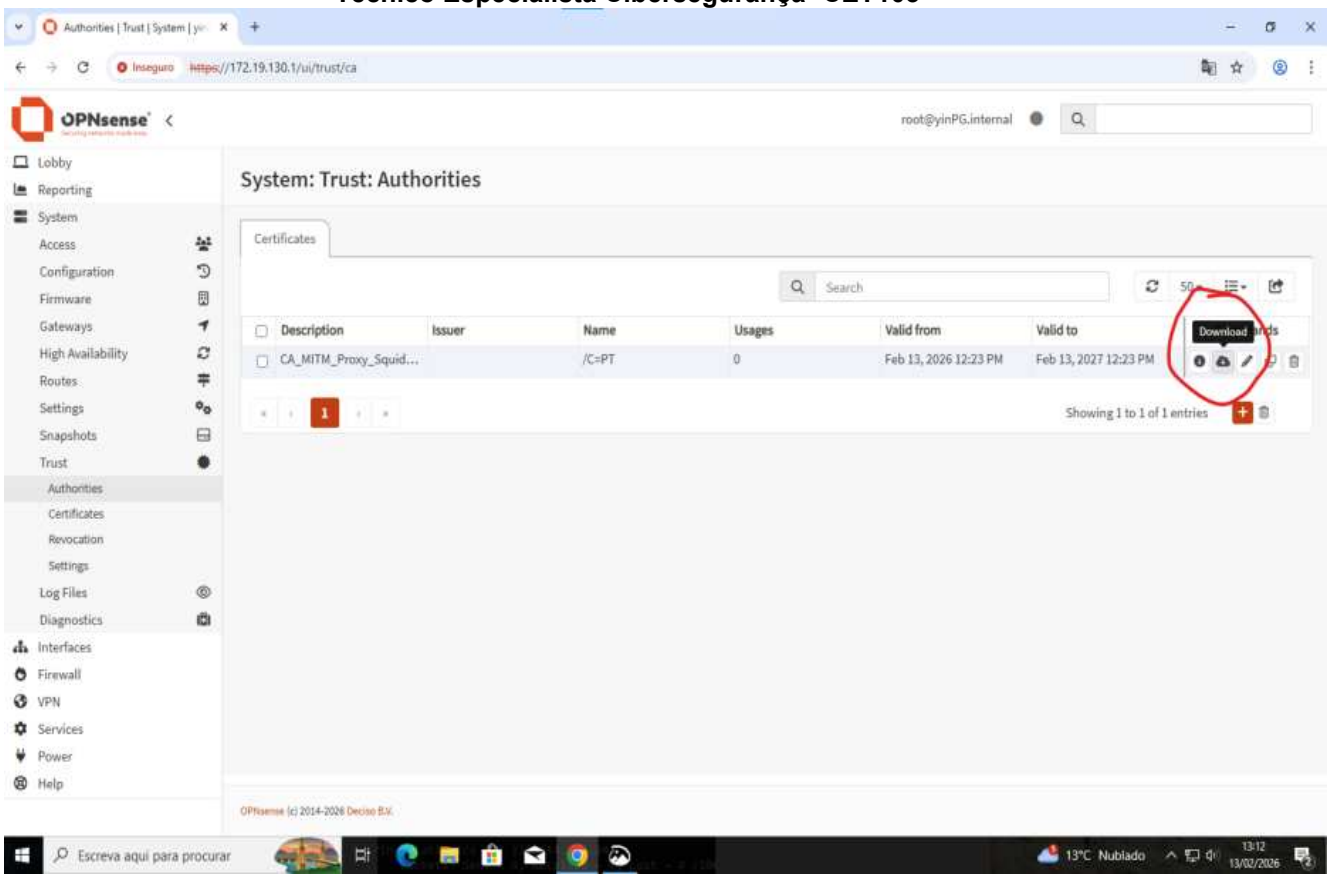
- Descrição: CA_MITM_Proxy_Squid_Yin
- Key length: 2048
- Lifetime: ex 3650
- Save



Resumo :



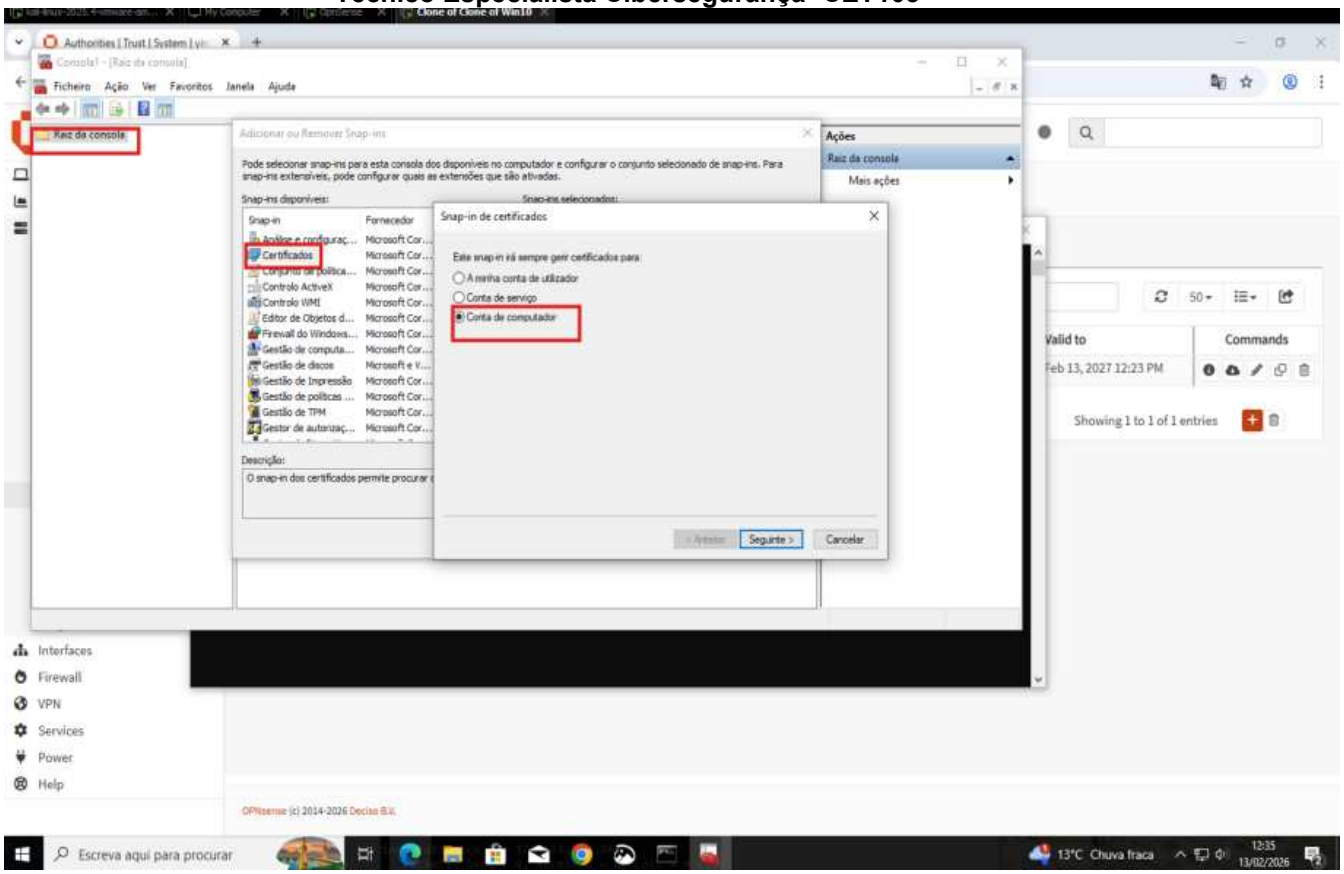
7.4 Download do certificado :



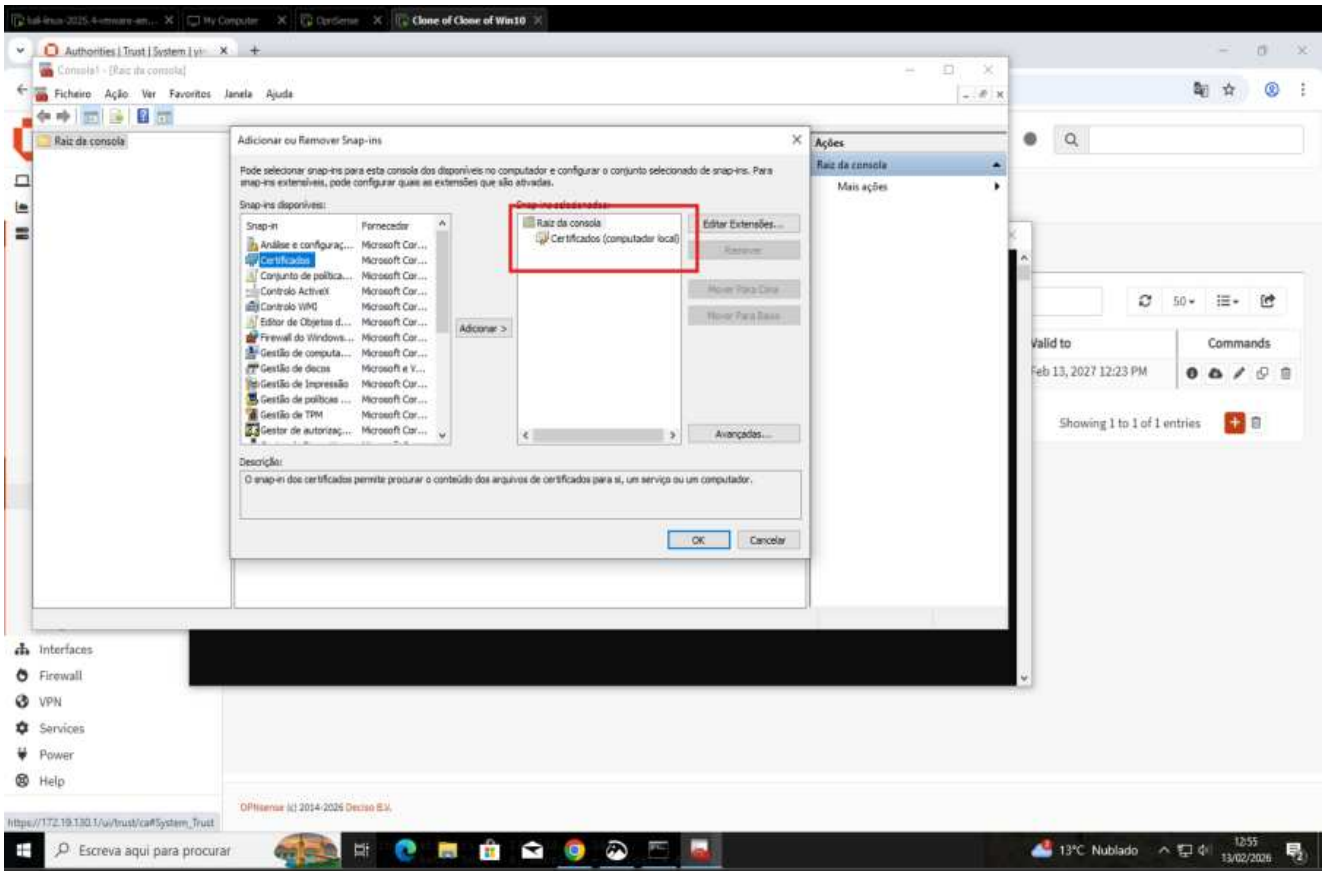
7.5 Exportar o certificado CA e instalar no Windows 10

System → Trust → Authorities → (CA_Squid_Inspection_yinPG) → Export (CRT)
No Windows 10:

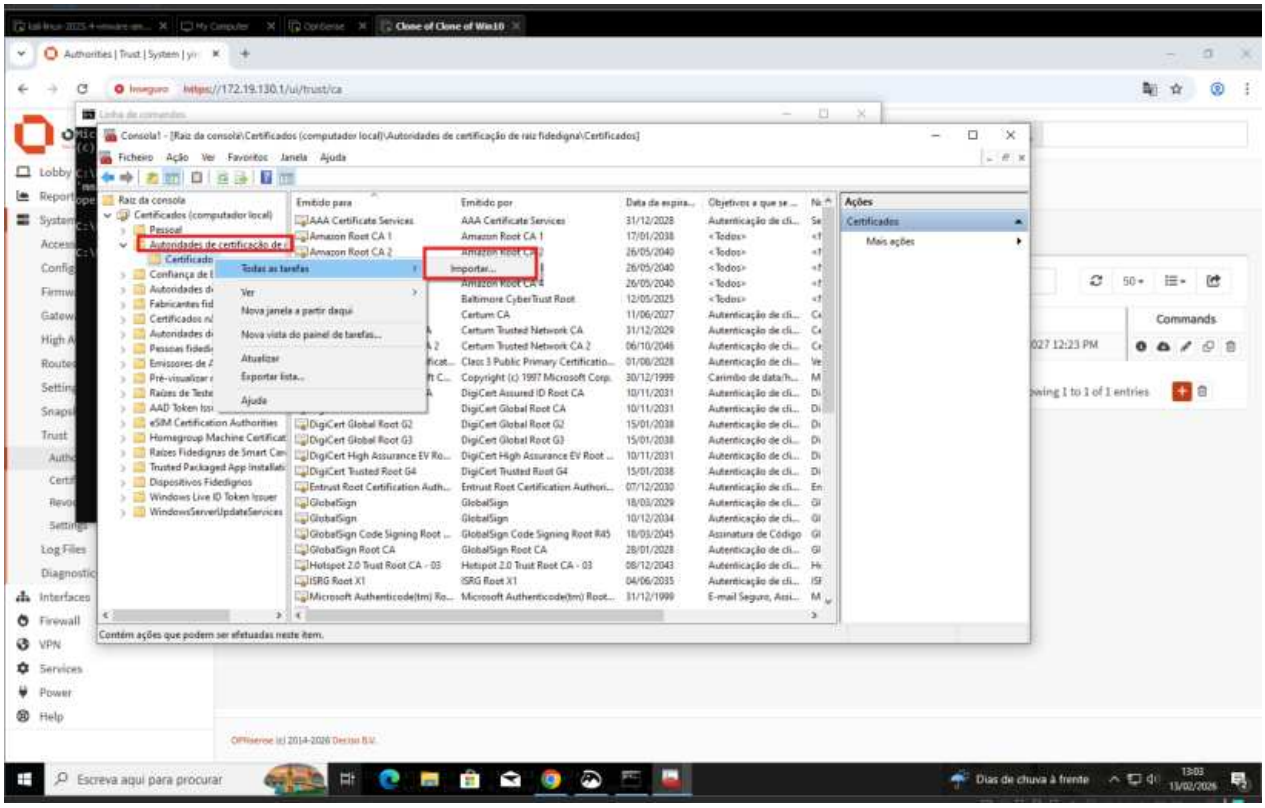
1. Win + R → mmc
2. File → Add/Remove Snap-in
3. Certificates → Add → **Computer account** → Local computer



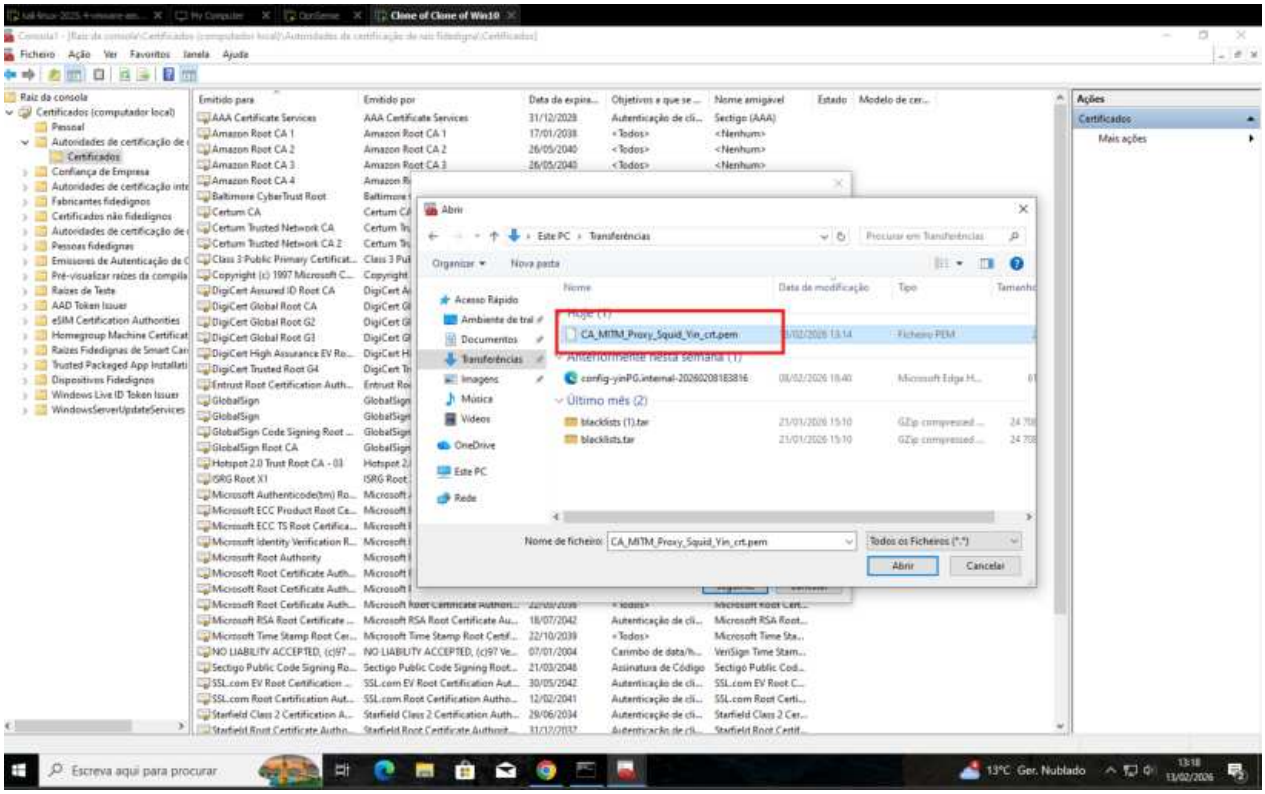
Continuação…



7.6 Trusted Root Certification Authorities → Certificates → Import



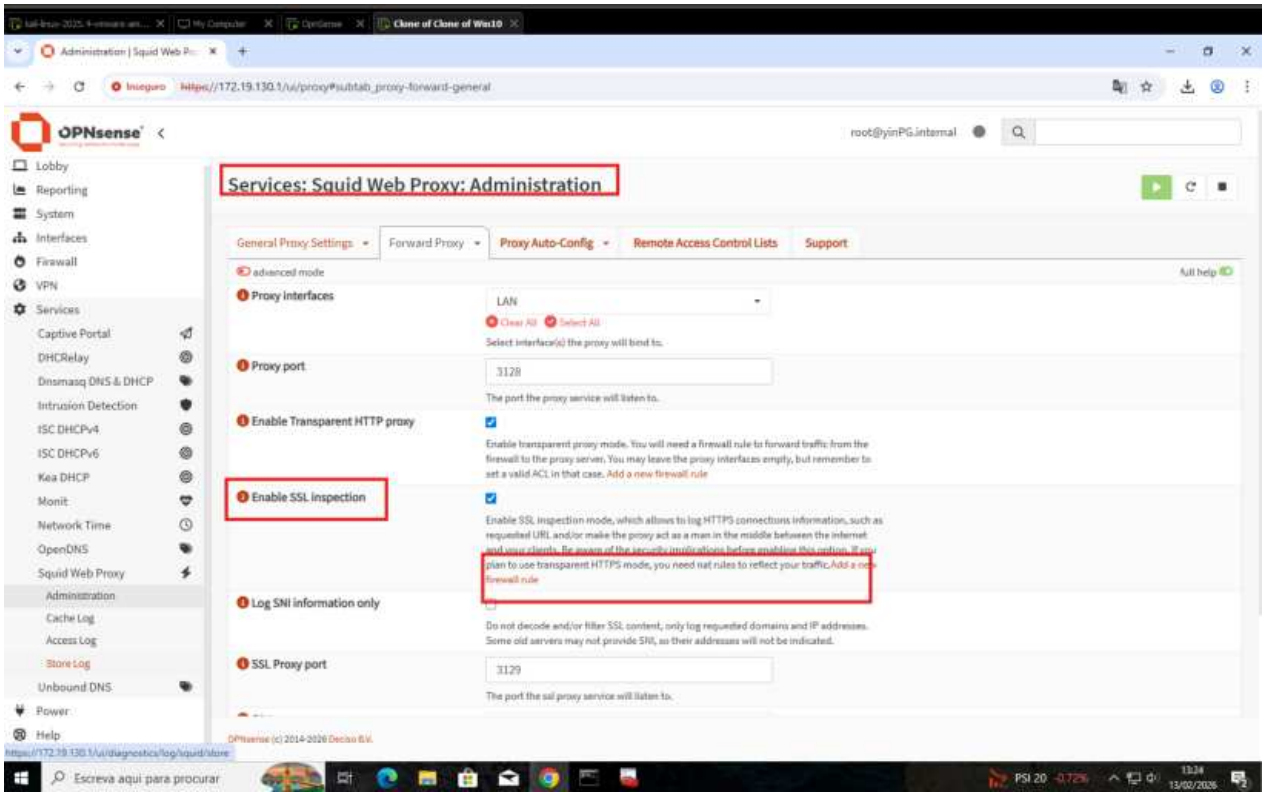
7.7 Importar o .crt do OPNsense.



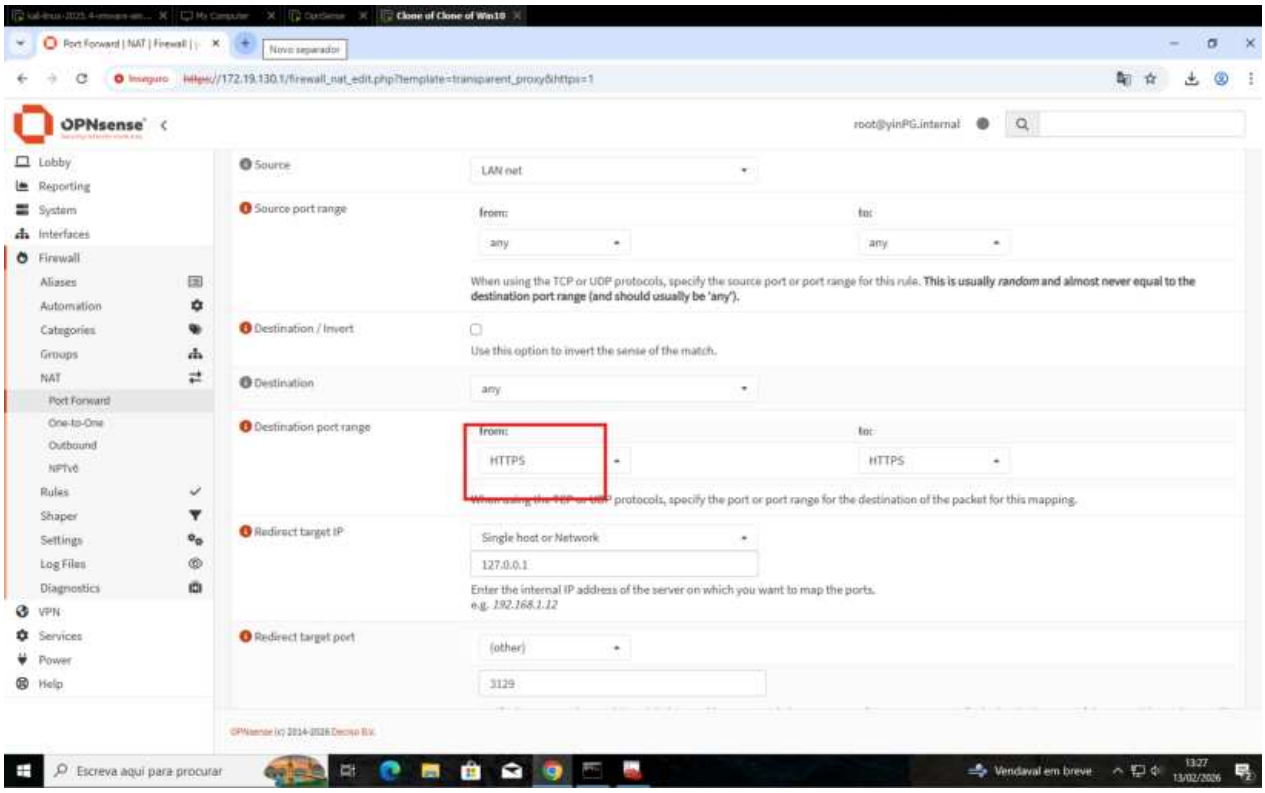
7.8 Ativar SSL inspection no Squid

Services → Web Proxy → Forward Proxy → SSL

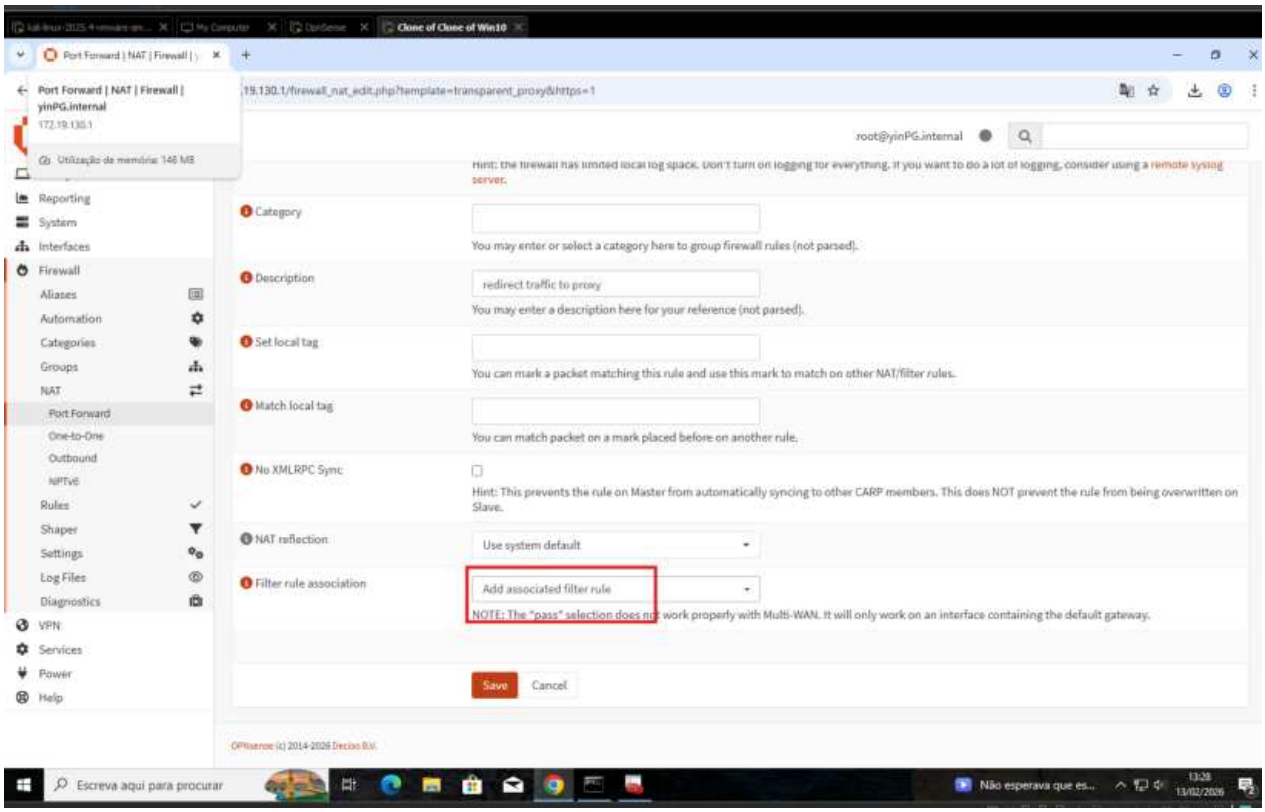
- Enable SSL inspection / HTTPS Intercept:



- SSL Port: 3129
- Intercept SSL on port 443

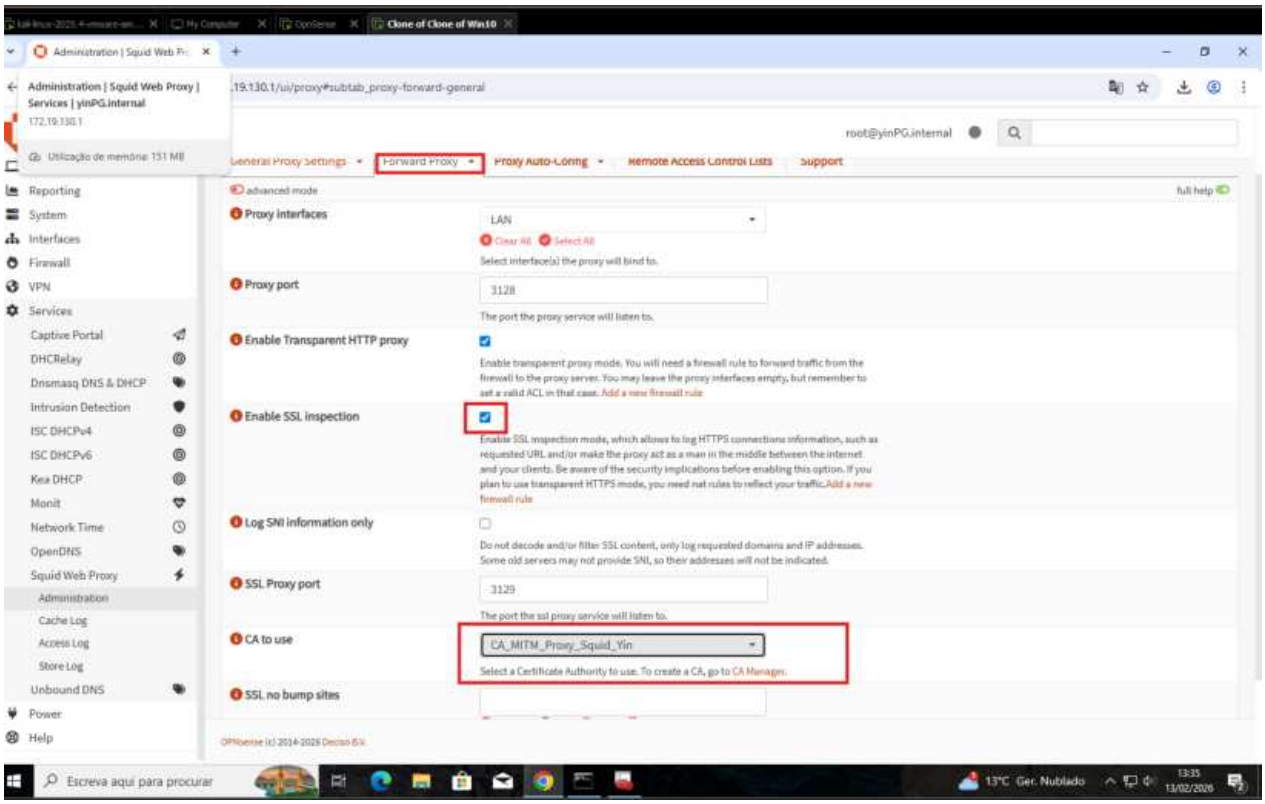


Continuação …

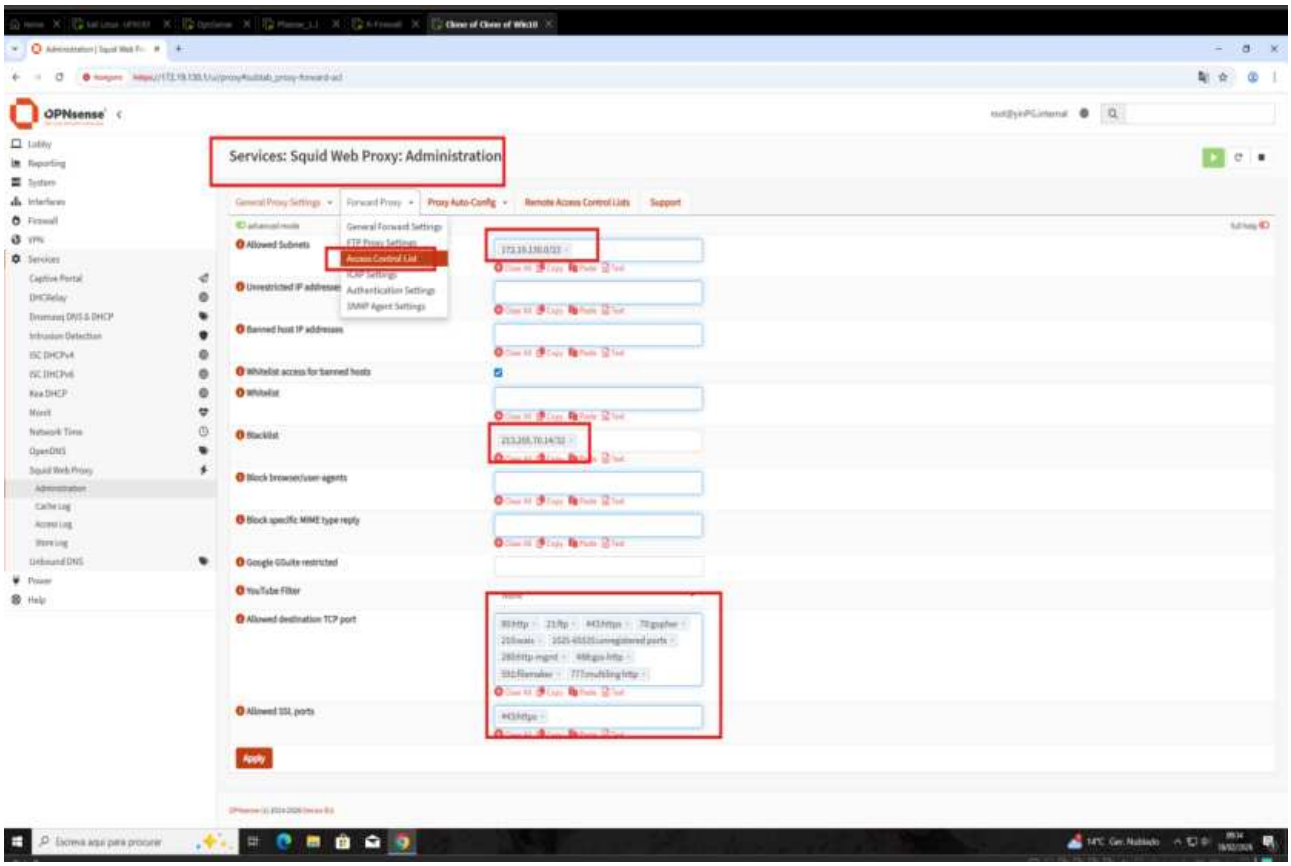


Apply + restart.

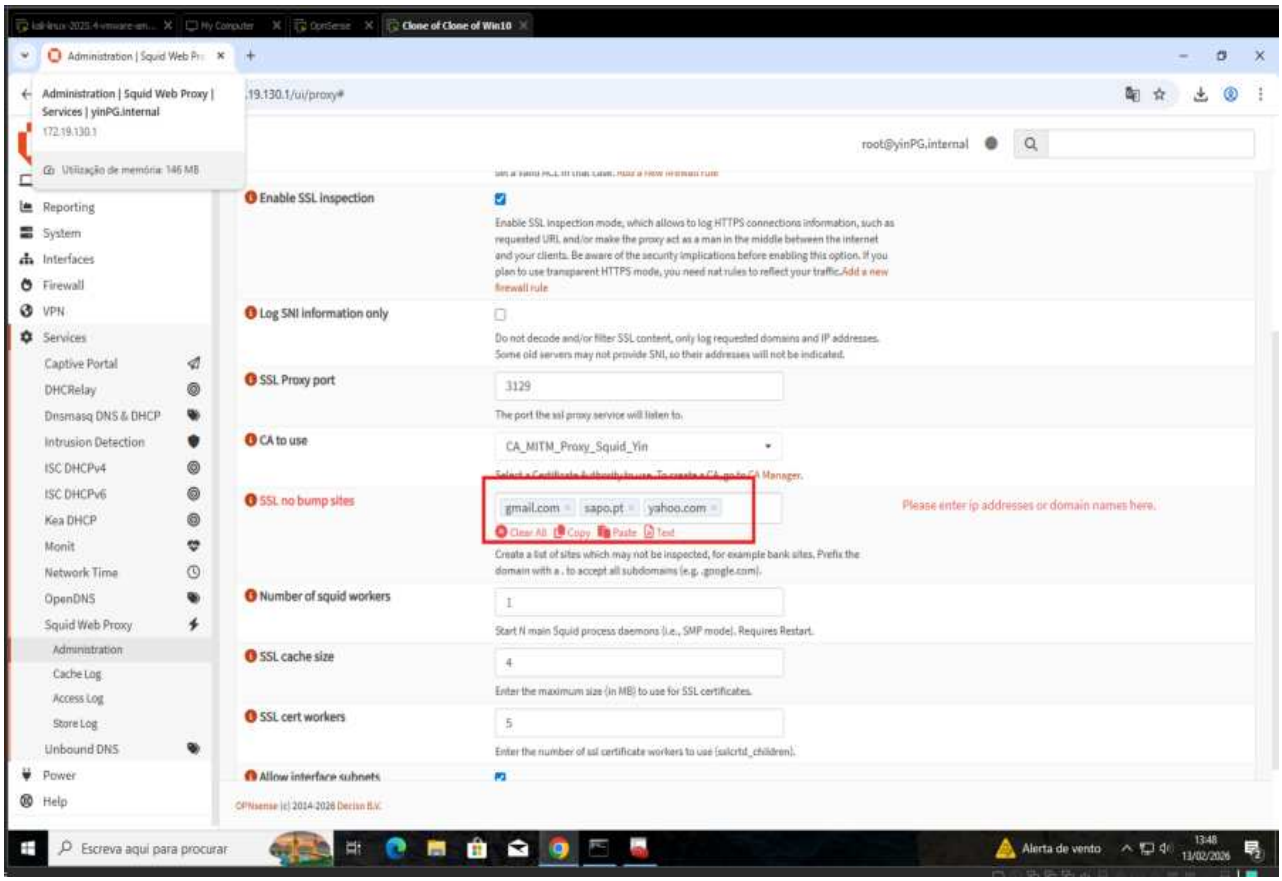
7.9 Squid Web Proxy Administation>Forward Proxy



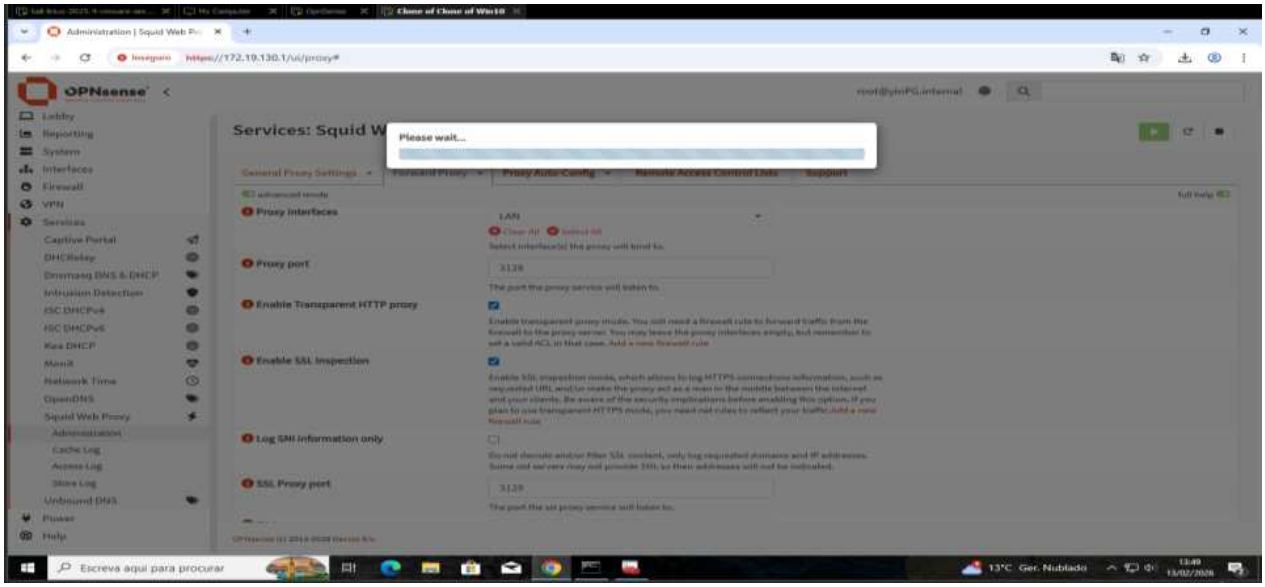
7.10 Forward Proxy > Access Control List



7.11Bloqueio : ssl no bump sites



7.12 Nota: nos proxies pode ser necessário aguardar alguns minutos após aplicar listas/regras.

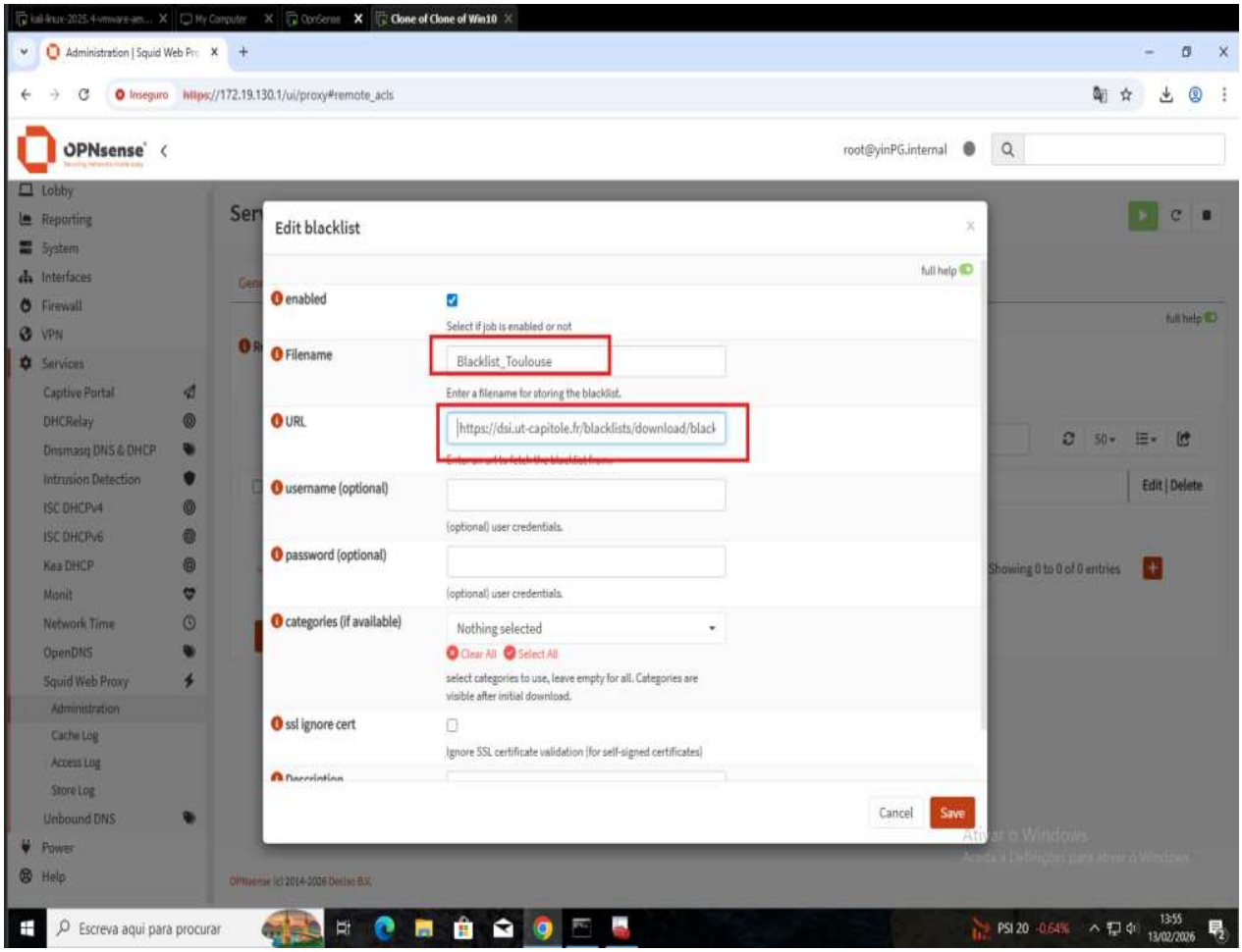


7.13 Blacklists (porn/chat/bitcoin/webmail)

Services → Web Proxy → Remote Access Control Lists

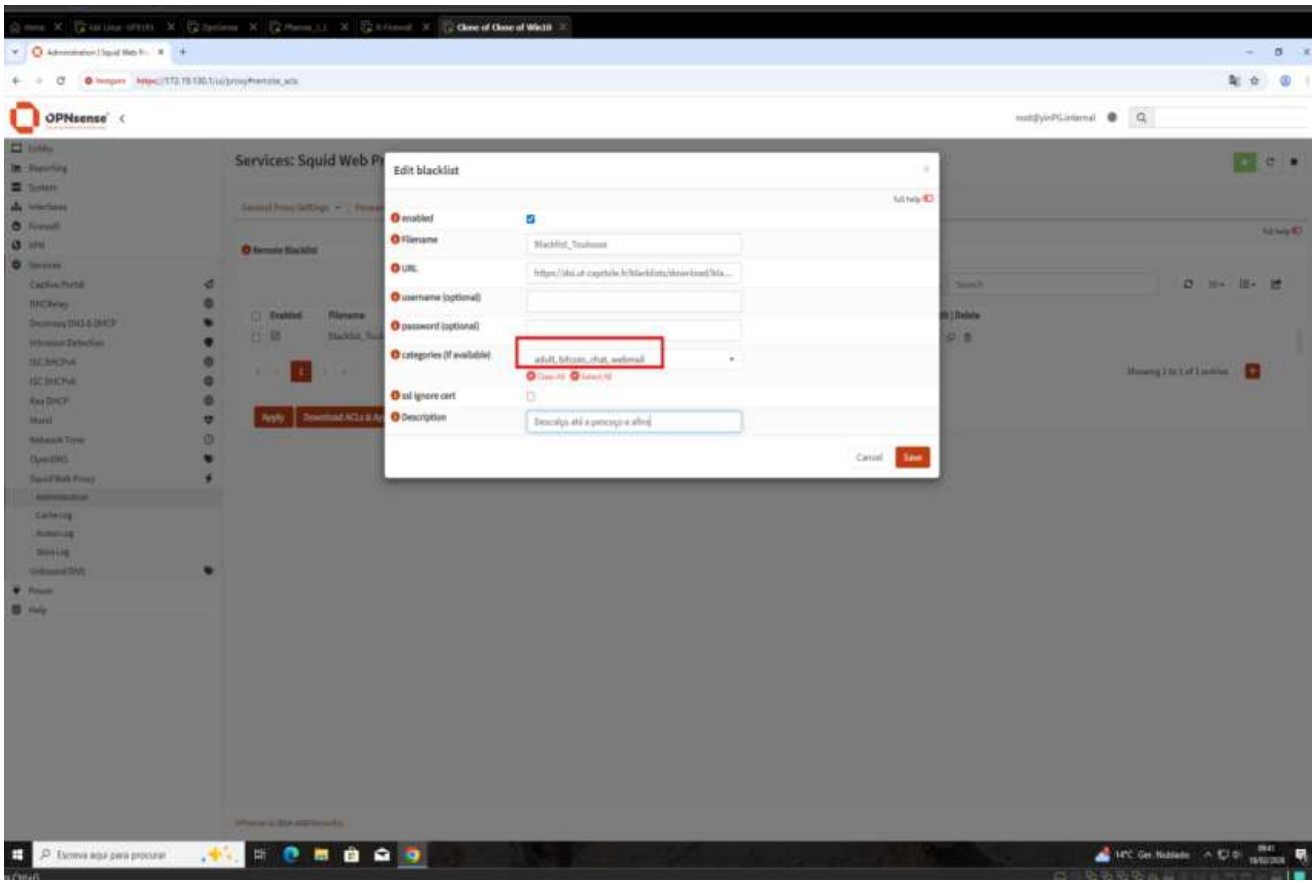
- URL das blacklists: **https://dsi.ut-capitole.fr/blacklists/download/blacklists.tar.gz**

1. Adicionar a fonte (URL acima)
2. Fazer download / apply

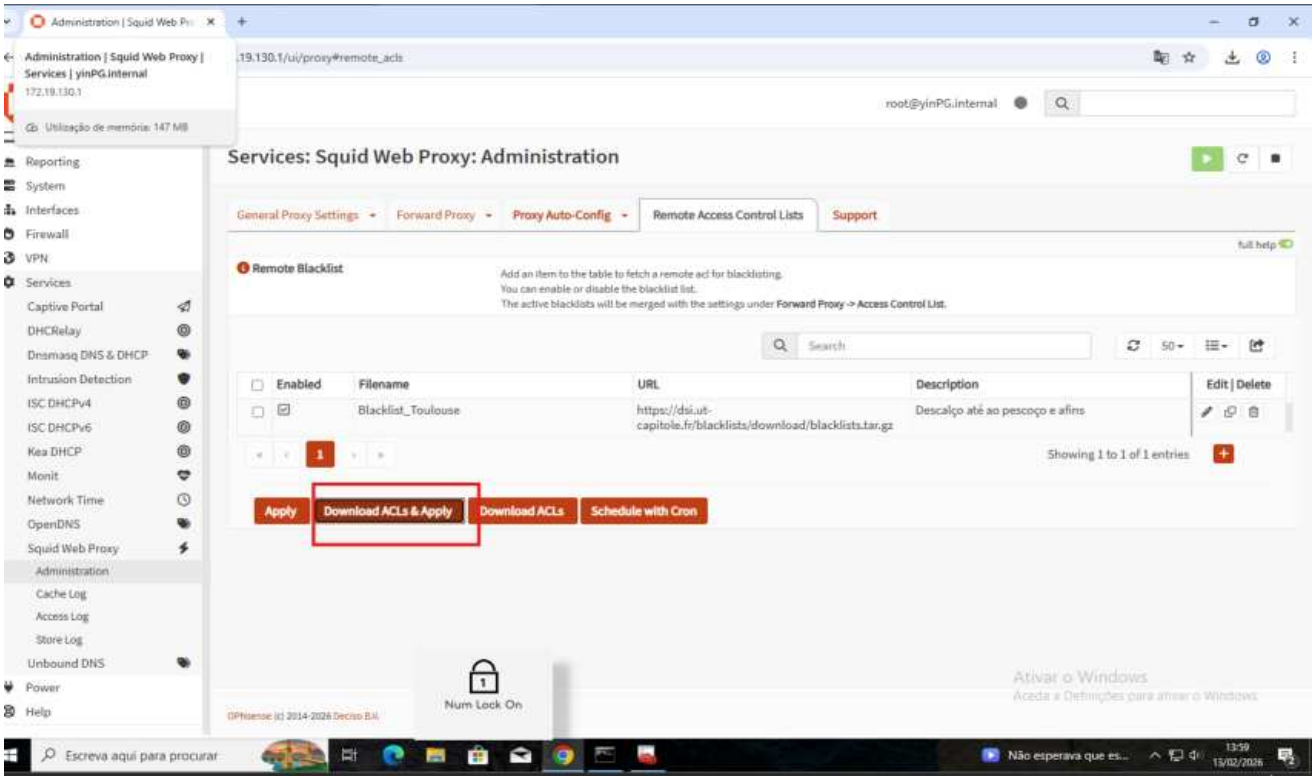


7.14 Selecionar categorias:

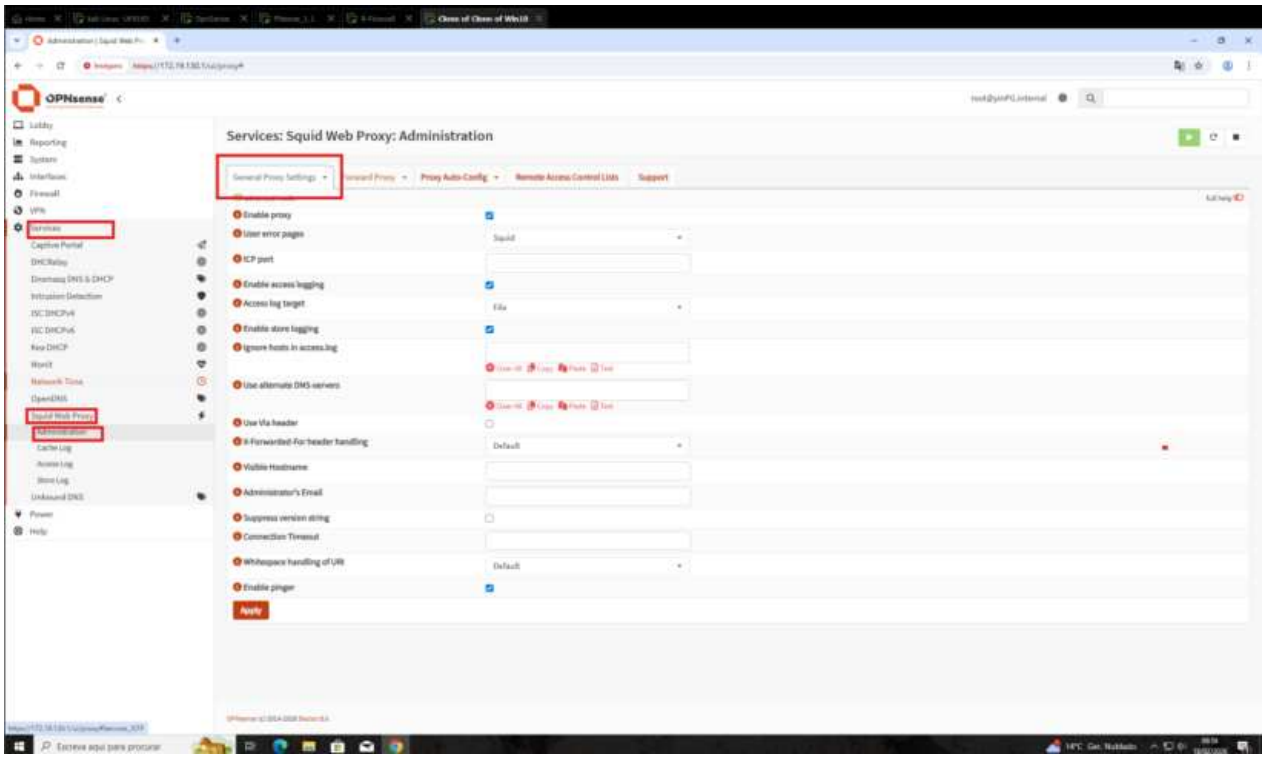
-adult (porn); chat ; bitcoinn ;webmail



7.15 Save + download ACLs + apply -- nota importante “apagar os cookies do browser primeiro “

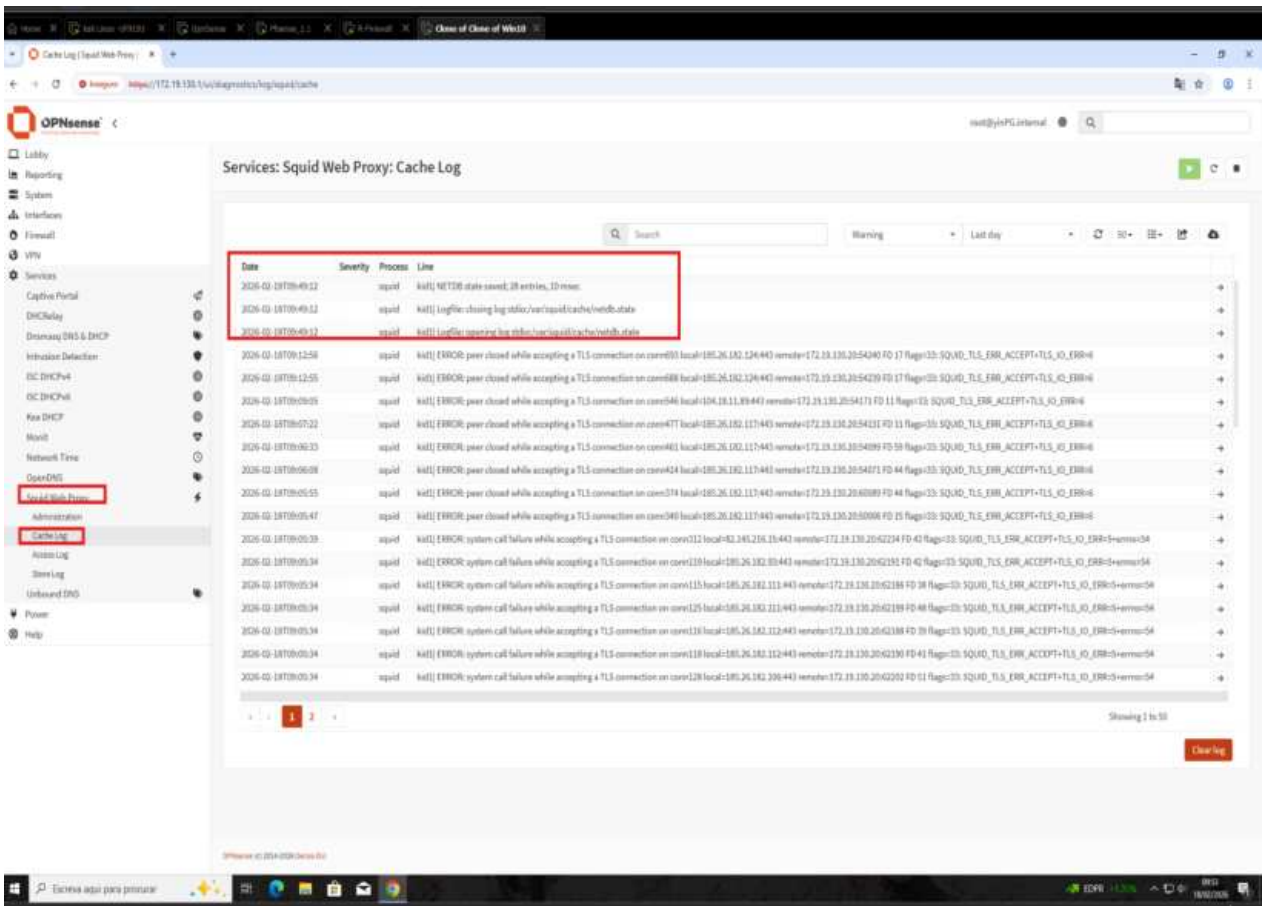


7.15 Resumo Geral :Squid Web Proxy > Administraton



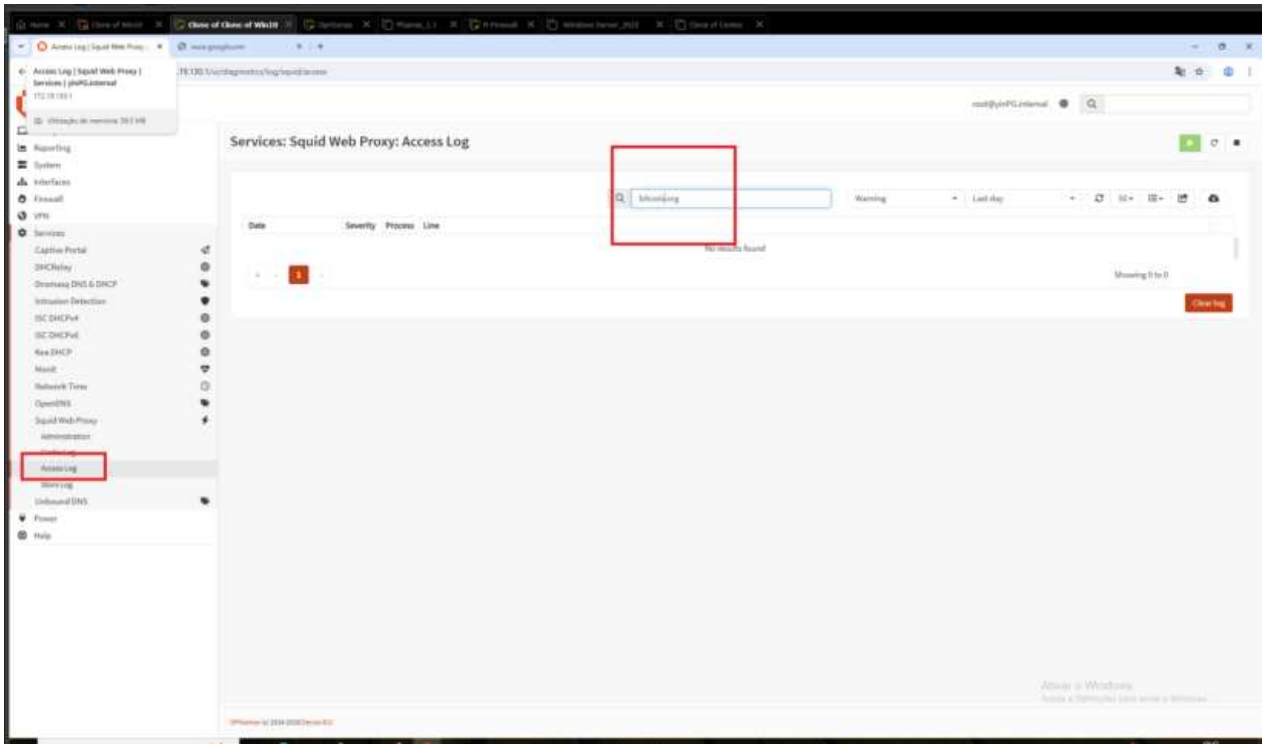
7.16 Restart serviço

- **Services → Squid Web Proxy → Cache Log (ou logs do serviço**



Services -> Squid Web Proxy -> Access Log

Dá erro (sem acesso)



d) A ferramenta de acesso remoto Chrome Remote Desktop deve de estar barrada para qualquer dia/hora

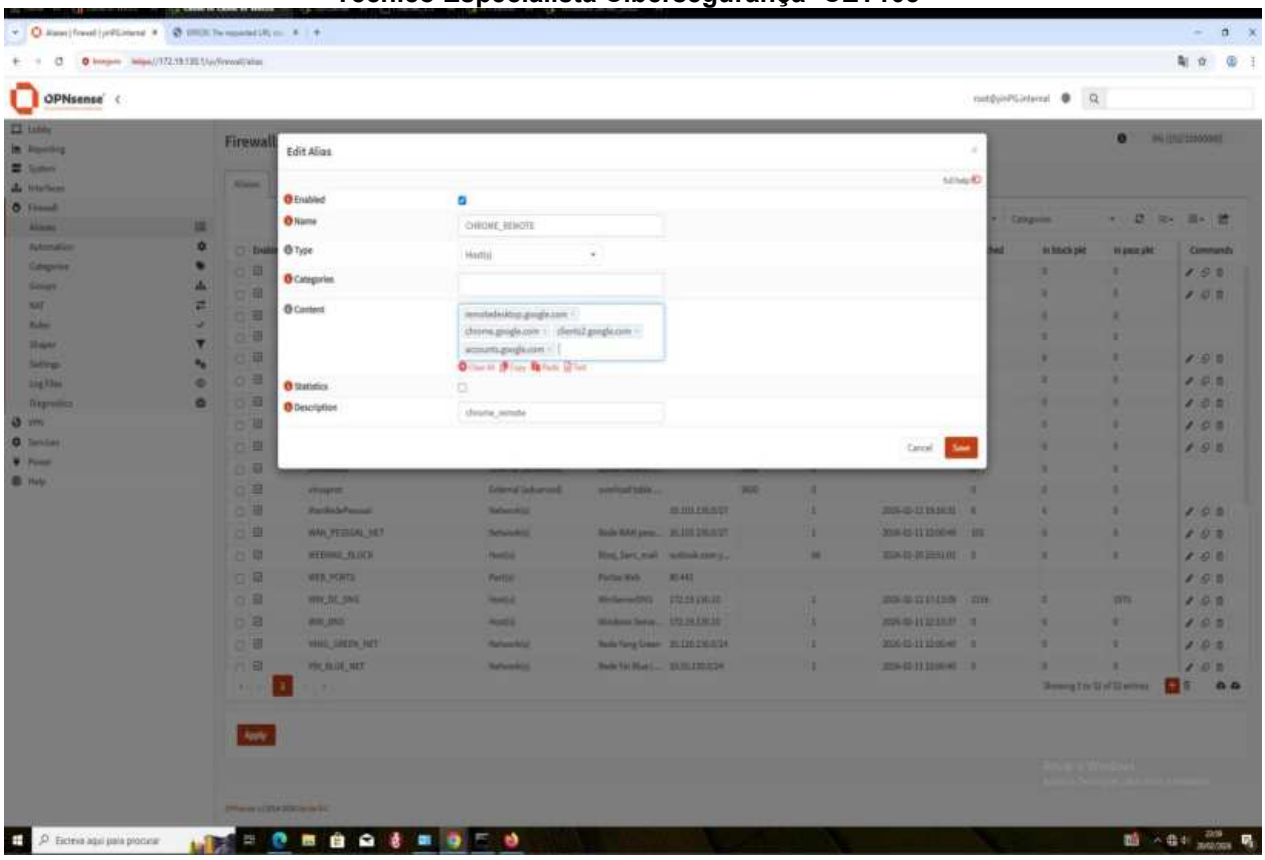
8. Bloqueios por domínio (DN + Chrome Remote Desktop + Webmail se necessário)
8.1 DN por horário

Firewall → Aliases → Add

Alias: CHROME_REMOTE

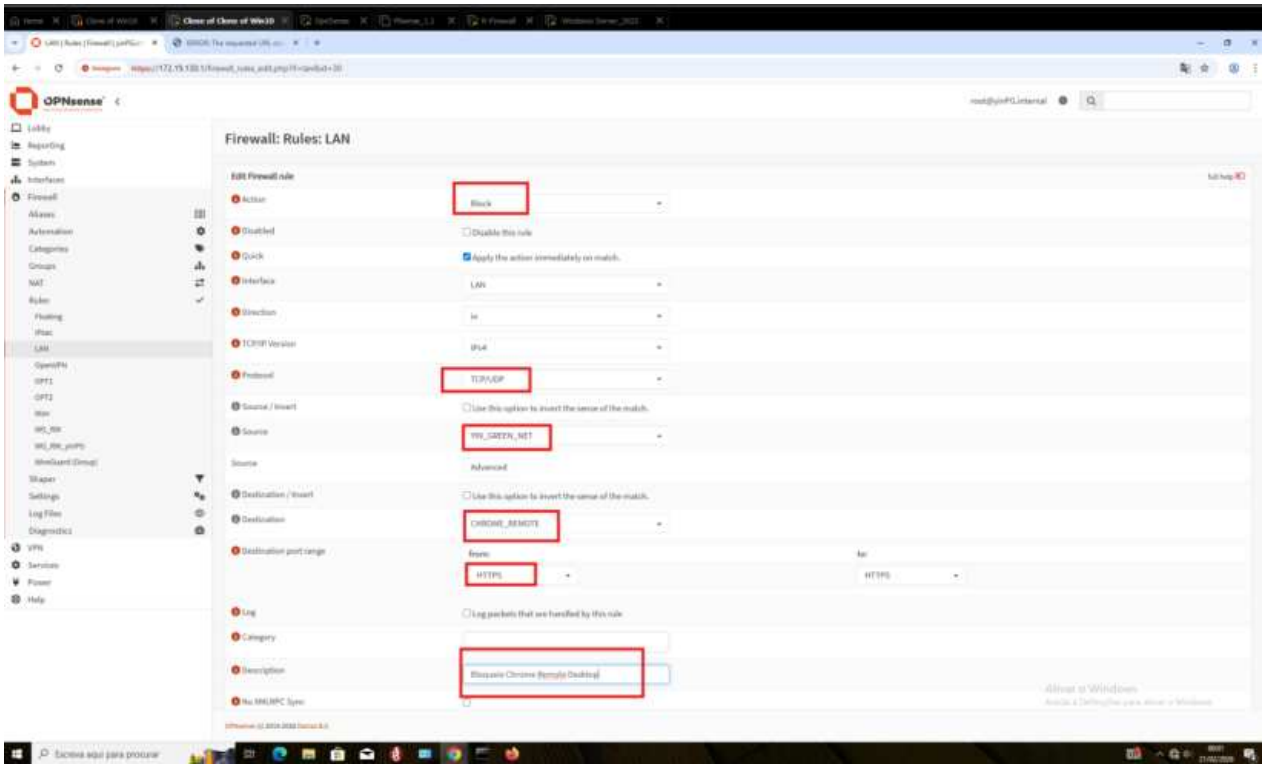
Adicionar:

remotedesktop.google.com
chrome.google.com
clients2.google.com
accounts.google.com



8.2 PASSO 2 — Criar regra na GREEN

Firewall → Rules → LAN

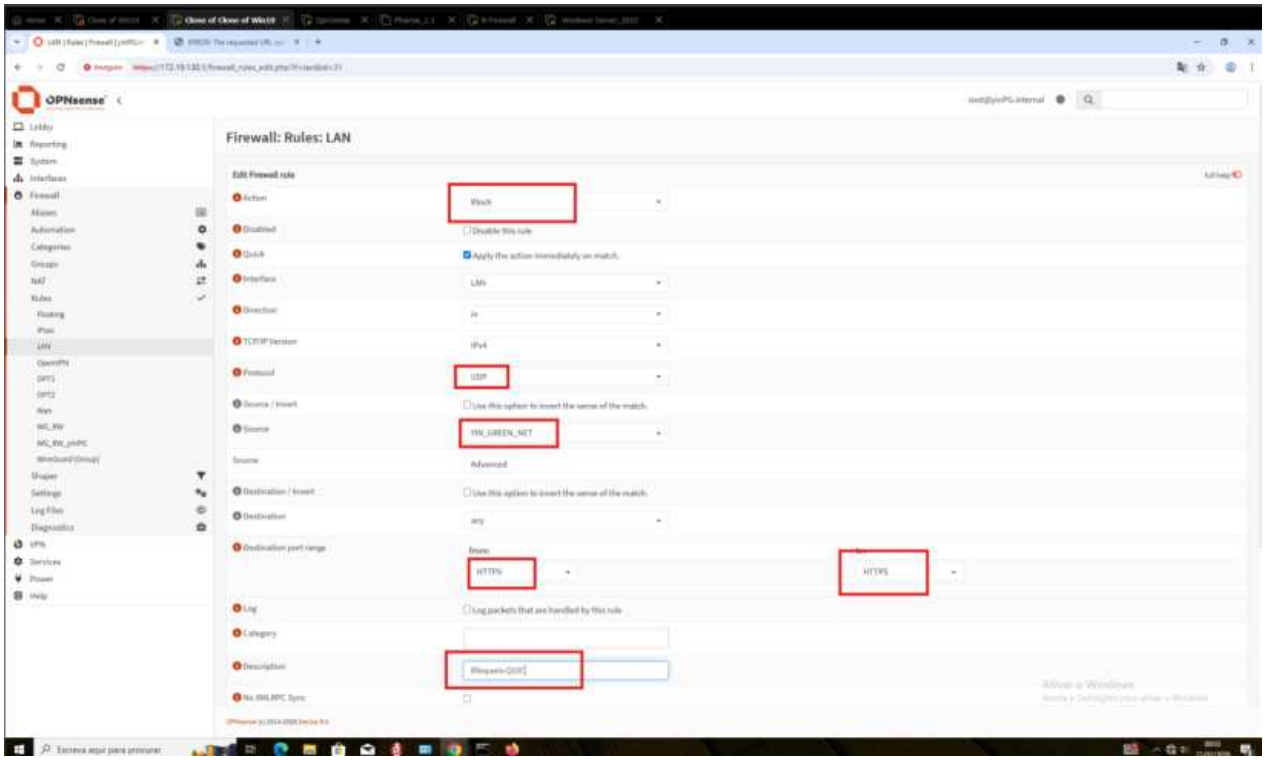


8.3 PASSO 3 — (Recomendado) Bloquear QUIC

Criar regra adicional:

Action: Block
Protocol: UDP
Source: YIN_GREEN_NET
Destination: any
Port: 443
Description: Bloqueio QUIC

Colocar acima da regra allow.



No cliente GREEN:

Abrir:

https://remotedesktop.google.com

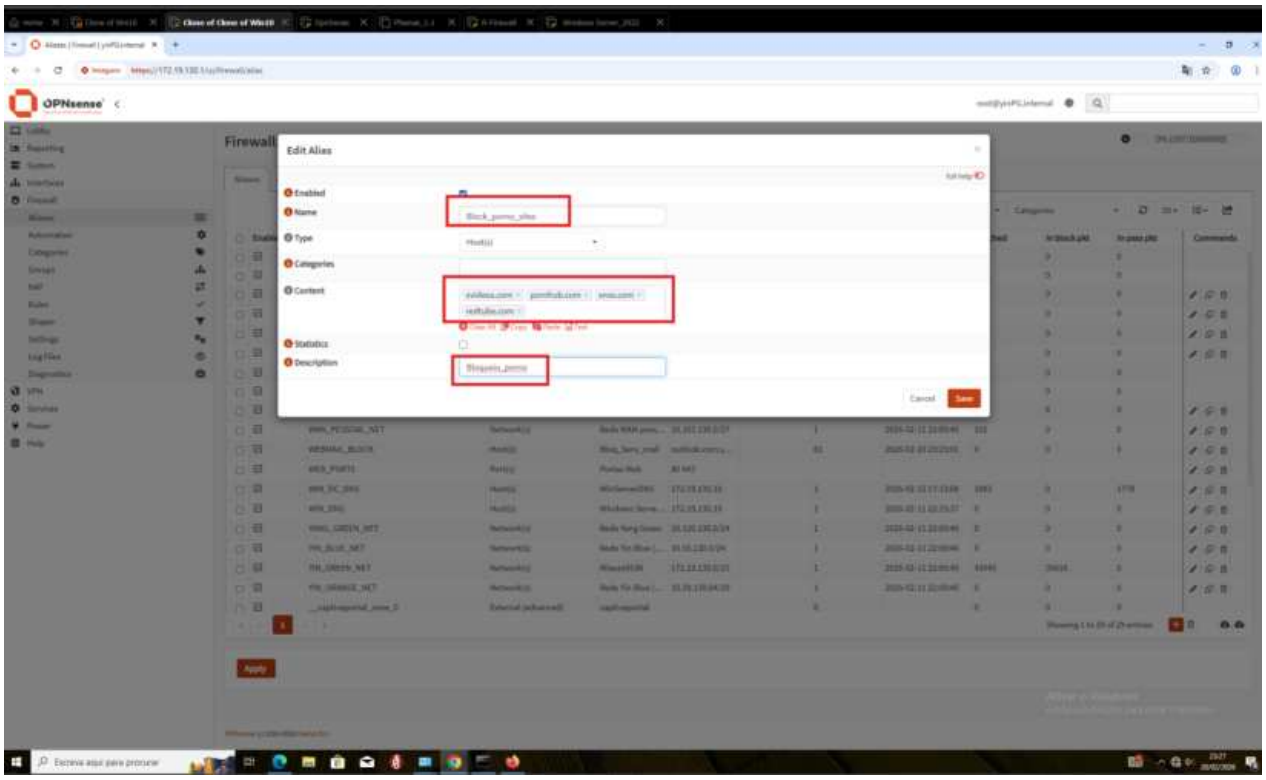
Deve falhar.

2- e) Sites considerados porn, chat e bitcoin, estão barrados

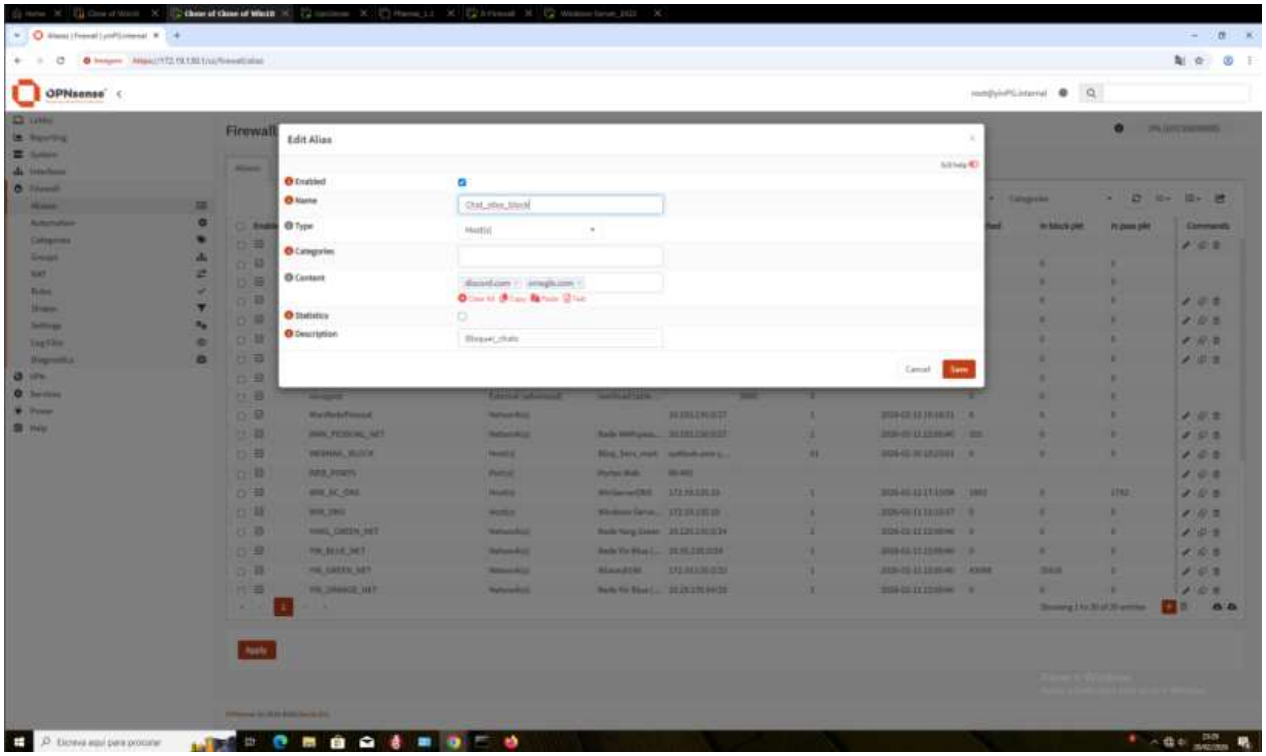
9 — Criar Alias

Firewall → Aliases → Add

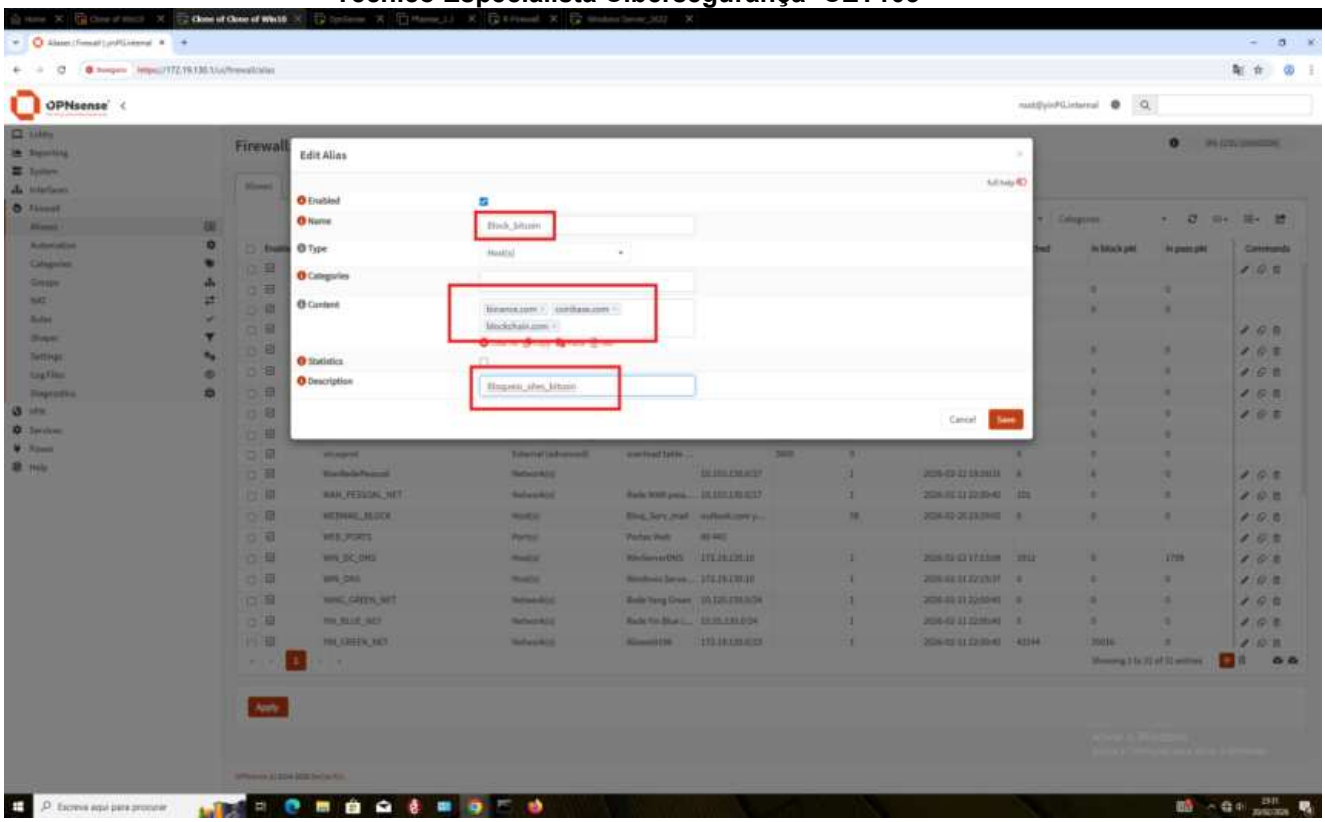
9.1 Block_porno



9.2 Block_chat

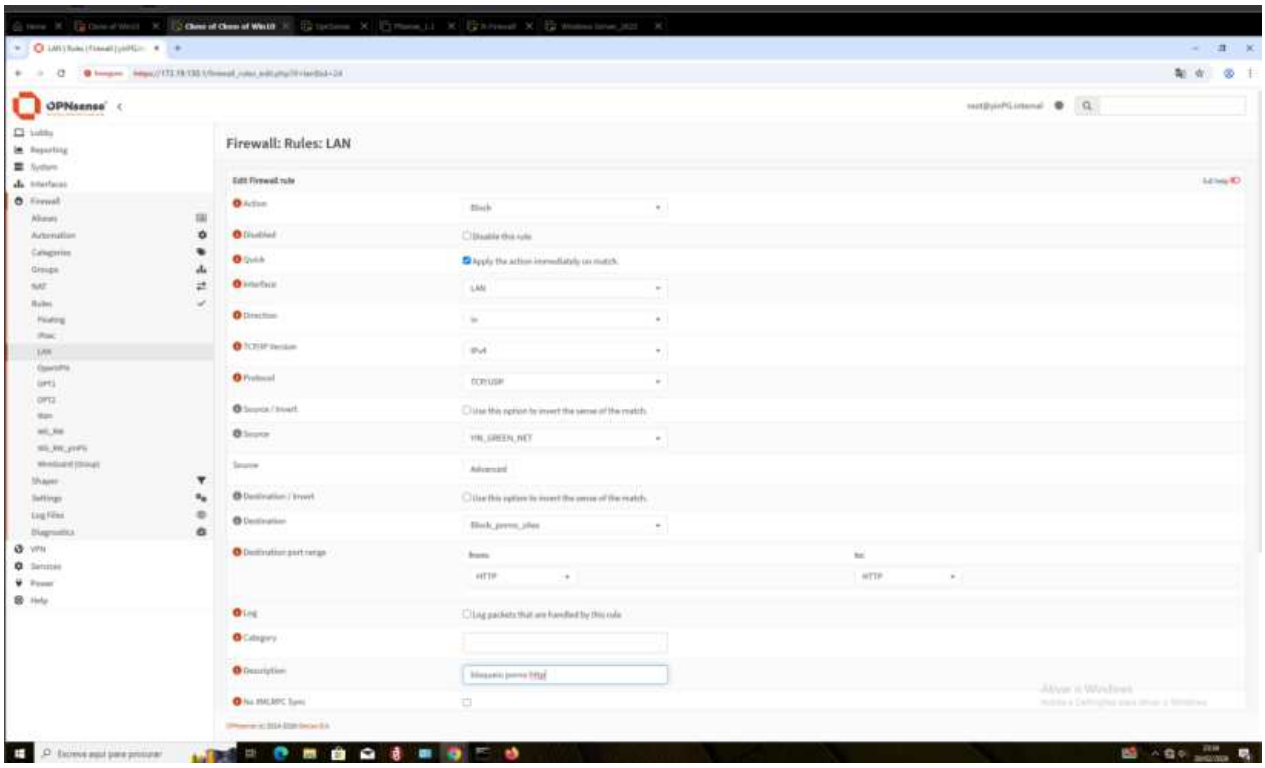


9.3 Block_bitcoin

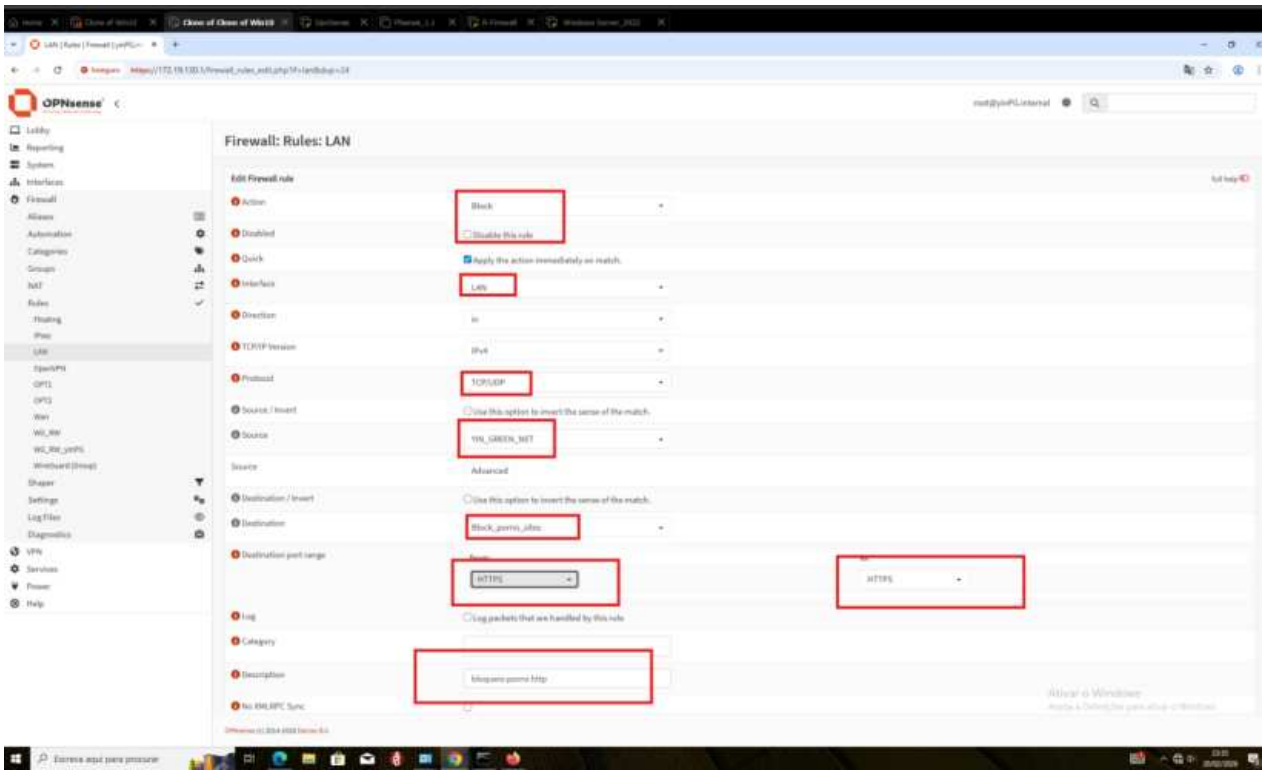


9.4 Firewall → Rules → LAN

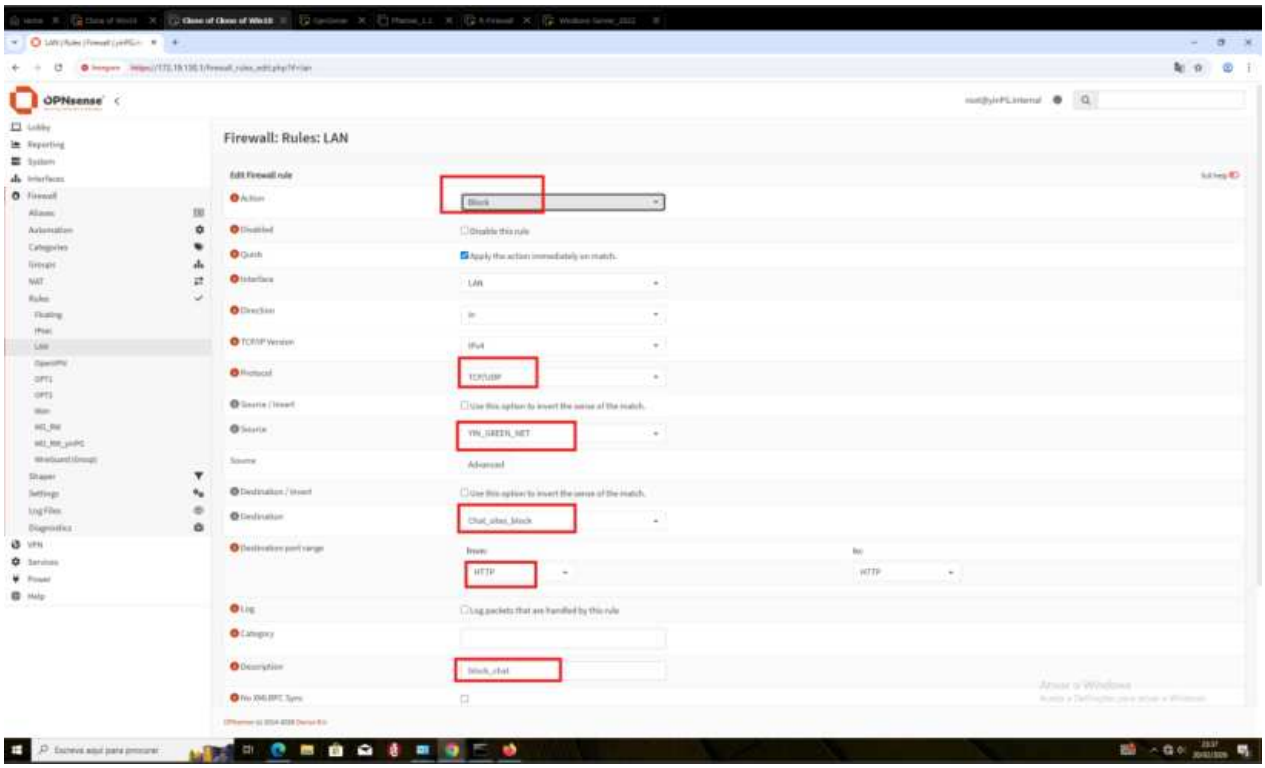
Bloqueio Sites_porno-http



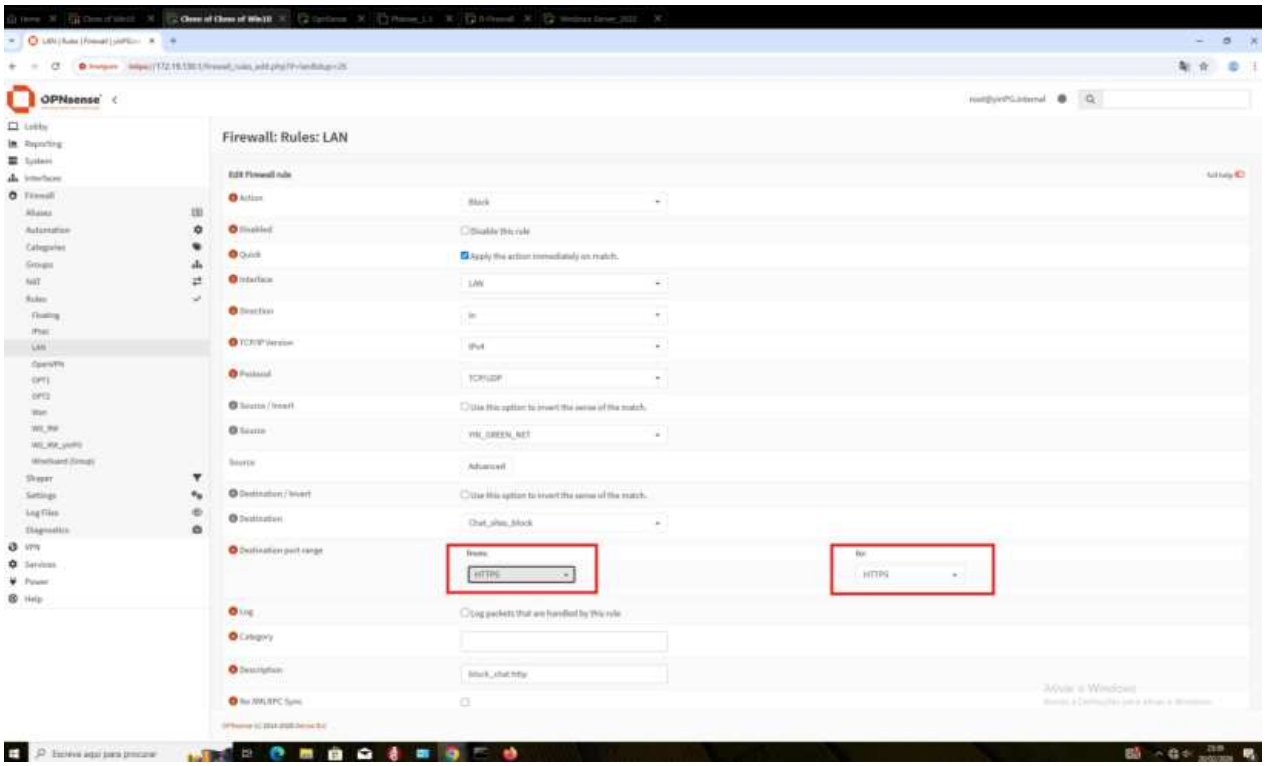
9.5 Bloqueio Sites_porno-https



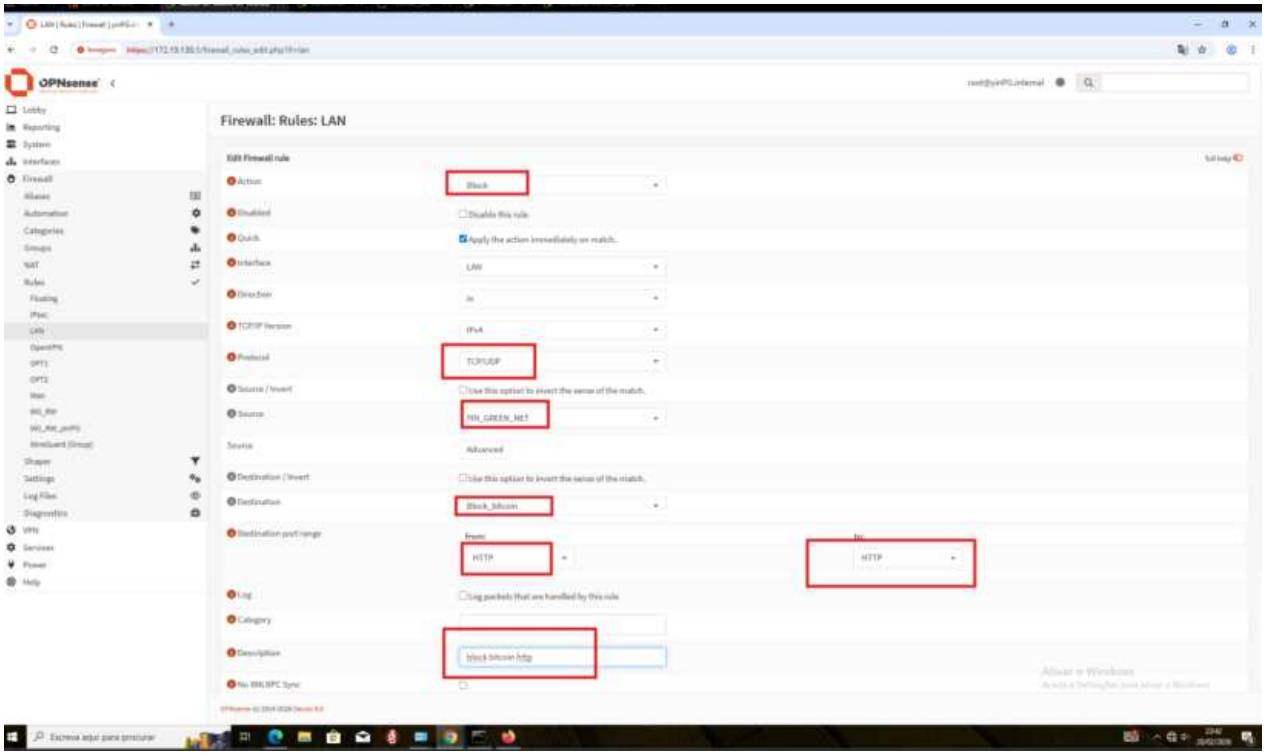
9.6 Bloqueio Sites_chat -http



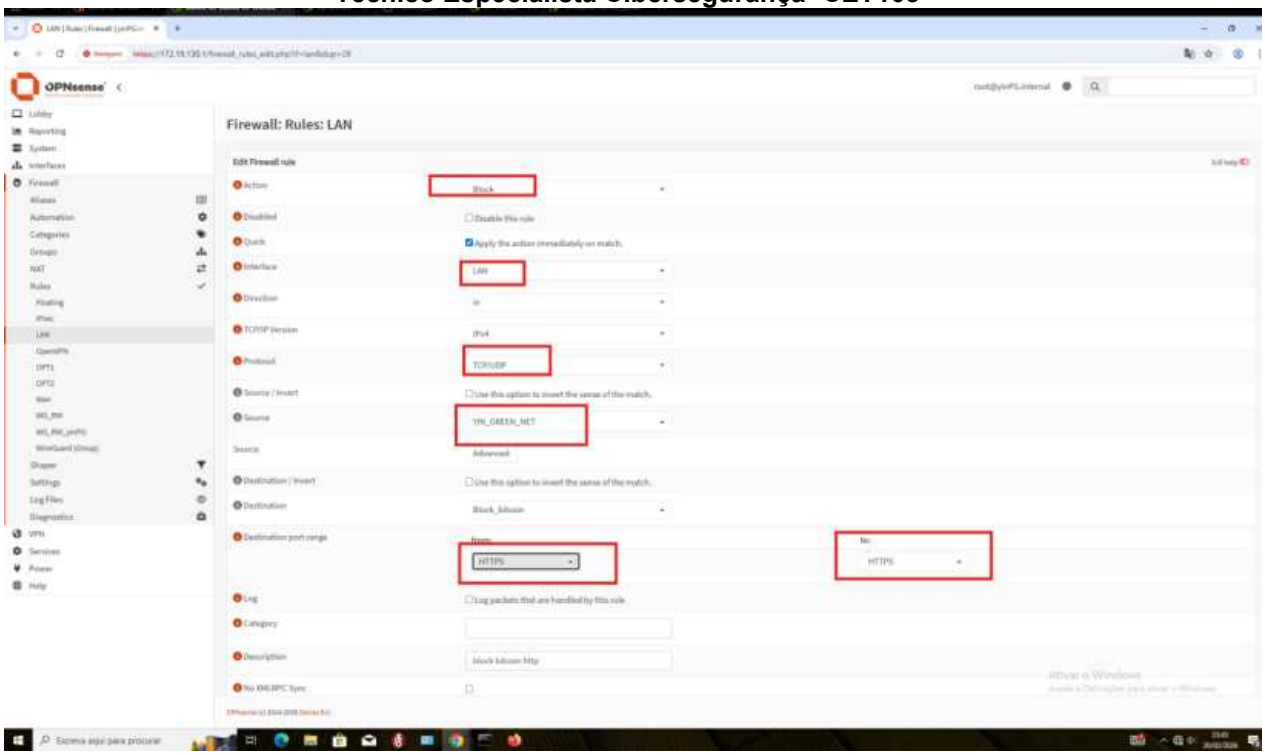
9.7 Bloqueio Sites_chat -https



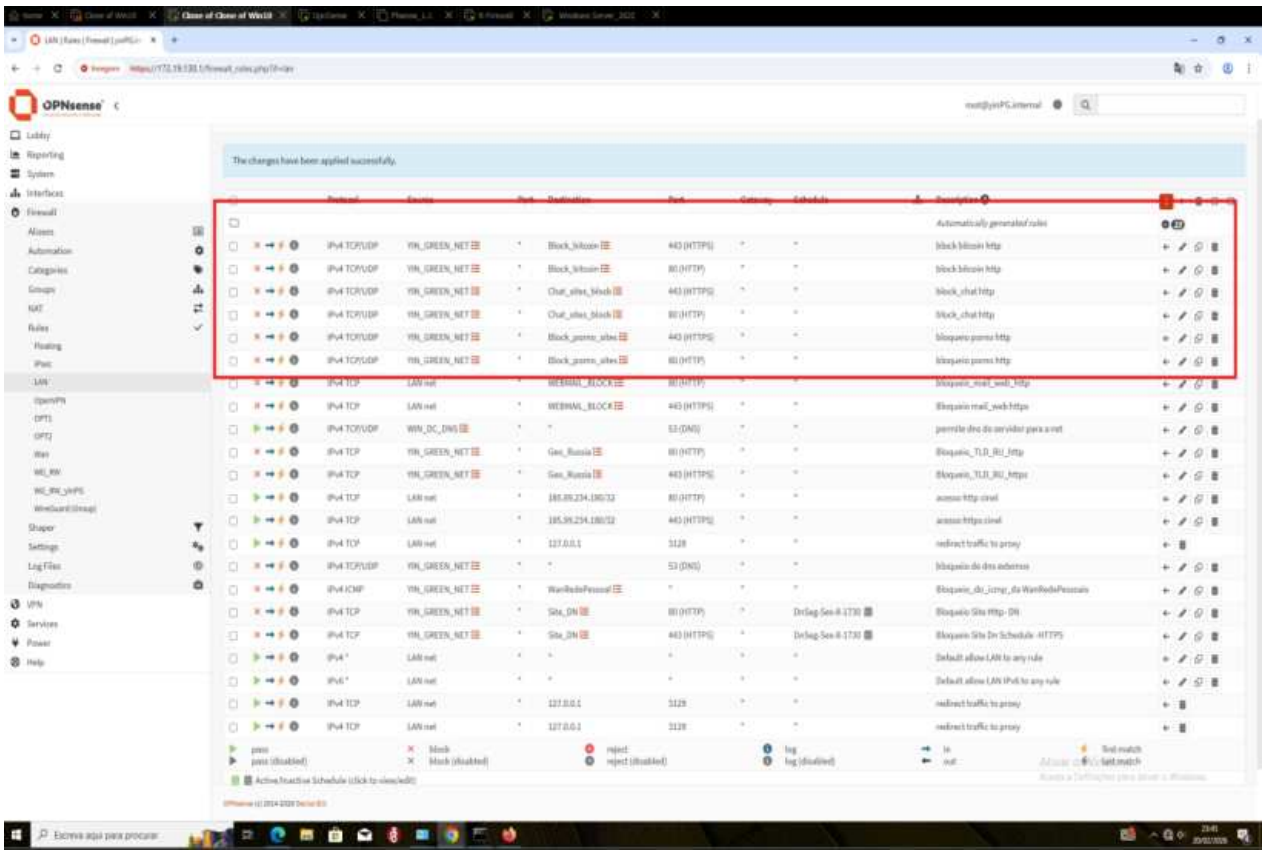
9.8 Block_sites_bitcoin http



Block_sites_bitcoin https



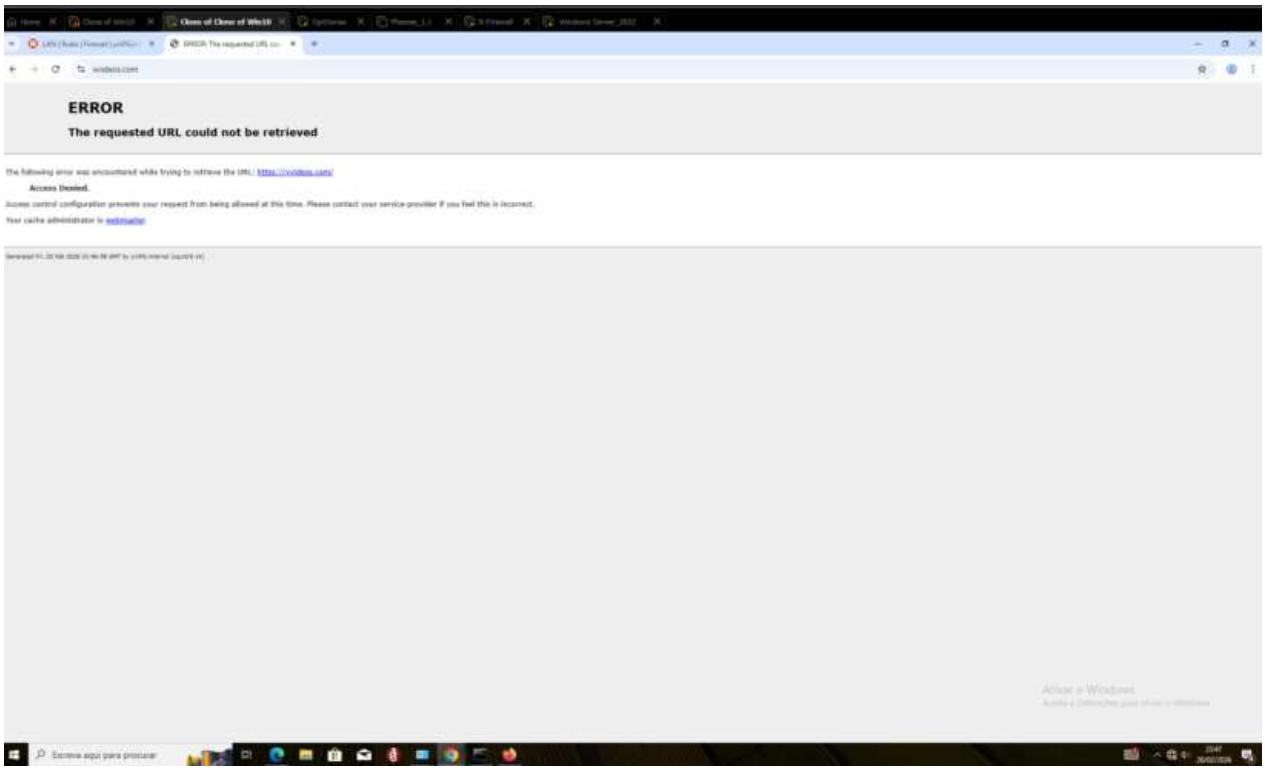
9.9 Resumo :



9.9.1

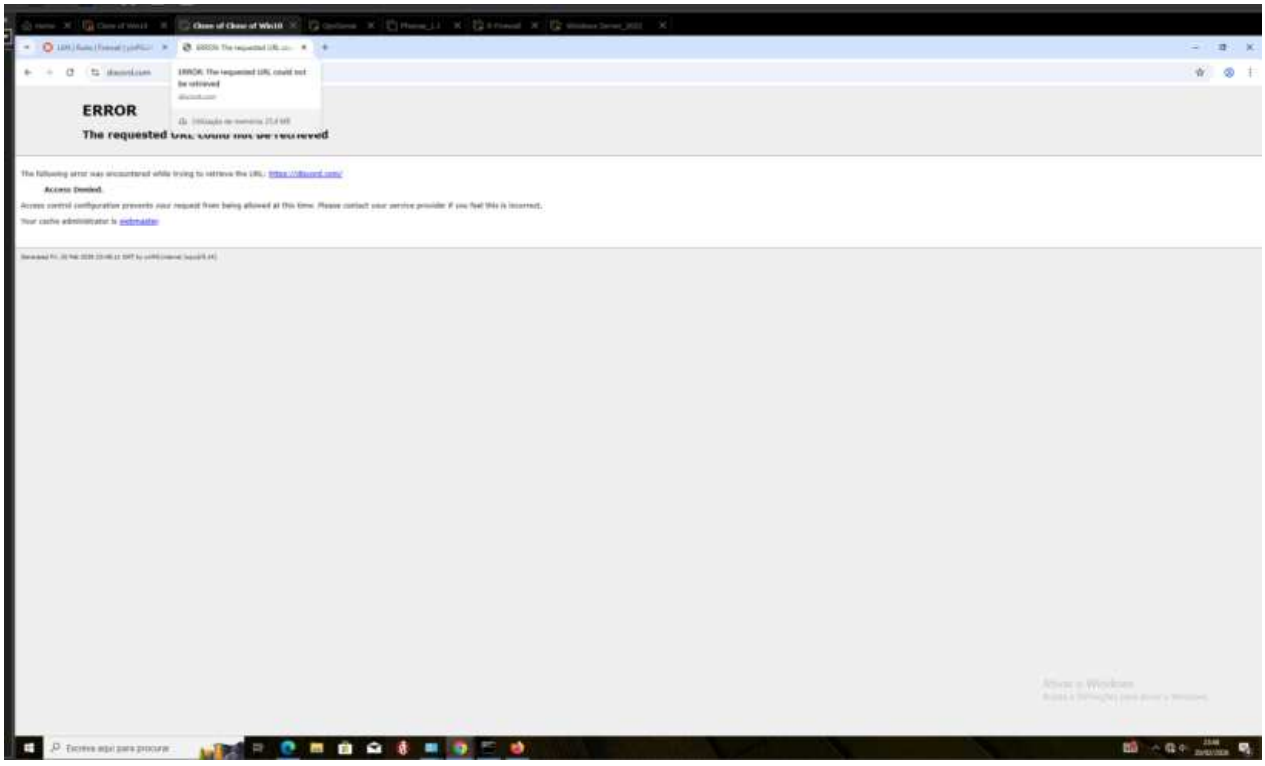
Testes :

Porno nota _ 1 exemplo



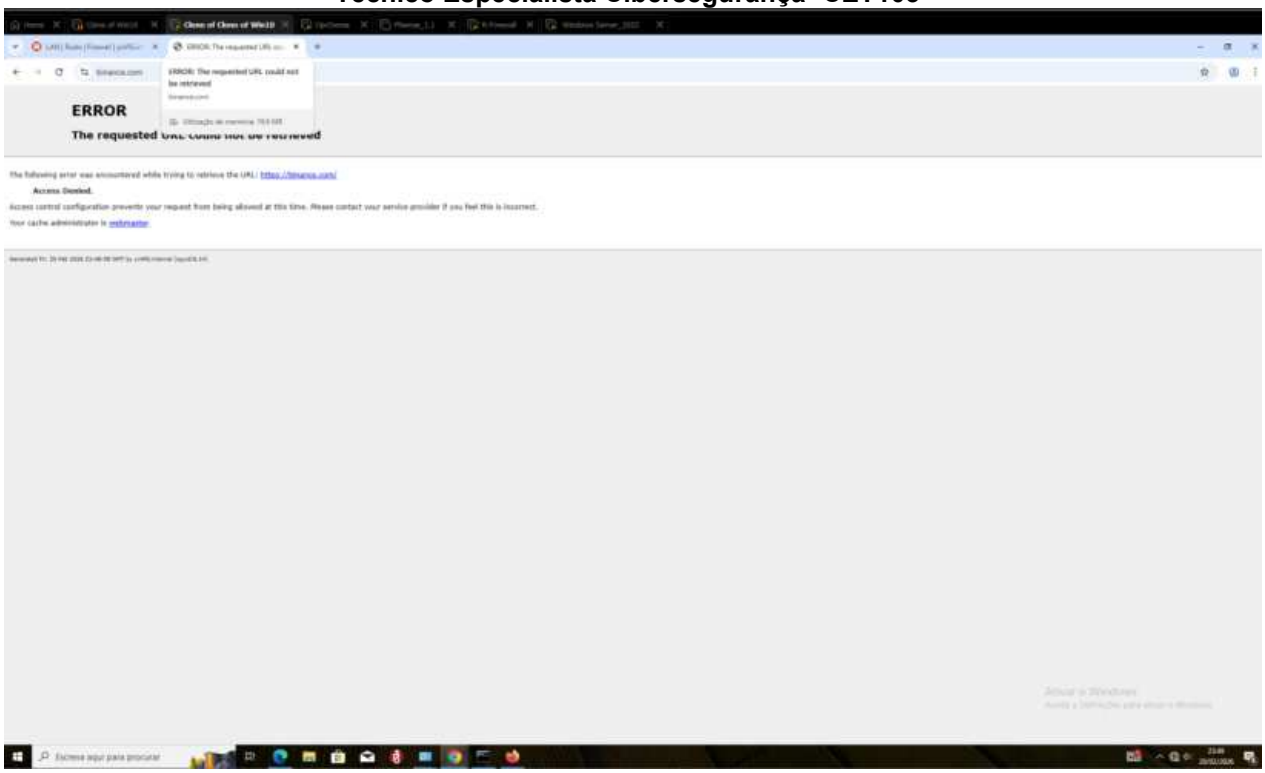
9.9.2

Chat -1 exemplo



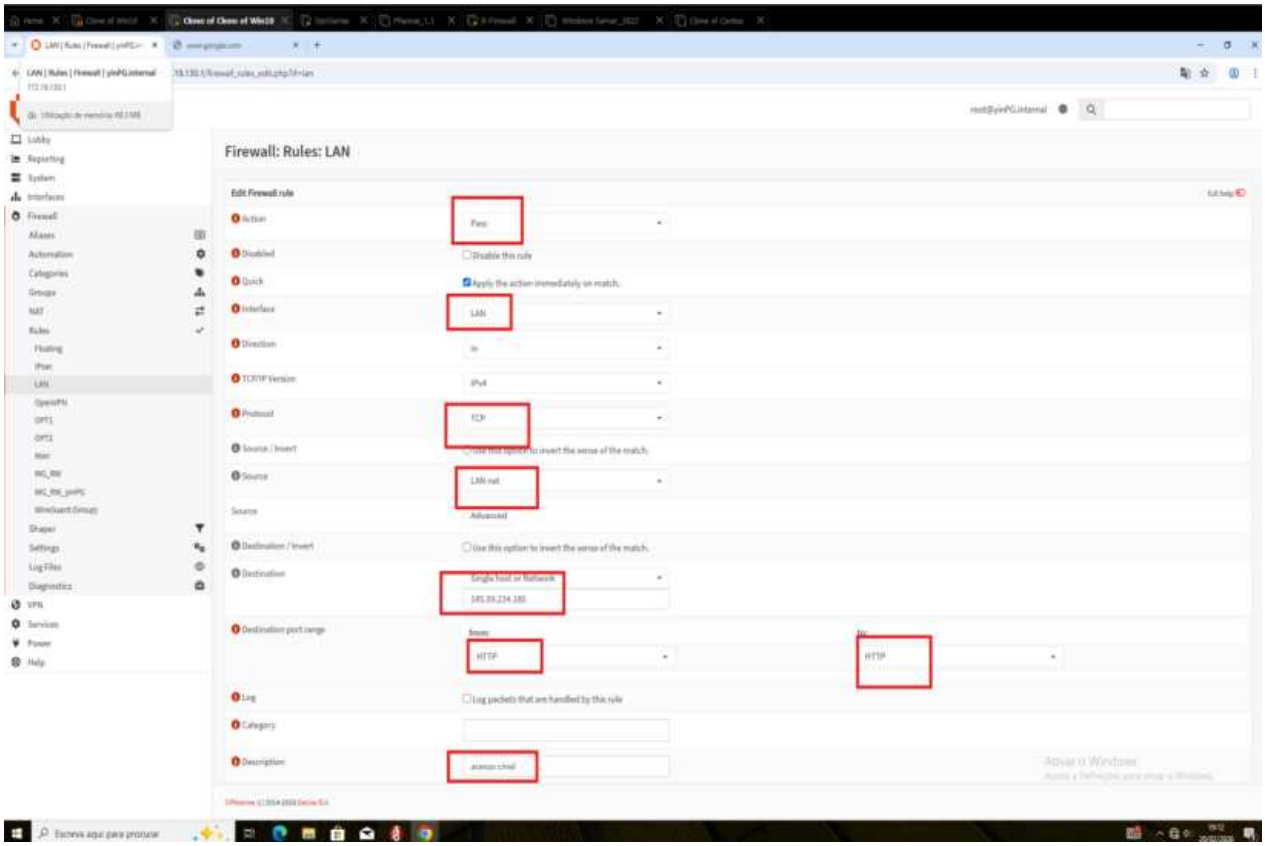
9.9.3

Bitcoin _1 exemplo

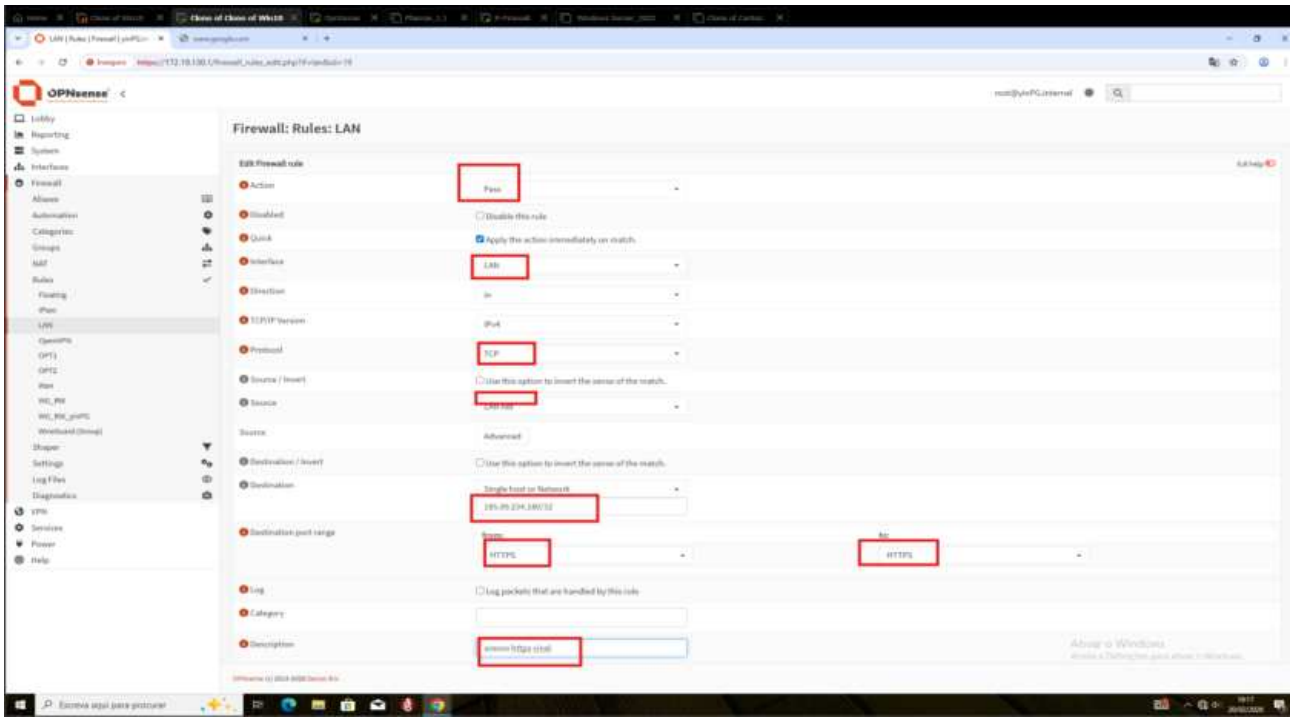


2-f) Os utilizadores não podem aceder a qualquer serviço de email baseado em Web (Gmail, Hotmail, mail.sapo.pt, Yahoo, etc) tanto em http, como em https. Mas podem aceder à páginas de https (ex: https://www.iefp.pt)

10 Acesso http



10.1Acesso https



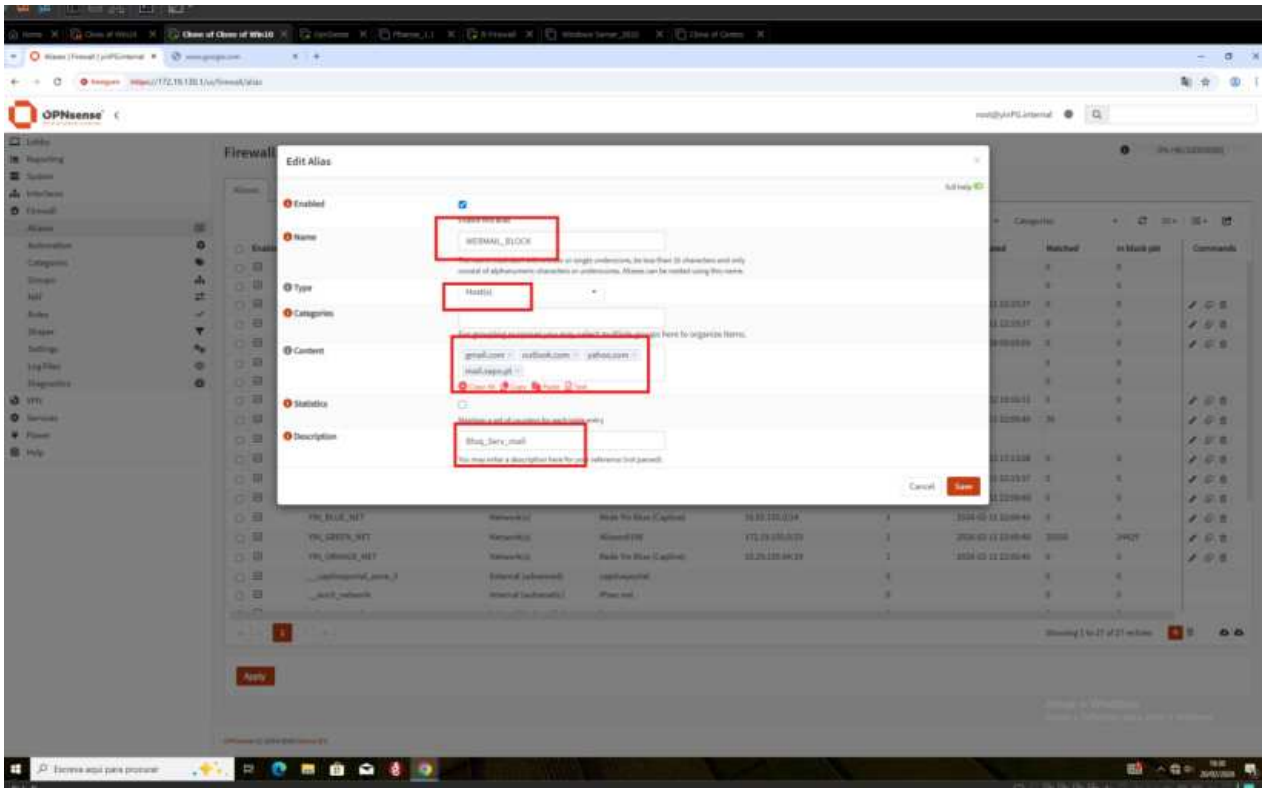


10.2 Bloqueio ao serviço de email baseado em Web (Gmail, Hotmail, mail.sapo.pt, Yahoo, etc) tanto em http, como em https

Serv**Firewall → Aliases → Add**

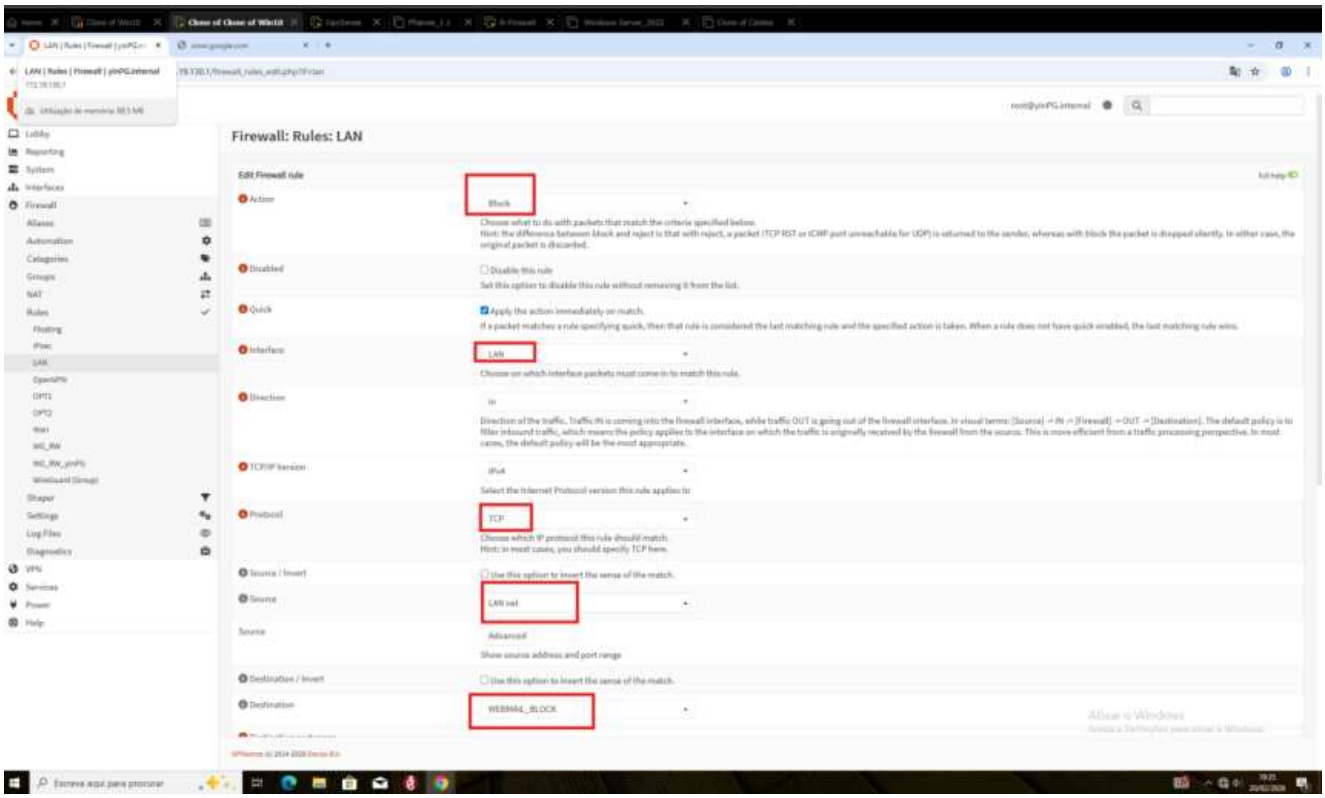
- Type: **host**

- Nome: WEBMAIL_BLOCK
- Inserir:gmail.com ;outlook.com ;yahoo.com ;mail.sapo.pt

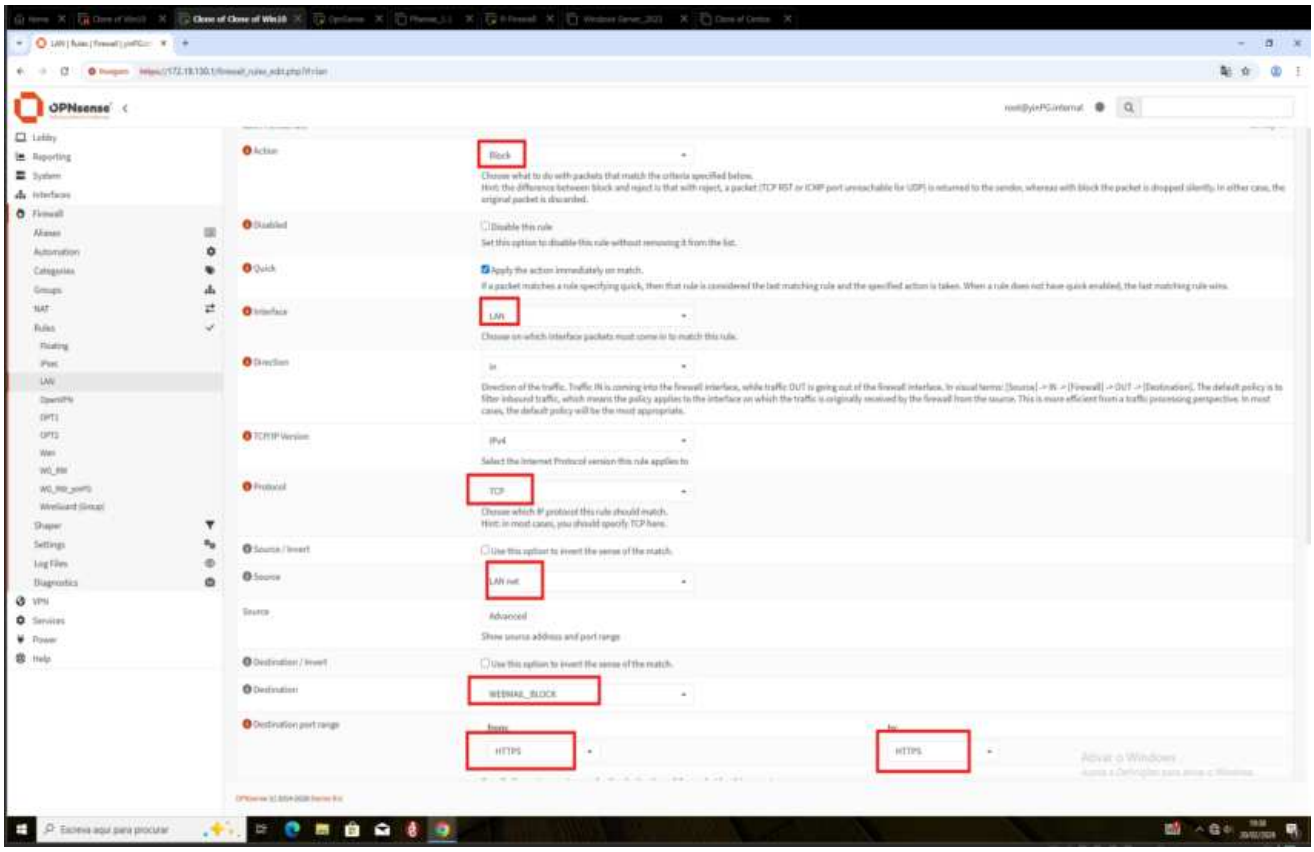


10.3 Firewall → Rules → LAN

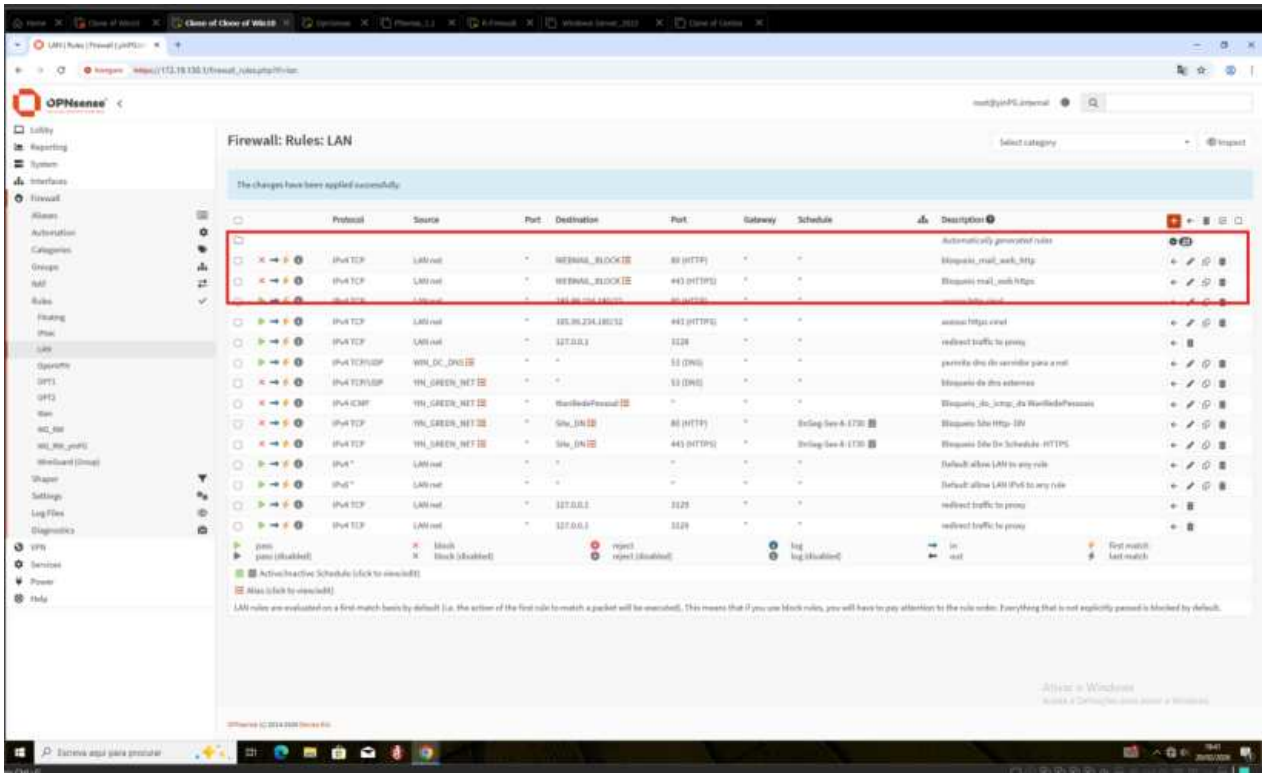
Bloqueio para http :



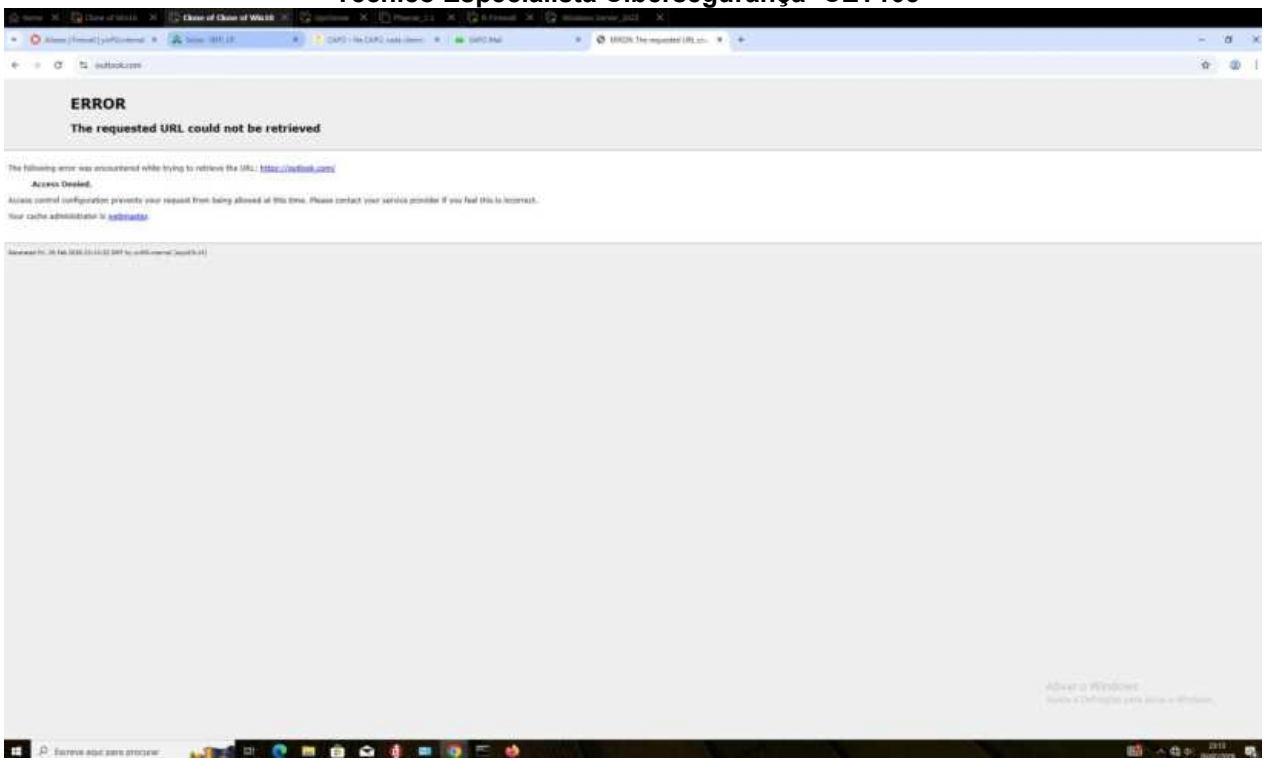
10.4 - Bloqueio para https



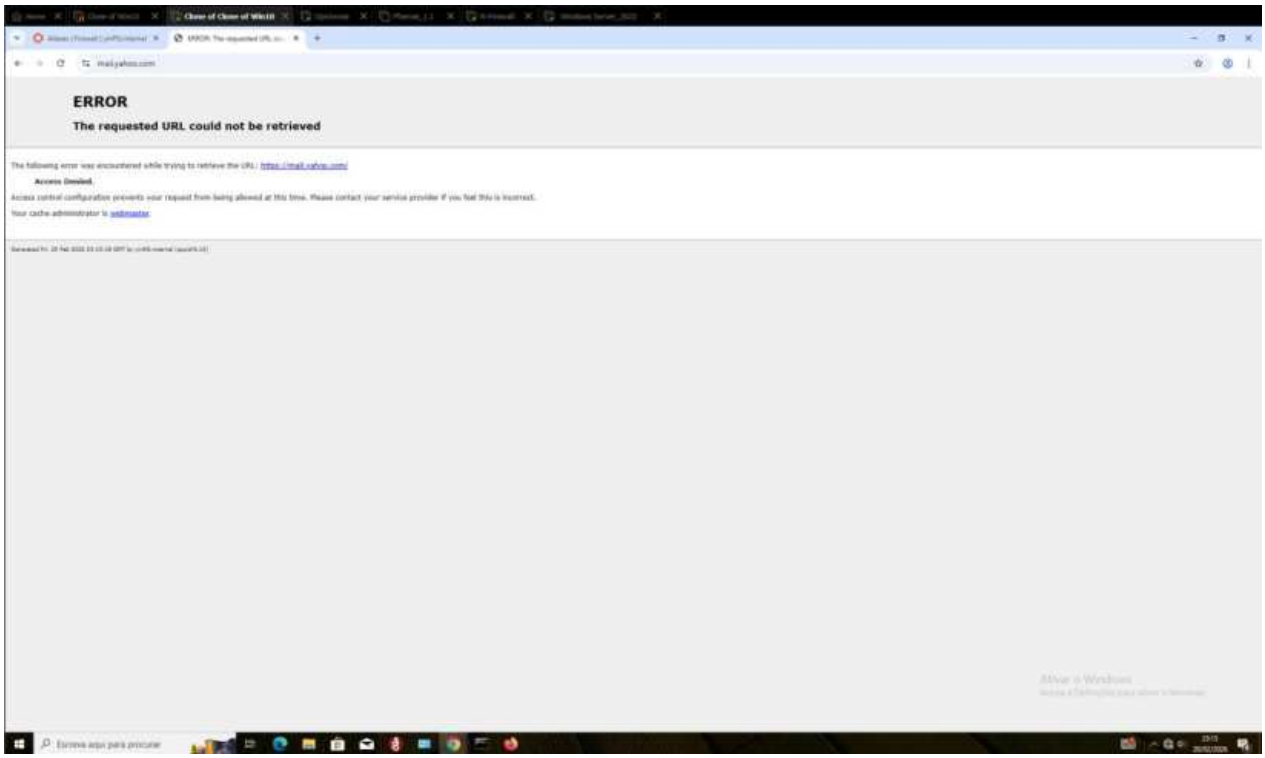
10.5 Resumo :



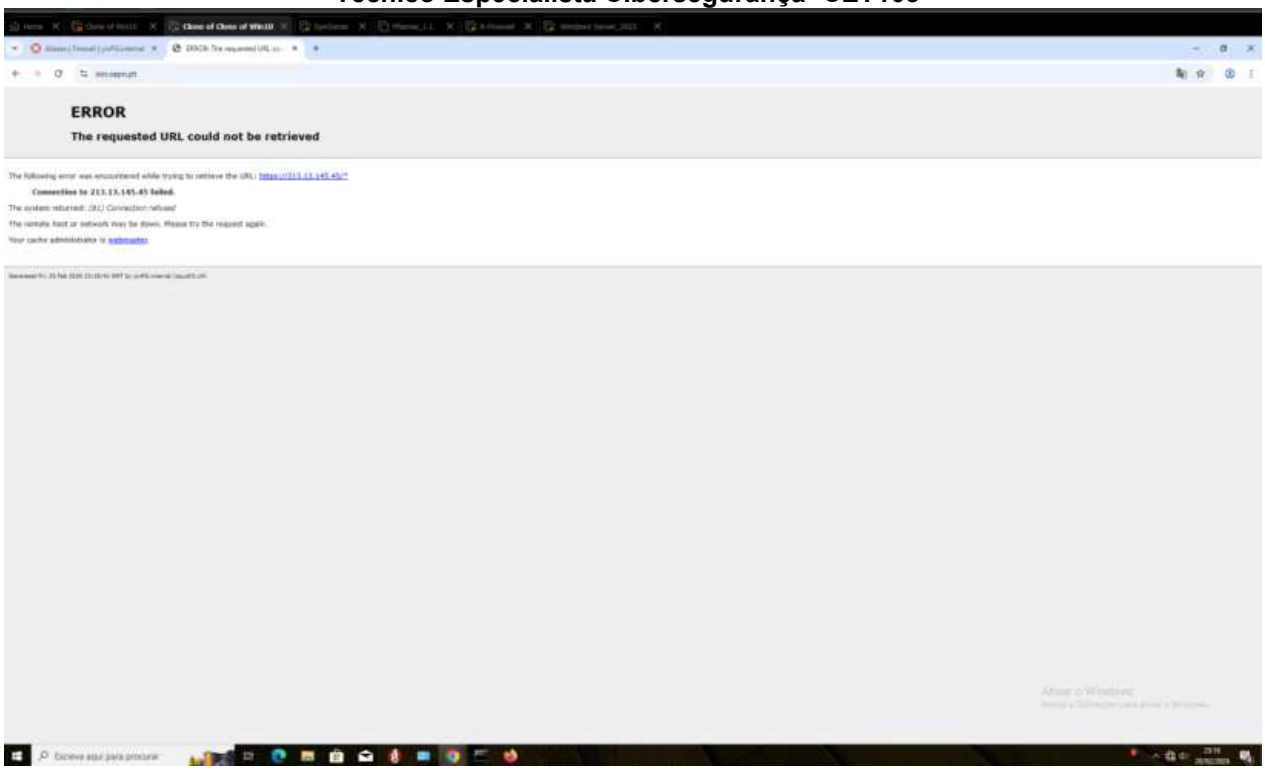
Bloqueio https://outlook.com



https://mail.yahoo.com



https://mail.sapo.pt



11 - Instalação pfSense

pfSense yangPG — Configuração base (WAN + LAN)
O enunciado define a yang como pfSense com 2 interfaces.

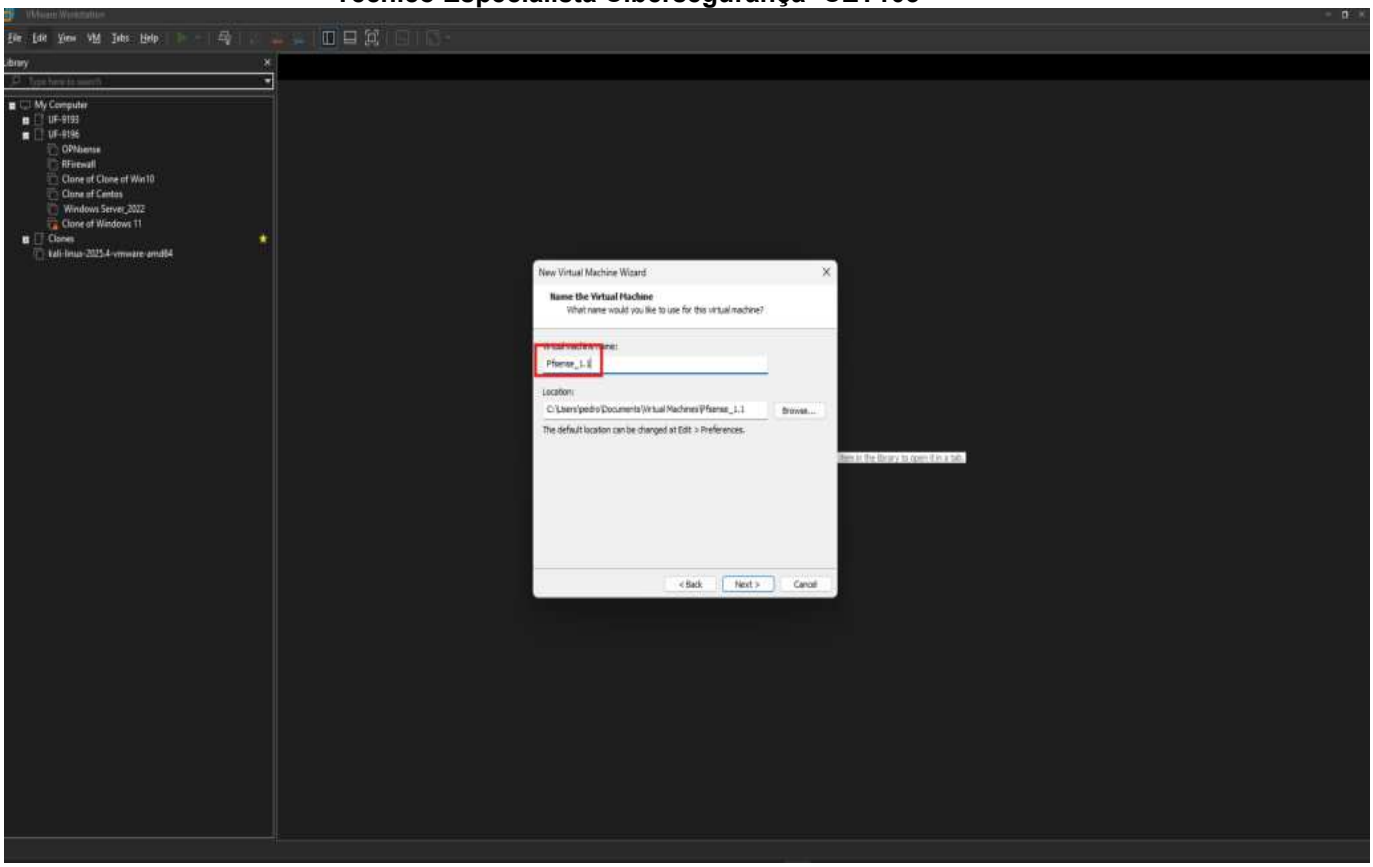
1-Interfaces

- WAN: 10.103.130.11/27 gateway 10.103.130.1
- LAN (yang green): 10.120.130.1/24

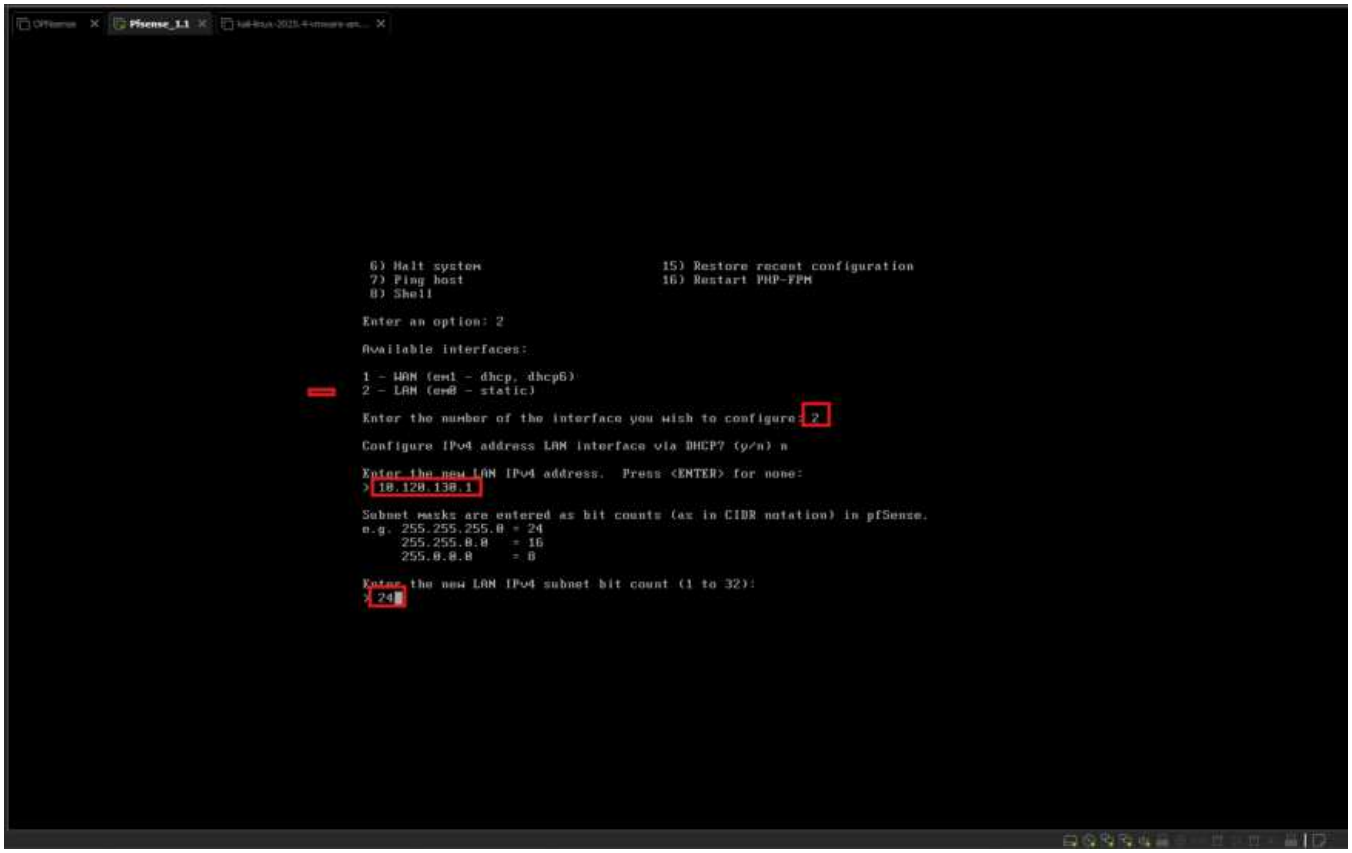
- NAT

Firewall → NAT → Outbound → Automatic.

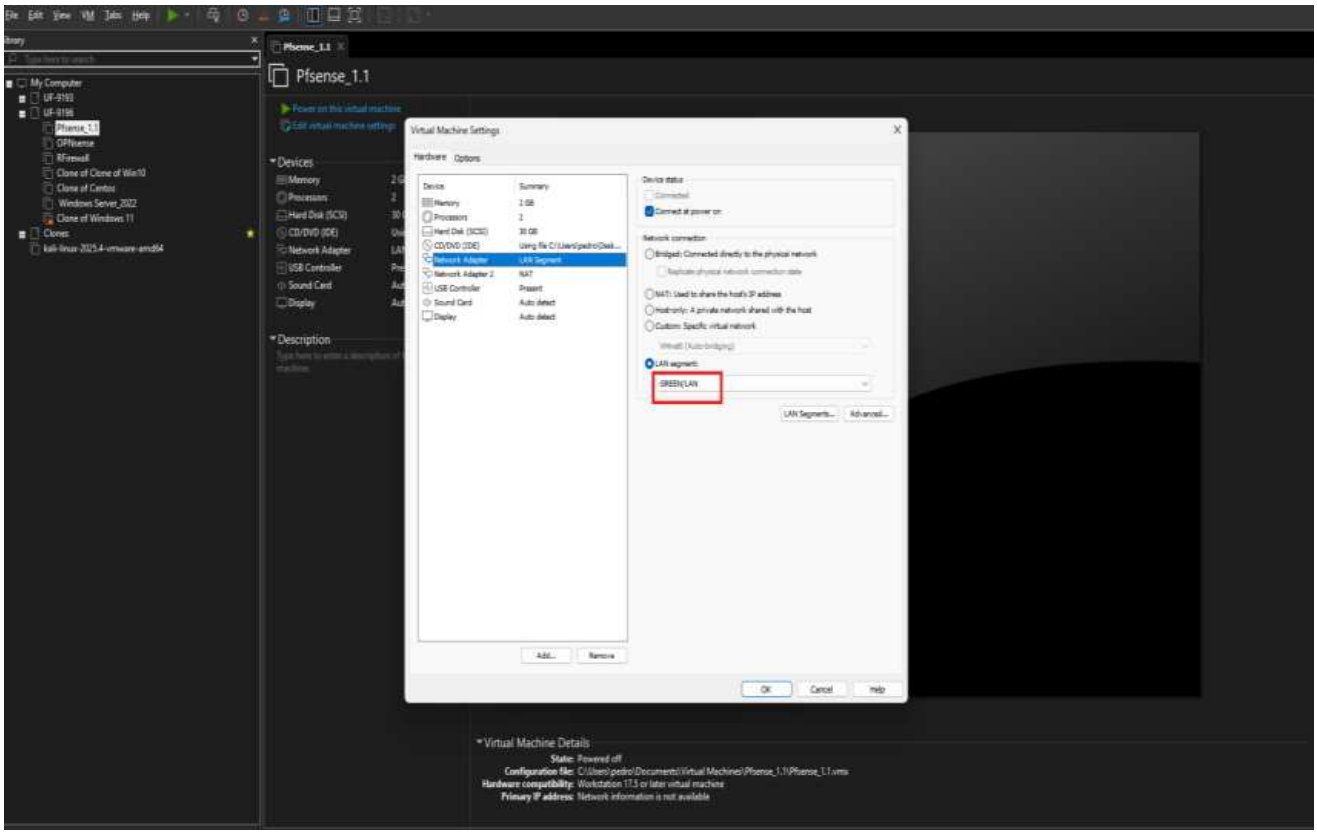
11.1 Configurações de Interfaces



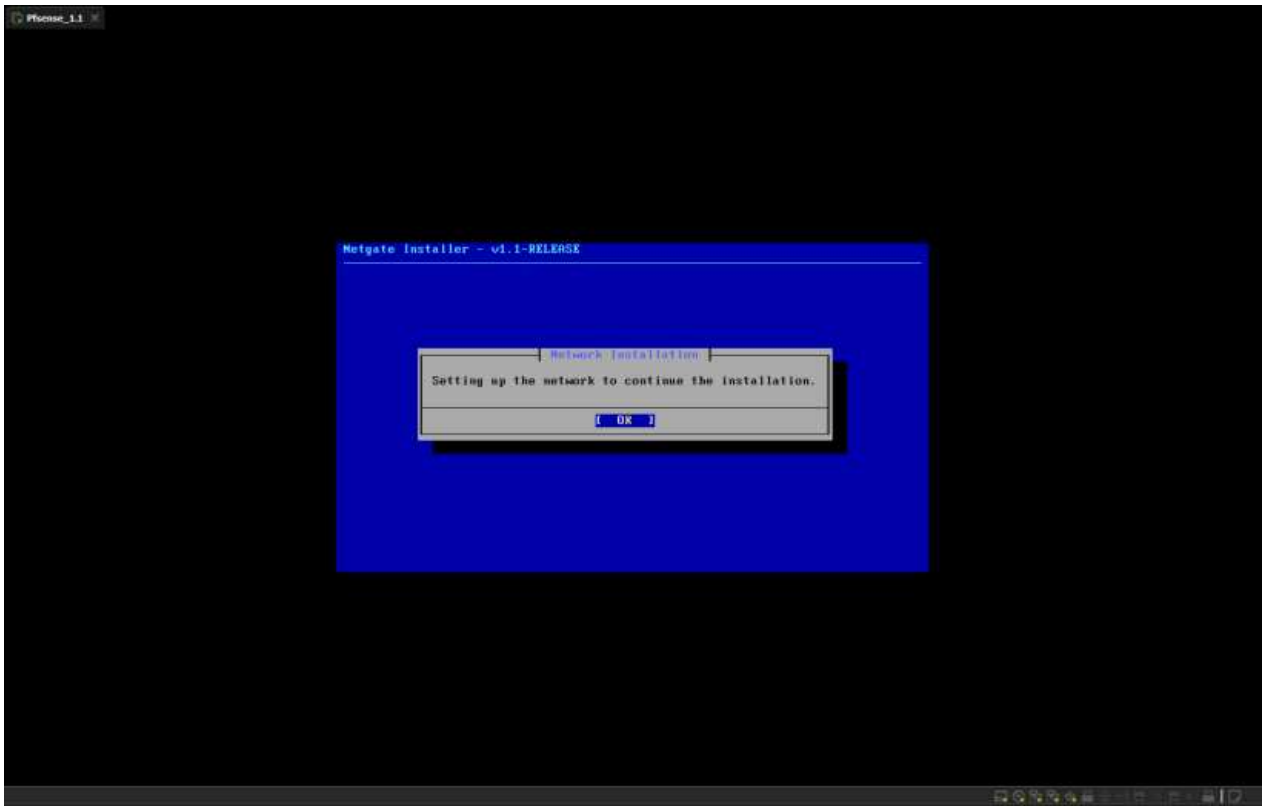
11.2 Configurações :



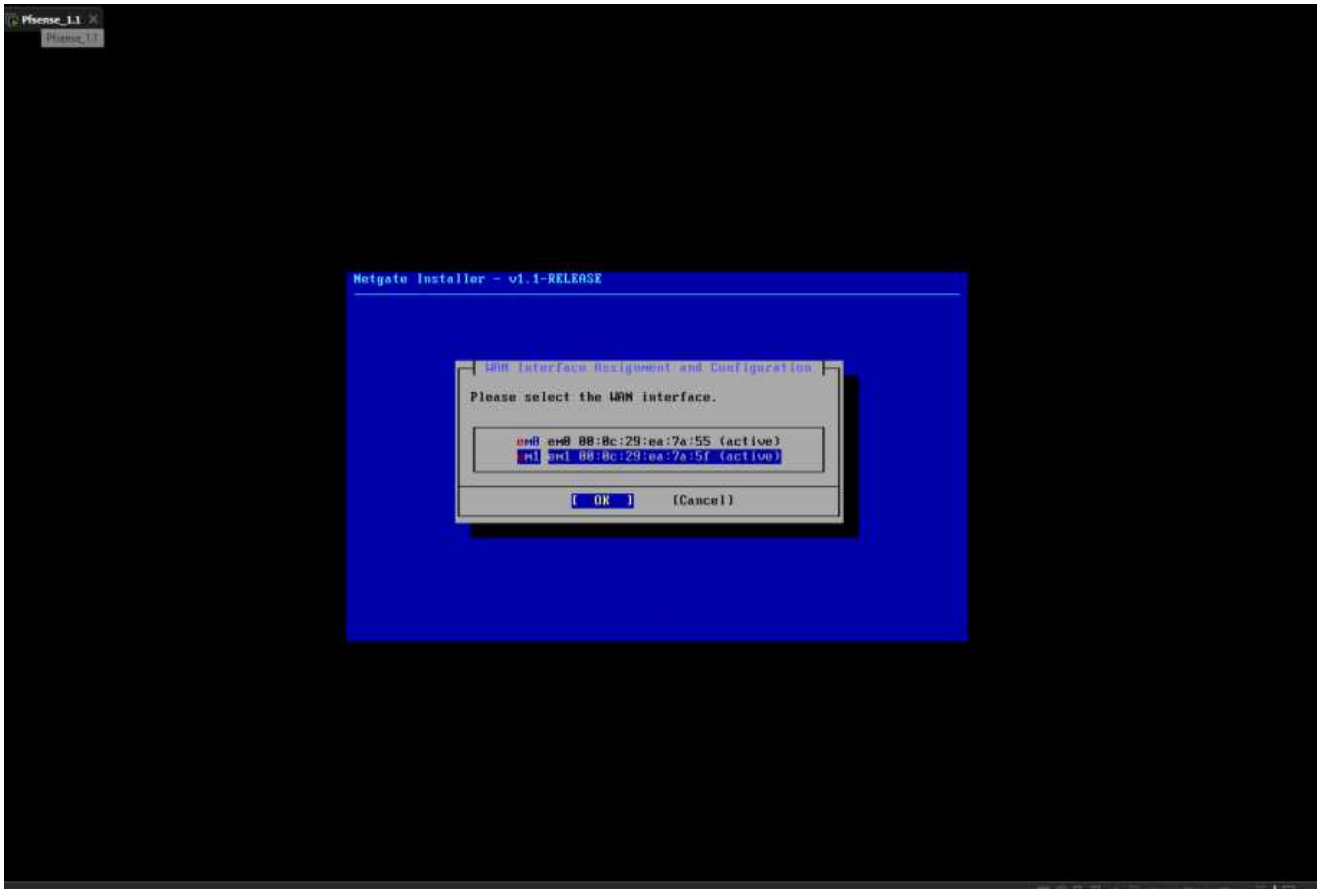
Atribuição de interfaces –"f2green "; NAT



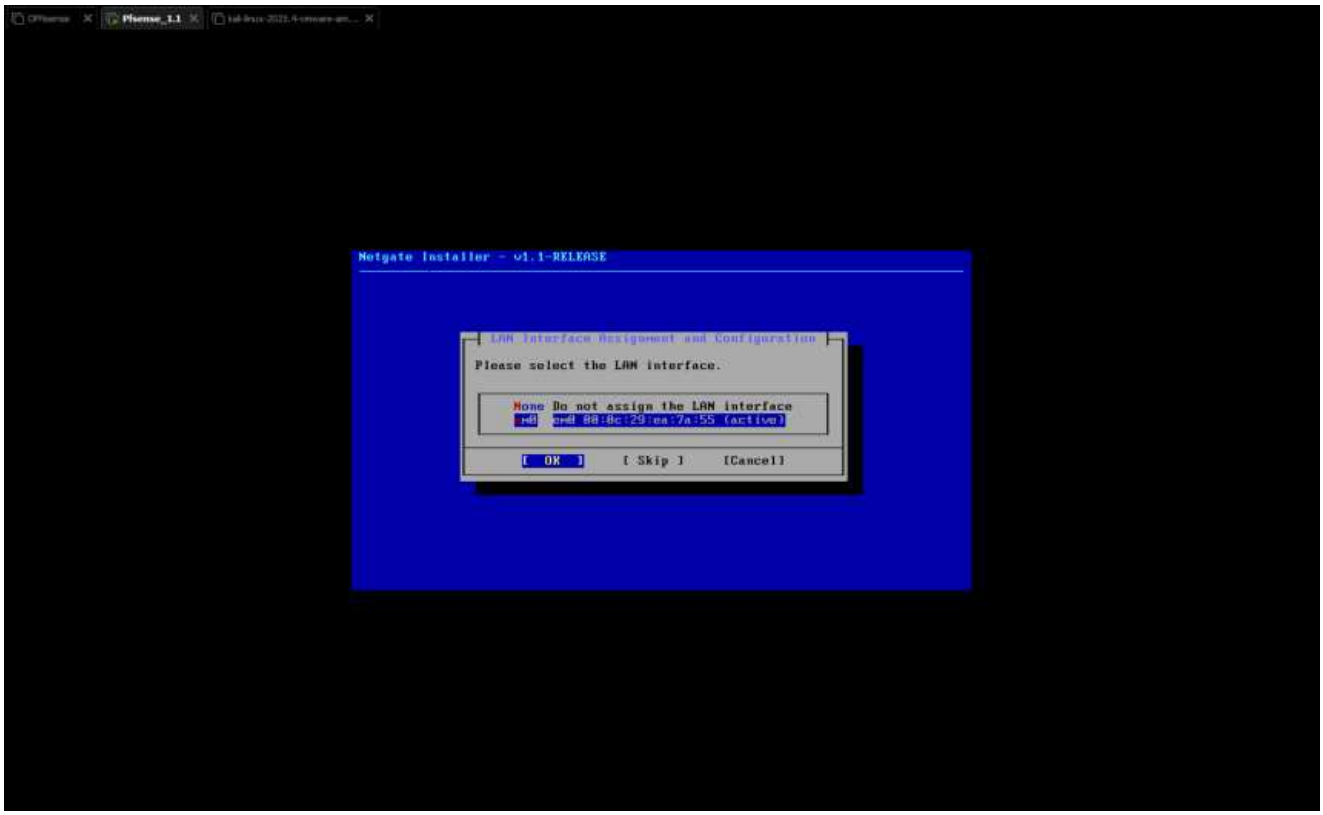
11.3 Instalação :



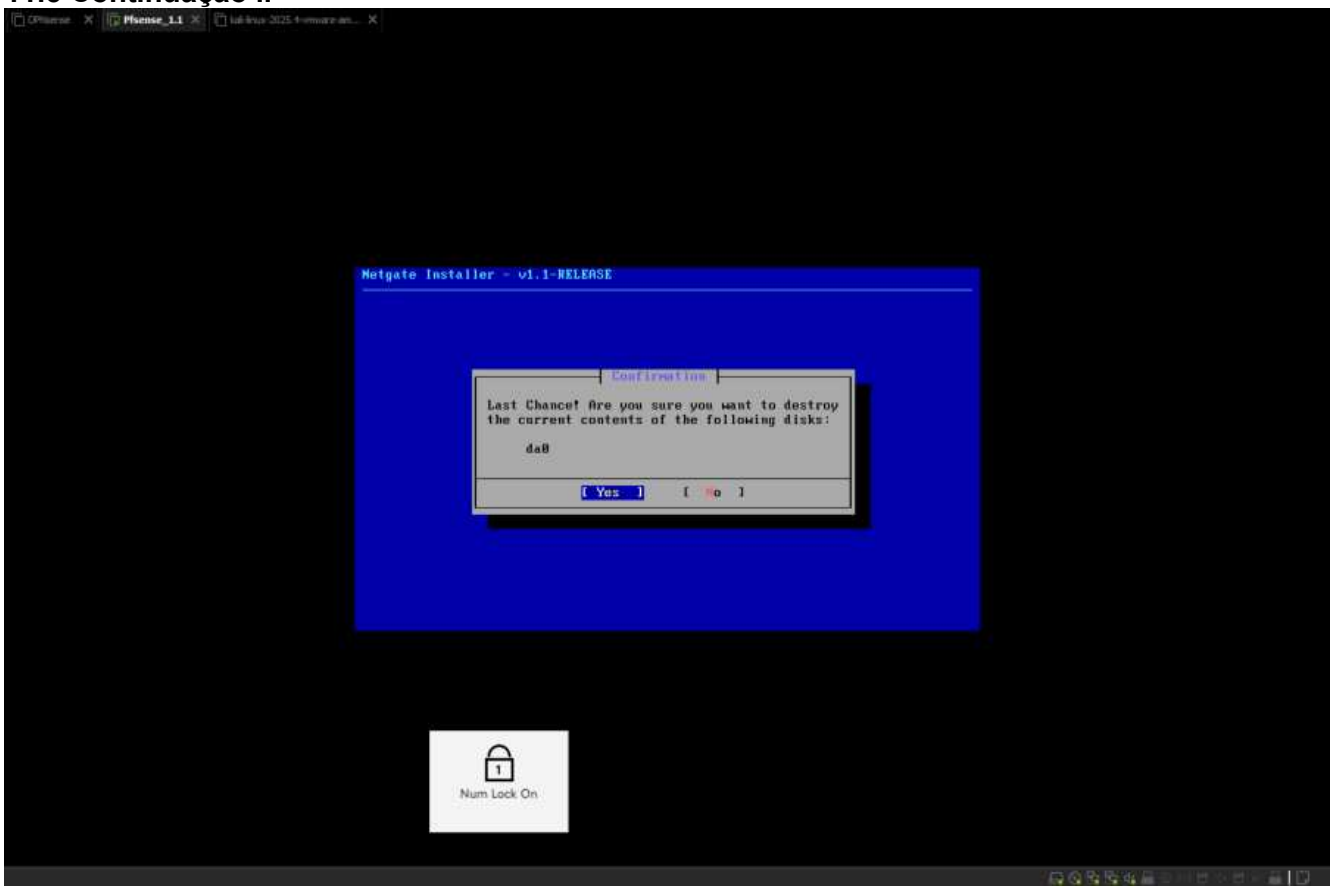
Configuração da **em1**



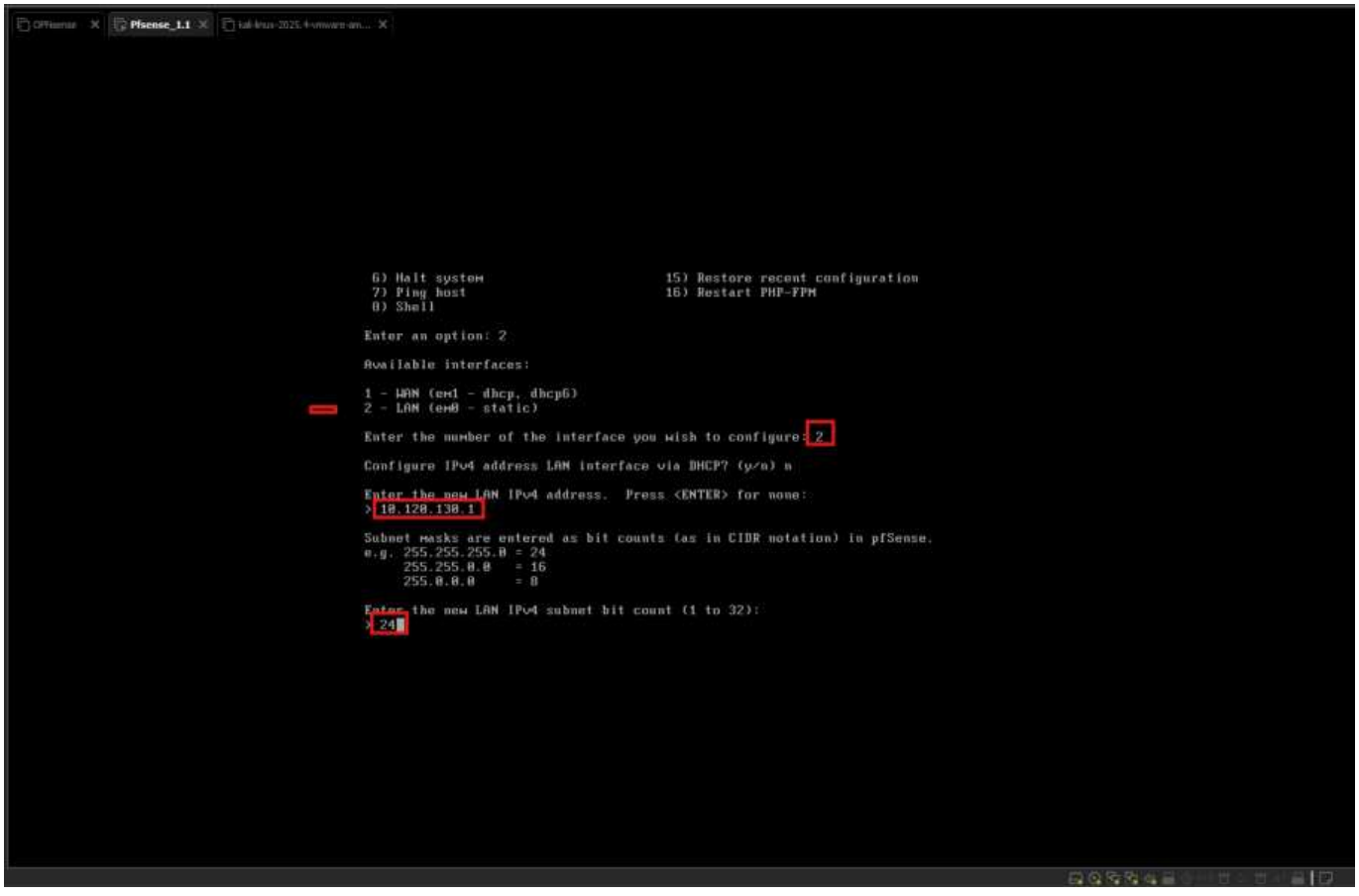
11.4 Continuação :"indicação que Wan é em1"→ importante



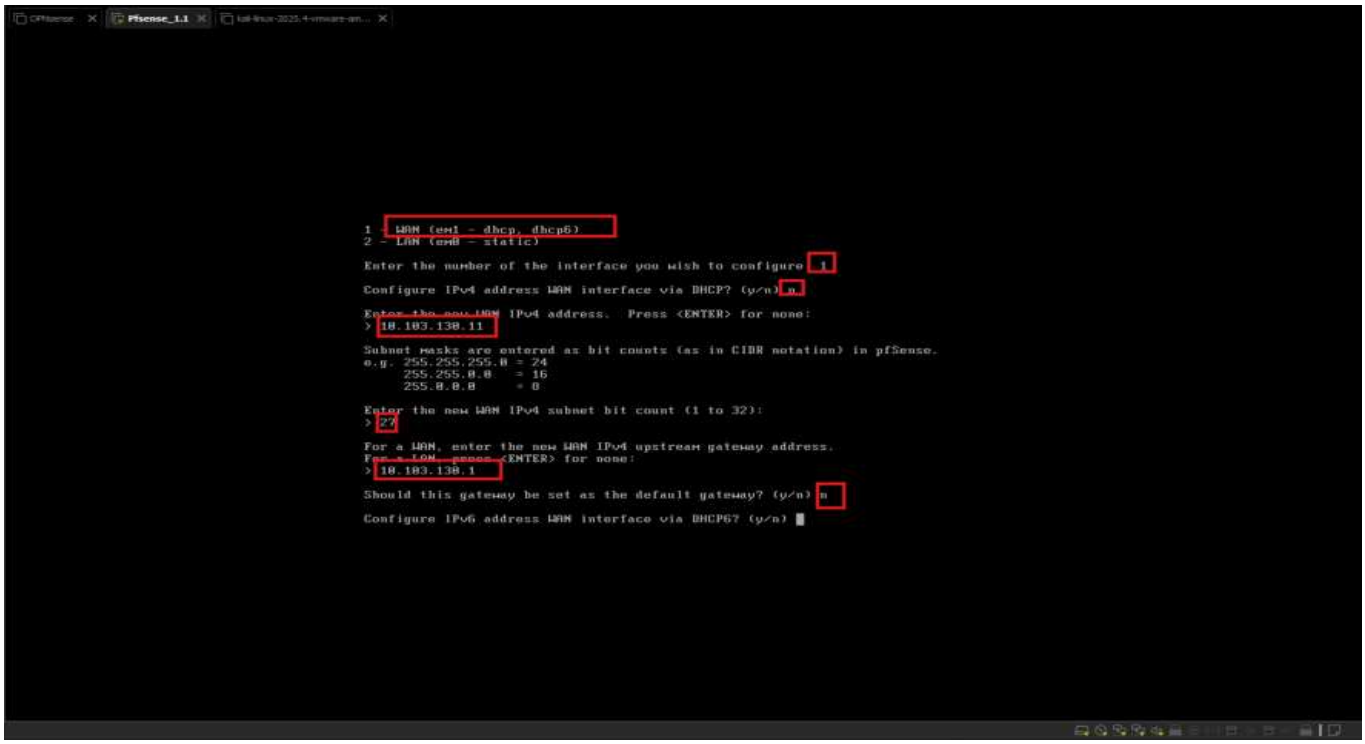
11.5 Continuação ..



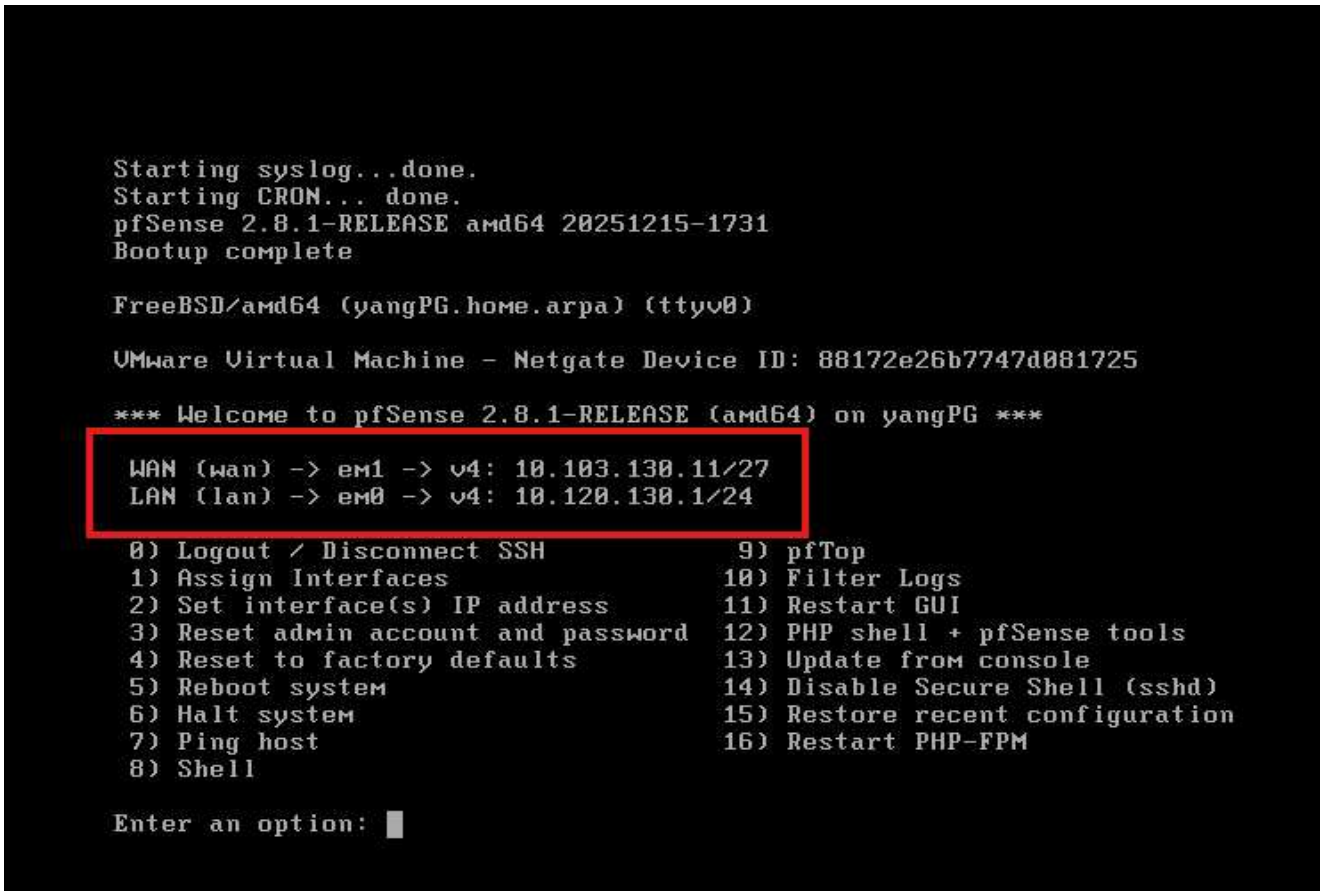
11.6 Configurações LAN



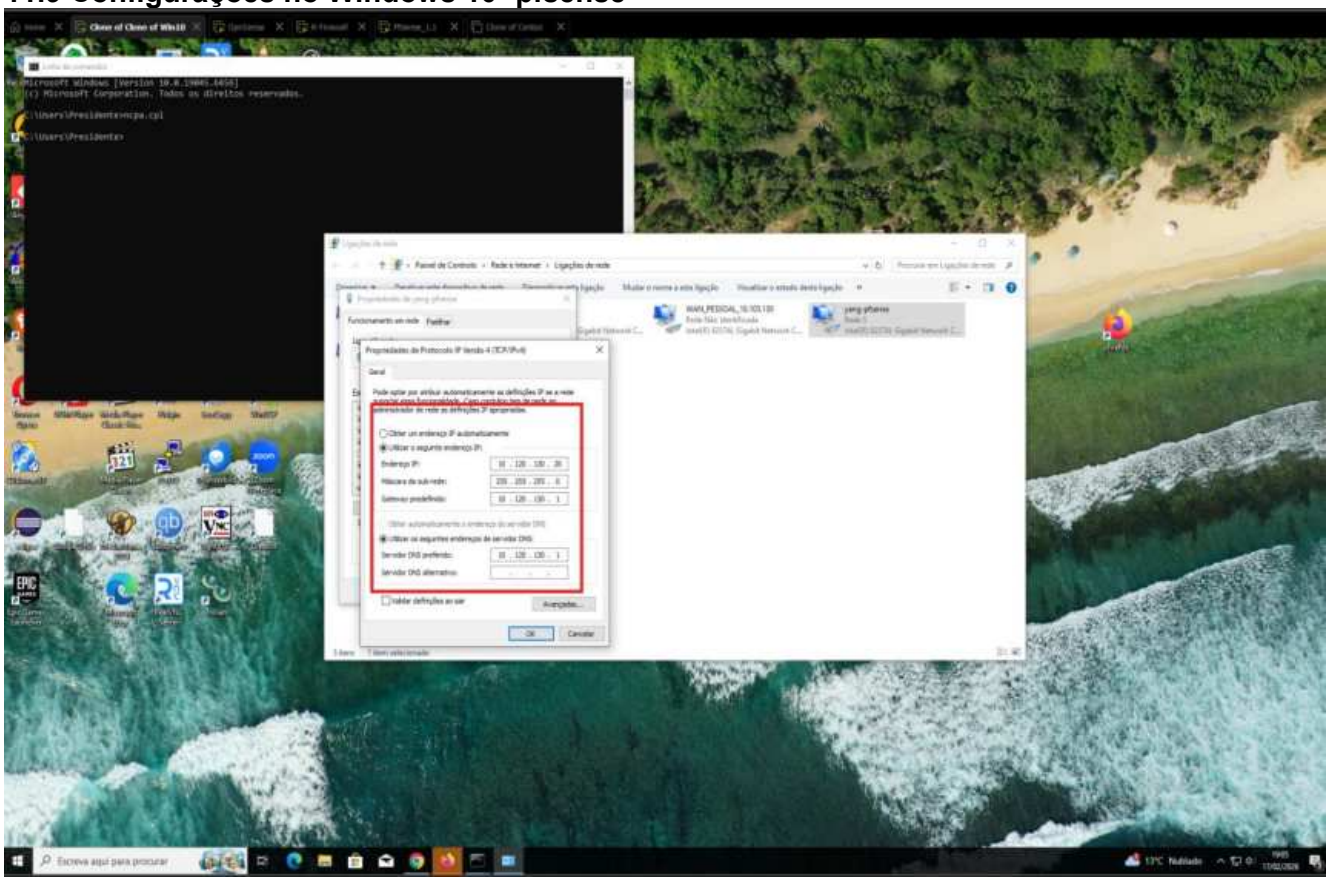
11.7 Configurações WAN :



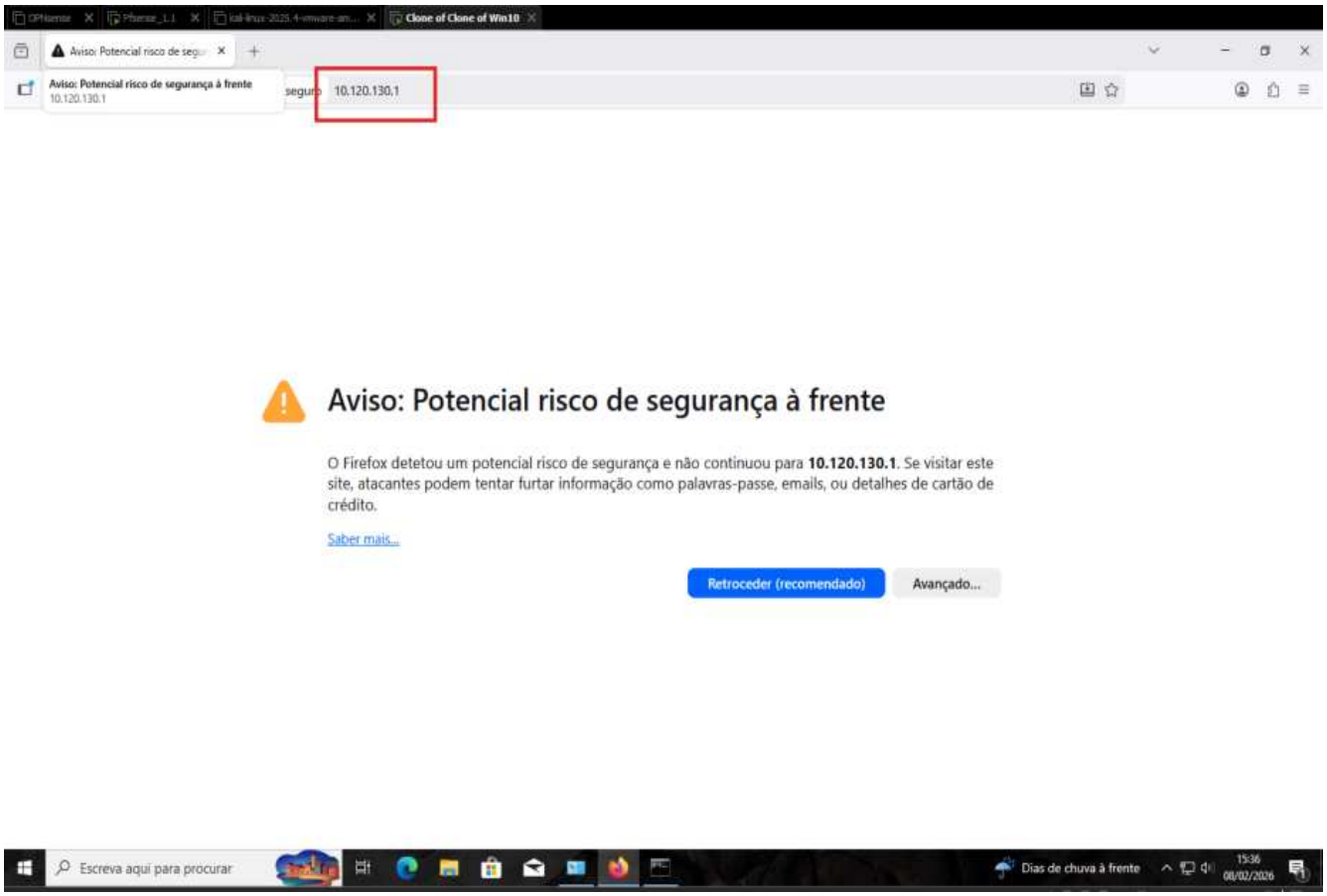
11.8 Resumo (resultado final)



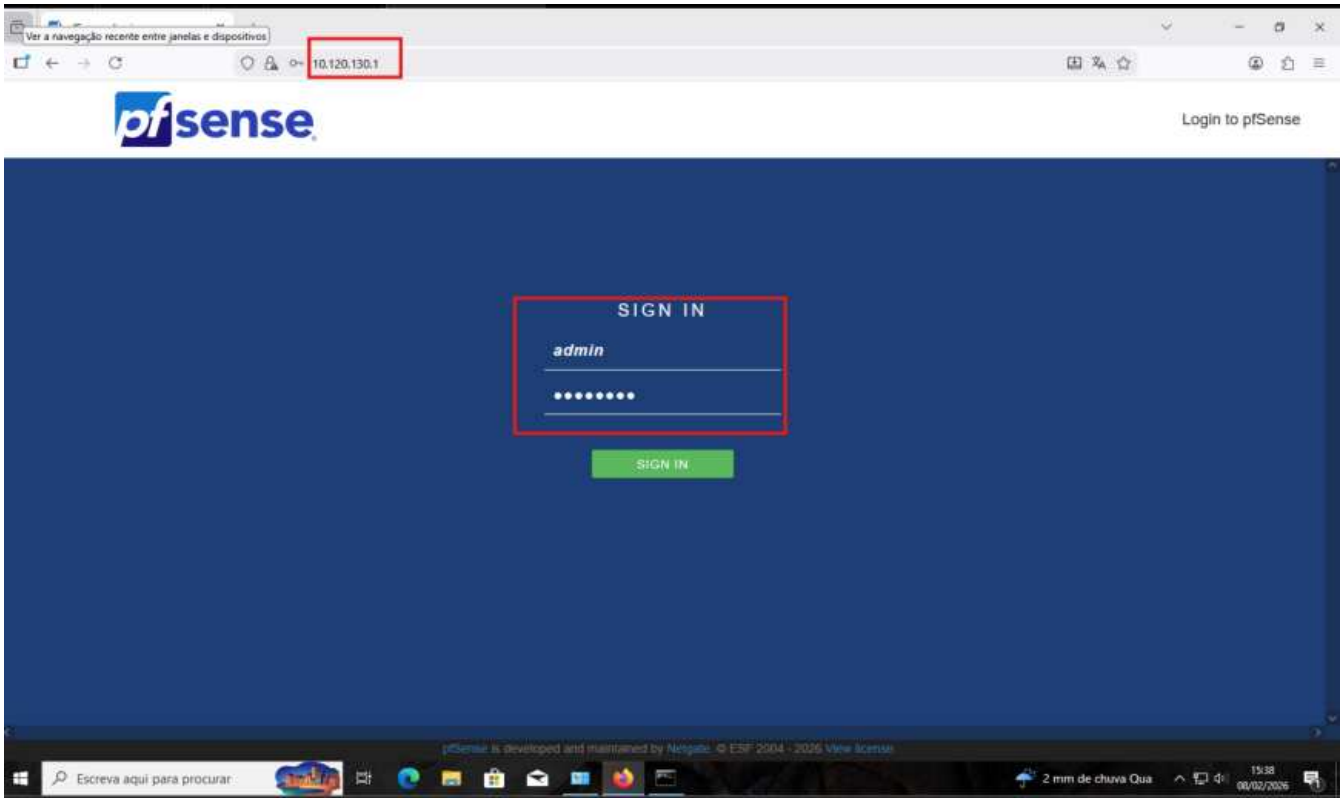
11.9 Configurações no Windows 10 -pfsense



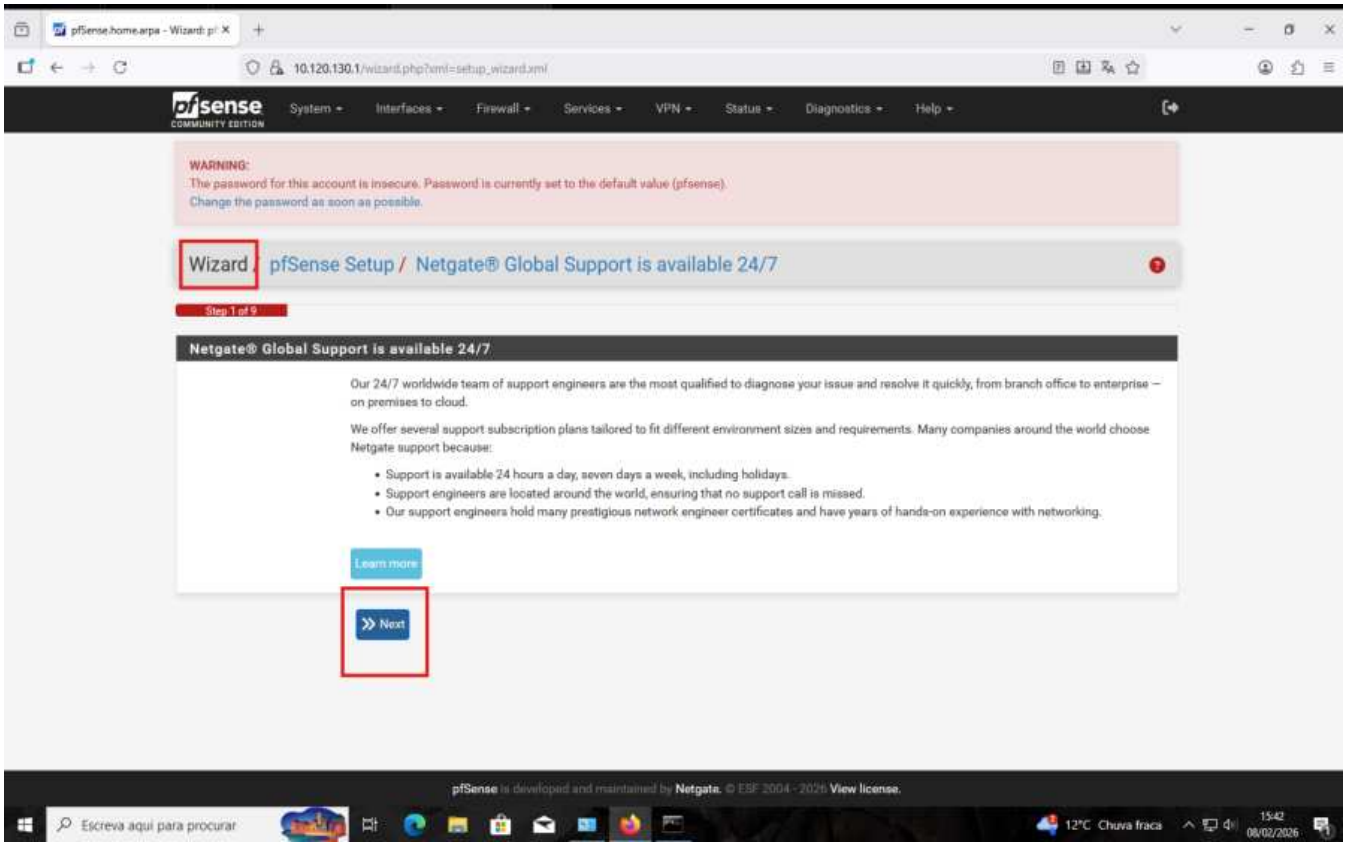
11.10 Acesso a pfsense -Browser Firefox



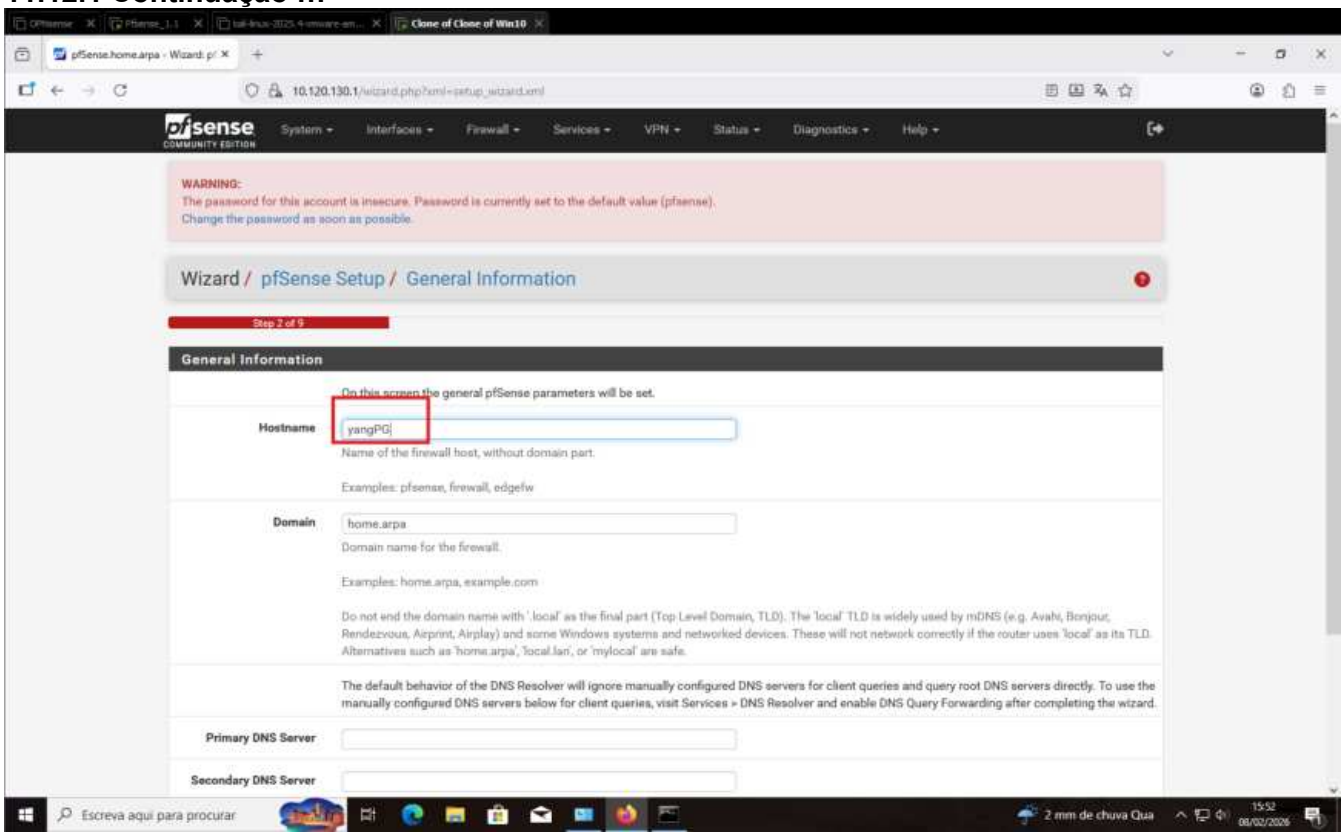
11.11Acesso : user : **admin** : Passwd : **pfsense**



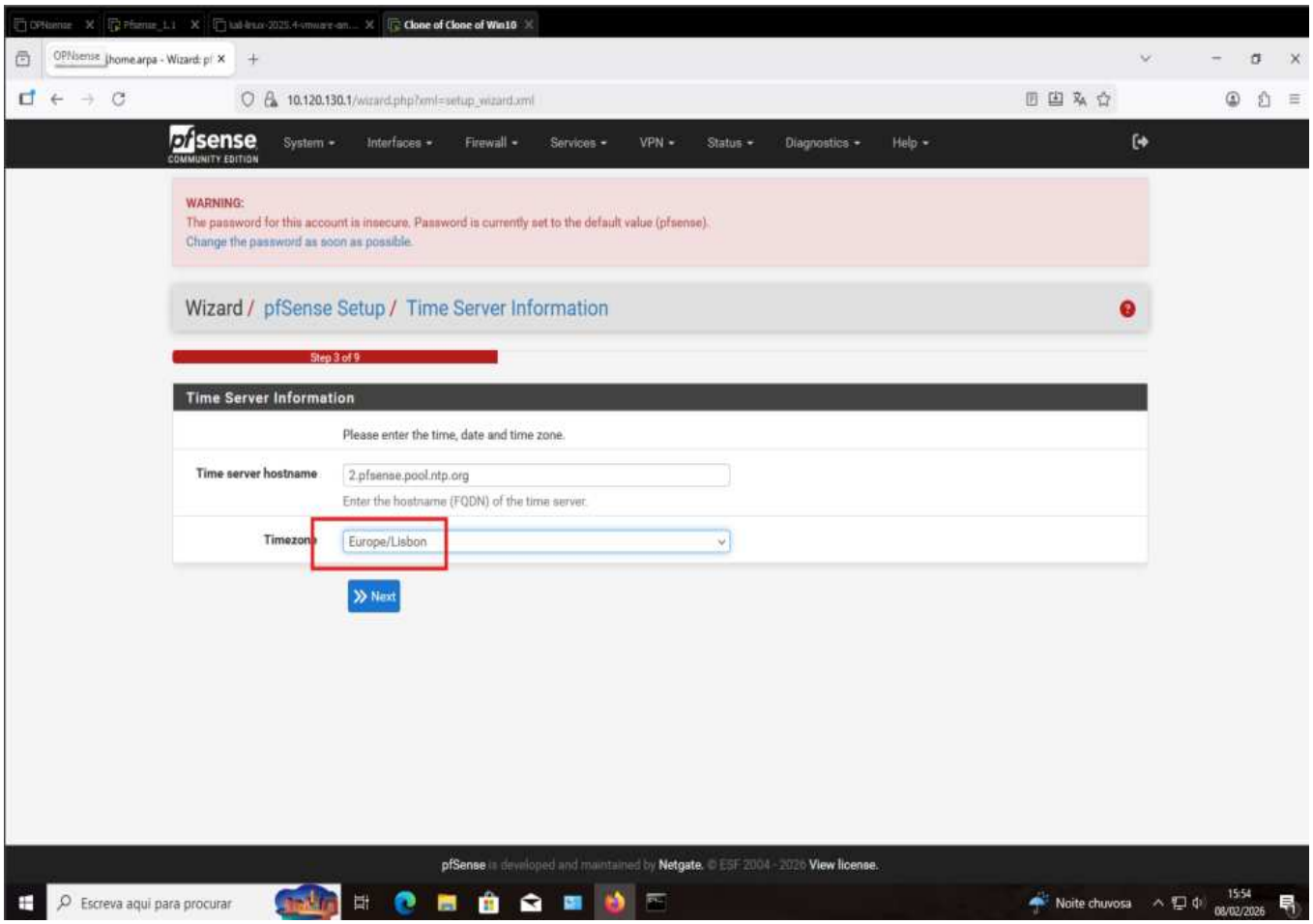
11.12 Correr o wizard inicial :



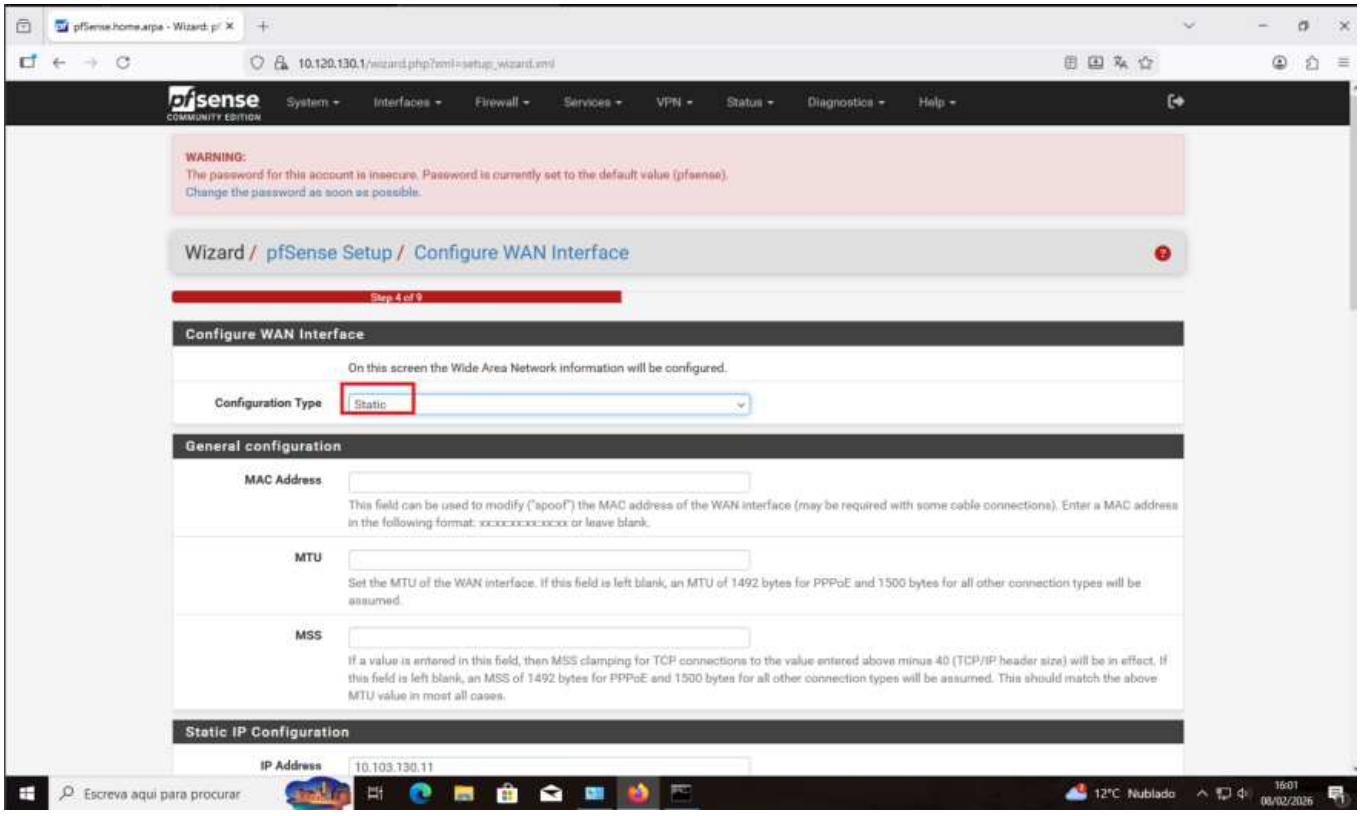
11.12.1 Continuação …



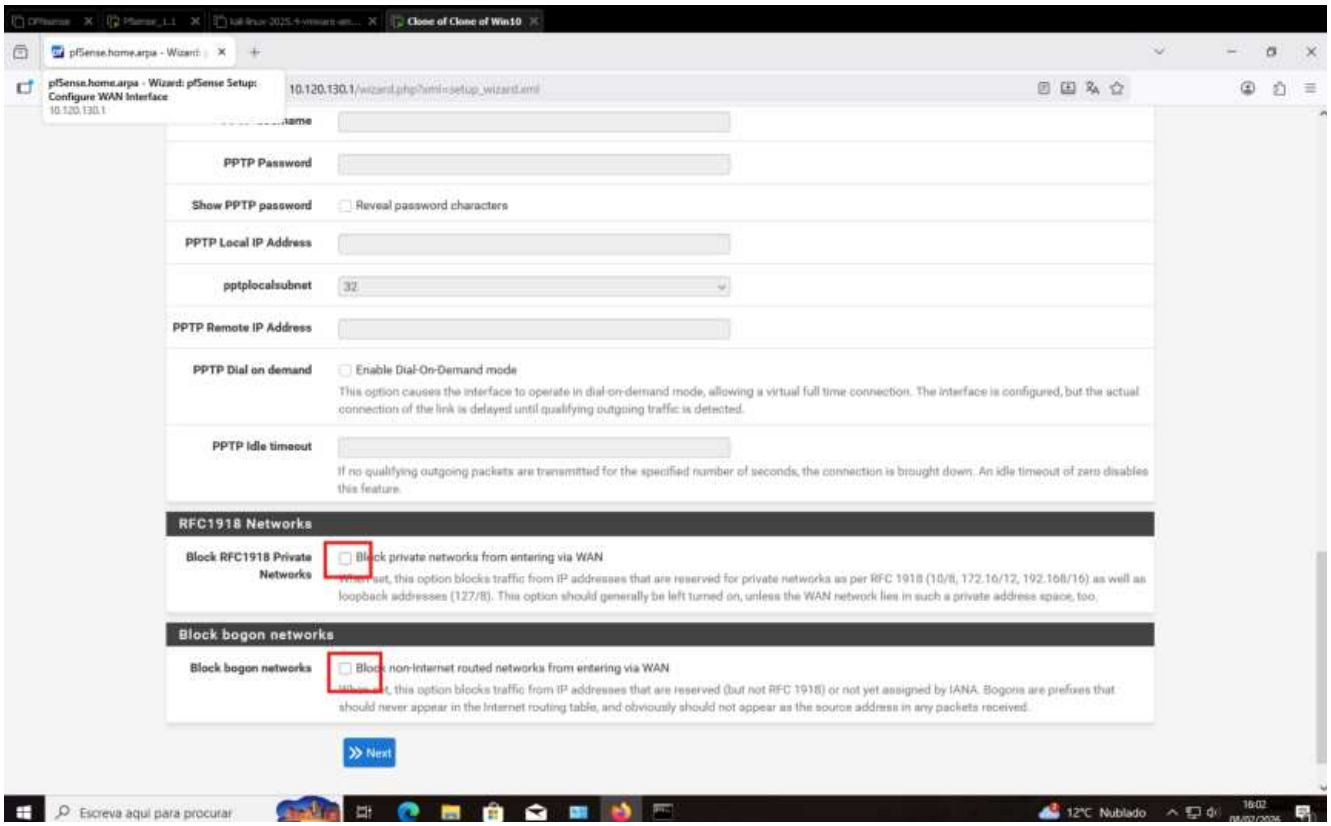
11.12.2 Continuação …



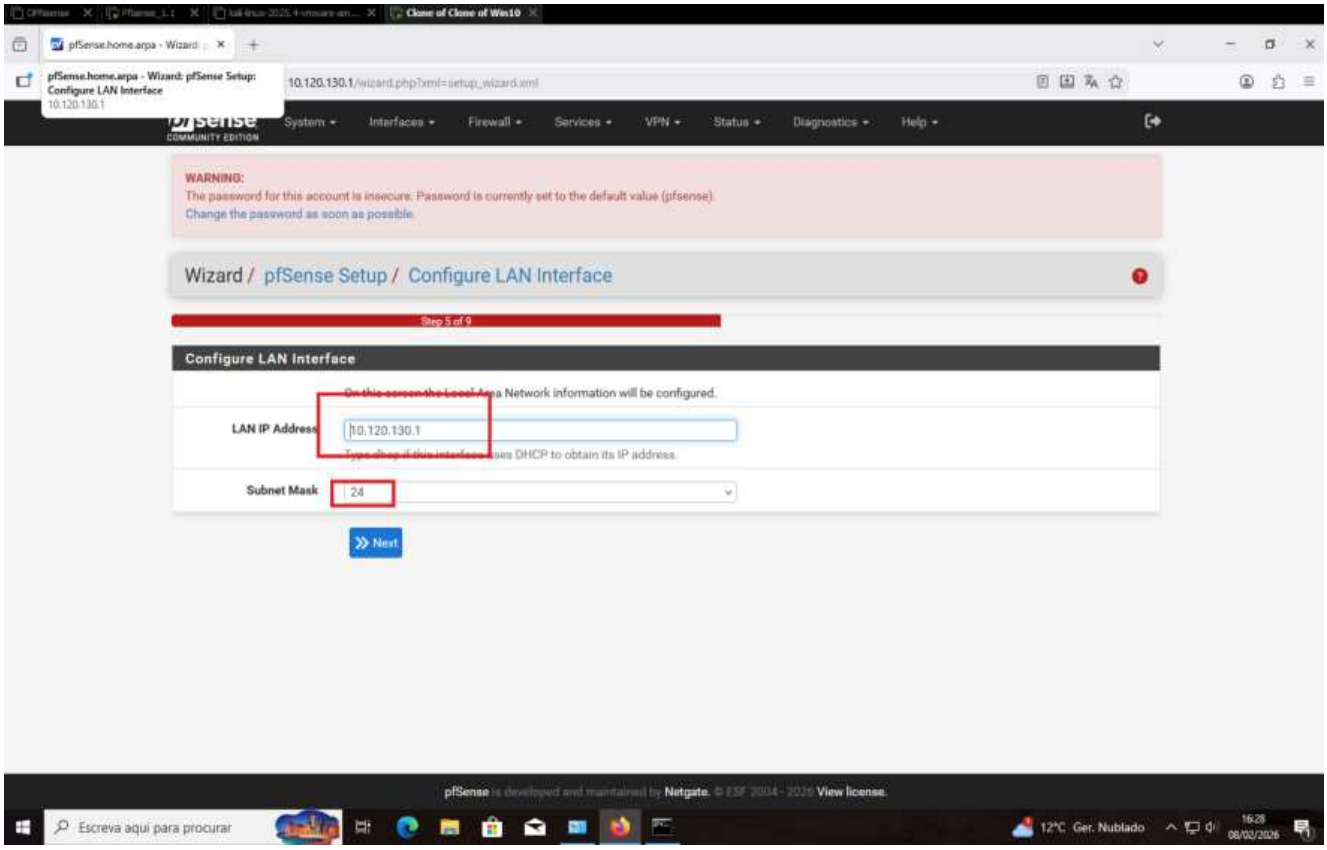
11.12.3 Continuação …



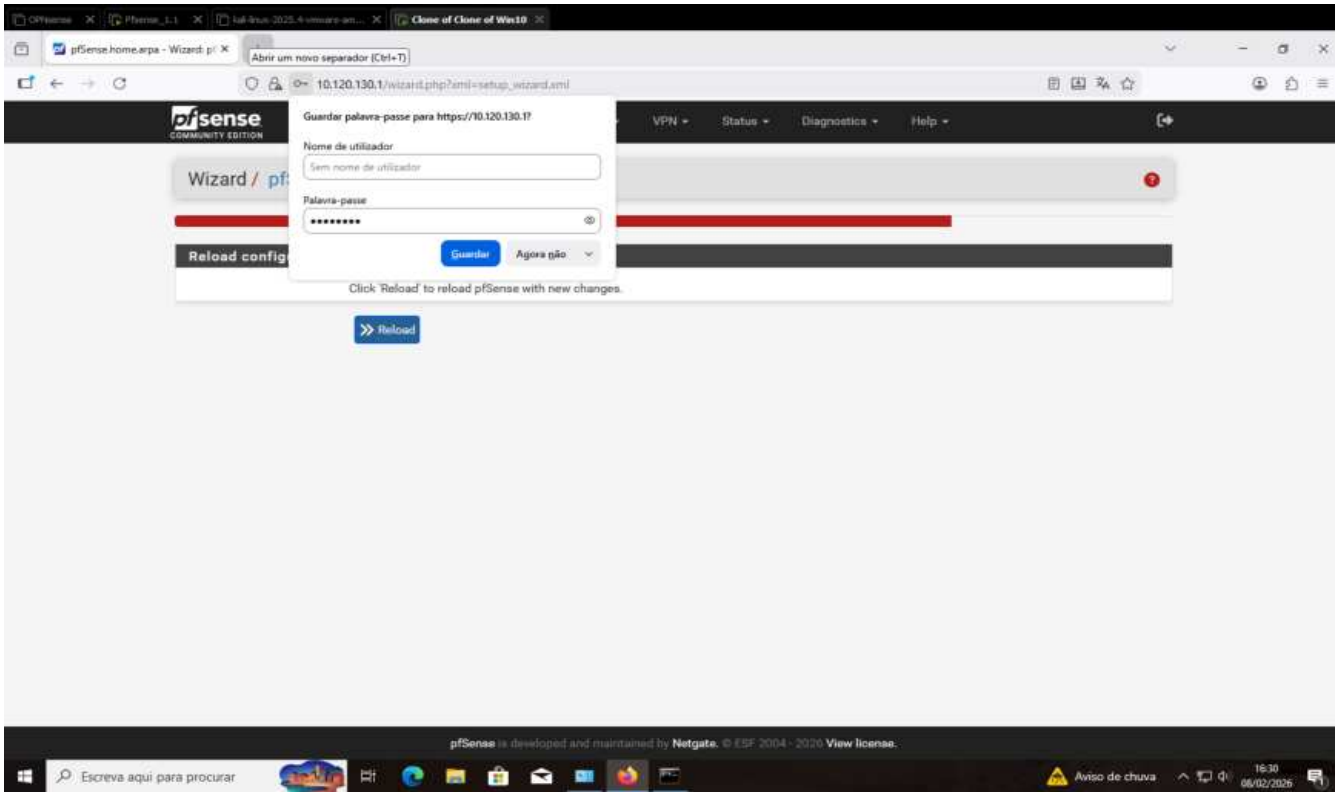
11.12.4Continuação…(desselecionar campos)



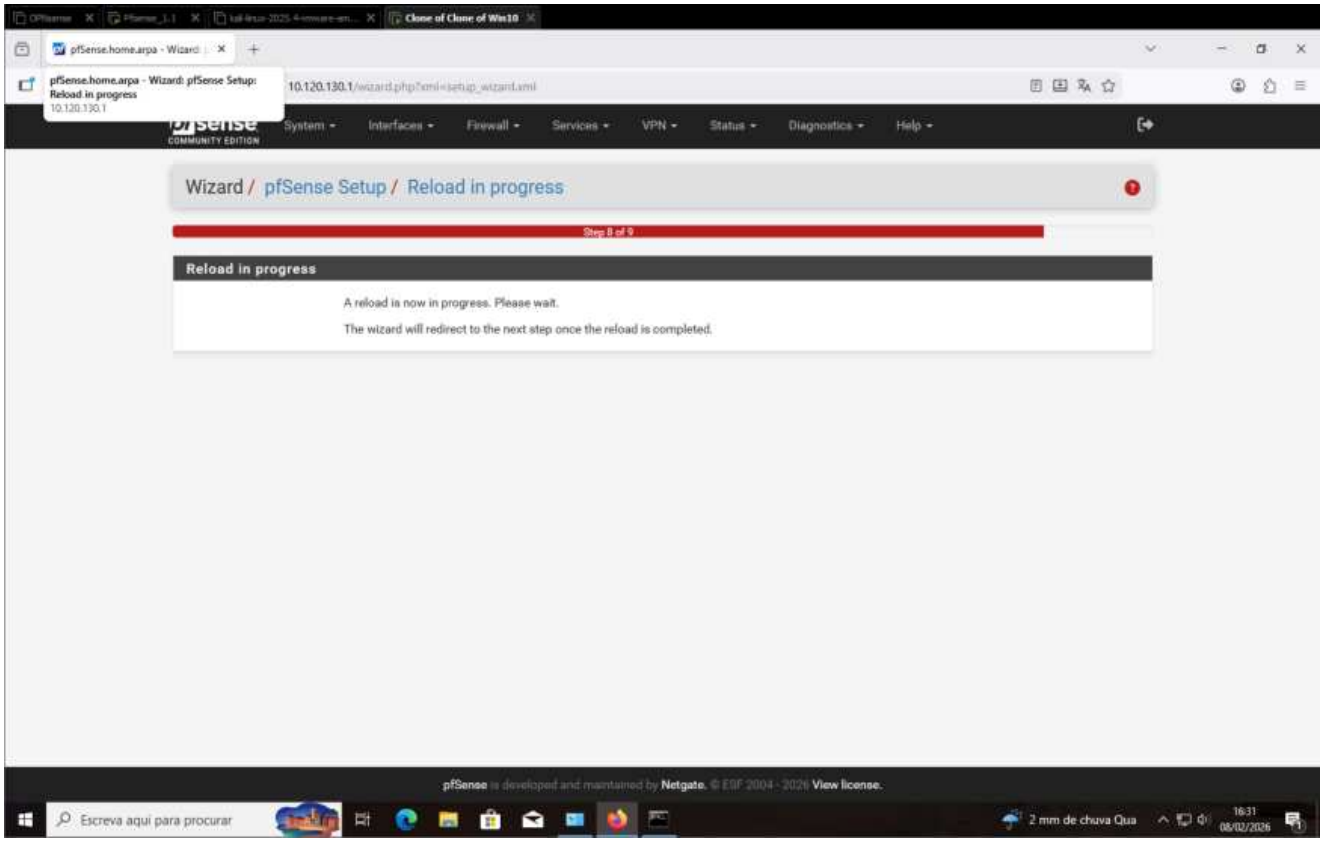
11.12.5 Continuação…



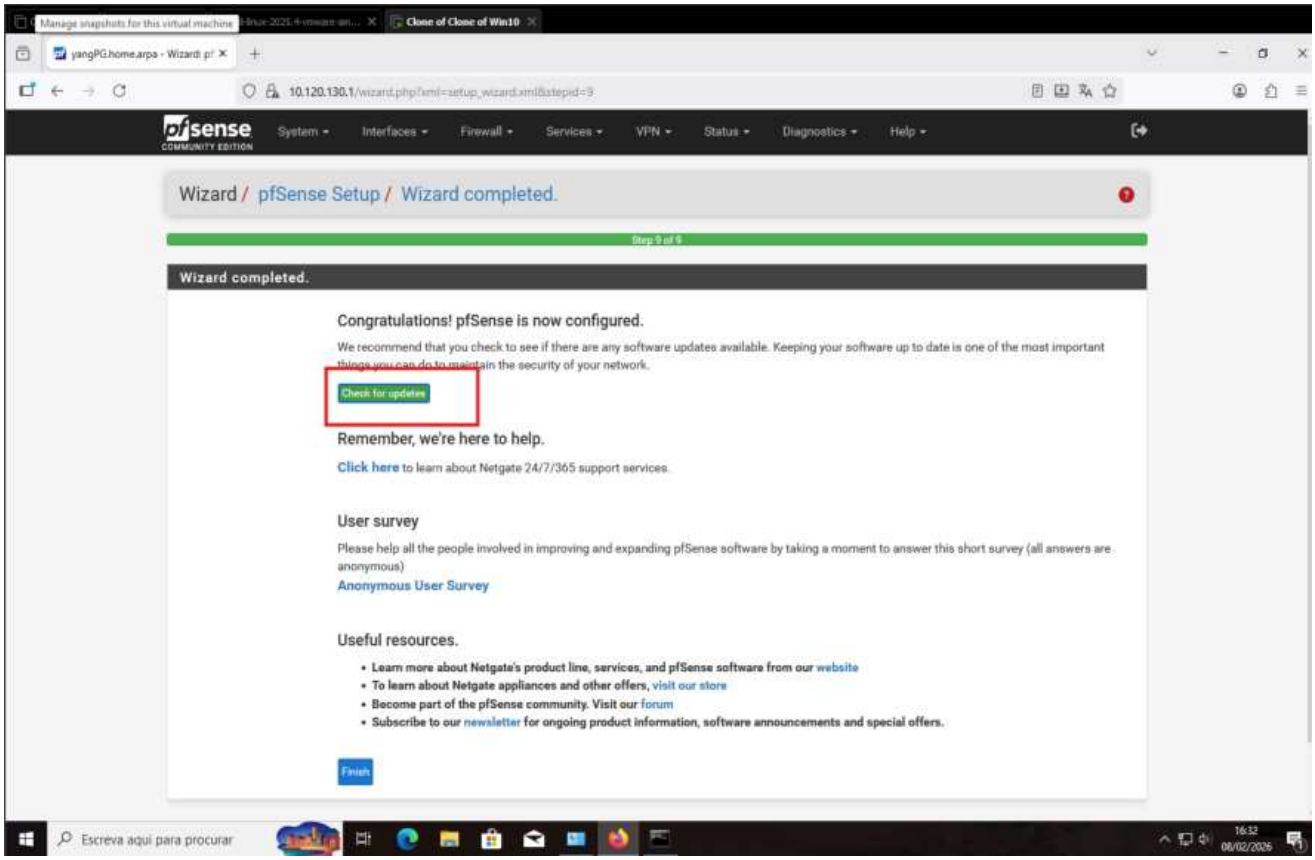
11.12.6 Seleção password : **Passw0rd**



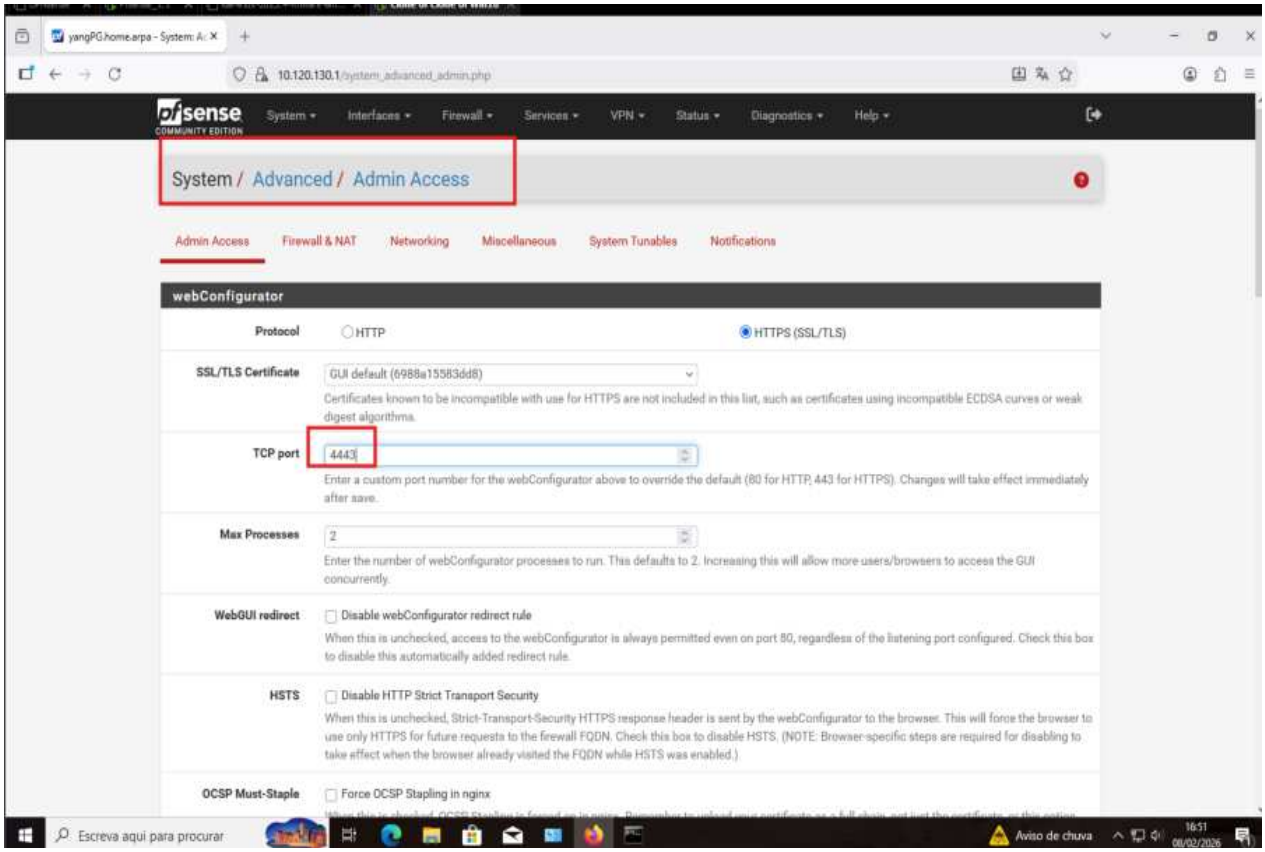
Continuação ...



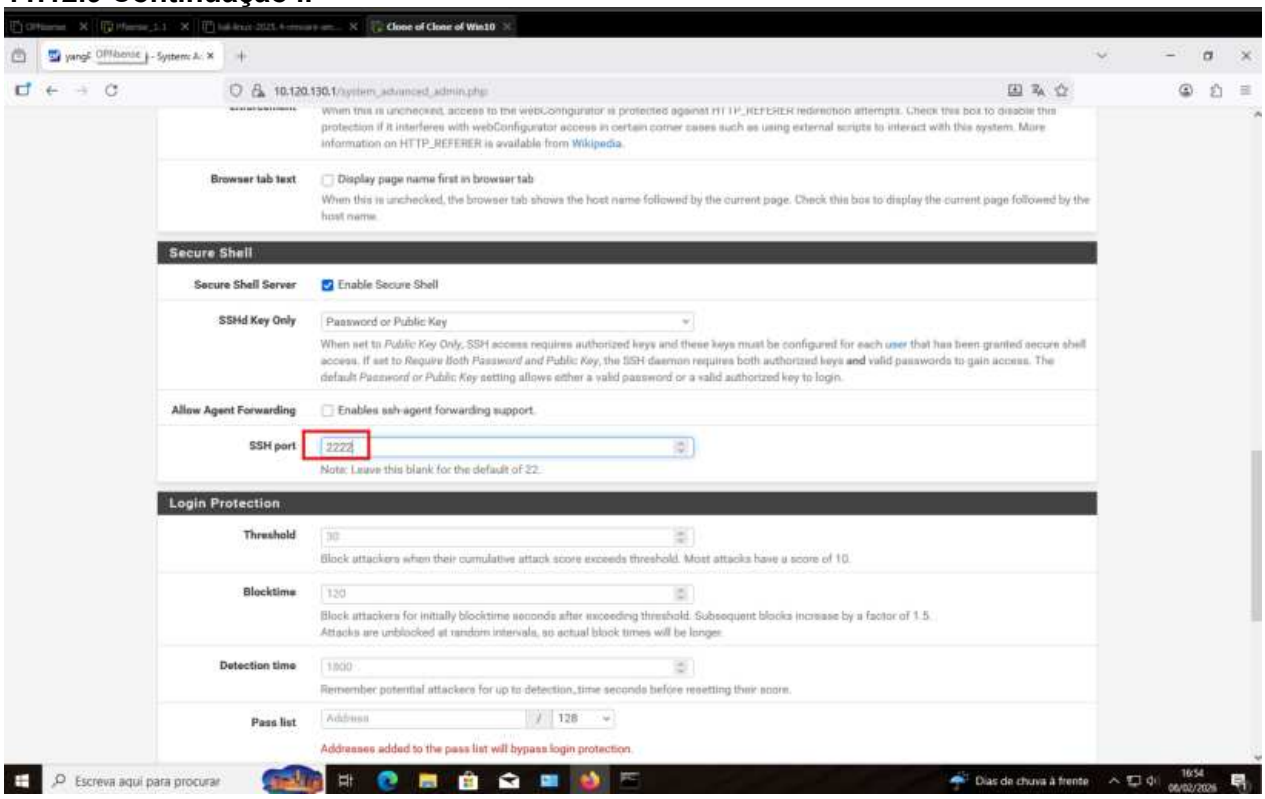
11.12.7 Updates :



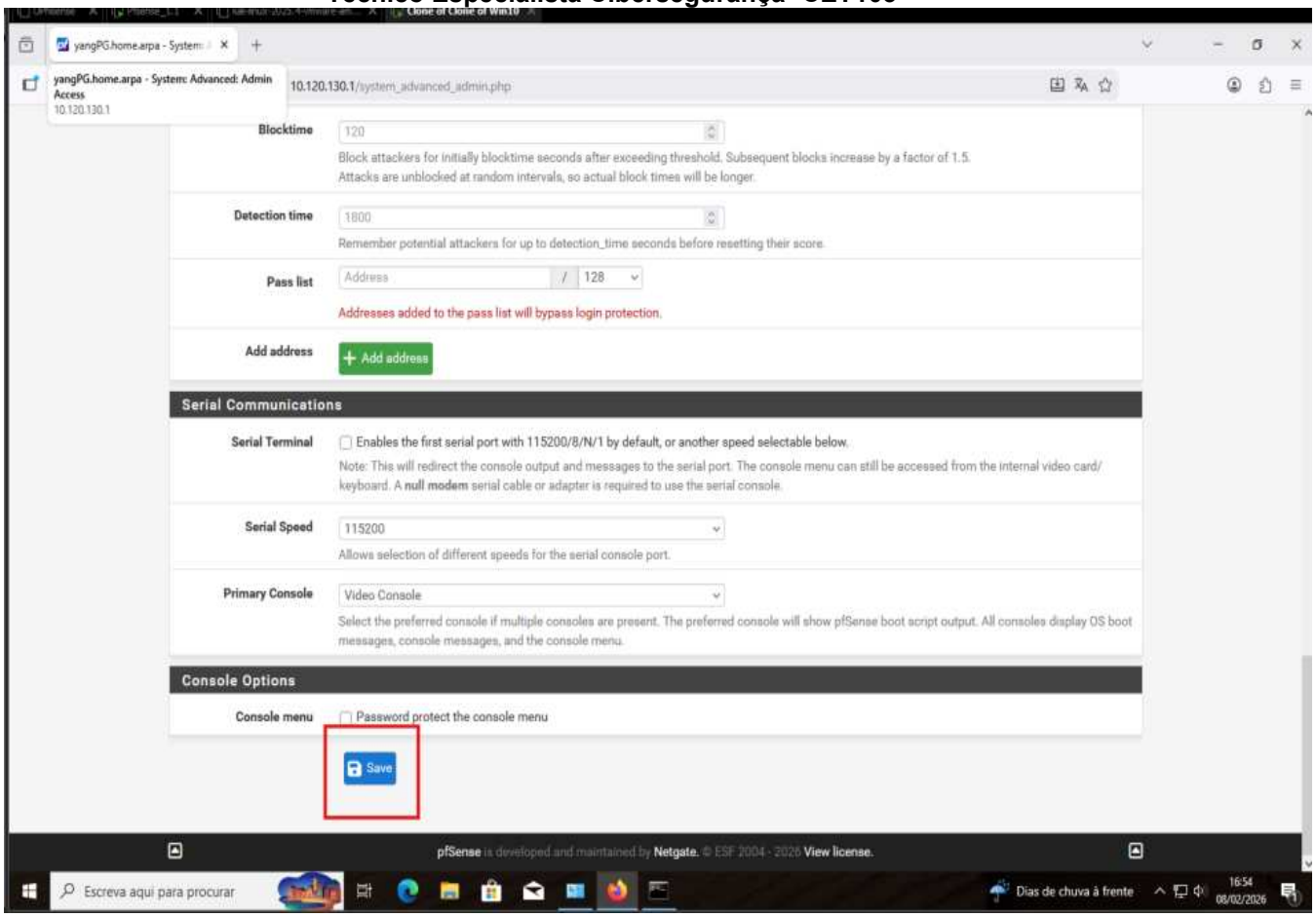
11.12.8 System/Advanced/Admin Access



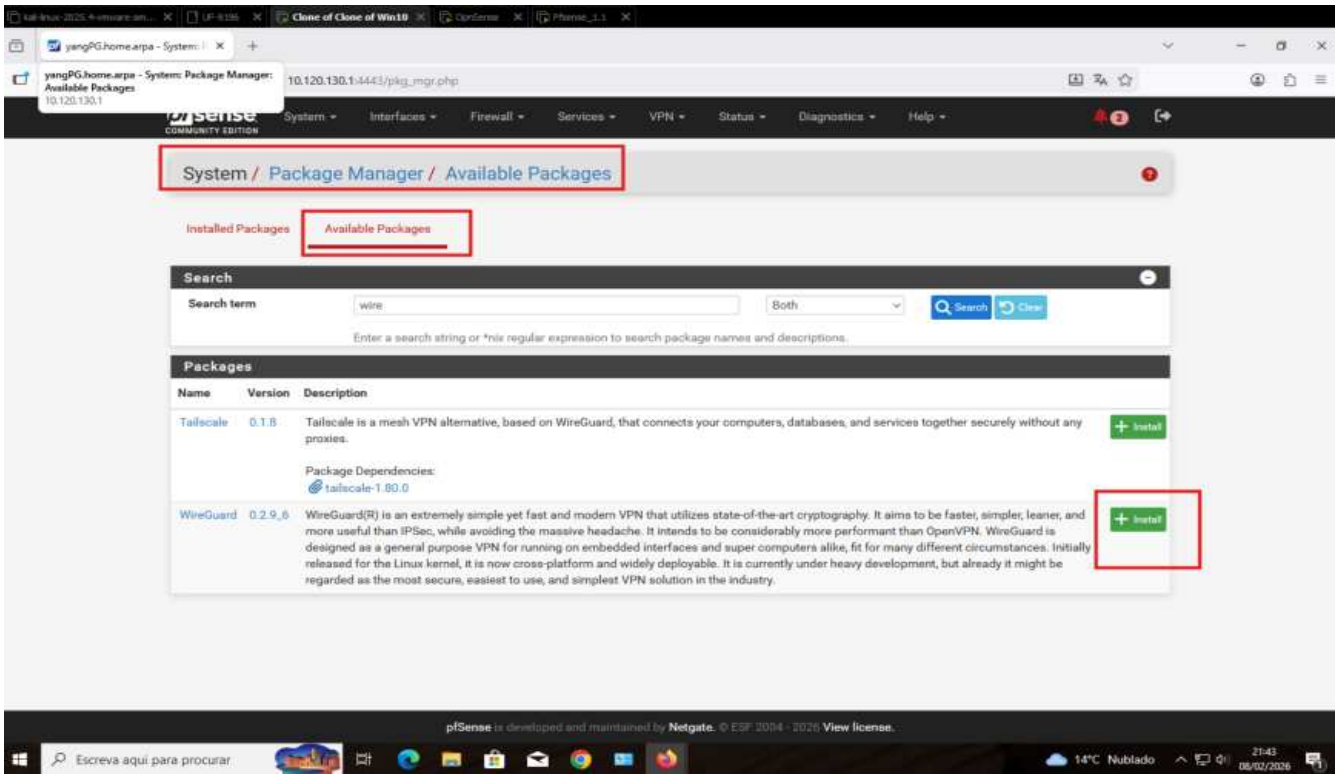
11.12.9 Continuação ..



11.12.10 Continuação…

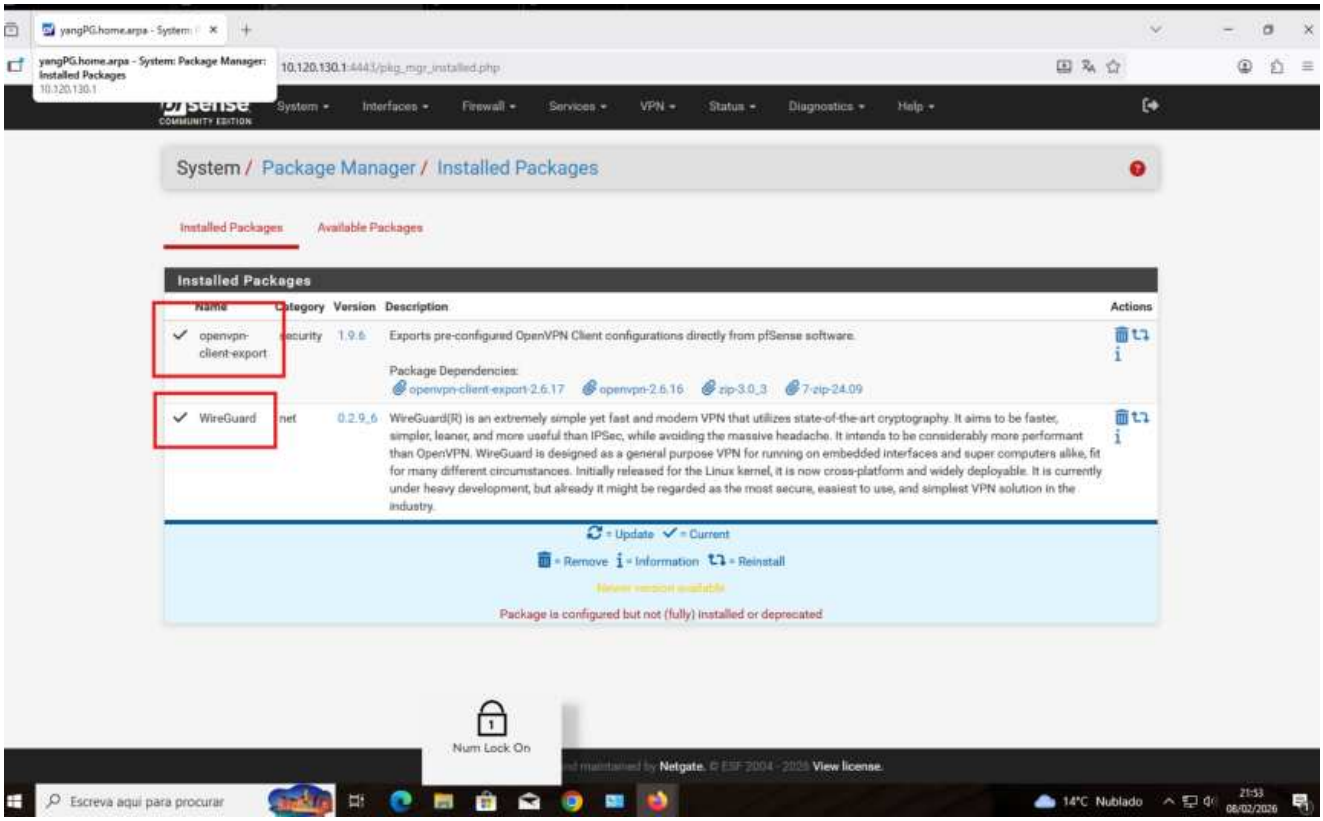


11.12.11 Atualização dos Packages :- instalação do Wireguard



Pacotes_instalados

:



12- VyOS (R1) — Internet + NAT para o WAN pessoal

Configuraçoes

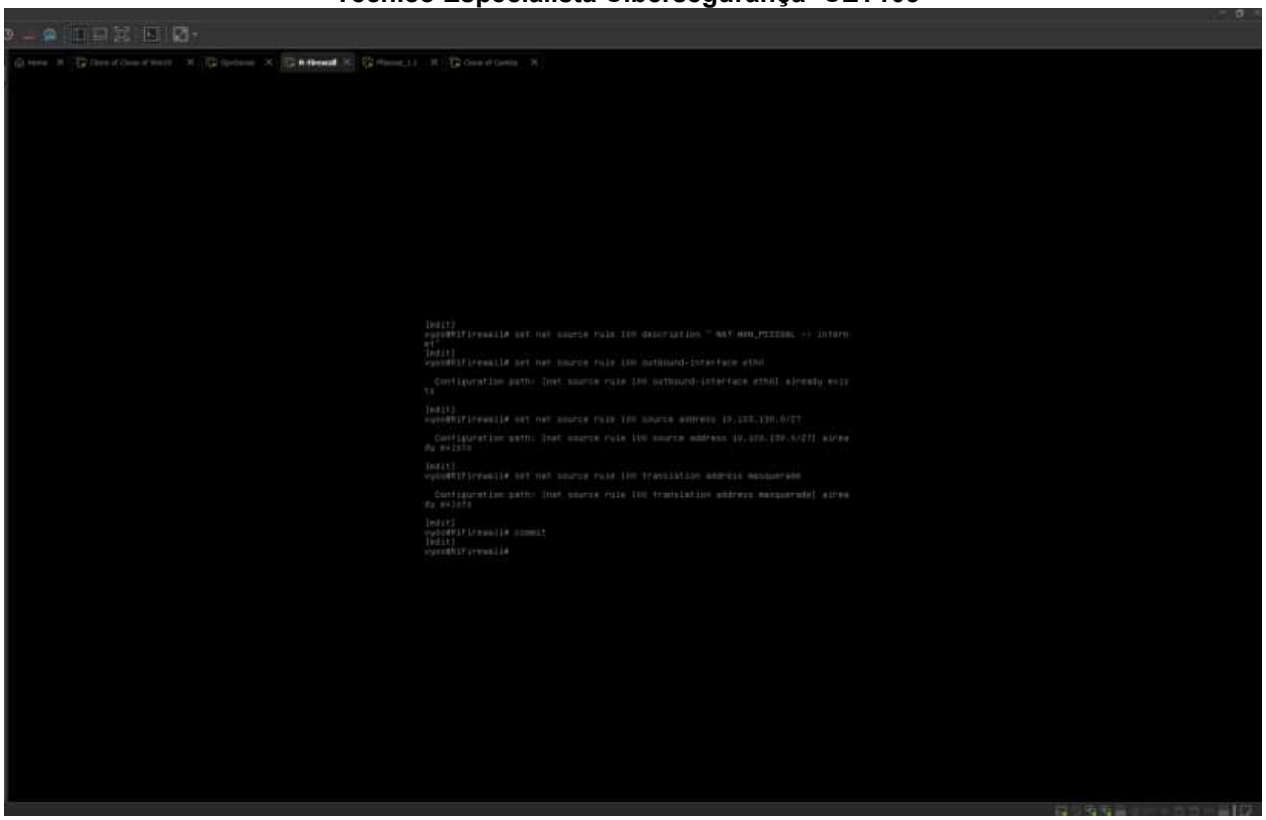
Comandos

“configure

set interfaces ethernet eth0 address dhcp
set interfaces ethernet eth1 address 10.103.130.1/27

set nat source rule 100 description 'NAT WAN_PESSOAL -> Internet'
set nat source rule 100 outbound-interface eth0
set nat source rule 100 source address 10.103.130.0/27
set nat source rule 100 translation address masquerade

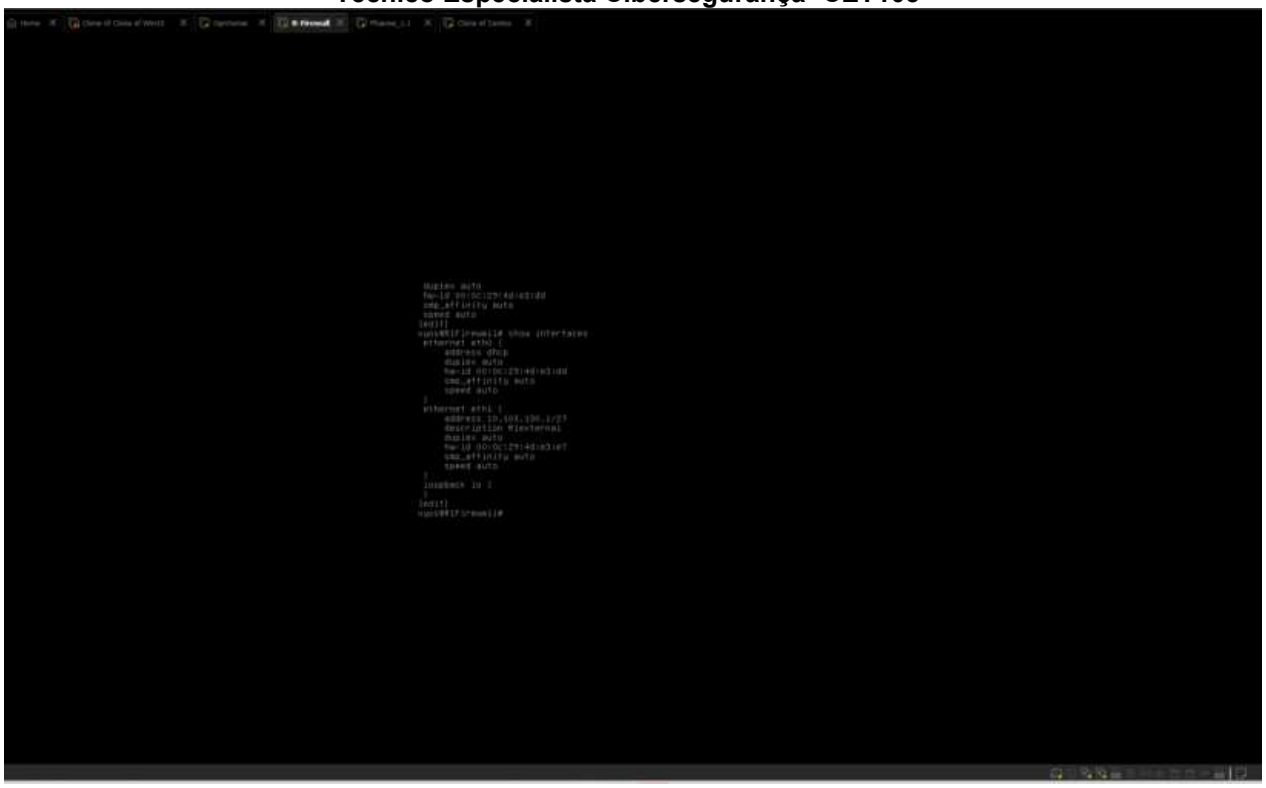
commit
save
exit”



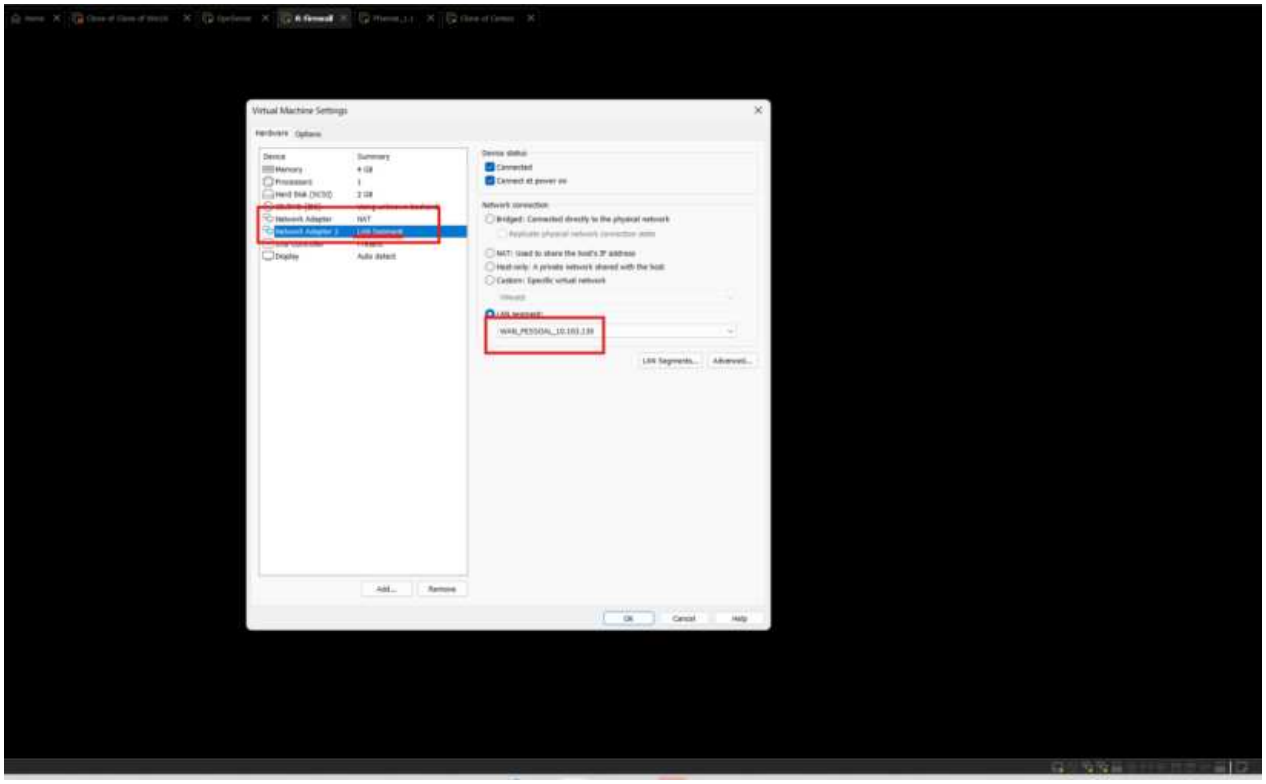
Nota:

*O enunciado define explicitamente a existência de "**uma única rede WAN pessoal**" correspondente ao bloco **10.103.130.0/27** (máscara 255.255.255.224).*
*Dentro deste segmento, o **VyOS assume o papel de gateway**, sendo configurado com o endereço **10.103.130.1**, tornando-se o ponto central de encaminhamento e controlo de tráfego da rede.*
Todos os equipamentos indicados no enunciado encontram-se no mesmo domínio de rede :
***yinPG** → 10.103.130.10 ; **yangPG** → 10.103.130.11; **Windows na WAN** → 10.103.130.20*
*Desta forma, todos pertencem à rede **10.103.130.0/27**, utilizando como gateway comum o endereço **10.103.130.1 (VyOS)**.*
Nota suplementar : Esta abordagem não foi ensinada em aula .

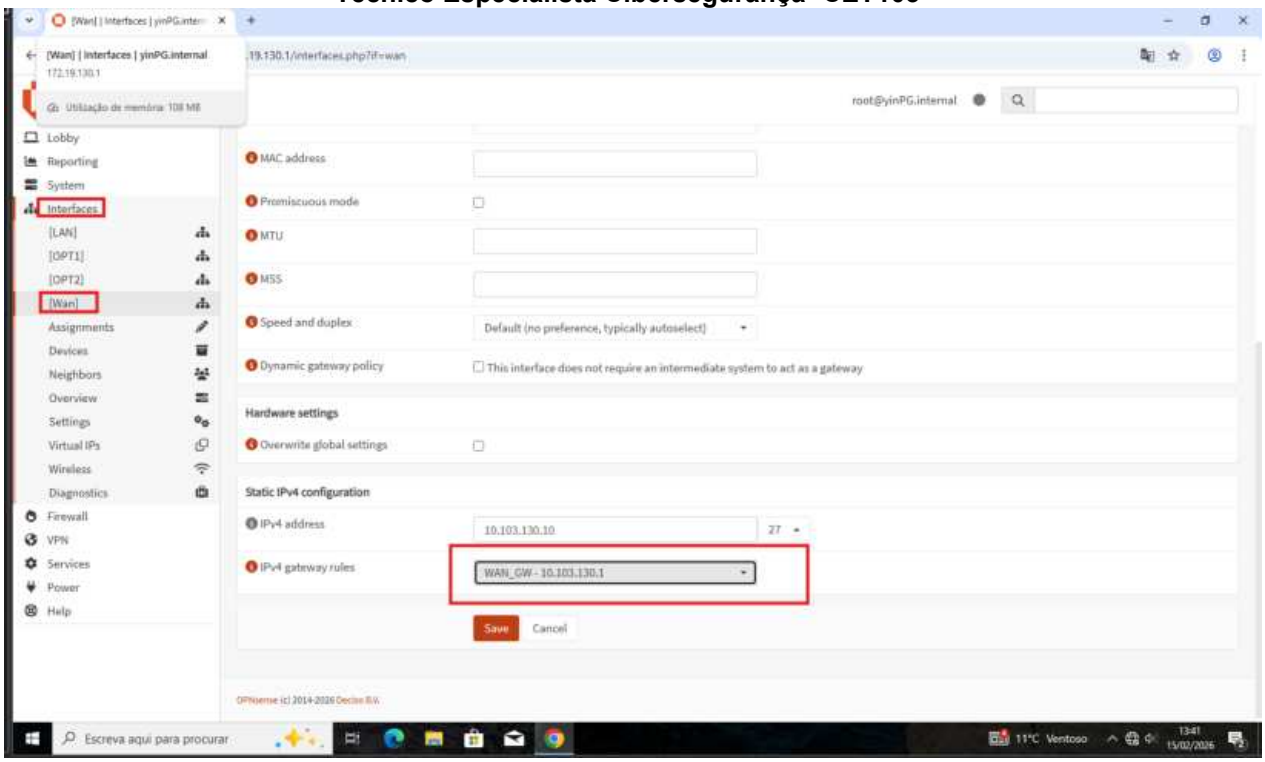
12.1 - Configurações de interface :



12.2 Lan-segments :

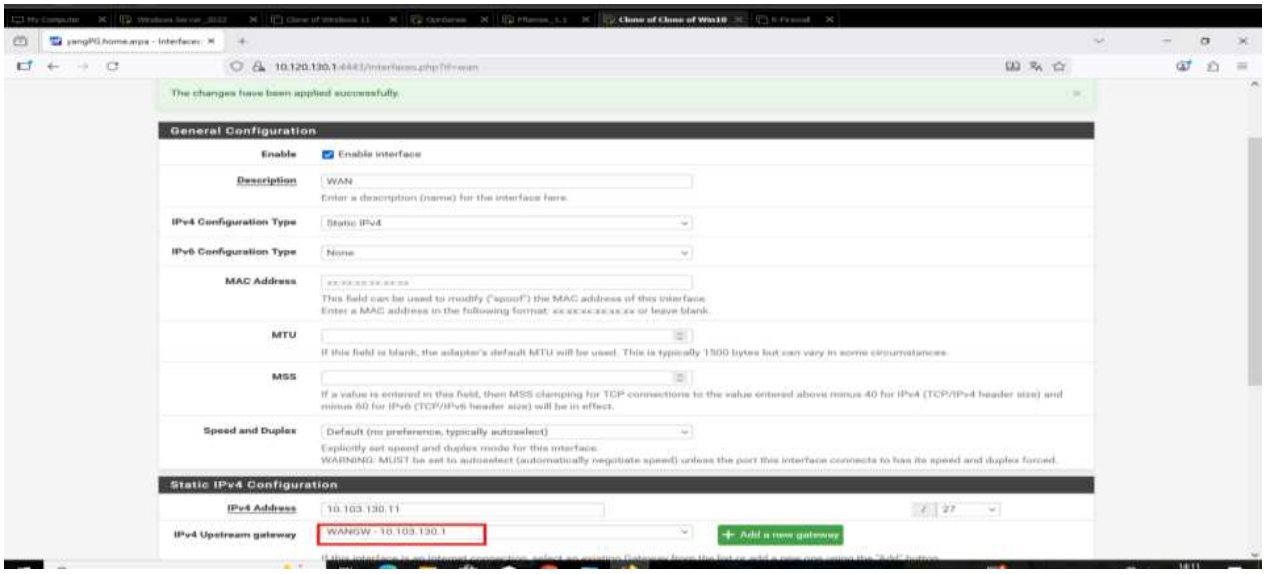


12.3 Atribuição de gateway – YinGreen -criada para " R1Vyos = WAN_GW " Interfaces→Wan



12.4 Atribuição de gateway – YanGreen –“ R1Vyos = WANGW “

Interfaces→ Wan



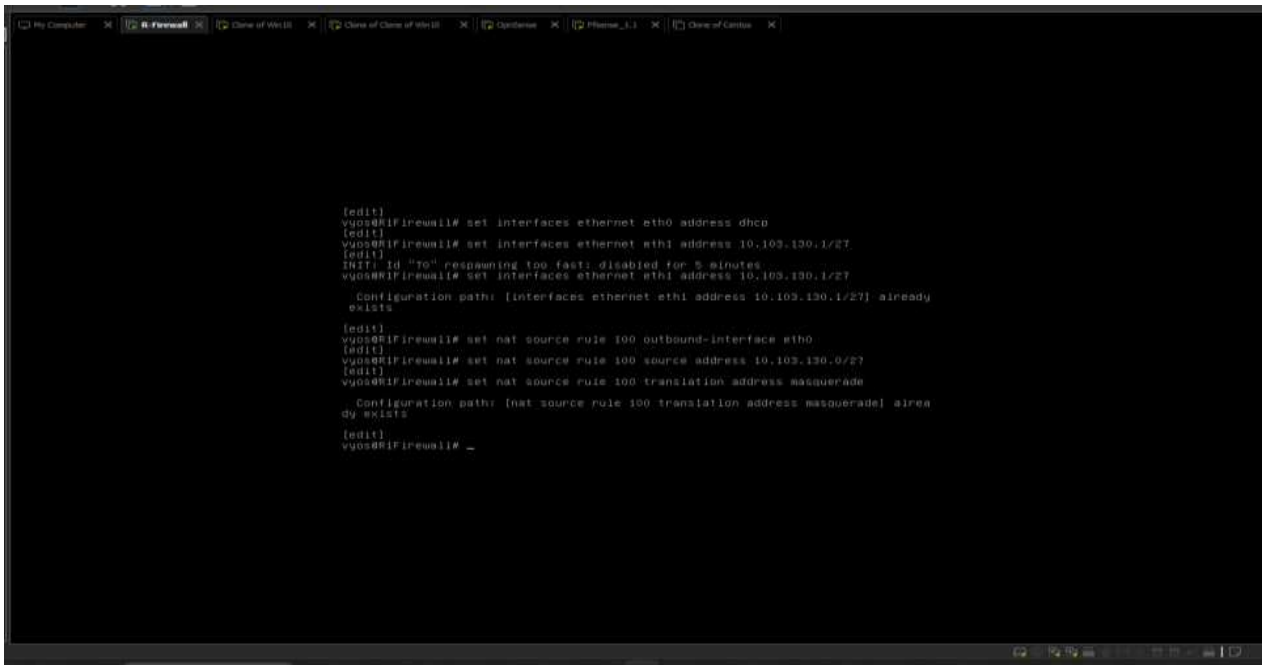
12.5 Nota importante >

- Em cada VM (VyOS, OPNsense, pfSense, Windows 10):
- NIC da WAN → LAN Segment “WAN_PESSOAL_10.103.130”

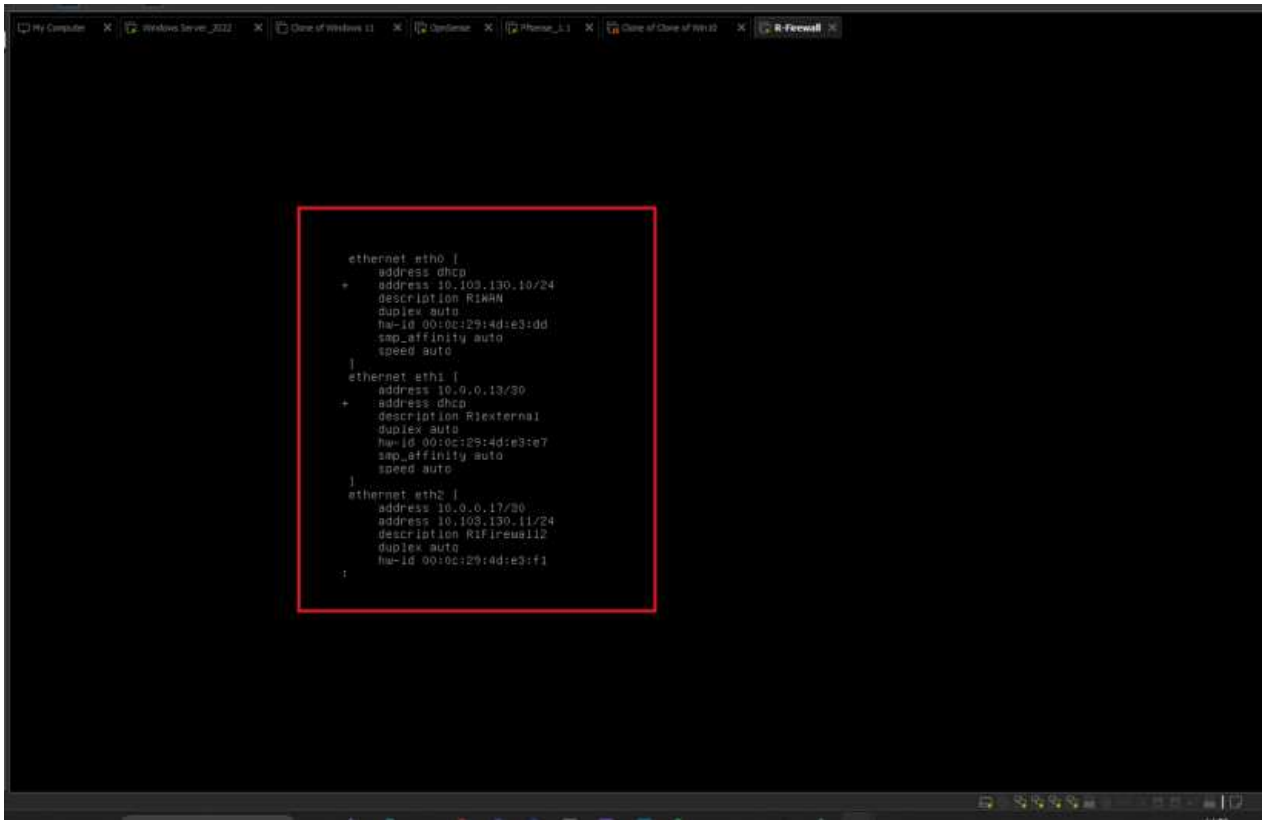
Configurações de interfaces e Default Route (Internet via NAT)

Continuação: NAT Masquerade (para sair para internet)

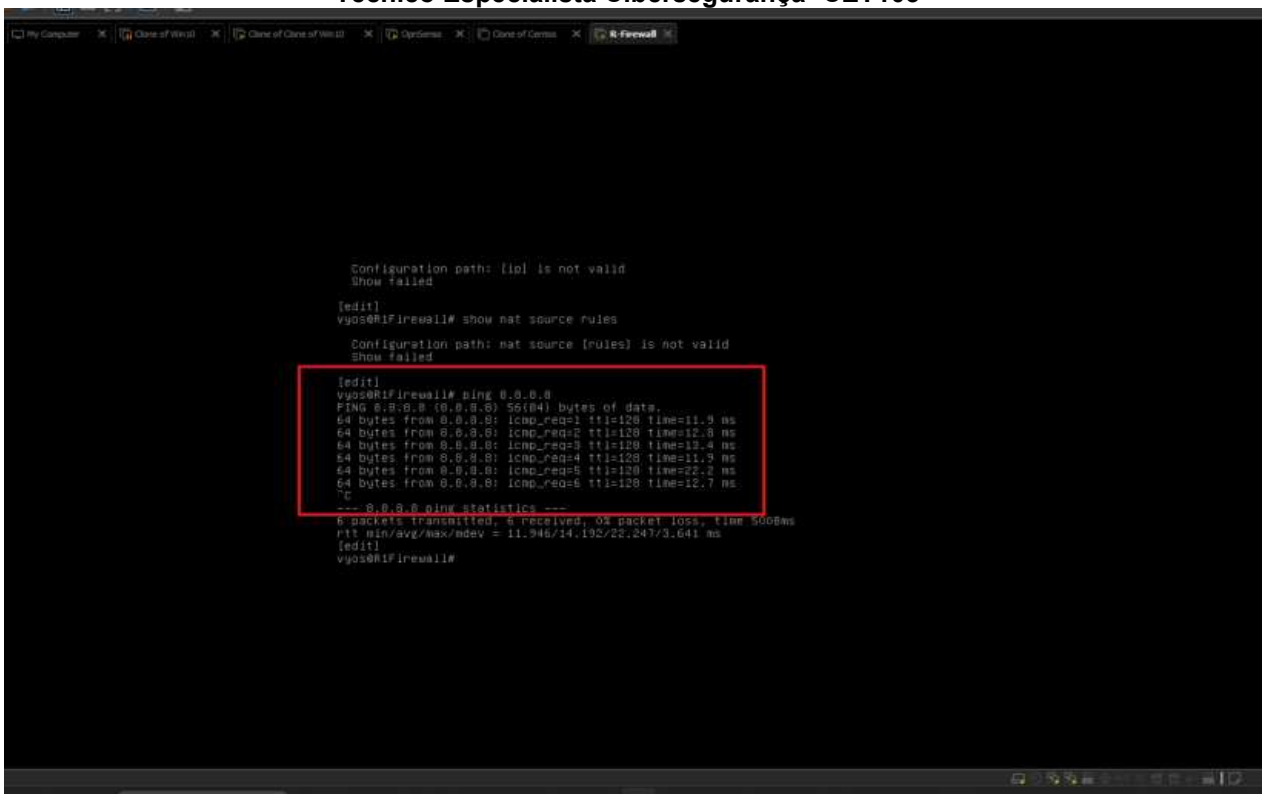
set nat source rule 100 outbound-interface eth0
set nat source rule 100 source address 10.103.130.0/27
set nat source rule 100 translation address masquerade



12.6 Resumo interfaces :

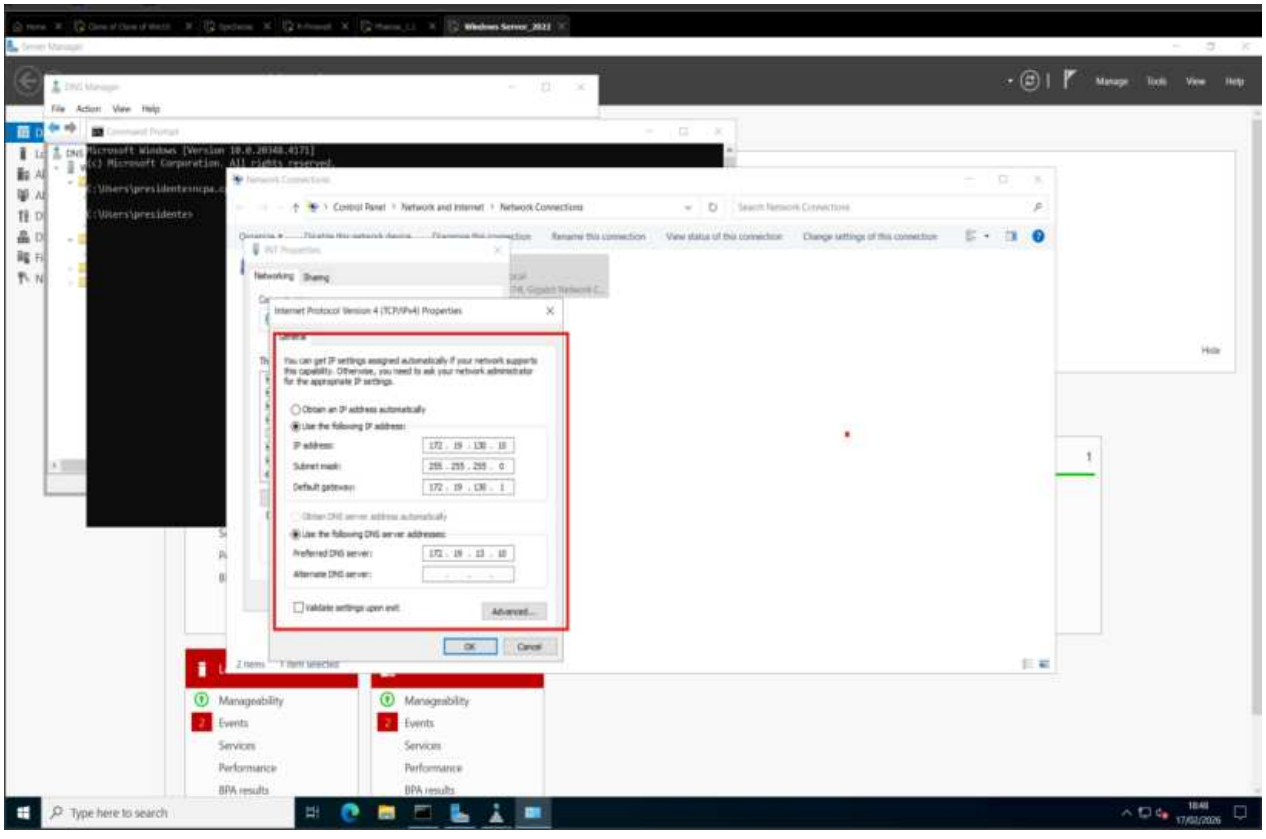


12.7 Testes: Ping



12.8 Windows Server 2022 — AD DS + DNS + DHCP (+ NPS)

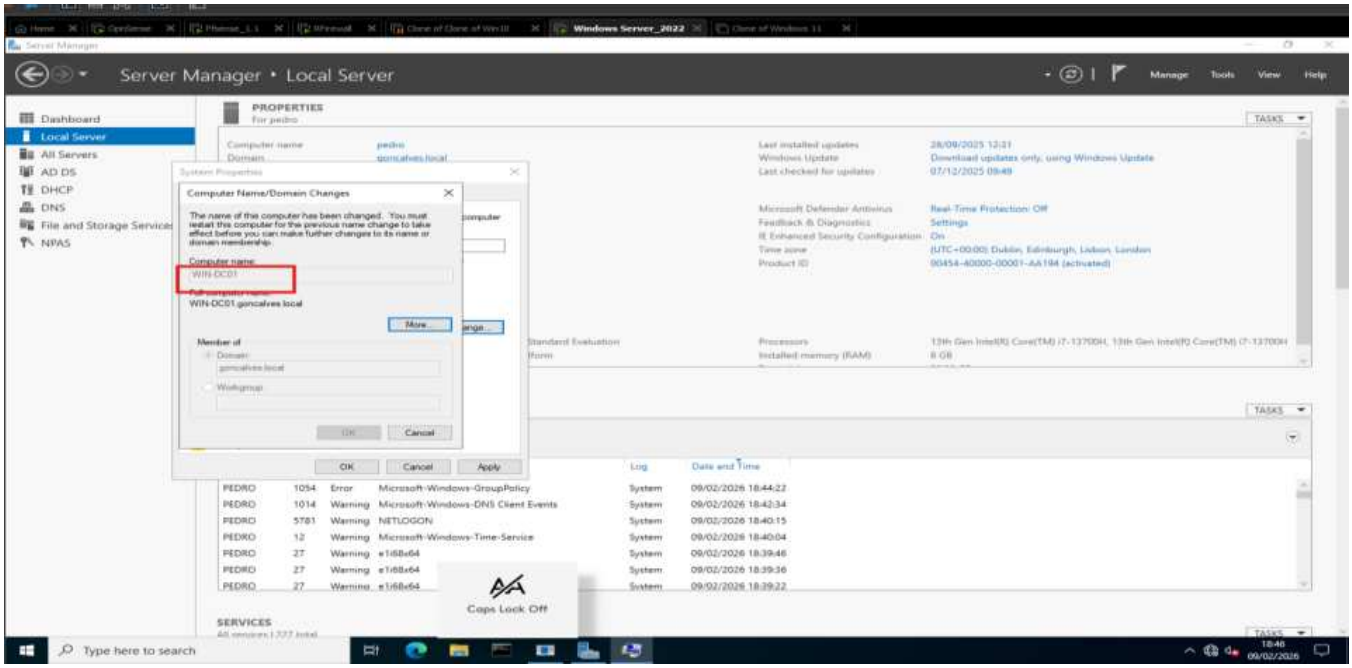
- IP: 172.19.130.10
- Mask: /23
- Gateway: 172.19.130.1
- DNS: 172.19.130.10



12.9 Nome do servidor

- Ex.: WIN-DC01

-Password padrão do projeto: **Passw0rd**



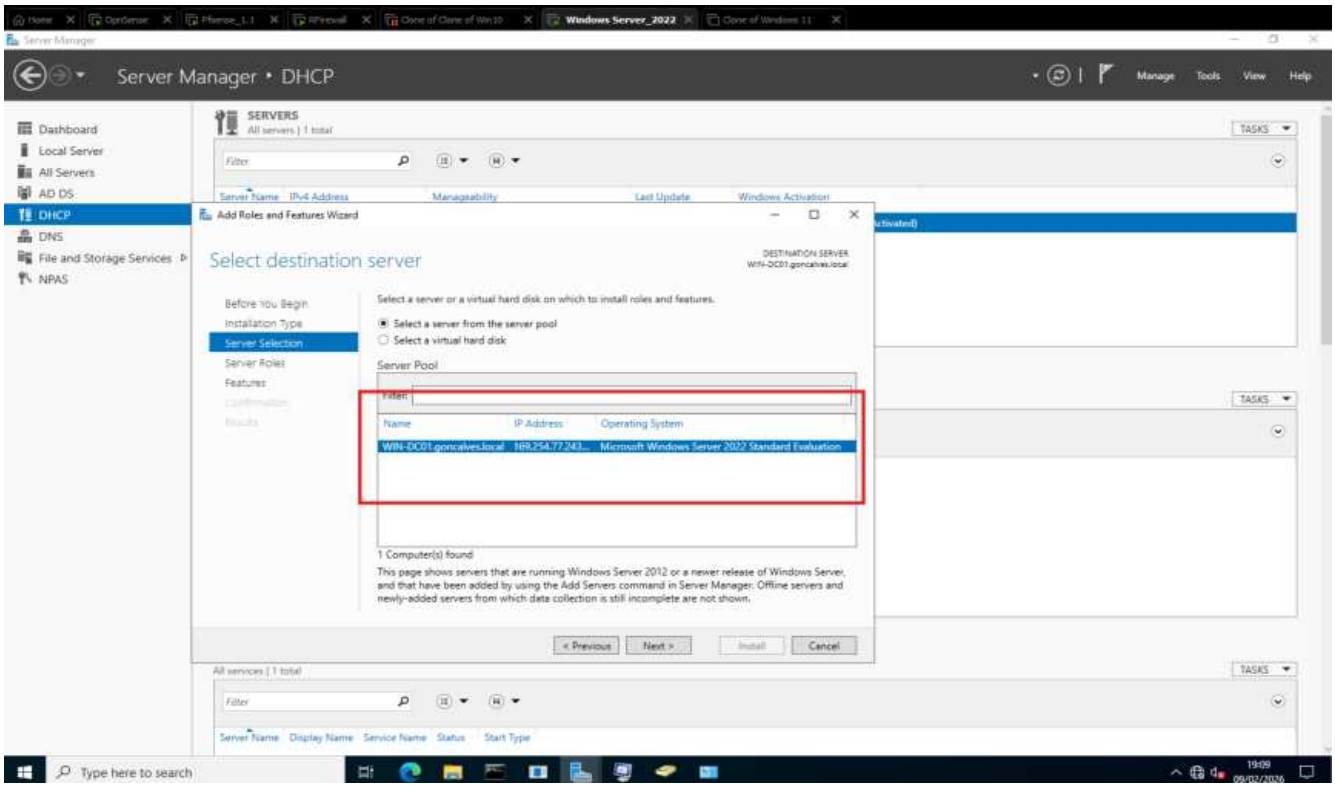
12.10 Instalar AD DS + DNS

Server Manager → Add Roles and Features

- Active Directory Domain Services
- DNS Server

Depois **Promote**:

- Novo domínio (se não existir): ex. pedrogoncalves.local (ou o domínio do trabalho **anterior)**



12.11 DNS (forwarders)
DNS Manager → Server Properties → Forwarders

- 8.8.8.8

Isto garante resolução externa "Windows → firewall → WAN".

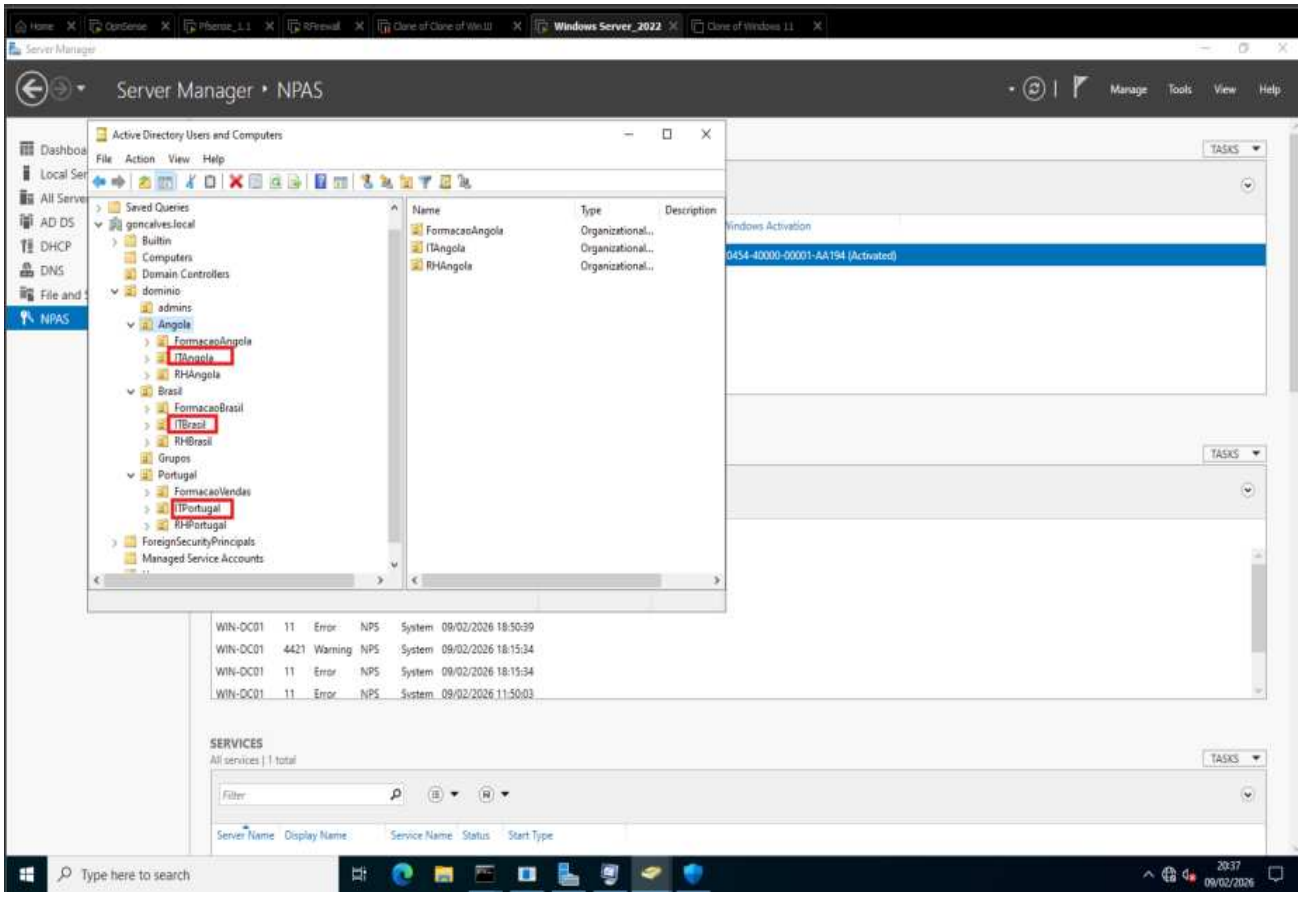
12.12 DHCP (GREEN)
Add Role → DHCP Server

- Scope: 172.19.130.0/23
- Range: 172.19.130.50 – 172.19.130.200
- Option 003 (Router): 172.19.130.1
- Option 006 (DNS): 172.19.130.10
- Autorizar no AD

Alinhar com o requisito: *DNS da GREEN só via Windows Server*.

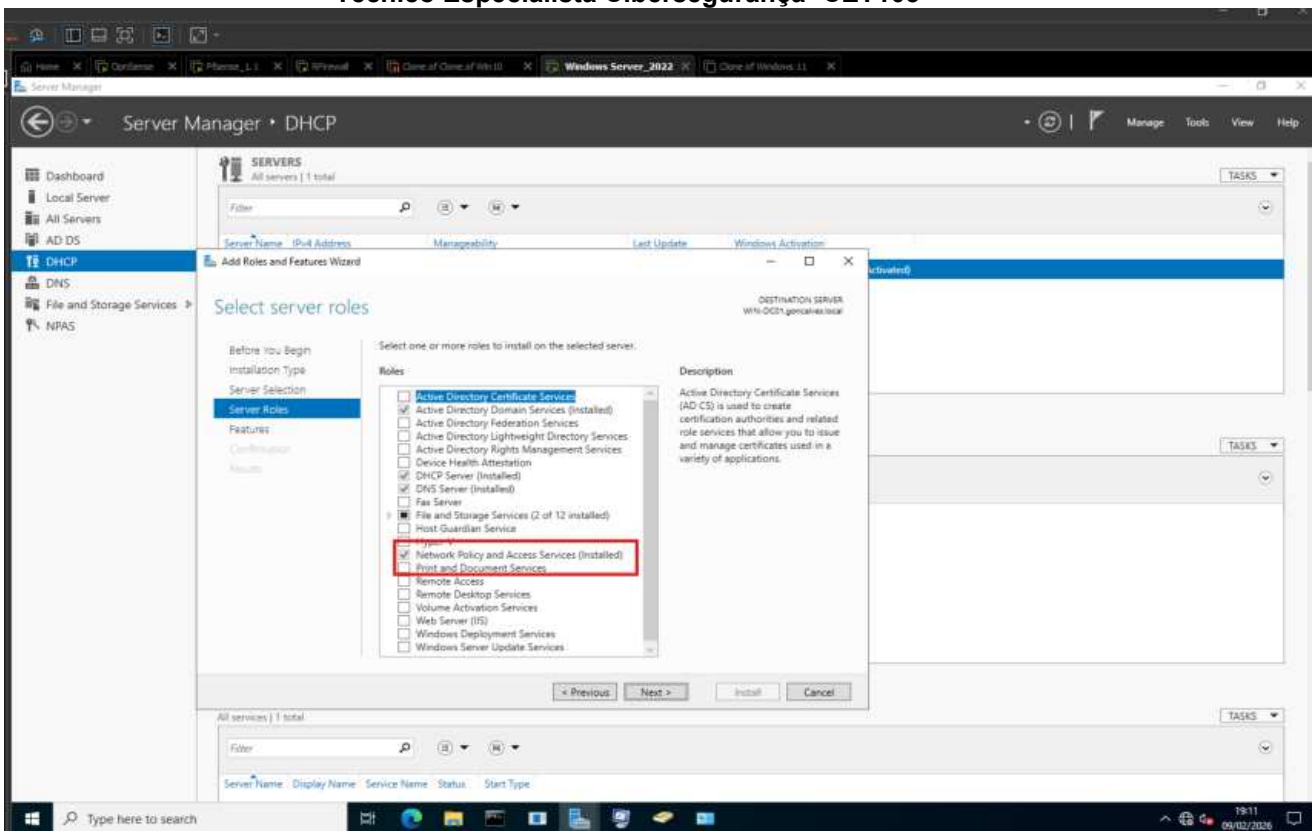
Criar grupo e utilizador "IT"

O enunciado pede user do **IT** com restrição de horário para OpenVPN.
Nota : grupos de IT já criados em ufcd anterior

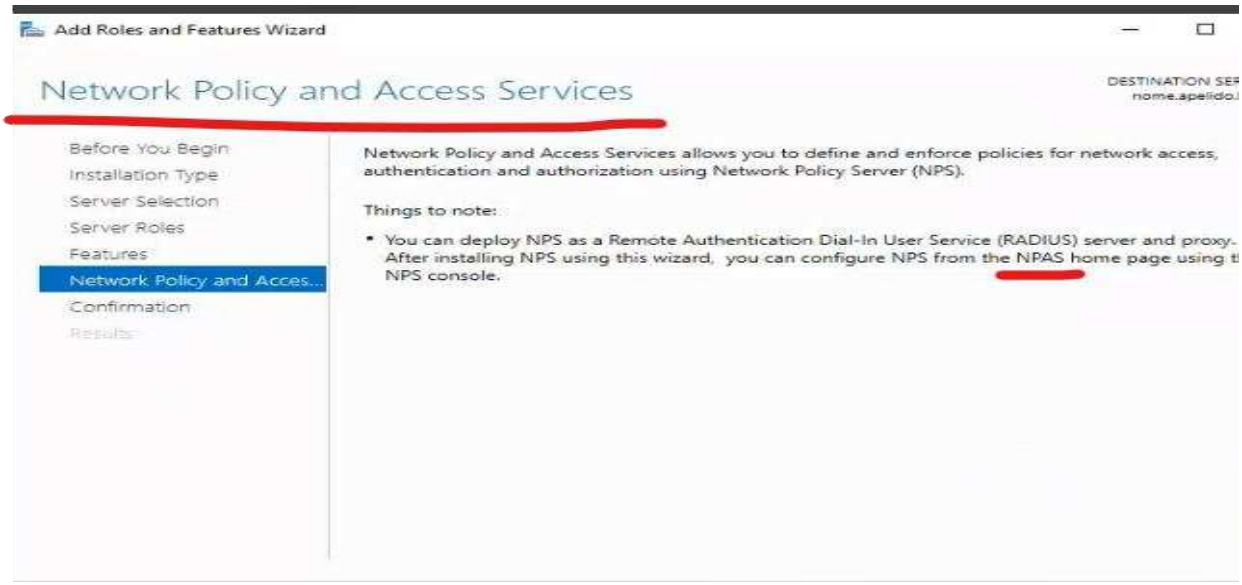


12.13 (Preparação) NPS/RADIUS

- Instalar **Network Policy and Access Services (NPS)**



Continuação…



12.14 Alternativa de instalação no comando do powershell

- Install-WindowsFeature -name NPAS -IncludeManagementTools

(Configuração seguir para OpenVPN)

Criar "RADIUS Client" para o OPNsense yinPG

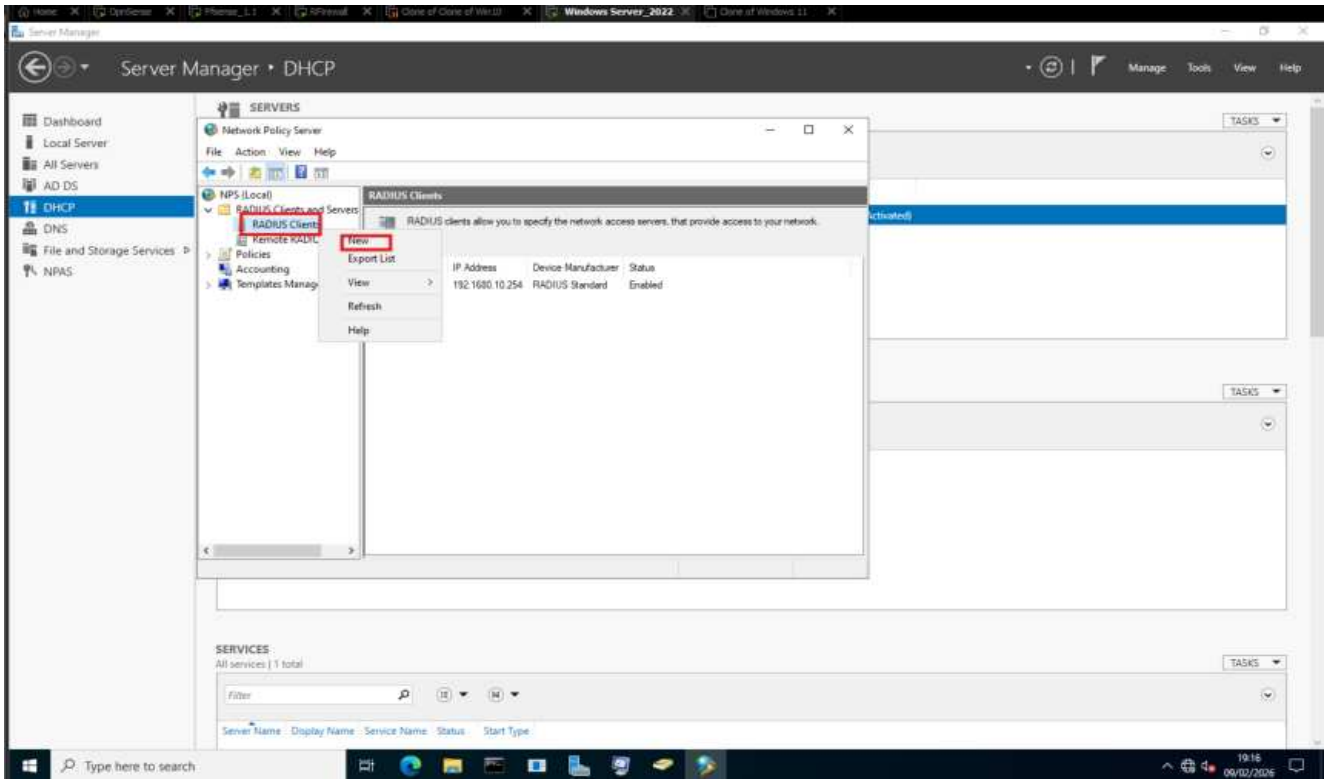
12.15 Tools→ Network Policy Server

Radius Cliente new

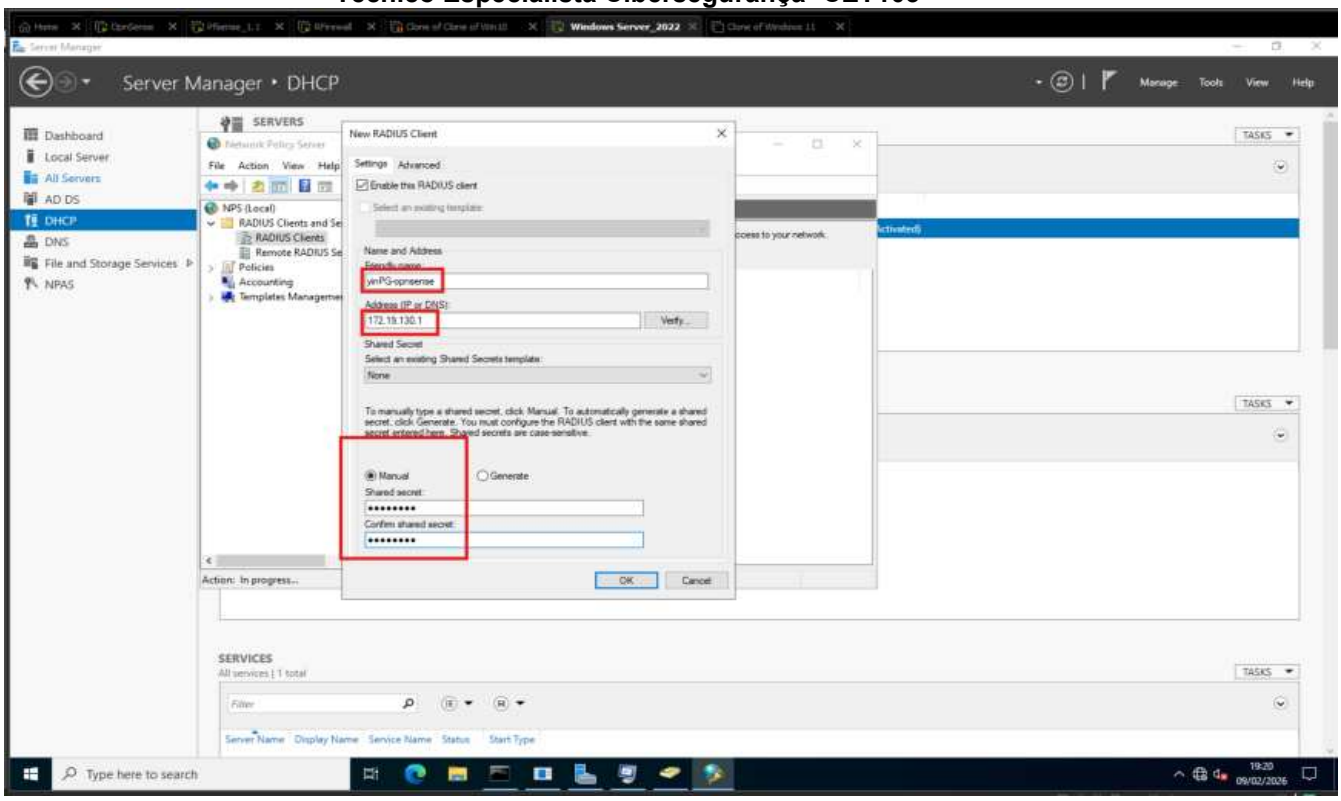
No NPS:

1. RADIUS Clients and Servers →RADIUS Clients
2. New

- Friendly name: yinPG-opnsense
- Address: 172.19.130.1 (IP da interface GREEN do yin)
- Shared secret: Passw0rd



Continuação :



12.16 Criar Network Policy com grupo IT e horário

No NPS:

1. Policies → Network Policies → New
2. Nome: OpenVPN_IT_8_20_SegSex
3. Conditions:

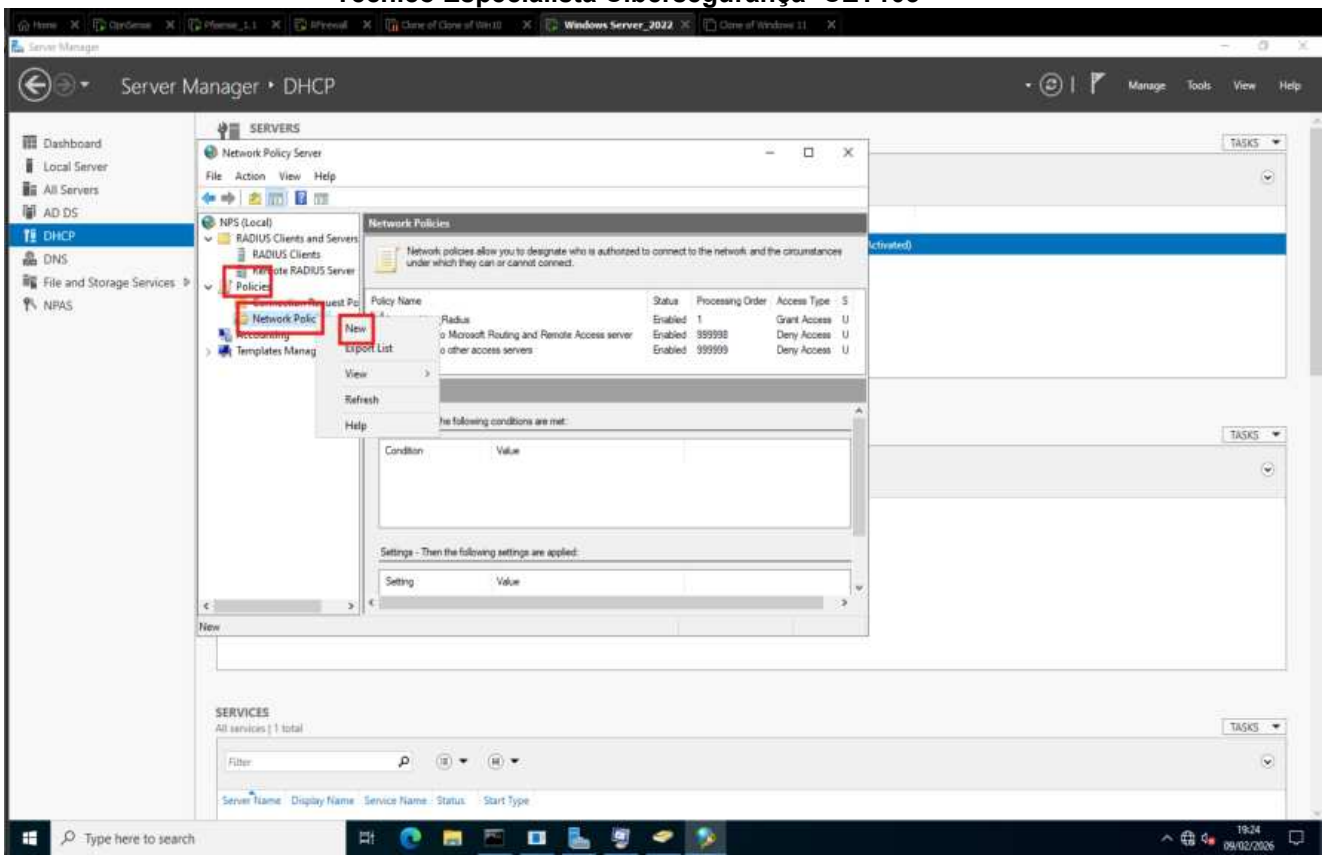
   -Windows Groups: IT- geral
4. Constraints:

   -Day and Time Restrictions: Seg–Sex, 08:00–20:00

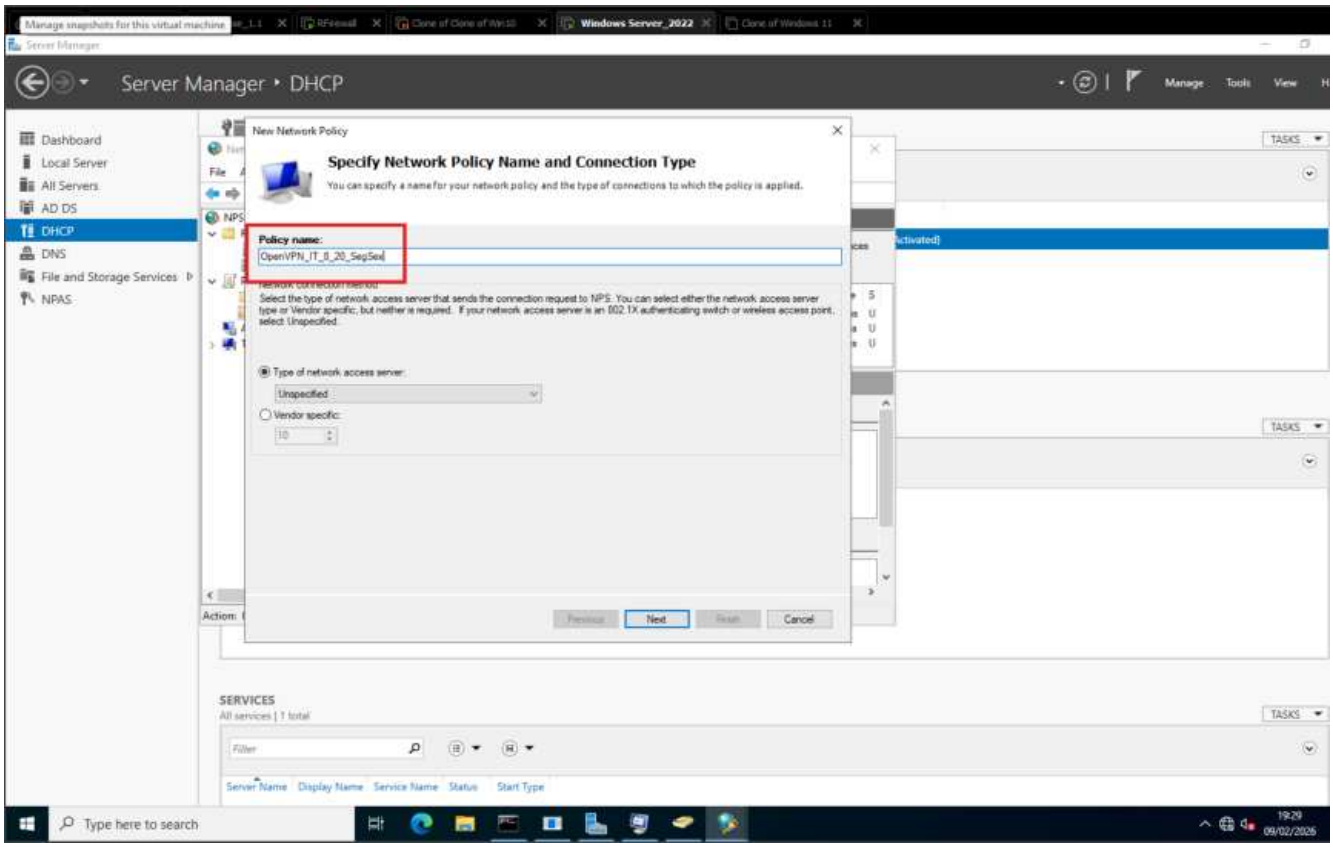
   - Authentication Methods: MS-CHAPv2
5. Settings: (manter defaults)
6. Assegurar que esta policy na primeira posição - (ordem importa).

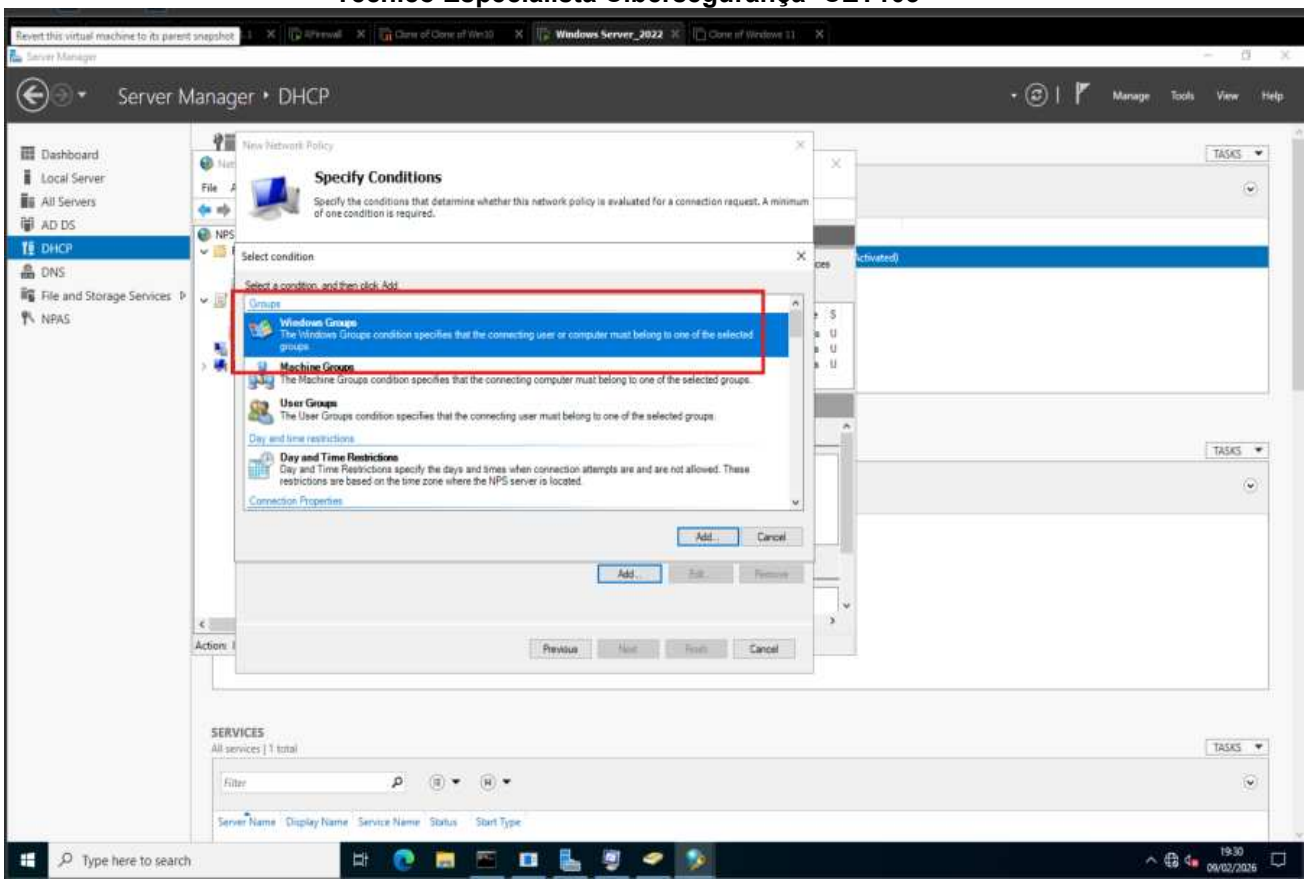
12.17 Policies Network Policies new

Policies→ Network Policies → New

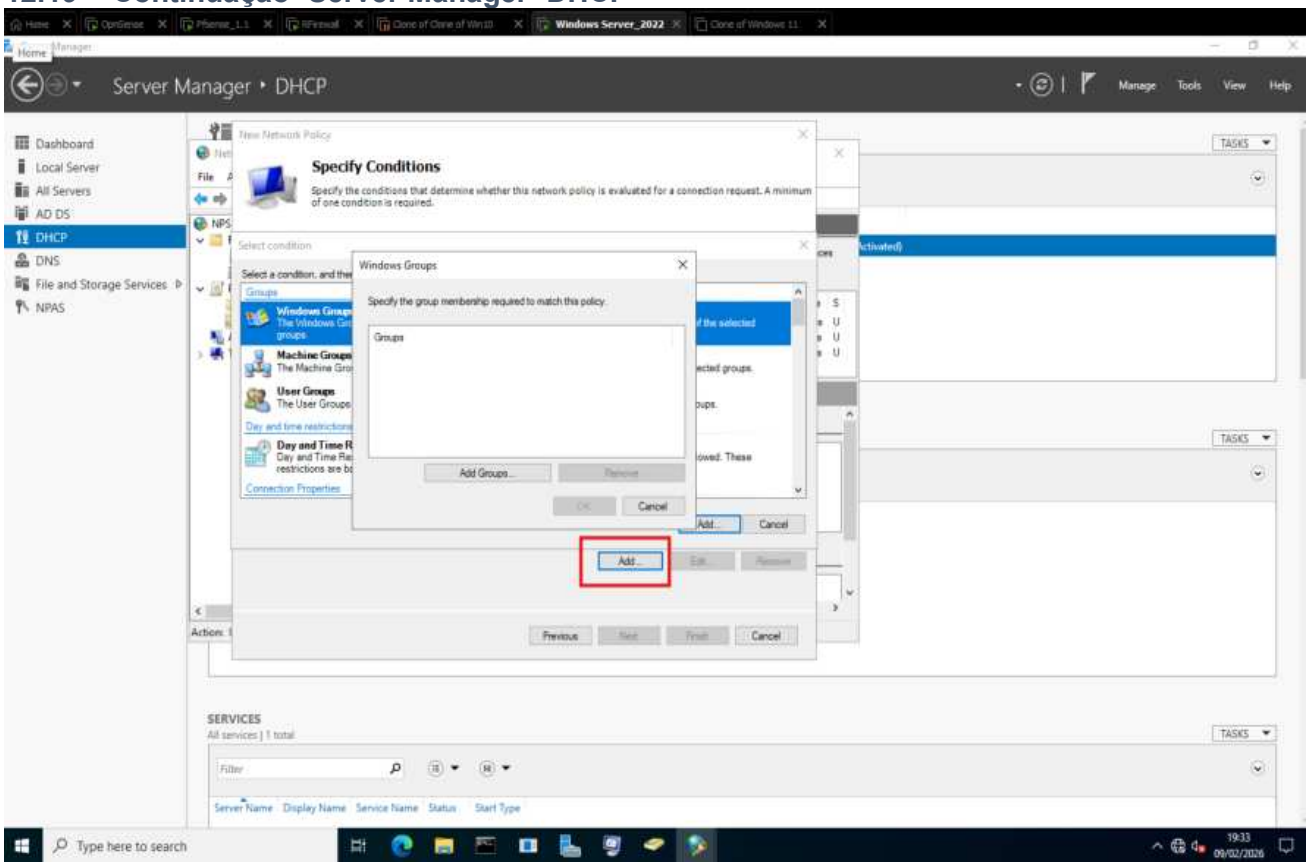


12.18 Continuação…Policy name “OpnVPN_IT_8_20SegSex”

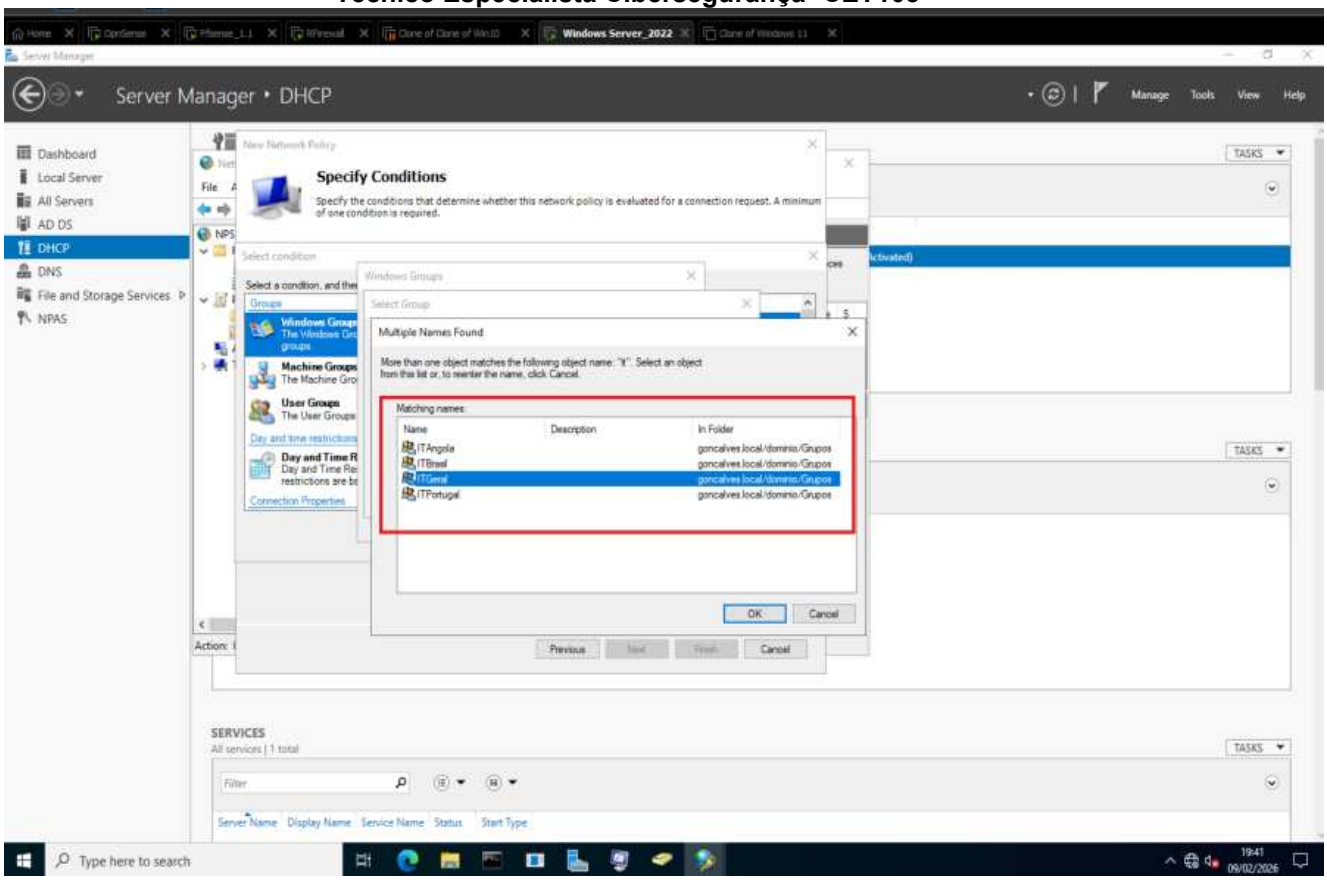




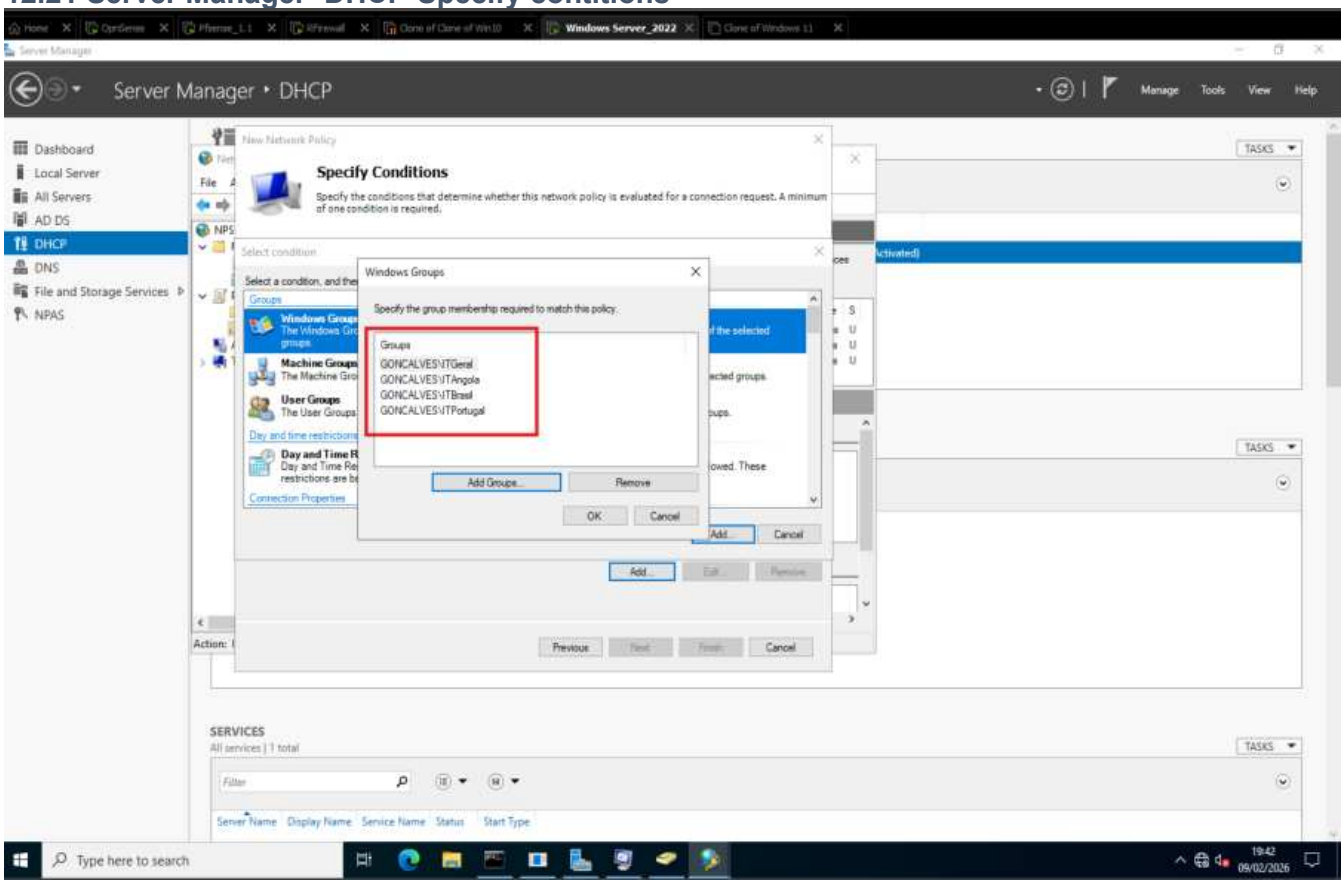
12.19 - Continuação- Server Manager -DHCP

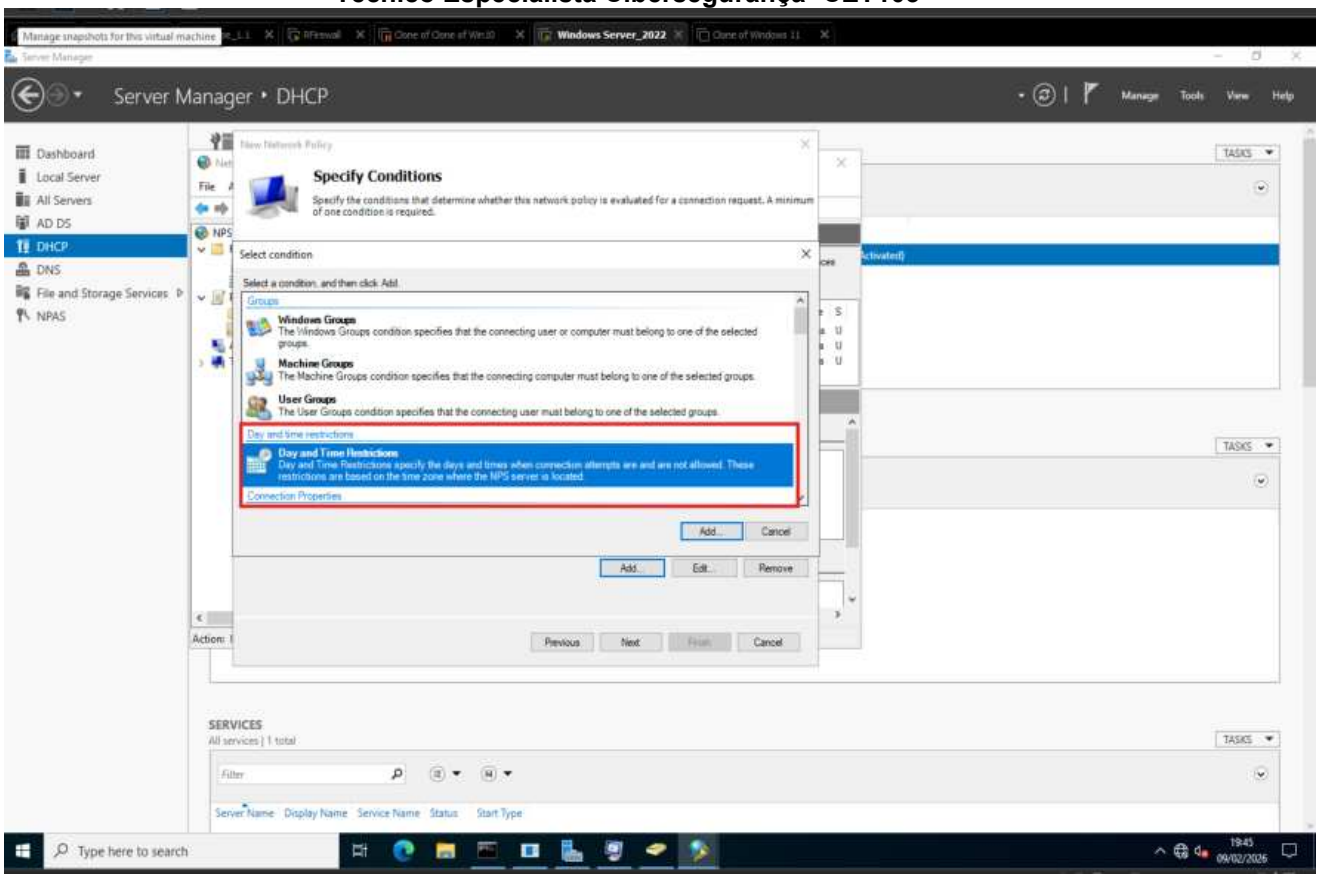


12.20 - Continuação…Server Manager - (Specify Conditions)

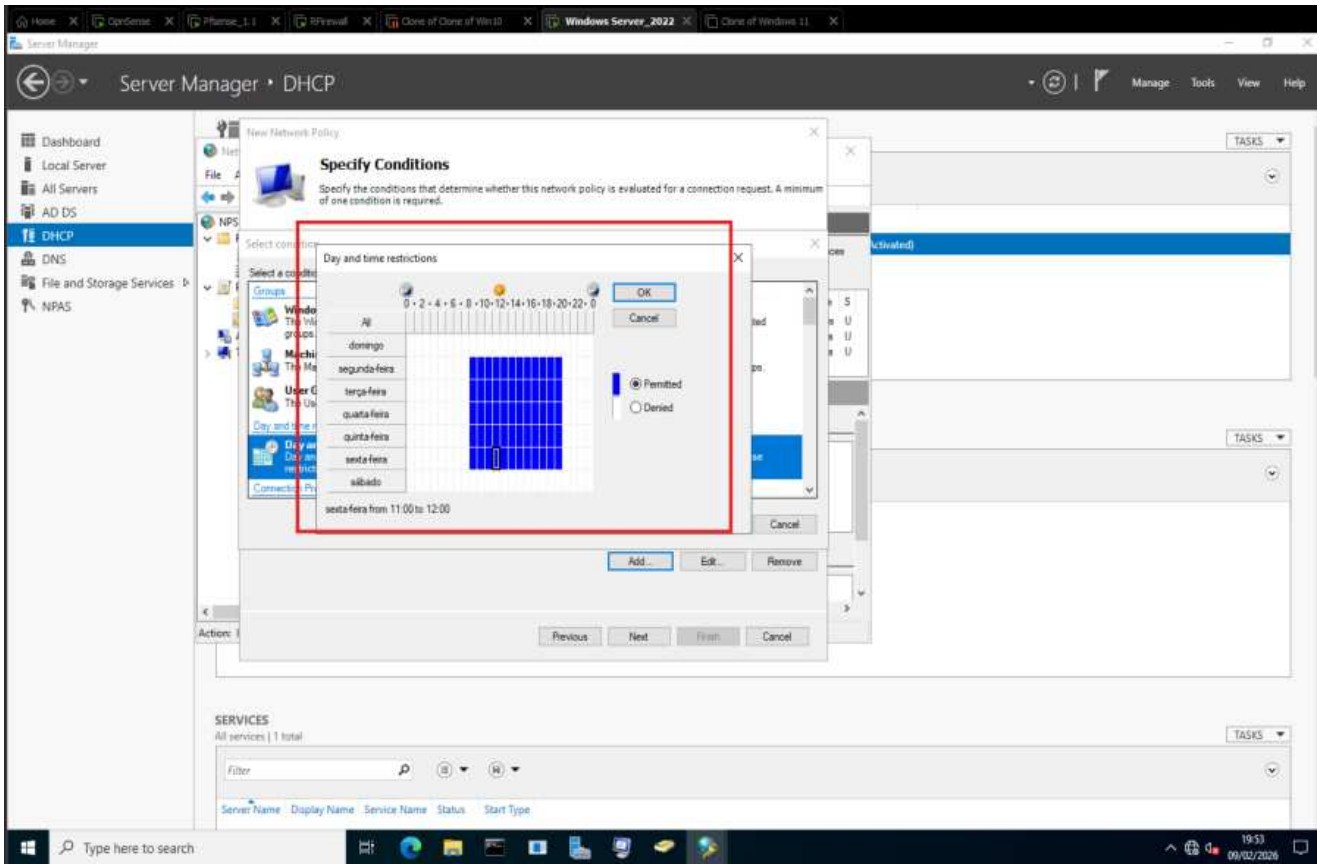


12.21 Server Manager -DHCP-Specify contitions

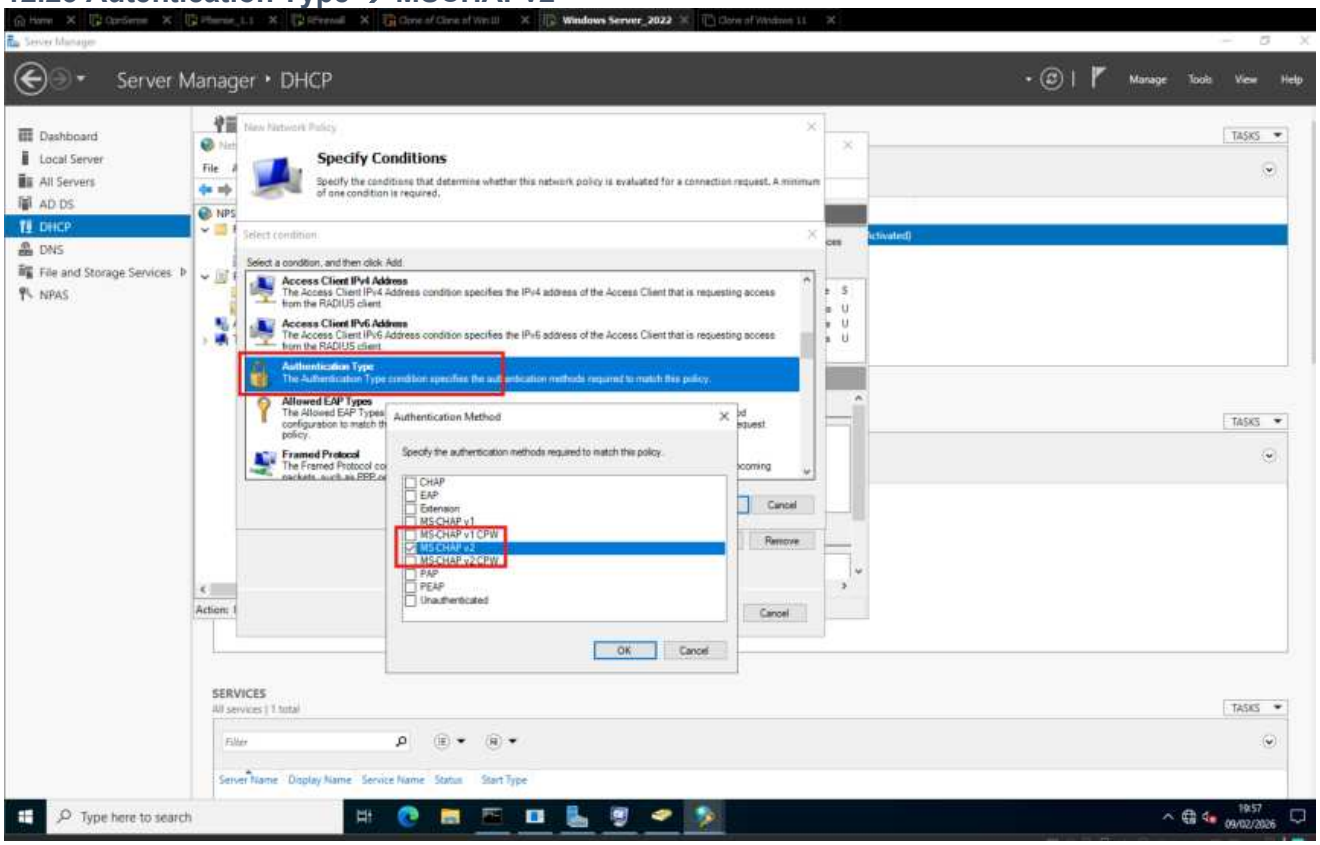




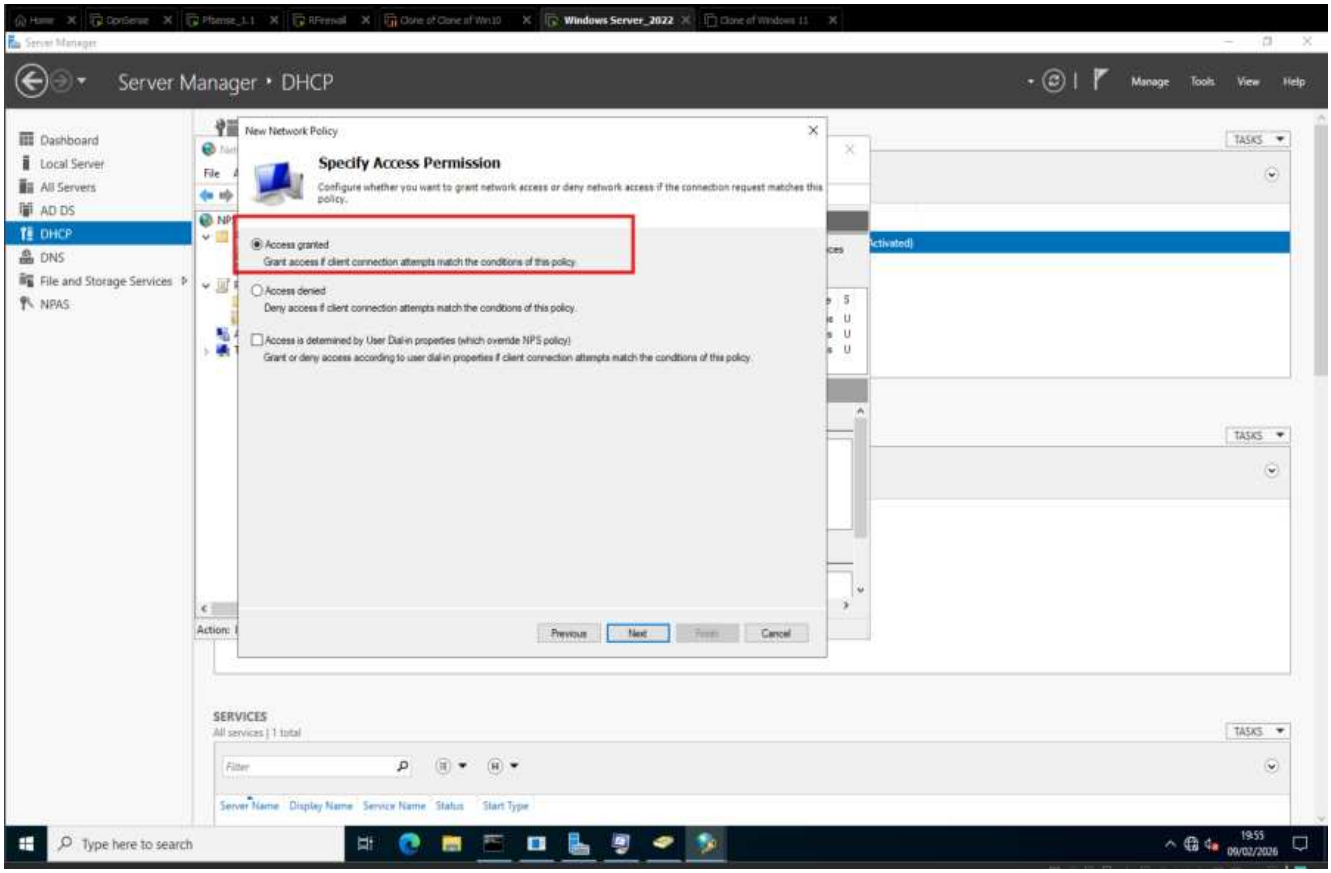
12.22 Day and time Restrictions- Specify Conditions



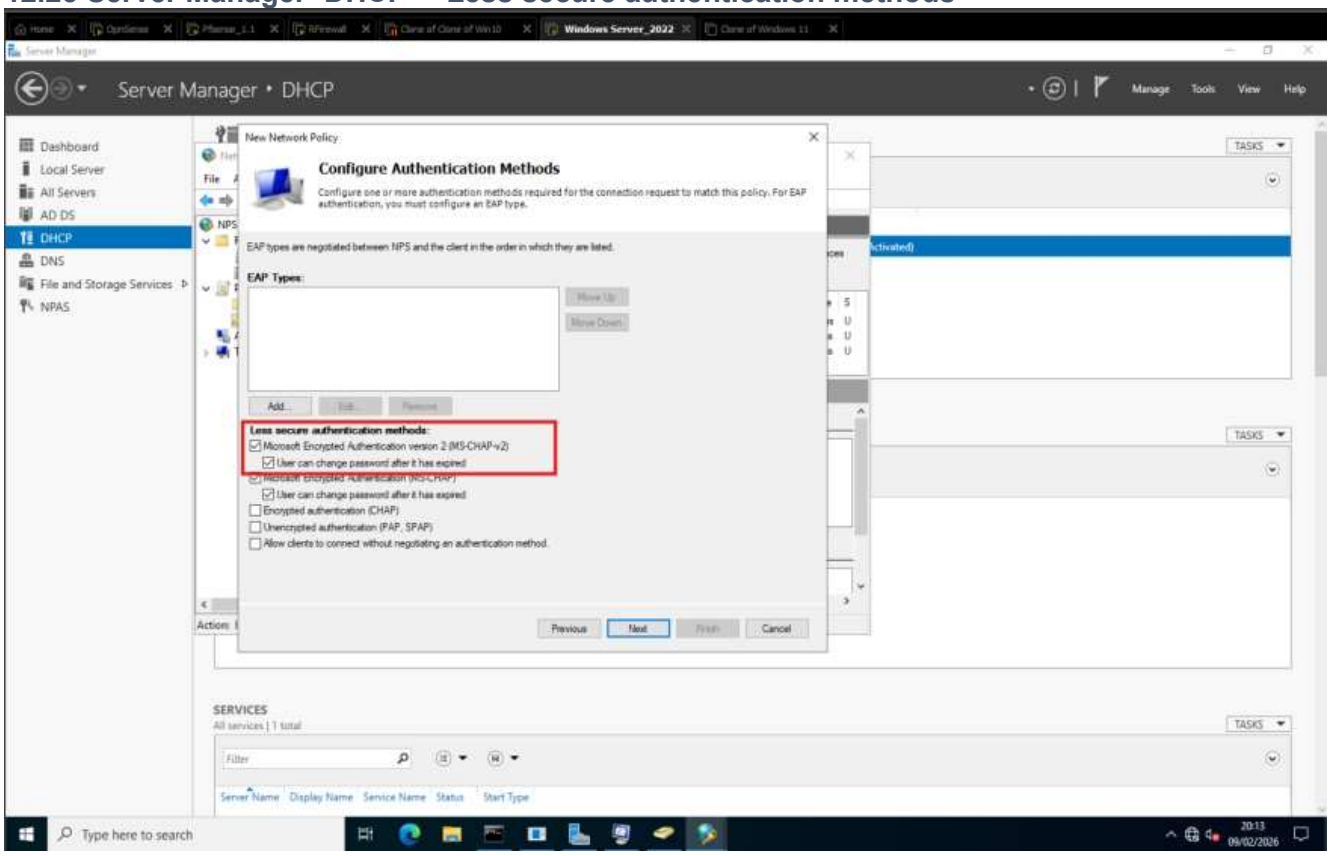
12.23 Autentication Type → MSCHAPv2



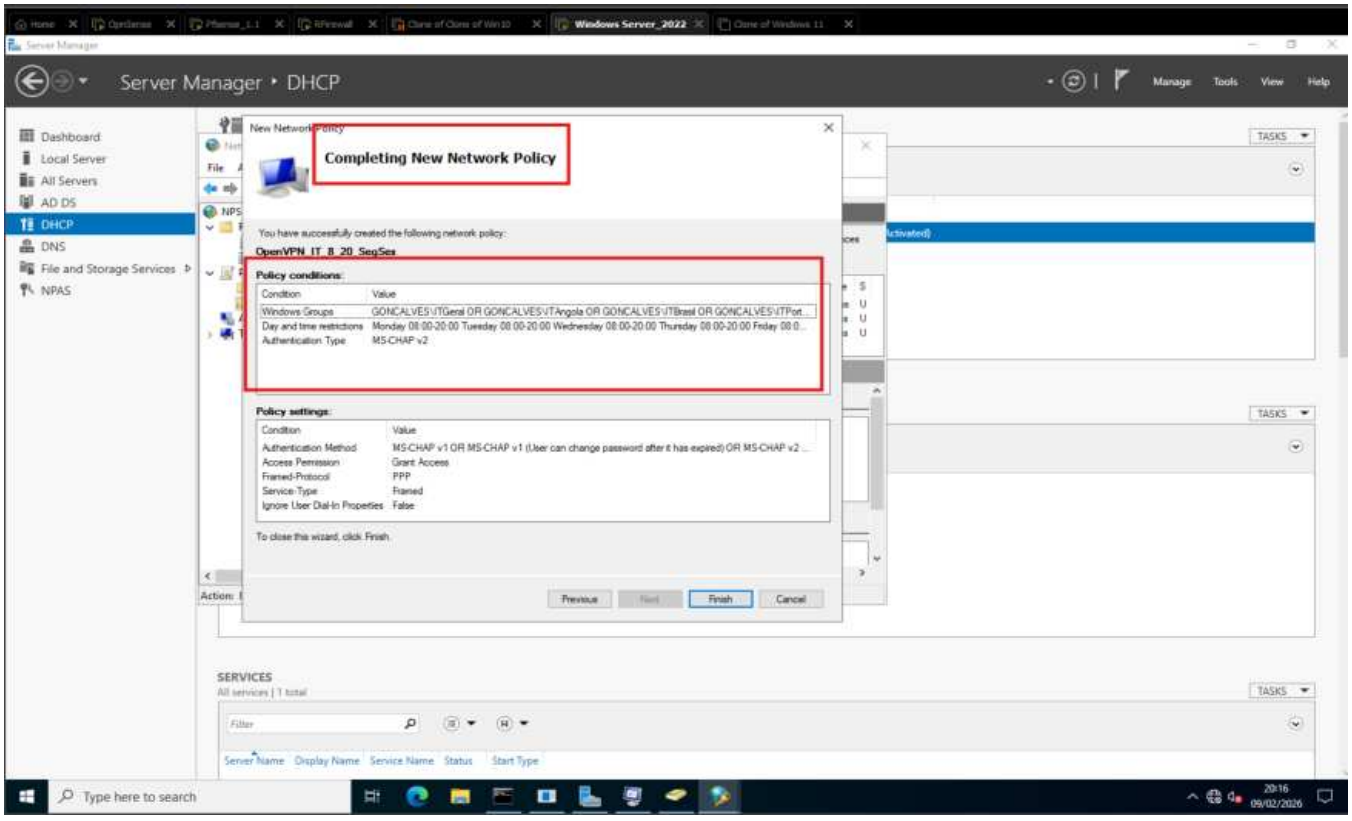
12.24 Specify Acess Permissions – “Acess granted “



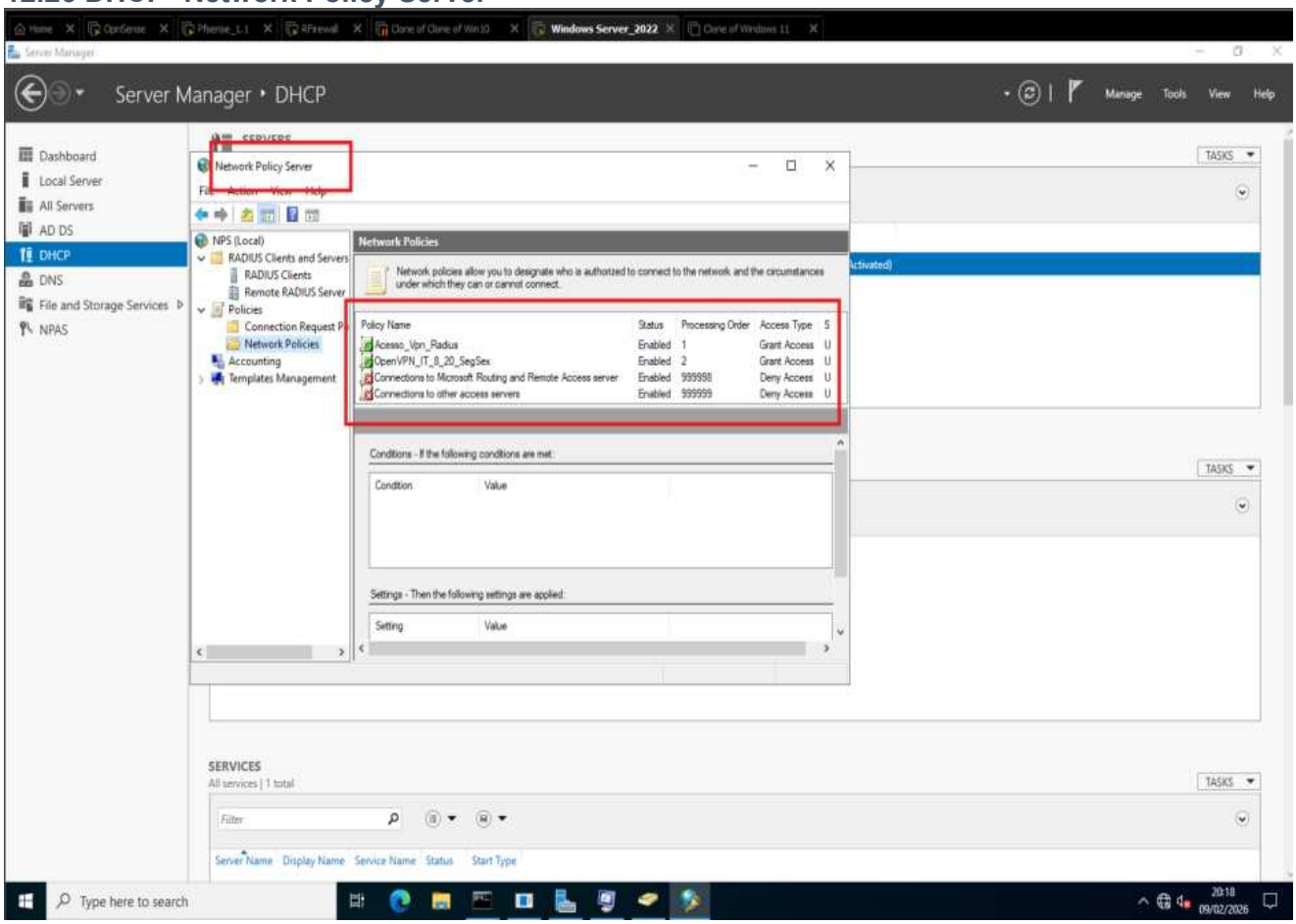
12.25 Server Manager -DHCP –"Less secure authentication methods"



12.26 Completing New Network Policy -OpenVPN IT a 20 segSex



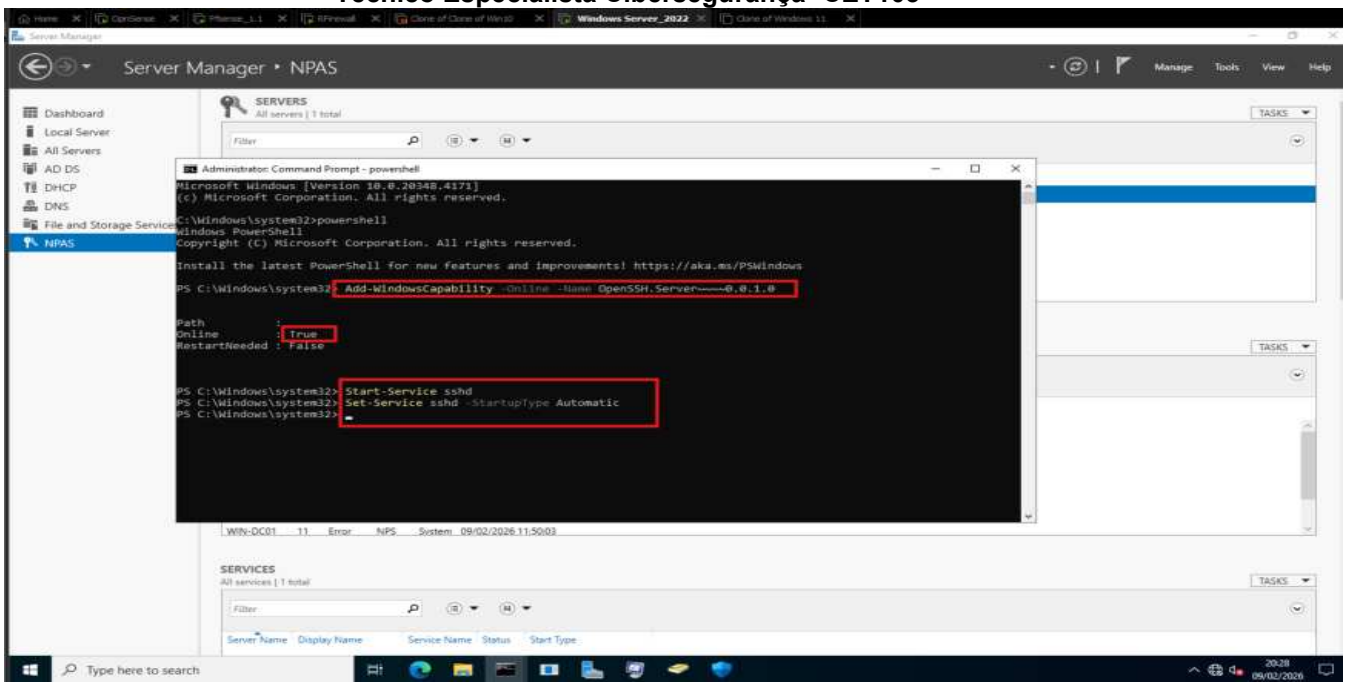
12.26 DHCP- Network Policy Server



12.27 Acesso remote

- **RDP**: Enable (System → Remote Desktop)
- **SSH** (para testes posteriores):

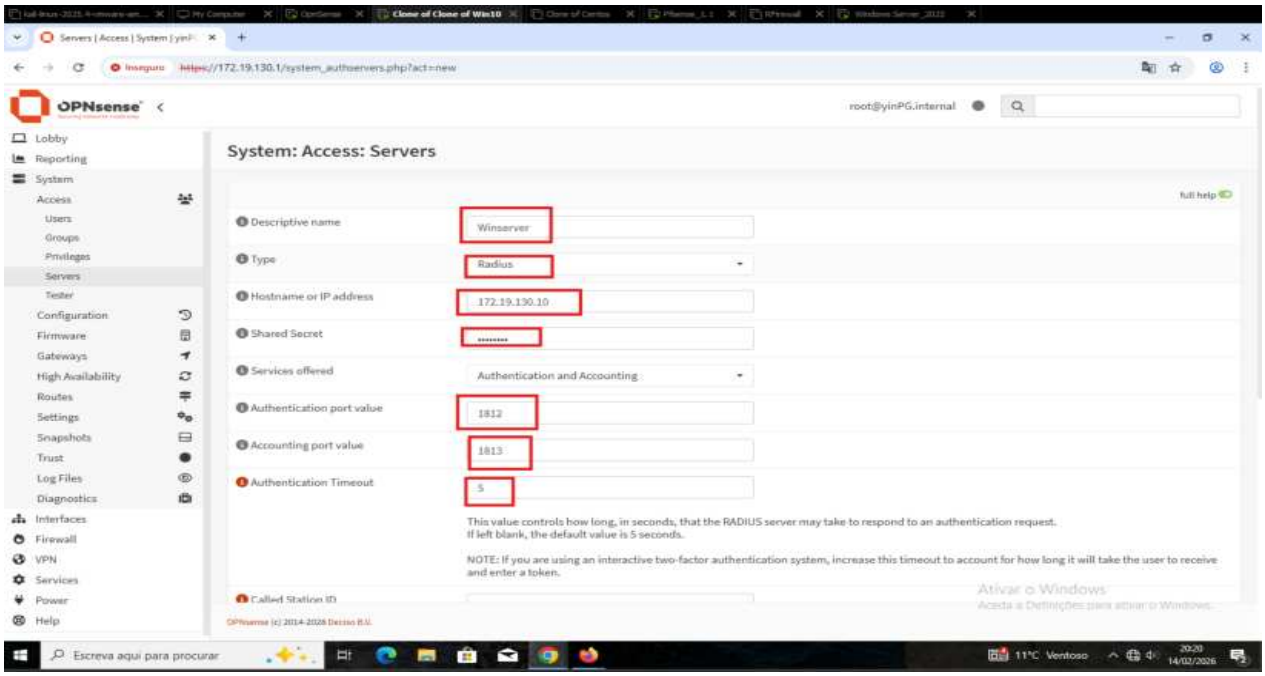
Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
Start-Service sshd
Set-Service sshd -StartupType Automatic



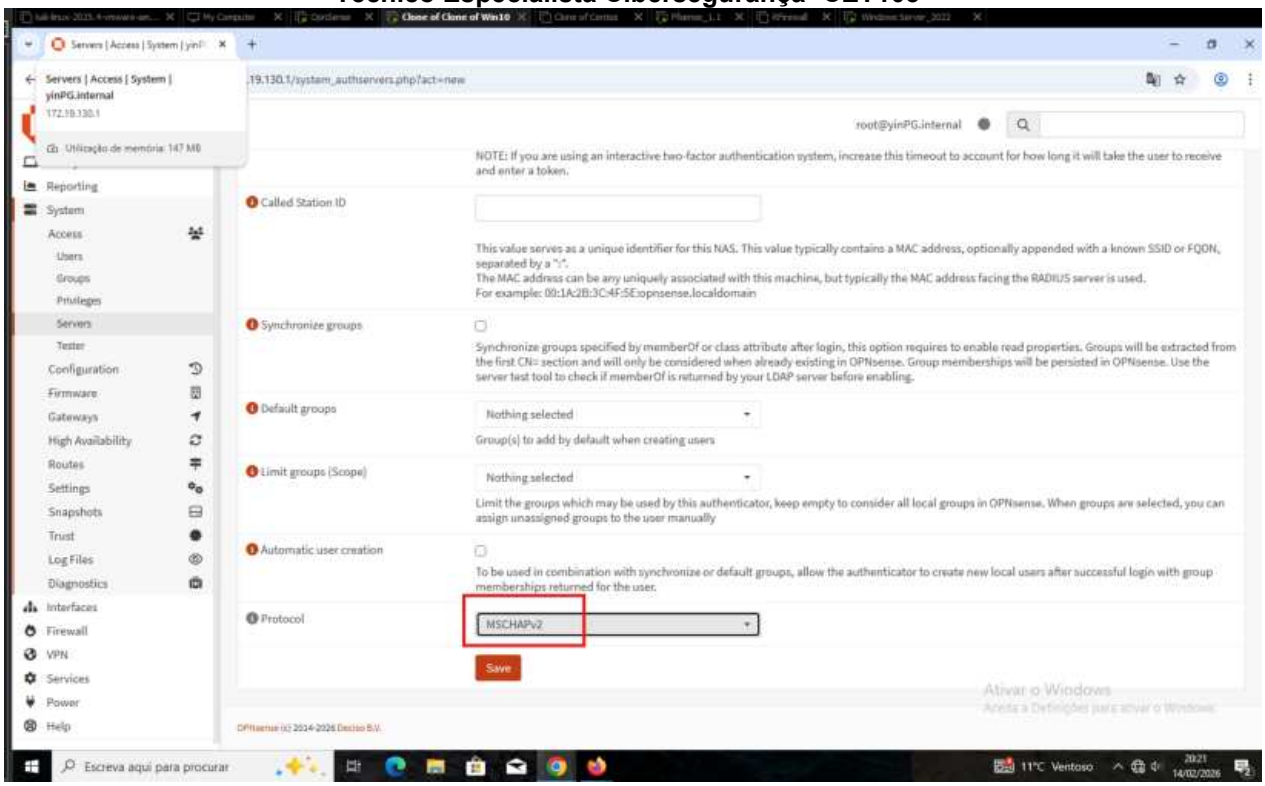
12.28 Configurações Radius – OpnSense para Windows Server

System → Access → Servers

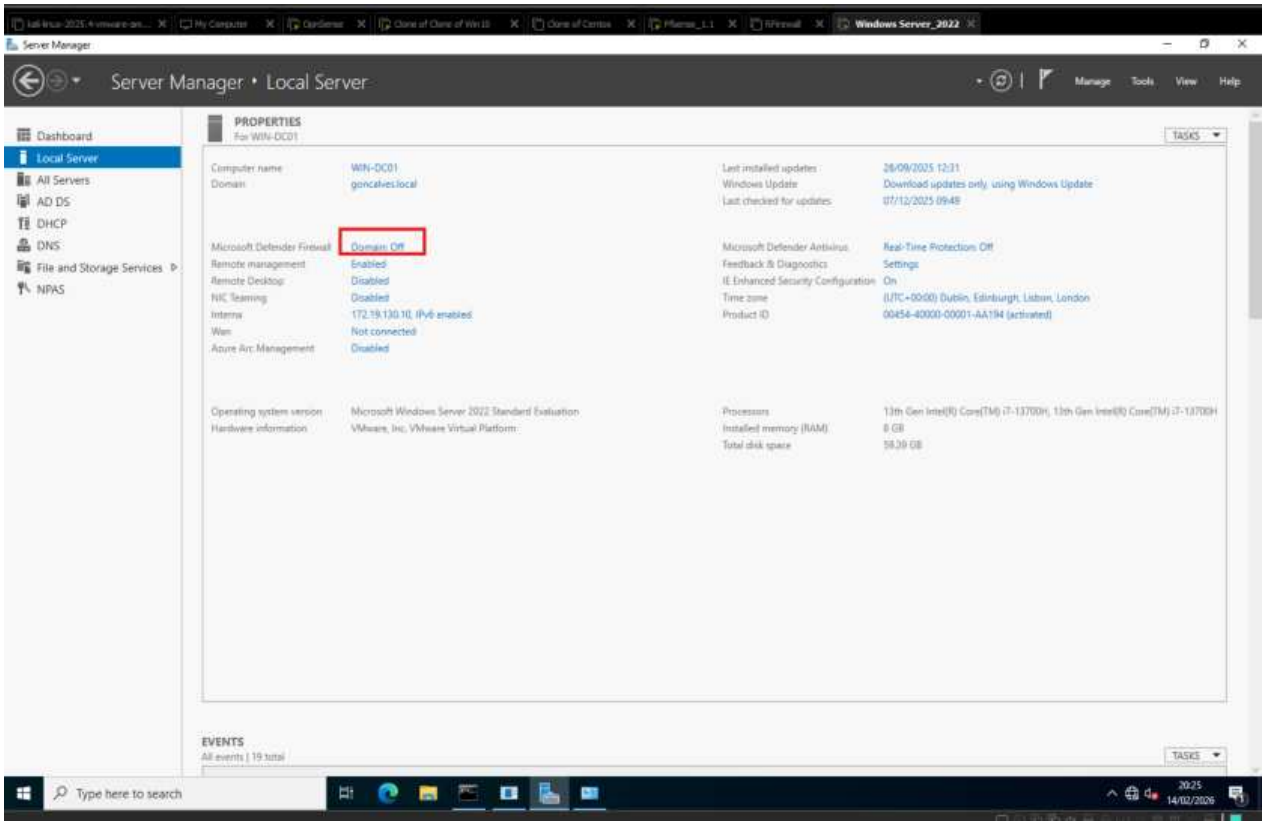
- **Descriptive name:** Winserver
- **Type:** Radius
- **Hostname or IP address:** 172.19.130.10
- **Shared Secret:** Passw0rd
- **Services offered:** Authentication and Accounting
- **Authentication port value:** 1812
- **Accounting port value:** 1813
- **Authentication Timeout:** 5
- **Protocol:** MSCHAPv2



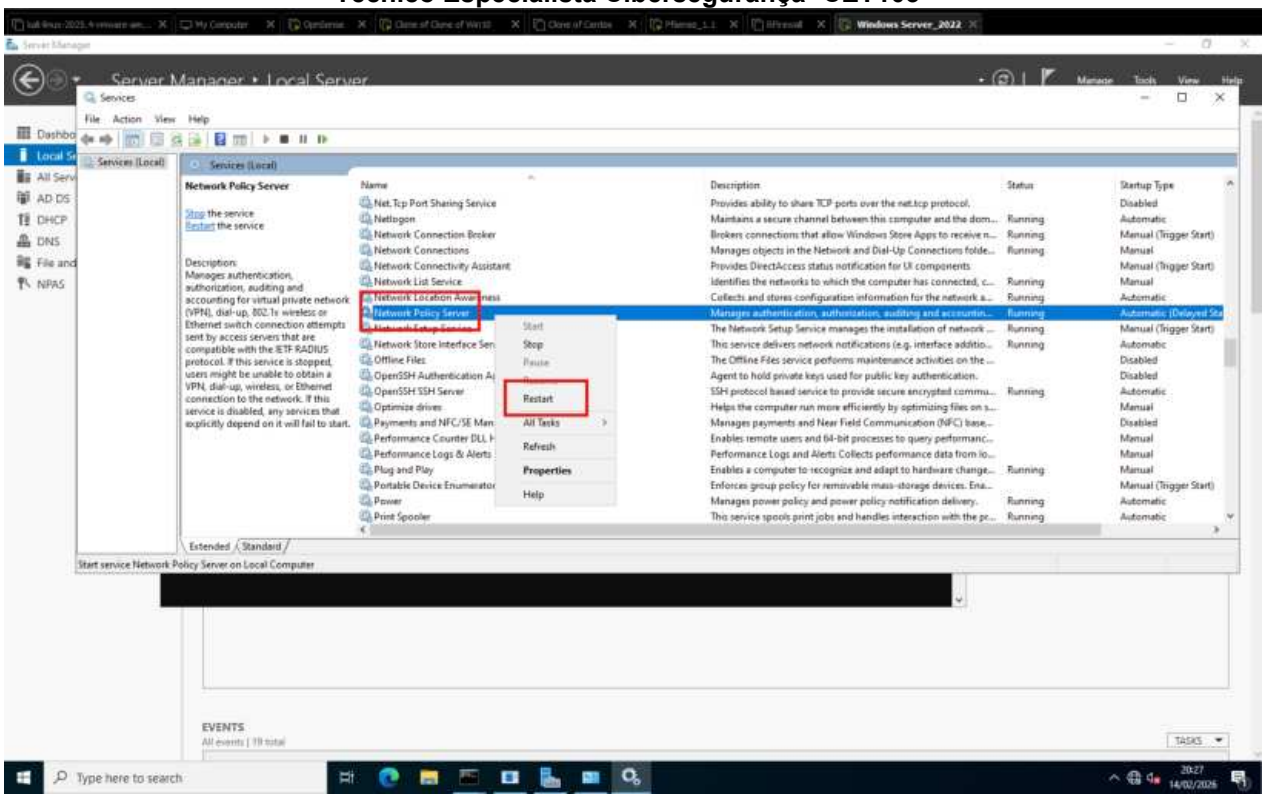
Continuação …



12.29 -Desabilitar a firewall



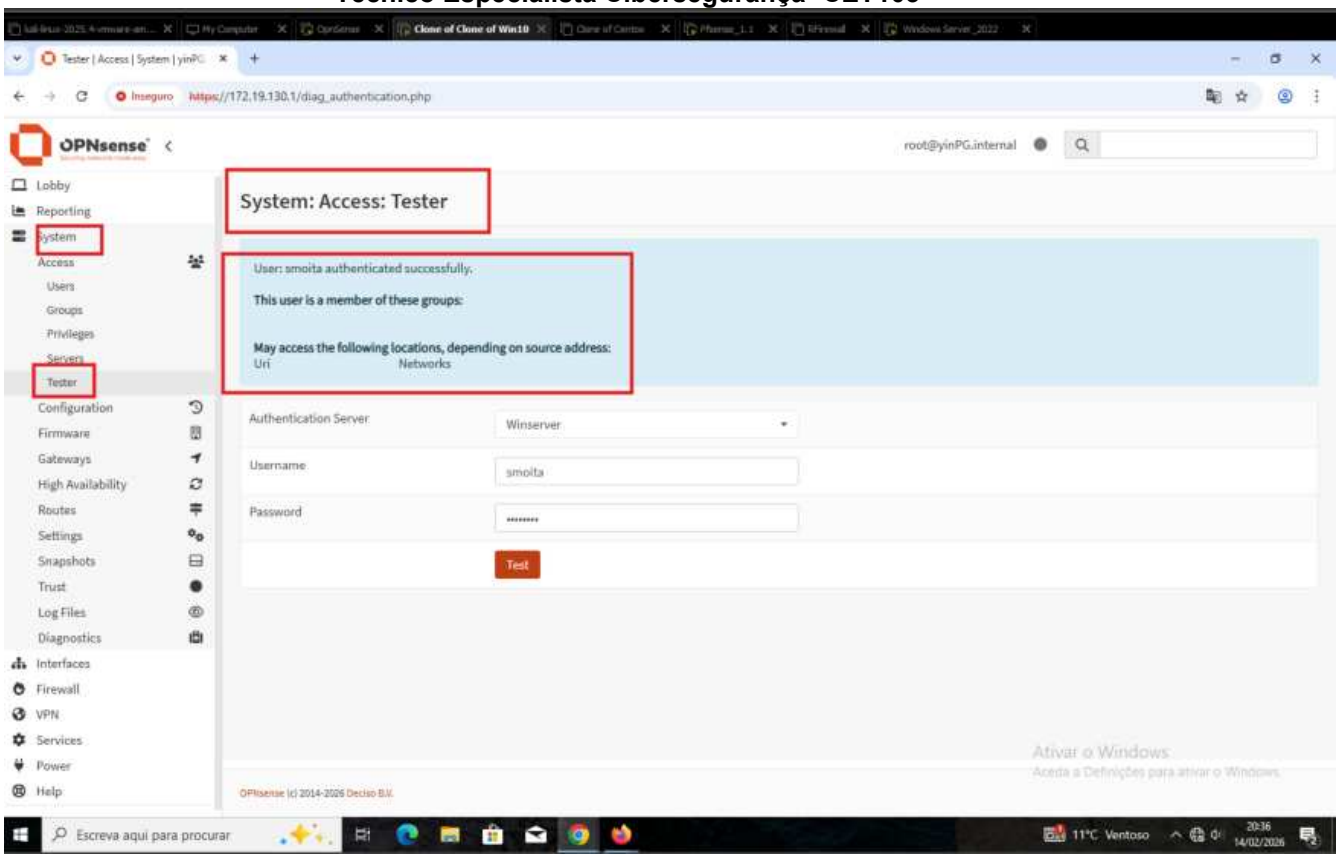
12.30 Restart de “Network Policy Server”- “services .smc”



12.31 Teste de Acesso – Bem sucedido

System → Access → Tester

- **Authentication Server:** Winserver
- **Username:** smoita
- **Password:** Passw0rd



13 CentOS 7 (DMZ) — Web + Auth + SSL/TLS + FTP

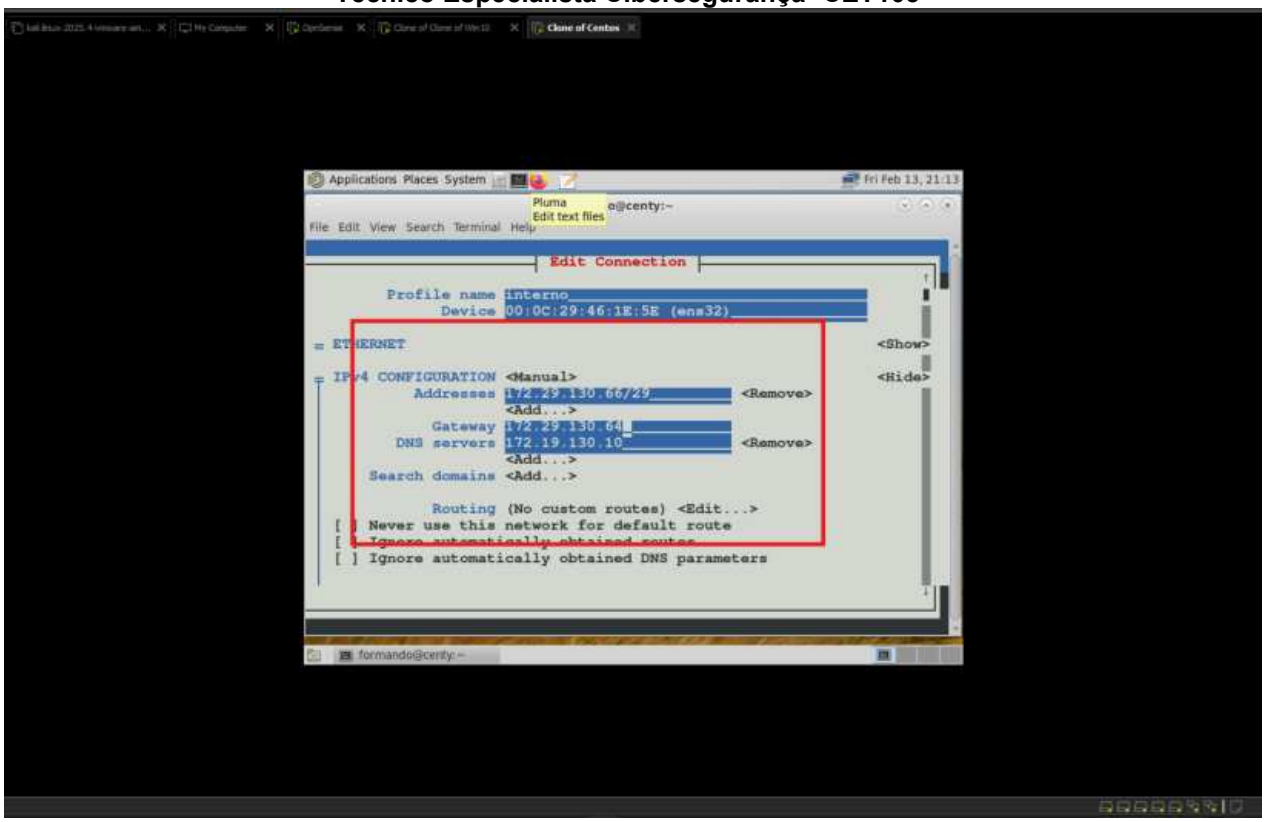
IP fixo

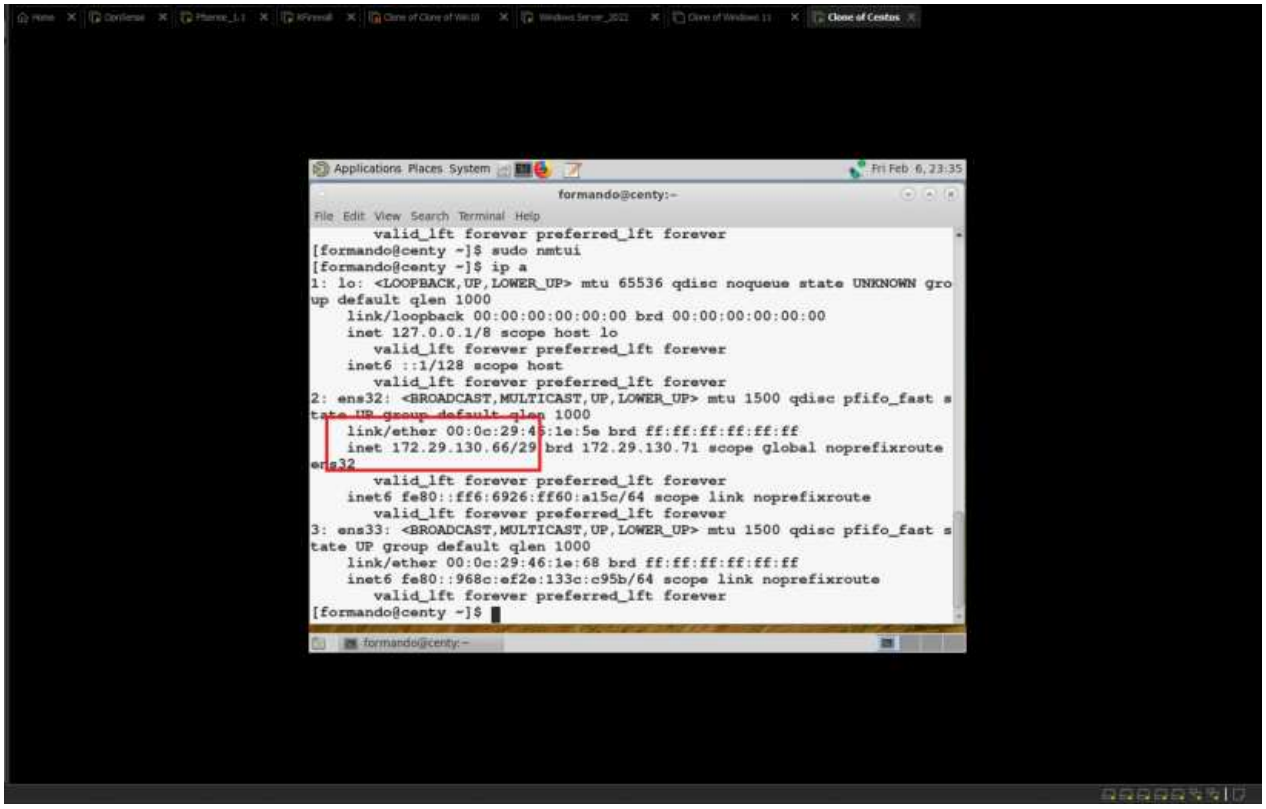
- IP: 172.29.130.66/29
- Gateway: 172.29.130.65
- DNS: 172.19.130.10

Nota suplementar :

“ Intervalo completo do bloco

- *Network (não utilizável): 172.29.130.64- yinPG -orange*
- *Primeiro host: 172.29.130.65*
- *Último host: 172.29.130.70*
- *Broadcast (não utilizável): 172.29.130.71”*





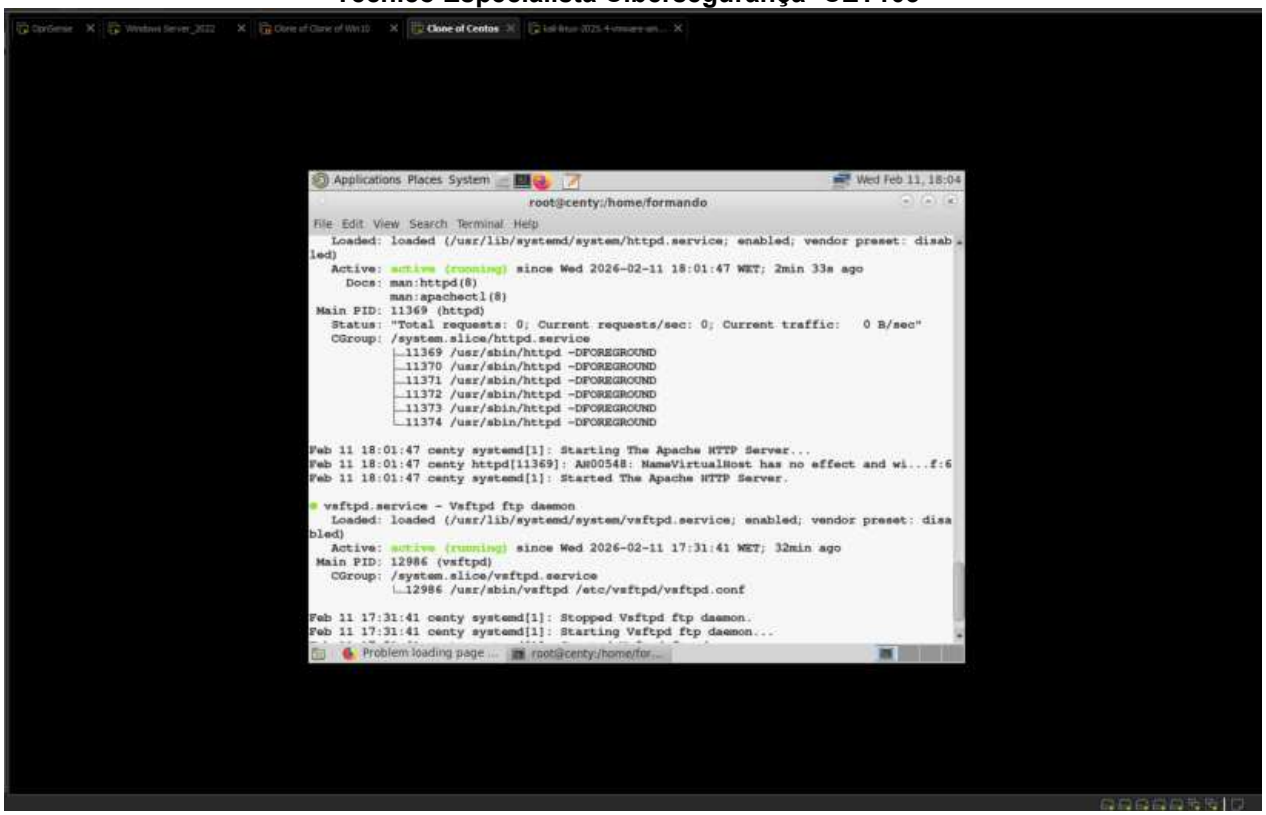
13.1 Atualizar e ferramentas

yum -y update

yum -y install epel-release httpd mod_ssl httpd-tools vsftpd policycoreutils-python

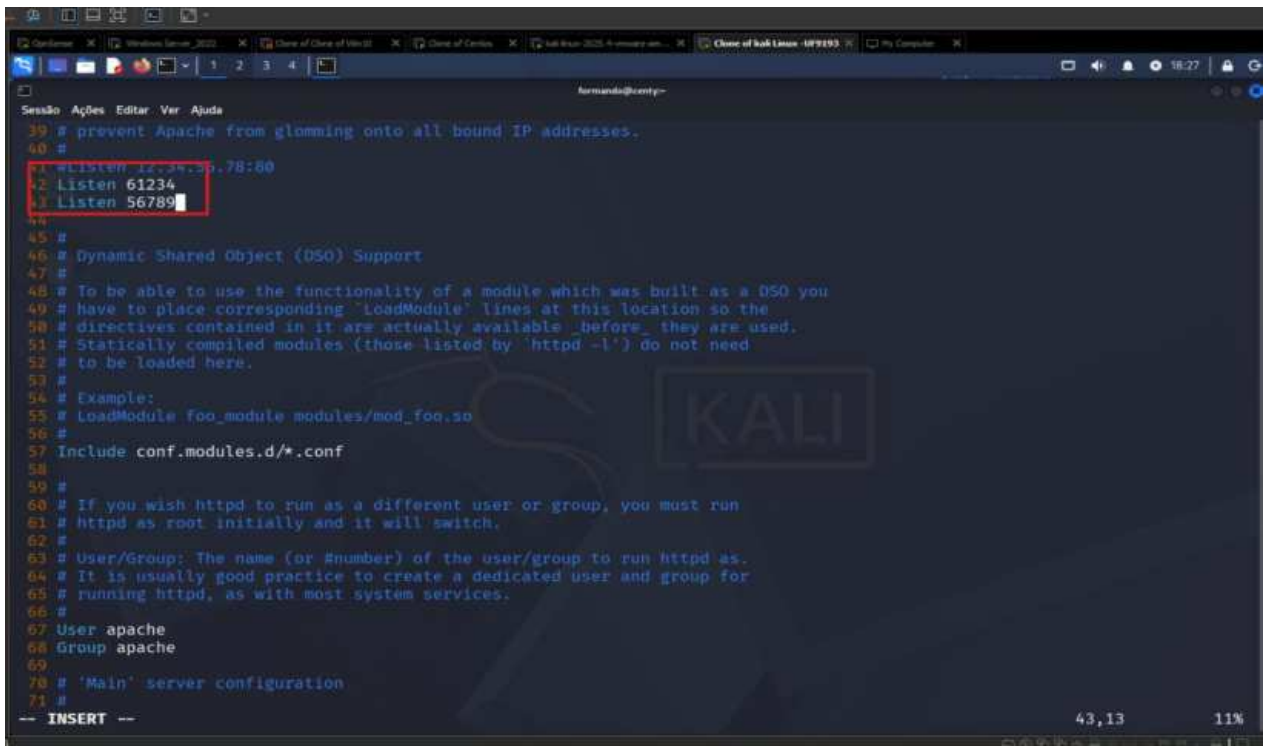
systemctl enable httpd vsftpd

systemctl start httpd vsftpd

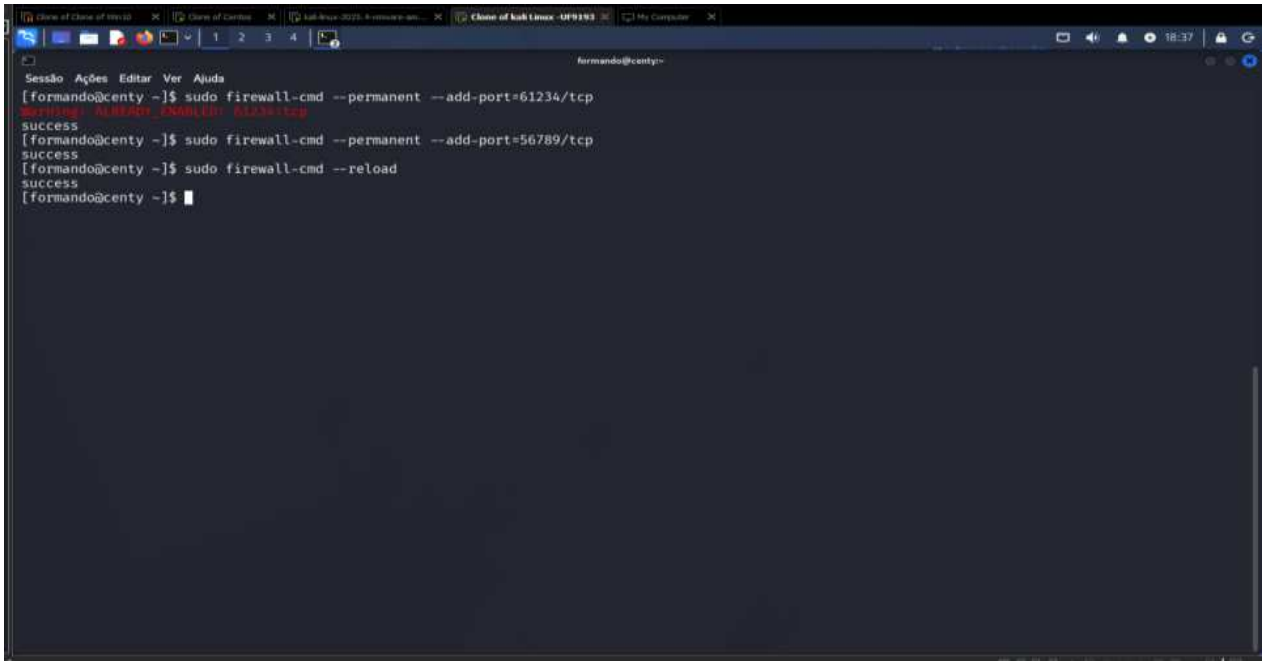


13.2 Apache — portas internas exigidas . Nota :Acesso por ssh ao centOS pelo kaliLinux

Editar /etc/httpd/conf/httpd.conf:
Listen 61234
Listen 56789
[**formando@centy ~]$ sudo vim /etc/httpd/conf/httpd.conf**

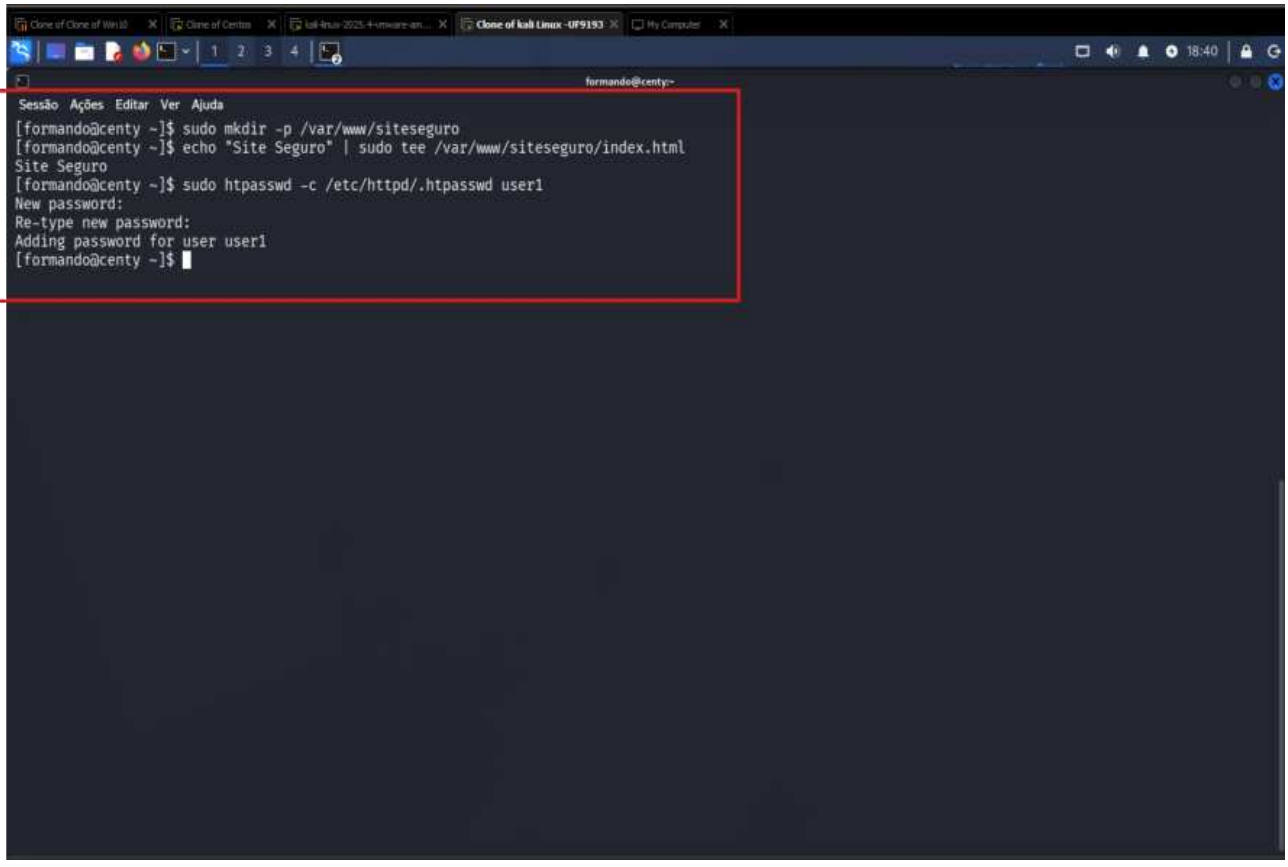


13.3 Abrir as portas na firewall



13.4 Site 1 — siteseguro.local (61234) com autenticação

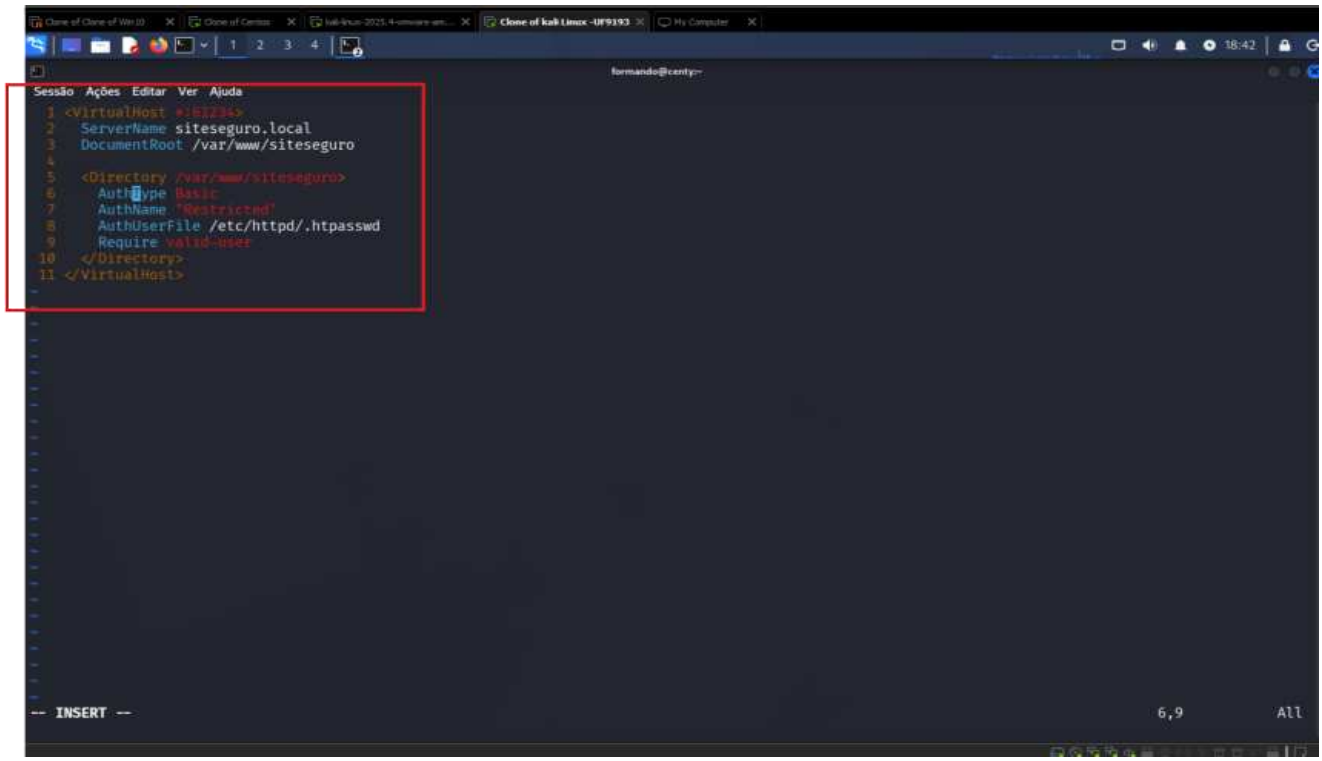
mkdir -p /var/www/siteseguro
echo "Site Seguro" > /var/www/siteseguro/index.html
htpasswd -c /etc/httpd/.htpasswd user1



/etc/httpd/conf.d/siteseguro.conf
<VirtualHost *:61234>
 ServerName siteseguro.local
 DocumentRoot /var/www/siteseguro
 <Directory /var/www/siteseguro>
 AuthType Basic
 AuthName "Restricted"
 AuthUserFile /etc/httpd/.htpasswd
 Require valid-user
 </Directory>
</VirtualHost>

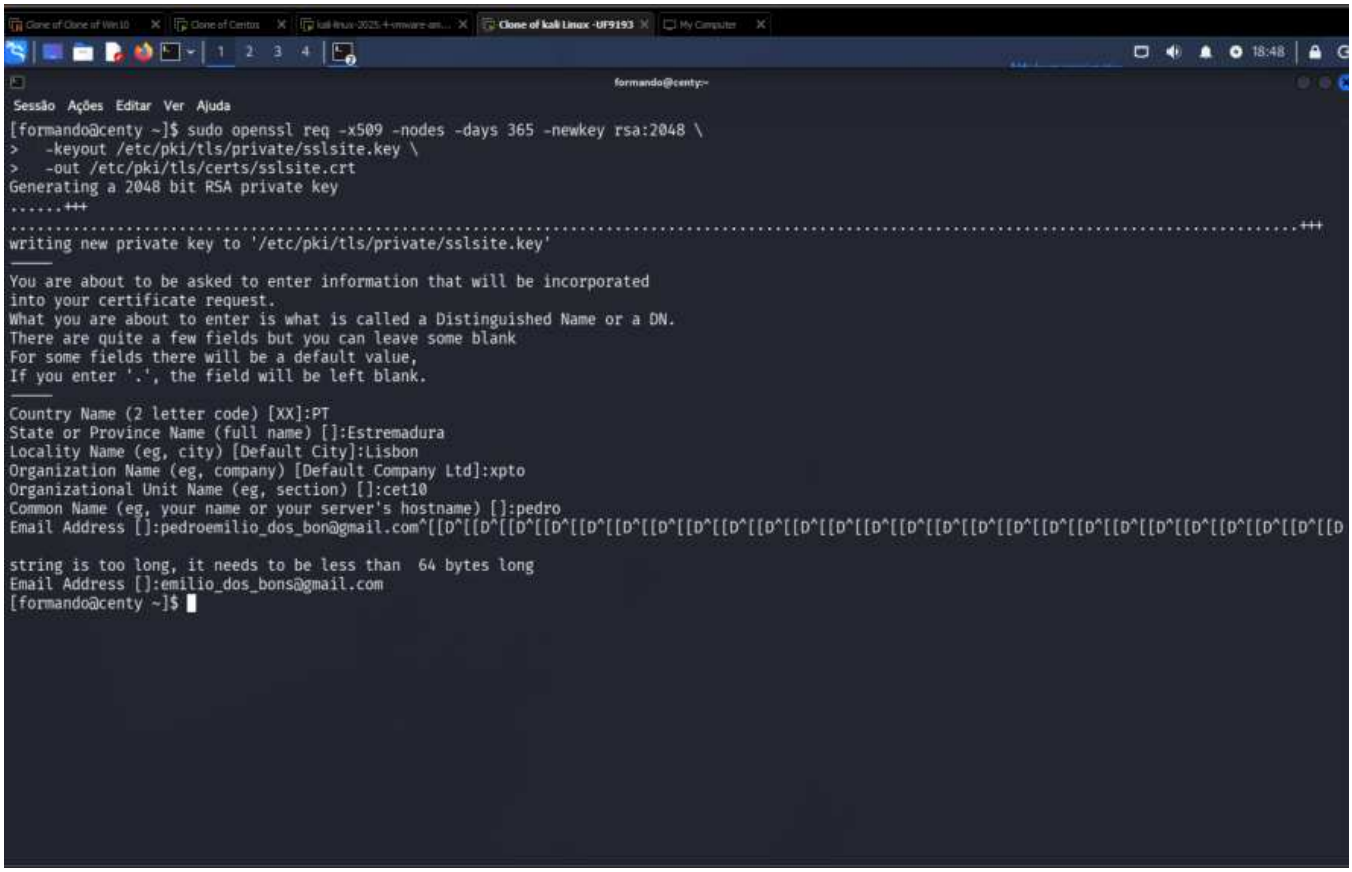
13.5 Criar o VirtualHost
Criar o ficheiro:

[formando@centy ~]$ **sudo vim /etc/httpd/conf.d/siteseguro.conf**



— sslsite.local (56789) HTTPS

13.6 Gerar certificado autoassinado
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
 -keyout /etc/pki/tls/private/sslsite.key \
 -out /etc/pki/tls/certs/sslsite.crt
Quando pedir comun name , colocar : sslsite.local



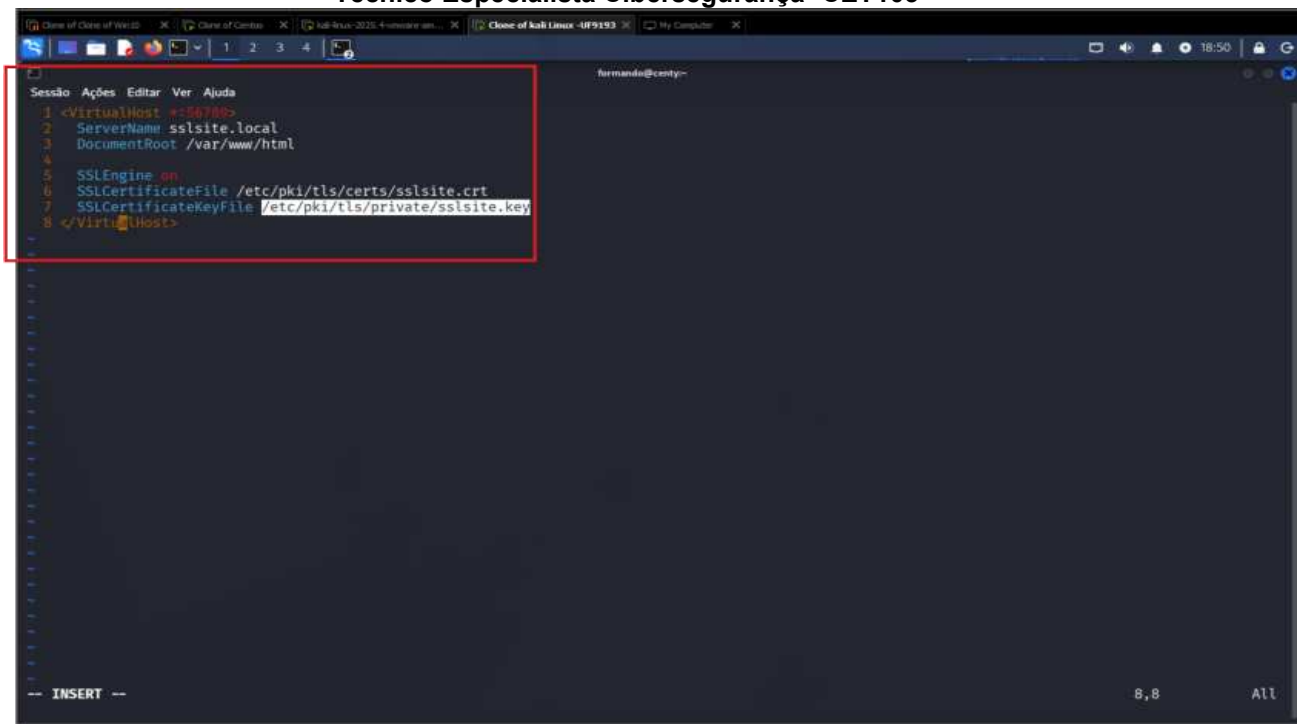
13.7 — sslsite.local (56789) HTTPS

openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
 -keyout /etc/pki/tls/private/sslsite.key \
 -out /etc/pki/tls/certs/sslsite.crt

/etc/httpd/conf.d/sslsite.conf

[formando@centy ~]$ **sudo vim /etc/httpd/conf.d/sslsite.conf**

<VirtualHost *:56789>
 ServerName sslsite.local
 DocumentRoot /var/www/html
 SSLEngine on
 SSLCertificateFile /etc/pki/tls/certs/sslsite.crt
 SSLCertificateKeyFile /etc/pki/tls/private/sslsite.key
</VirtualHost>

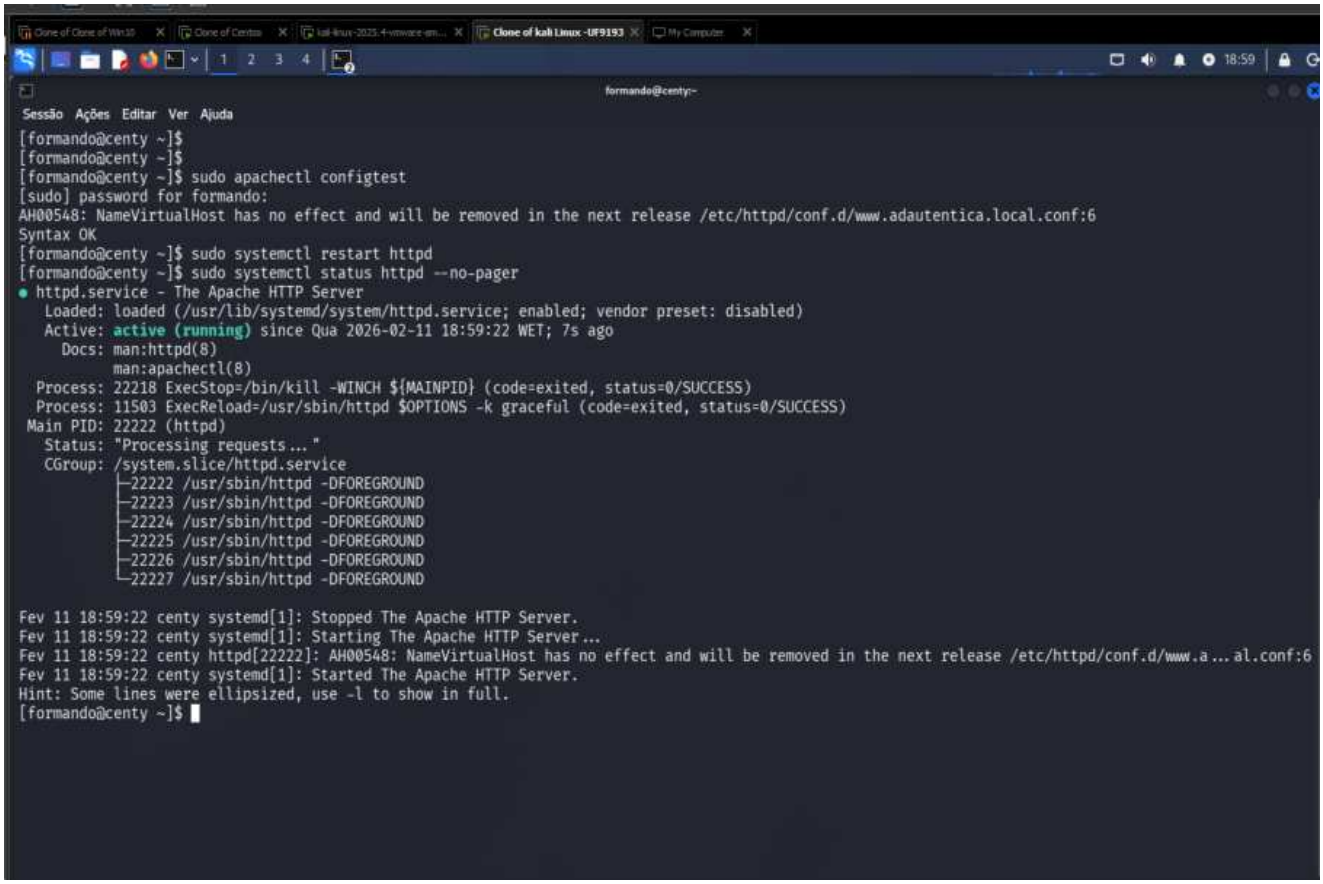


13.8 Teste de sintaxe

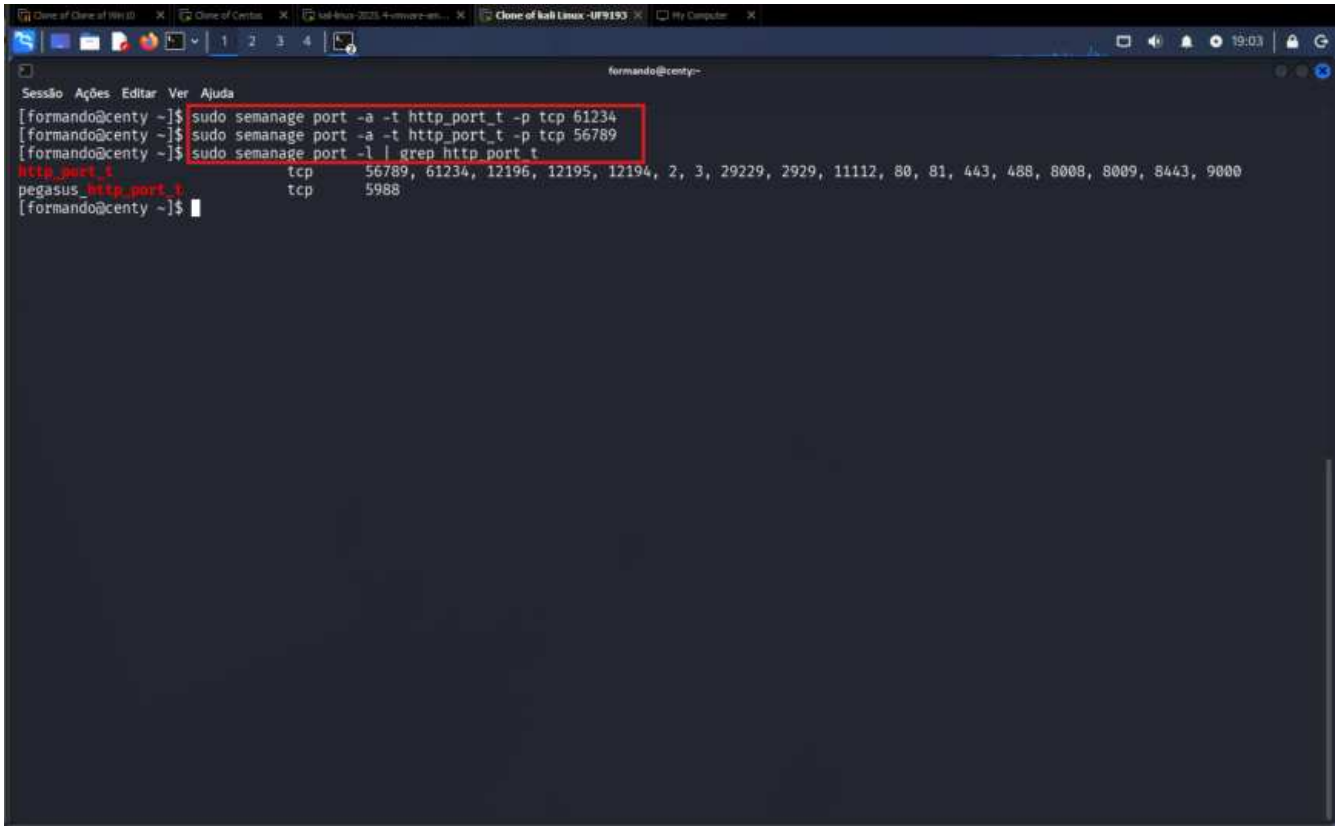
sudo apachectl configtest

13.9 Reiniciar

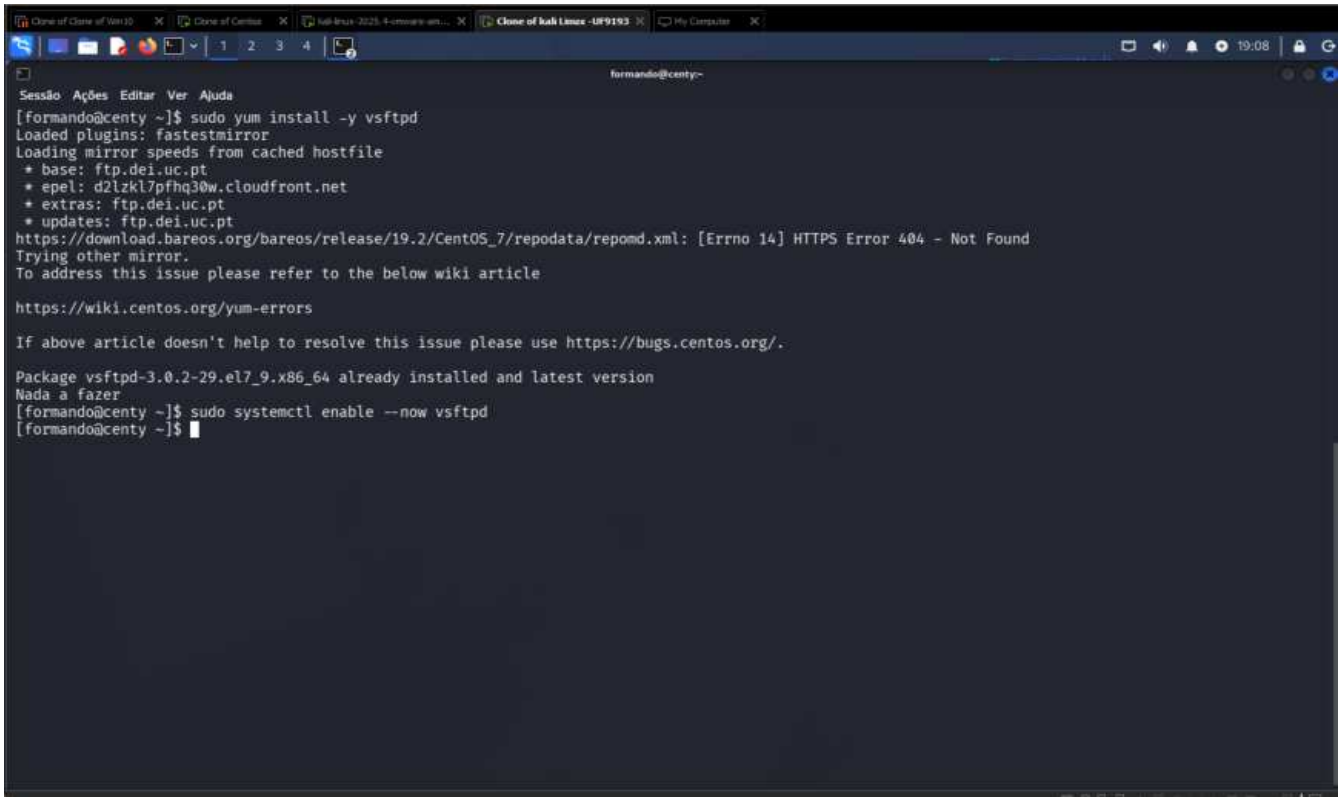
sudo systemctl restart httpd
sudo systemctl status httpd --no-pager



13.10 SELinux ():
semanage port -a -t http_port_t -p tcp 61234
semanage port -a -t http_port_t -p tcp 56789



13.11 Instalar vsftpd



13.12 FTP (vsftpd)

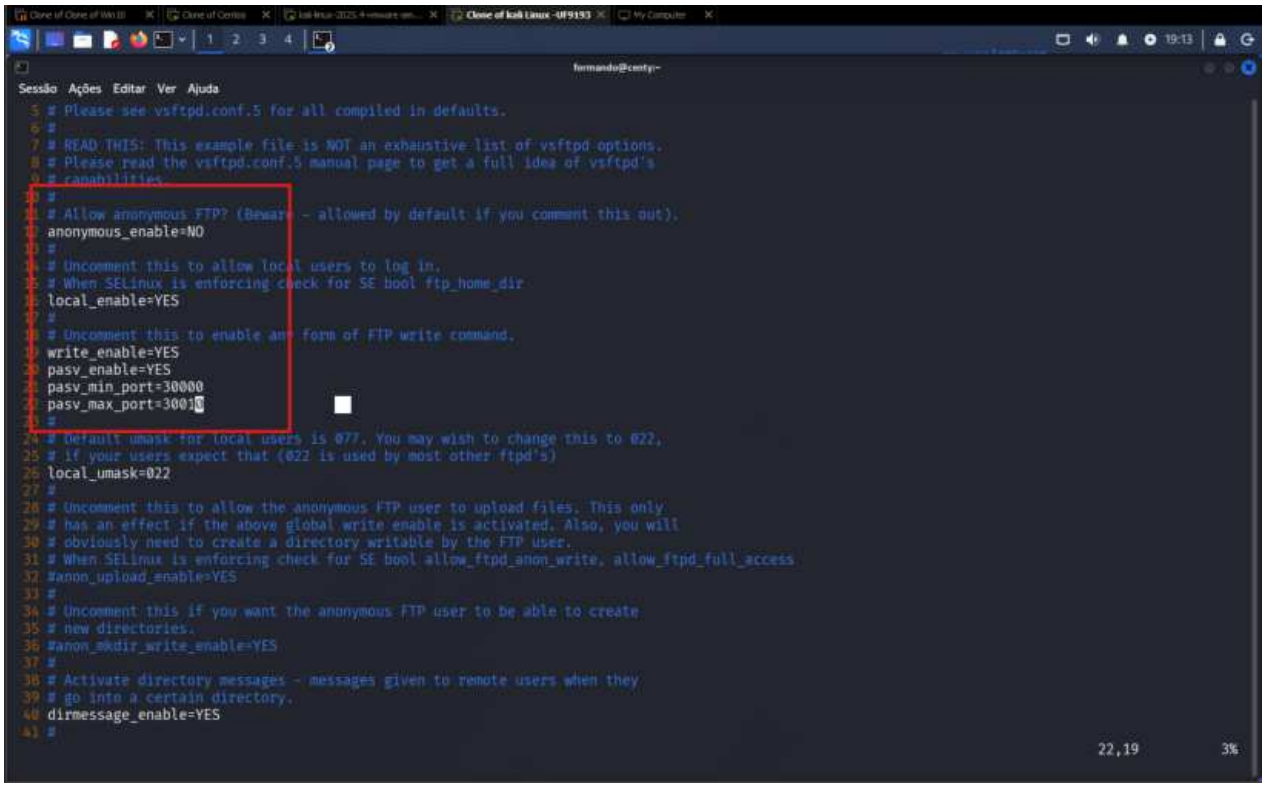
Editar

/etc/vsftpd/vsftpd.conf:

[formando@centy ~]$ **sudo vim /etc/vsftpd/vsftpd.conf**

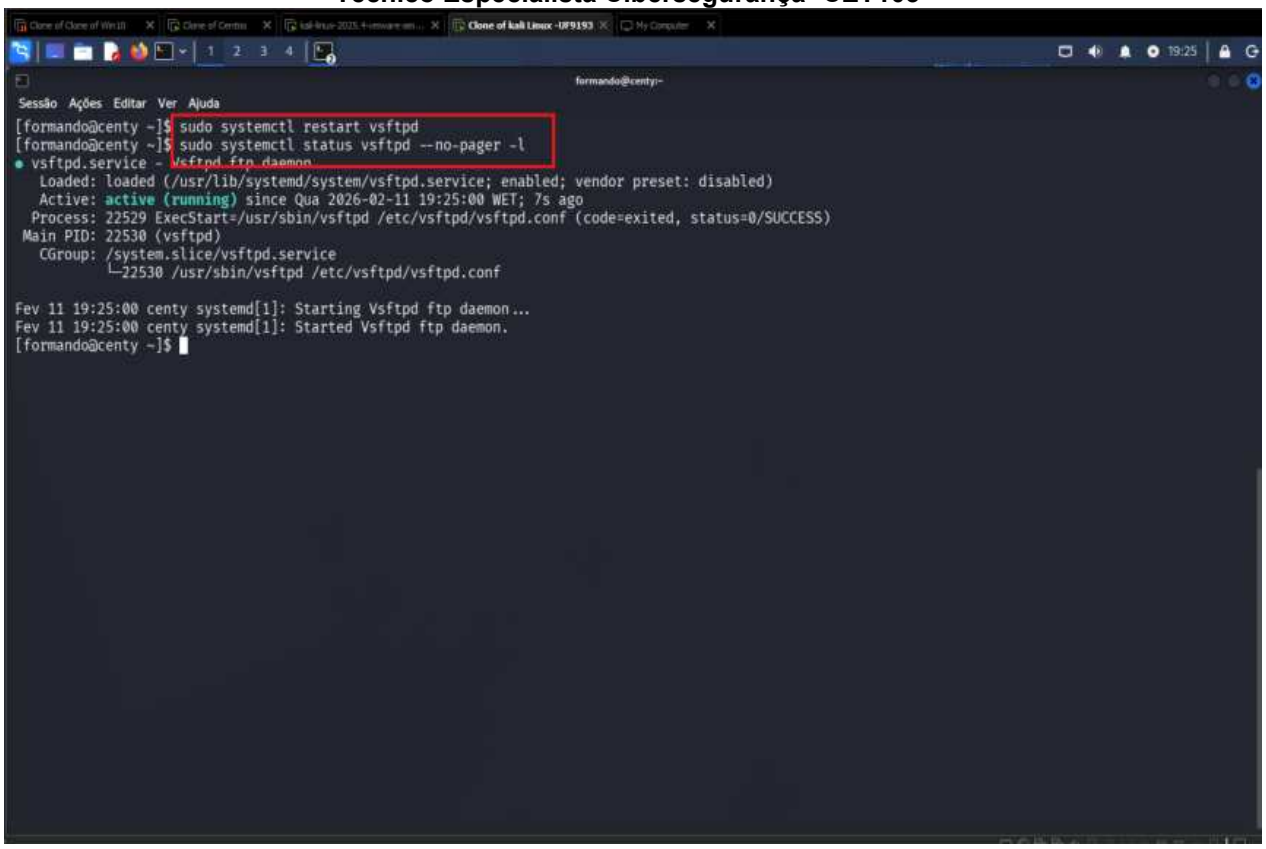
Adicionar linhas

anonymous_enable=NO
local_enable=YES
write_enable=YES
pasv_enable=YES
pasv_min_port=30000
pasv_max_port=30010



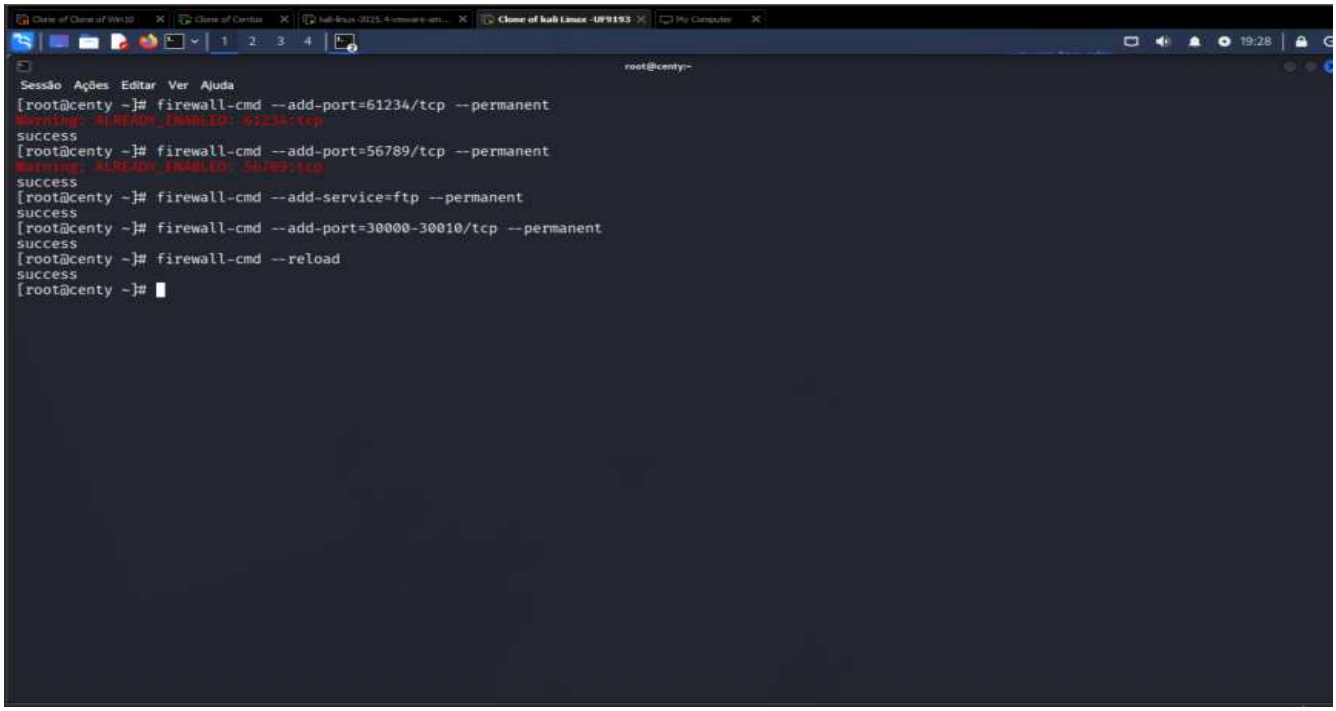
13.13 Inicializar Serviço

systemctl restart vsftpd



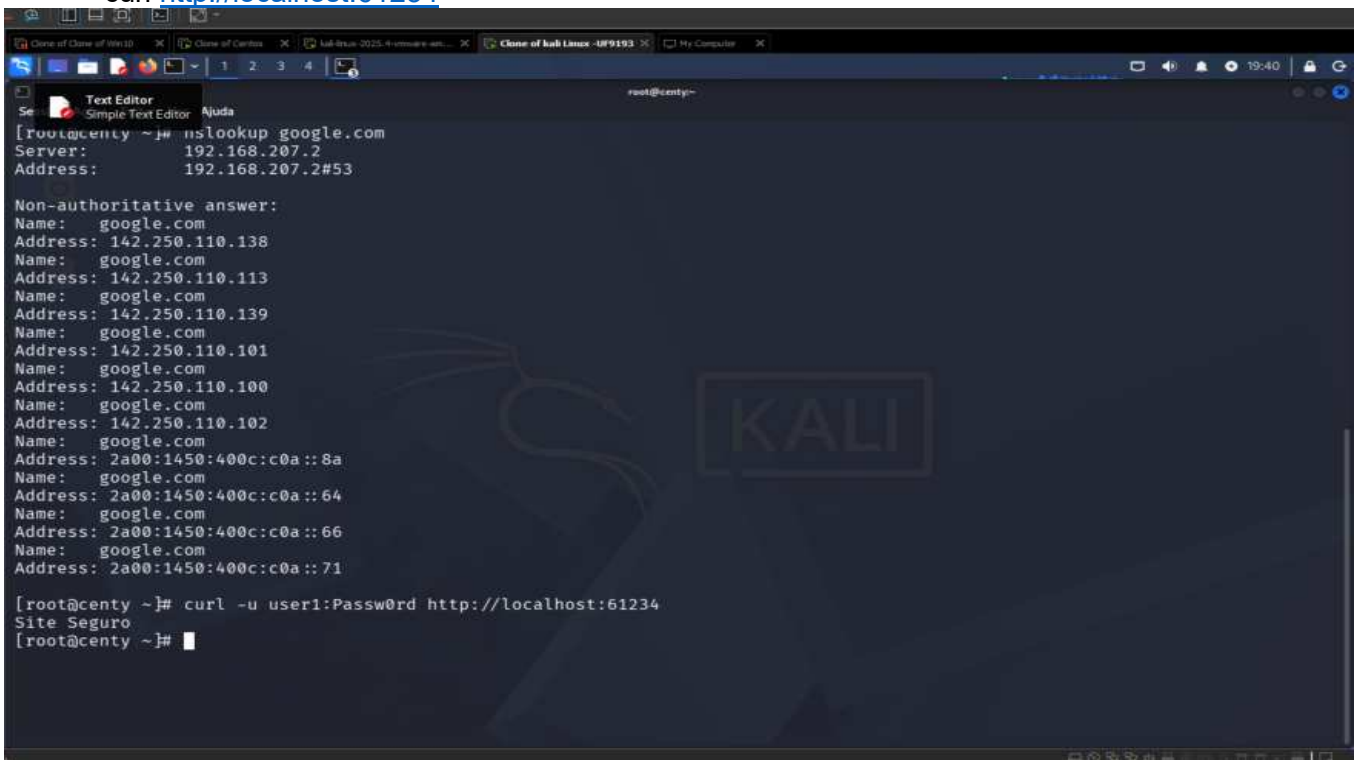
13.14 Firewalld

firewall-cmd --add-port=61234/tcp --permanent
firewall-cmd --add-port=56789/tcp --permanent
firewall-cmd --add-service=ftp --permanent
firewall-cmd --add-port=30000-30010/tcp --permanent
firewall-cmd --reload



13.15 Check rápido (antes das regras do OPNsense)

- **Do CentOS**:
- nslookup google.com (DNS via Windows)
- curl http://localhost:61234



- **Do Windows Server**:
  nslookup google.com
- **Do R1 (VyOS)**:
  ping 8.8.8.8

2 g) Situado com o cliente na “Wan pessoal” aceda aos seguintes sites (pode colocar no ficheiro de hosts do cliente IP FQDN para evitar a utilização de mais uma VM com o DNS). Sem utilizar VPN! ♣ http://www.siteseguro.local:22222 site com autenticação em que o webserver escuta no porto 61234 ♣ https://www.sslsite.local:42424 site em SSL/TLS em que o webserver escuta no porto 56789

14) OPNsense yinPG — NAT Port Forward para cumprir o acesso desde a WAN (sem VPN)

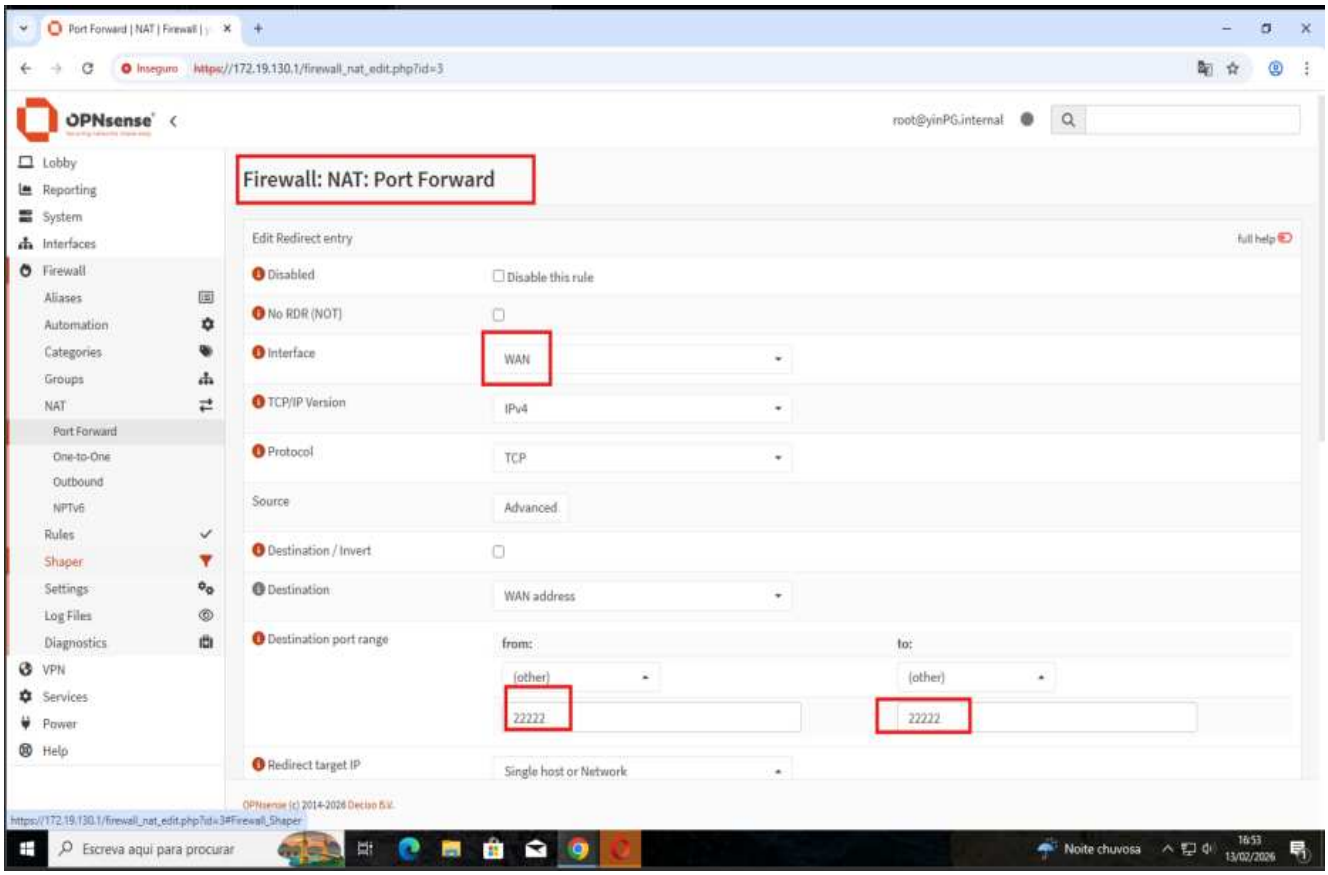
Requisito: na WAN pessoal aceder a:

- http://www.siteseguro.local:22222 → webserver escuta 61234
- https://www.sslsite.local:42424 → webserver escuta 56789

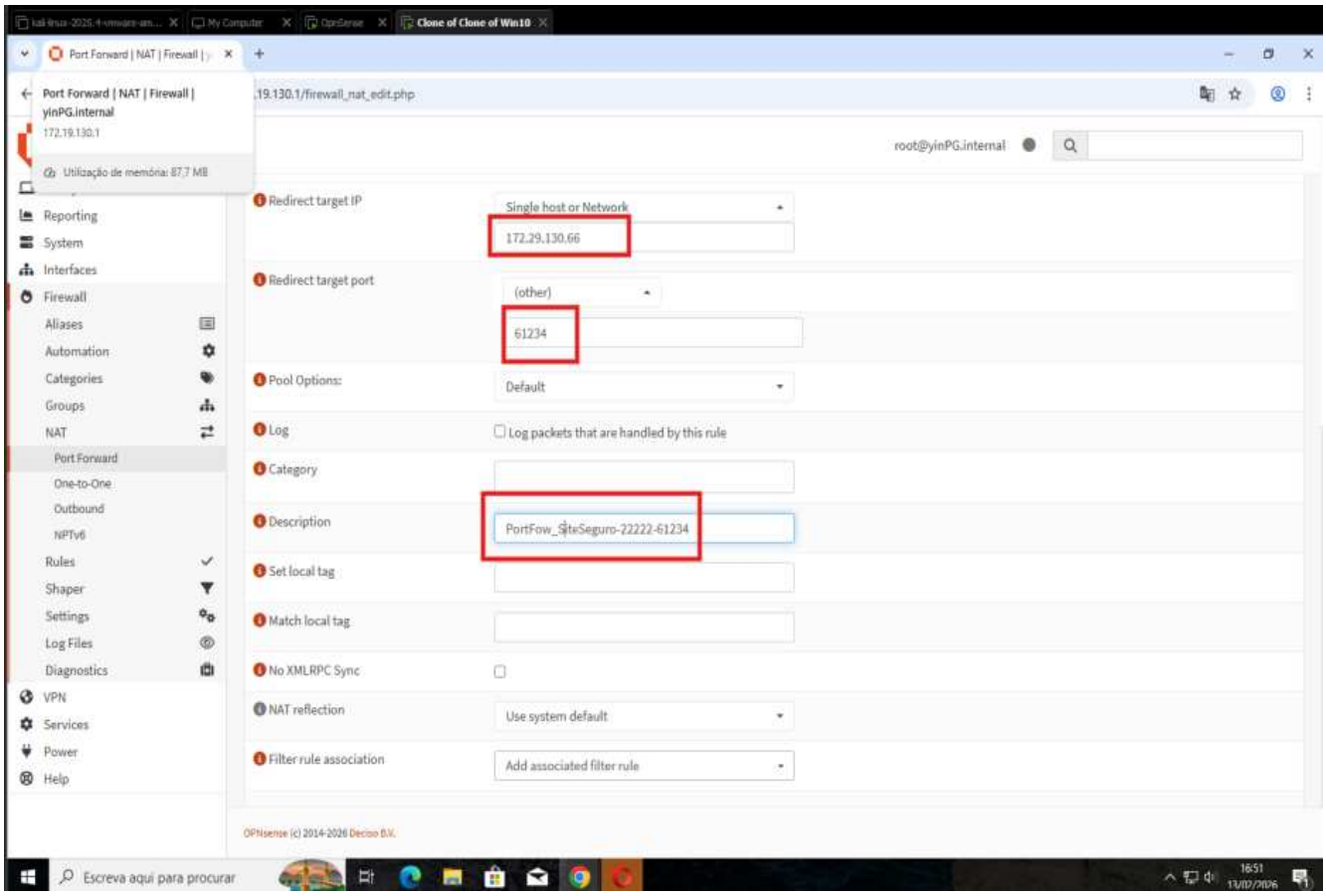
14.1 Port Forward 22222 → 61234

Firewall → NAT → Port Forward → +Add

- Interface: WAN
- Protocol: TCP
- Destination: WAN address
- Destination port: **22222**
- Redirect target IP: 172.29.130.66
- Redirect target port: **61234**
- Filter rule association: Add associated rule
- Description: PortFow_SiteSeguro-22222-61234

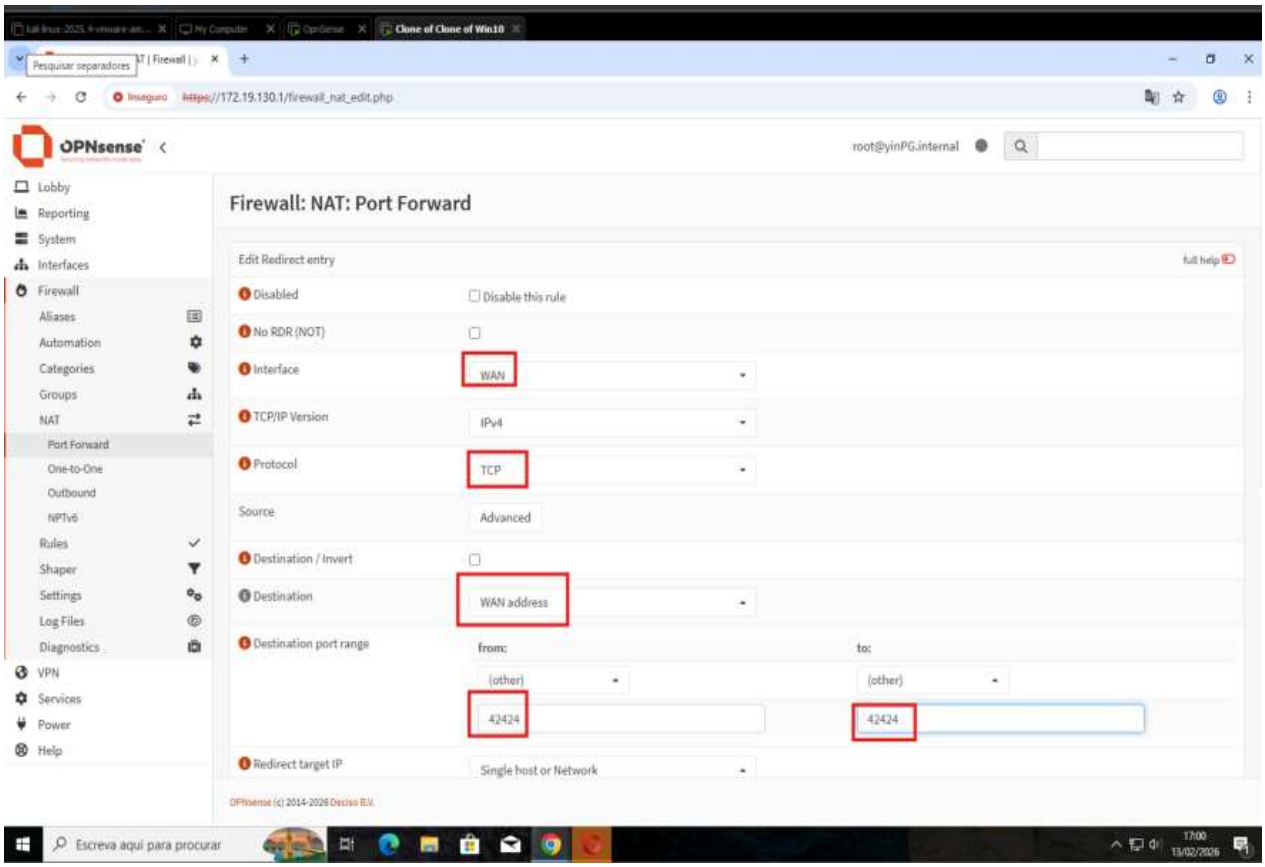


14.2 Continuação…

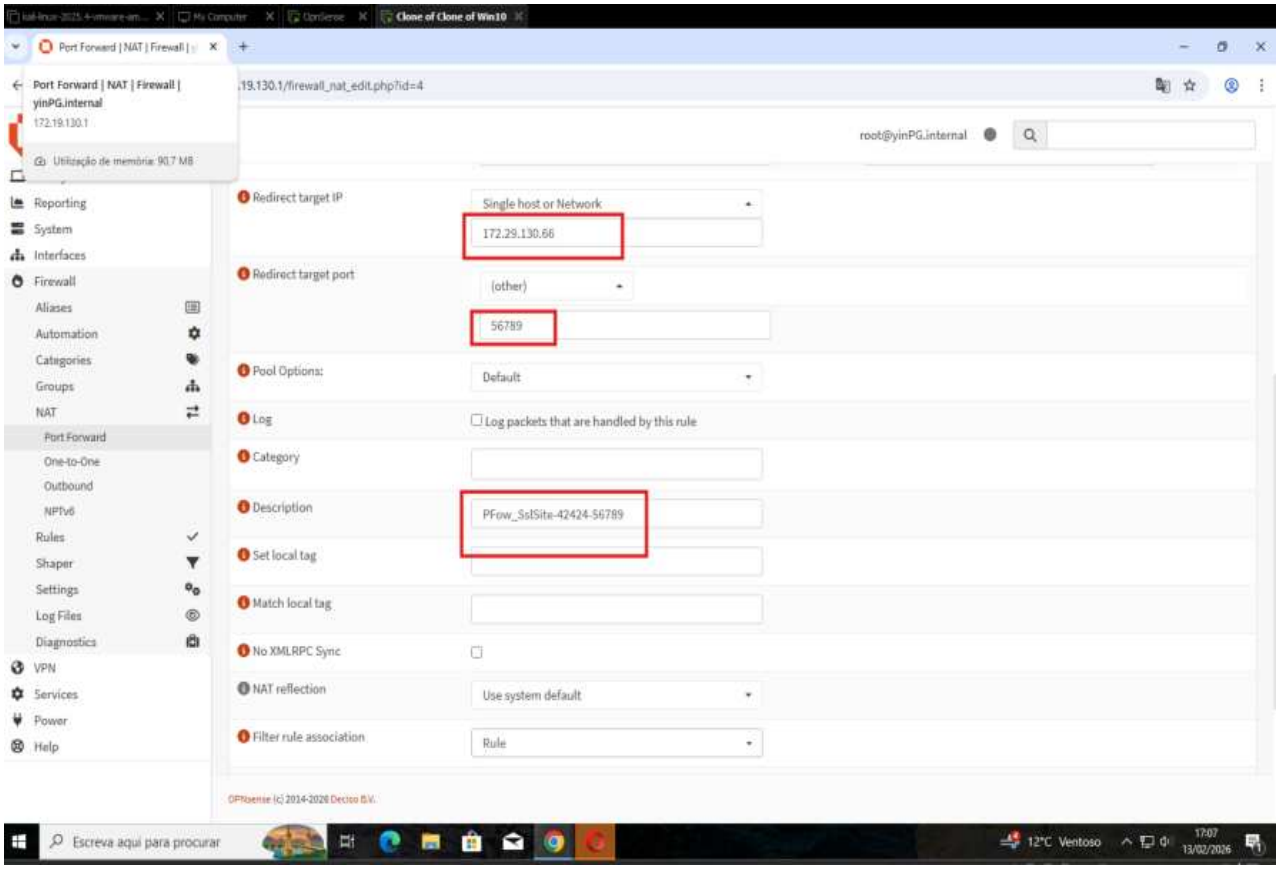


14.3 Port Forward 42424 → 56789

- Interface: WAN
- Protocol: TCP
- Destination: WAN address
- Destination port: **42424**
- Redirect target IP: 172.29.130.66
- Redirect target port: **56789**
- Filter rule association: rule
- Description: PFow_SslSite-42424-56789

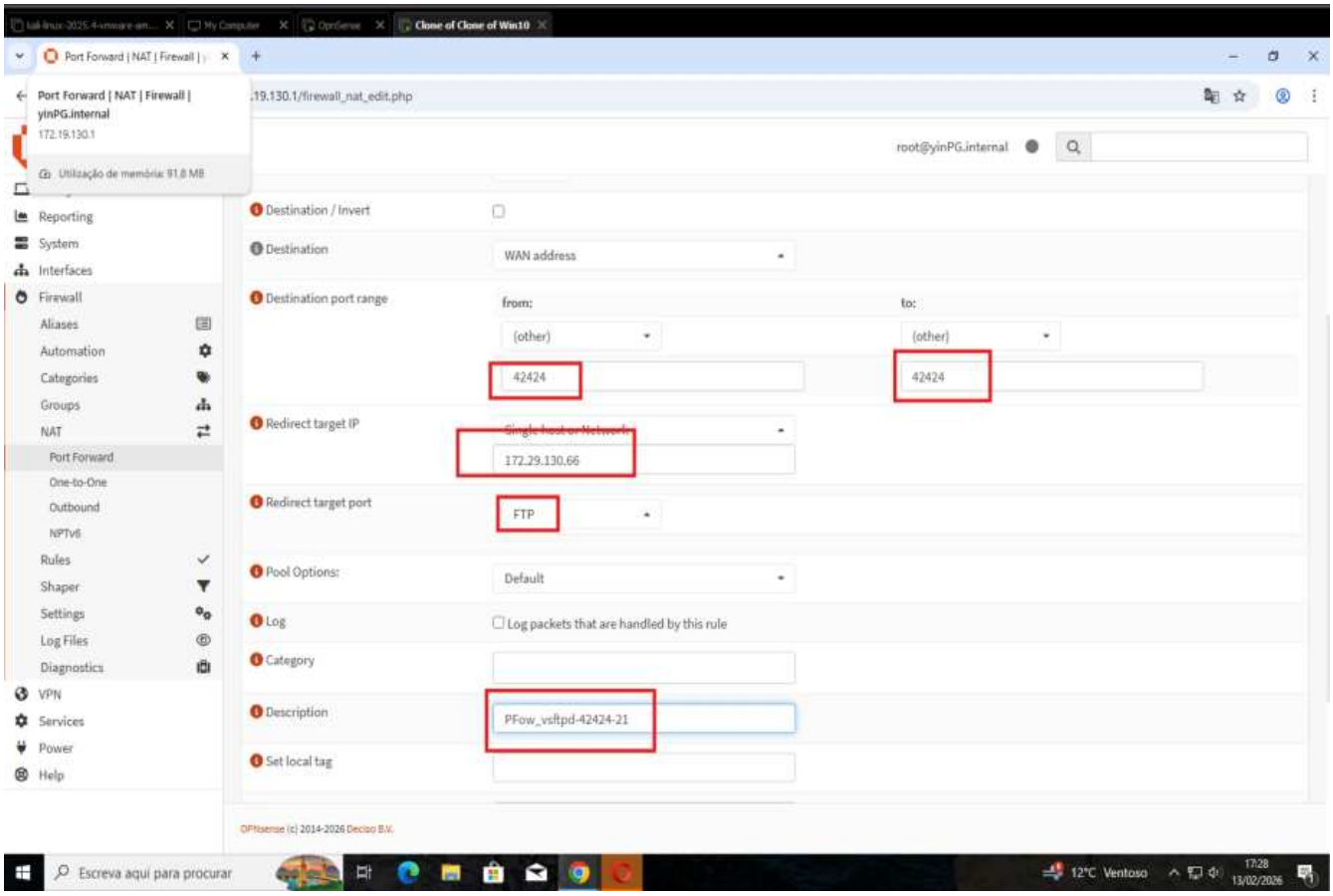


14.4 Continuação…



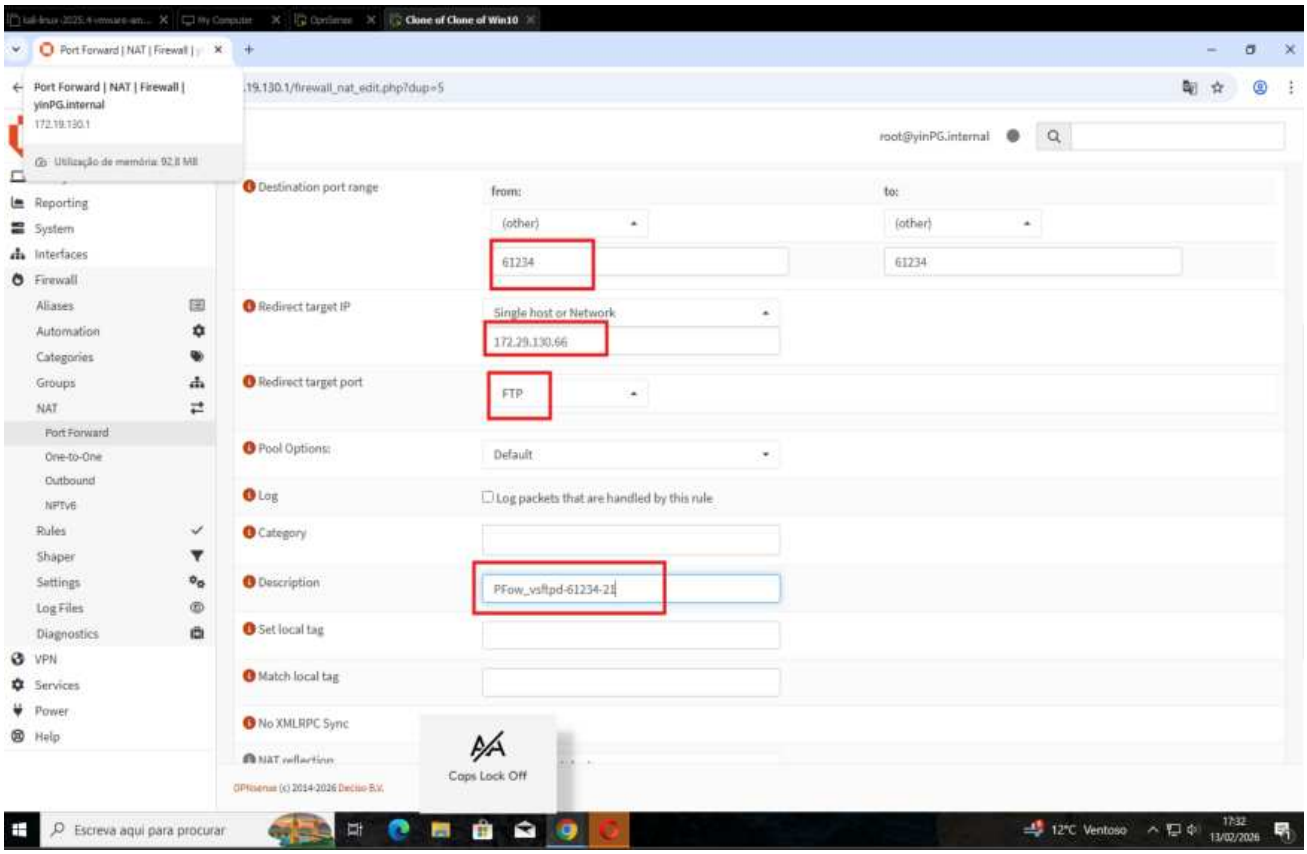
14.4 (Opcional) Port Forward FTP-- Destination port: 42424

- Interface: WAN
- Protocol: TCP
- Destination: WAN address
- Destination port: **42424**
- Redirect target IP: 172.29.130.66
- Redirect target port:21- **FTP**
- Description: PFow_vsftpd-42424-21

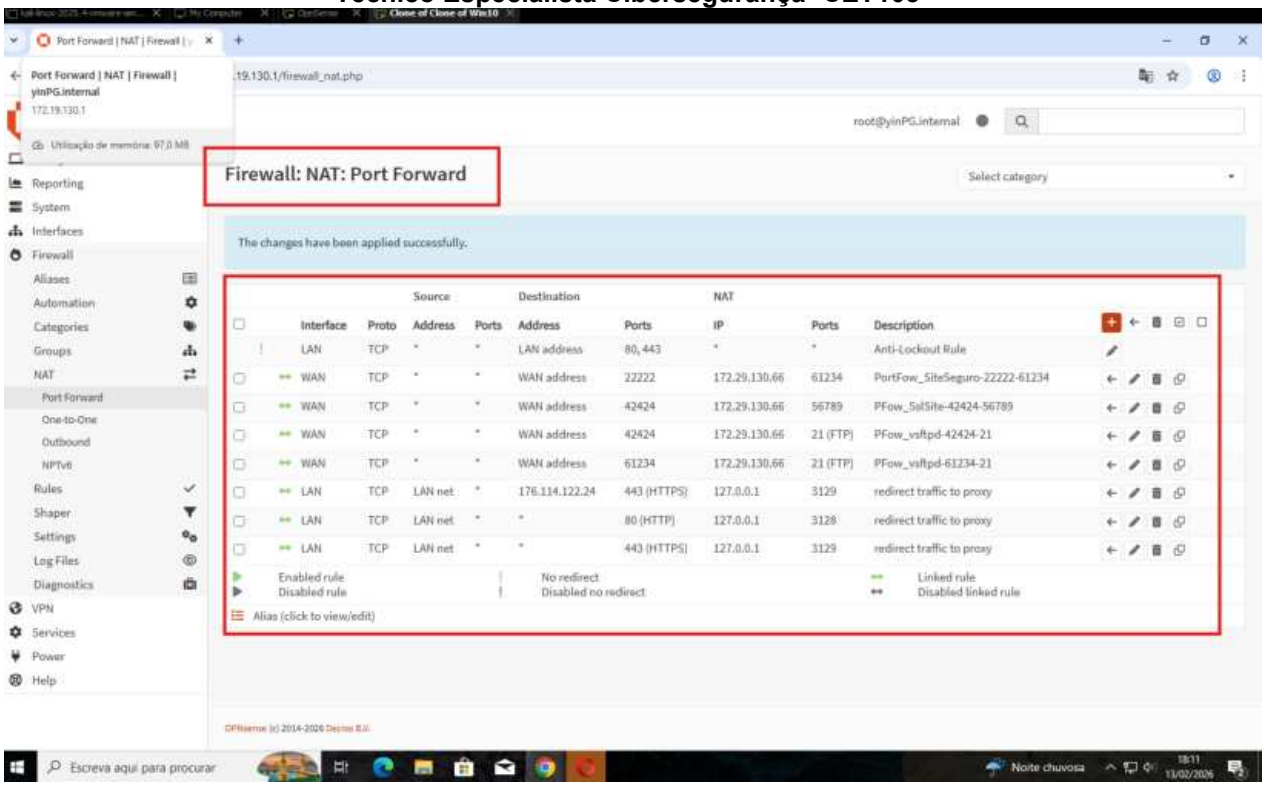


14.5 (Opcional) Port Forward FTP-- Destination port: **61234**

- Interface: WAN
- Protocol: TCP
- Destination: WAN address
- Destination port: **61234**
- Redirect target IP: 172.29.130.66
- Redirect target port: **21-FTP**
- Filter rule association: Add associated rule
- Description: PFow_vsftpd-61234-21



Resumo :



14.5 Hosts file no Windows 10 (WAN)

O enunciado permite usar hosts para FQDN.

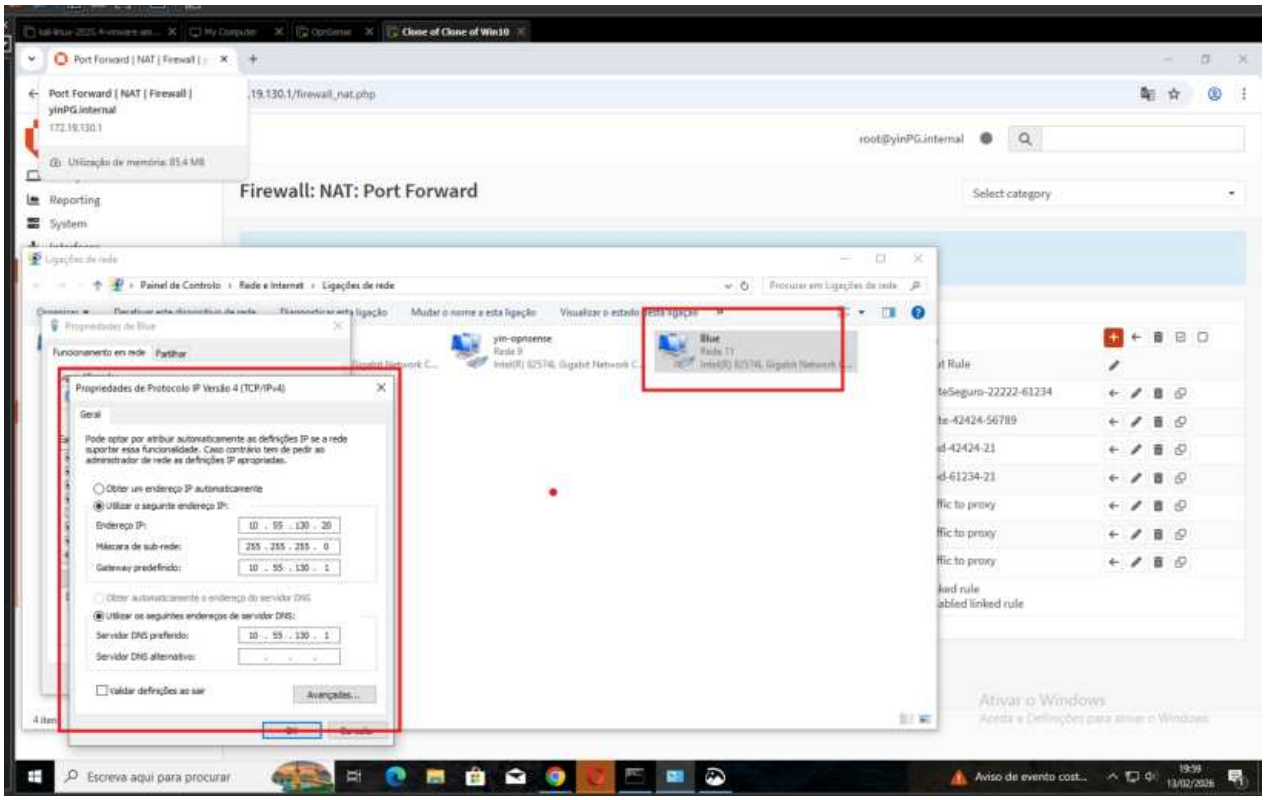
Configurações de placas de rede, resumo :

a) Windows 10 (Cliente de testes)

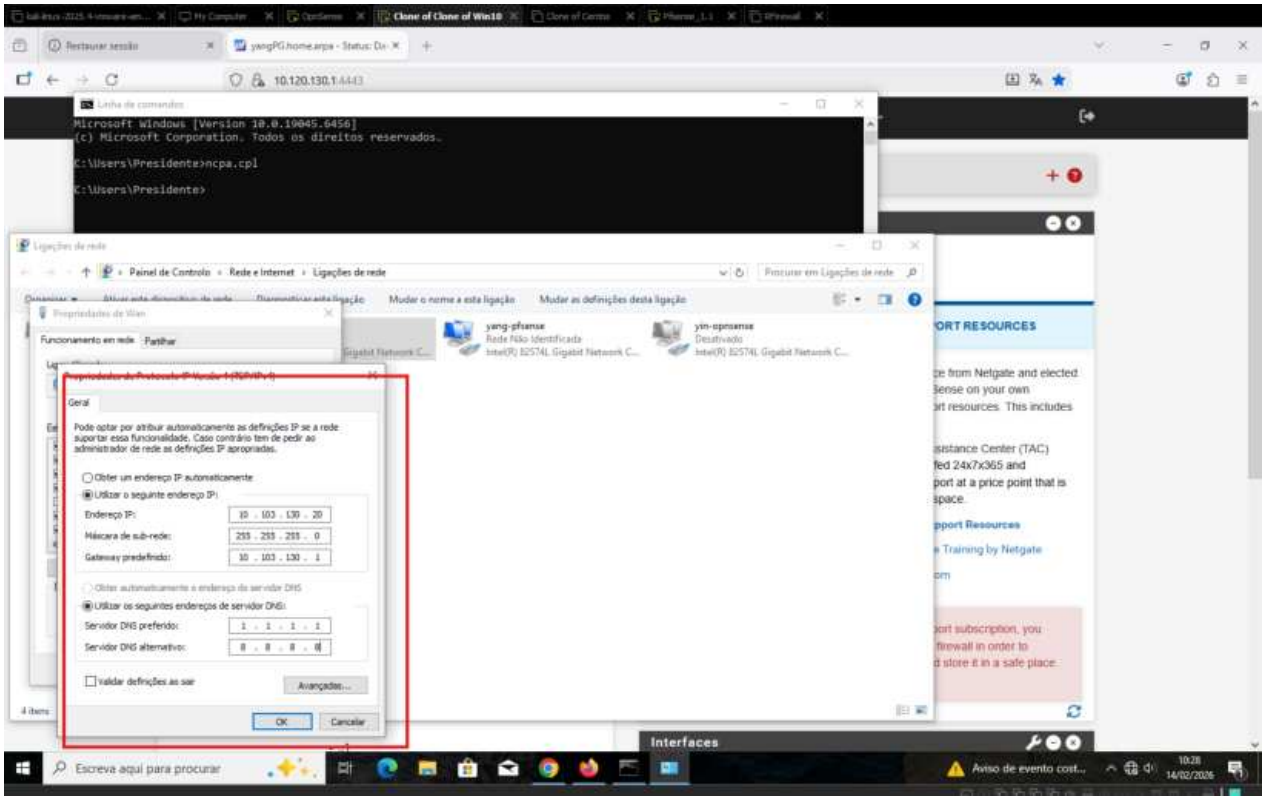
- NIC yin green: 172.19.130.20/23 GW 172.19.130.1
- NIC yang green: 10.120.130.20/24 GW 10.120.130.1

b) Interfaces ainda não documentados (print Screens)

NIC yin blue (para captive): 10.55.130.20/24 GW 10.55.130.1



c) NIC WAN: 10.103.130.20/27 GW 10.103.130.1
d) DNS Predefinido 1.1.1.1
e) Dns Alternativo: 8.8.8.8



14.6 No Windows 10 (na WAN):

Editar como admin:

C:\Windows\System32\drivers\etc\hosts

Adicionar:

10.103.130.10 www.siteseguro.local
10.103.130.10 www.sslsite.local

10.5 Teste sem VPN (Windows 10 na WAN)

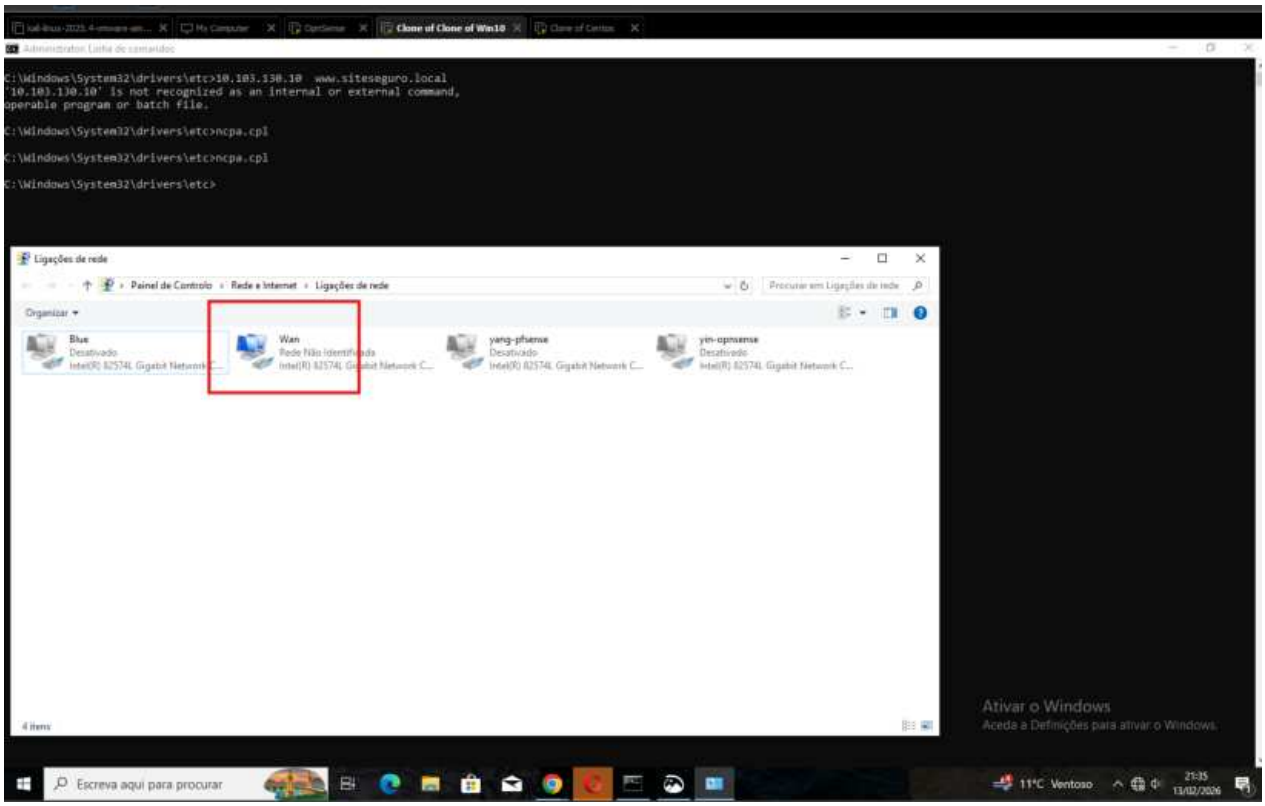
- Abrir: http://www.siteseguro.local:22222
    - Deve pedir user/pass (user1 / Passw0rd)
- Abrir: https://www.sslsite.local:42424
    - Vai dar warning de certificado (self-signed) — aceitável no lab.

14.7 Ficheiro hosts no Windows 10 (na WAN) —

O enunciado indica permite resolver os FQDNs via ficheiro hosts no Windows para ser preciso de outra VM só para DNS.

E os testes (sem VPN) sãos solicitados os sites/portas :22222 e :42424.

Nota : desativar temporariamente as NICs que não são WAN – ideal para testes



14.8 Abrir o Notepad/Bloco de Notas como Administrador

1. Start → escrever "**notepad**"
2. Botão direito → **Run as administrator**
3. File → Open
4. Ira a: C:\Windows\System32\drivers\etc\

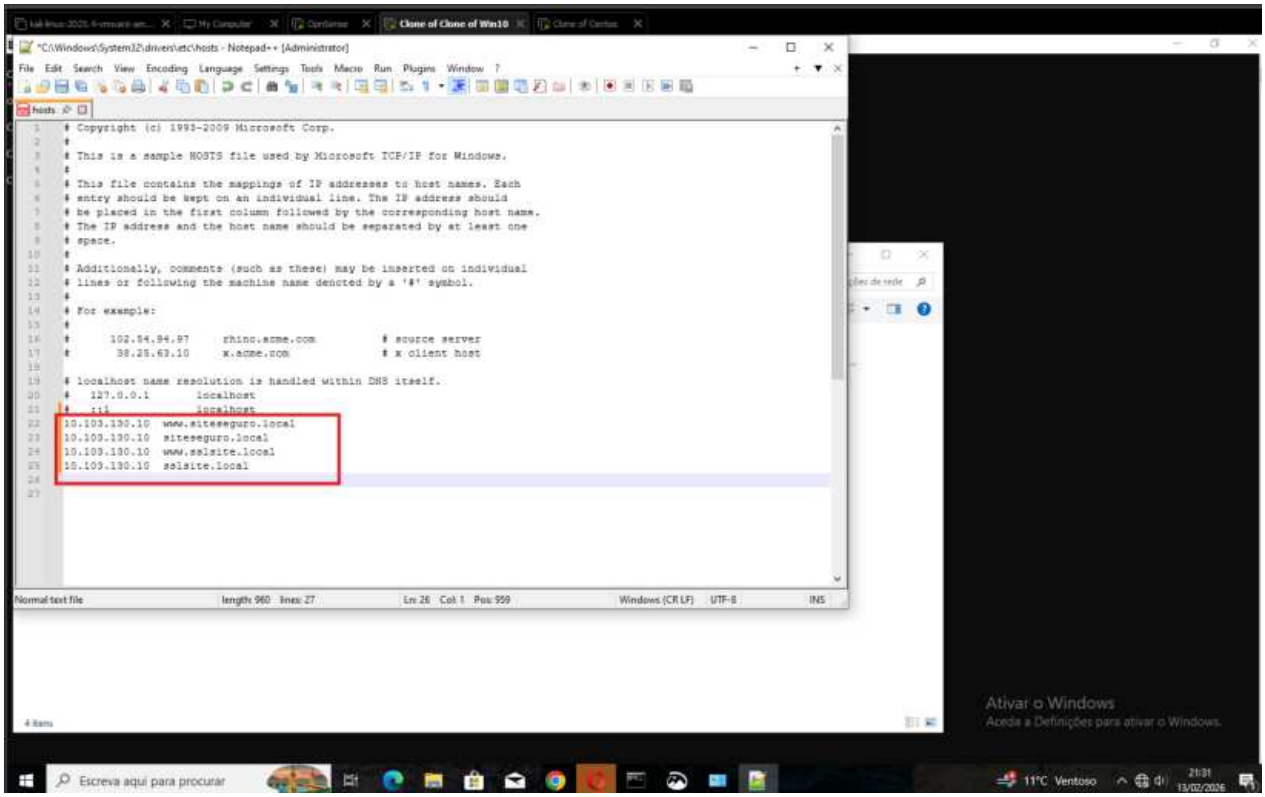
14.10 Adicionar as linhas :

10.103.130.10 www.siteseguro.local

10.103.130.10 siteseguro.local

10.103.130.10 www.sslsite.local

10.103.130.10 sslsite.local



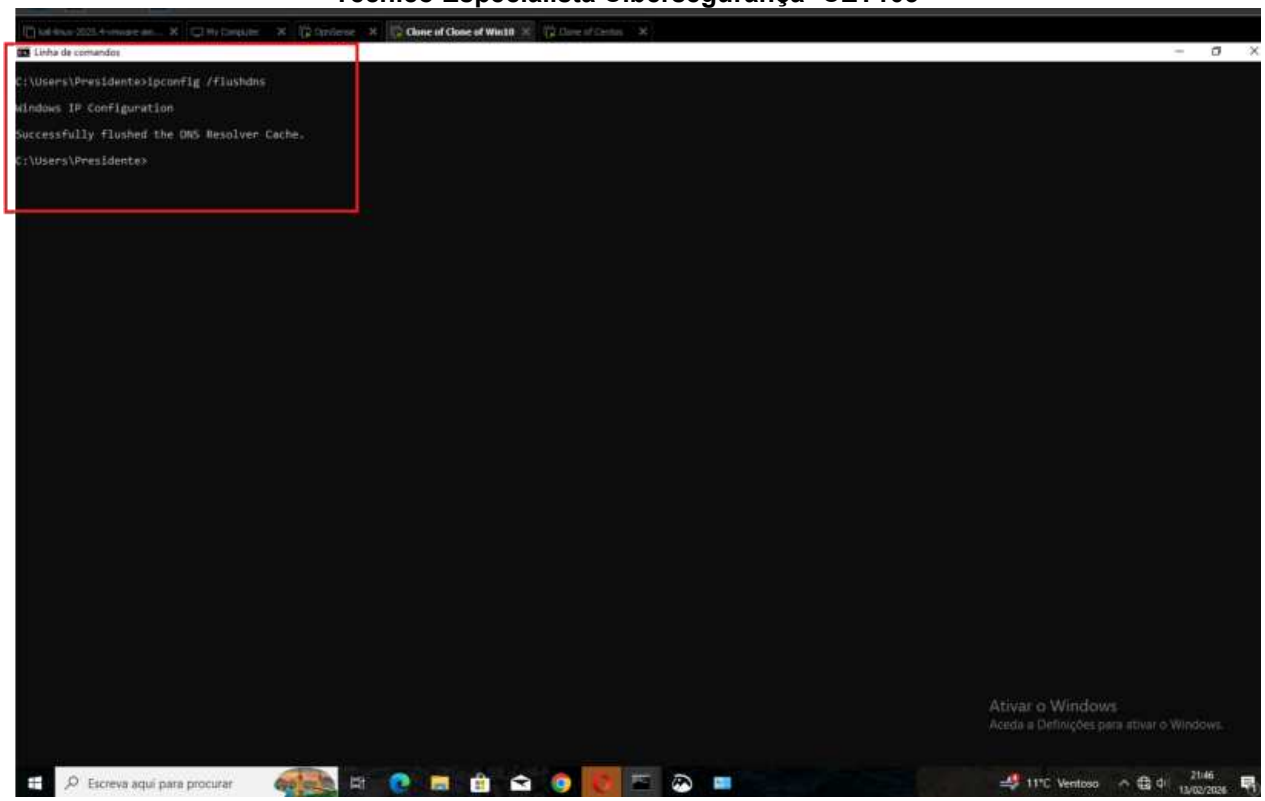
Guardar (Ctrl+S).

.

15- Limpar cache DNS do Windows

No CMD:

ipconfig /flushdns

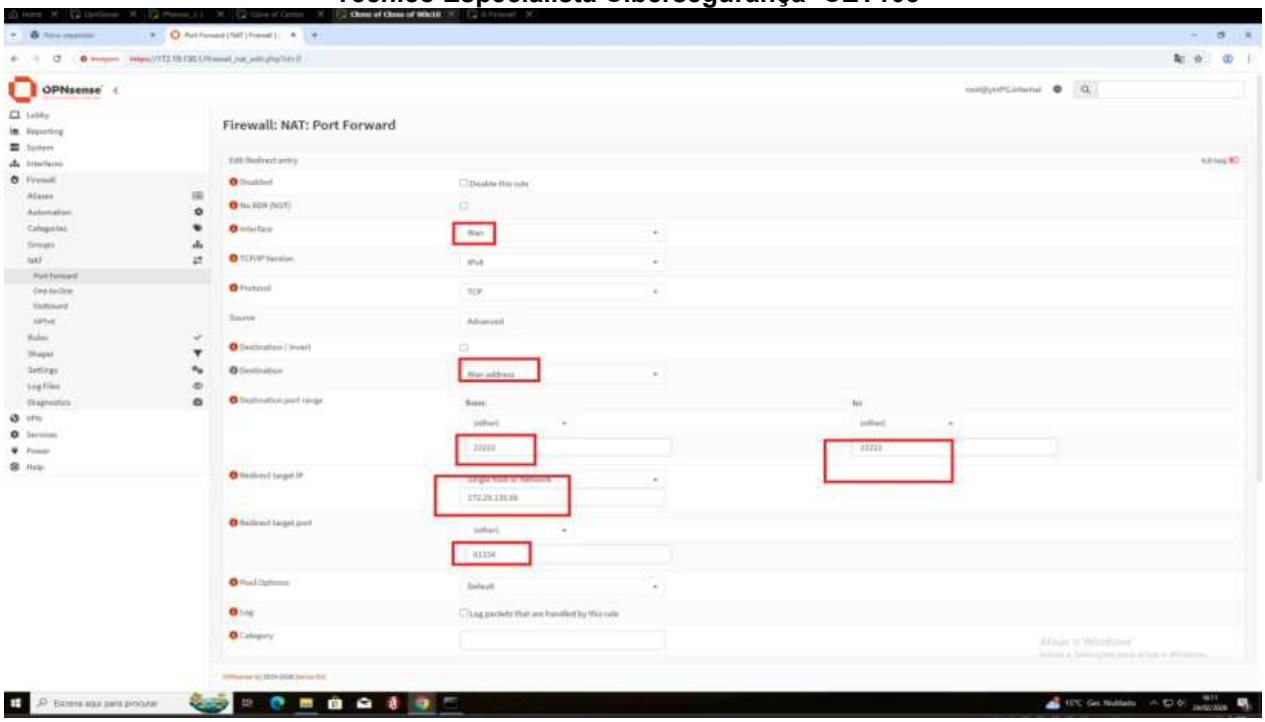


Criar a interface NAT

15.1 Firewall → Rules → WAN

Regra para 22222

- **Interface:** WAN
- **Protocol:** TCP
- **Destination:** WAN address
- **Destination Port:** 22222
- **Redirect target IP:** 172.29.130.66
- **Redirect target port:** 61234



15.2 Regra para 42424

-Interface: WAN

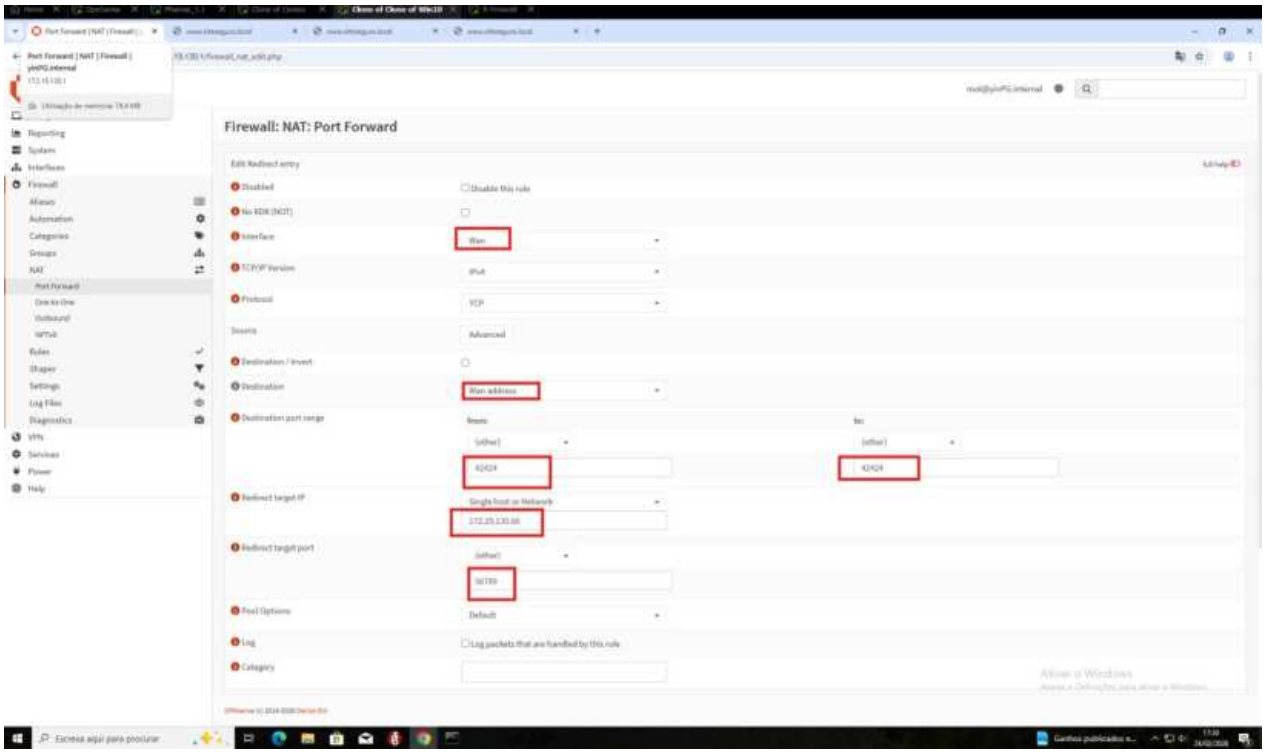
- **Protocol:** TCP

- Destination: WAN address

- Destination Port: 42424

- Redirect target IP: 172.29.130.66

- Redirect target port: 56789

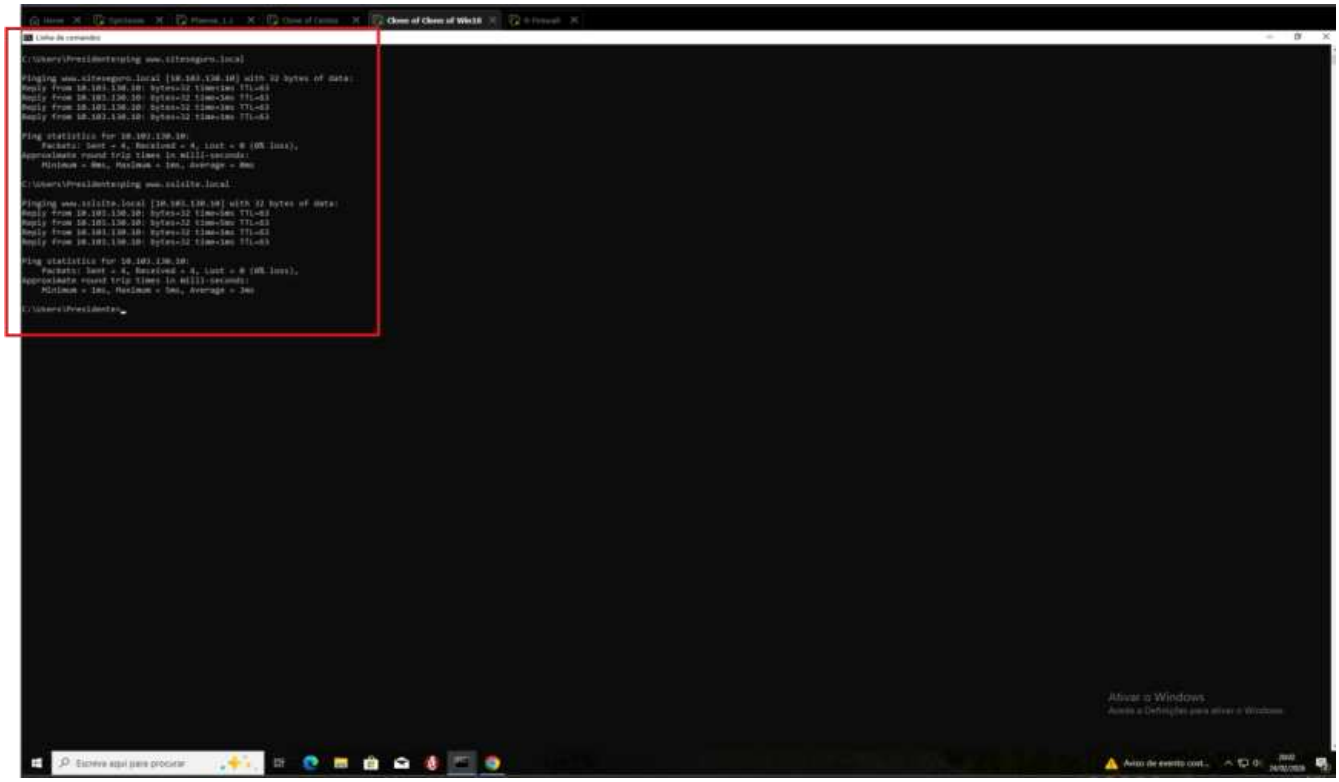


Firewall → Rules → WAN

15.3 Confirmar que o hosts está a ser usado

ping www.siteseguro.local

ping www.sslsite.local

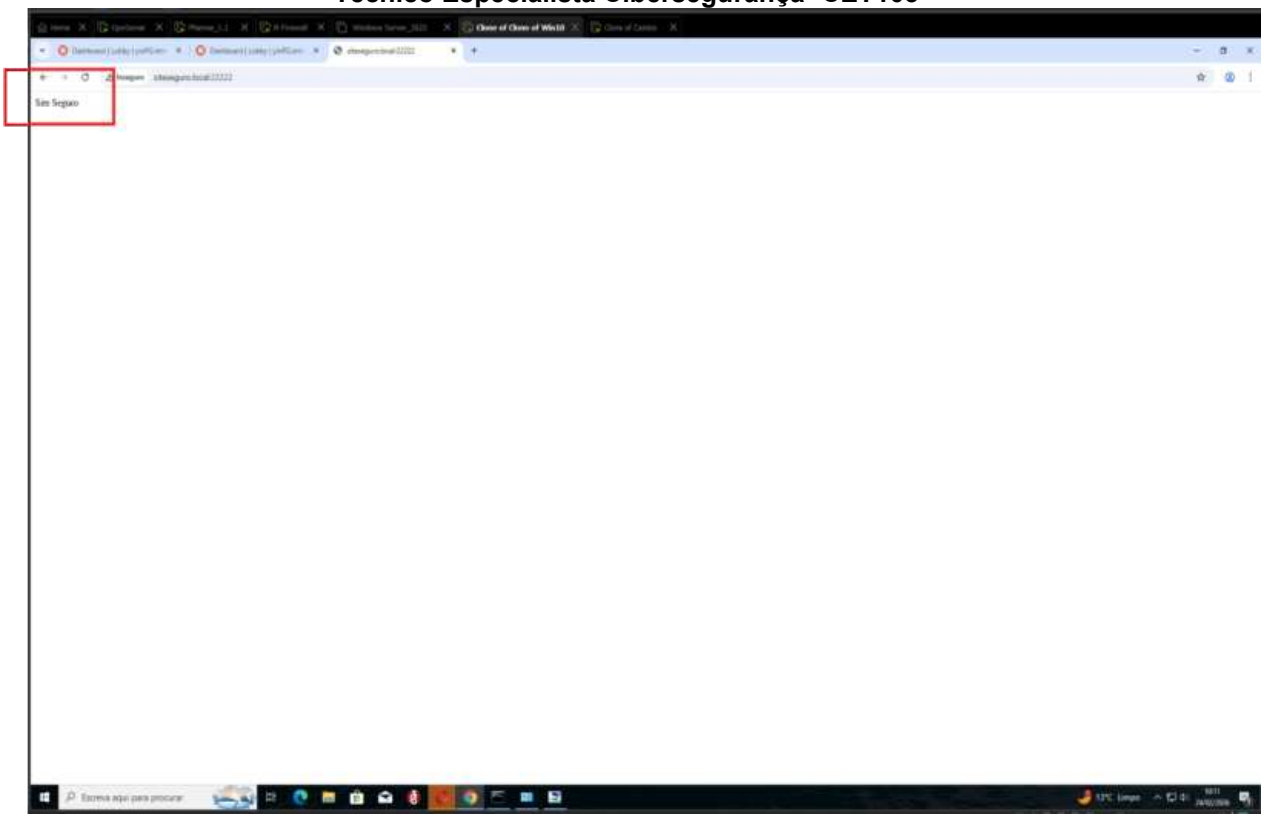


15.4 Teste sem VPN (Windows 10 na WAN)

Teste 1 — HTTP com autenticação

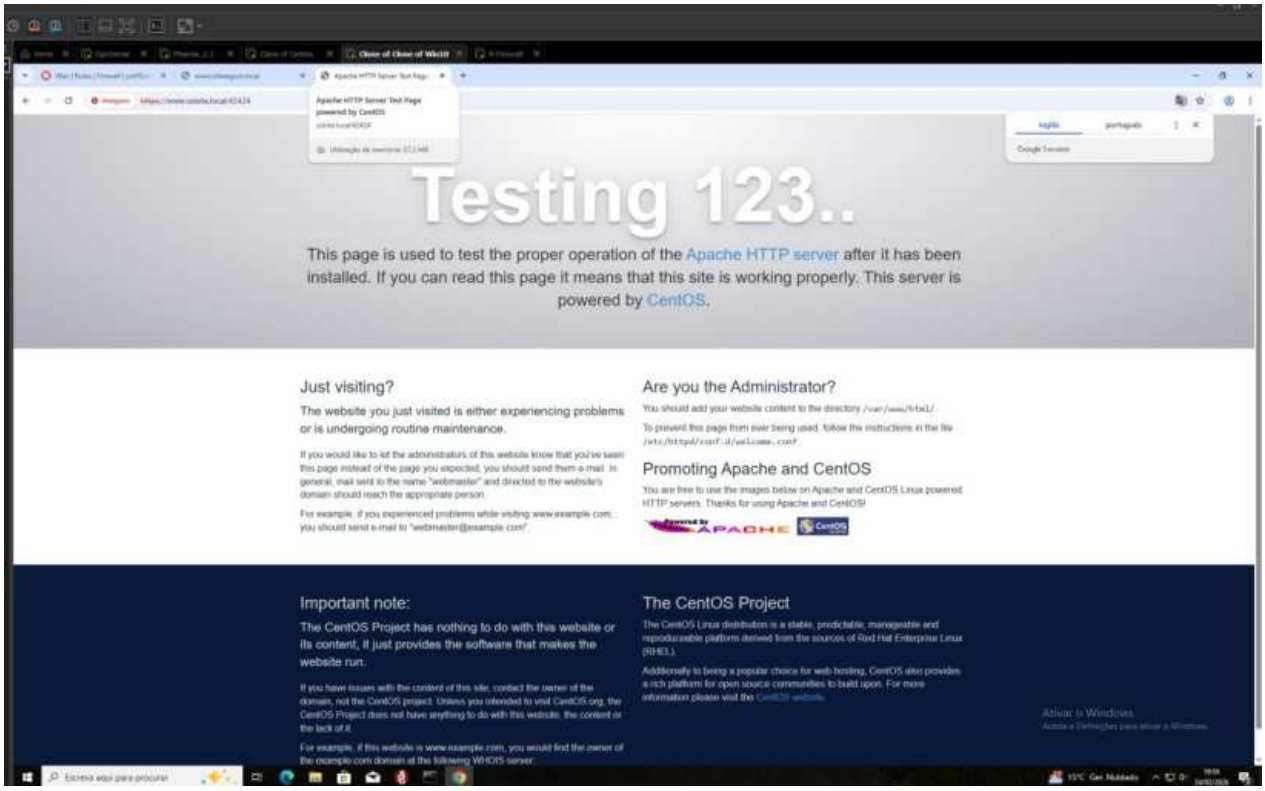
No browser (Chrome/Firefox):

- **Abrir: http://www.siteseguro.local:22222**
- **user: user1**
- pass: **Passw0rd**



15.5 No browser (Chrome/Firefox):

- **Abrir:** https://www.sslsite.local:42424
- **user: user1**
- pass: **Passw0rd**



3-VPNS----------

Enunciado:

“ 3) VPNs a) Crie uma ligação VPN IPSEC entre os firewall yinNA e yangNA Demonstre que tem acesso, com o cliente na zona Green do yangNA ao servidor Windows na zona green do outro ponto da VPN (ping é insuficiente !!!). Parâmetros da VPN (encriptação, algortimos, comprimentos de chaves etc, ao seu critério)”

16 VPN 3.a — IPsec Site-to-Site entre yinPG (OPNsense) e yangPG (pfSense)

16.1Definições

- Endpoint yinPG (WAN): 10.103.130.10

- Endpoint yangPG (WAN): 10.103.130.11

- Rede local yin: 172.19.130.0/23

- Rede local yang: 10.120.130.0/24

- PSK: Passw0rd

Apresentação Sumarizada

16.2 OPNsense yinPG — IPsec Pre-Shared Key

VPN → IPsec → Pre-Shared Keys → +Add

- Identity: MEUenderecoIP

  PSK: Passw0rd

16.3 Connection (Phase 1 equivalente)

VPN → IPsec → Connections → +Add

- Remote gateway: 10.103.130.11

- Authentication:

PSK

- Local: 10.103.130.10

- IKEv2

Local network: 172.19.130.0/23

16.4 Regras(resumo) na WAN (IKE/ESP)

Firewall → Rules → WAN

- Allow UDP 500 (ISAKMP/IKE)

- Allow UDP 4500 (NAT-T)

- Allow ESP (Protocol ESP)

16.5 Regras (Resumo)no IPsec

Firewall → Rules → IPsec

16.6 OPNsense yinPG — IPsec (Site-to-Site) com pfSense yangPG

Objetivo :

- **Local (yinPG)**: 172.19.130.0/23 (YIN_GREEN_NET)
- **Remoto (yangPG)**: 10.120.130.0/24 (YANG_GREEN_NET)
- **Endpoint yinPG (WAN)**: 10.103.130.10
- **Endpoint yangPG (WAN)**: 10.103.130.11
- **Autenticação**: PSK = **Passw0rd**

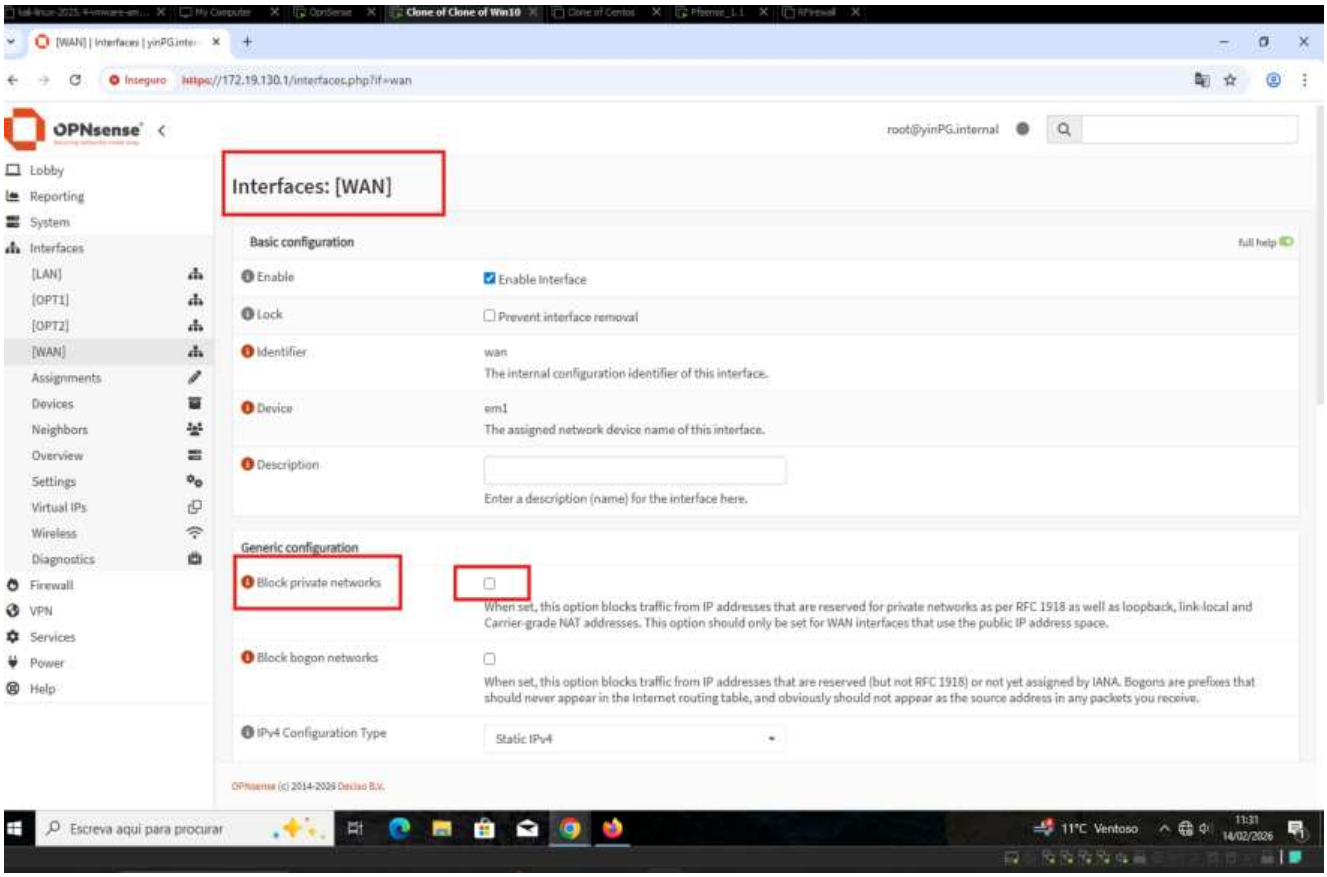
16.7 Pré-check Nota :muito importante antes de criar a VPN

A) Confirmar que a WAN é rede privada e não está a ser "bloqueada"

Como a rede WAN , garantir que o OPNsense **não está a bloquear redes privadas na WAN**.

1. Ir a **Interfaces → WAN**
2. **DESATIVAR** (se estiver ativo):

   Block private networks
3. **Save** e **Apply**



Nota : Se isto estiver ativo, o yinPG pode bloquear logo à entrada o tráfego do yangPG (10.103.130.11) e a VPN .

B) Confirmação de IPs

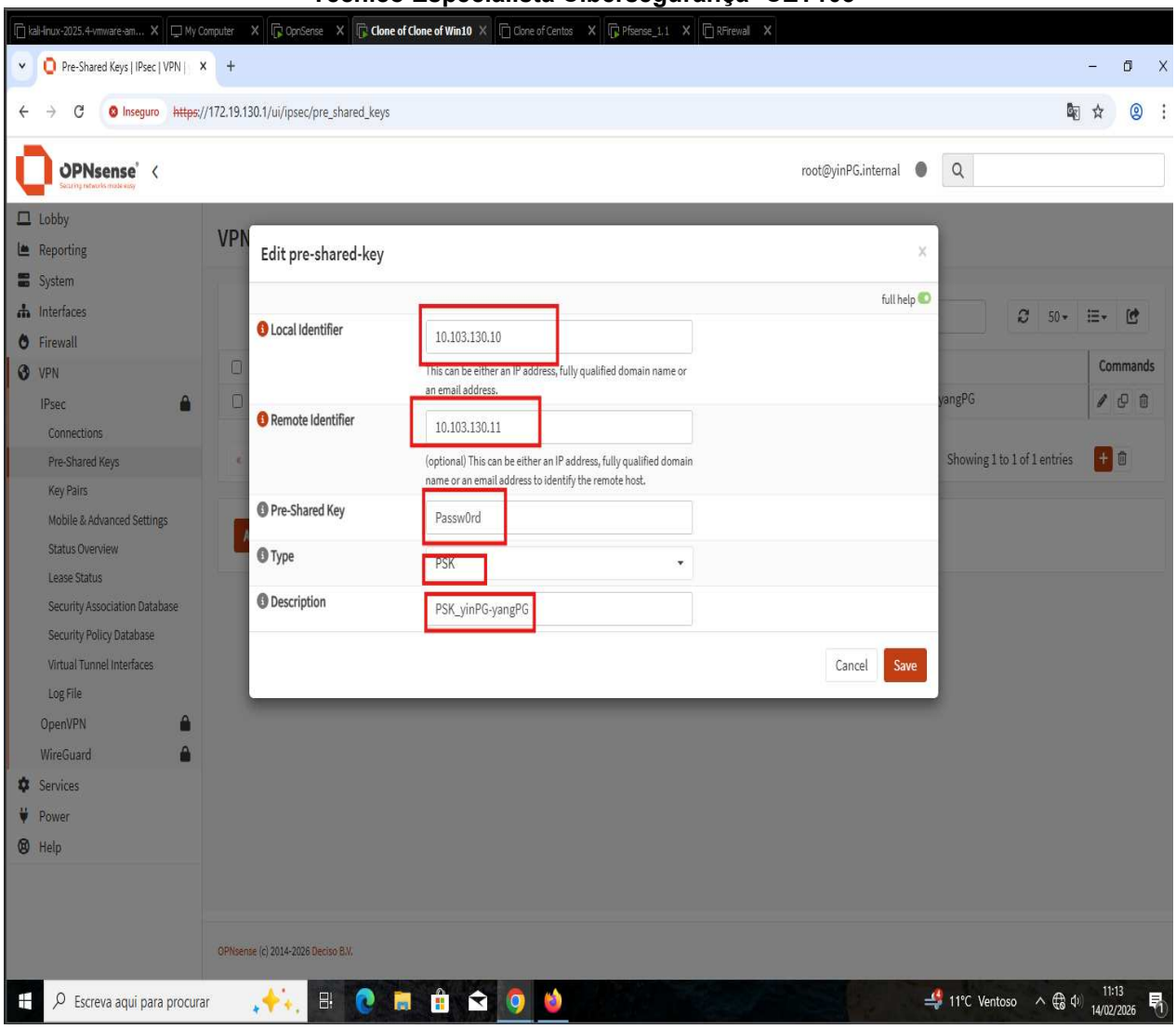
Resumos:

- yinPG WAN = 10.103.130.10/27 GW 10.103.130.1
- yangPG WAN = 10.103.130.11/27 GW 10.103.130.1

16.8- Criar a Pre-Shared Key (PSK)

VPN → IPsec → Pre-Shared Keys → +Add

- Local Identifier -10.103.130.10
- Remote Identifier – 10.103.130.11
- Secret: Passw0rd
- Description: PSK_yinPG-yangPG



16.9 Criar a "Connection" (equivalente à Phase 1)

VPN → IPsec → Connections → +Add

Secção: Connection (Phase1)

A) Geral

- Enabled:

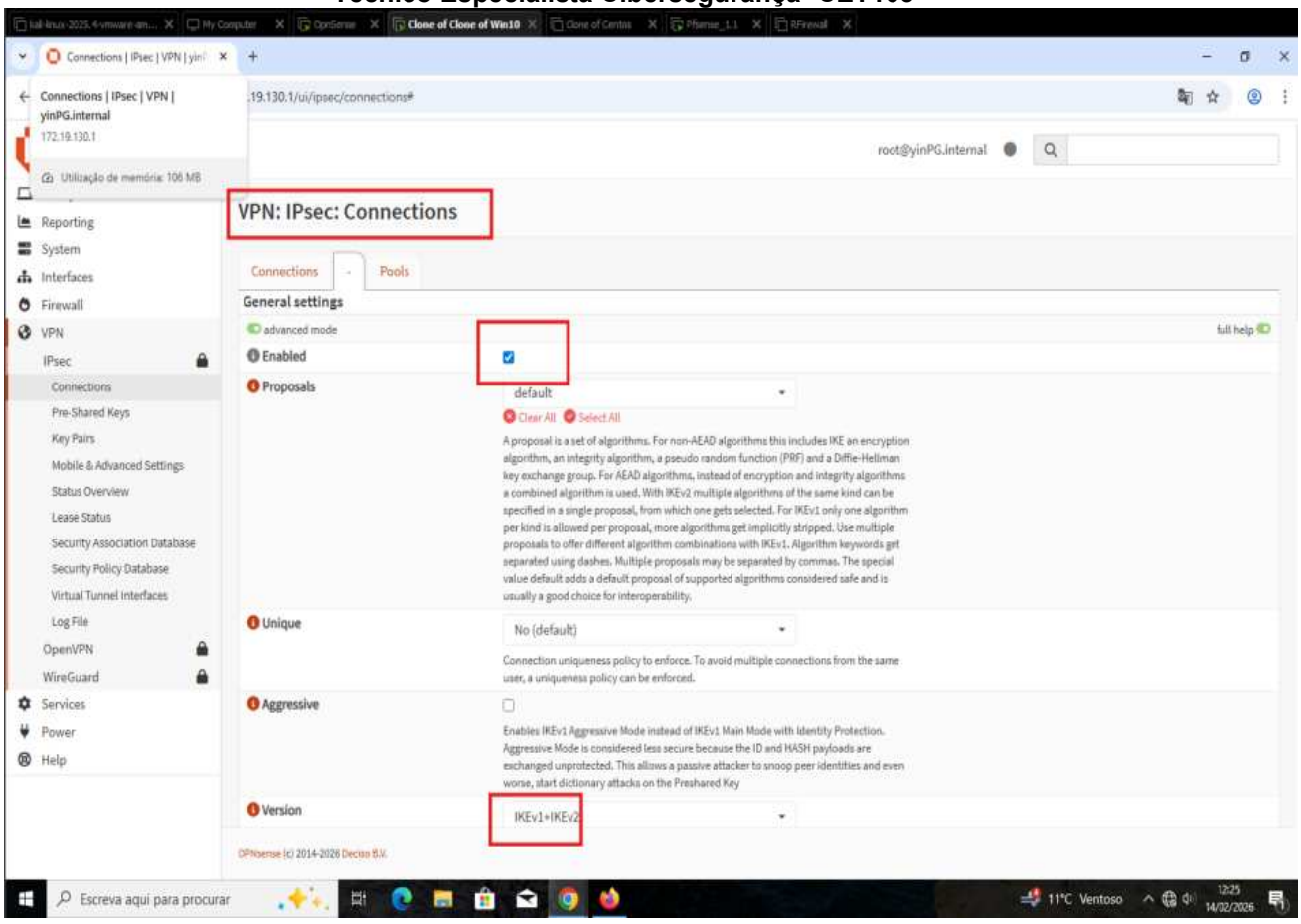
- Proposals: aes256-sha256-ecp521 [DH21, NIST EC]

- Key Exchange version: IKEV2

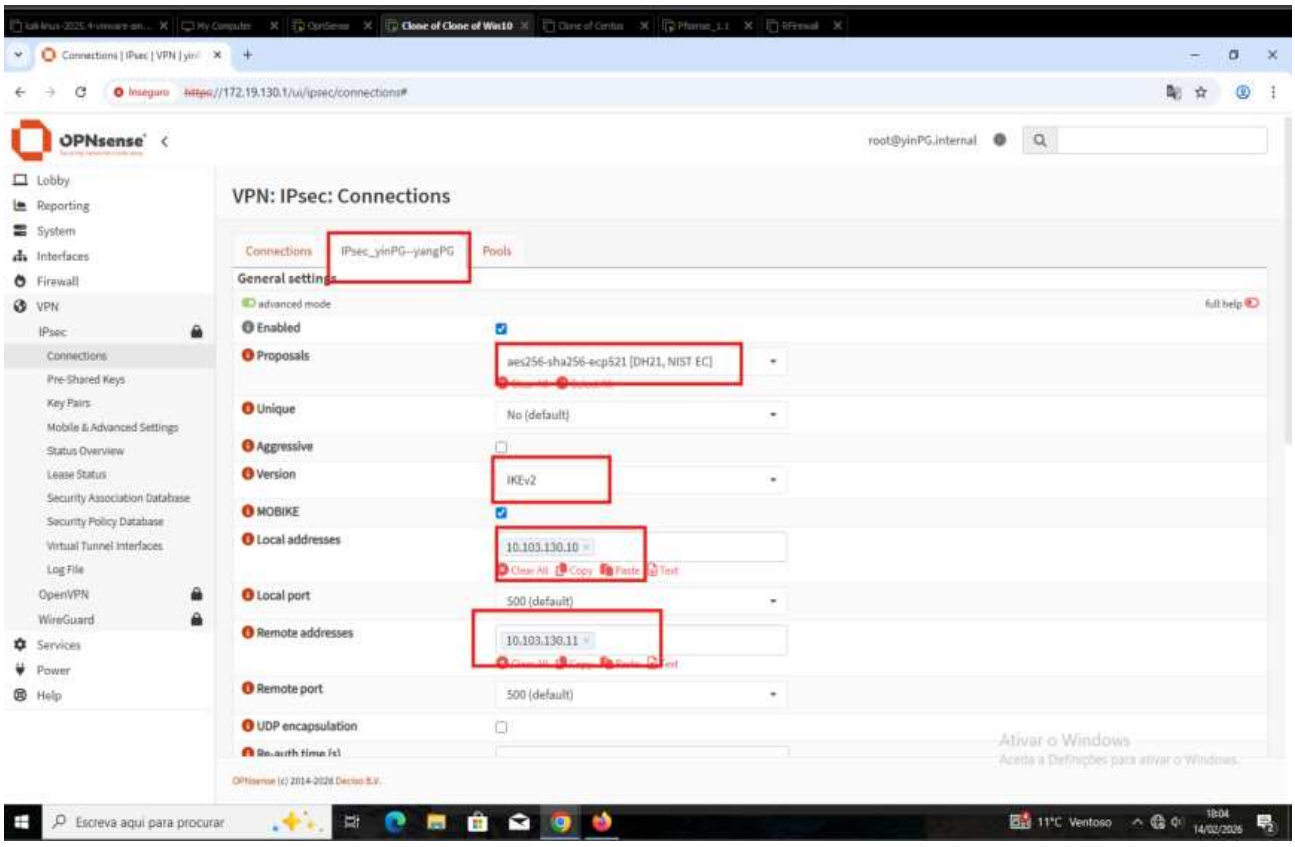
- Local address 10.103.130.10

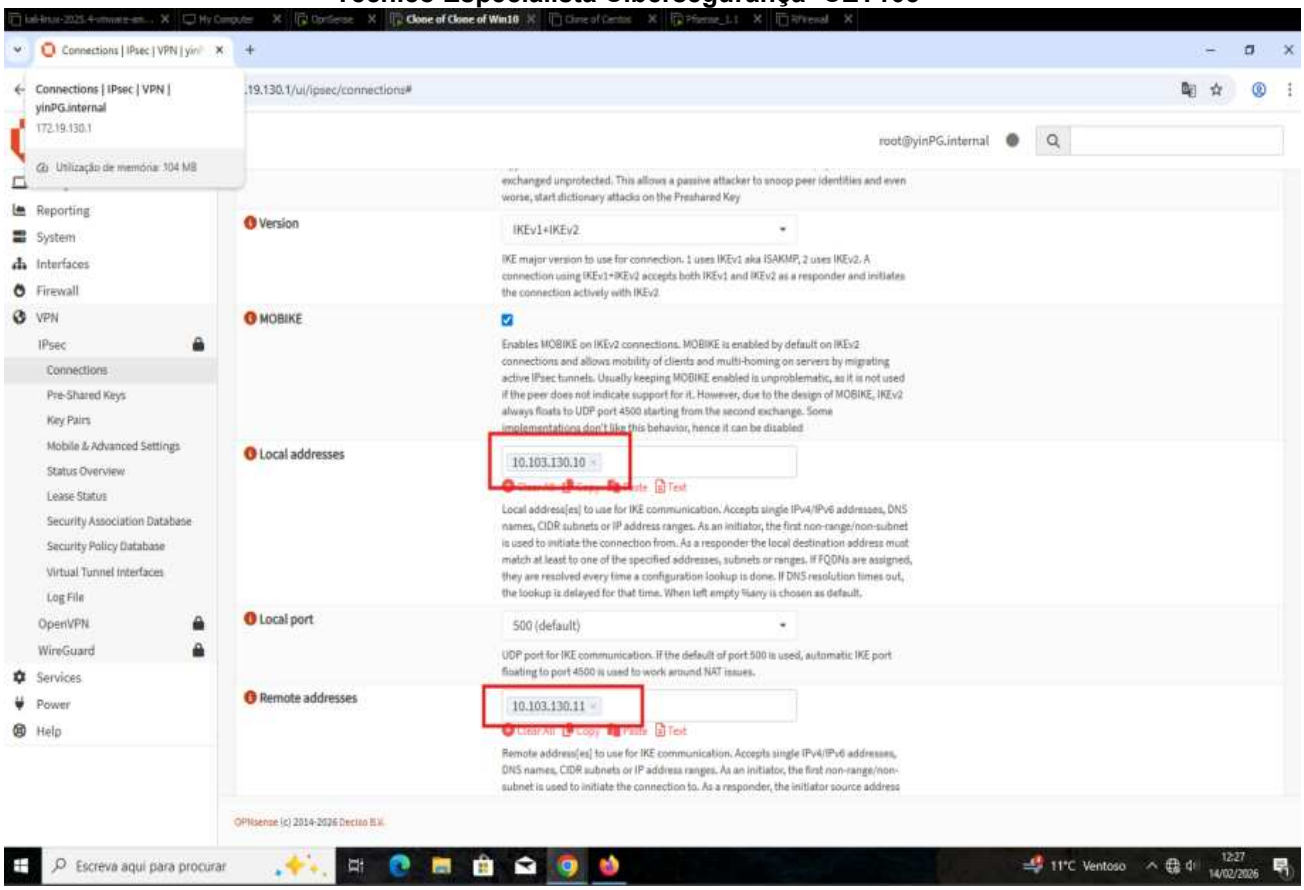
- Remote addresses 10.103.130.11

- Description: IPsec_yinPG—yangPG

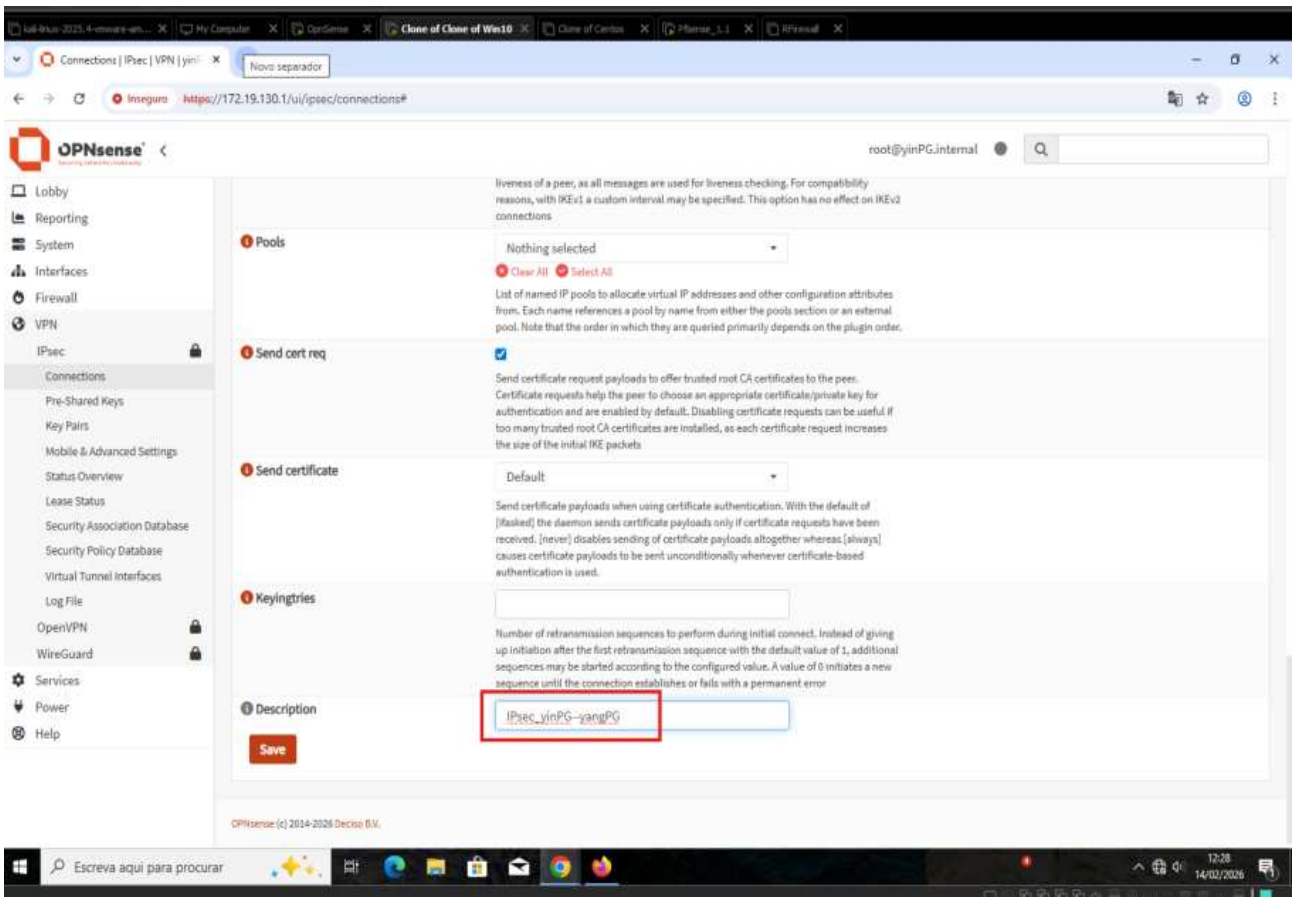


Continuação..





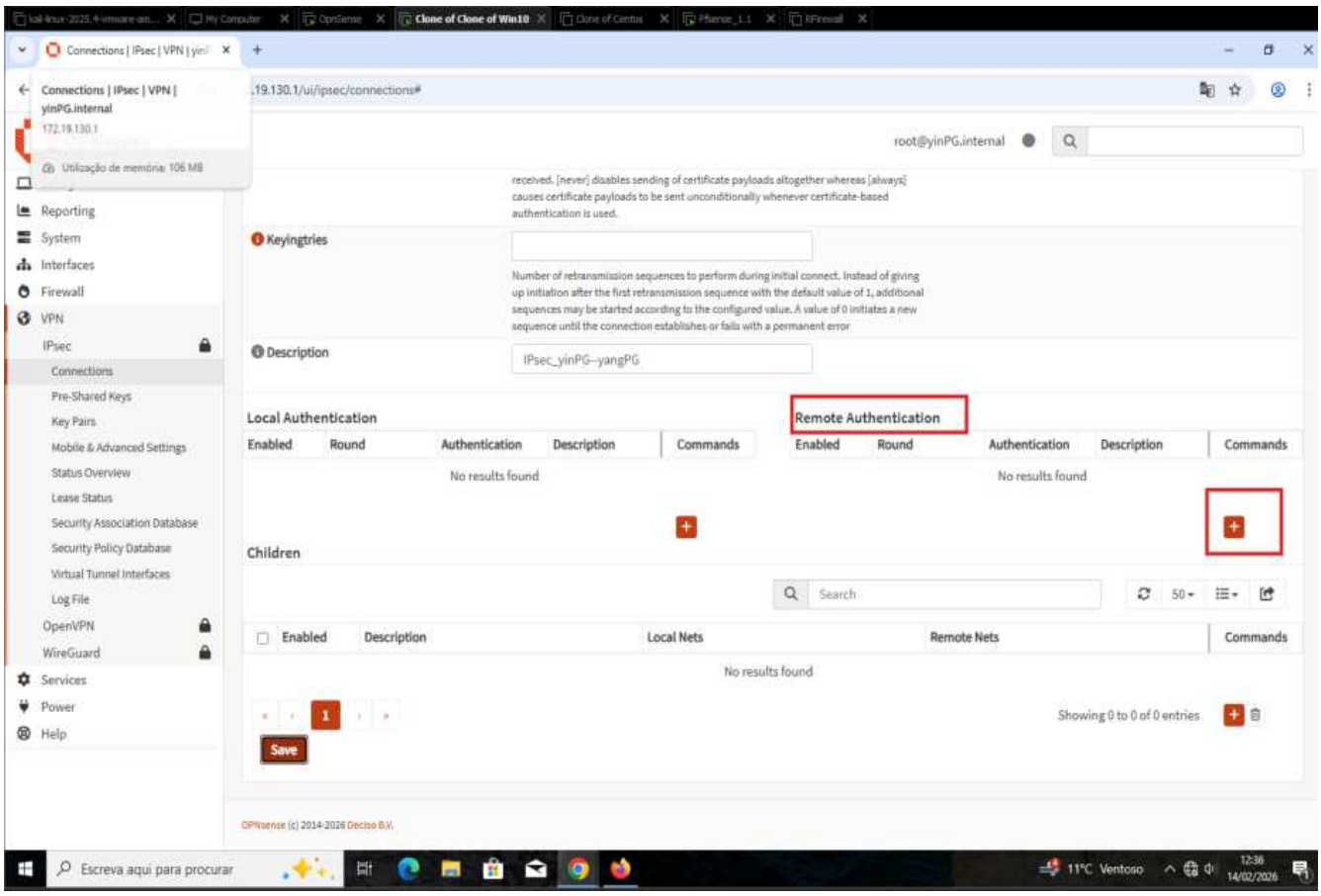
Continuação…



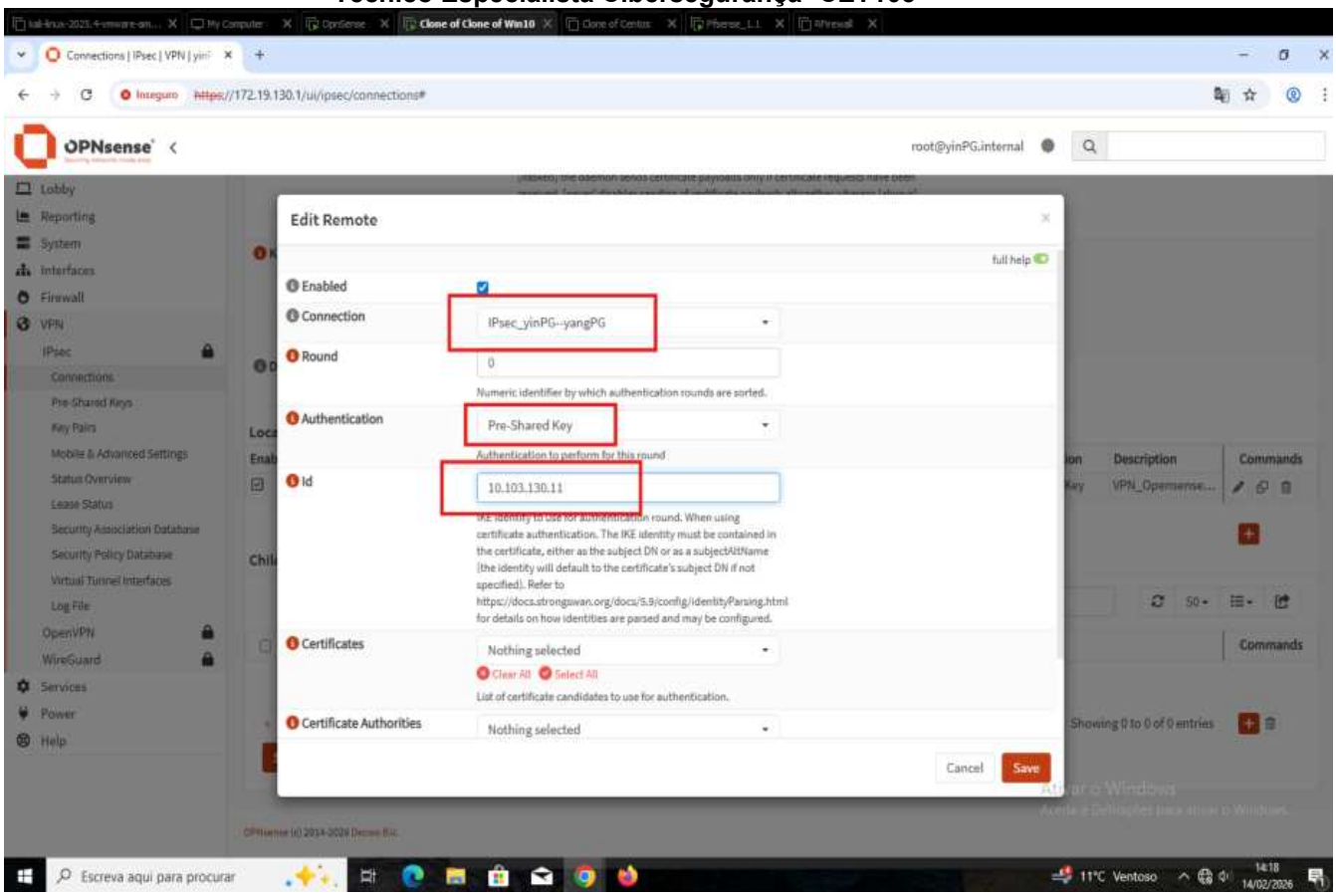
16.10 -Edit -Remote Authentication

Edit Remote – Configuração

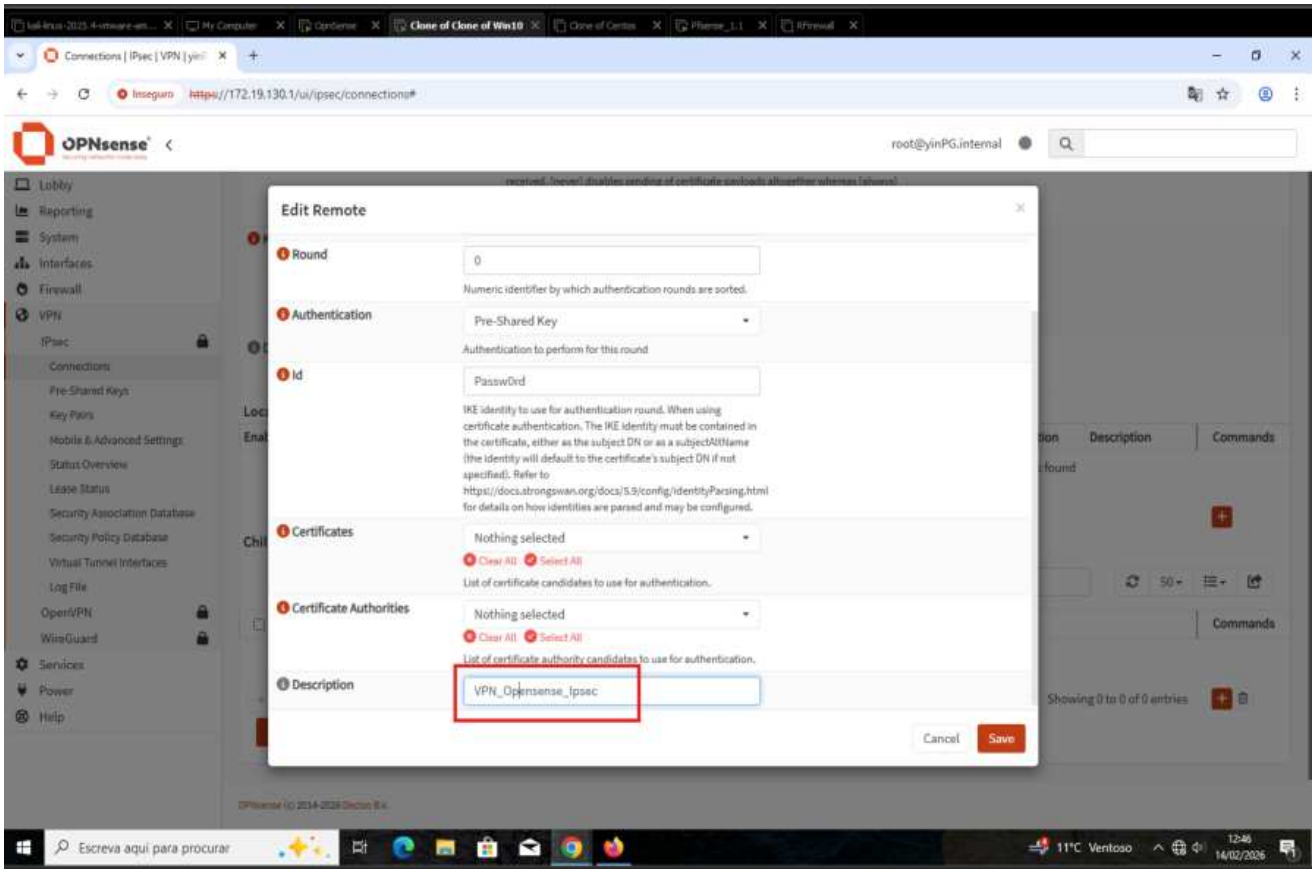
- **Enabled:** Ativado
- **Connection:** IPsec_yinPG--yangPG
- **Round: 0**
- **Authentication:** Pre-Shared Key
- **Id: 10.103.130.11**
- **Certificates:** Nothing selected
- **Certificate** Authorities: Nothing selected
- **Description:** VPN_Opnsense_Ipsec_Remote



Continuação…



Continuação …



16.11- Edit - Local Authentication

- Enabled: ativo

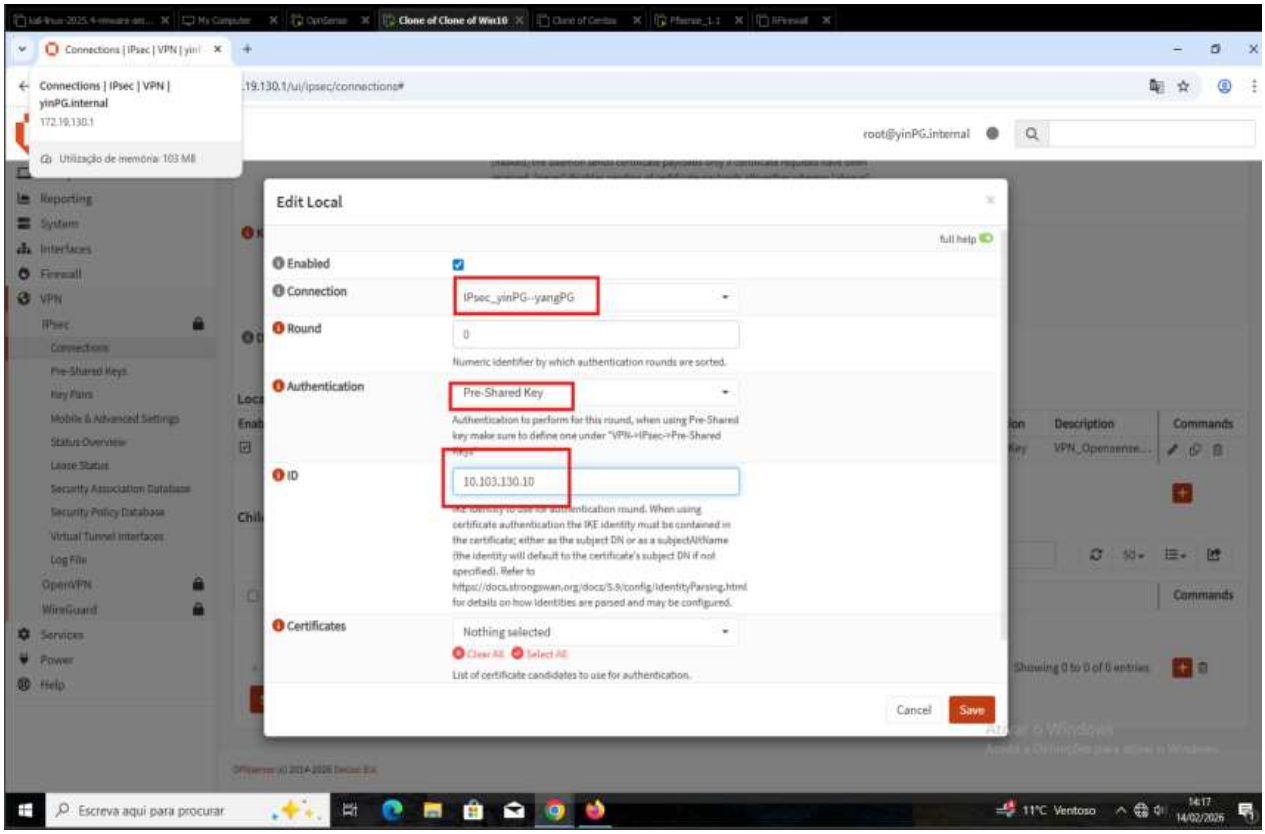
- Connection: IPsec_yinPG--yangPG

- Round: 0

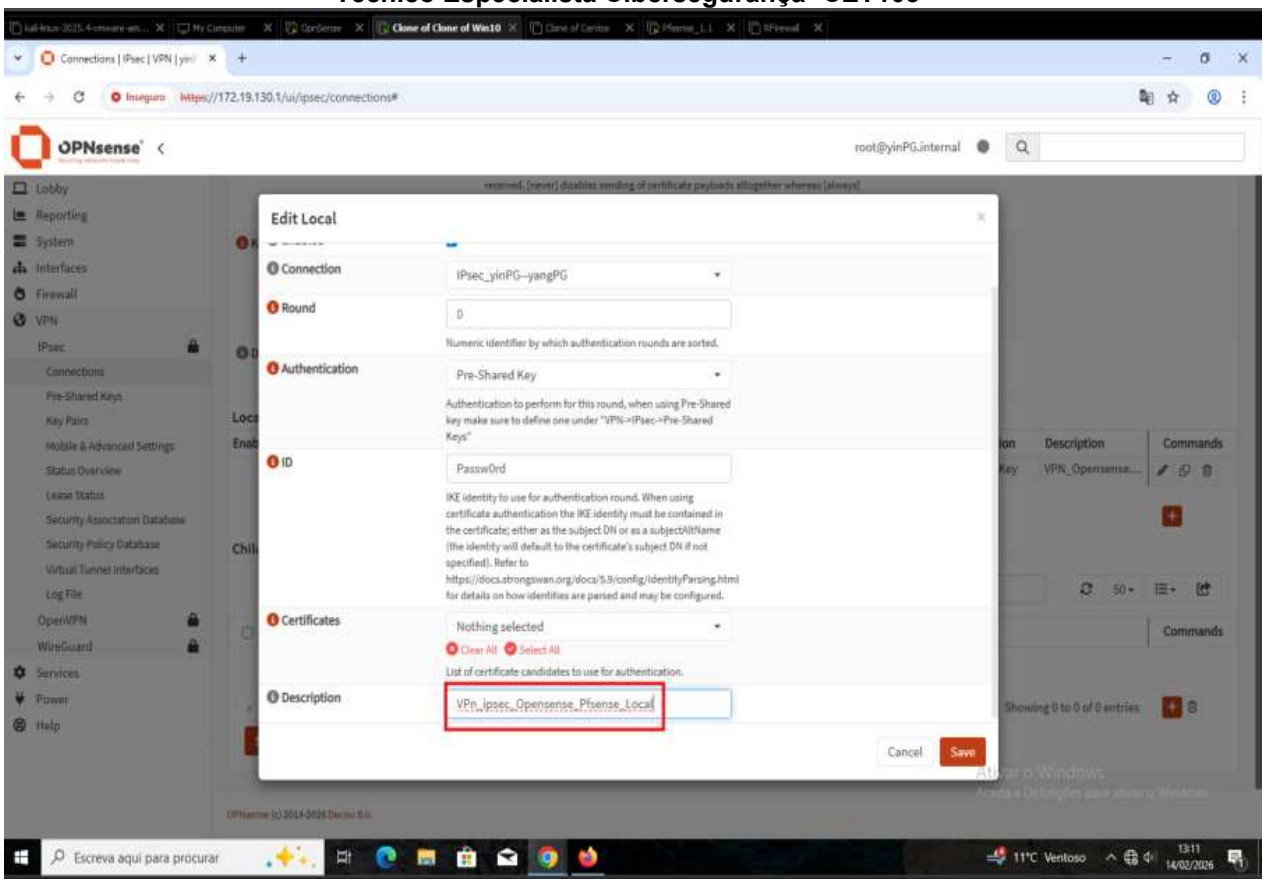
- Authentication: Pre-Shared Key

- ID: 10.103.130.10

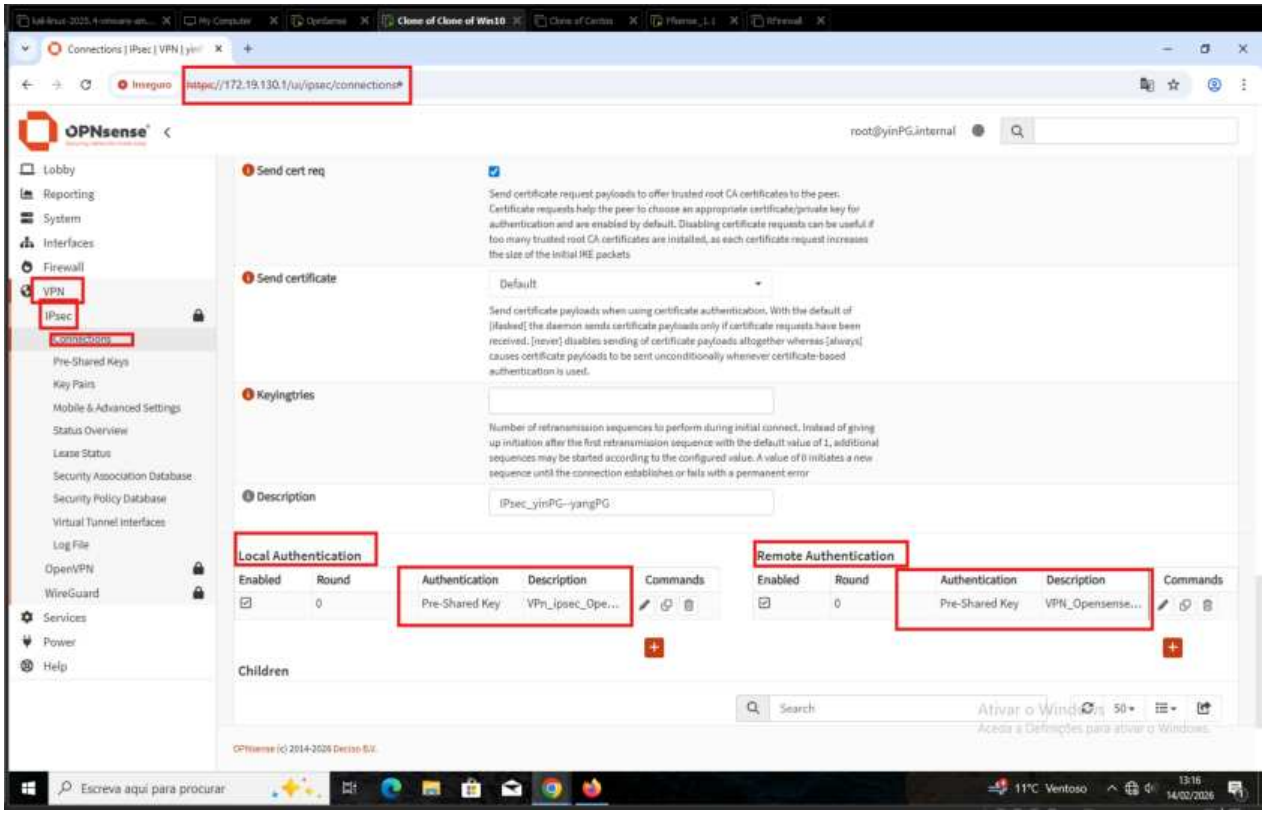
- Connection: IPsec_yinPG—yangPG



Continuação…


Edit Local
Connection
IPsec_yinPG--yangPG
Round
0
Numeric identifier by which authentication rounds are sorted.
Authentication
Pre-Shared Key
Authentication to perform for this round, when using Pre-Shared key make sure to define one under "VPN->IPsec->Pre-Shared Keys"
ID
PassW0rd
IKE identity to use for authentication round. When using certificate authentication the IKE identity must be contained in the certificate; either as the subject DN or as a subjectAltName (the identity will default to the certificate's subject DN if not specified). Refer to https://docs.strongswan.org/docs/5.9/config/identityParsing.html for details on how identities are parsed and may be configured.
Certificates
Nothing selected
Clear All
Select All
List of certificate candidates to use for authentication.
Description
VPn_ipsec_Opensense_Pfsense_Local
Cancel
Save

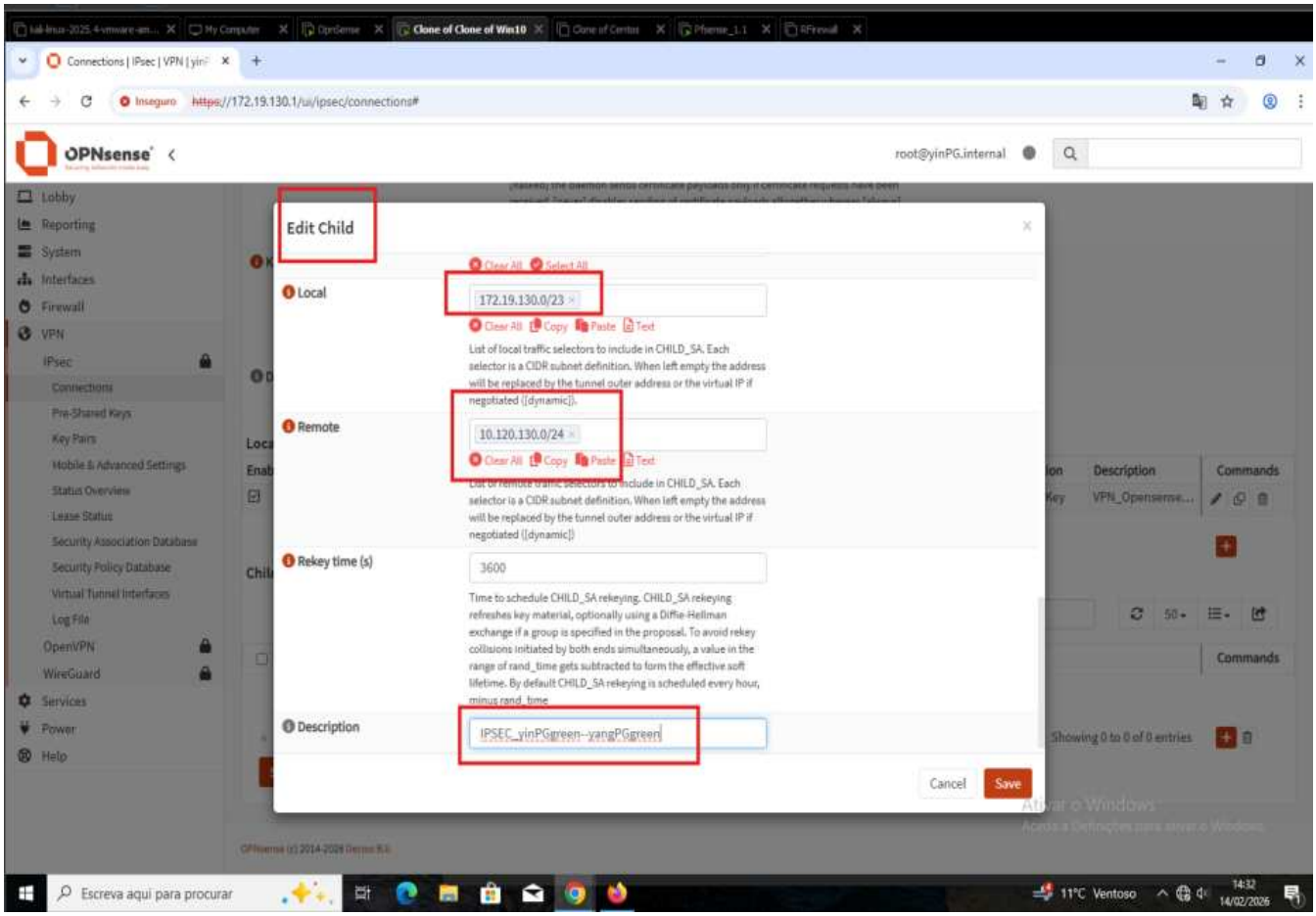
Resumo :


Send cert req
Send certificate request payloads to offer trusted root CA certificates to the peer. Certificate requests help the peer to choose an appropriate certificate/private key for authentication and are enabled by default. Disabling certificate requests can be useful if too many trusted root CA certificates are installed, as each certificate request increases the size of the initial IKE packets
Send certificate
Default
Send certificate payloads when using certificate authentication. With the default of [Hashed] the daemon sends certificate payloads only if certificate requests have been received, [never] disables sending of certificate payloads altogether whereas [always] causes certificate payloads to be sent unconditionally whenever certificate-based authentication is used.
Keyingtries
Number of retransmission sequences to perform during initial connect. Instead of giving up initiation after the first retransmission sequence with the default value of 1, additional sequences may be started according to the configured value. A value of 0 initiates a new sequence until the connection establishes or fails with a permanent error
Description
IPsec_yinPG--yangPG
Local Authentication
Enabled
Round
Authentication
Description
Commands
Pre-Shared Key
VPn_ipsec_Ope...
Remote Authentication
Pre-Shared Key
VPN_Opensense...
Children
Search

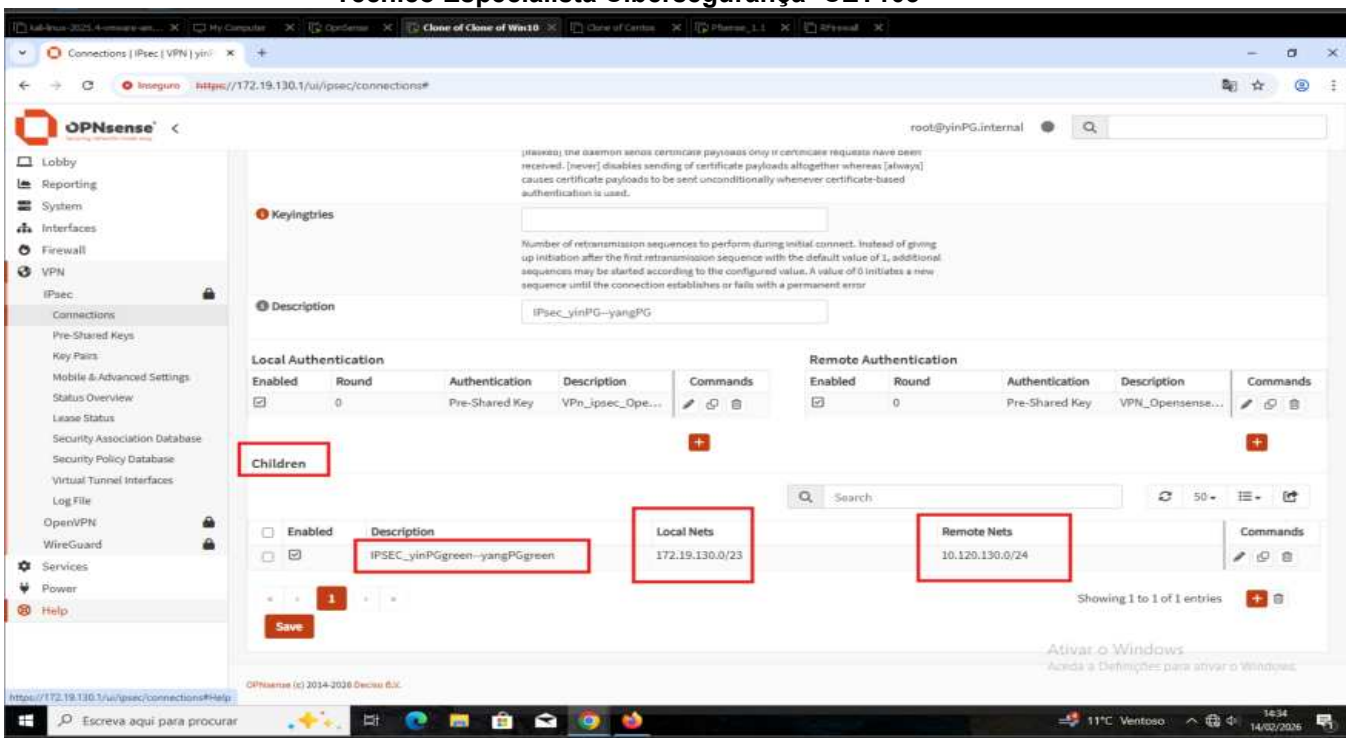
16.12 Adicionar o “Child” (equivalente à Phase 2)

Children → +Add

- **Enabled**:
- **Connection:** IPsec_yinPG—yangPG
- **Use sha256_96 :** ativo
- **Mode**: Tunnel
- **Policies**: ativo
- **Start action**: Start
- **ESP proposals**: default
- **Local**: 172.19.130.0/23
- **Remote**: 10.120.130.0/24
- **Description**: IPSEC_yinPGgreen--yangPGgreen



Resumo :

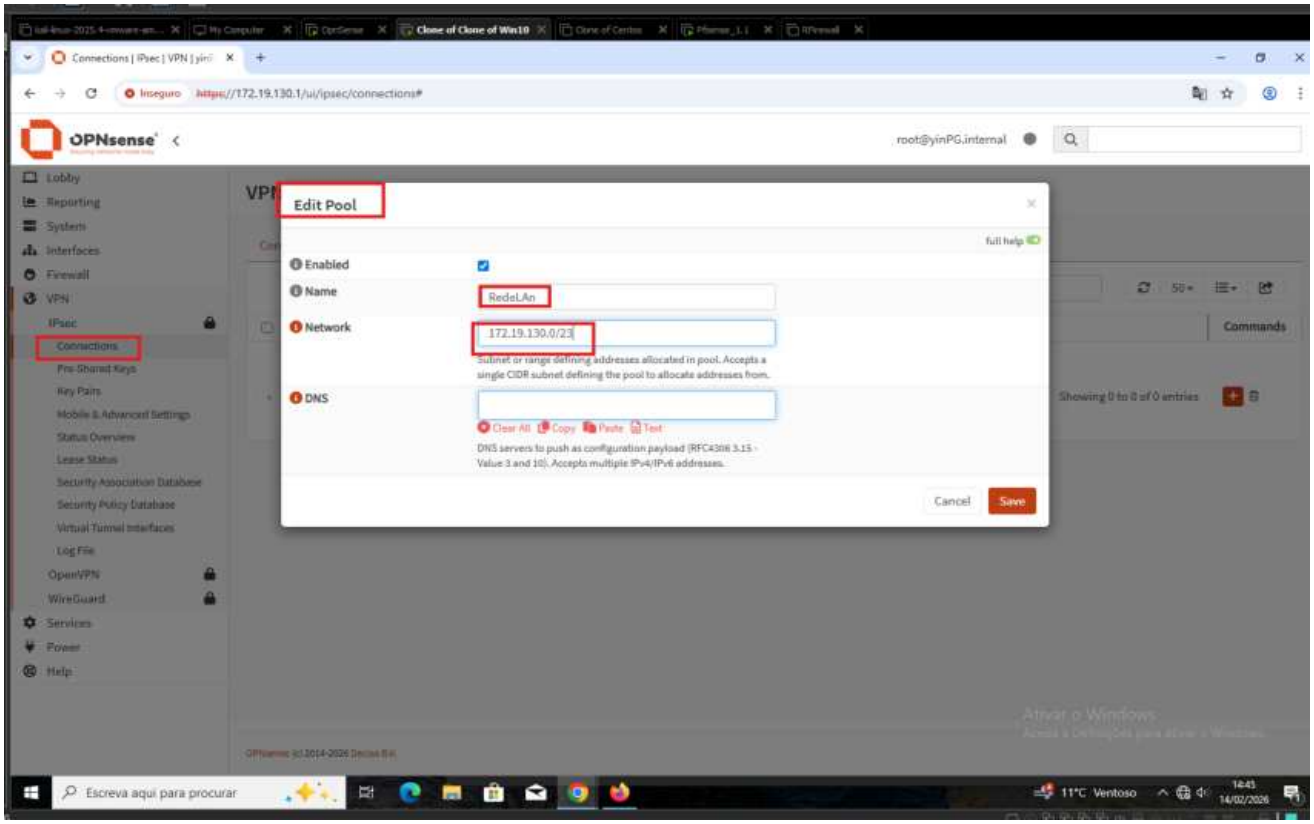


16.13- IPsec → Connections → Pools → Edit Pool

- Enabled:

- Name: RedeLAN

- Network: 172.19.130.0/23

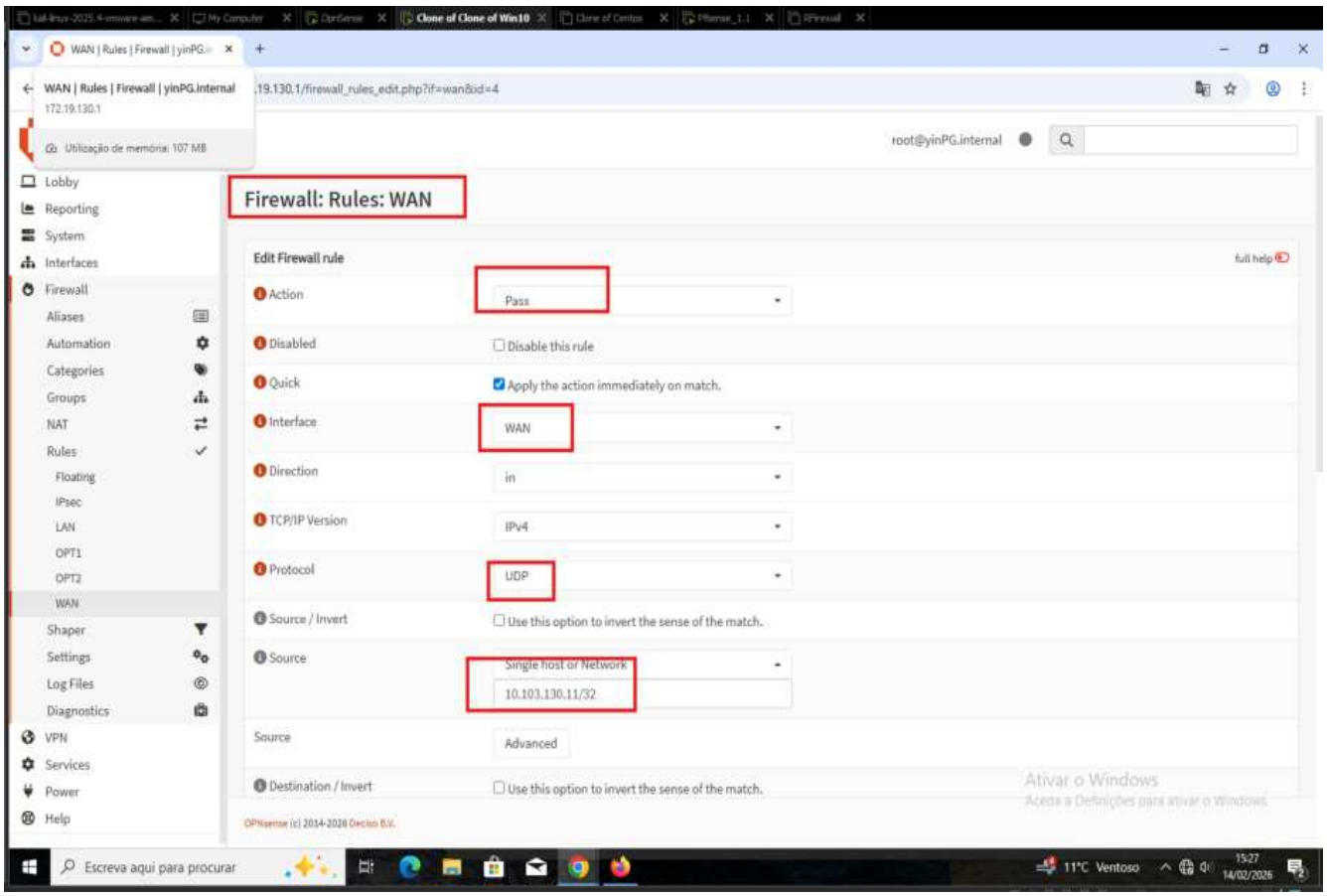


16.14 Regras na WAN do yinPG (IKE/ESP)

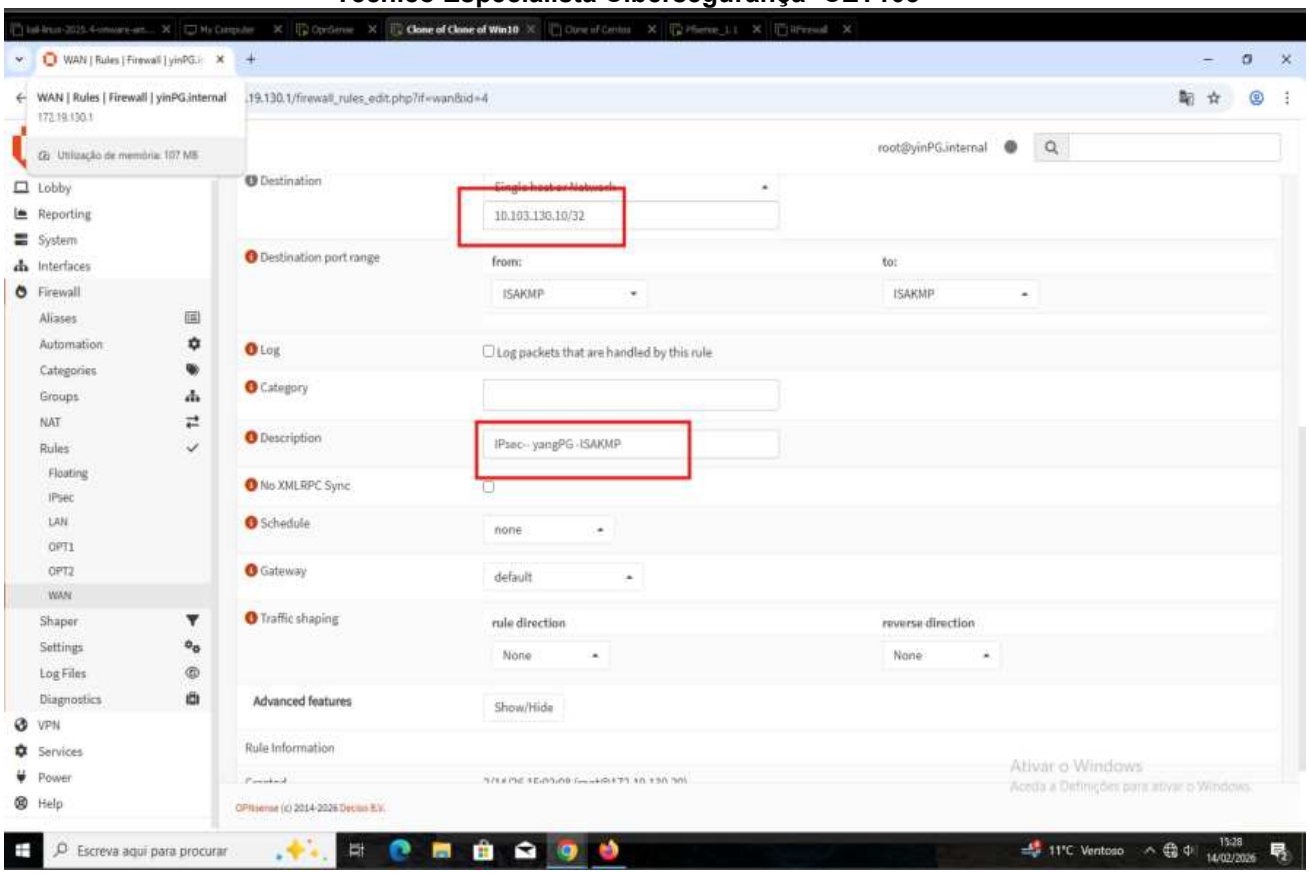
Firewall → Rules → WAN → +Add

Regra WAN #1 — IKE (UDP 500)

- **Action**: Pass
- **Interface**: WAN
- **Direction**: in
- **Protocol**: UDP
- **Source**: 10.103.130.11 (yangPG WAN)
- **Destination**: WAN address (10.103.130.10)
- **Destination port**: **500- ISAKMP**
- **Description**: IPsec-- yangPG -ISAKMP

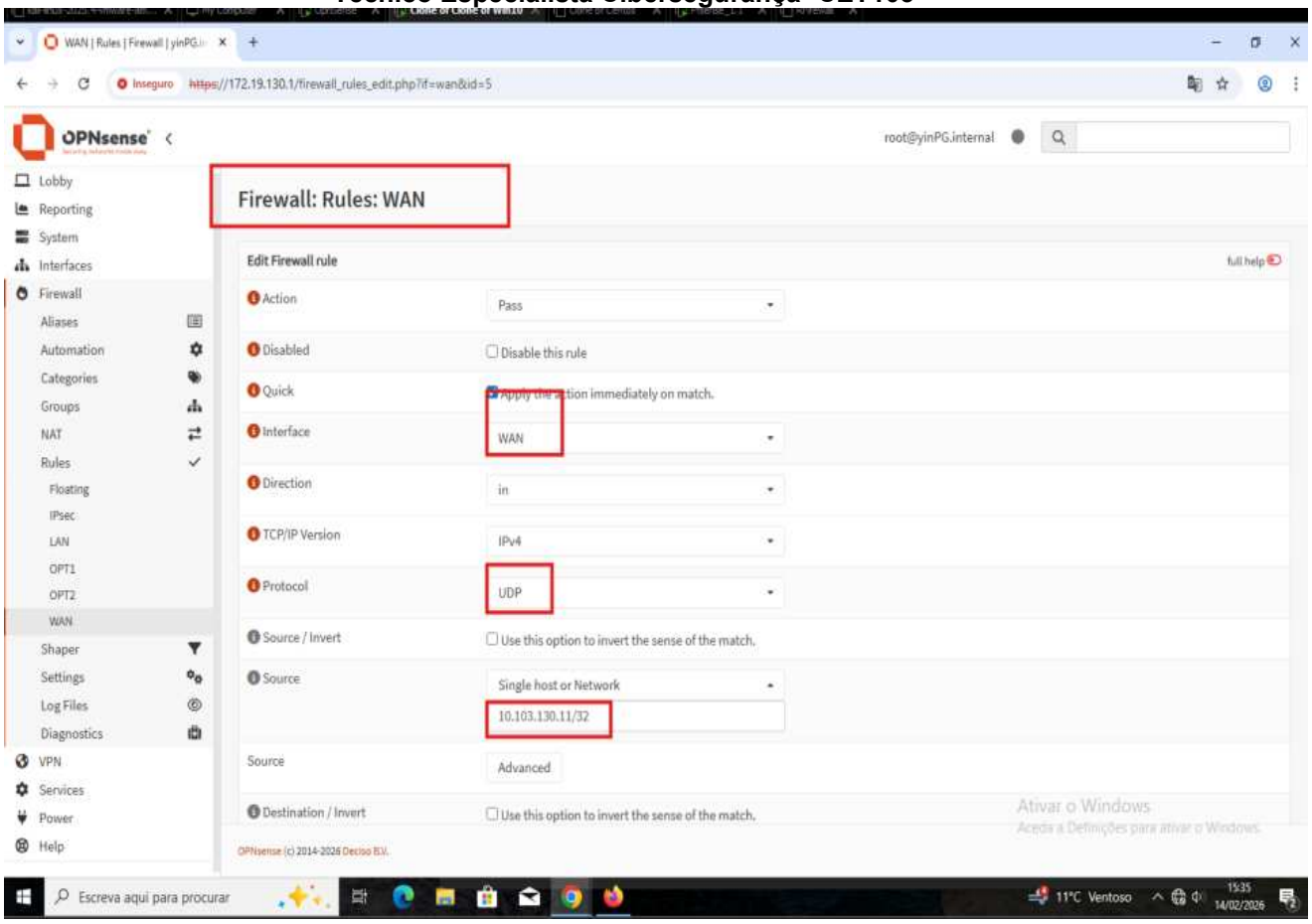


Continuação …

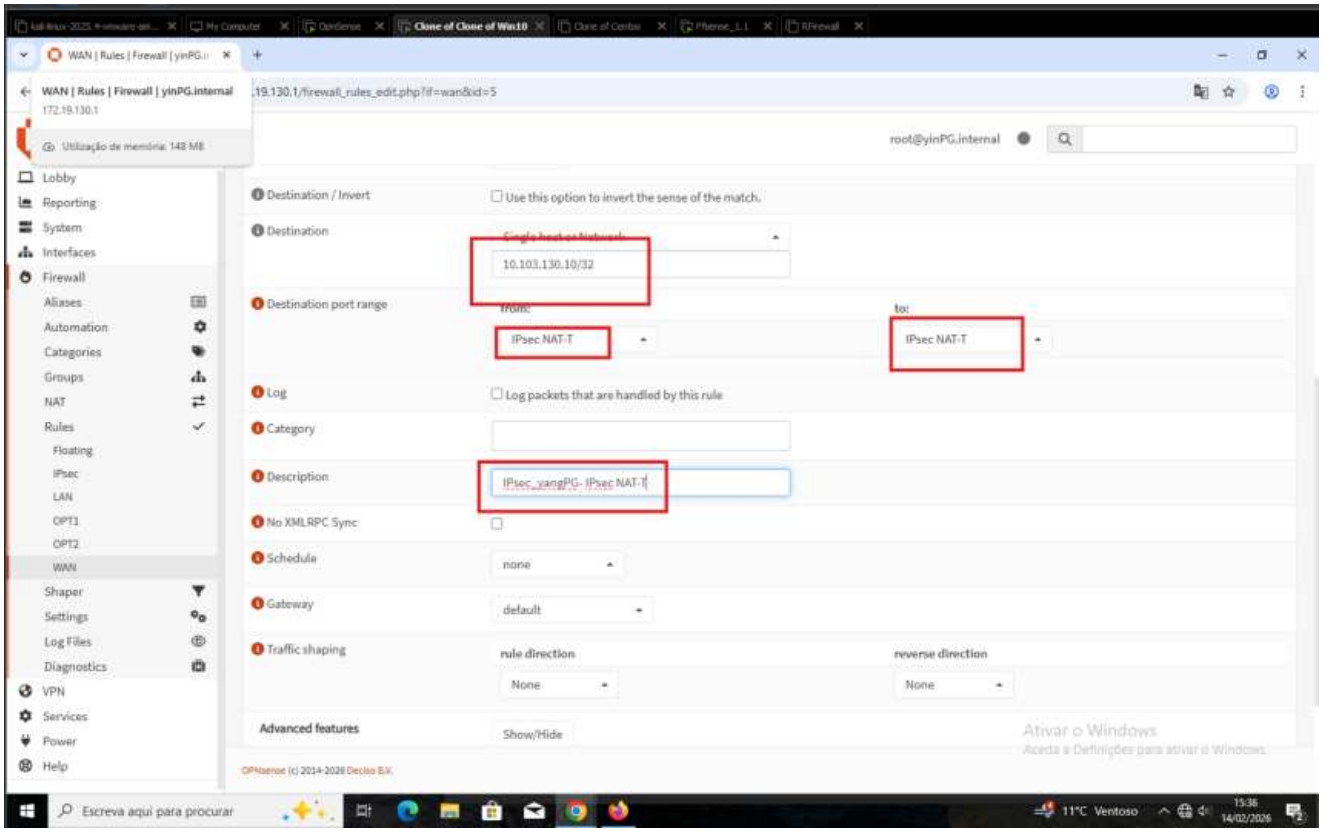


16.15 Regra WAN #2 — NAT-T (UDP 4500)

- **Action**: Pass
- **Interface**: WAN
- **Direction**: in
- **Protocol**: UDP
- **Source**: 10.103.130.11 (yangPG WAN)
- **Destination**: WAN address (10.103.130.10)
- **Destination port**: **4500- IPsec NAT-T**
- **Description**: IPsec_yangPG- **IPsec NAT-T**

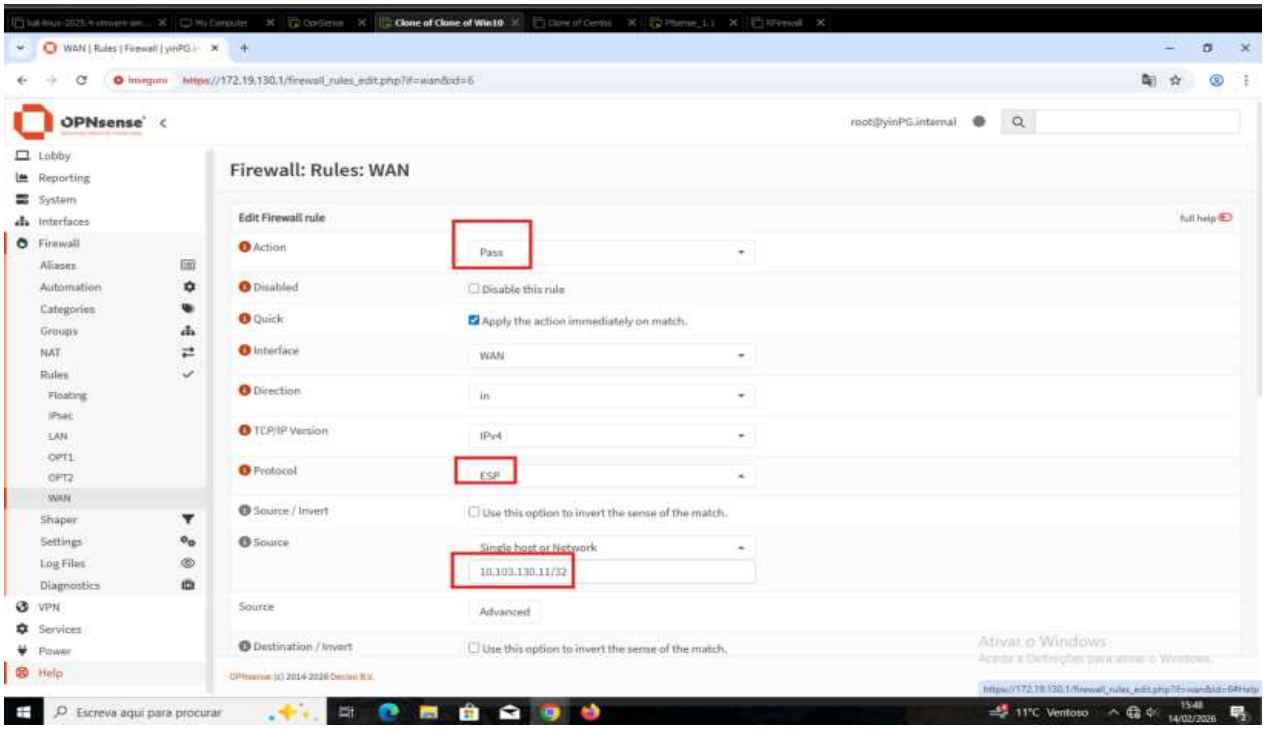


Continuação…

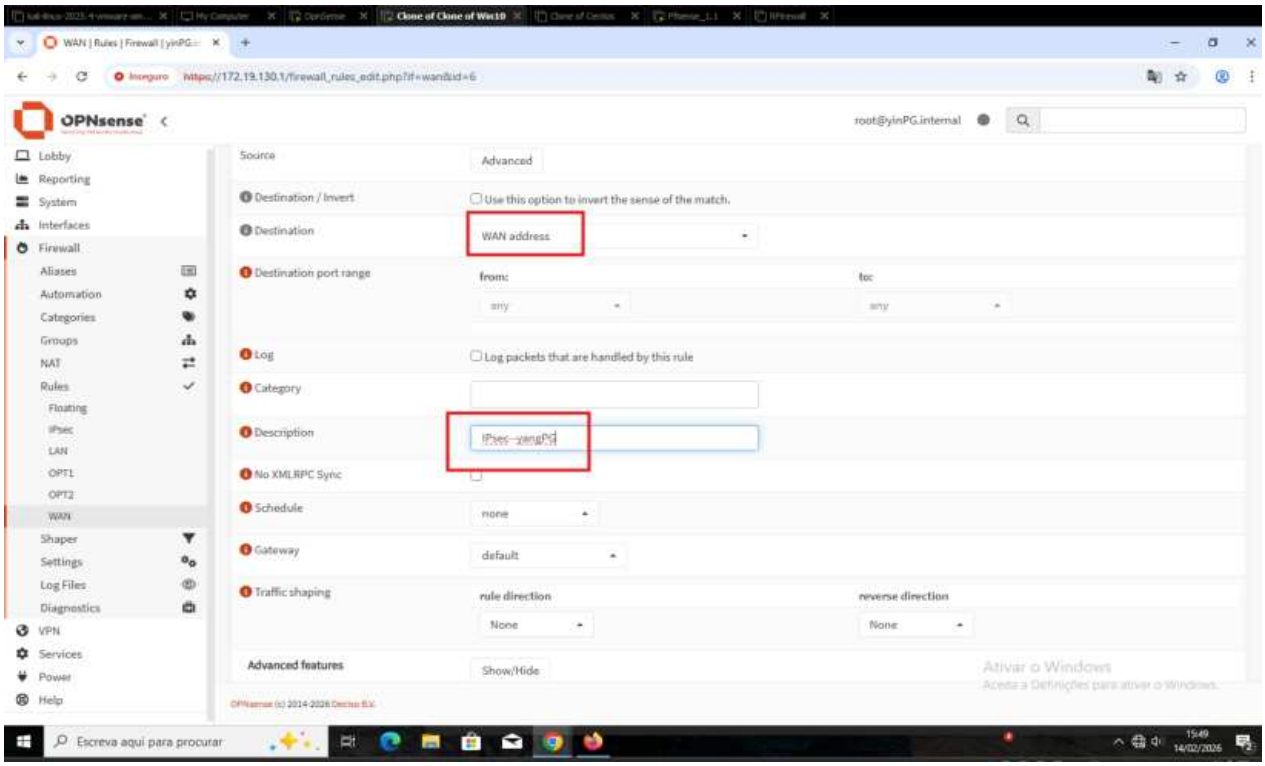


16.16 Regra WAN #3 — ESP (protocol 50)

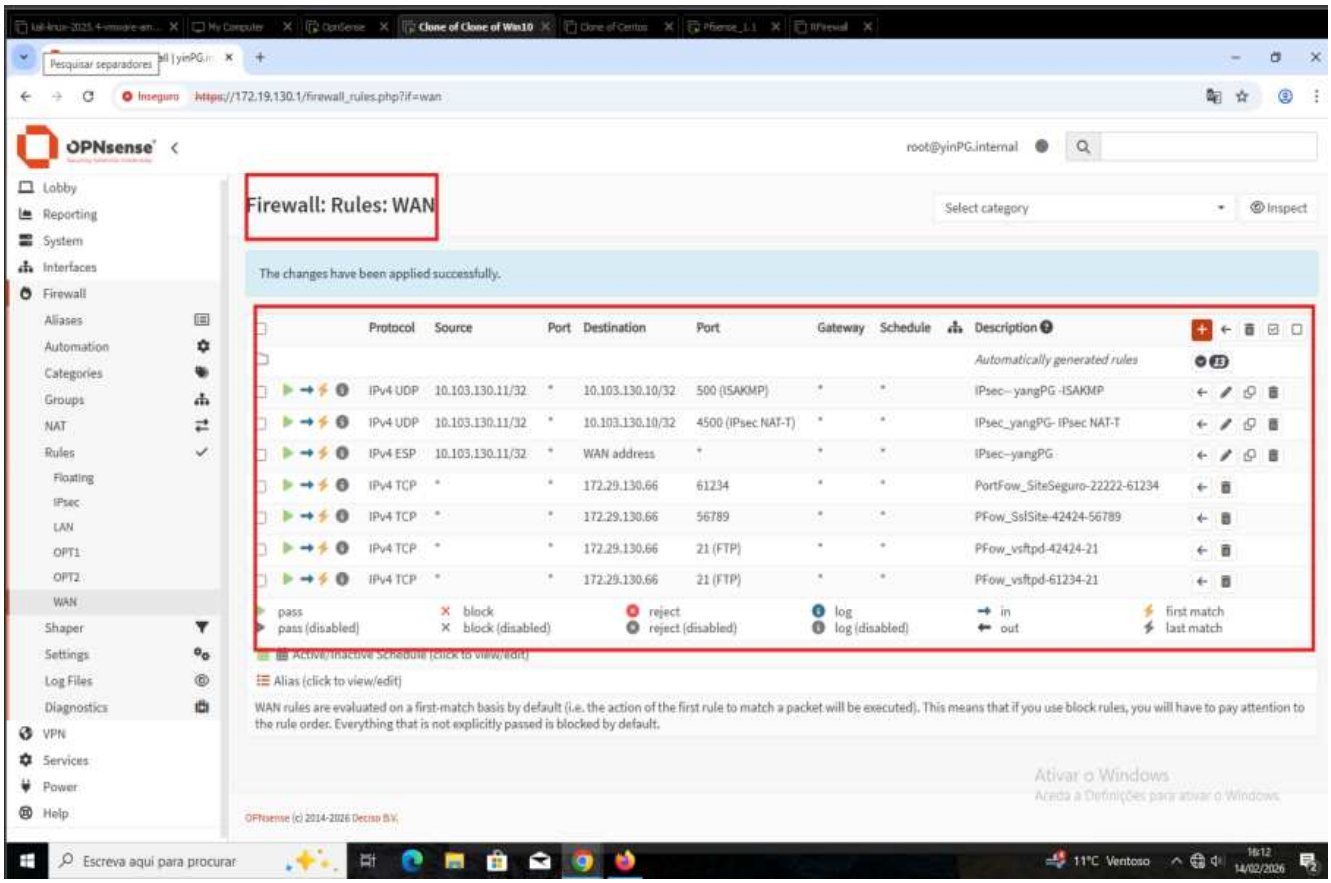
- **Action**: Pass
- **Protocol**: **ESP**
- **Source**: 10.103.130.11
- **Destination**: WAN address
- **Description**: IPsec—yangPG



Continuação ...



Resumo :



3-VPNS----------

Enunciado:

“ b) Situado com o Windows cliente, na zona Green da yangNA, comprove que tem acesso por SSH ao servidor Windows Server. Mas não consegue aceder por RDP a este servidor. (Necessário habilitar o acesso remoto no Windows server e instalar o serviço SSH no mesmo) Não esquecer de prints das regras!”

17 Regras no separador IPsec do yinPG (o tráfego dentro do túnel)

Regra IPsec #1 — BLOQUEAR RDP

-Action: Block

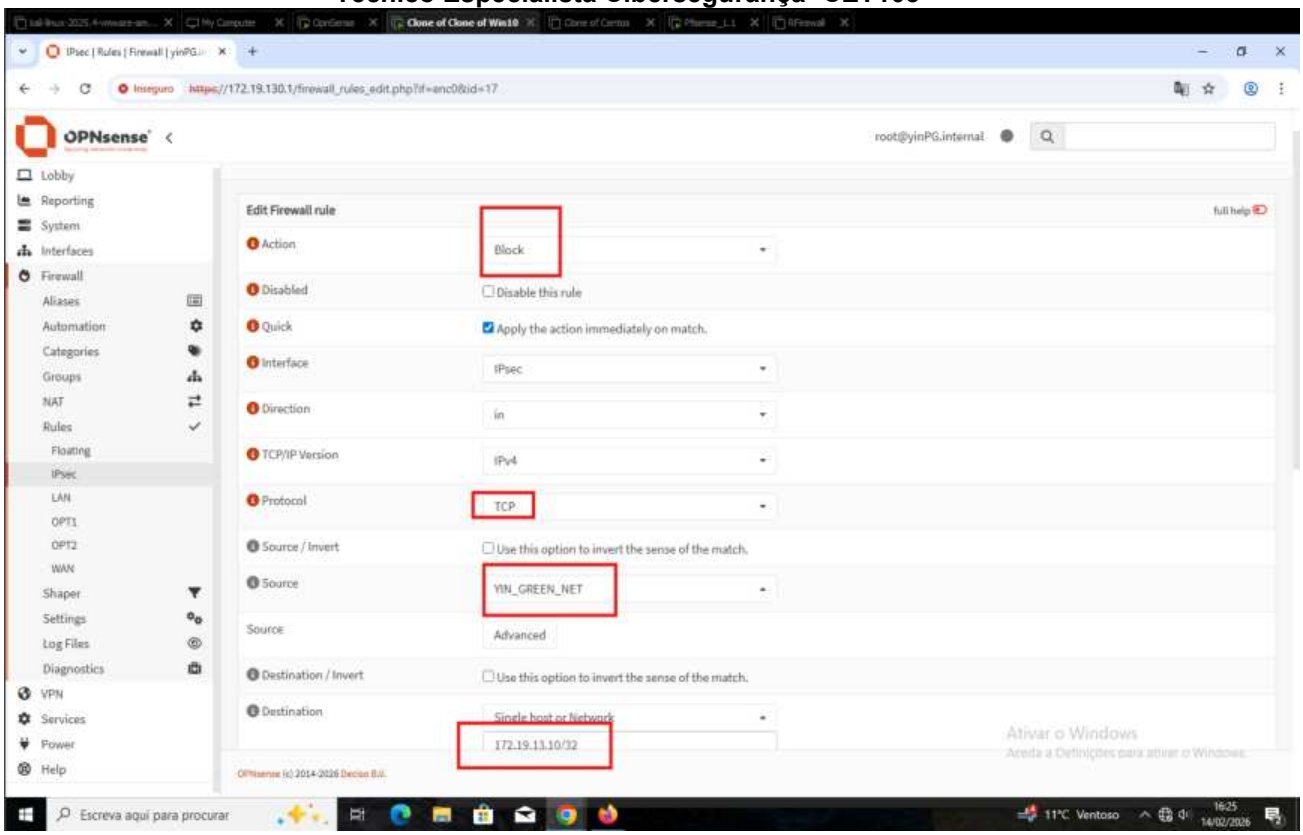
- Protocol: TCP

- Source: 10.120.130.0/24 - YANG_GREEN_NET

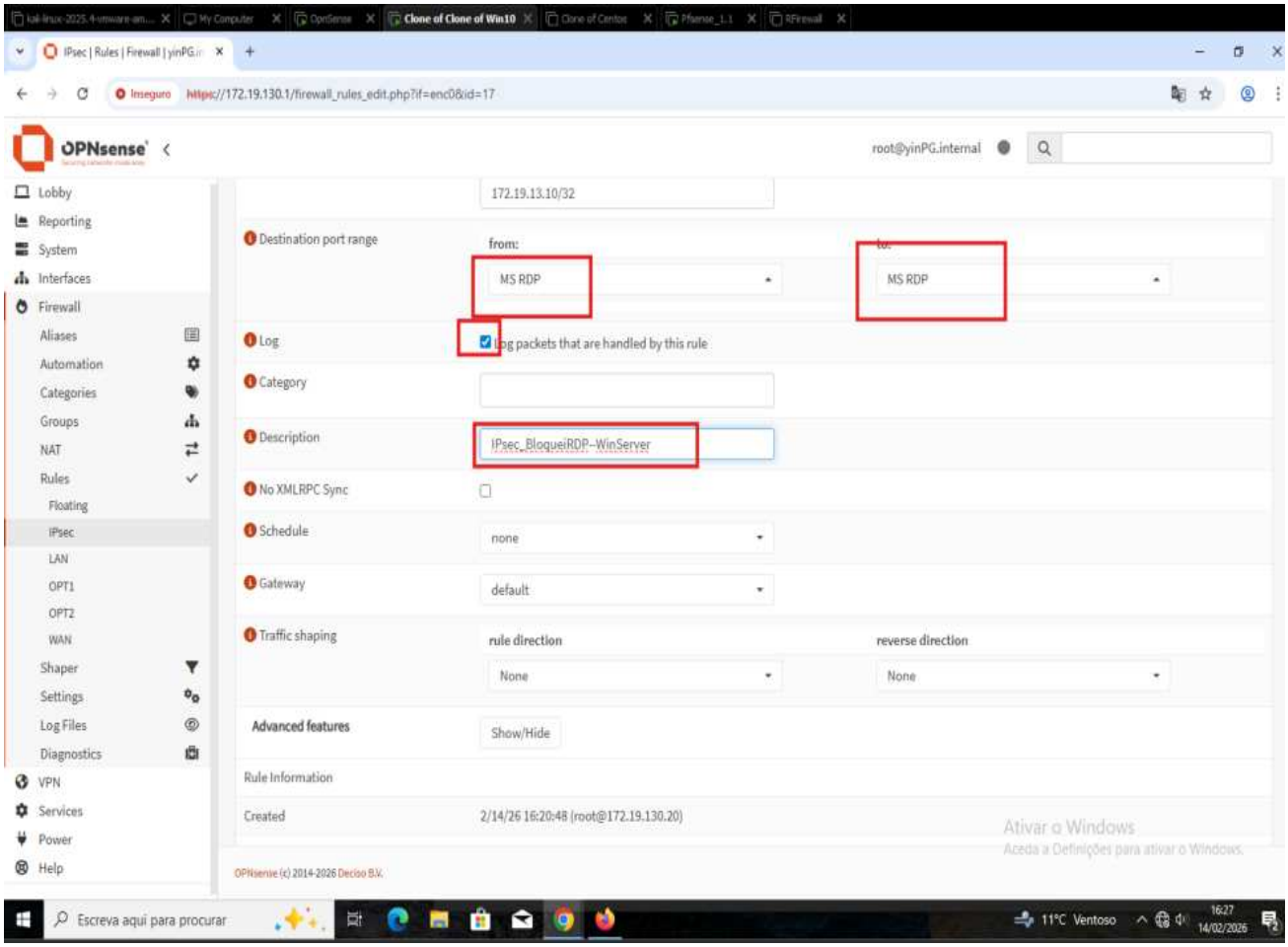
- Destination: 172.19.130.10 (Windows Server)

- Destination port: **3389- MS RDP**

-Description: IPsec_BloqueiRDP--WinServer

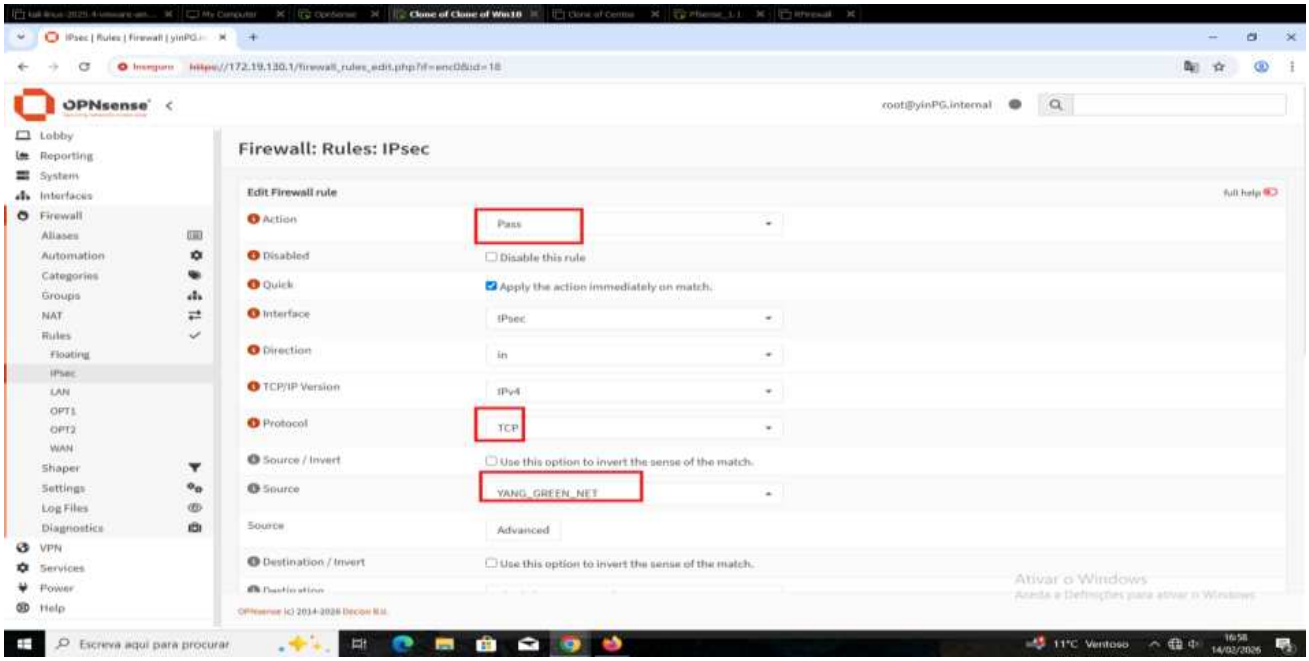


Continuação ..

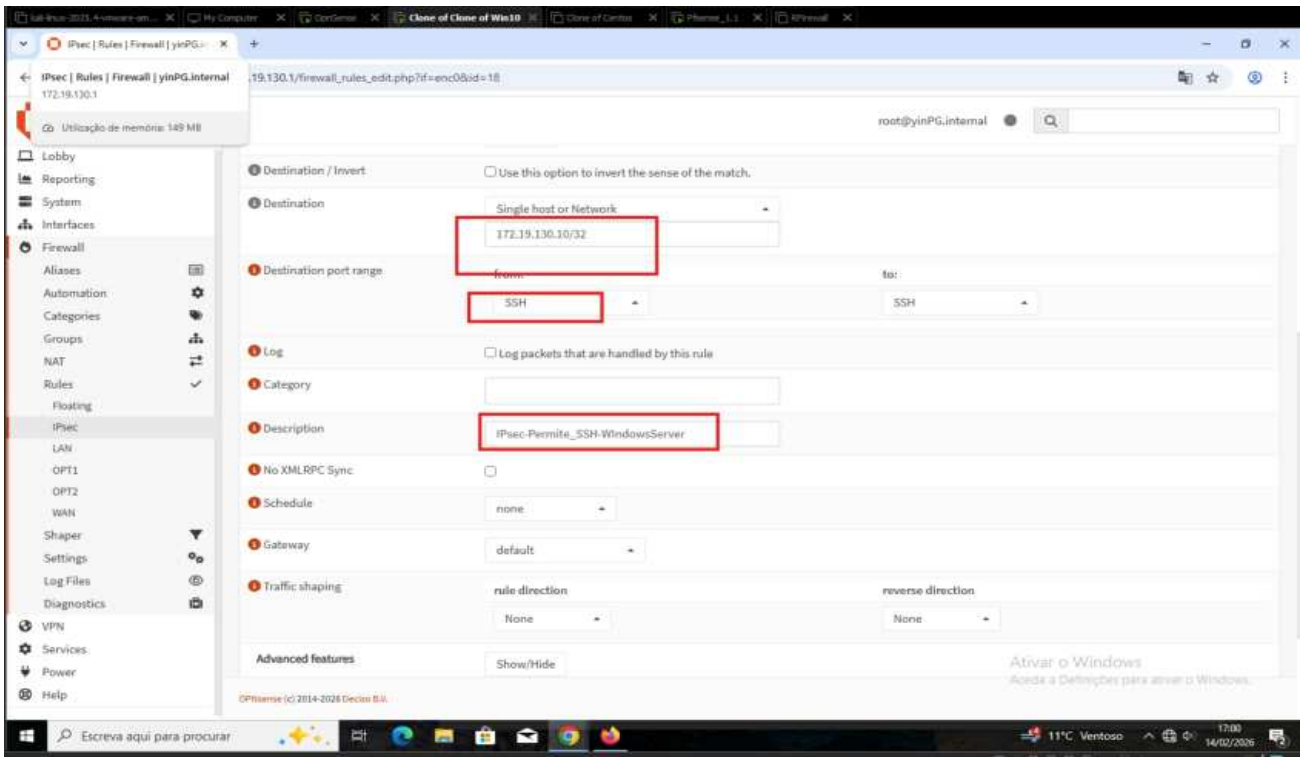


17.1 Regra IPsec #2 — PERMITIR SSH

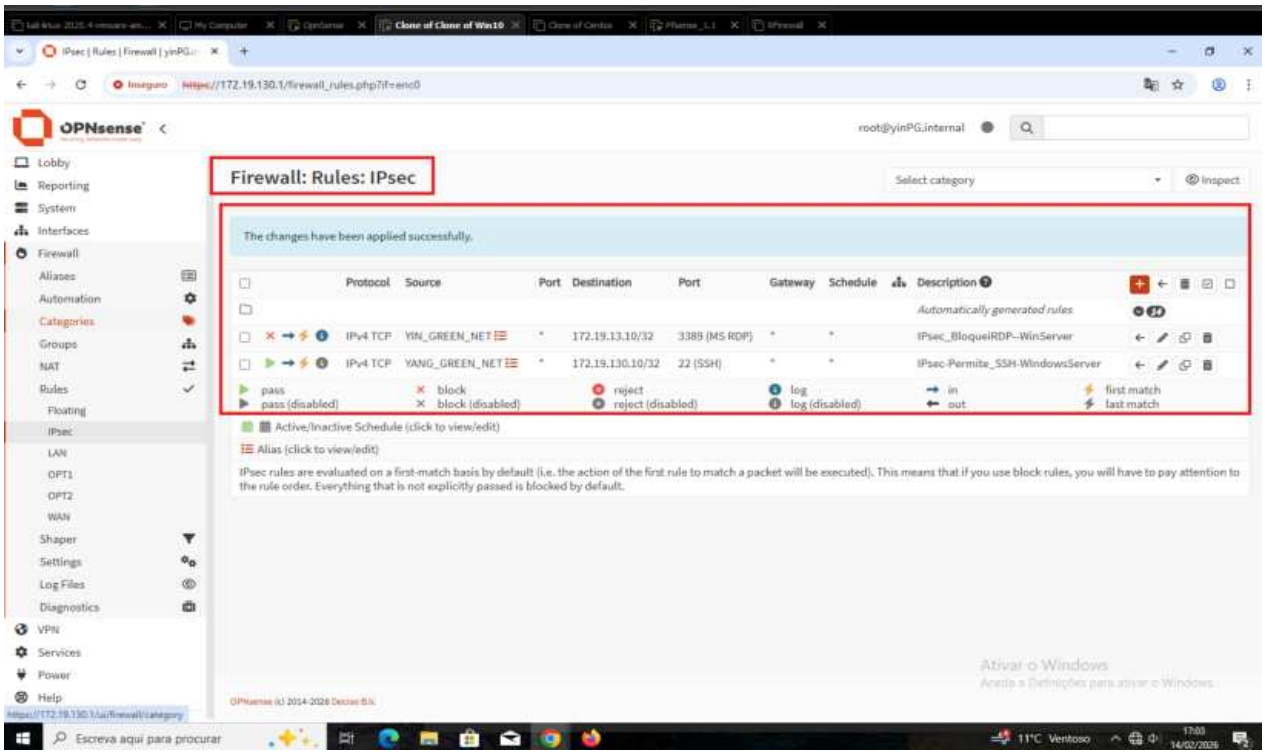
- **Action**: Pass
- **Protocol**: TCP
- **Source**: 10.120.130.0/24 YANG-GREEN NET
- **Destination**: 172.19.130.10
- **Destination port**: **22**
- **Description**: IPsec-Permite_SSH-WIndowsServer



Continuação …



Resumo :



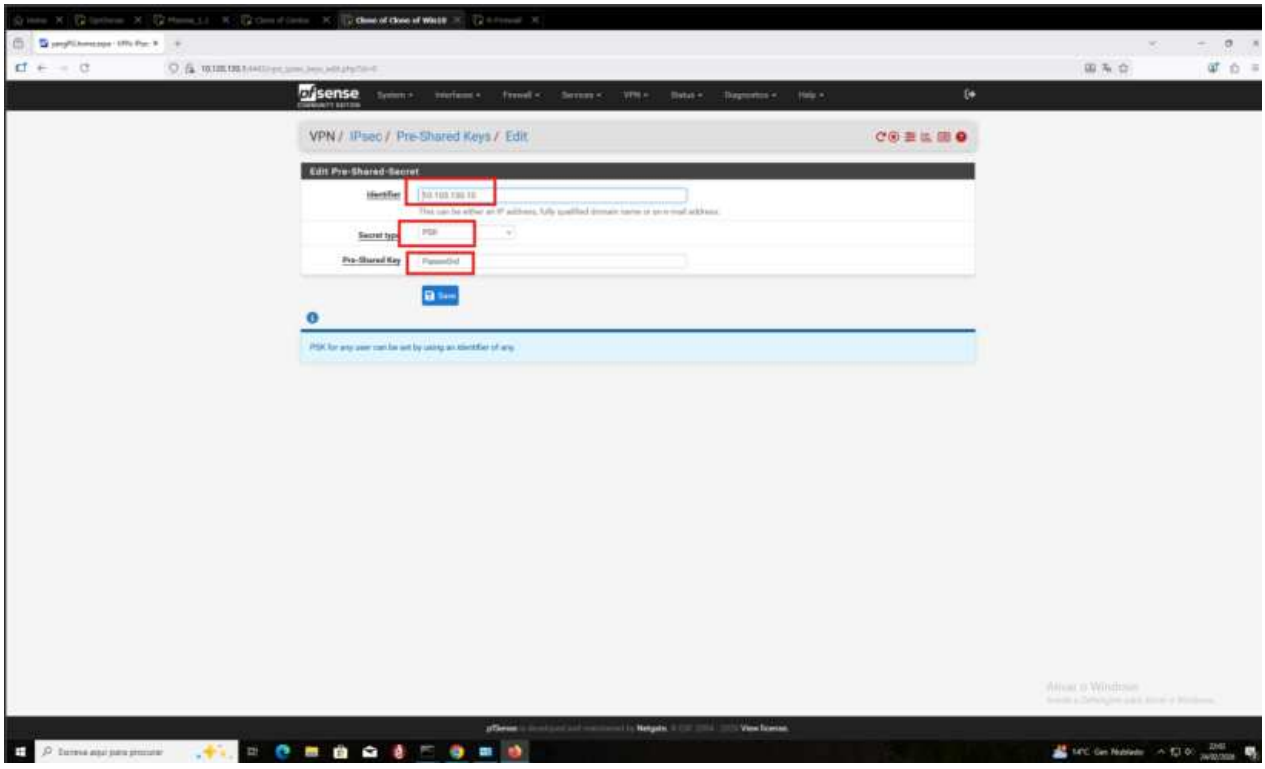
17.2 pfSense yangPG — IPsec

Pre-Shared Key

VPN → IPsec → Pre-Shared Keys → +Add

- Identity: 10.103.130.10

  PSK: **Passw0rd**



17.3 pfSense yangPG — IPsec

pfSense: criar Phase 1 (IKE)

1. **VPN → IPsec → Tunnels**
2. Clicar **Add P1** (Add")

Phase 1 – General Information

- Description: ipsec--yinPG

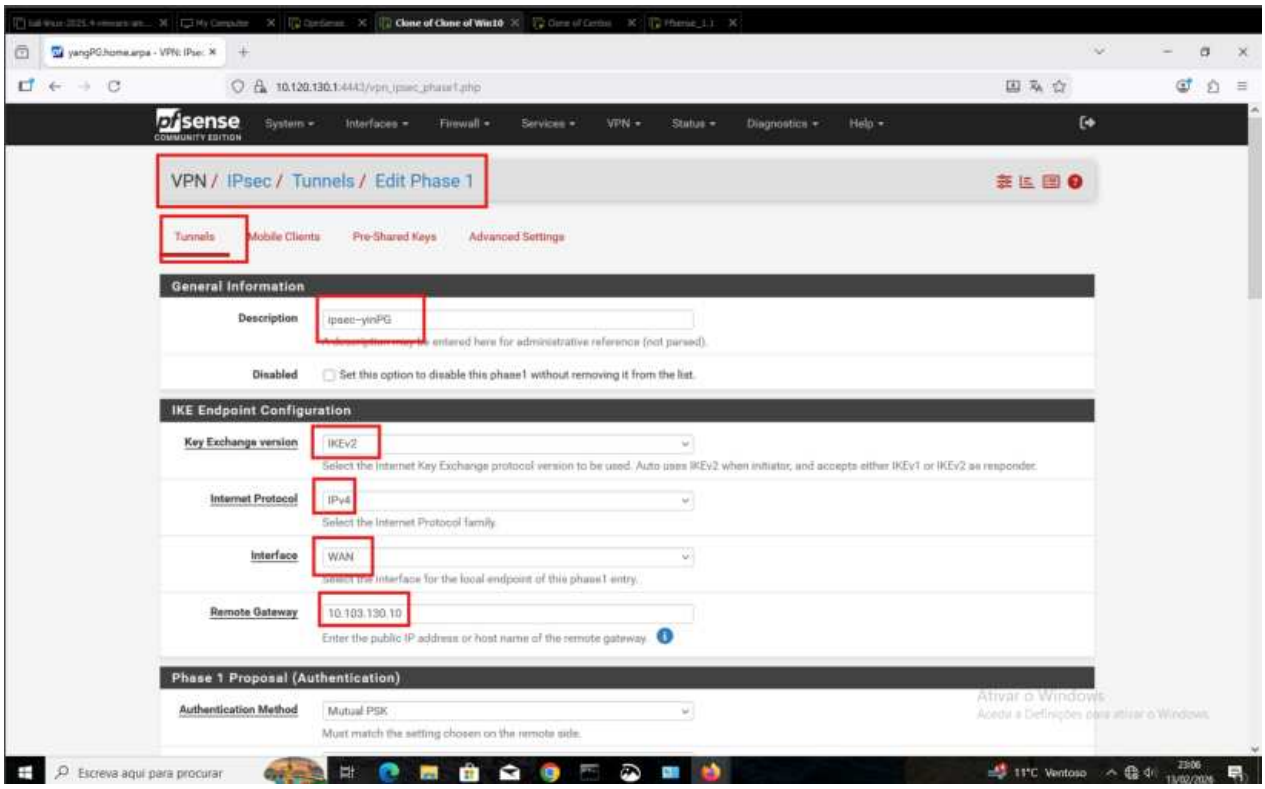
- Key Exchange version: IKEv2

- Internet Protocol: IPv4

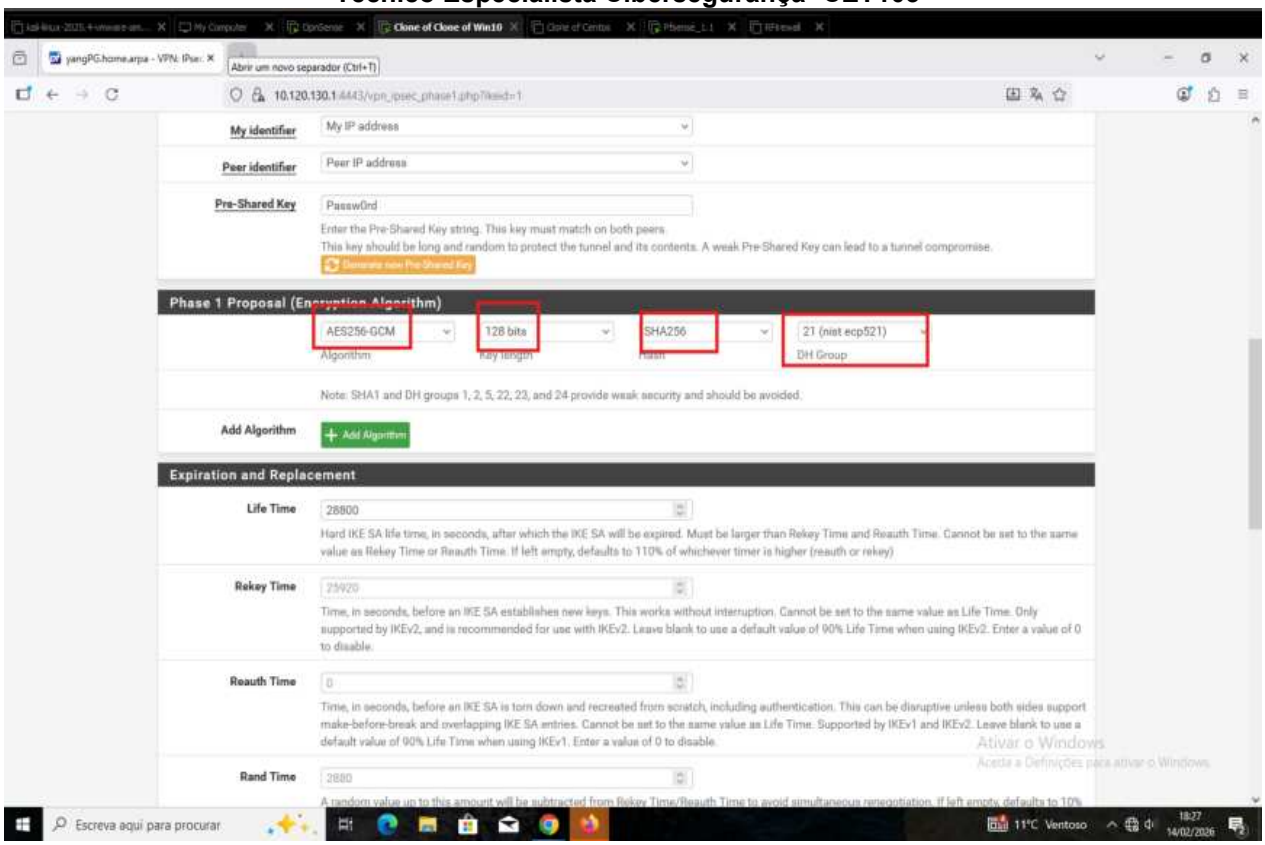
- Interface: WAN

- Remote Gateway: 10.103.130.10

- PSK: Passw0rd



- **Protocol: ESP**
- **Encryption: AES-256-GCM**
- **Hash: SHA256**
- **PFS key group: 21**



17.4 pfSense: criar Phase 2 (Child SA)

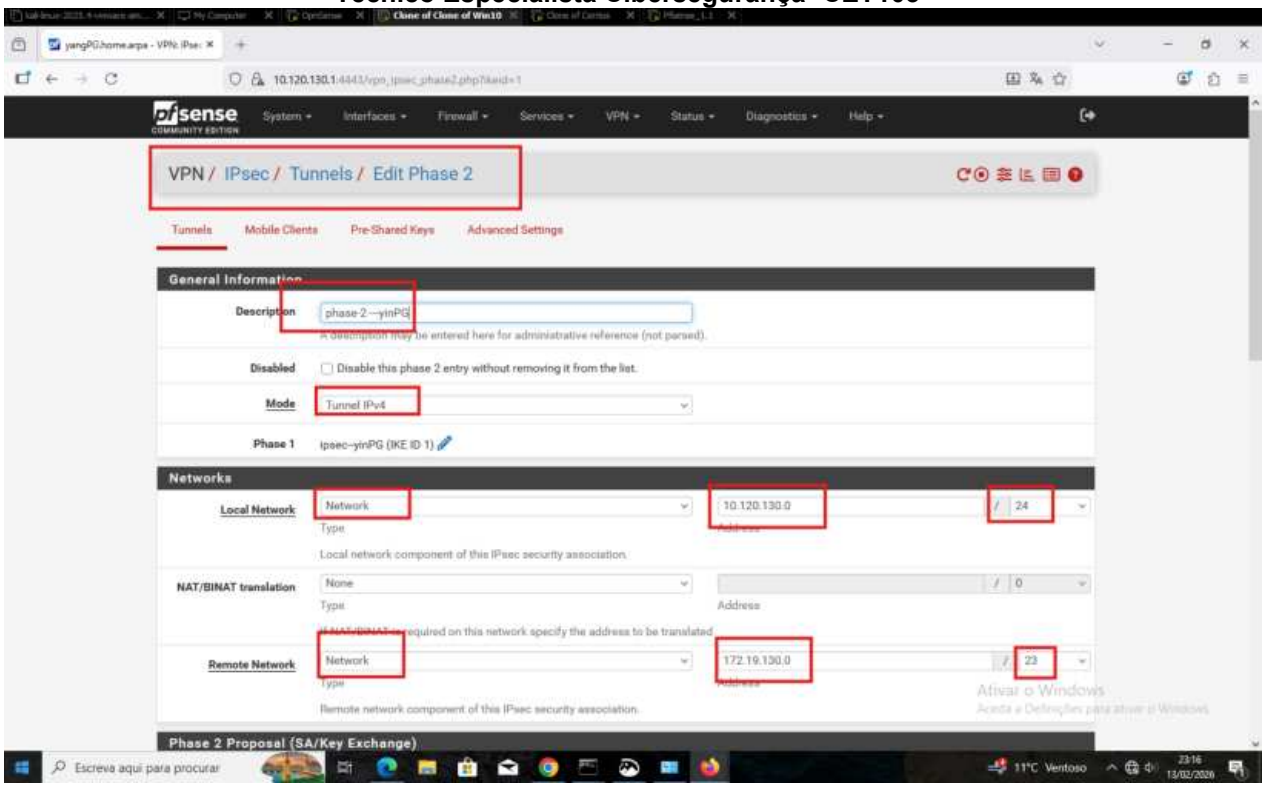
Ainda em **VPN → IPsec → Tunnels**,

Clicar em **Show Phase 2 Entries (Expand)**

1. Clicar **Add P2**

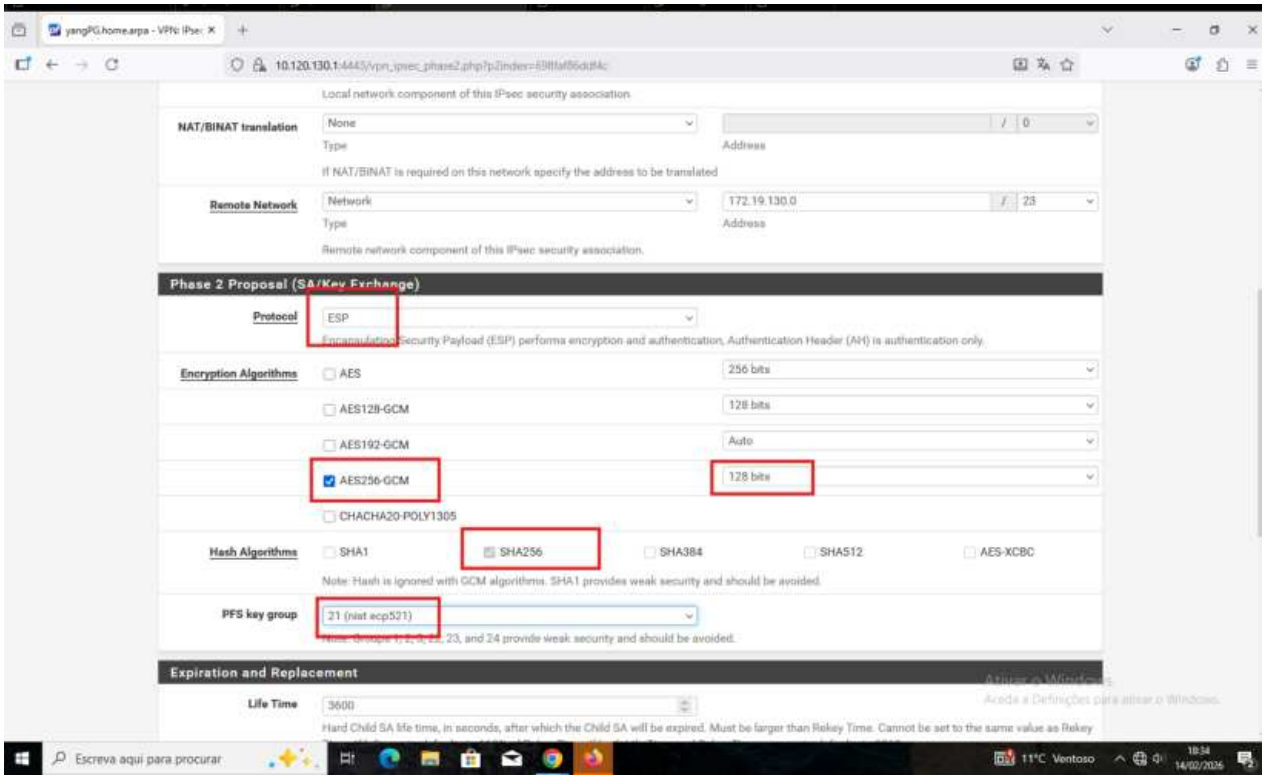
Phase 2 – General Information

- **Mode**: Tunnel IPv4
- **Local Network**: Network → 10.120.130.0/24
- **Remote Network**: Network → 172.19.130.0/23



Phase 2 – Phase 2 Proposal (ESP)

- **Protocol**: ESP
- **Encryption**: AES-256-GCM
- **Hash**: SHA256
- **PFS key group**: 21



17.5 Regras de Firewall no pfSense

Firewall → Rules → WAN
- Allow UDP 500, UDP 4500, ESP

A) Regras na WAN do yangPG

Ira a: Firewall → Rules → WAN

Regra WAN #1 — IKE

- Action: Pass

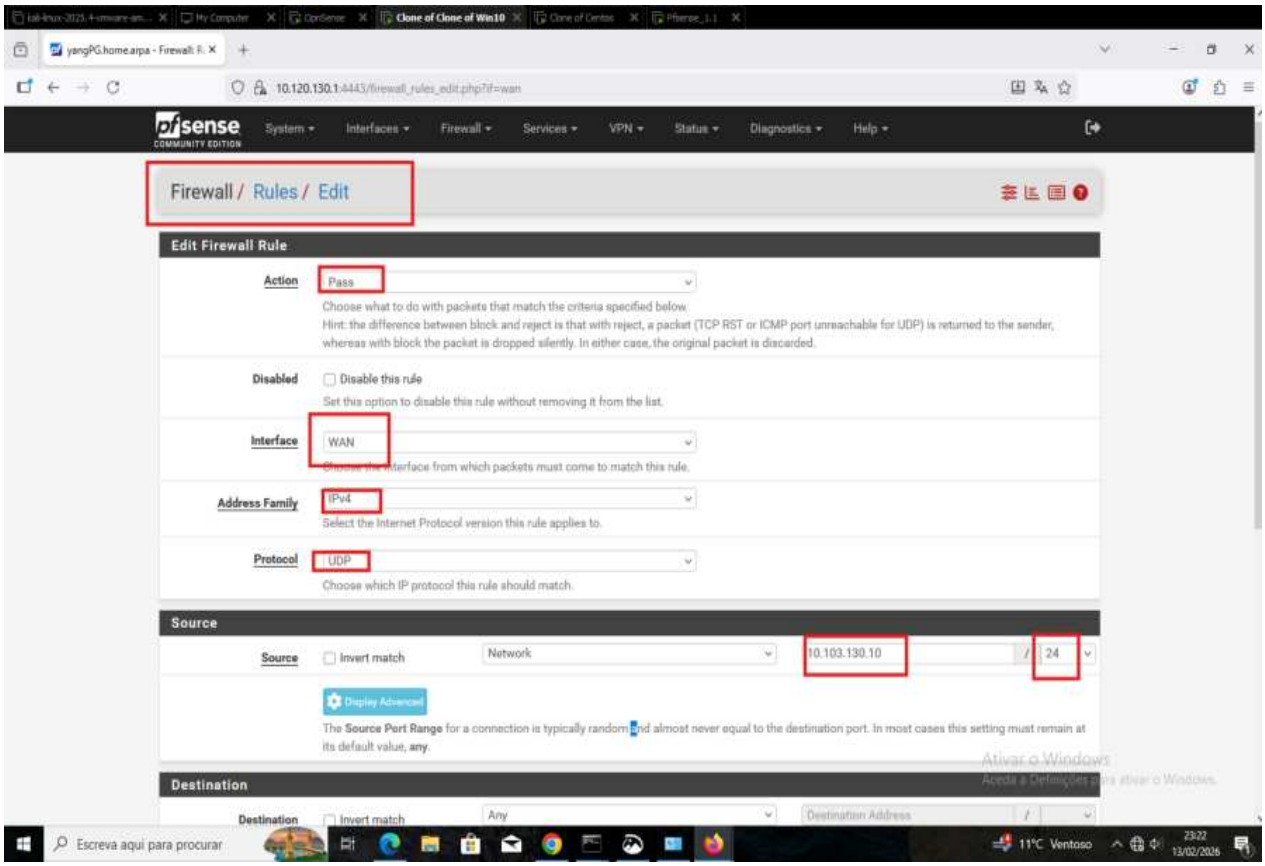
- Protocol: UDP

- Source: 10.103.130.10

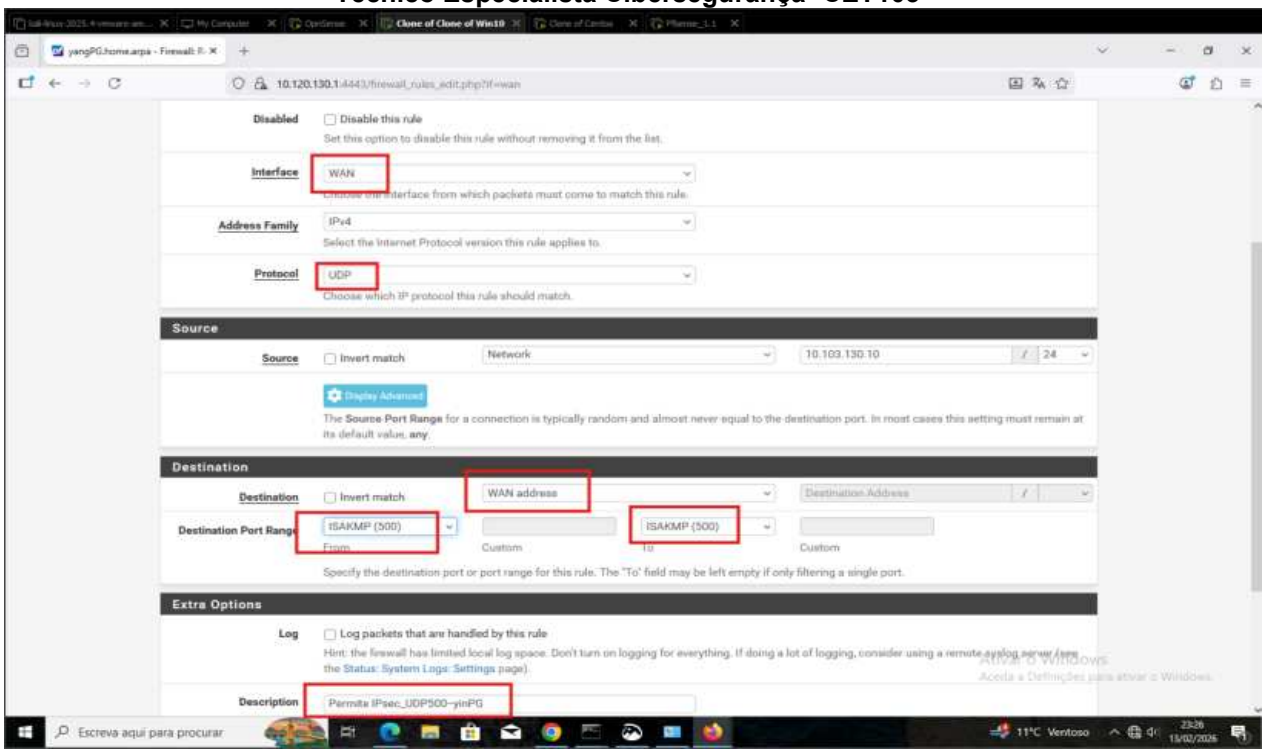
- Destination: WAN address

- Destination port: 500 (ISAKMP)

- Description: Permite IPsec_UDP500—yinPG

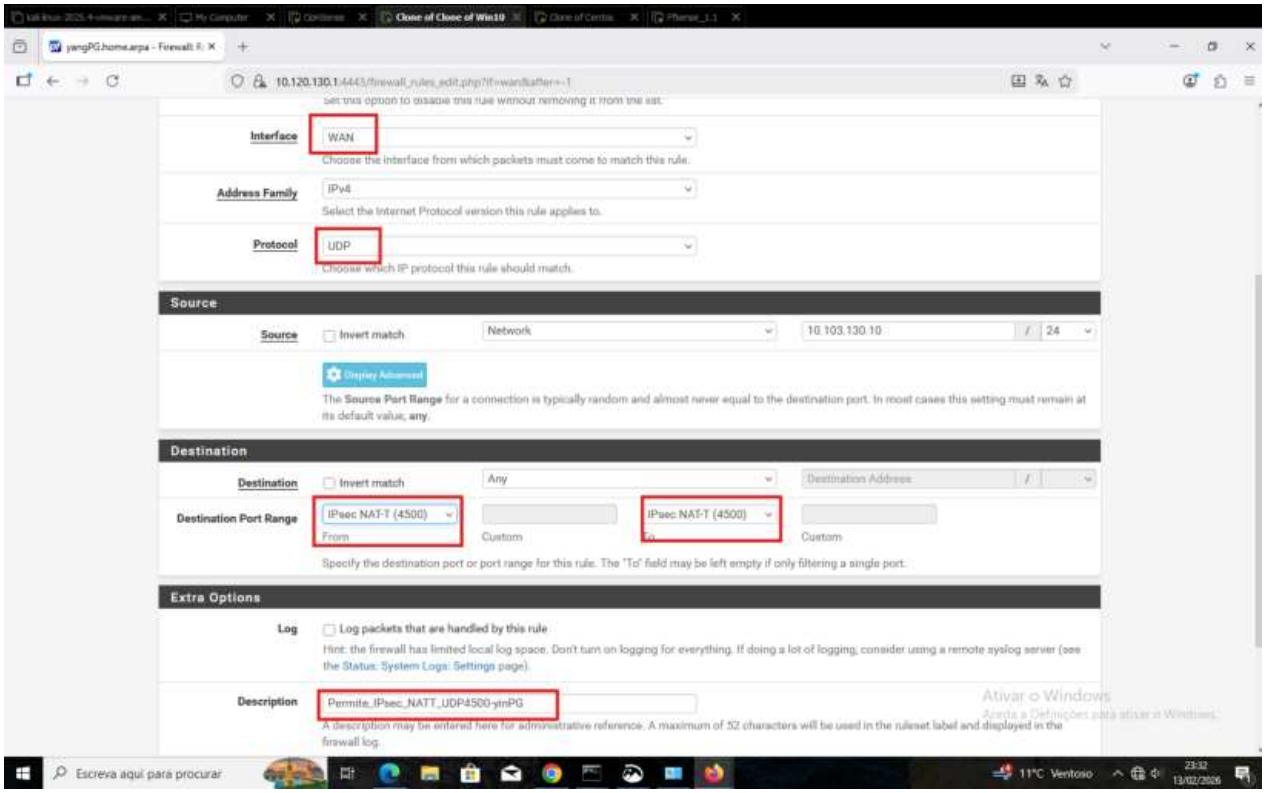


Continuação …



17.6 Regra WAN — NAT-T

- Protocol: UDP
- Source: 10.103.130.10
- Destination: WAN address
- Port: 4500
- Description: Permite_IPsec_NATT_UDP4500-yinPG



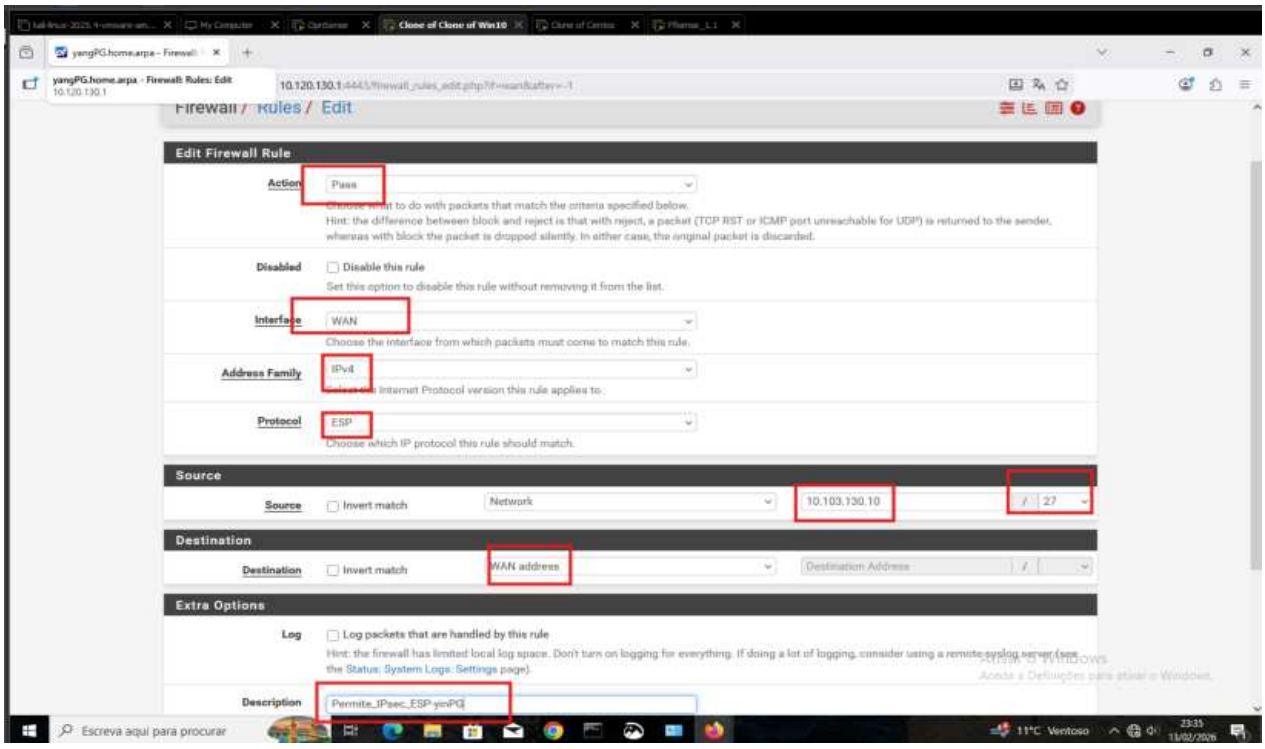
17.7 Regra WAN #3 — ESP

- Protocol: ESP

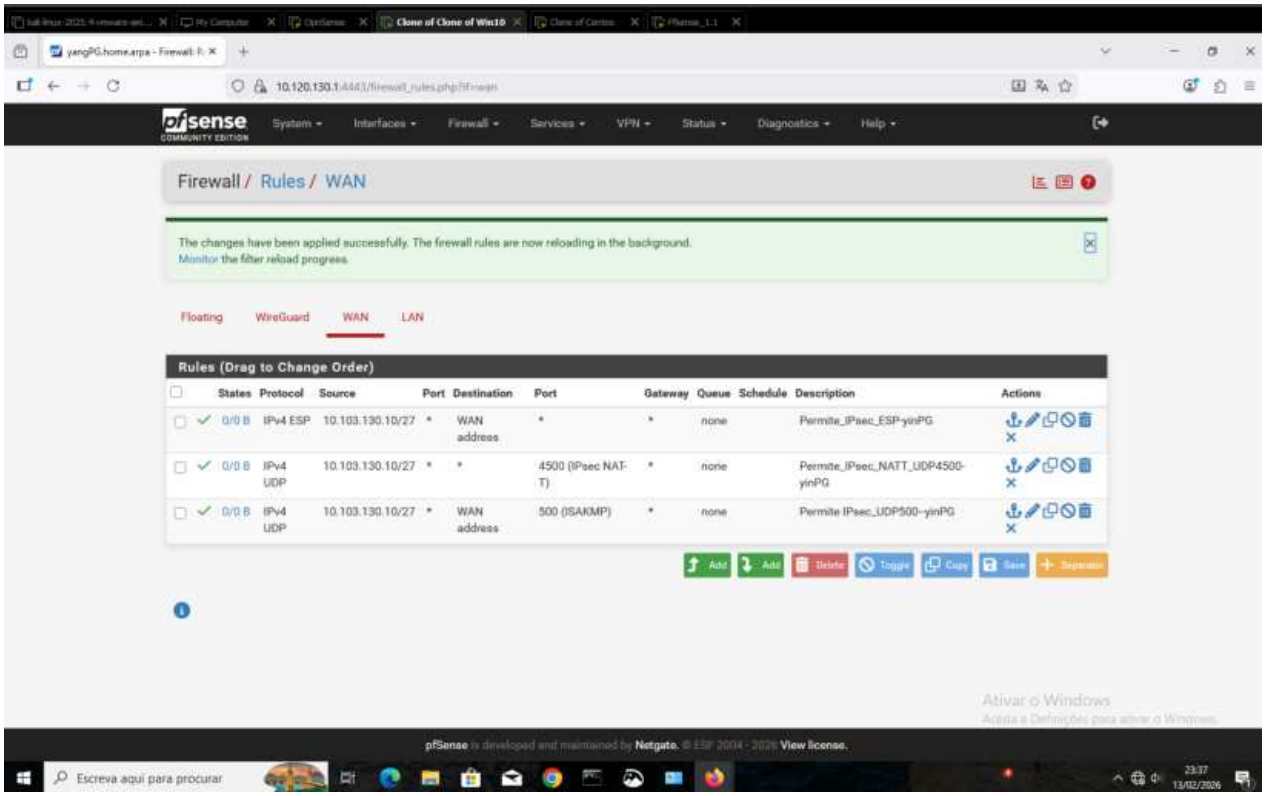
- Source: 10.103.130.10

- Destination: WAN address

- Description: Permite_IPsec_ESP-yinPG



Resumo :

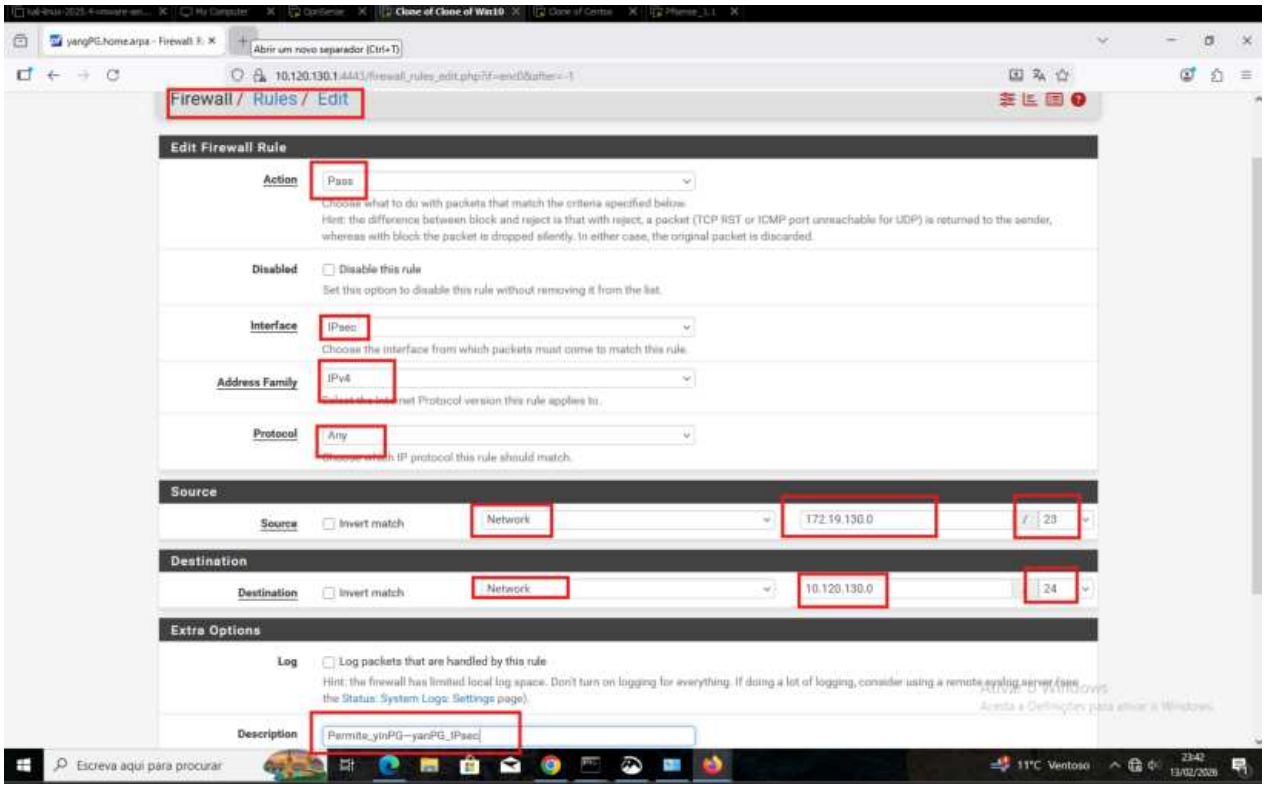


17.8 Regras na tab IPsec (tráfego dentro do túnel)

Firewall → Rules → IPsec

Regra IPsec #1 — permitir tráfego entre redes

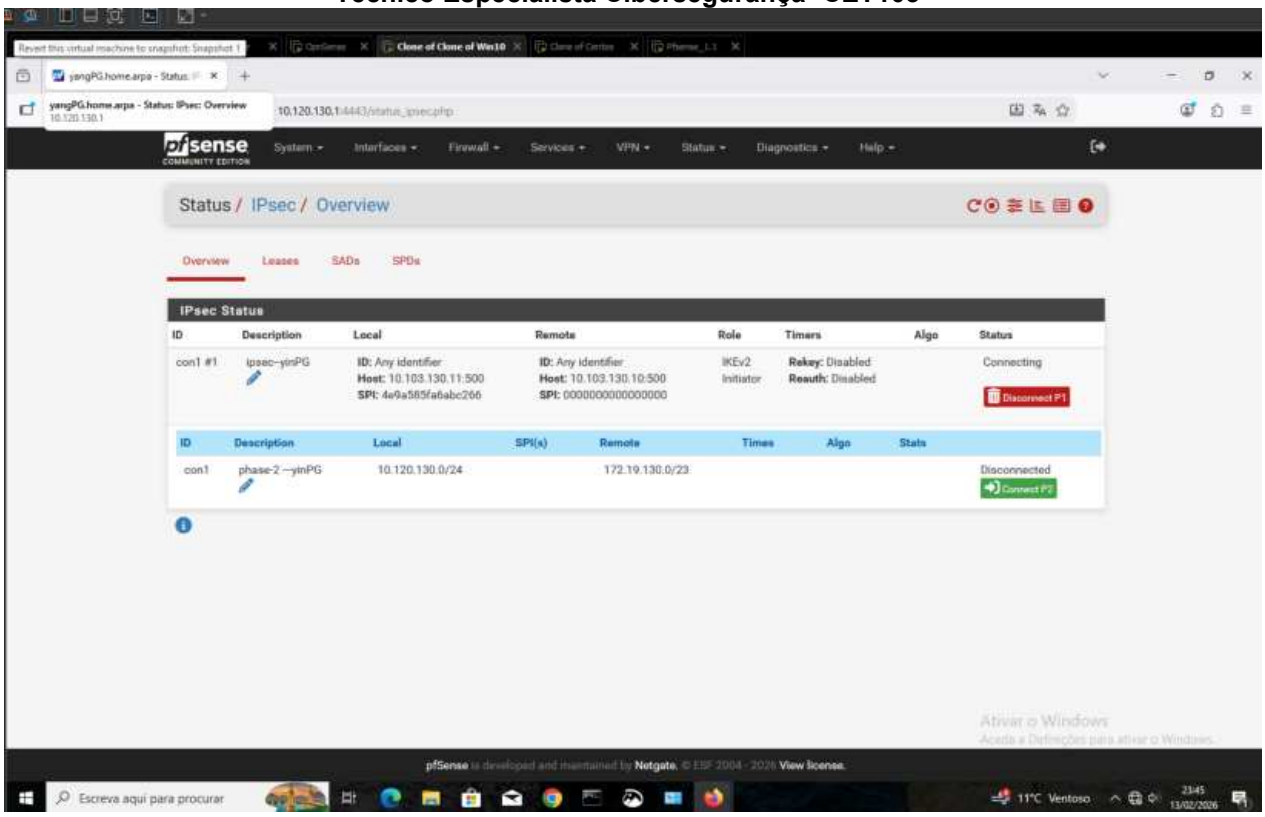
- Action: Pass
- Protocol: any
- Source: 172.19.130.0/23
- Destination: 10.120.130.0/24
- Description: Permite_yinPG—yanPG_IPsec



17.9 Confirmar que o IPsec está UP

No pfSense:

- **Status → IPsec**



3) VPNs----------

Enunciado :

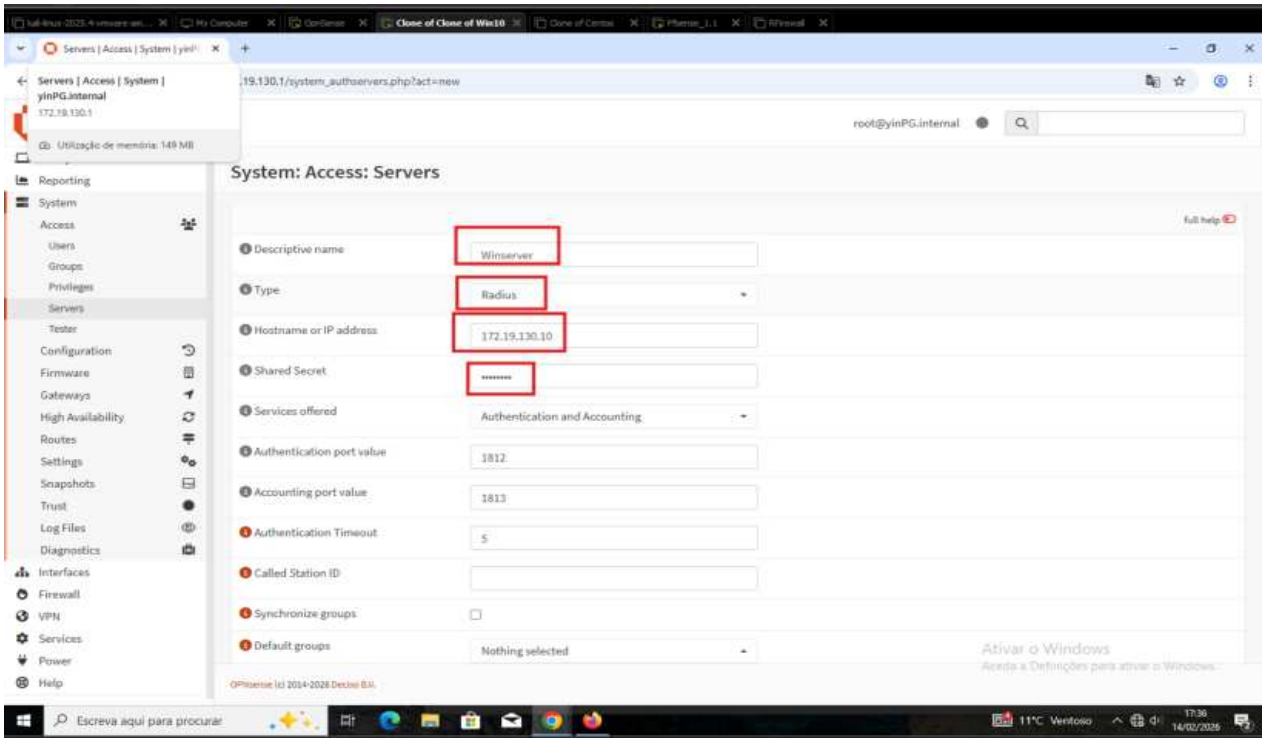
" c) Crie uma ligação VPN em OpenVPN (SSL VPN) para um Roadwarrior, para o cliente identificado na Figura 2. A autenticação deste cliente, será para um user **pertencente ao IT da AD do windows Server, durante o periodo 8h-20h e de segunda a sexta-feira. (O servidor Windows pode ser o que utilizou em trabalhos anteriores, ou outro que respeite a estrutura de OUs, users e grupos que temos utilizado como exemplo) Esse servidor está situado na zona green do yinNA. Assim esta autenticação será baseada em Radius Server em Windows"**

18- VPN 3.c — OpenVPN Roadwarrior (SSL) no yinPG com autenticação RADIUS (IT, Seg–Sex, 8–20)

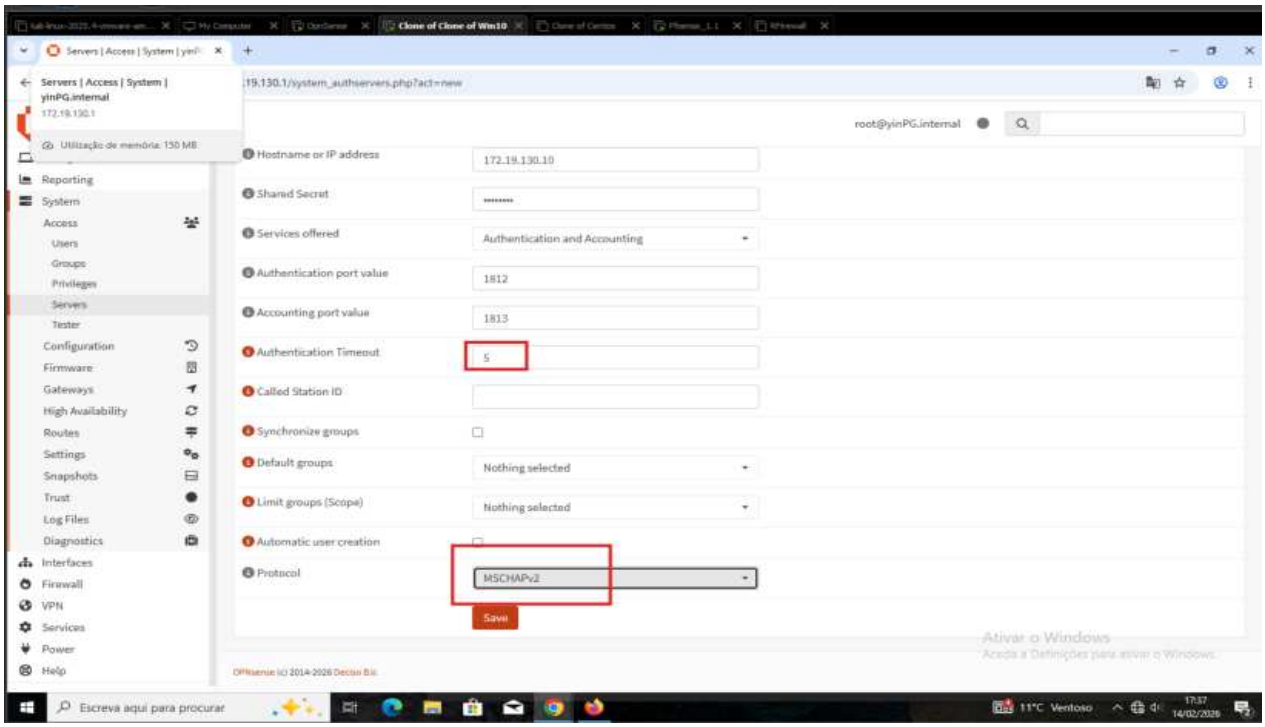
18.1 OPNsense — adicionar servidor RADIUS

System → Access → Servers → +Add

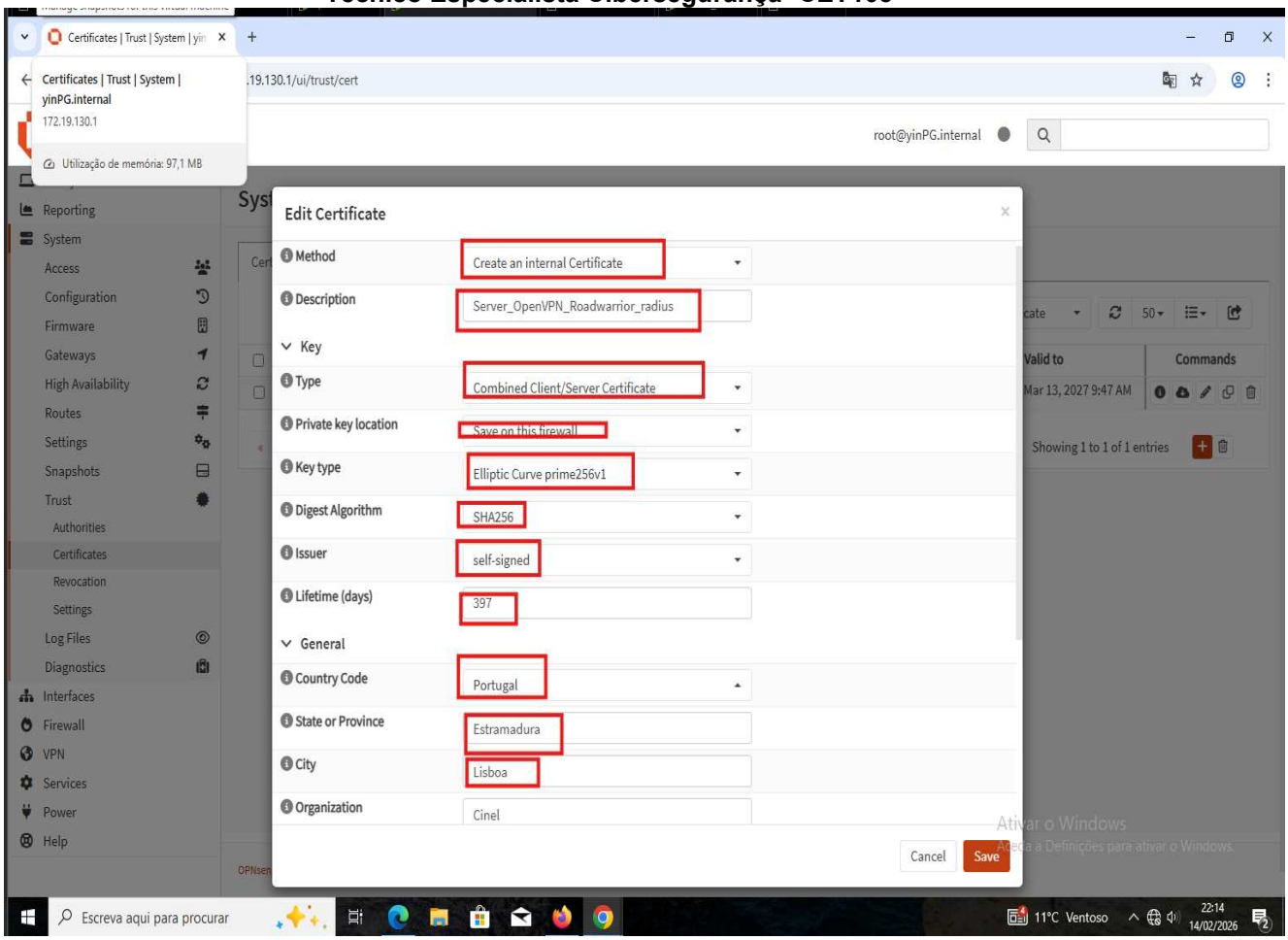
Descriptive name: Winserver
Type: Radius
Hostname or IP address: 172.19.130.10
Shared Secret: Passw0rd



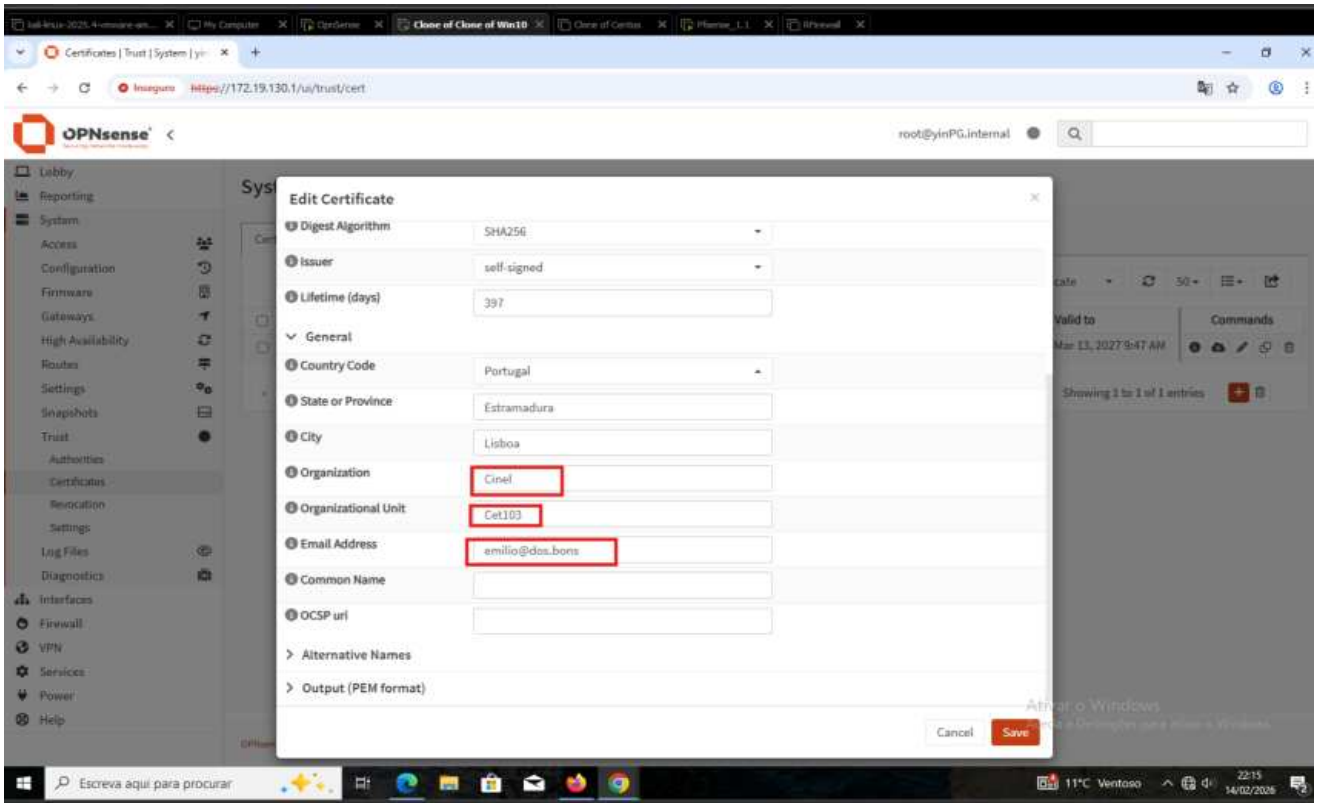
Continuação…



18.2 OPNsense — criar CA e Certificado do servidor OpenVPN - System → Trust →Certificates → +Add (Create internal Certificate)



Continuação …



System → Trust → Authorities → +Add (Create internal CA)

Method

- **Create an internal Certificate Authority**

Description

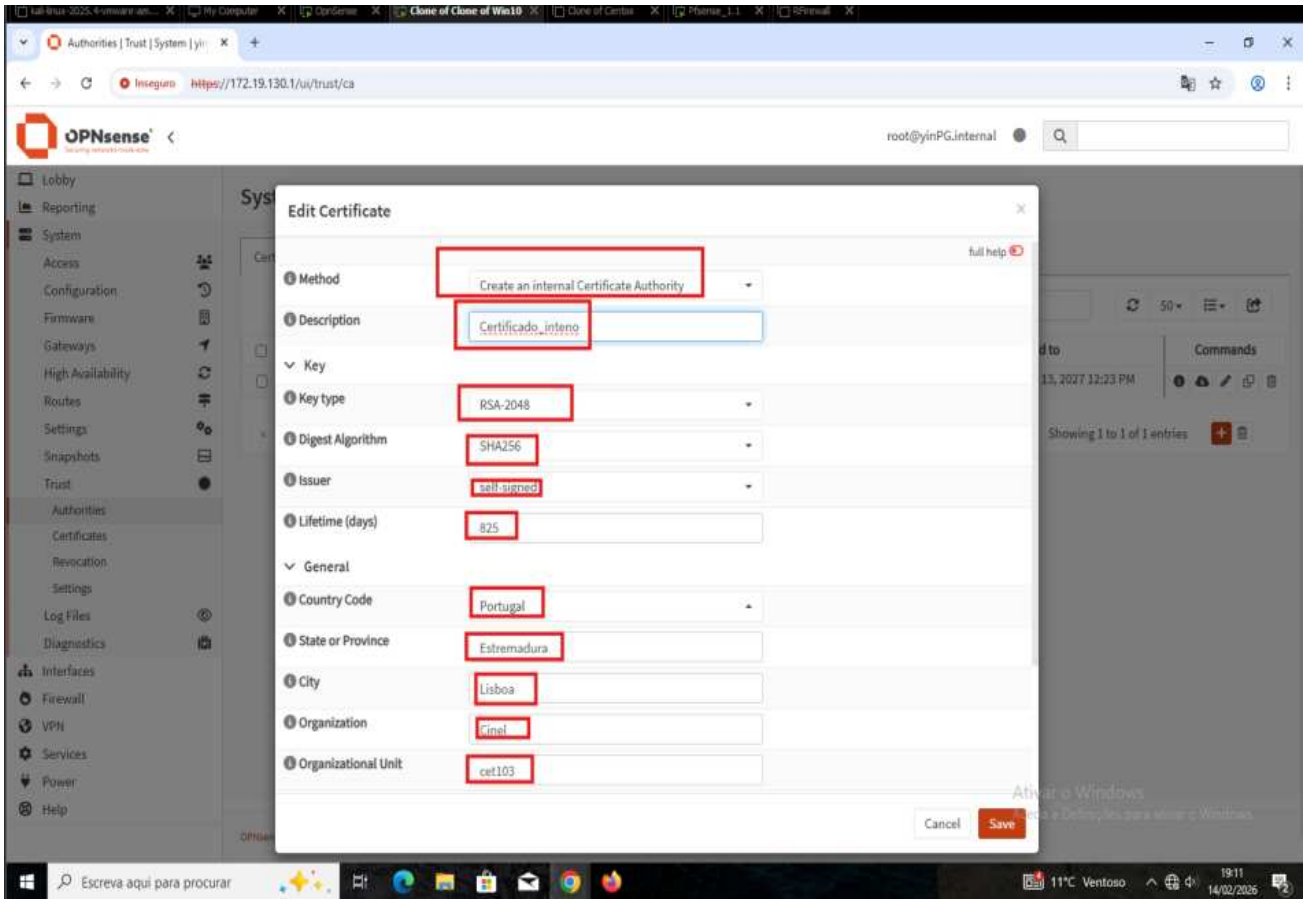
- **Certificado_interno**

Key

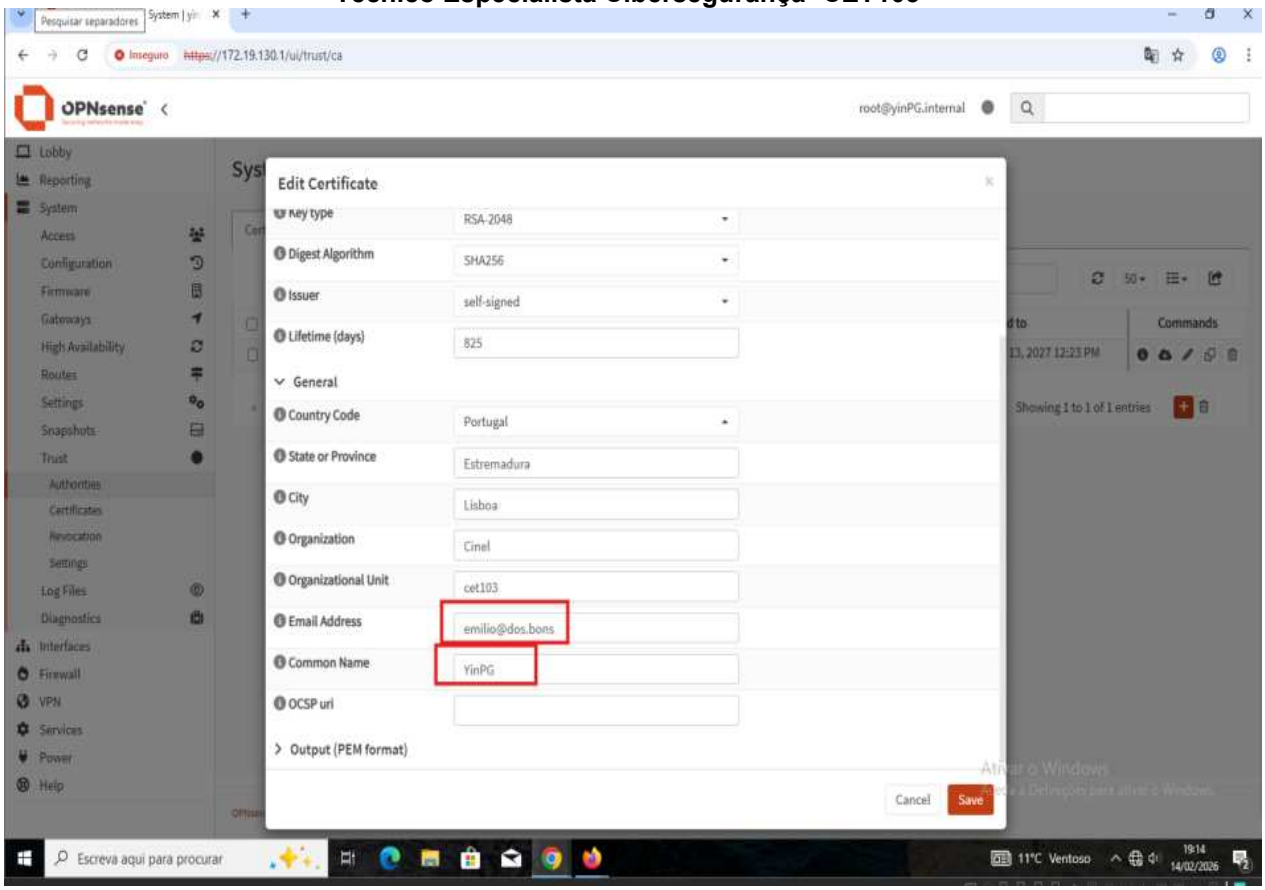
- **Key type:** RSA-2048
- **Digest Algorithm:** SHA256
- **Issuer:** self-signed
- **Lifetime (days):** 825

General

- **Country Code:** Portugal
- **State or Province:** Estremadura
- **City:** Lis
- **Organization:**
- **Organizational Unit:** cet103
- **Common Name:** YinPG

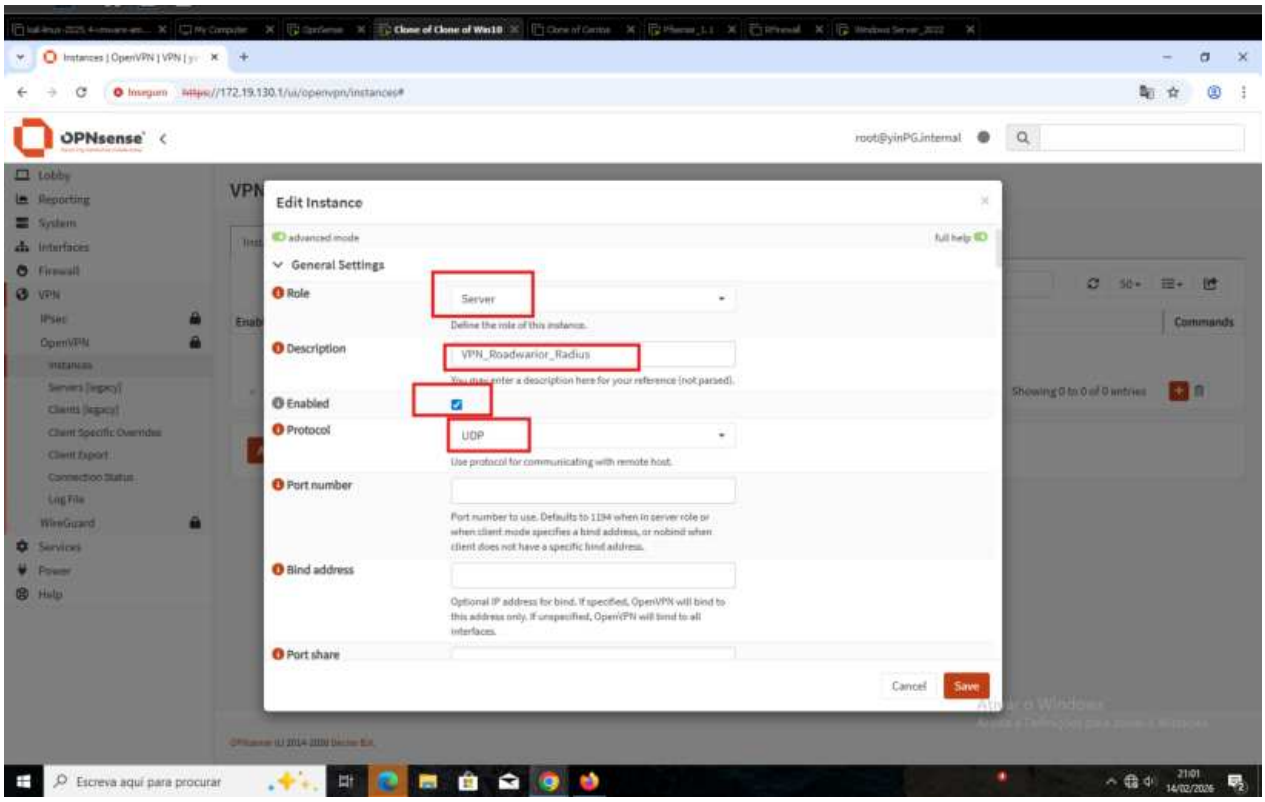


Continuação…

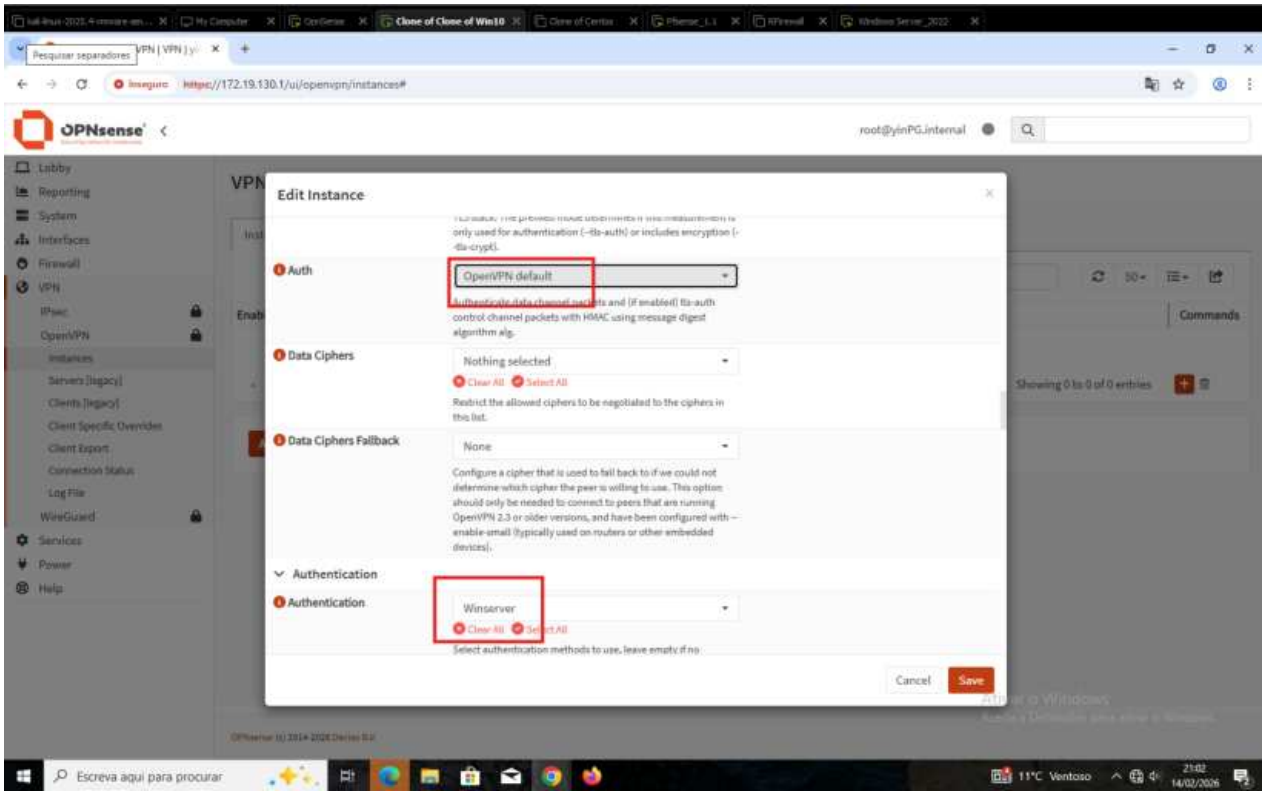


18.3 OPNsense — criar OpenVPN Server Instance→ Nota : referente ao Windows Serv22

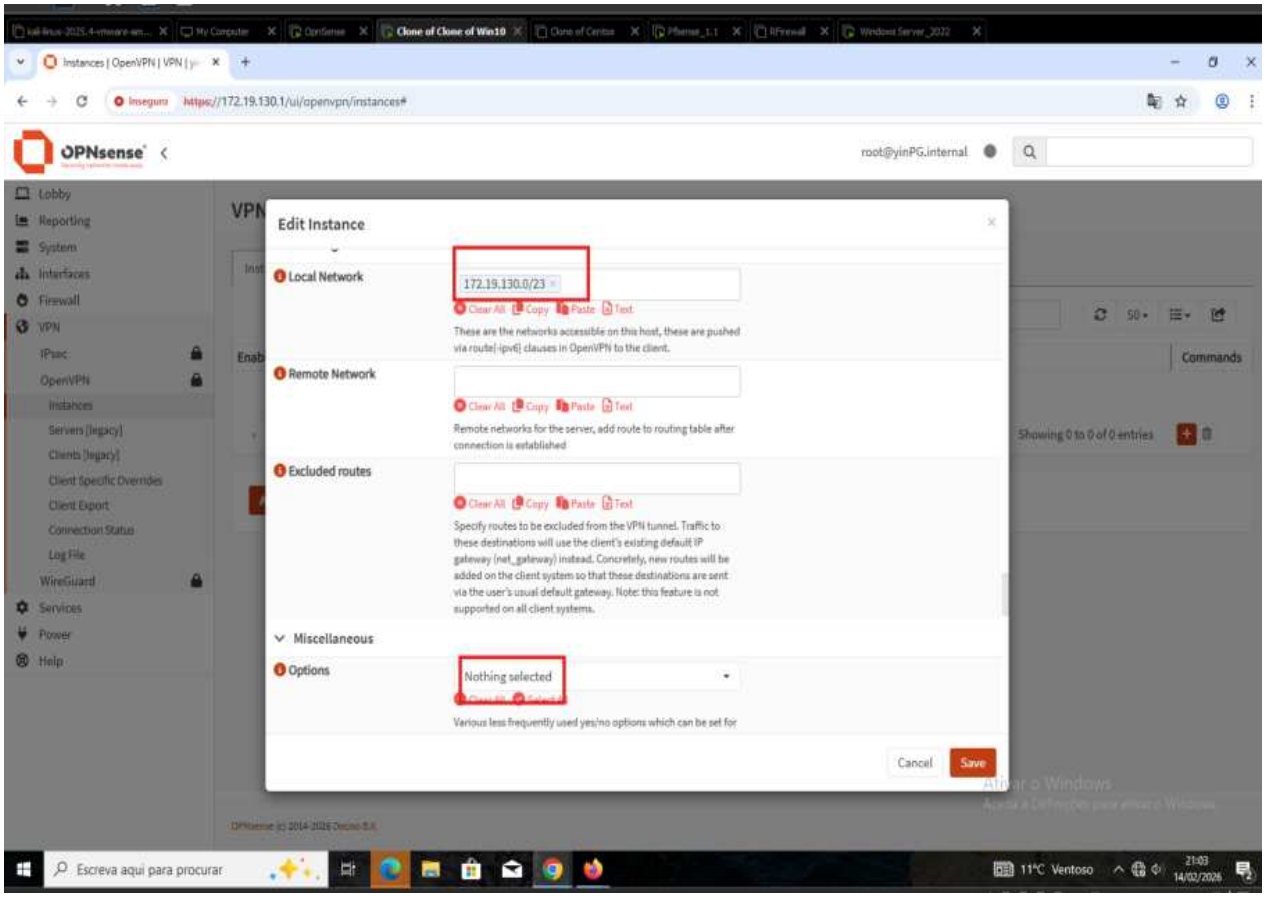
VPN → OpenVPN → Instances → +Add



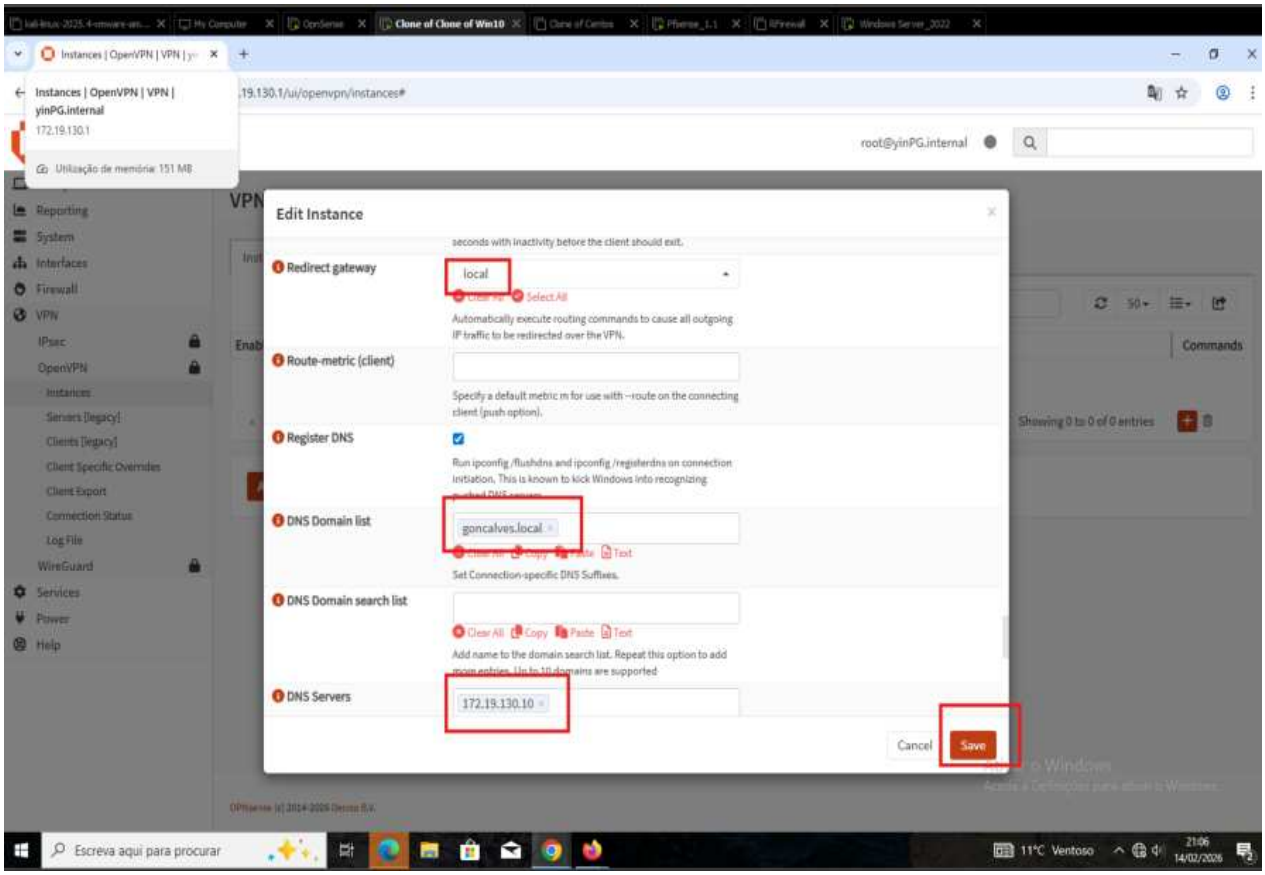
Continuação…



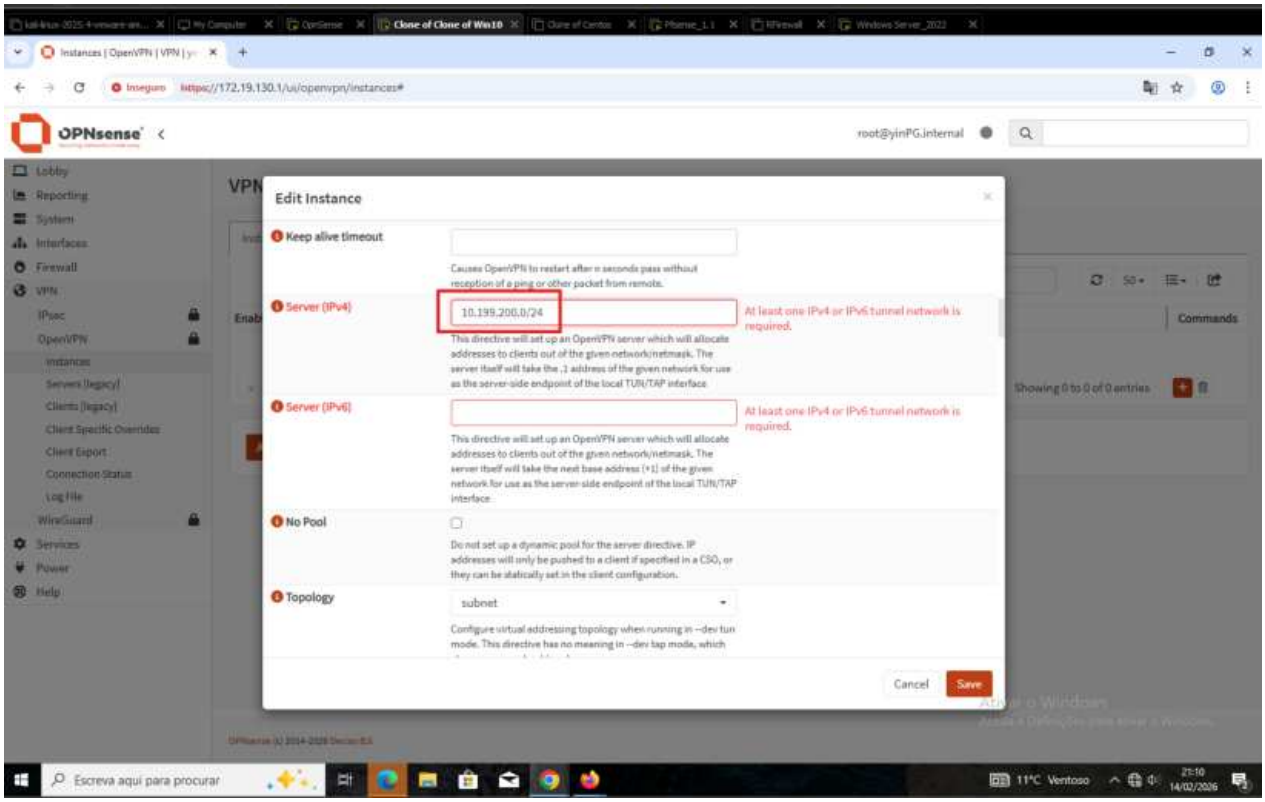
Continuação...



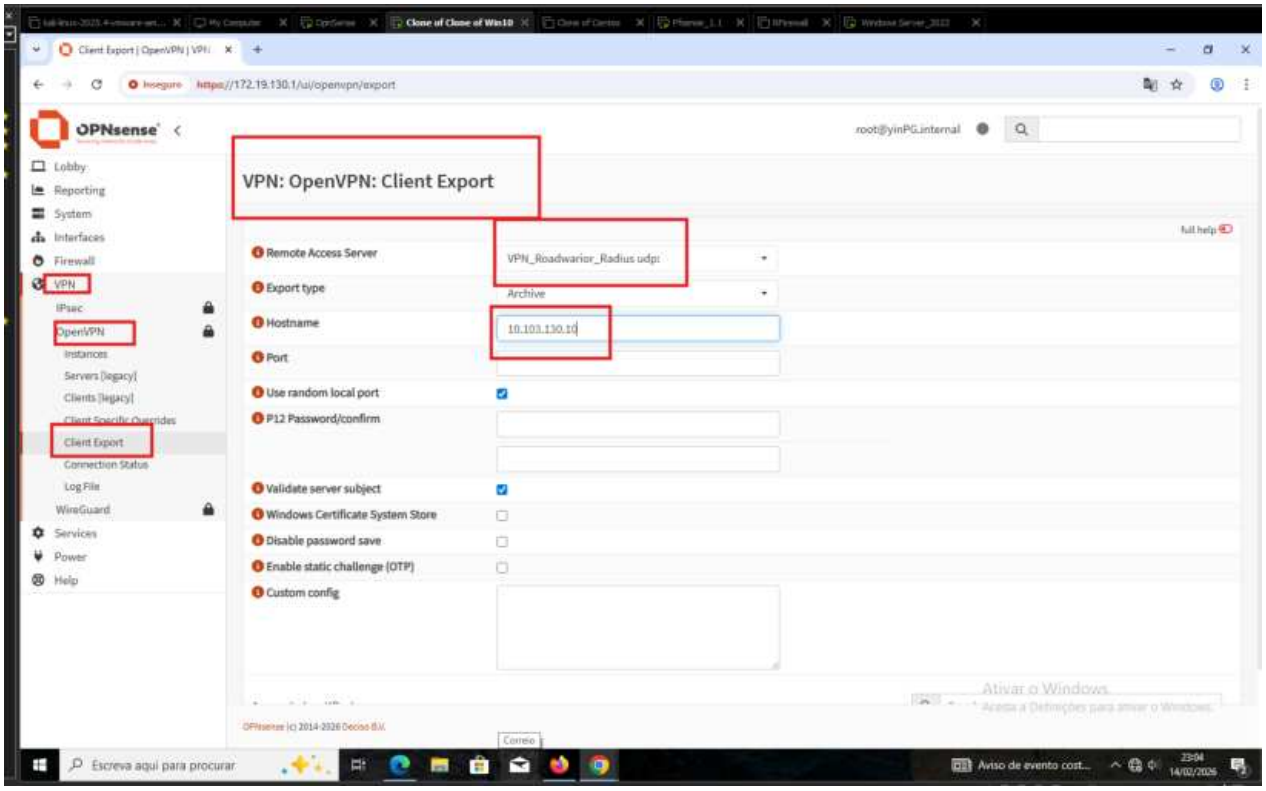
Continuação …



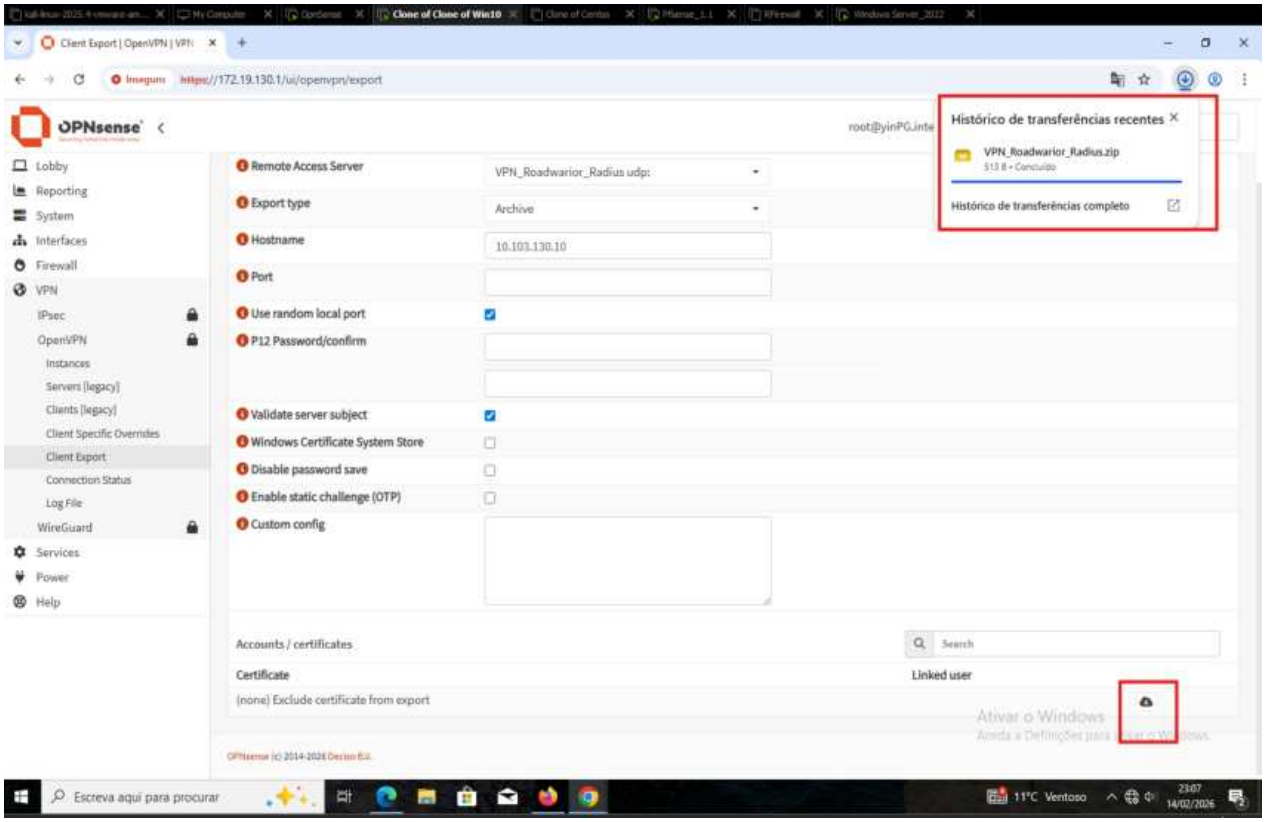
Continuação..



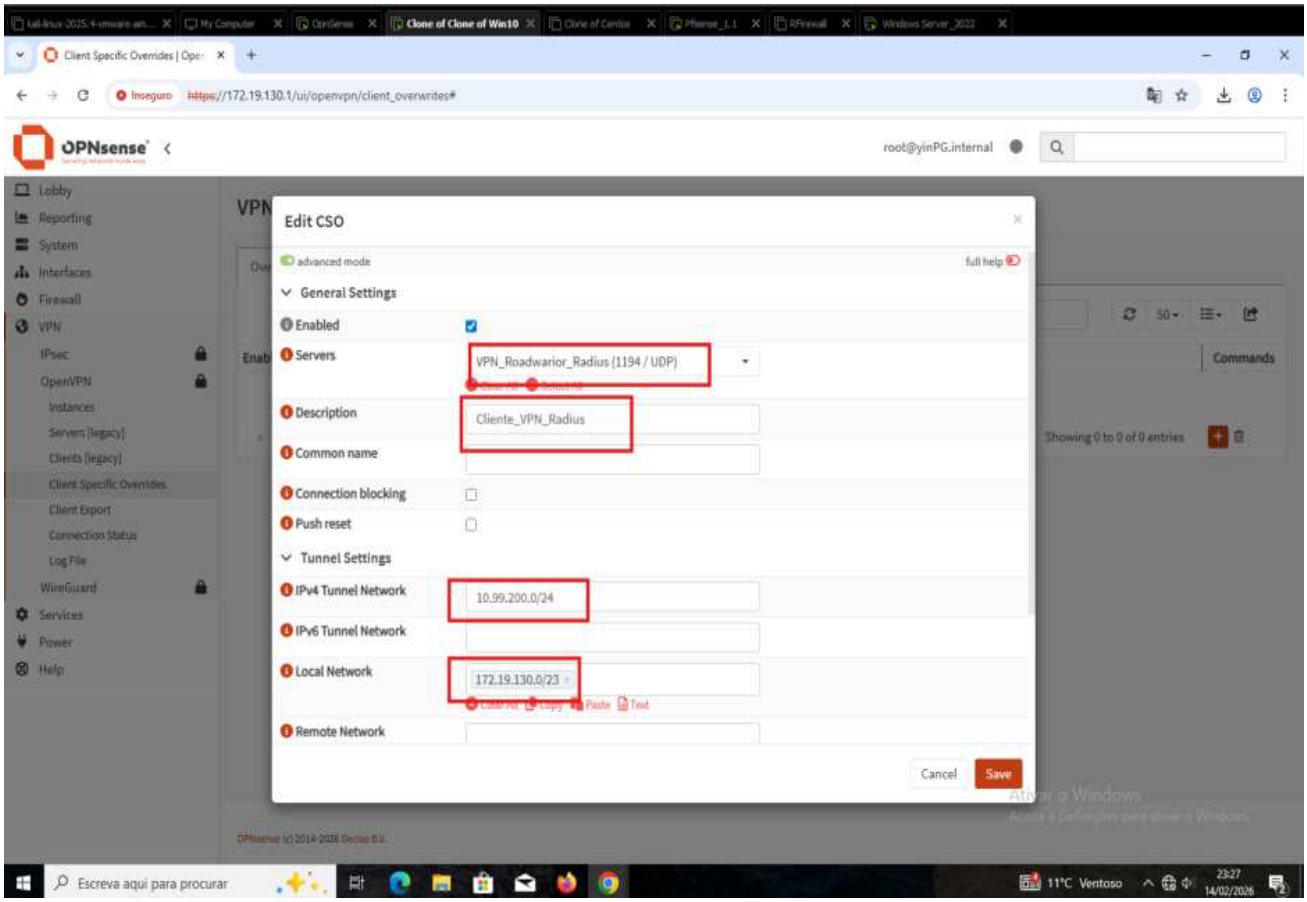
18.4 VPN→ OpenVPN → Client Export



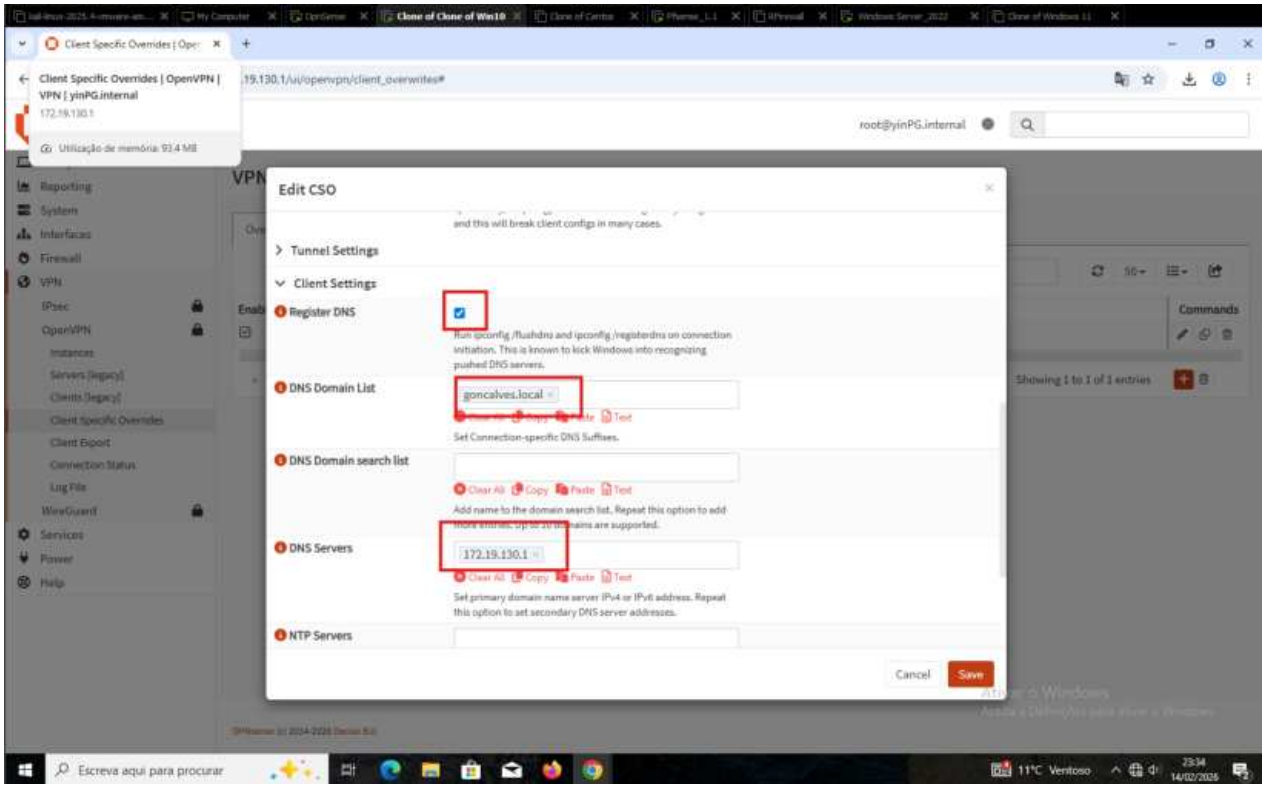
18.5 Download do ficheiro :"VPN_Roadwarrior_Radius.zip"



18.6 VPNOpenVPN Client Specific Overrides



Editar :



18.7 Regras firewall para OpenVPN (OPNsense yinPG)

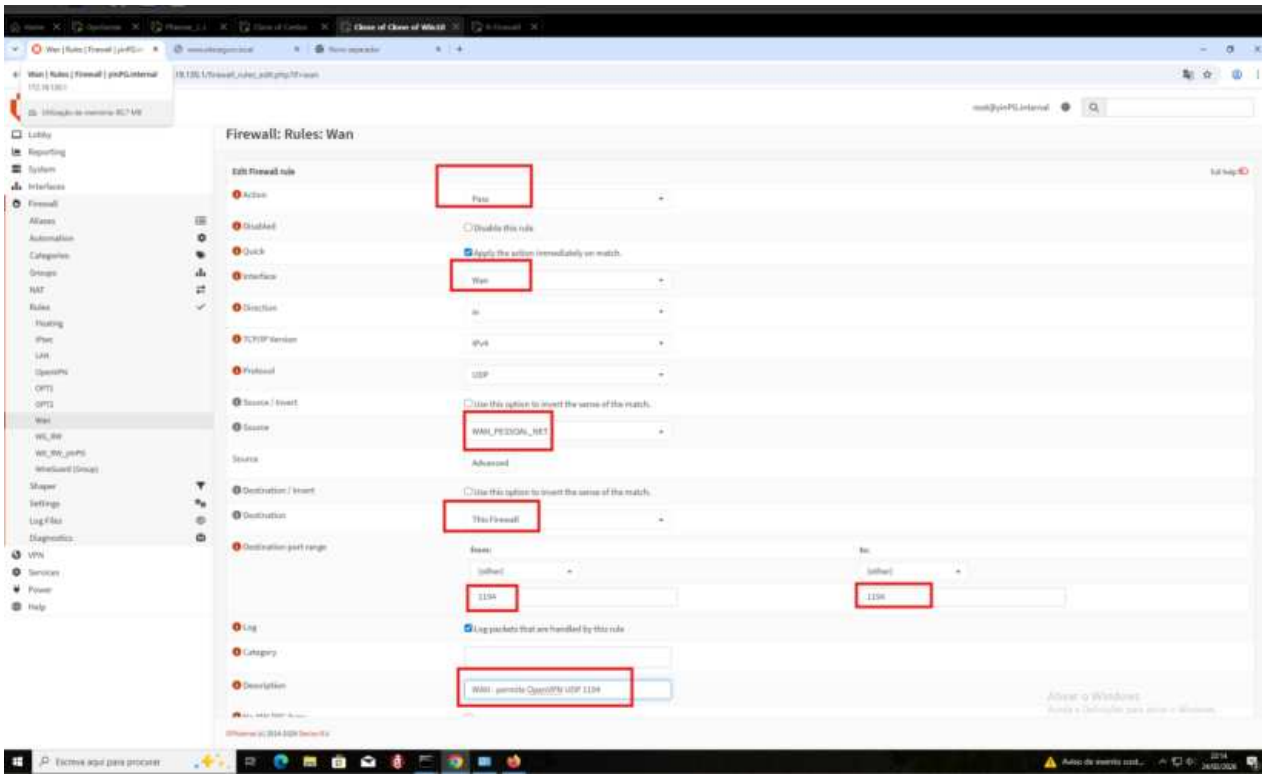
18.7.1 Regra na WAN (abrir UDP 1194 para o OPNsense)

Firewall → Rules → WAN → (+ Add)

- **Action:** Pass
- **Interface:** WAN
- **Direction:** in
- **TCP/IP Version:** IPv4
- **Protocol:** UDP

Source WAN_PESSOAL_NET
Destination: *This Firewall*

- **Destination:** WAN address
- **Destination port range:** 1194 (OpenVPN)



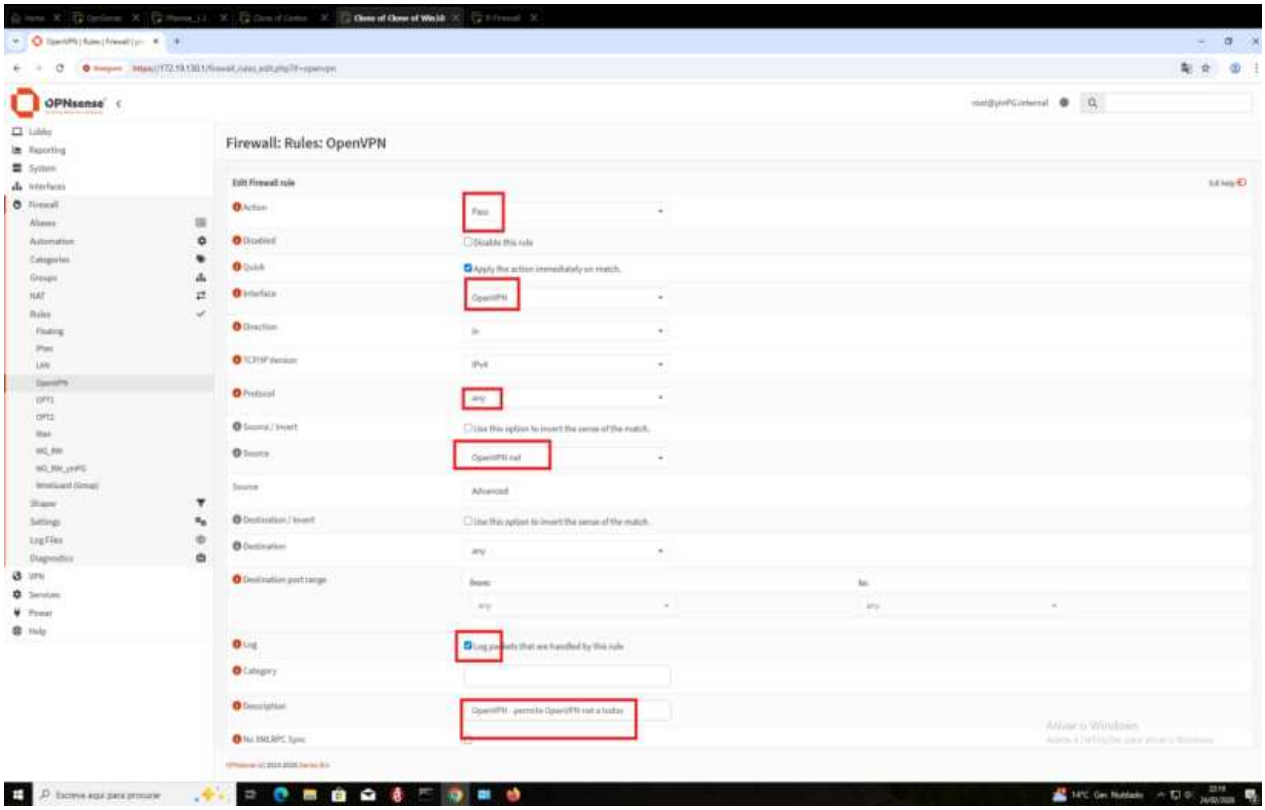
18.7.2 Regra na tab OpenVPN (permitir tráfego do túnel)

Firewall → Rules → OpenVPN → (+ Add)

- **Action:** Pass
- **Interface:** OpenVPN
- **Direction:** in
- **TCP/IP Version:** IPv4
- **Protocol:** any

Source: OpenVPN net
Destination: any

Description: OpenVPN - OpenVPN- permite net a todos

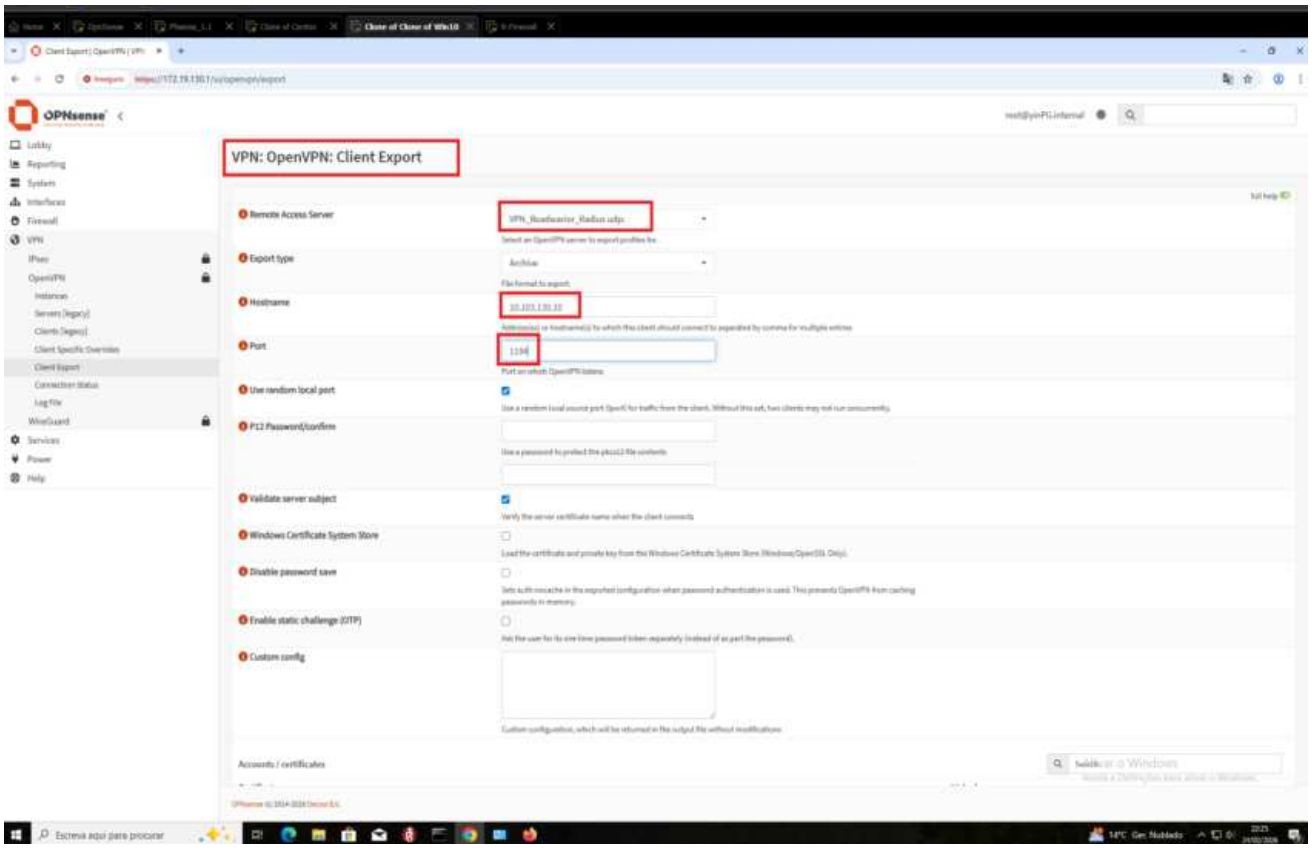


18.7.3 Export do cliente e configuração no Windows 10 (WAN)

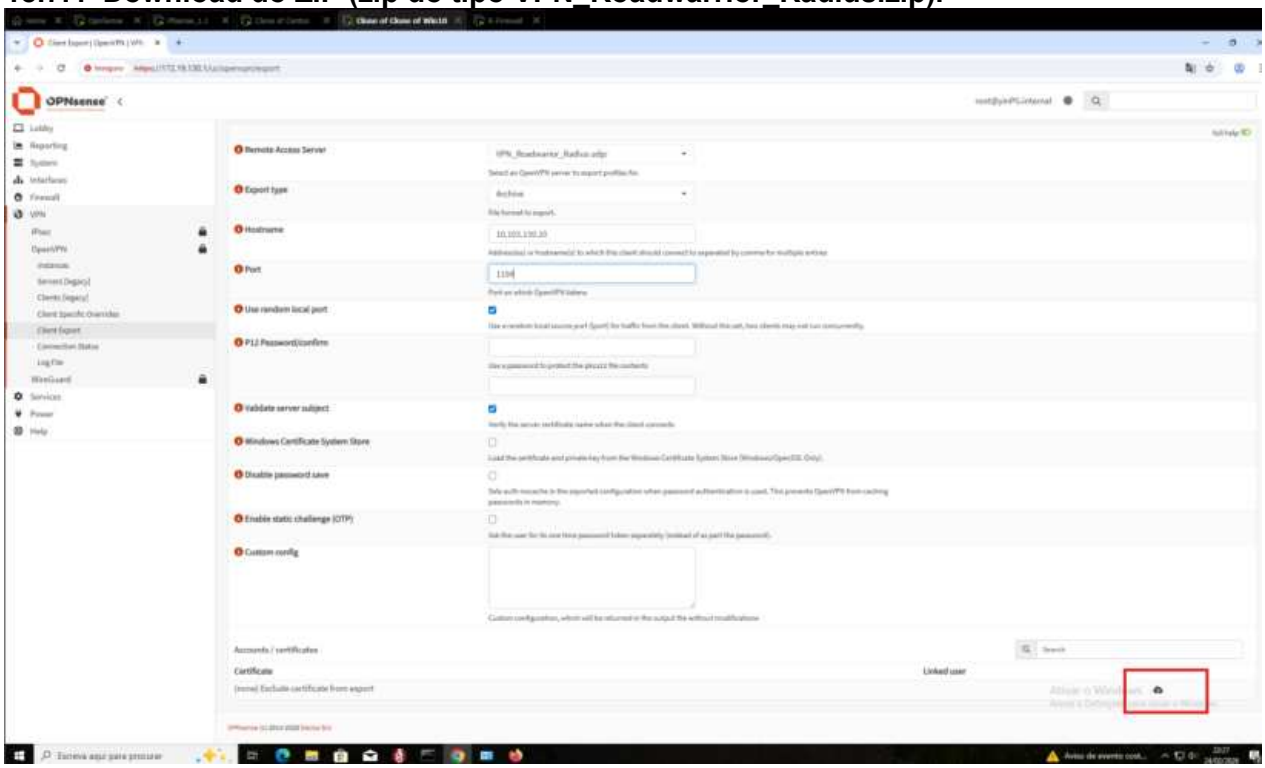
Export no OPNsense

Caminho:

VPN → OpenVPN → Client Export



18.7.4 Download do ZIP (zip do tipo VPN_Roadwarrior_Radius.zip).



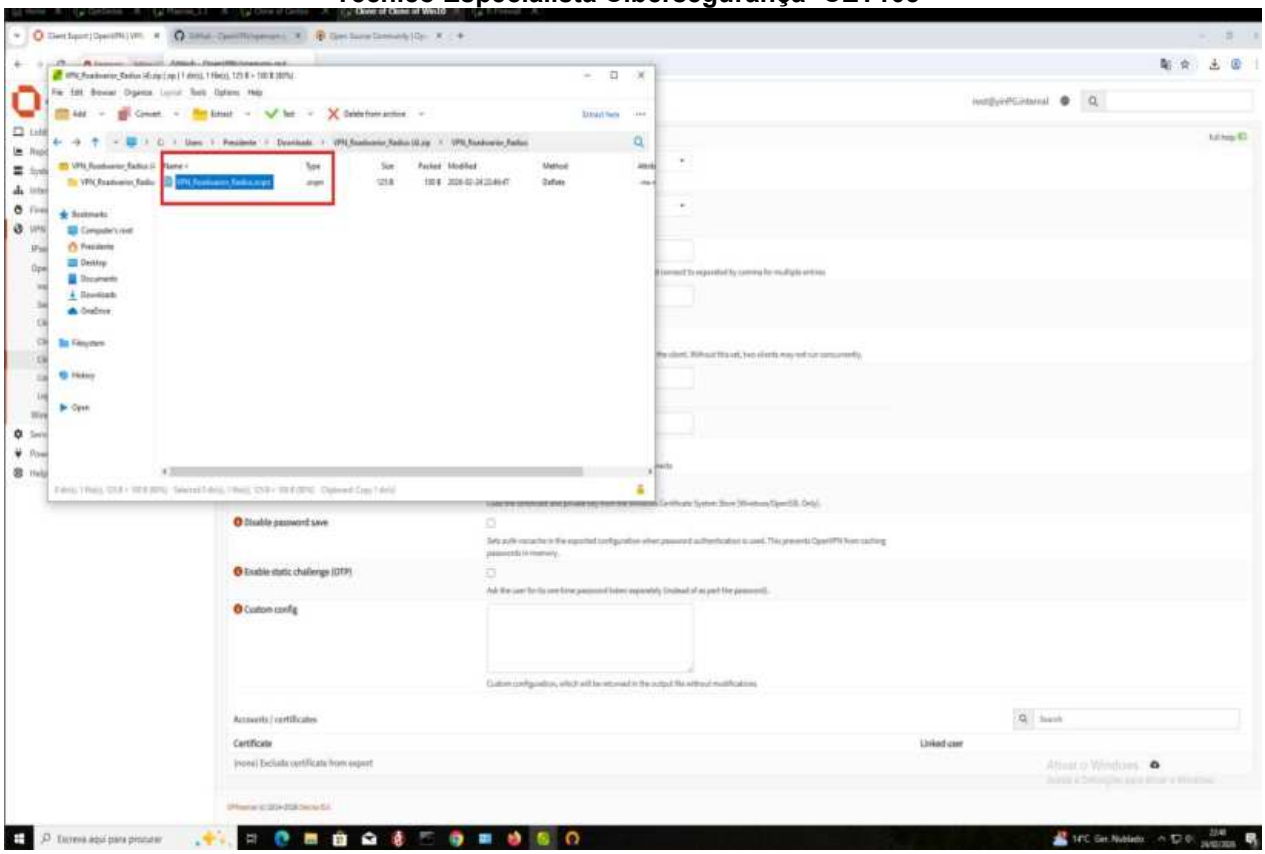
18.7.5 Windows 10: instalar e ligar

No Windows 10 (WAN):

1. **Instalar OpenVPN GUI**

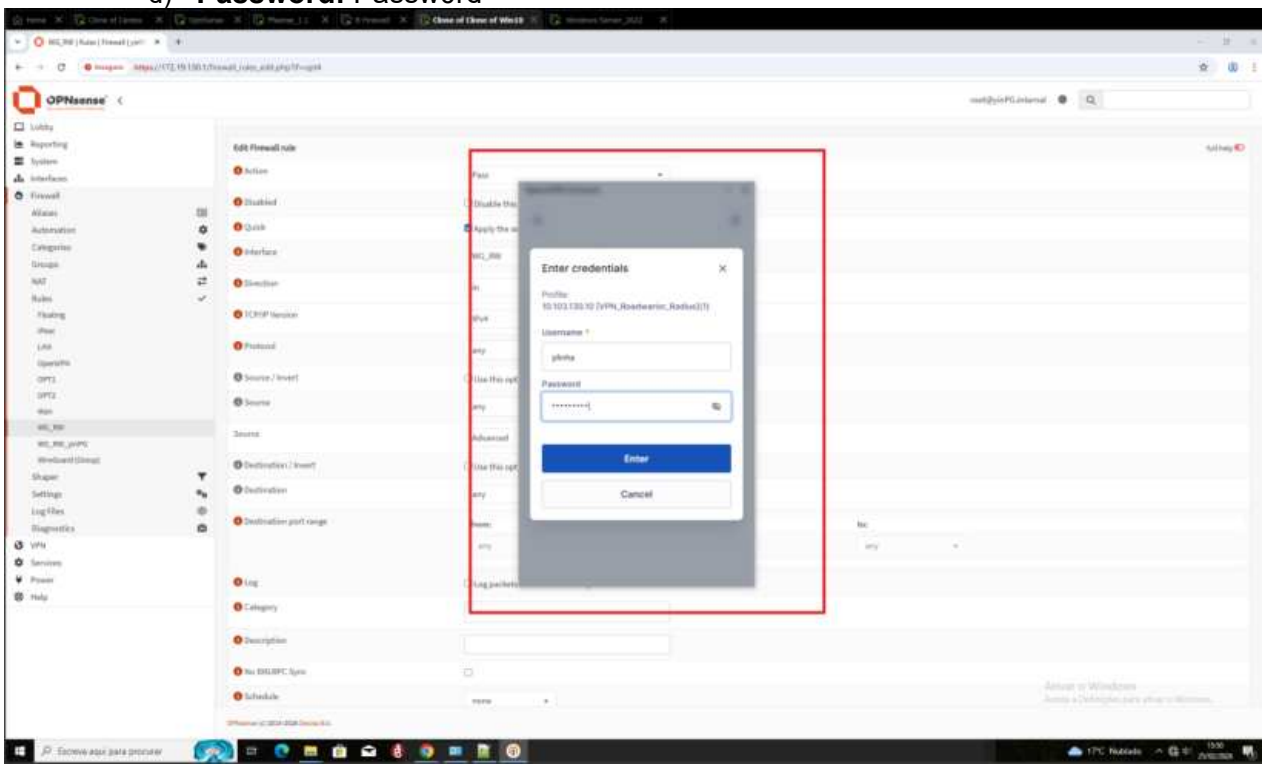
Instalação do OpenVPN GUI :





C:\Program Files\OpenVPN\config\

- Abrir o **OpenVPN GUI**:
    a) botão direito → **Run as Administrator**
    b) ícone junto ao relógio (system tray) → botão direito → **Connect**
- credenciais:
    c) **Username:** plinha
    d) **Password:** Passw0rd



3 VPNs----------

"j) Crie uma VPN Wireguard entre as duas firewall yangNA e yinNA. Não pode passar trafego para acesso a SSH ou RDP nesta VPN. Mas qualquer outro tráfego é possível. Não se esqueça das imagens das regras! (Ping não é suficiente!!!!)"

19) VPN 3.h/3.i — WireGuard Roadwarrior no yangPG (pfSense) + full tunnel mas sem ping a 8.8.8.8/1.1.1.1

19.1) pfSense — instalar WireGuard

19.2- IPs base do yangPG

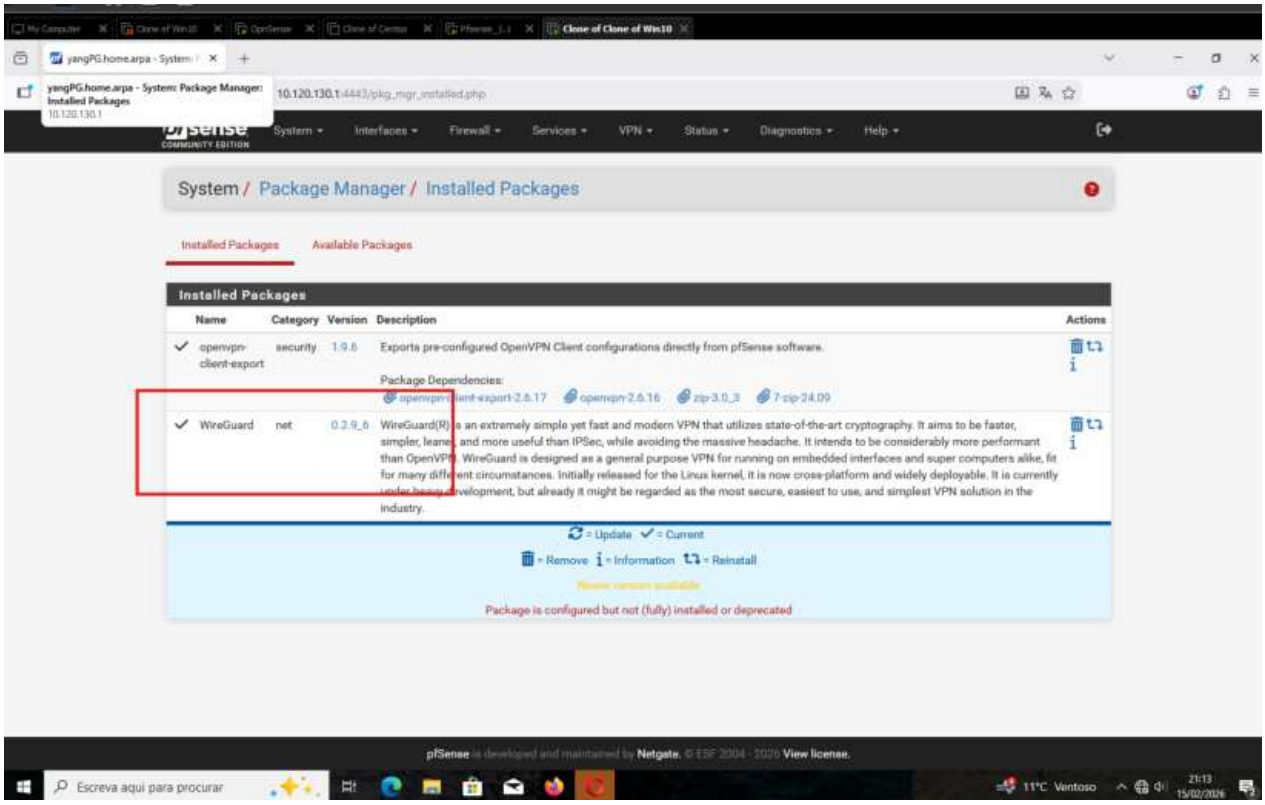
- **WAN yangPG**: 10.103.130.11/27
- **Gateway WAN**: 10.103.130.1
- **LAN/Green yangPG: 10.120.130.1/24**

System -> Package Manager → Available Packages → WireGuard → Install

System → Package Manager → Available Packages

a) Procurar **wireguard**

b) **Install**



19.3) — Criar o "Tunnel" (Servidor WG) no pfSense

VPN → WireGuard → Tunnels → Add

Geral

a) **Enable**: yes

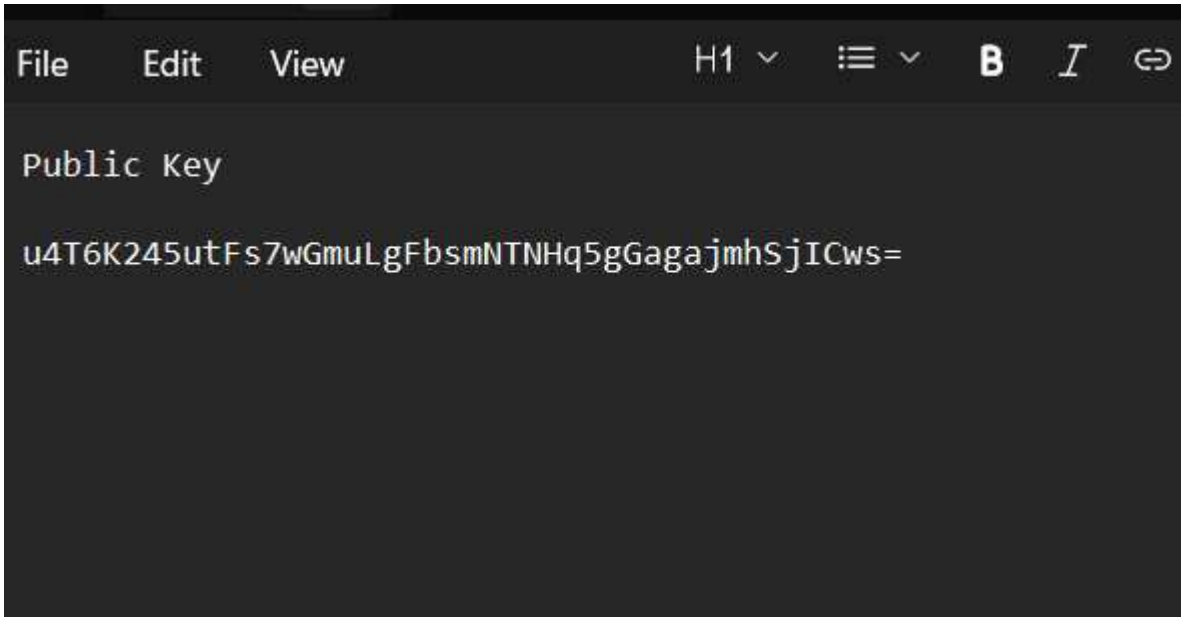
b) **Description**: WG_RW_yangPG

c) **Listen Port**: 51820

Keys

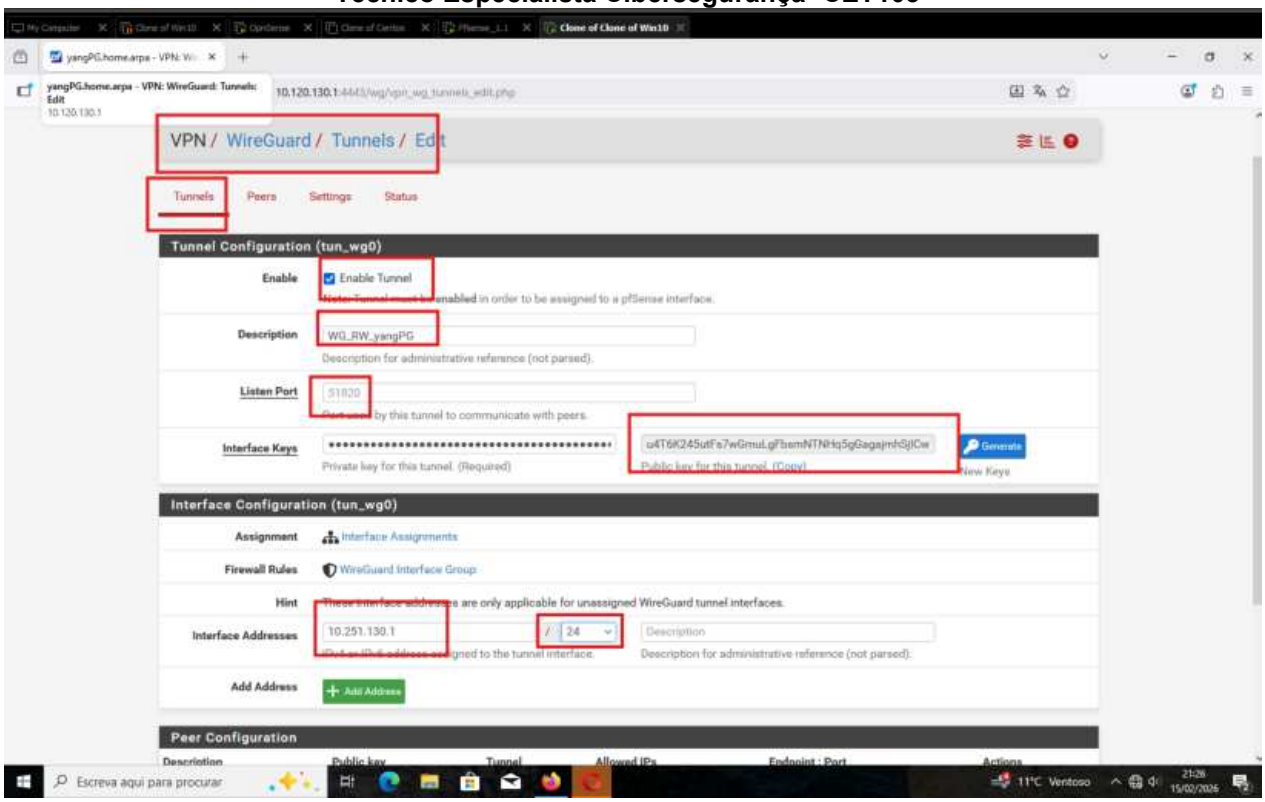
a) Carregar em **Generate** (para criar Private/Public key do servidor no pfSense)

b) Guardar **Public Key** (será necessária no cliente Windows)



19.4) Interface Address (rede do túnel)

a) Interface Address: 10.251.130.1/24 ---Esta é a “rede virtual” do túnel



19.5) — Criar "Peer" (o cliente Windows Roadwarrior)

VPN → WireGuard →Peers →Add

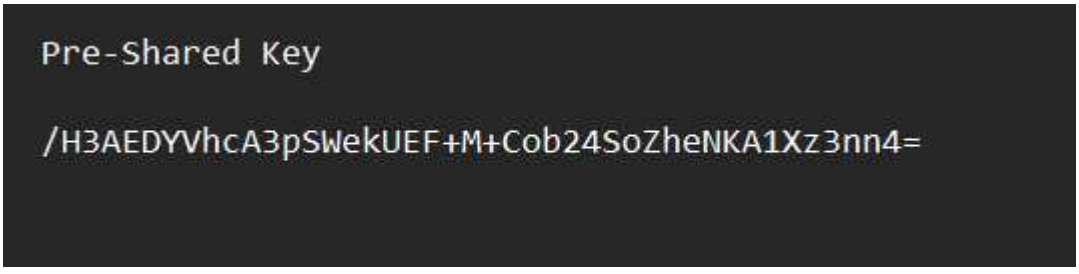
- Enable: sim

- Tunnel: seleciona WG_RW_yangPG

- Description: Peer_Win10_WAN

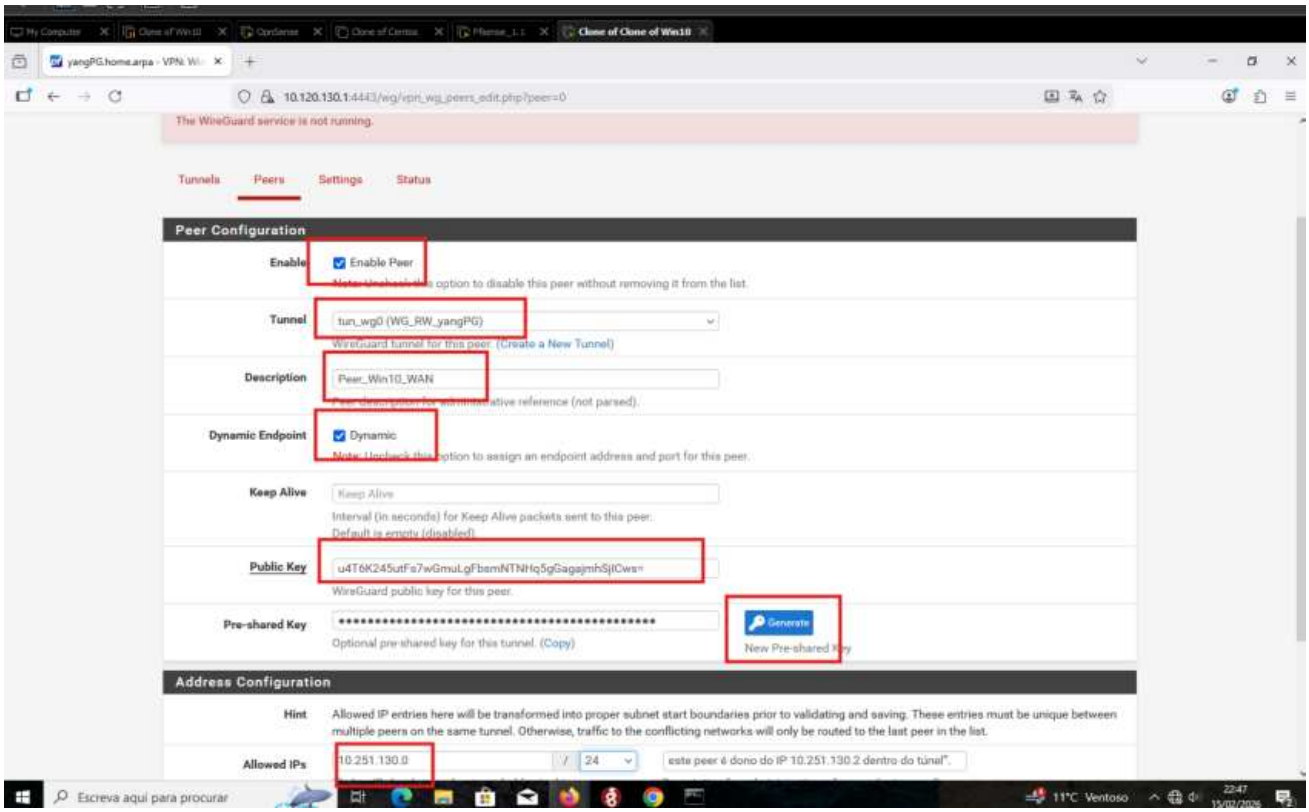
Public Key

- Colar a **Public Key do Windows**
- Pre-Shared Key > gera/define uma PSK
- Guardar a PSK para porteriormente meter também no Windows
-

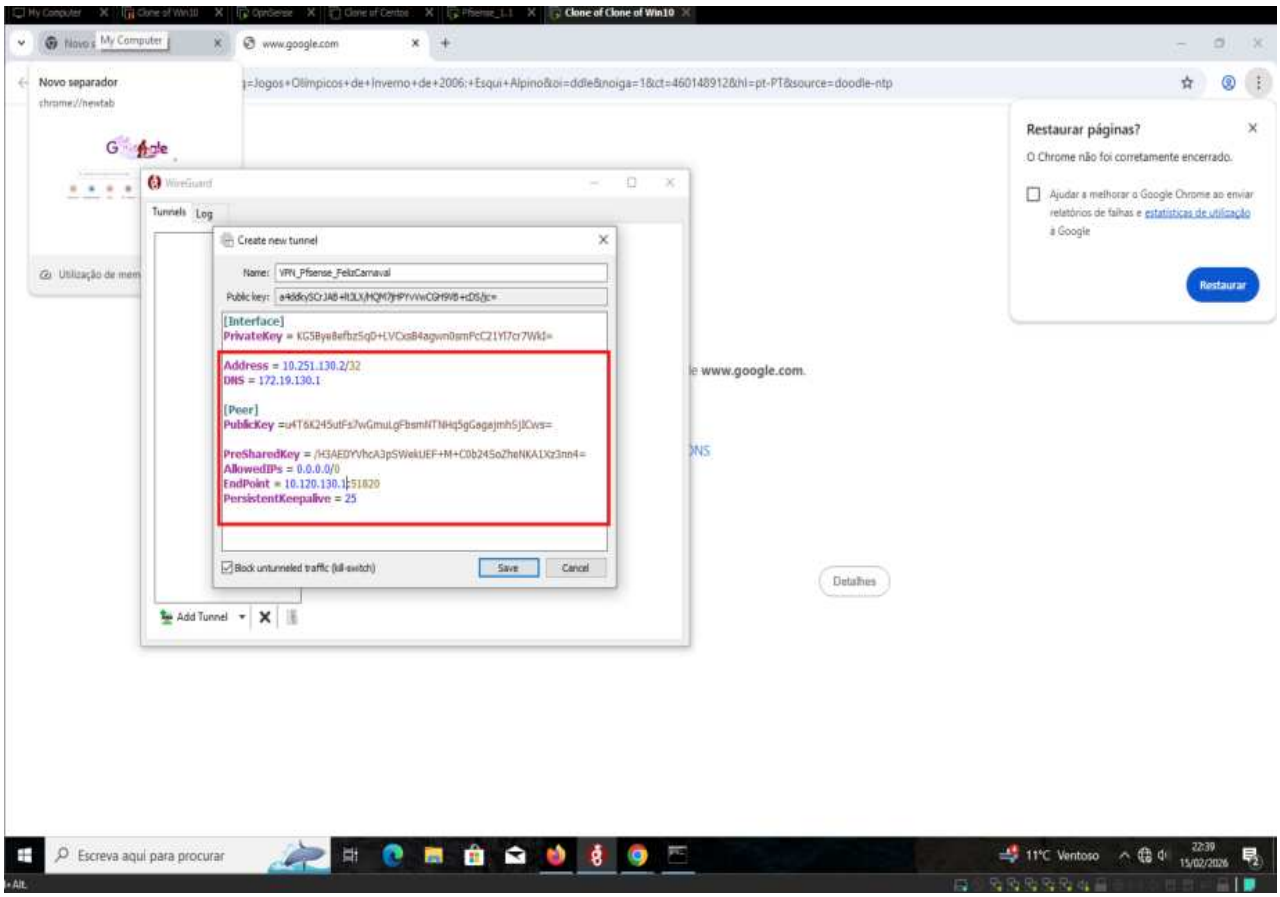


Allowed IPs

Allowed IPs: 10.251.130.0/24

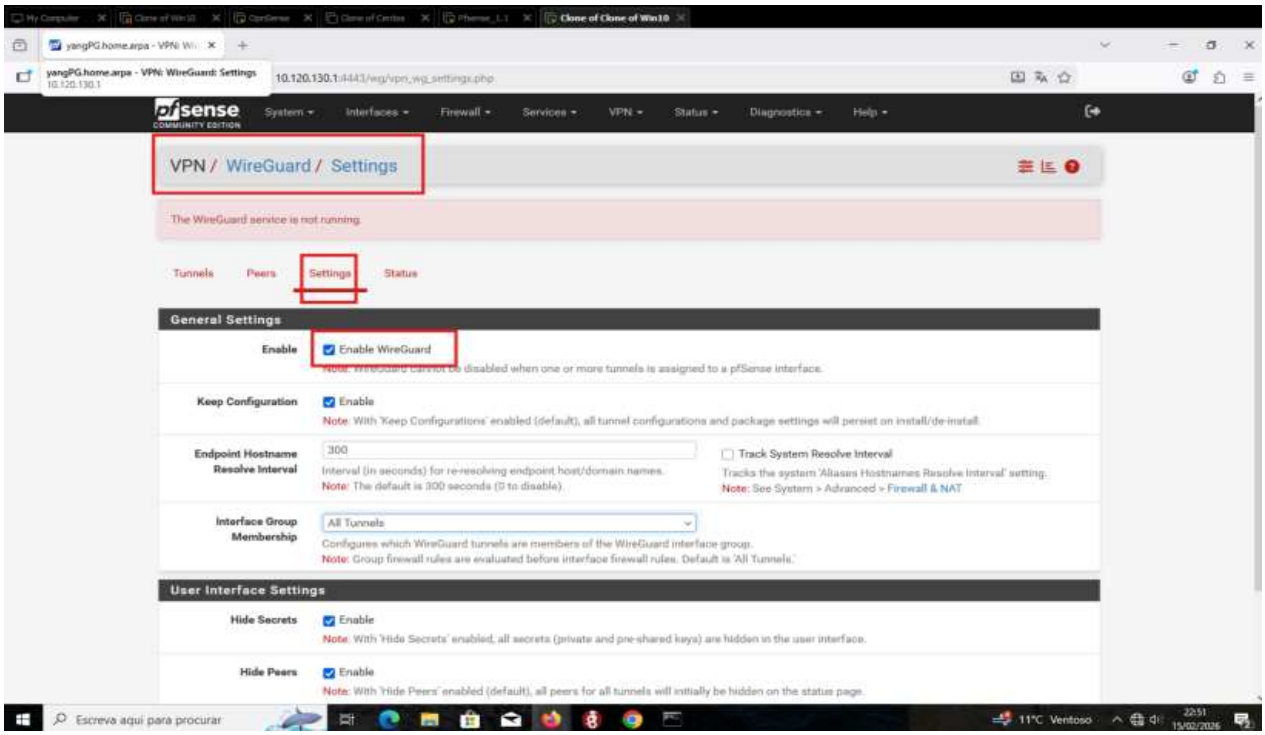


19.6) -Editar no wireguard



19.7 -Vpn Wireguard Settings

“enable Wireguard “



19.8 — Regra na WAN do pfSense para permitir WireGuard (UDP 51820)

Nota- esta regra é opcional porque é usado wireguard do Windows – software previamente instalado

Firewall → Rules → WAN → Add (Pass)

Action: Pass

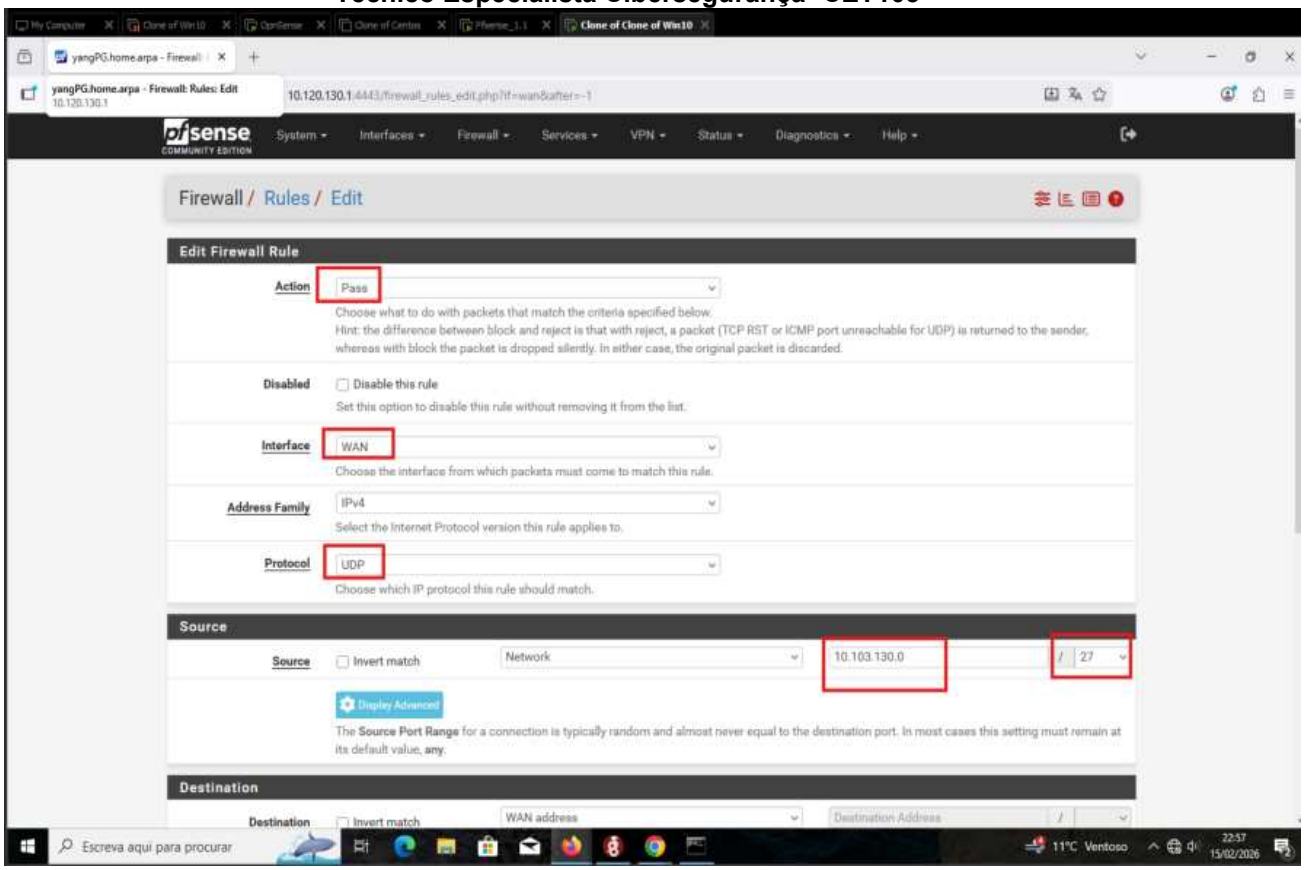
Protocol: UDP

Source: 10.103.130.0/27

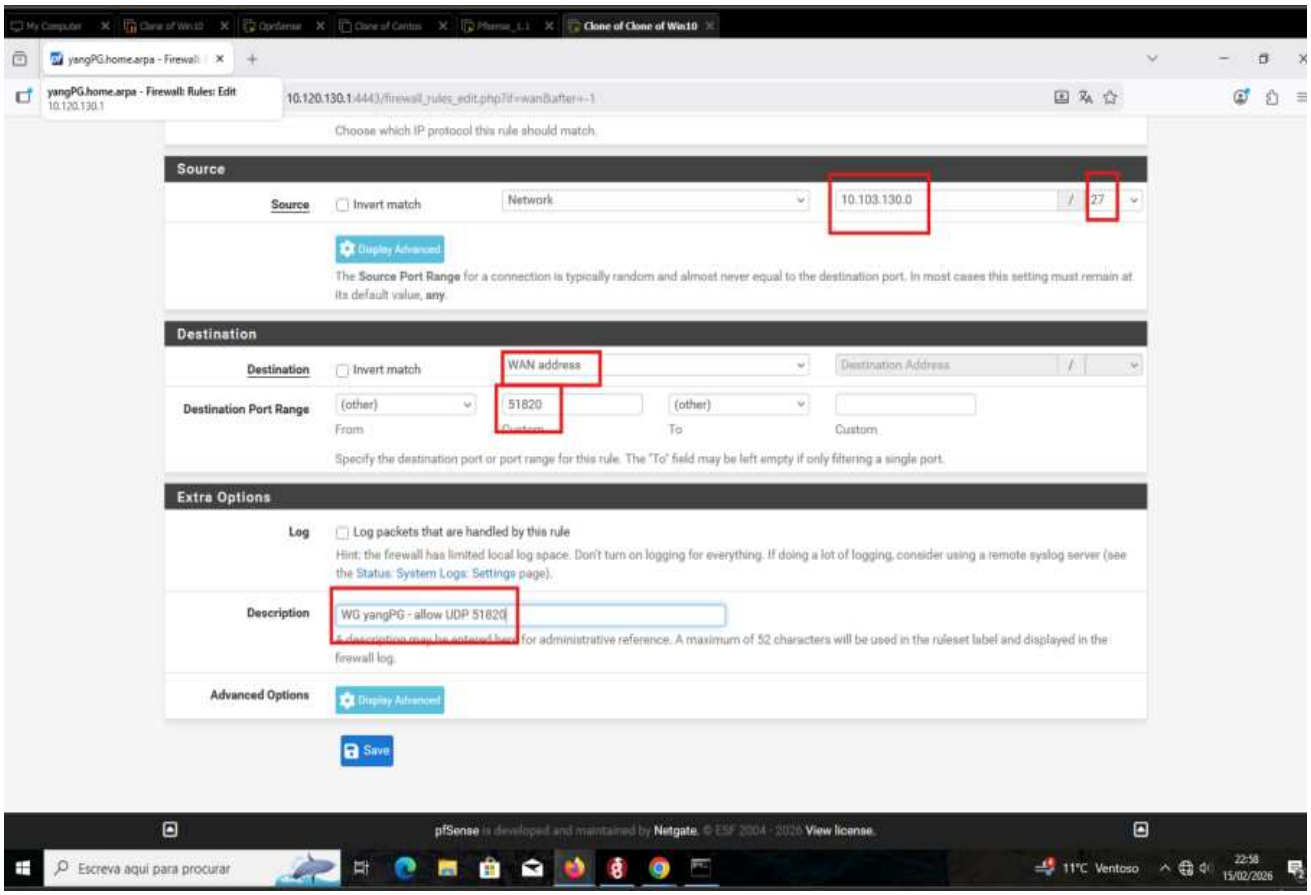
Destination: **WAN address** (do pfSense yang)

Destination port: 51820

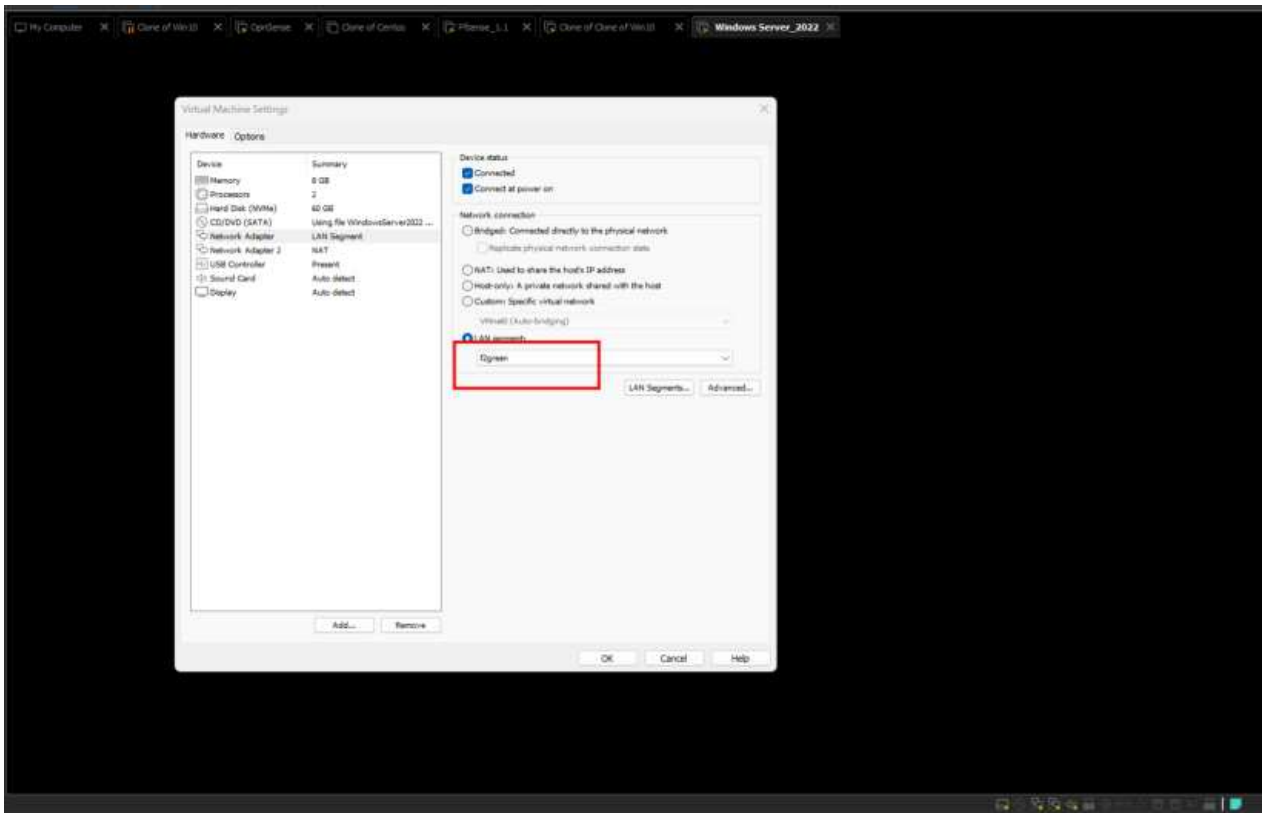
Description: WG yangPG – permite UDP_51820



Continuação..



19.9) Colocar A interface do Windows Server em f2 Green -pfsense

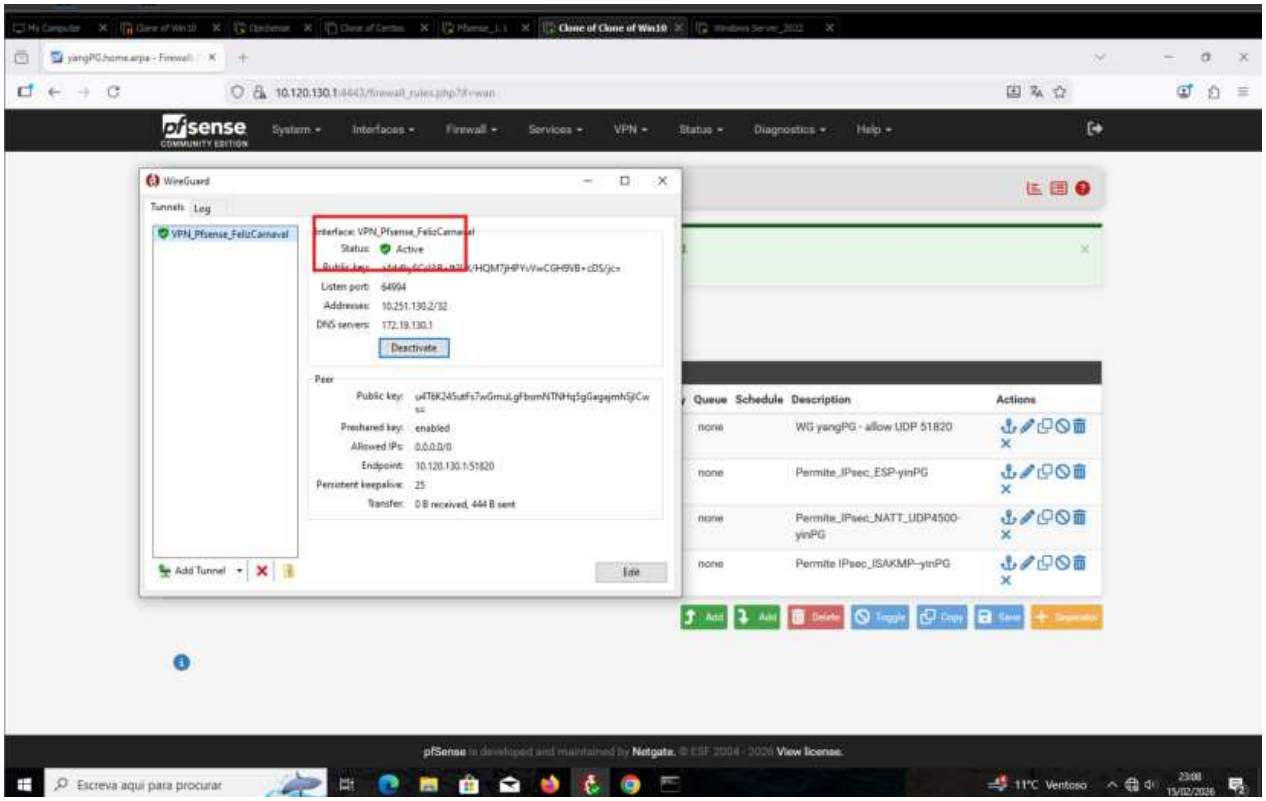


19.10) Ativação de túnel wireguard

VPN liga

a) WireGuard: “Activate”

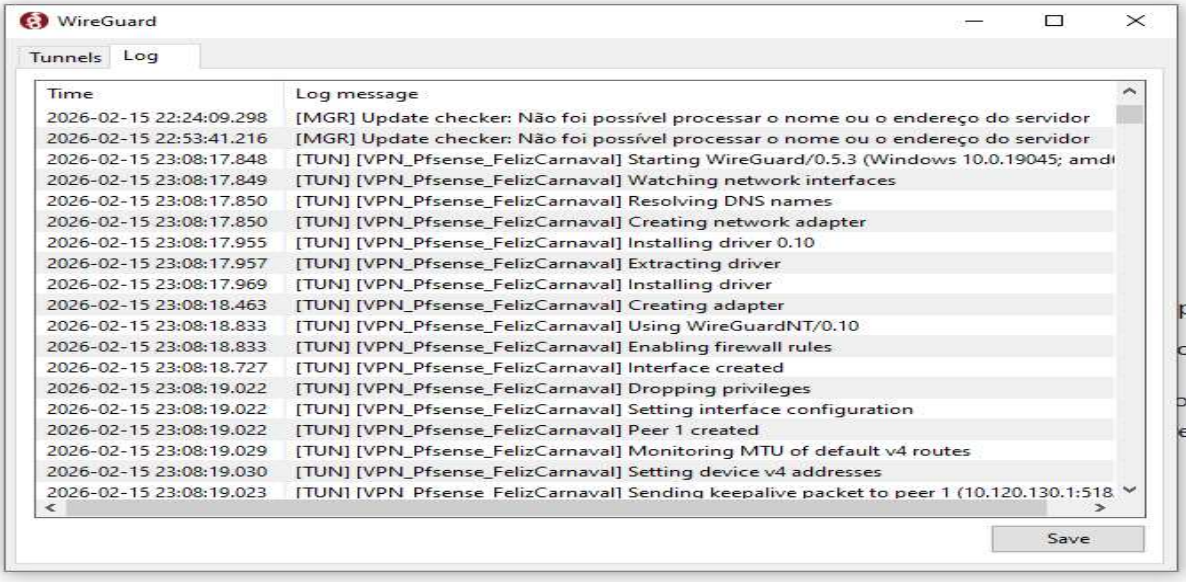
b) Print do estado “Active”



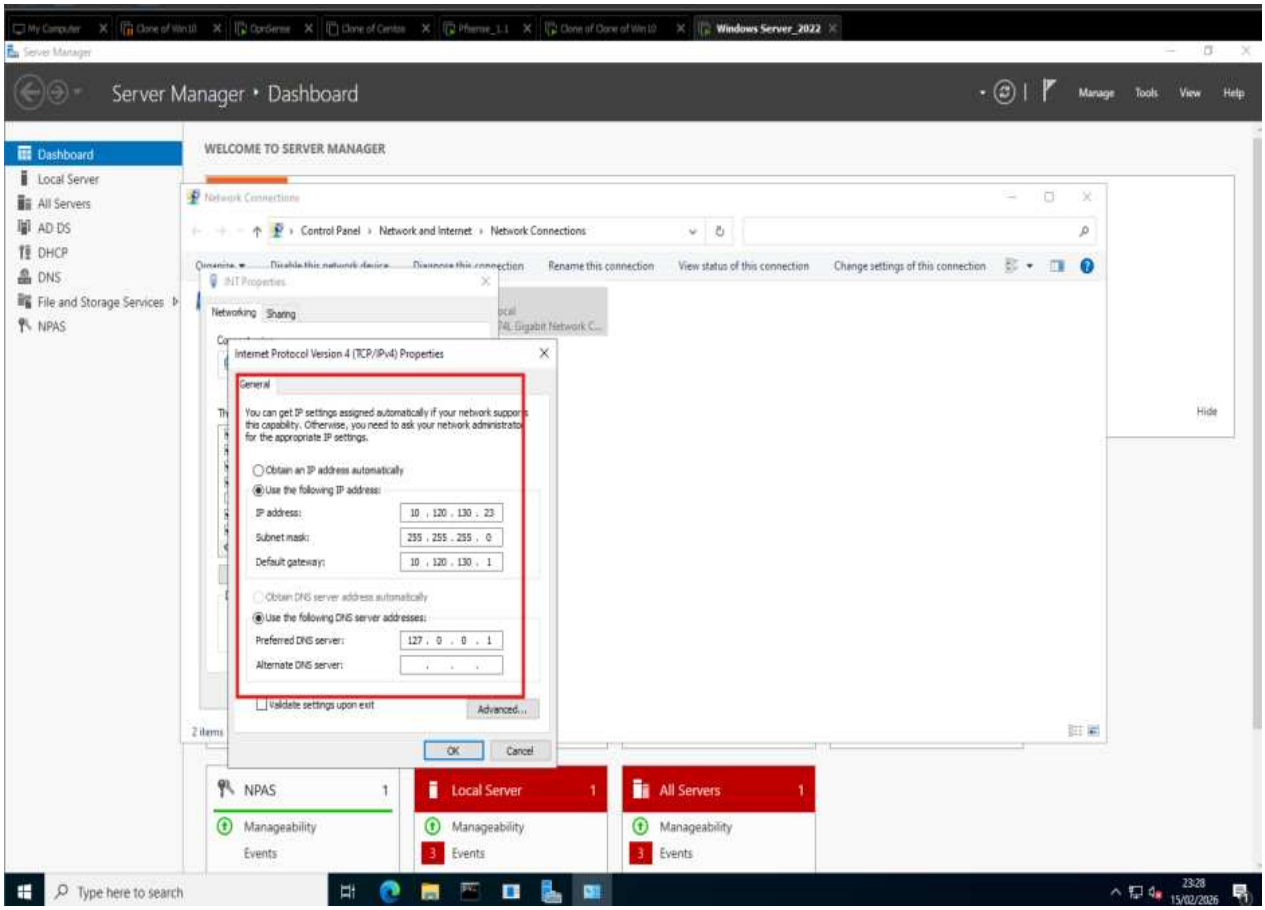
19.11 — Testes

O enunciado obriga validação no **Windows cliente** e diz que **ping é insuficiente**.

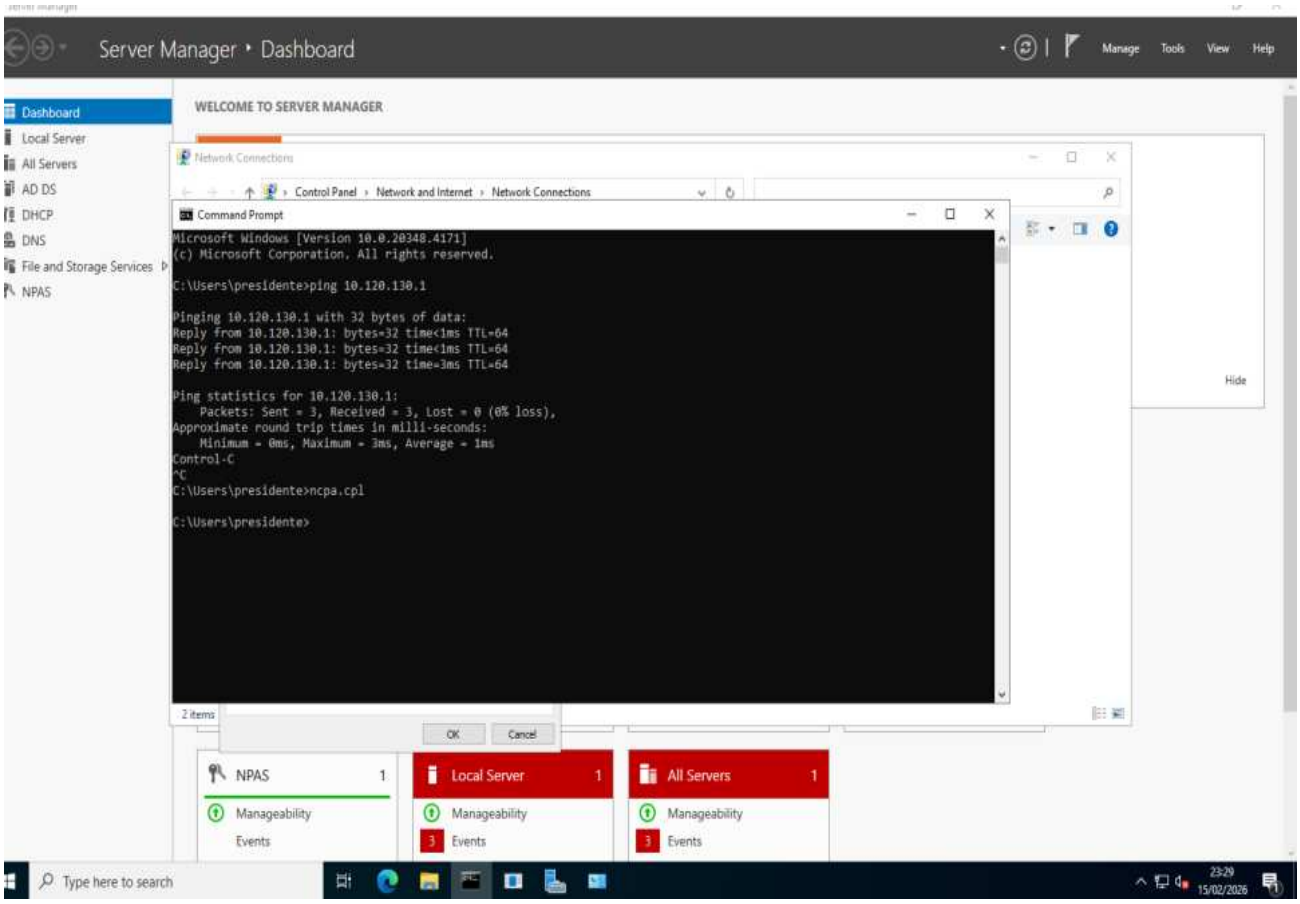
- **WireGuard** -Logs



19.12 -Alteração de interfaces NIC-pf sense no Windows server



19.13 -Teste ping no Windows Server



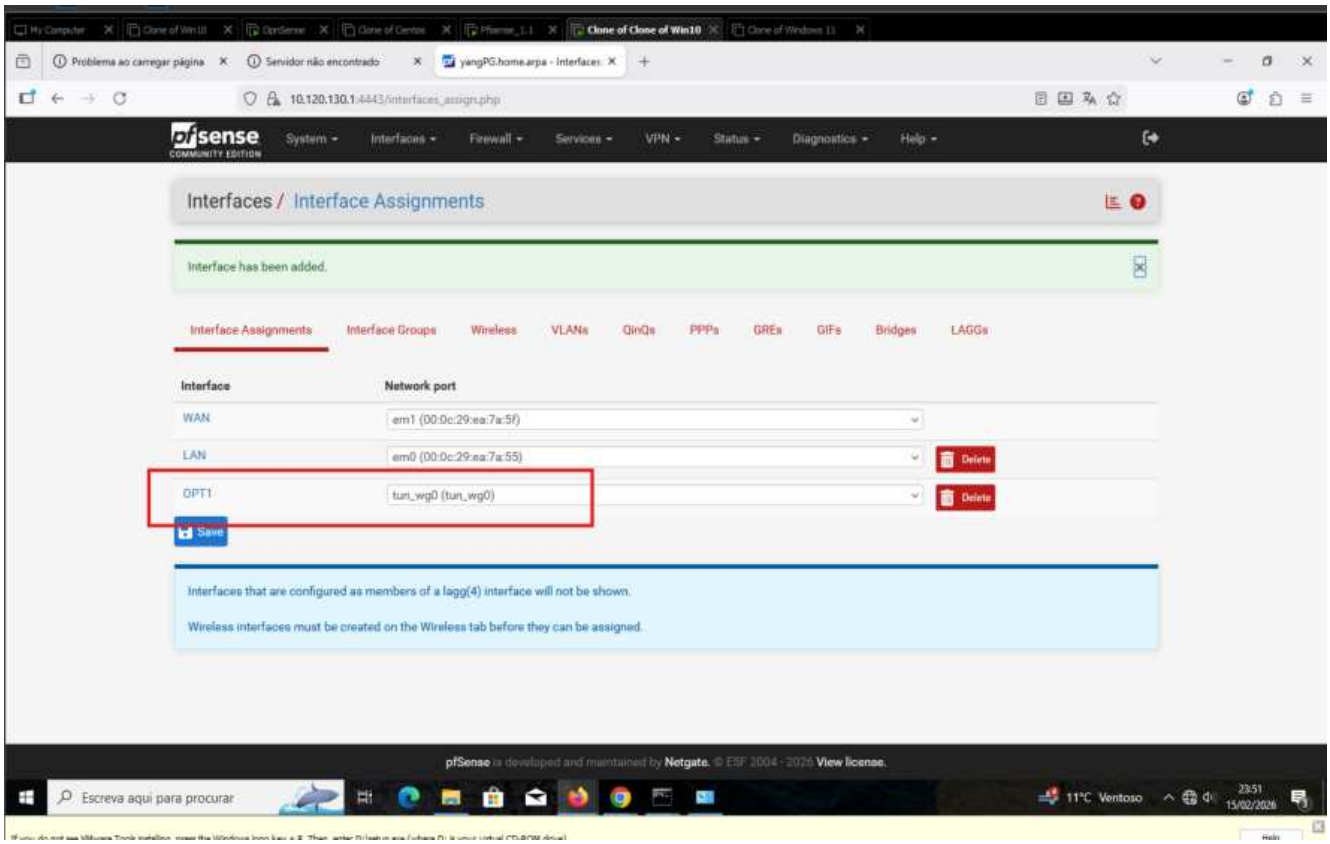
19.14 — Atribuir Interface WG e Regras

19.14.1 Assign interface

Interfaces → Assignments

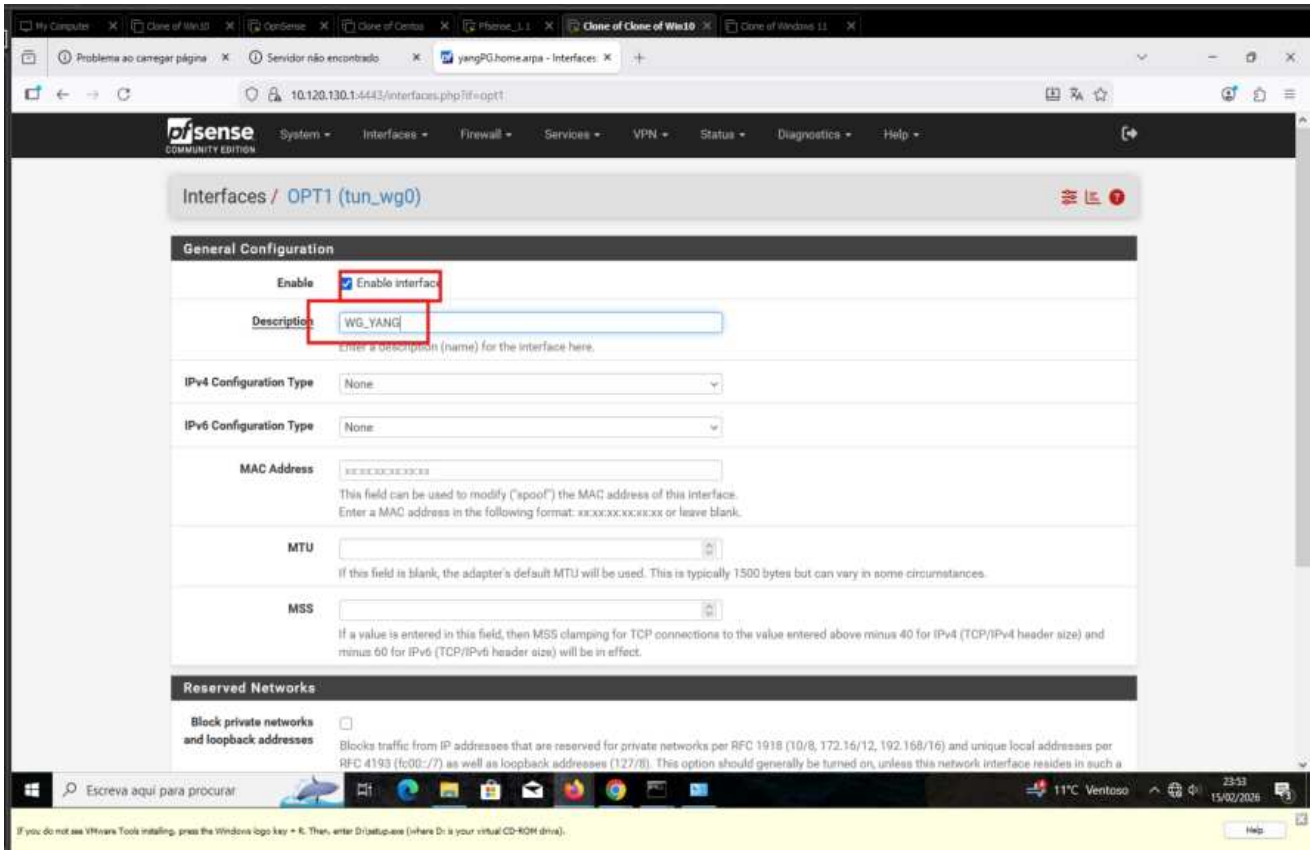
a) Em “Available network ports” escolher OPT1

b) Clicar na nova interface (OPT1



c) **Enable** yes

d) **Description**: WG_YANG

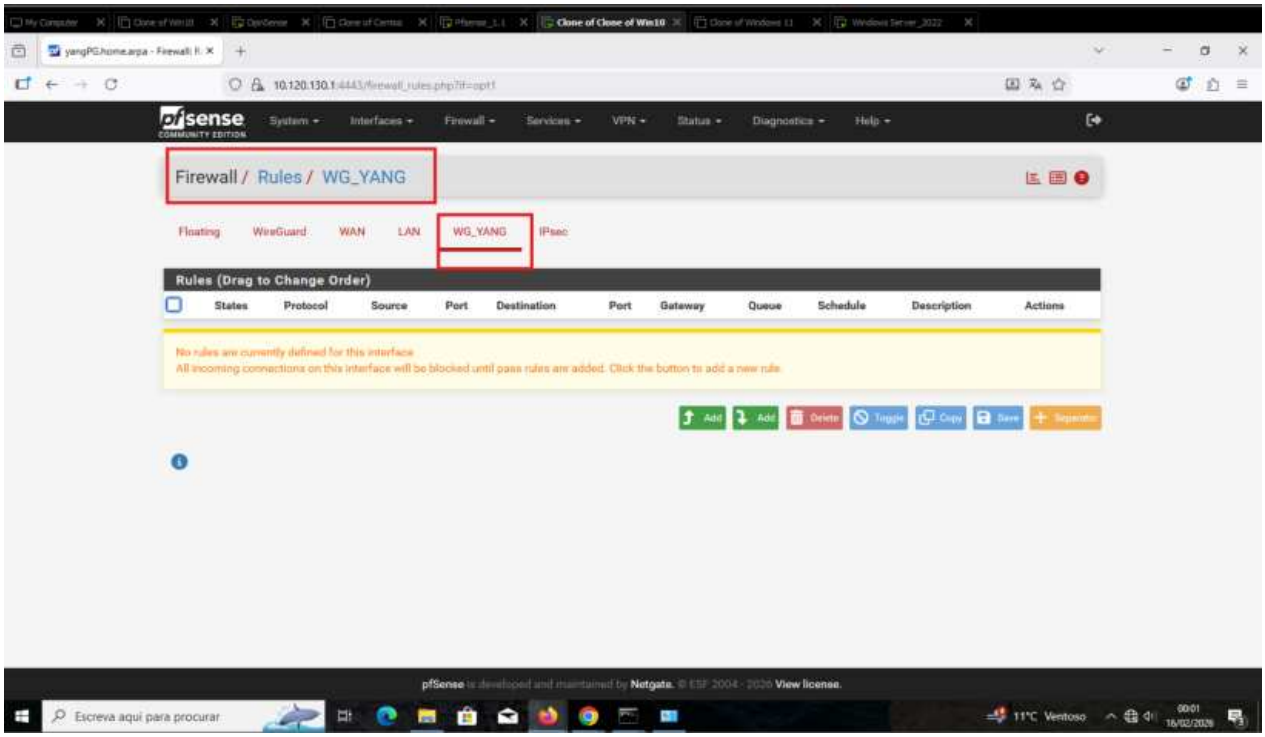


3-VPNs----------

“i) Demostre que todo o tráfego desta VPN passa pelo túnel da VPN, quando o cliente acede à internet, mas o cliente não consegue pingar o IP 8.8.8.8 ou o 1.1.1.1. Pode utilizar o OpenDNS ou transfira o Windows Server para a zona green do yangNA para permitir a resolução de DNS e assim possibilitar navegar na internet “

20) — Regras no Interface WG_YANG

Firewall → Rules → WG_YANG



20.1-Regra 1 — BLOQUEAR ping a 8.8.8.8

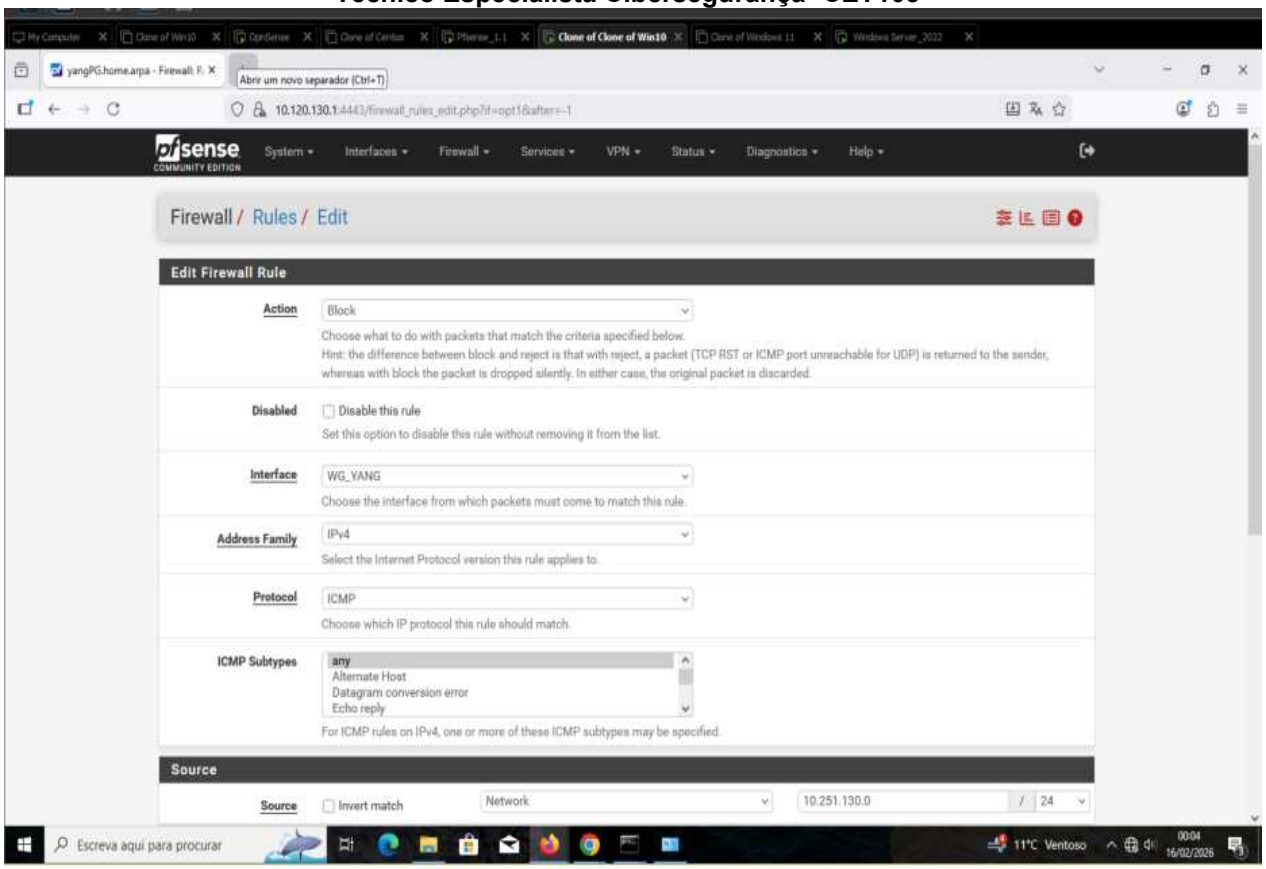
- Action: Block

- Protocol: ICMP

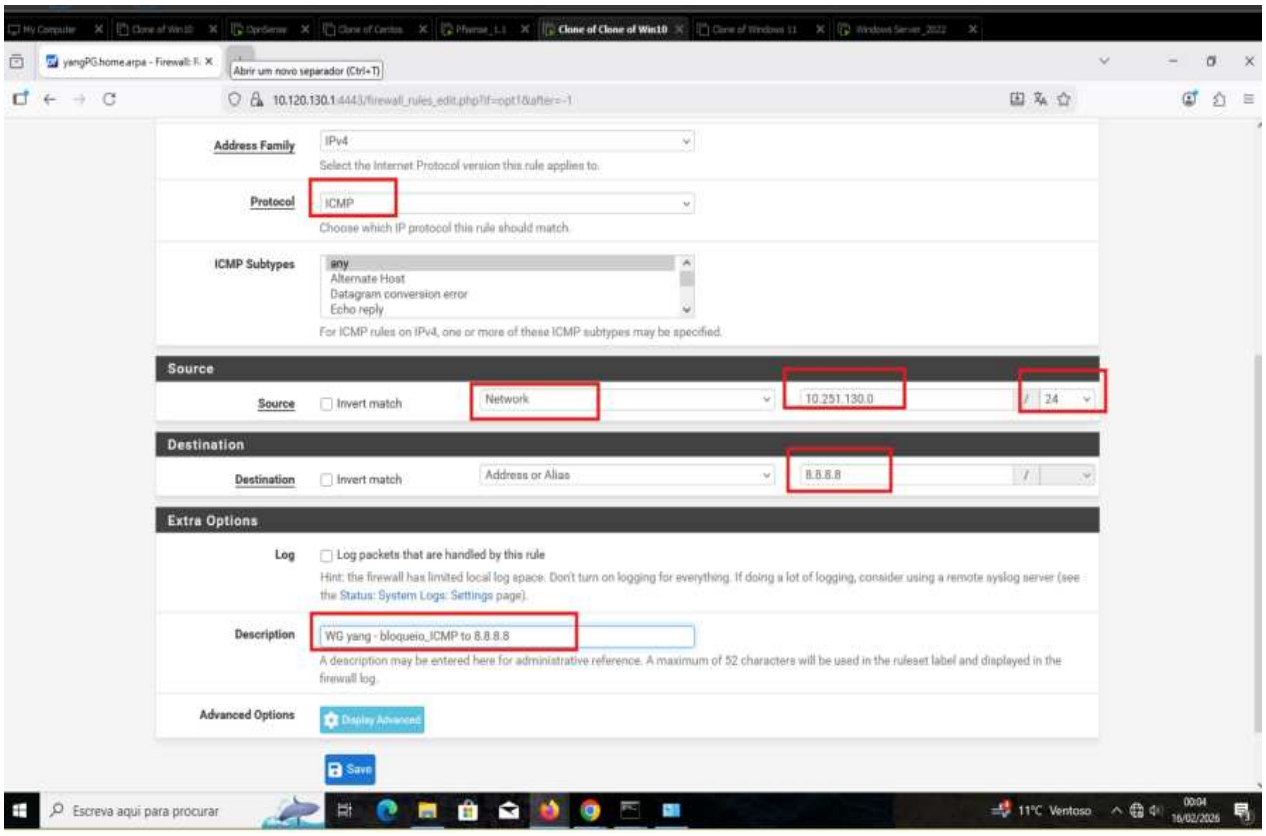
- Source: 10.251.130.0/24

- Destination: 8.8.8.8

- Description: WG yang – bloqueio_ICMP to 8.8.8.8



Continuação …



20.2 Regra 2 — BLOQUEAR ping a 1.1.1.1

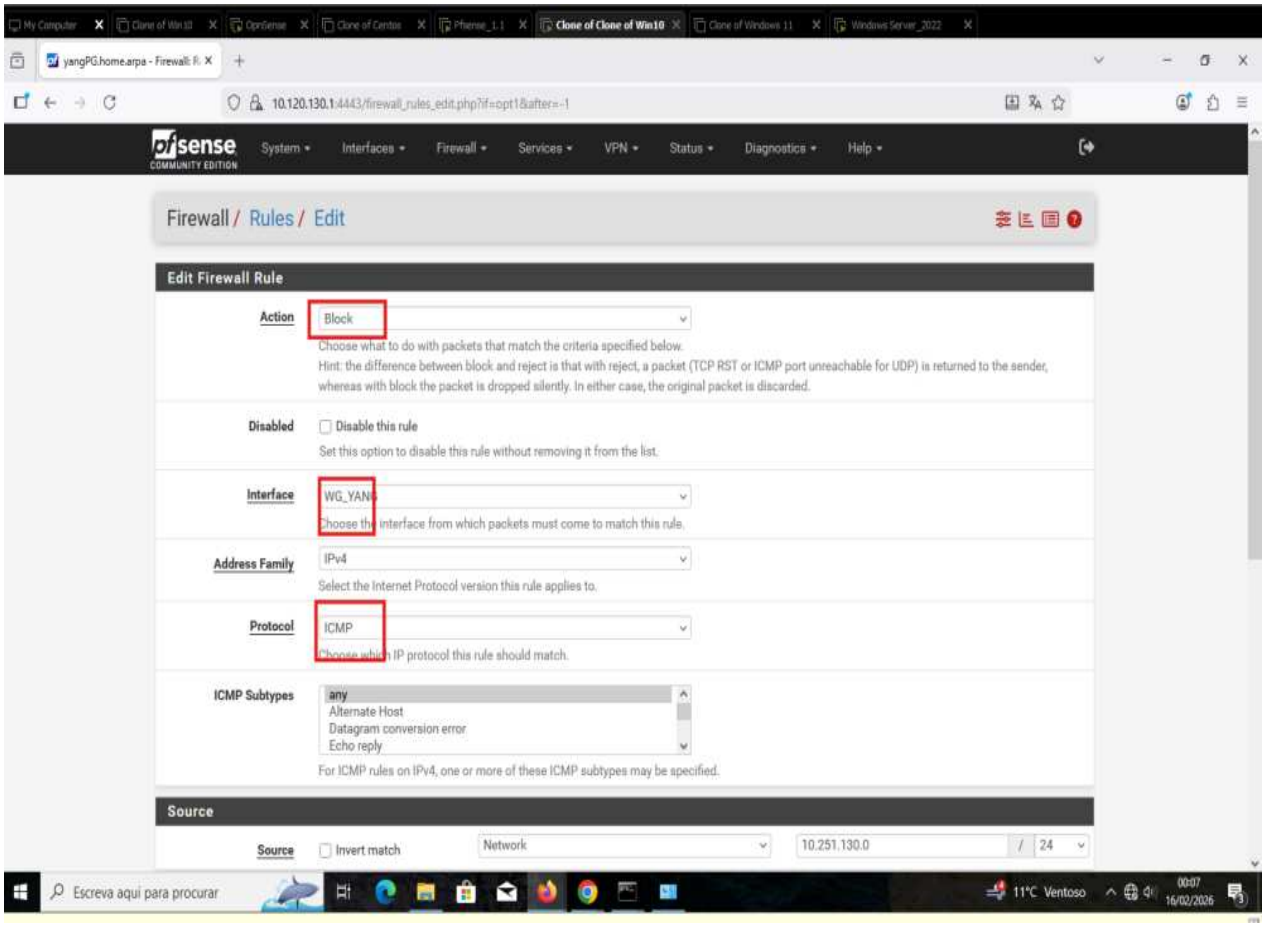
- Action: Block

- Protocol: ICMP

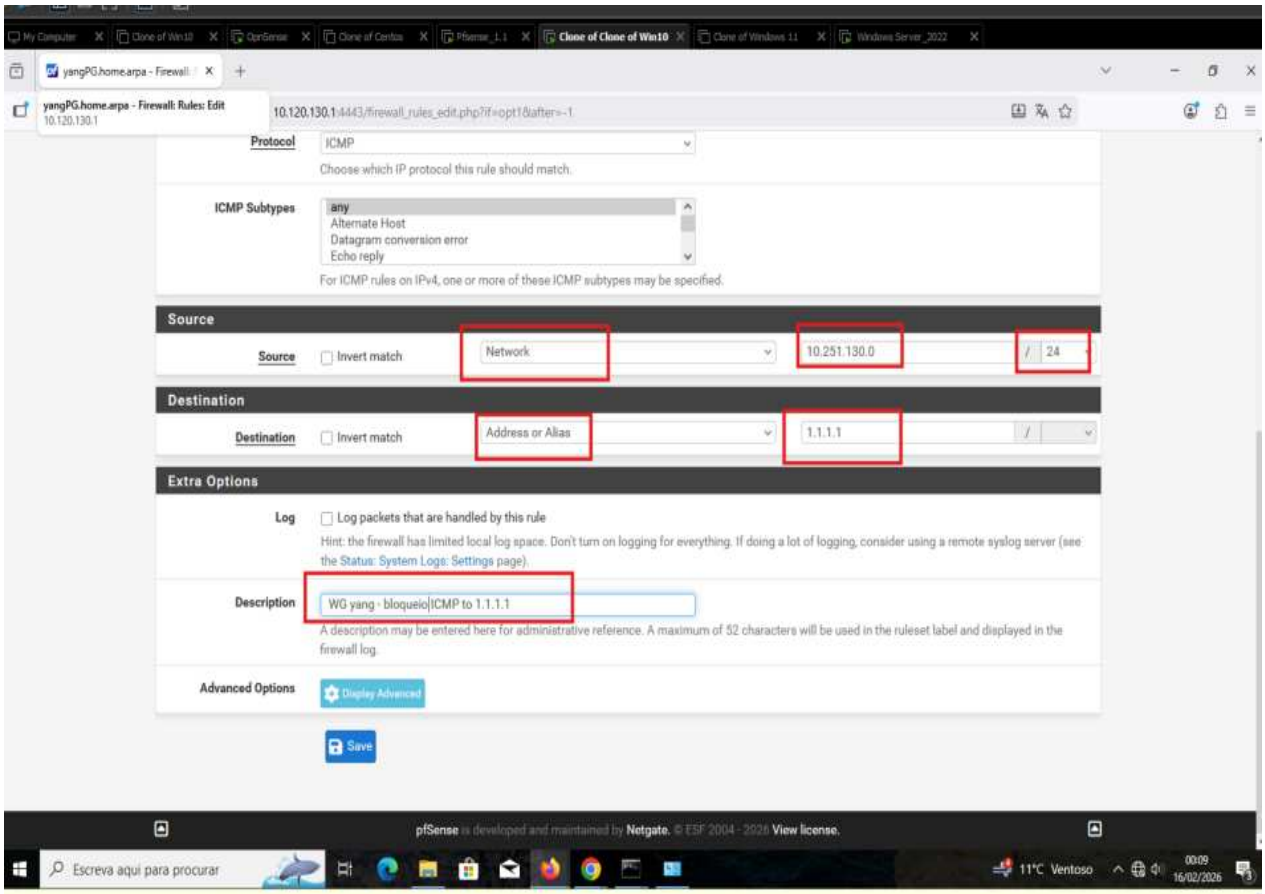
- Source: 10.251.130.0/24

- Destination: 1.1.1.1

- Description: WG yang - bloqueio ICMP to 1.1.1.1



Continuação …



20.3 - Regra 3 — Permitir acesso à INTERFACE do yang (prova "ping é insuficiente")

O enunciado pede **prova real** (não ping). A prova mais limpa é **abrir o WebGUI do pfSense**.

- Action: Pass

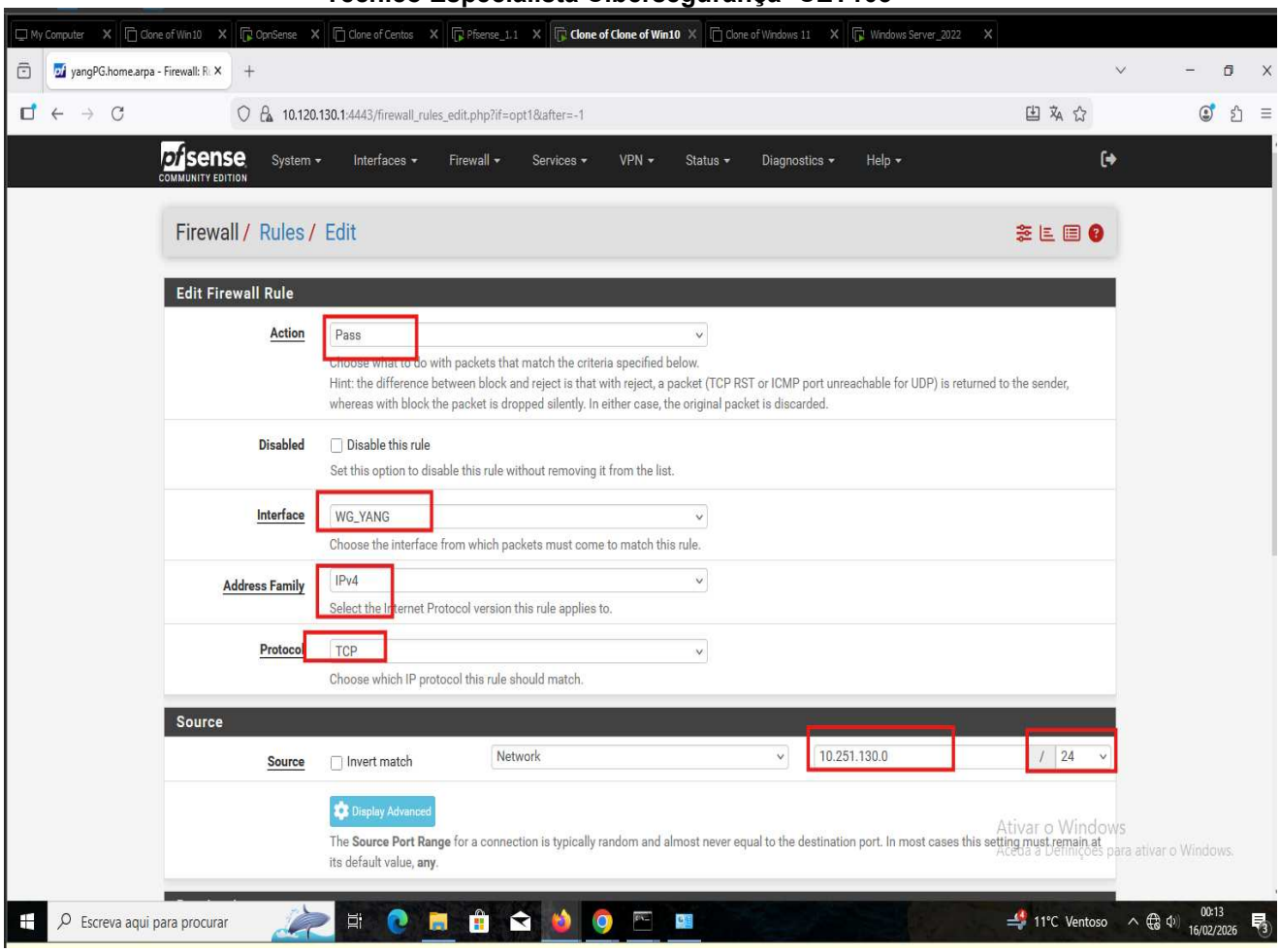
- Protocol: TCP

- Source: 10.251.130.0/24

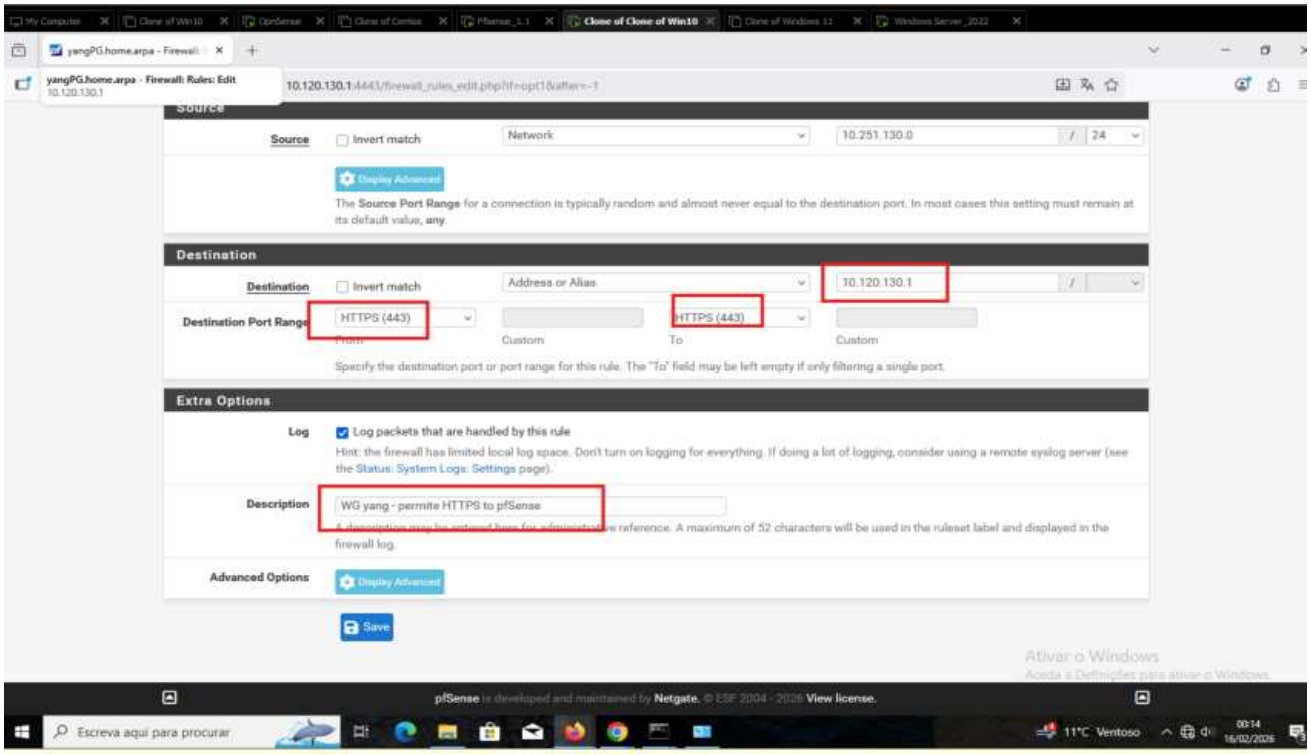
- Destination: **Single host** 10.120.130.1

- Destination port: 443 (HTTPS)

- Description: WG yang – permite HTTPS to pfSense GUI (10.120.130.1)



Continuação…



20.3 Regra -4 — Permitir DNS para OpenDNS (para conseguir navegar)

- Action: Pass

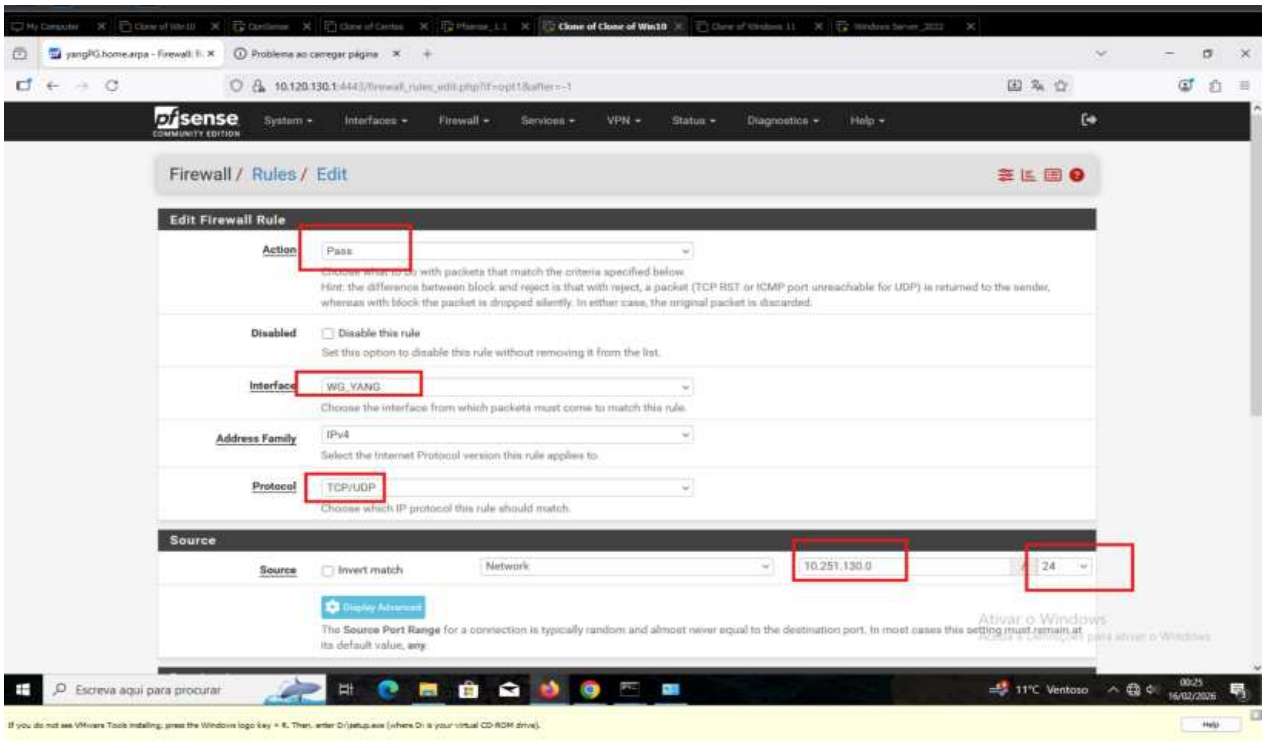
- Protocol: TCP/UDP

- Source: 10.251.130.0/24

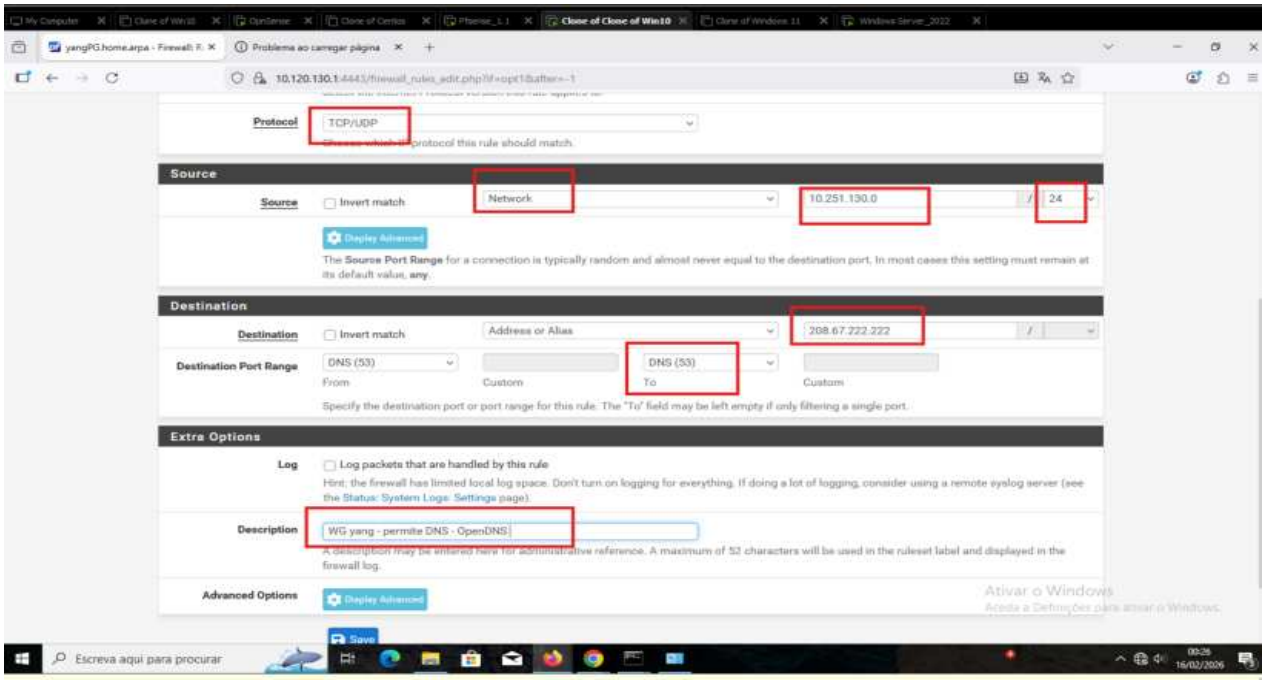
- Destination: 208.67.222.222 (OpenDNS)

- Destination port: 53 (DNS)

Description: WG yang - permite DNS-OpenDNS



Continuação...



20.4 - Regra 5 — Permitir Internet (web) para o cliente navegar (full tunnel)

- Action: Pass

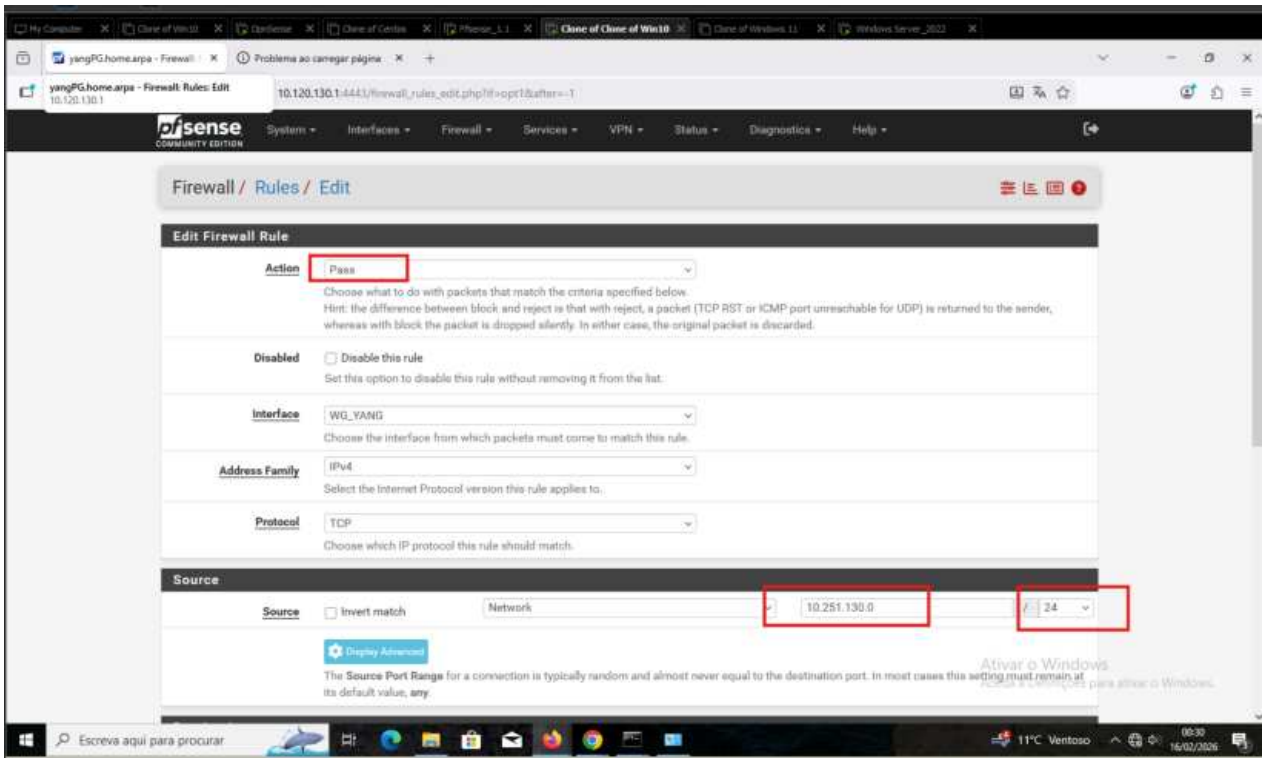
- **Protocol**: TCP

- **Source**: 10.251.130.0/24

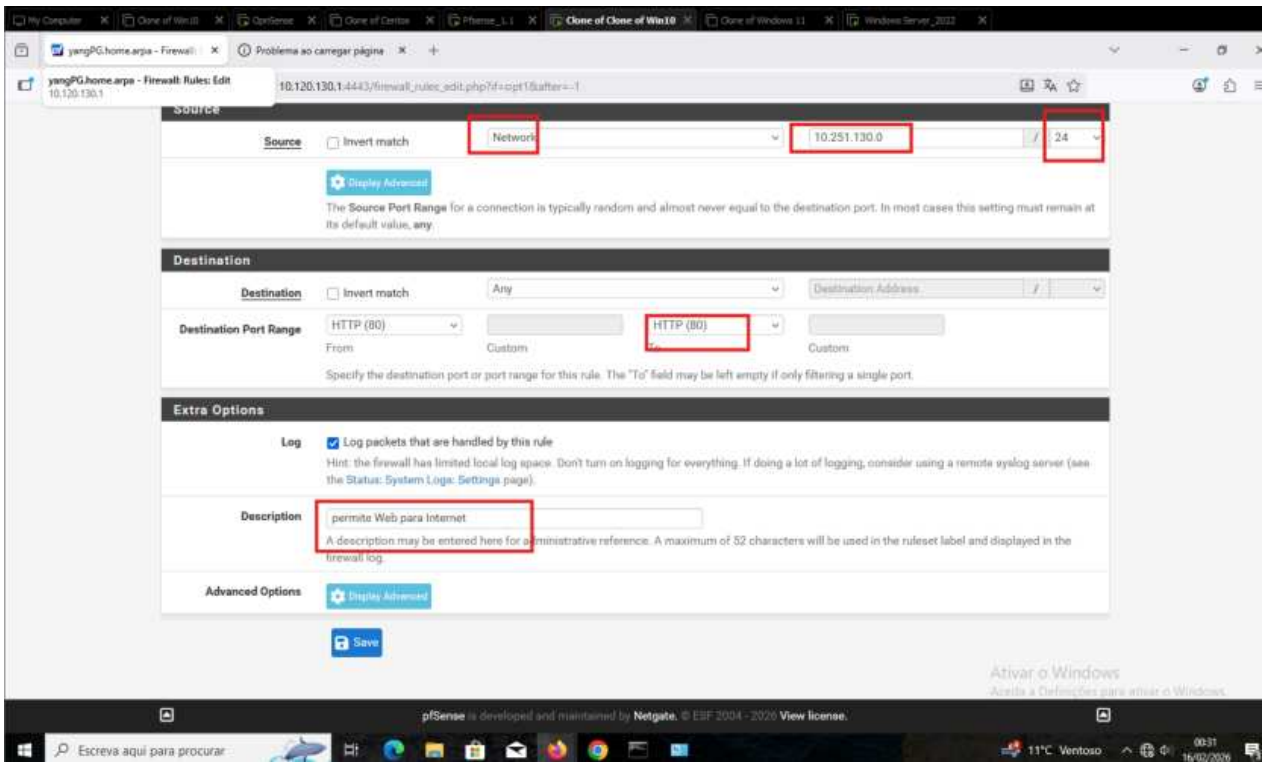
- **Destination**: any

- **Destination port**: 80 (HTTP)

- **Description**: permite Web para Internet



Continuação…



20.5-Regra 5 — Permitir Internet (web) para o cliente navegar (full tunnel)

- Action: Pass

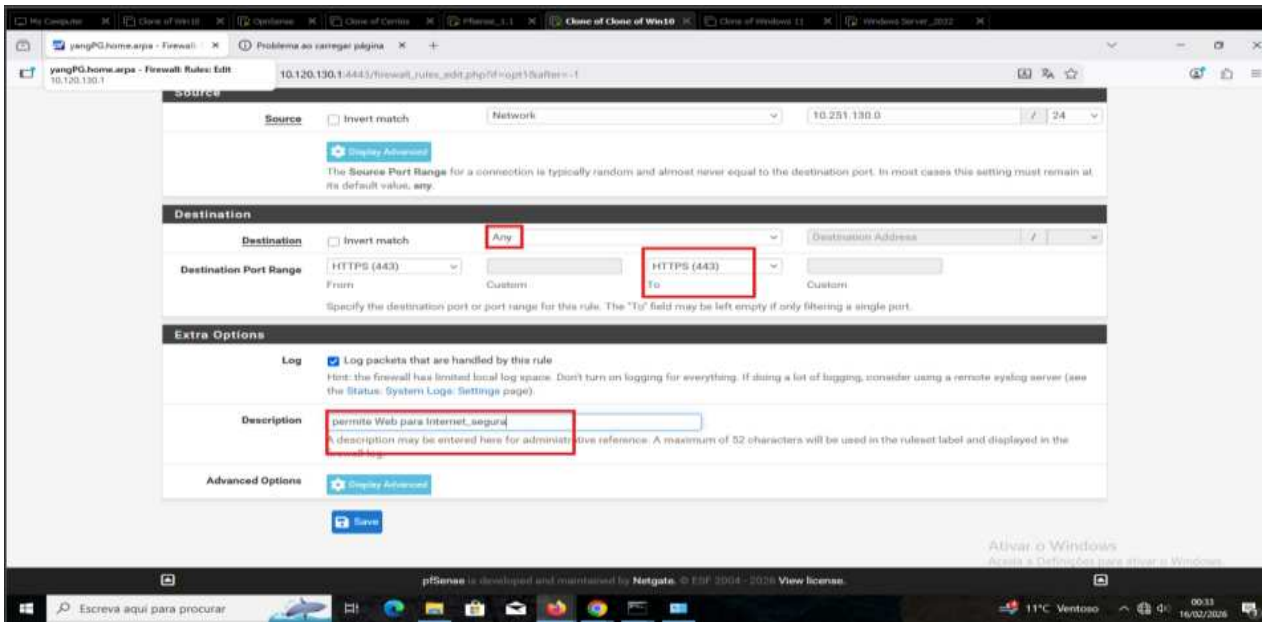
- Protocol: TCP

- Source: 10.251.130.0/24

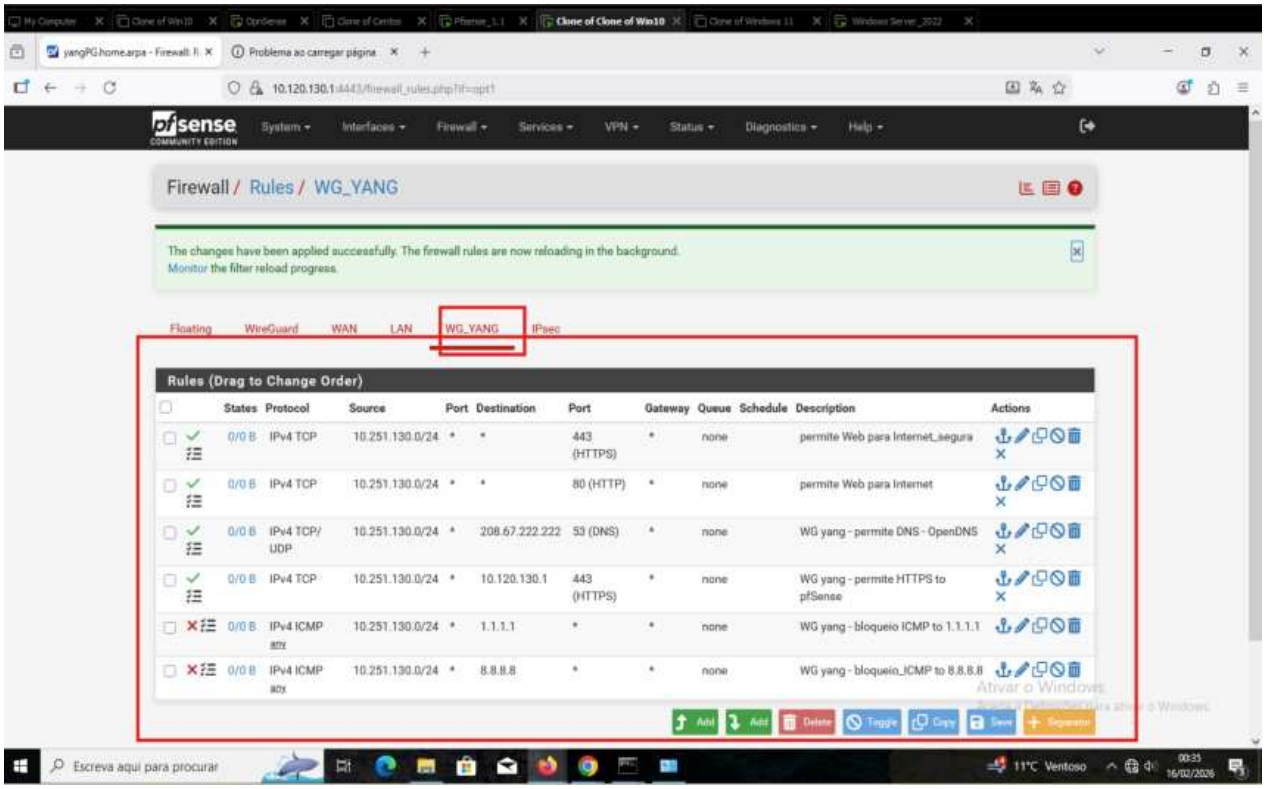
- Destination: any

- Destination port: 443 (HTTPS)

Description: permite Web para Internet



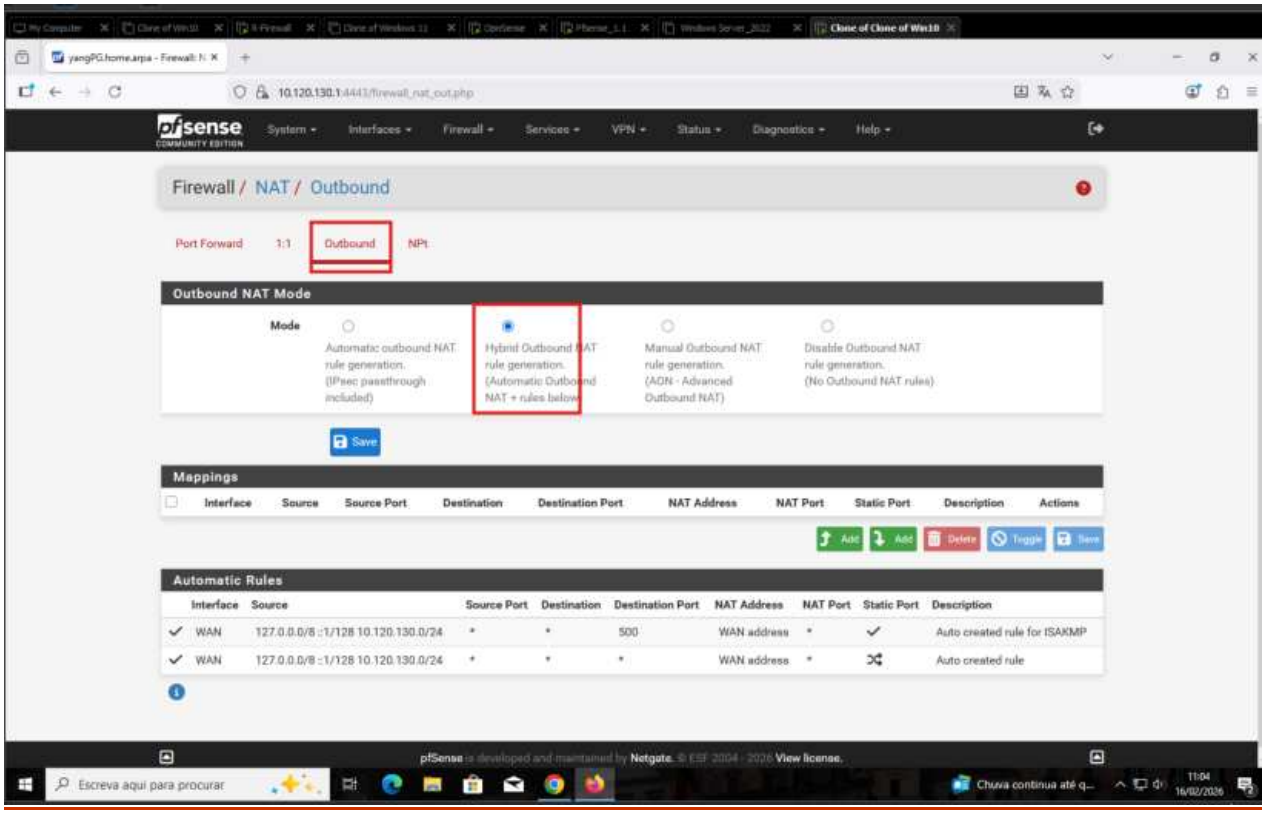
Resumo :



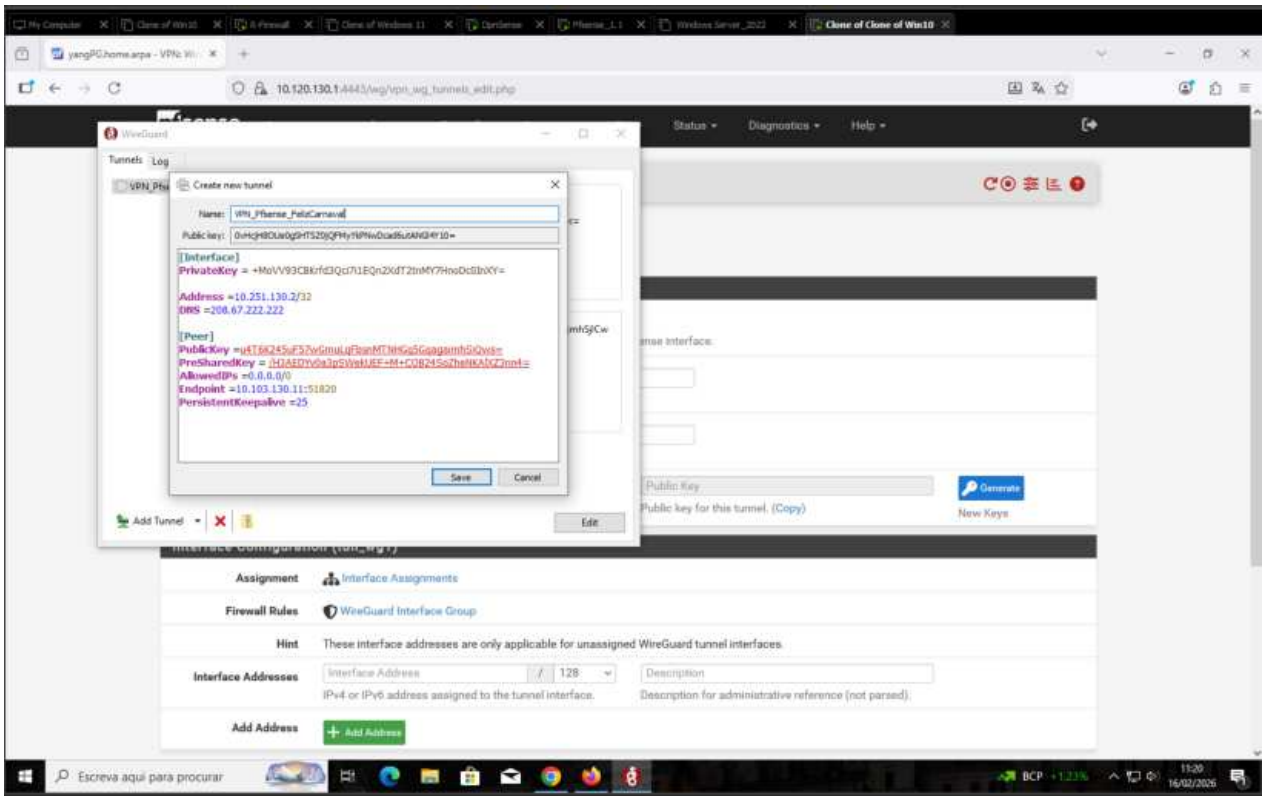
20.6) 16.5/16.6 — NAT de saída ()

Firewall → NAT → Outbound

Escolher: **Hybrid Outbound NAT**



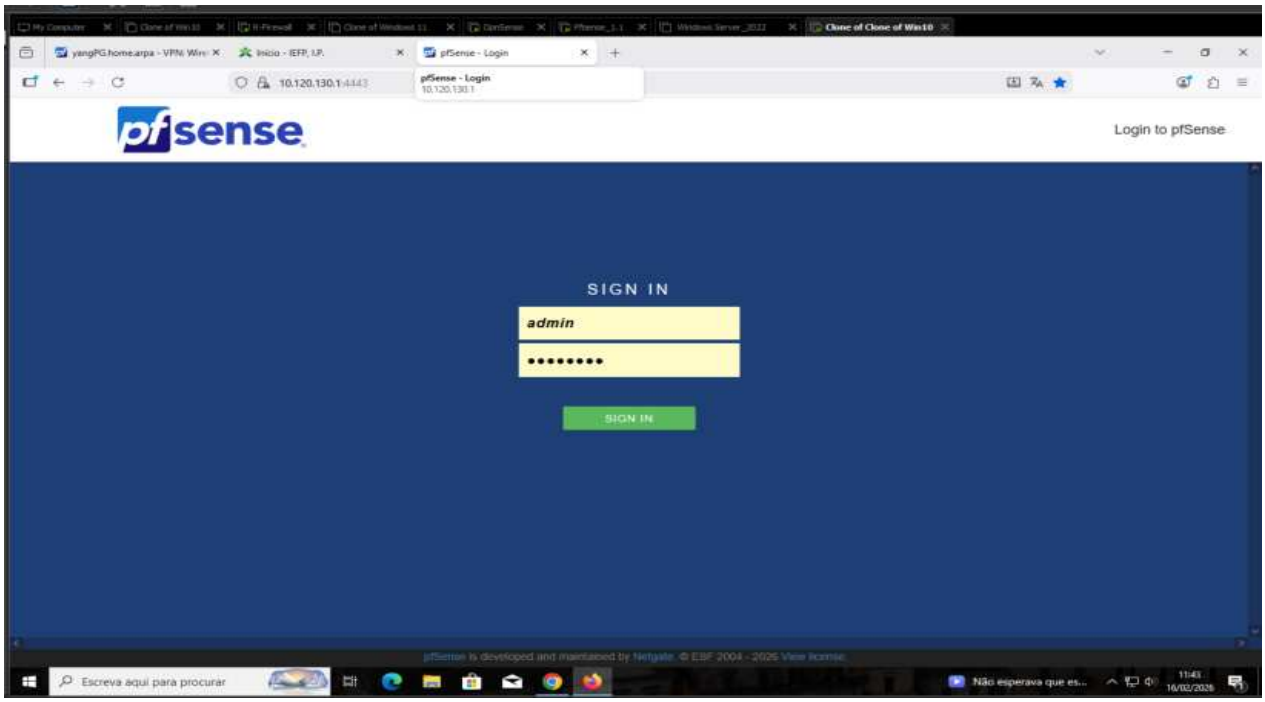
20.7 Ficheiro de configuração



20.8 Teste — "ping insuficiente": provar acesso real à interface do yang

No browser do Windows:

- Abrir: https://10.120.130.1
- Deve abrir o login do pfSense



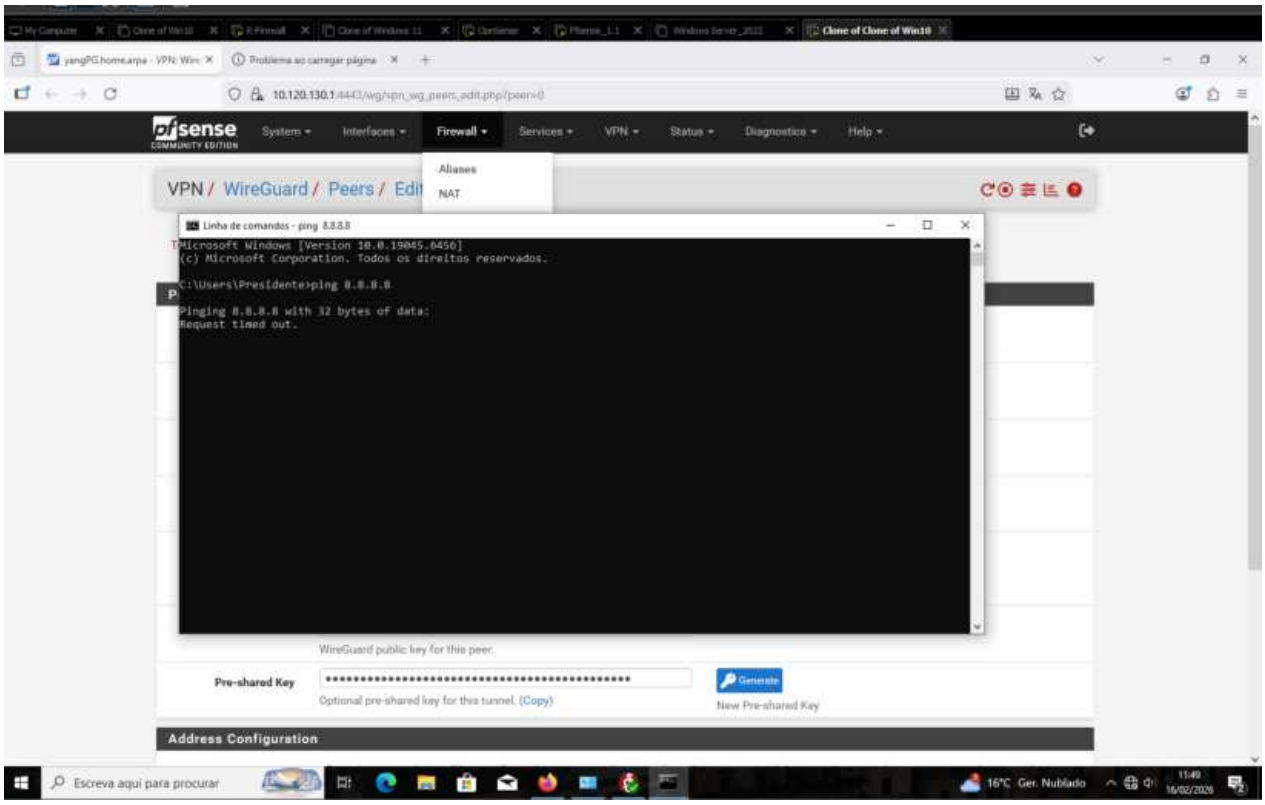
20.9 Teste — Internet funciona

- Abrir sites HTTPS (ex: https://www.iefp.pt)

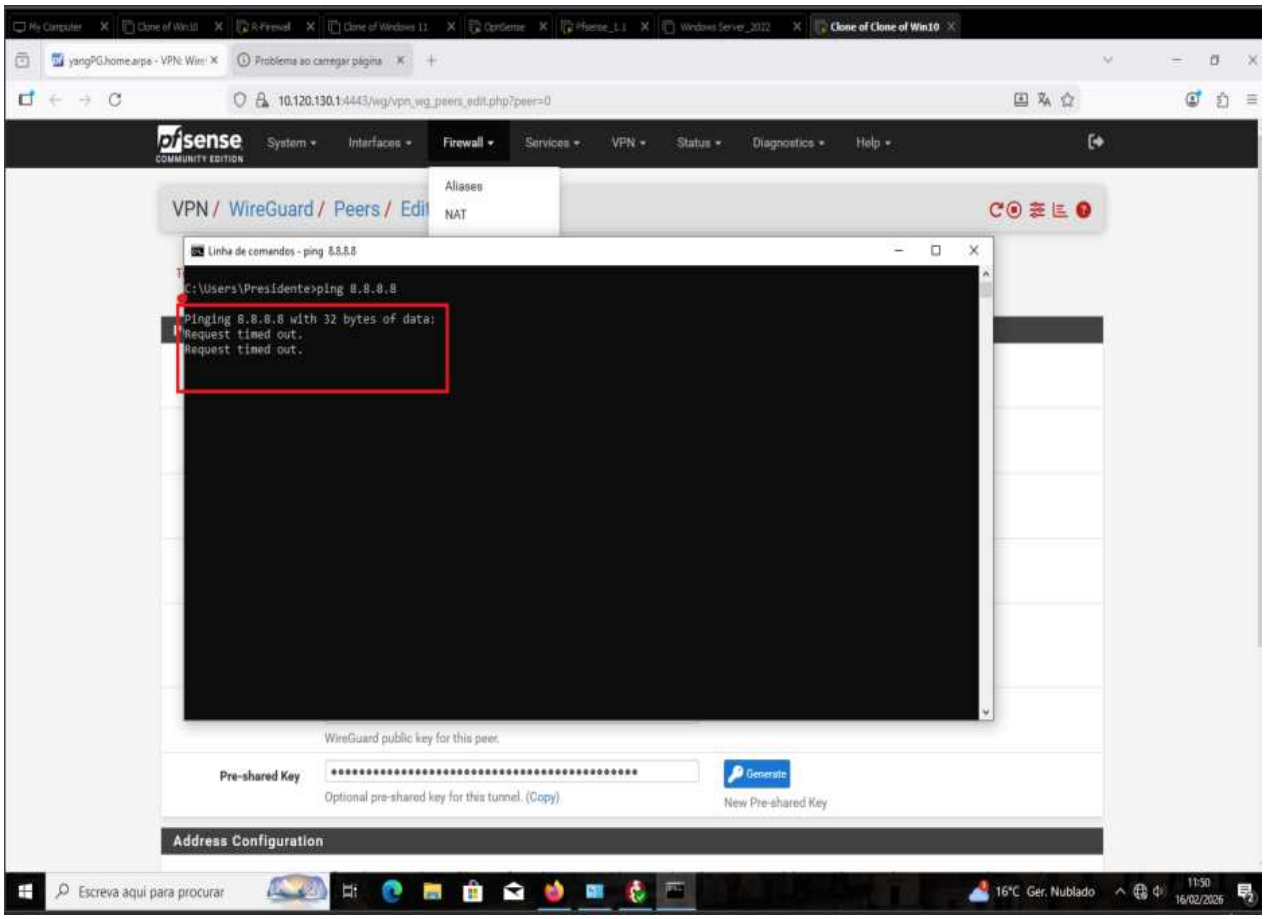


20.10 Teste E — Não consegue pingar 8.8.8.8 e 1.1.1.1

a) ping 8.8.8.8

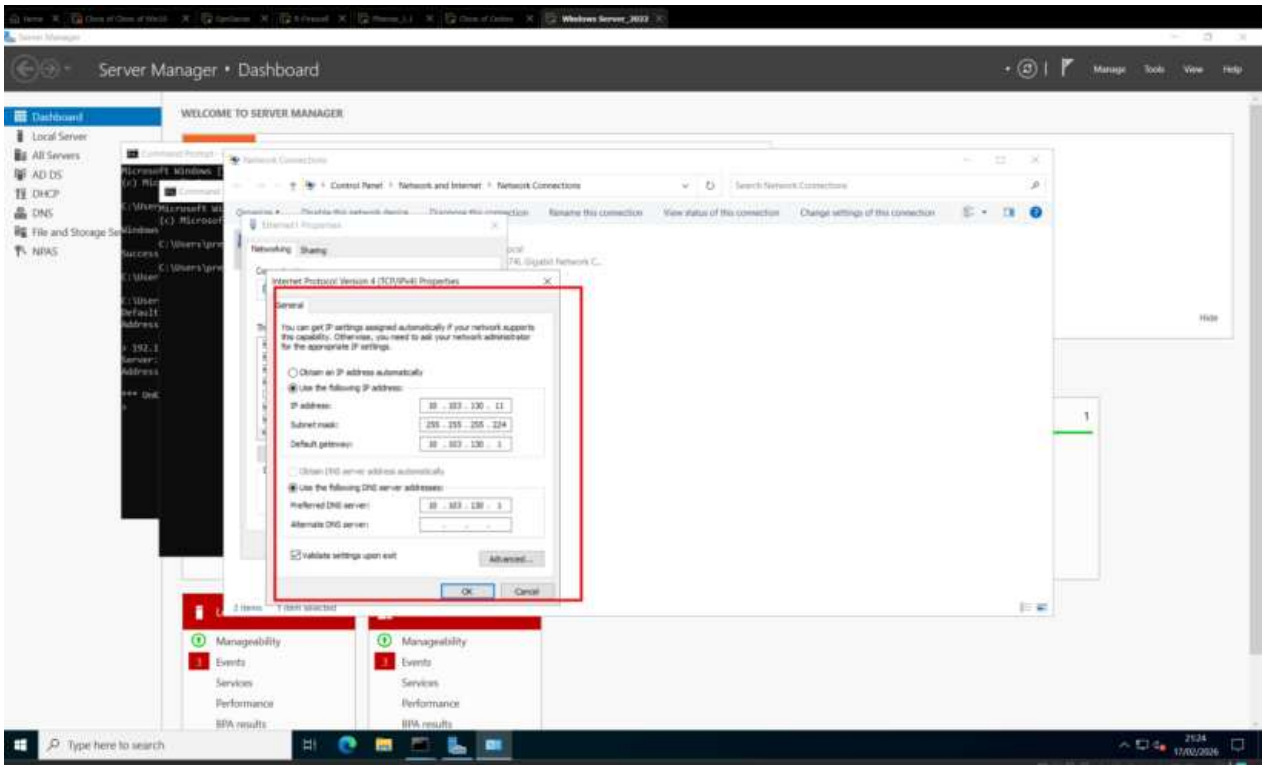


b)ping 1.1.1.1



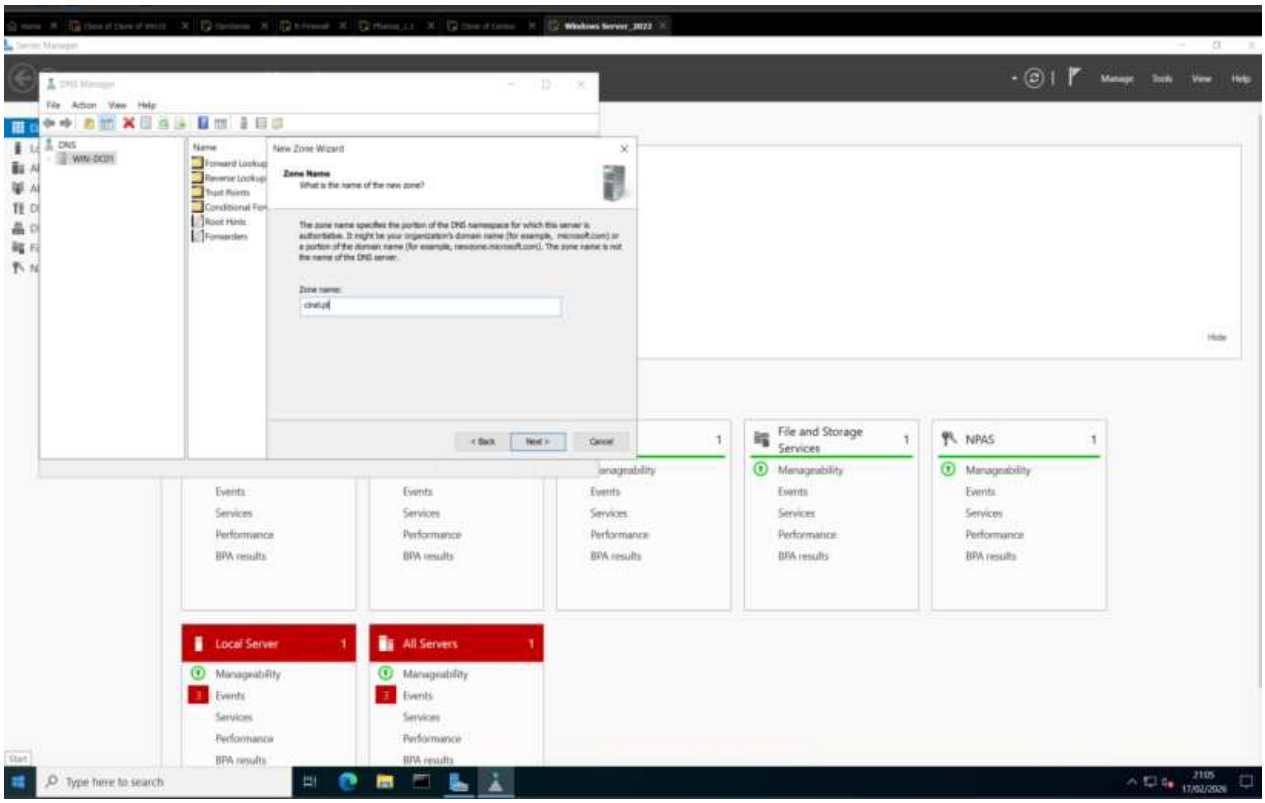
20.11Testes no windows Server :

a) Colocação na interface WAN_PESSOAL

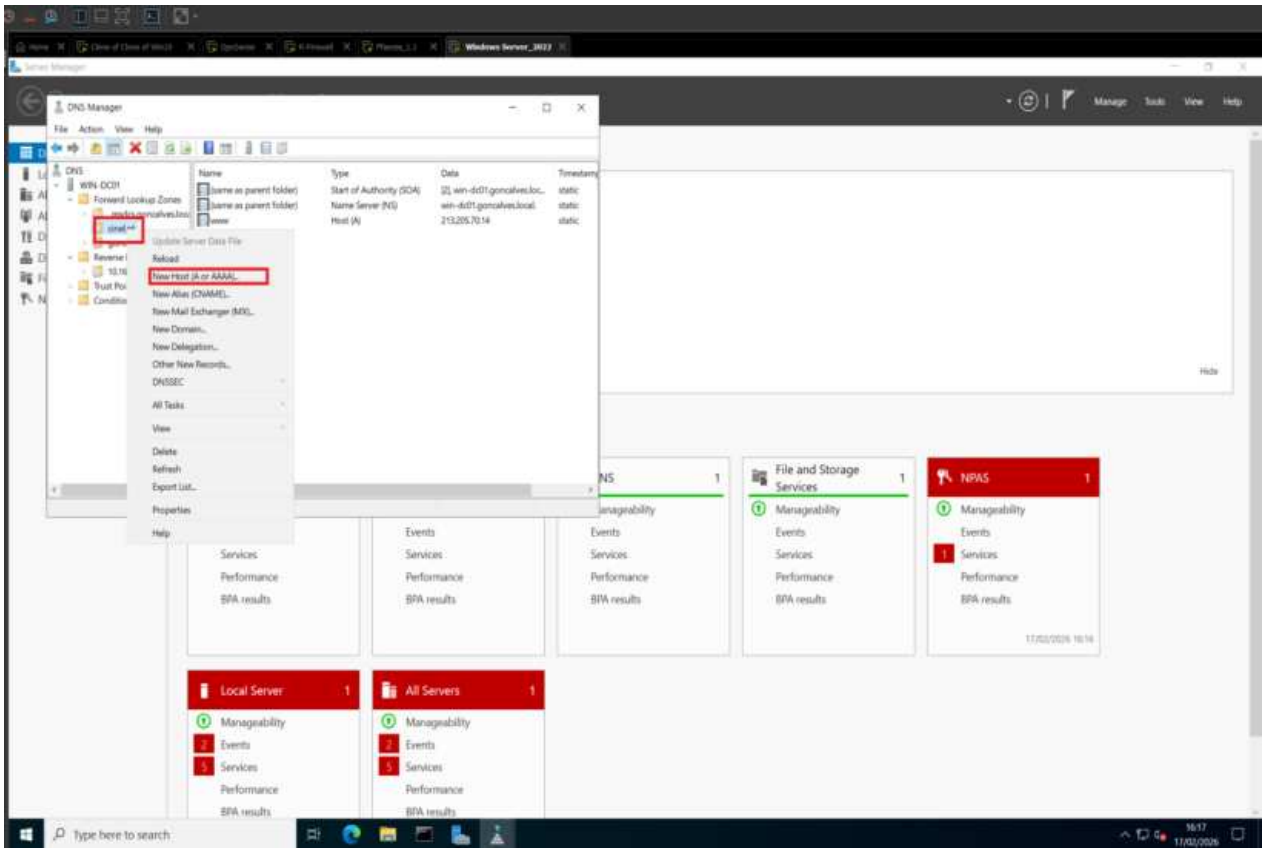


b) Criação de **Forward Lookup Zone** chamada **.pt** no servidor DNS do Windows.

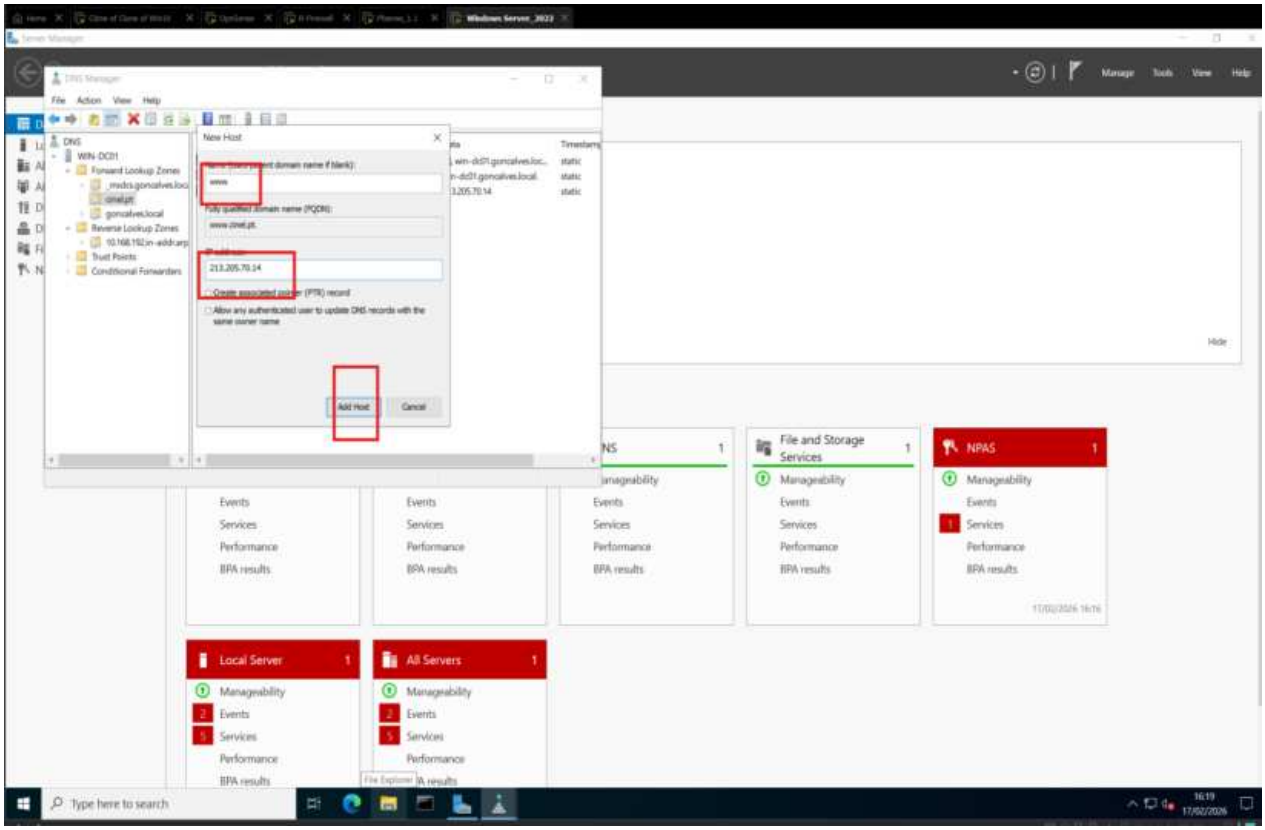
Isso significa que este servidor DNS vai passar a:Resolver nomes do tipo: www. .pt



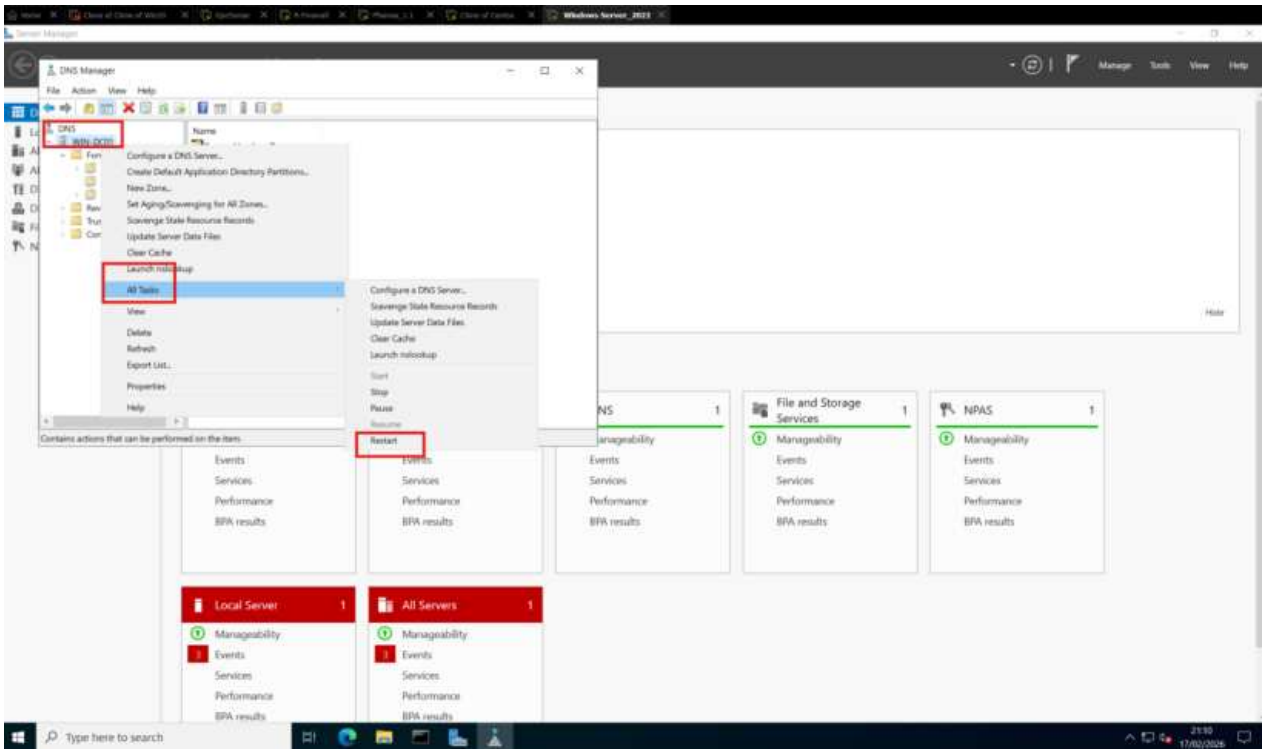
Continução...



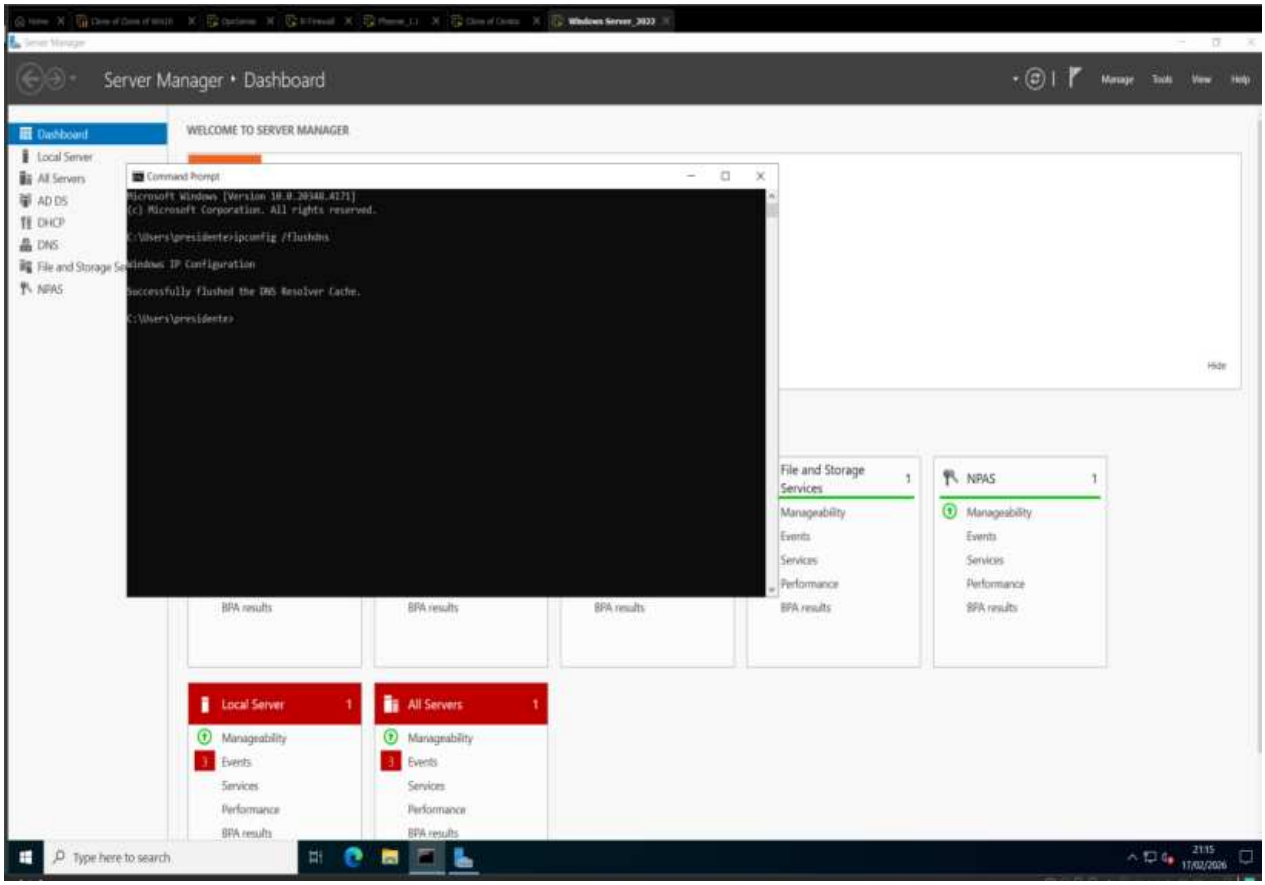
Continuação...



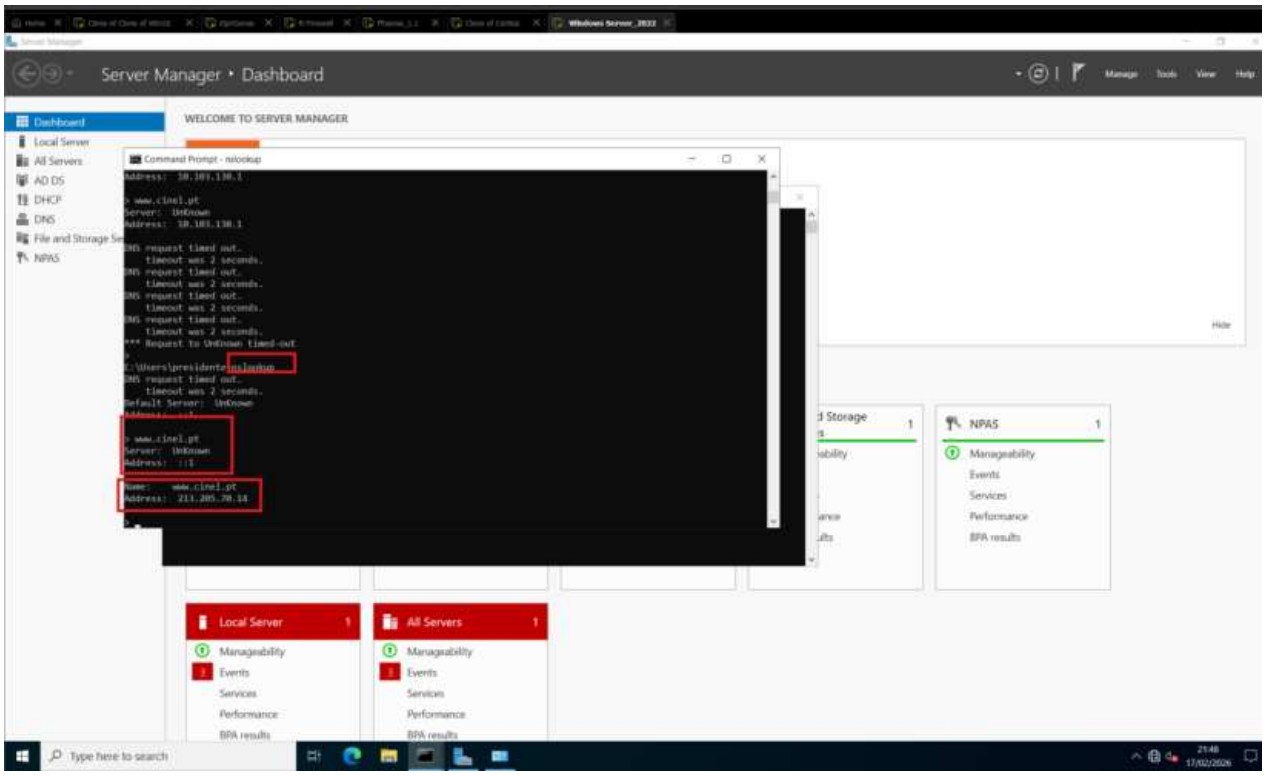
A) WIN-DC01> All tasks > Restart



b) Limpeza de chache do windows Server

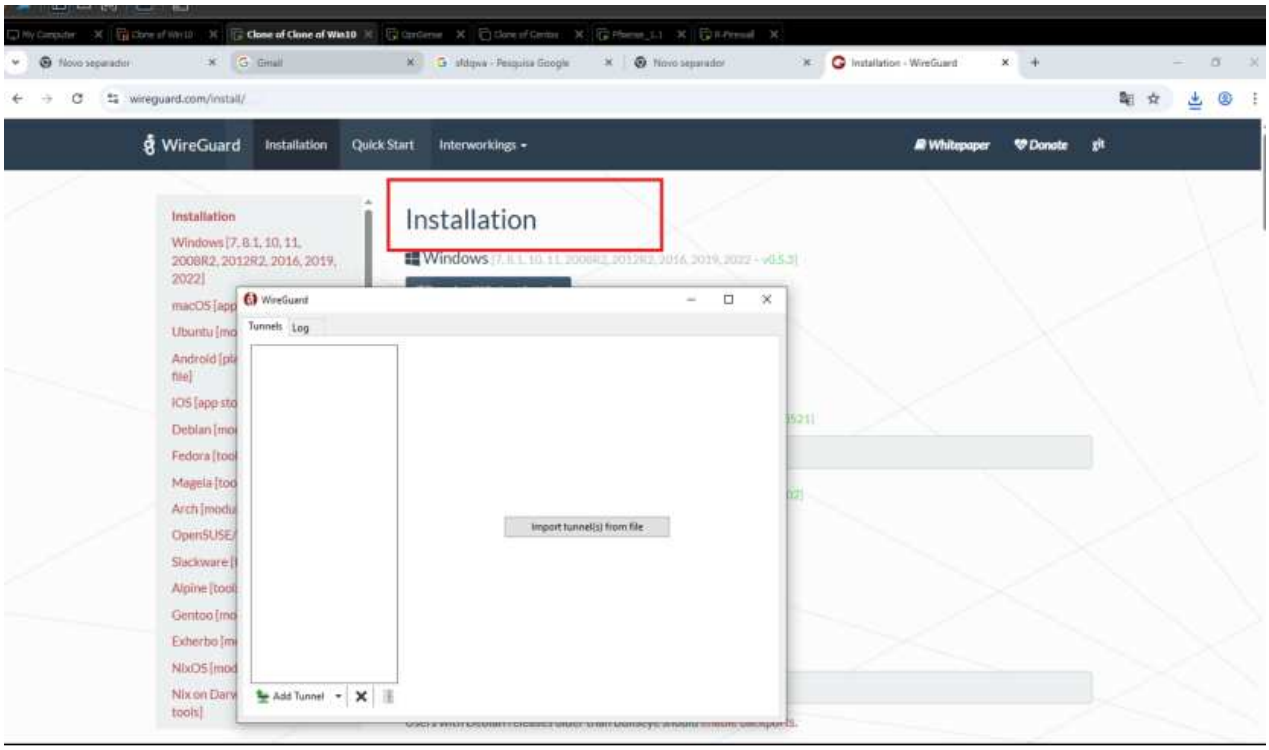


c)Resolução DNS com o comando nslookup no Windows Server.



20-11 VPN 3.e/3.f/3.g — WireGuard Roadwarrior no yinPG (RDP permitido, SSH bloqueado, full tunnel + DNS do DC)

Donwload Wireguard :



21 WireGuard Roadwarrior no yinPG (OPNsense)

Objetivos para Aavalição :

1. **O cliente "na WAN pessoal" liga por WireGuard ao yinPG.**
2. **RDP funciona para o Windows Server (172.19.130.10), mas SSH falha (bloqueado).**
3. **Full tunnel: o cliente manda tudo pela VPN (não é só redes internas).**
4. **O DNS usado pelo cliente é o do DC/DNS (172.19.130.10) e não DNS externos.**
5. **"Ping insuficiente" → a prova principal é RDP (mstsc) e evidências (logs / prints).**

21.1 Escolher rede do túnel (WireGuard)

Usar uma rede só para WireGuard (não pode colidir com as redes GREEN/BLUE/WAN).

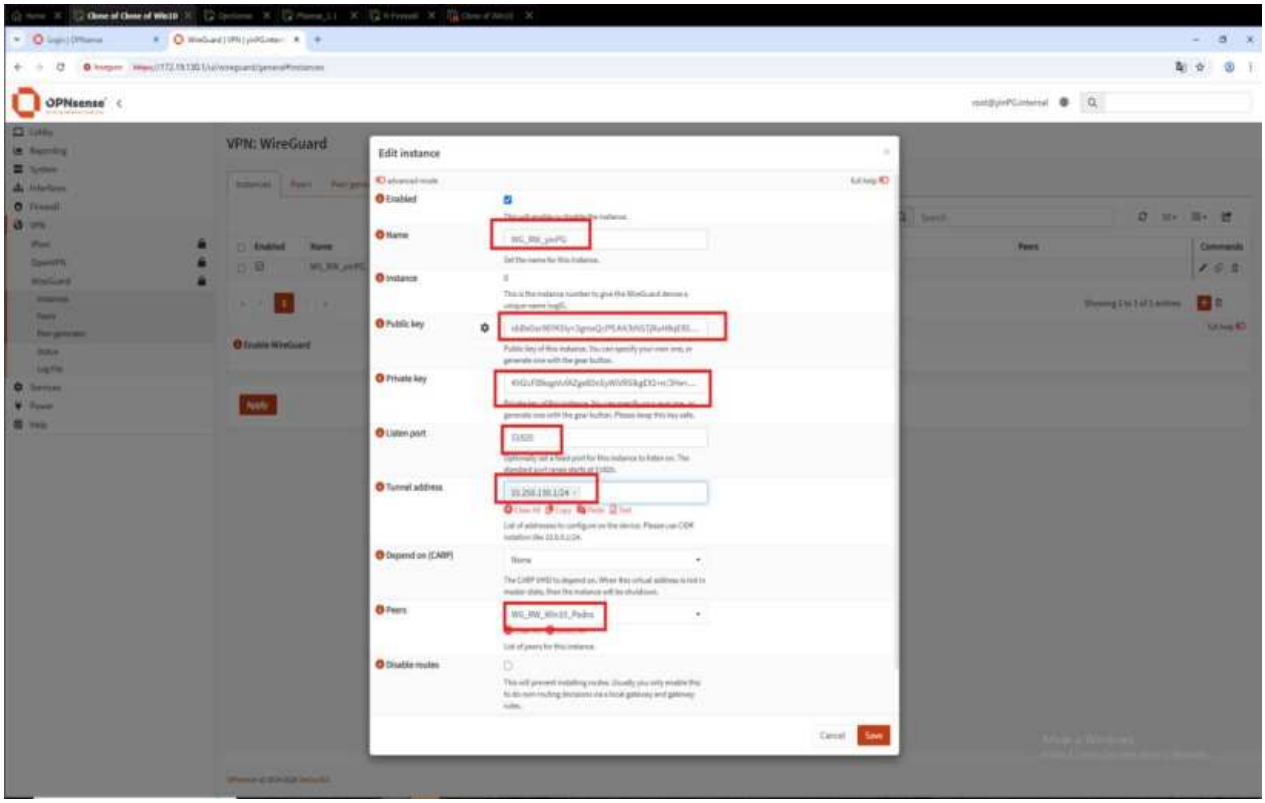
- **WG_TUN_NET = 10.250.130.0/24**
- **yinPG (server) = 10.250.130.1/24**
- **Windows (cliente) = 10.250.130.2/32**
-

21.2 OPNsense (yinPG) — criar a instância WireGuard (Local/Instance)

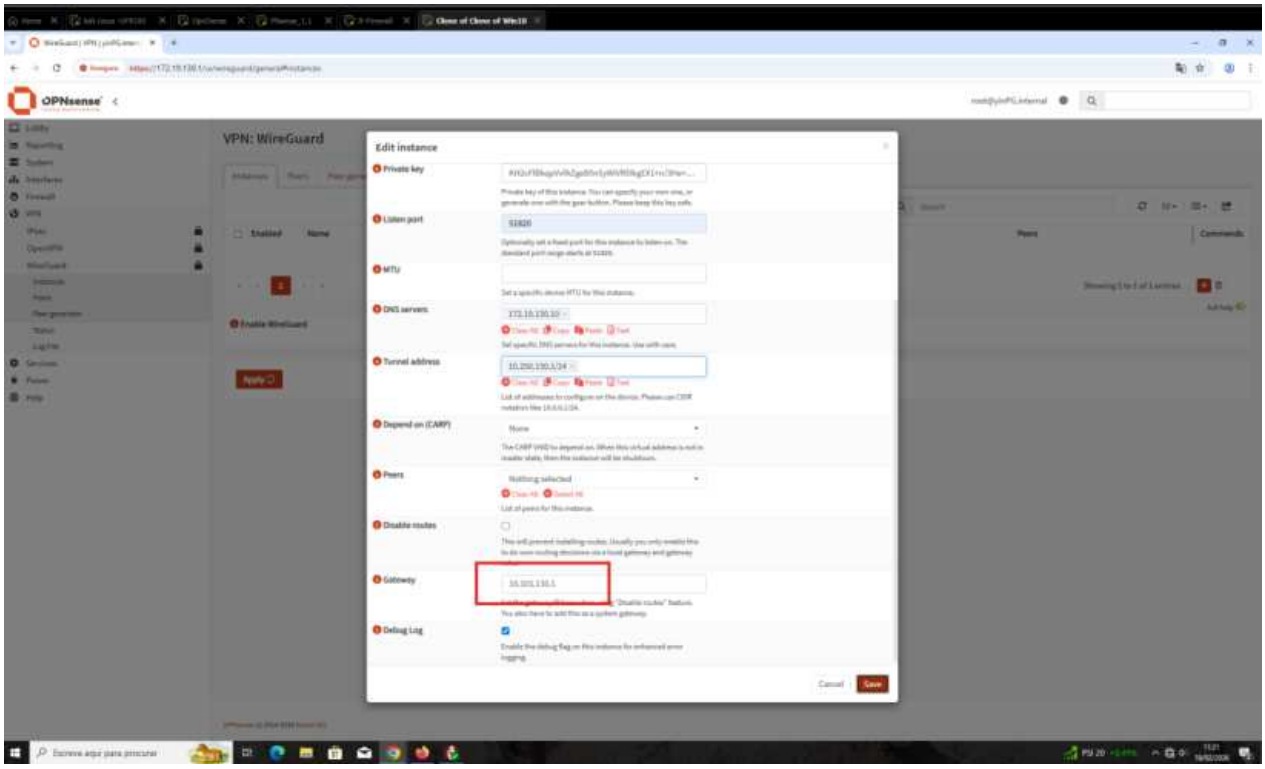
A) Criar "Instance / Local"

1. **VPN → WireGuard → Instances**

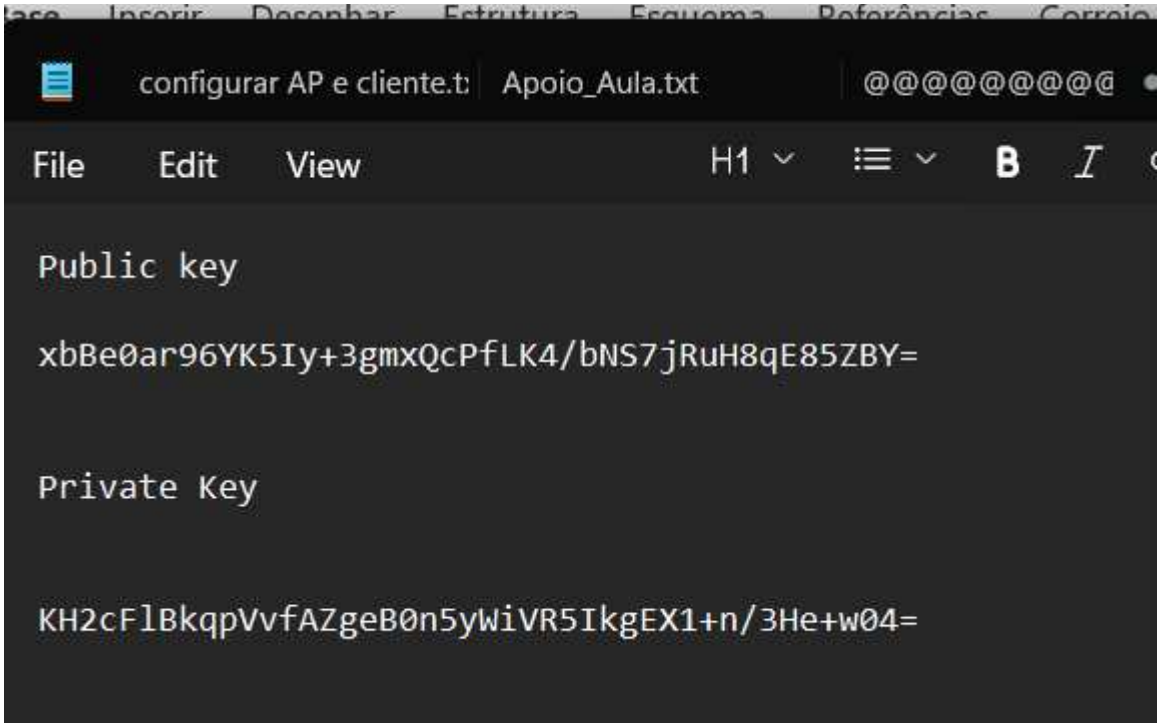
2. **Preencher:**
   - **Enabled: yes**
   - **Name/Description: WG_RW_yinPG**
   - **Listen Port: 51820**
   - **DNS servers : 172.19.130.10**
   - **Tunnel Address / Interface Address: 10.250.130.1/24**
   - **Gateway : 10.103.130.1**



Continuação ..



21.3 -Guardar Public key e private key em notepad



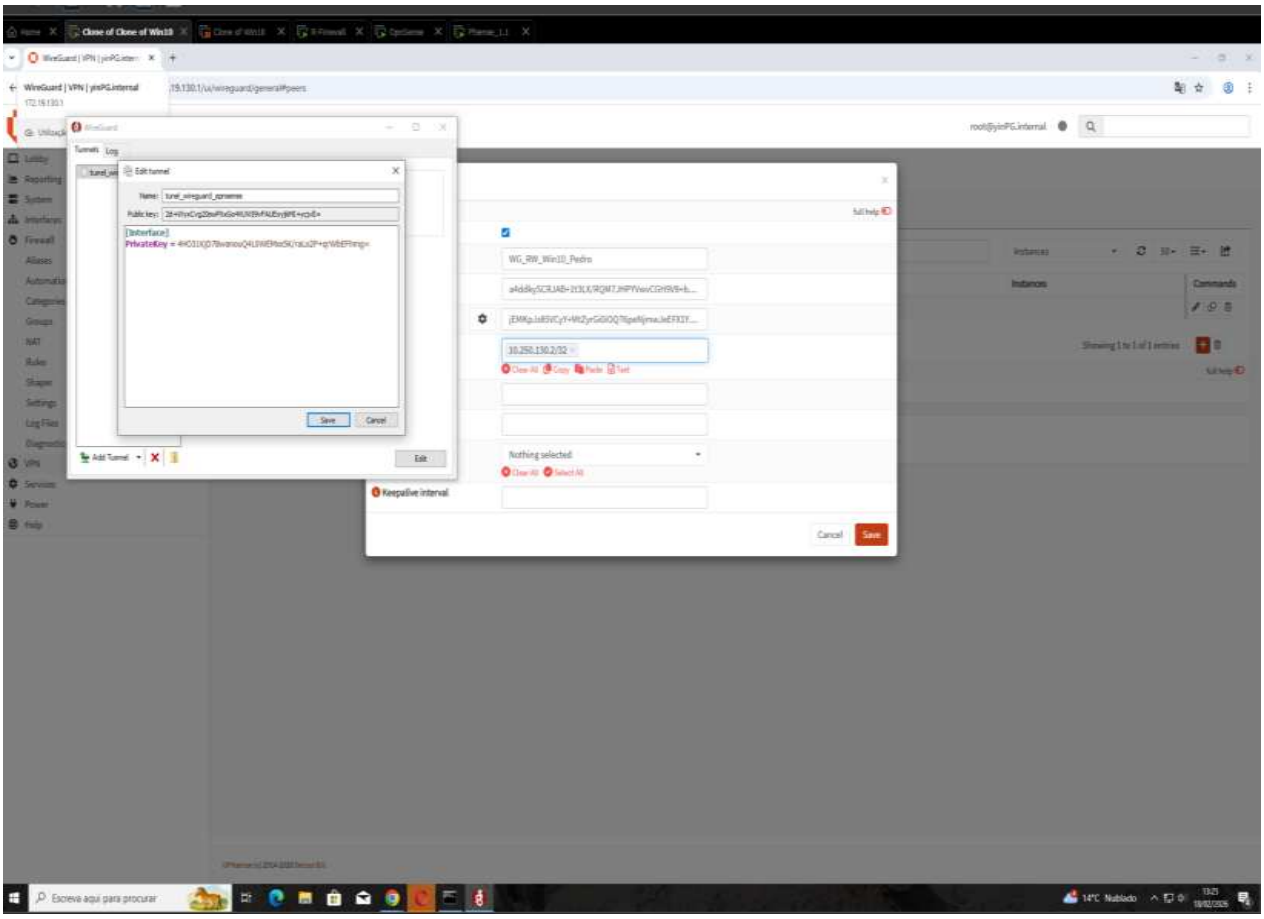
21.4 OPNsense — criar o Peer/Endpoint (cliente Roadwarrior)

- **VPN → WireGuard → Peers**

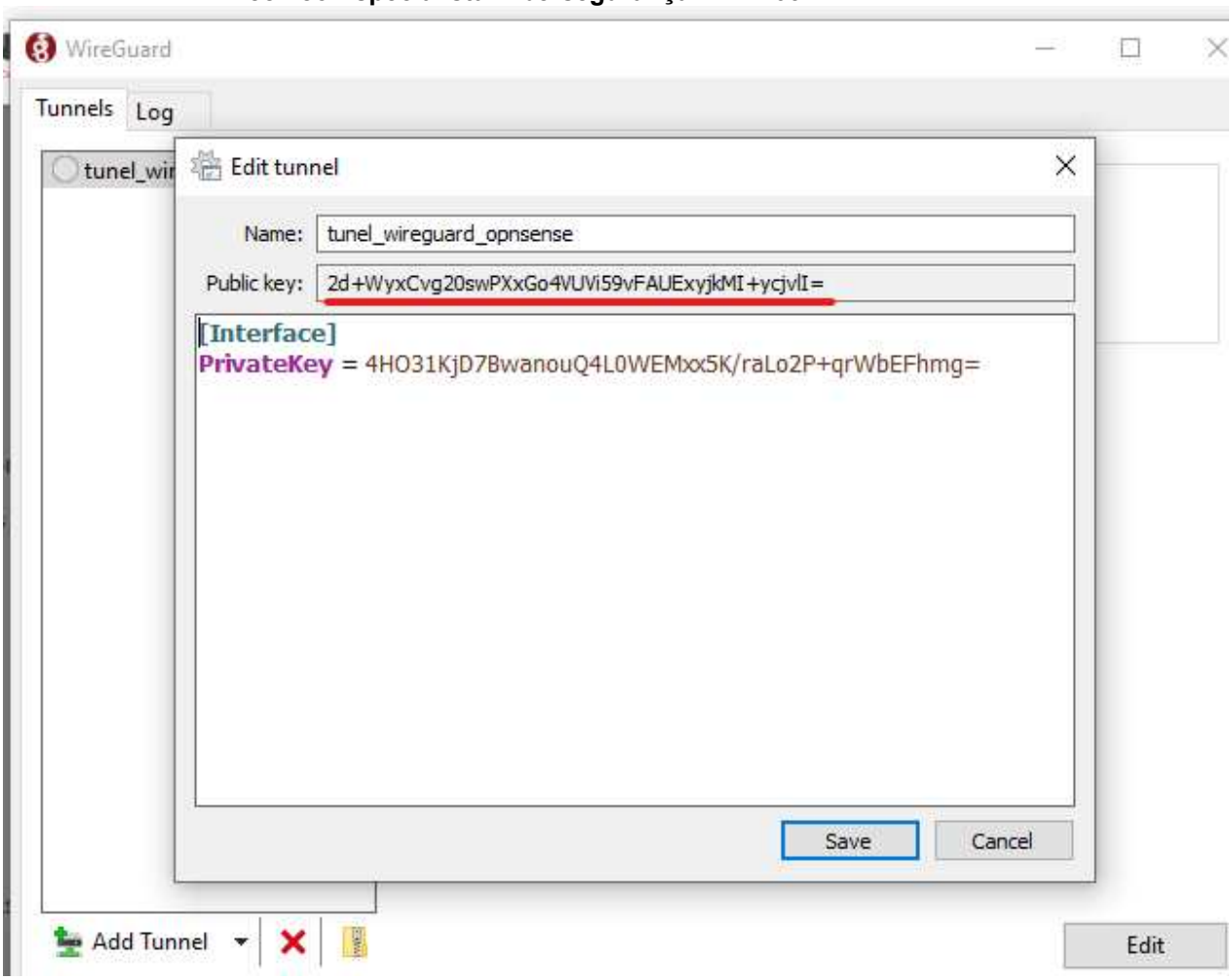
Nota :No Windows (wireguard), copiar a Public Key do cliente que será utilizada Public Key do cliente.(Opnsense)

B) Criar o Peer no OPNsense

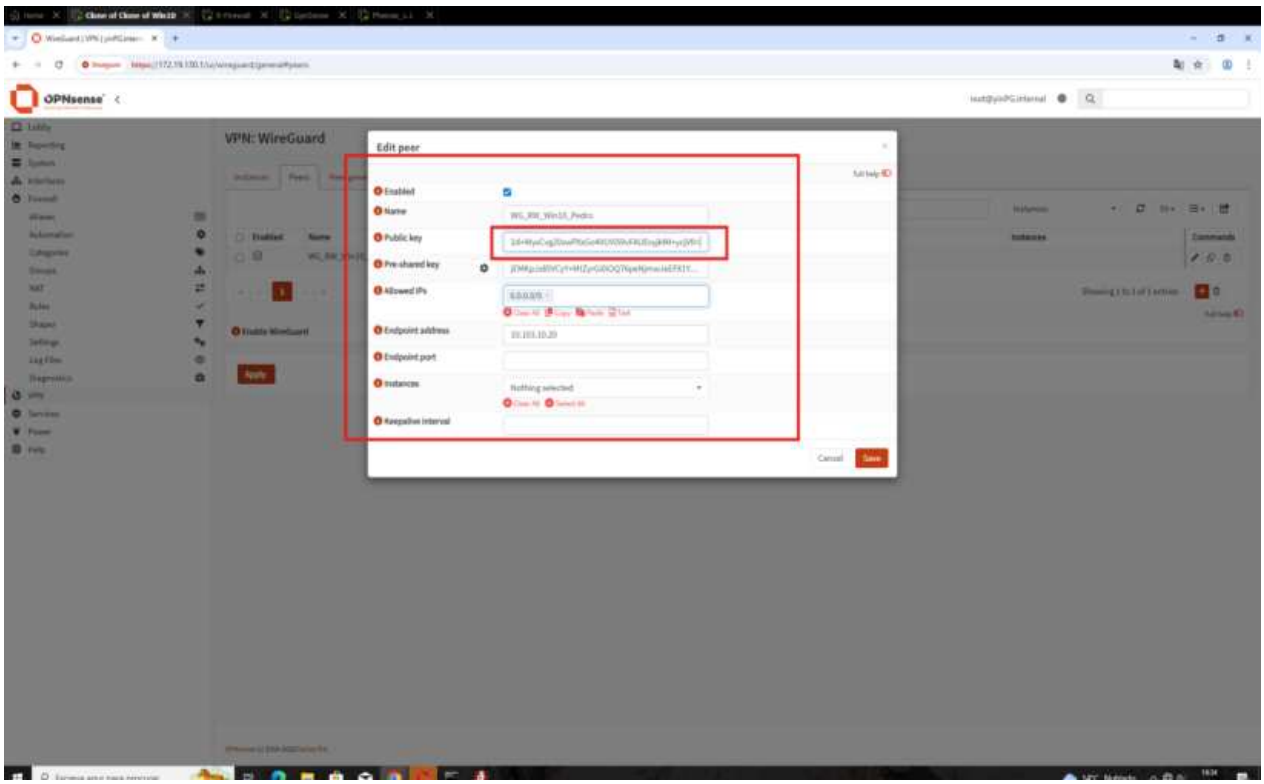
1. **VPN → WireGuard → Peers→ +Add**
2. **Preenche:**
   - Enabled:
   - **Name/Description: WG_RW_Win10_Pedro**
   - **Public Key:** ***(a do Windows WireGuard client)***
     d+WyxCvg20swPXxGo4VUVi59vFAUExyjkMl+ycjVIl=



Amplificado :

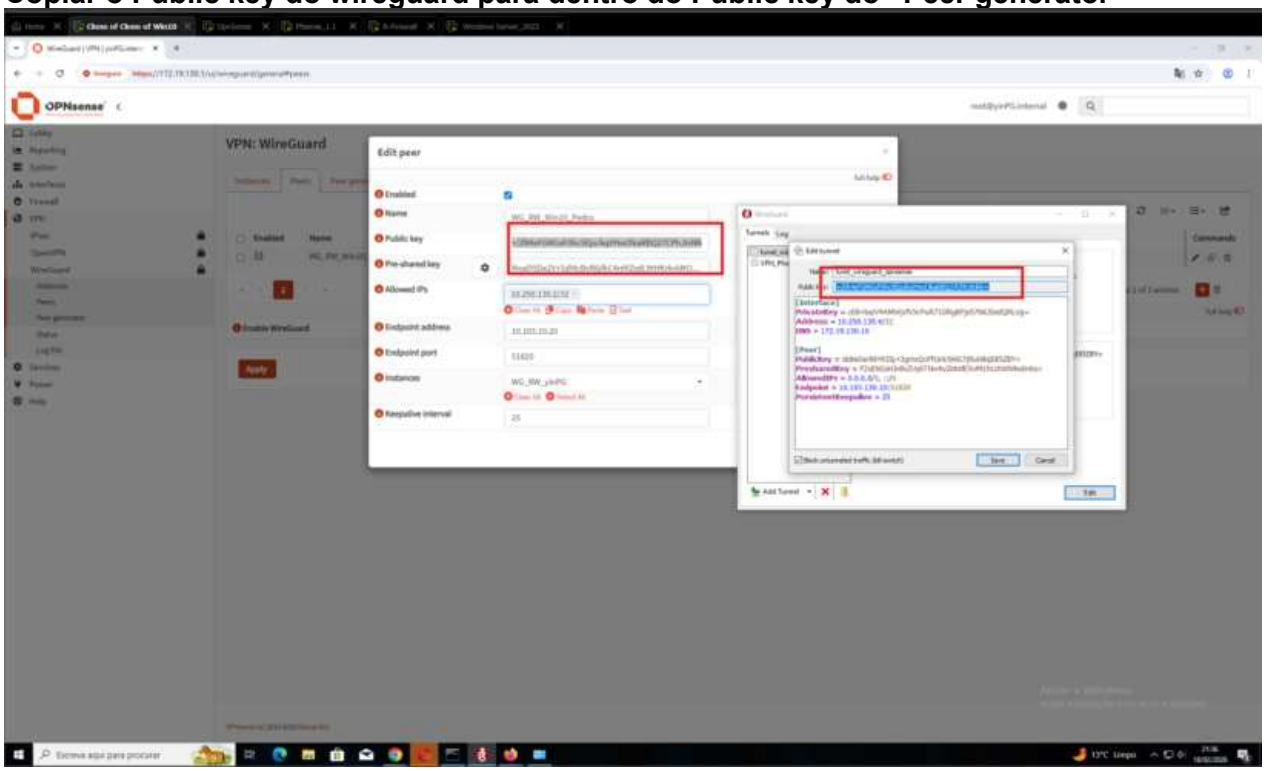


Pre-Shared Key :- botão automatic(generate new psk)
Allowed IPs :0.0.0.0/0

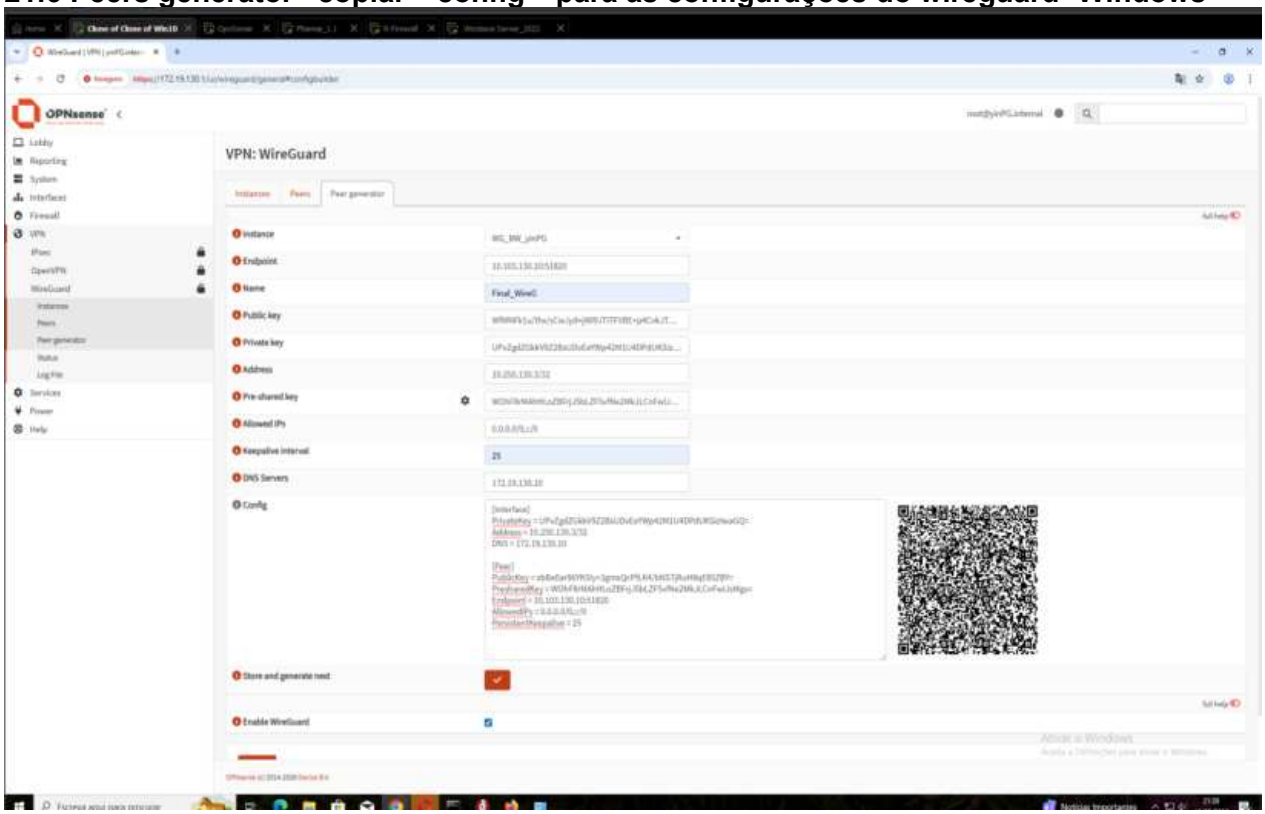


21.5 VPN >WireGuard> peers generator

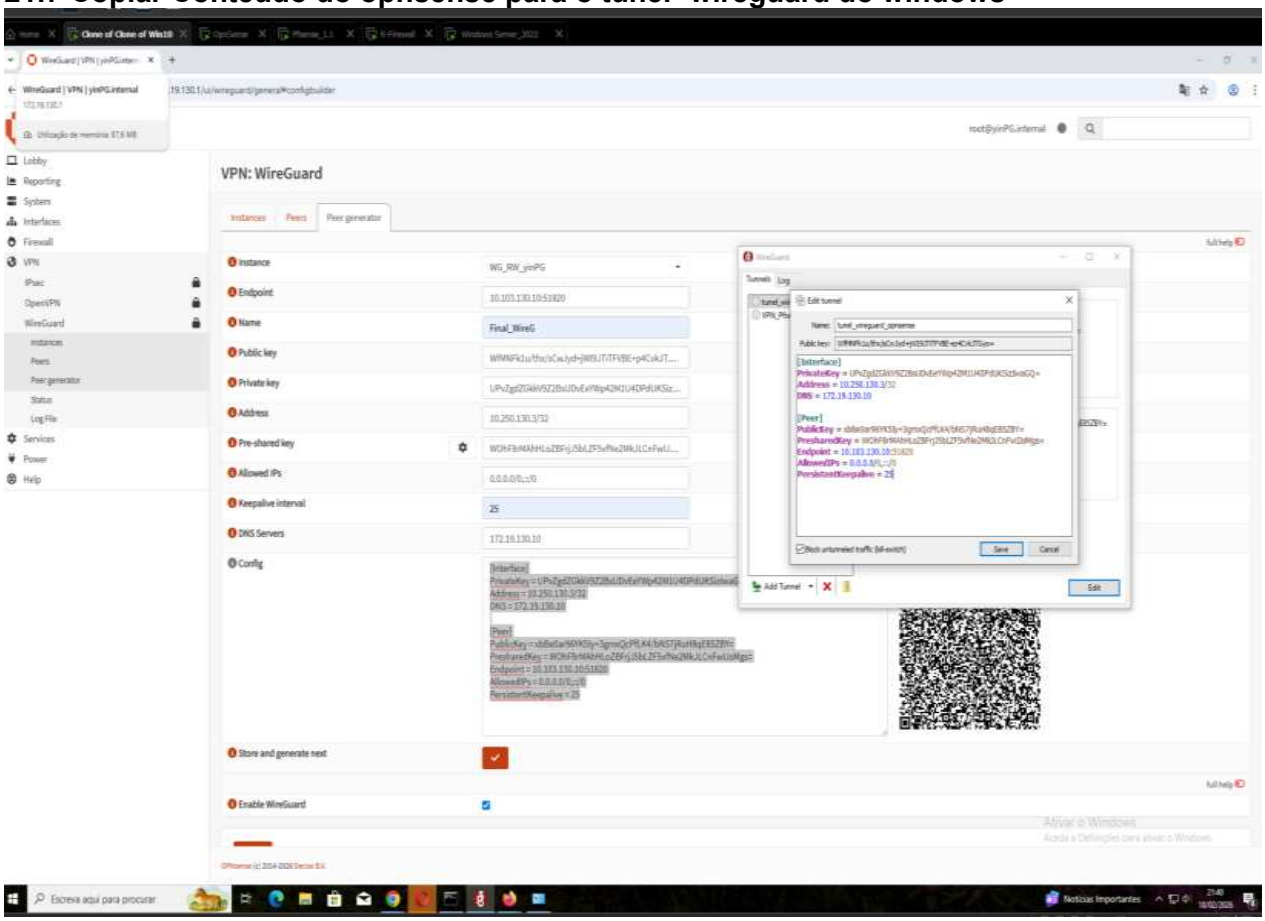
Copiar o Public key do wireguard para dentro do Public key do “Peer generator”



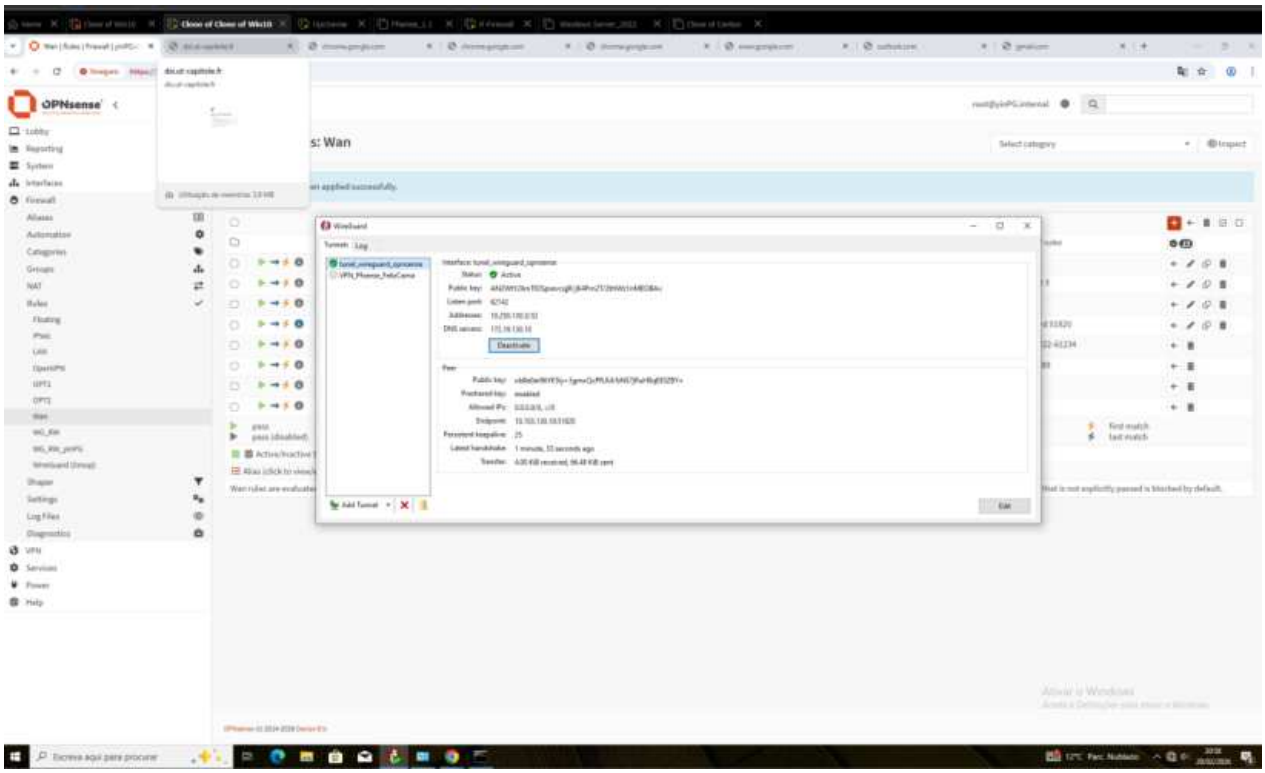
21.6 Peers generator –copiar “ config “ para as configurações do wireguard -Windows



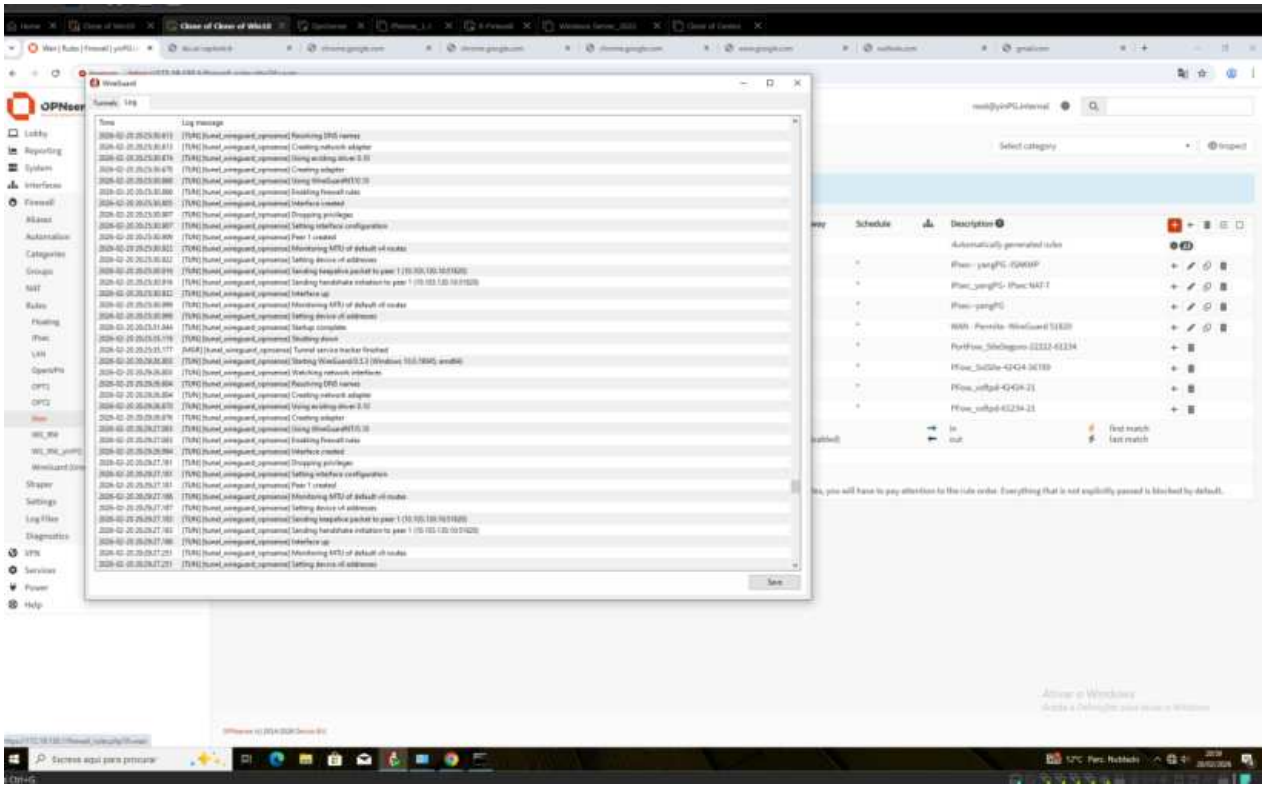
21.7 Copiar Conteudo do opnsense para o tunel -wireguard do windows



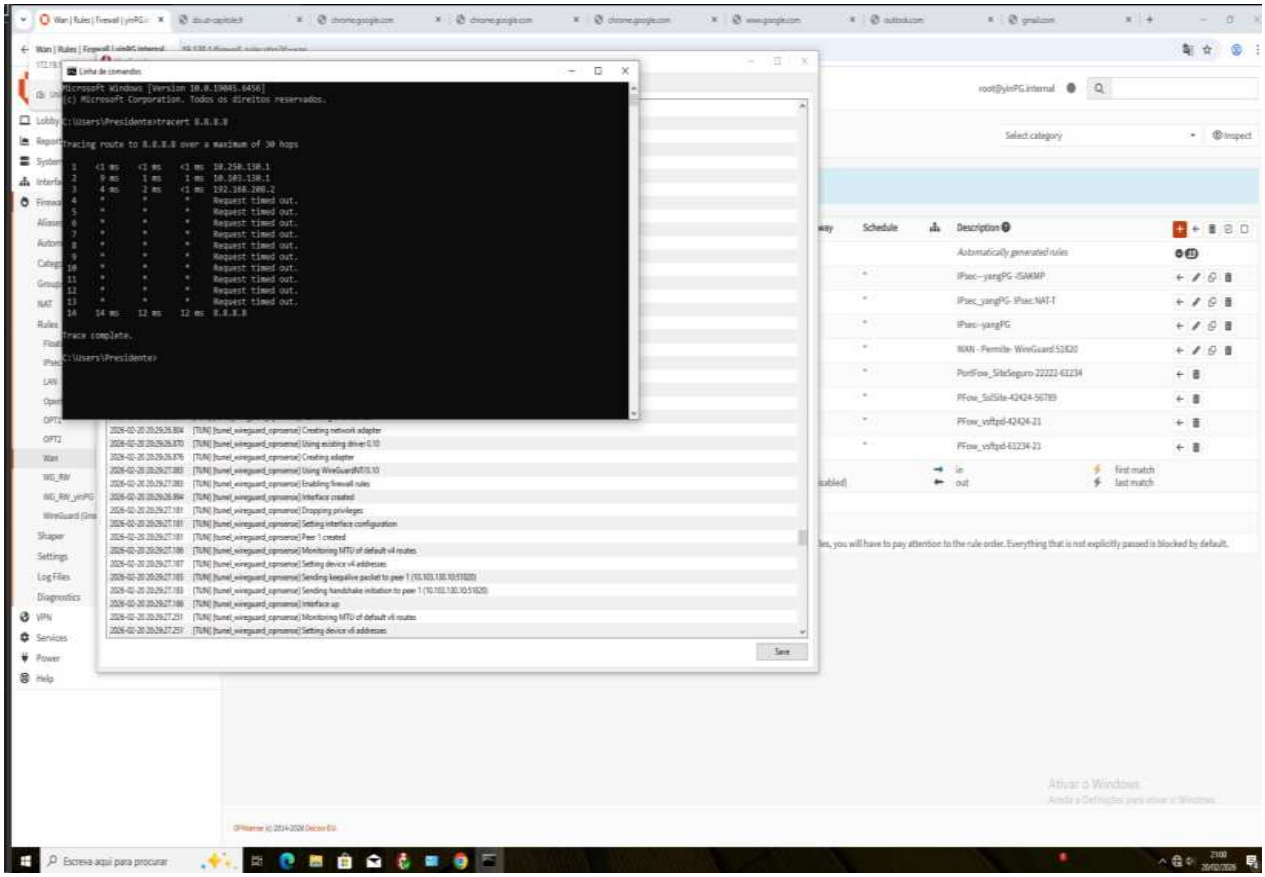
21.8 Ativação do wireguard :



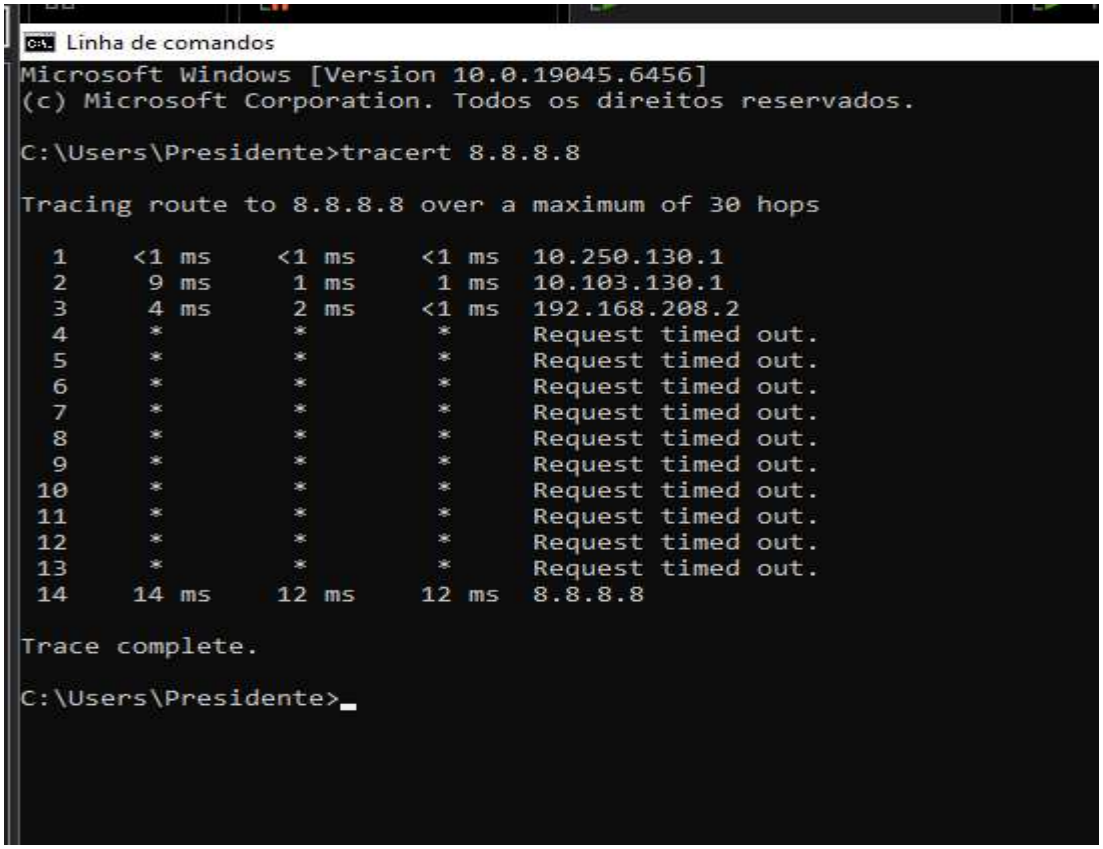
Logs :



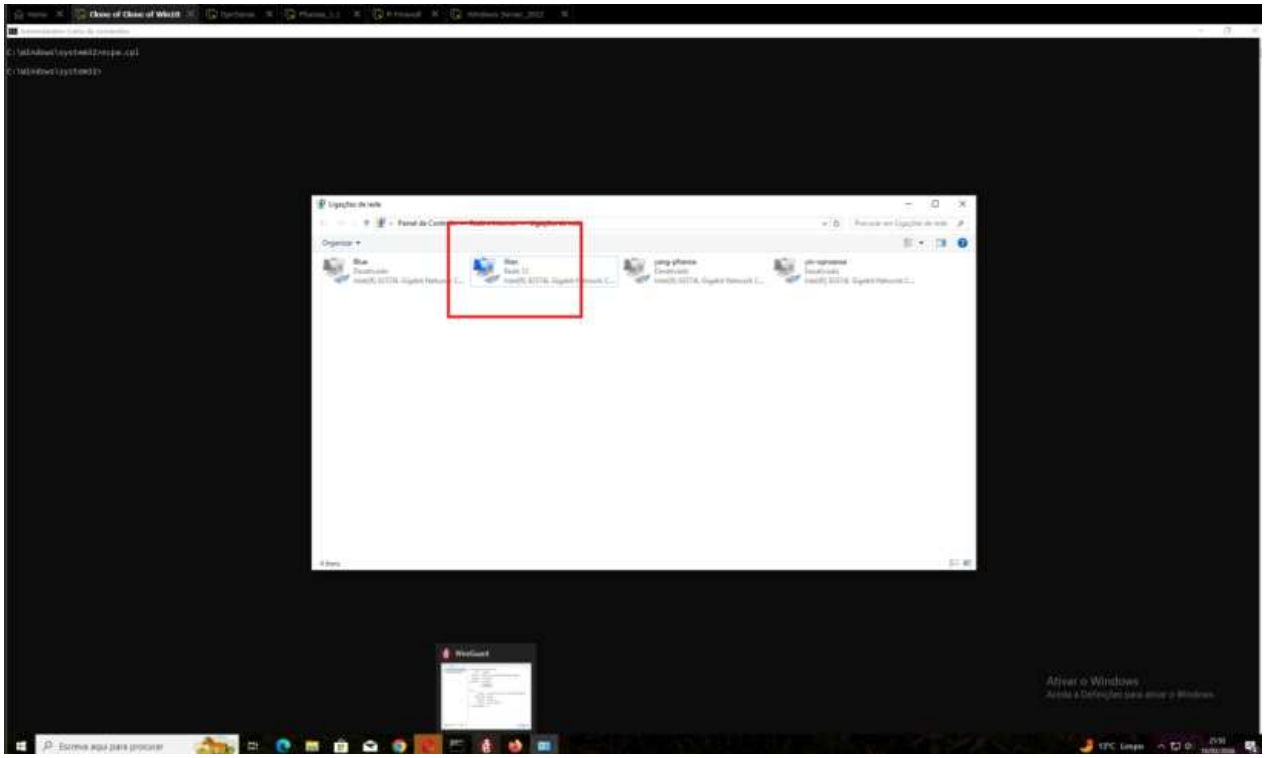
21.9 Teste : tracert 8.8.8.8



21.10 Amplificado :

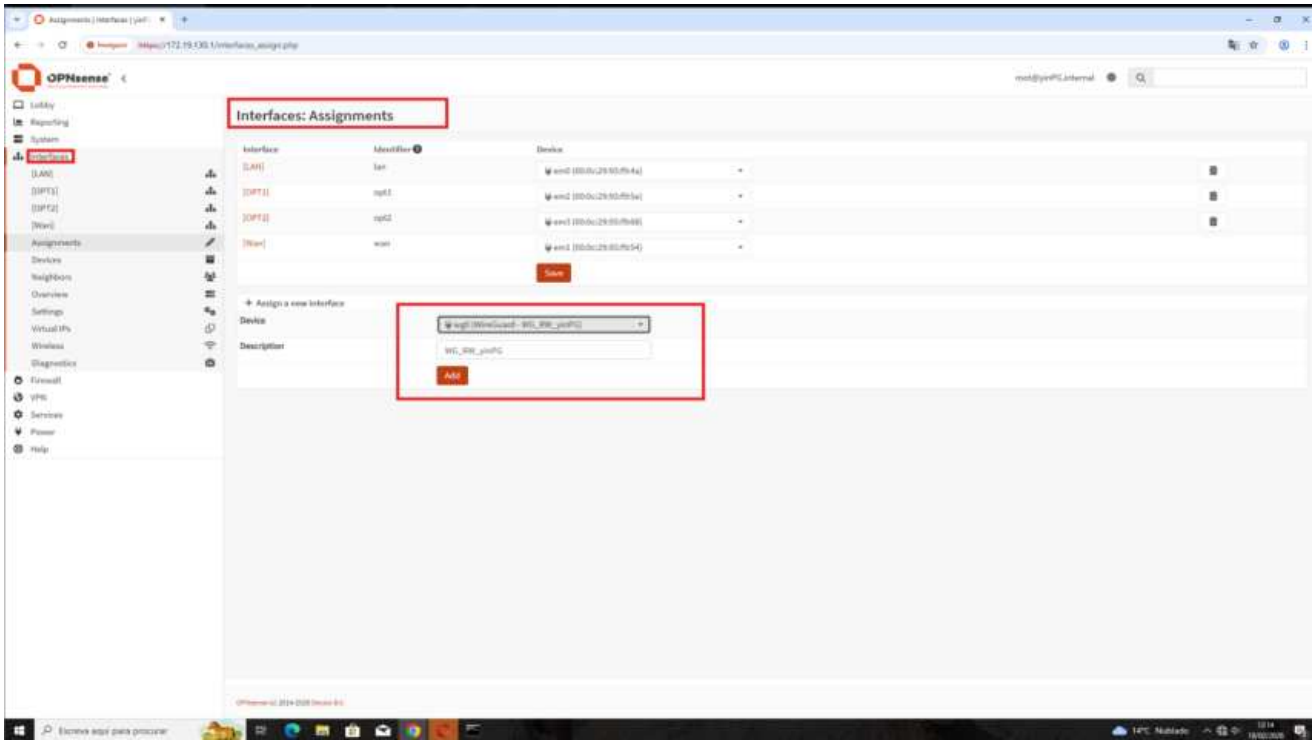


21.11 Testes : ligar apenas a rede Wan (desligar as outras)

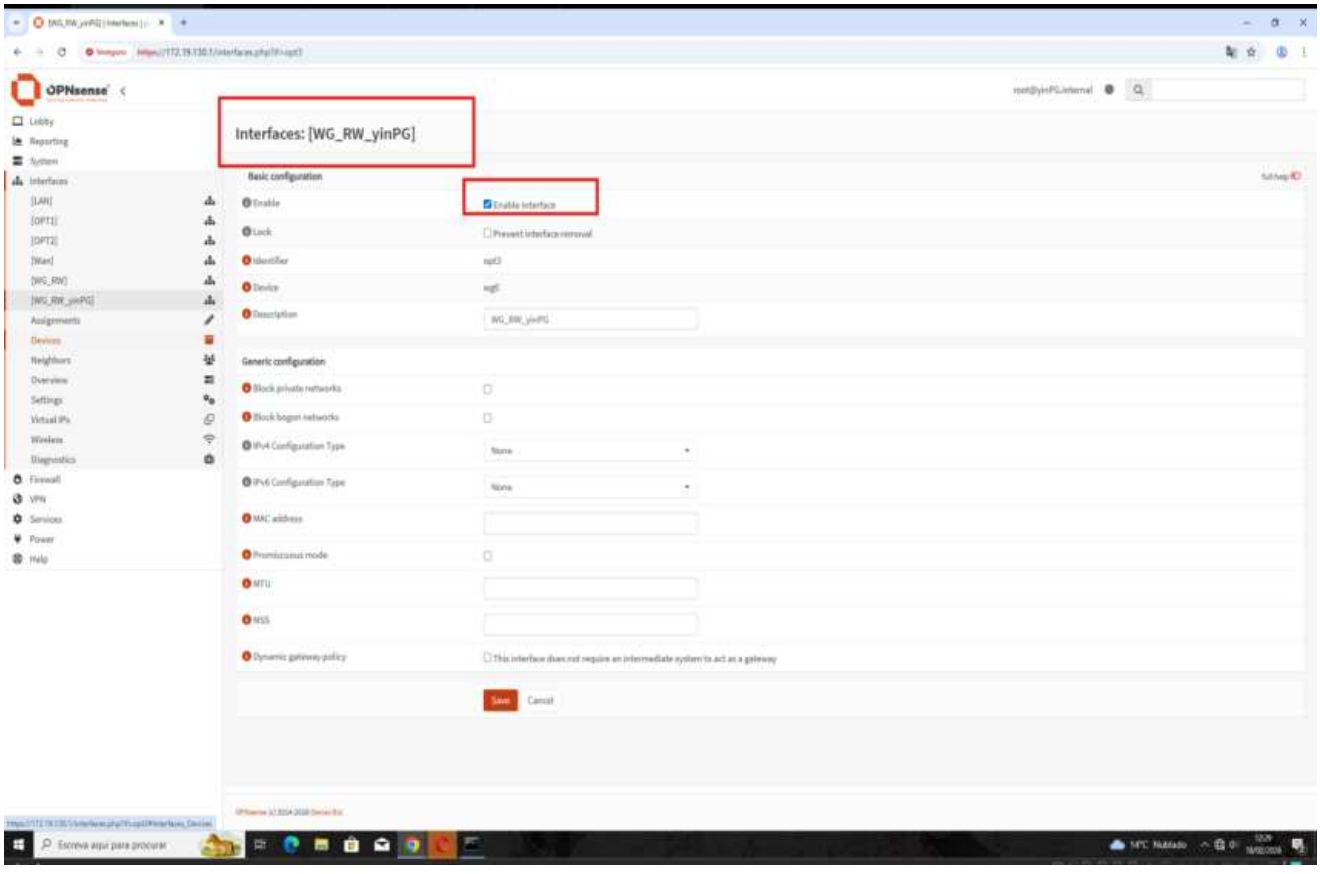


21.12 OPNsense — atribuir interface (para ter "Firewall Rules" dedicadas)

1. Ir a Interfaces → Assignments
2. No dropdown, escolher : WG_RW_yinPG

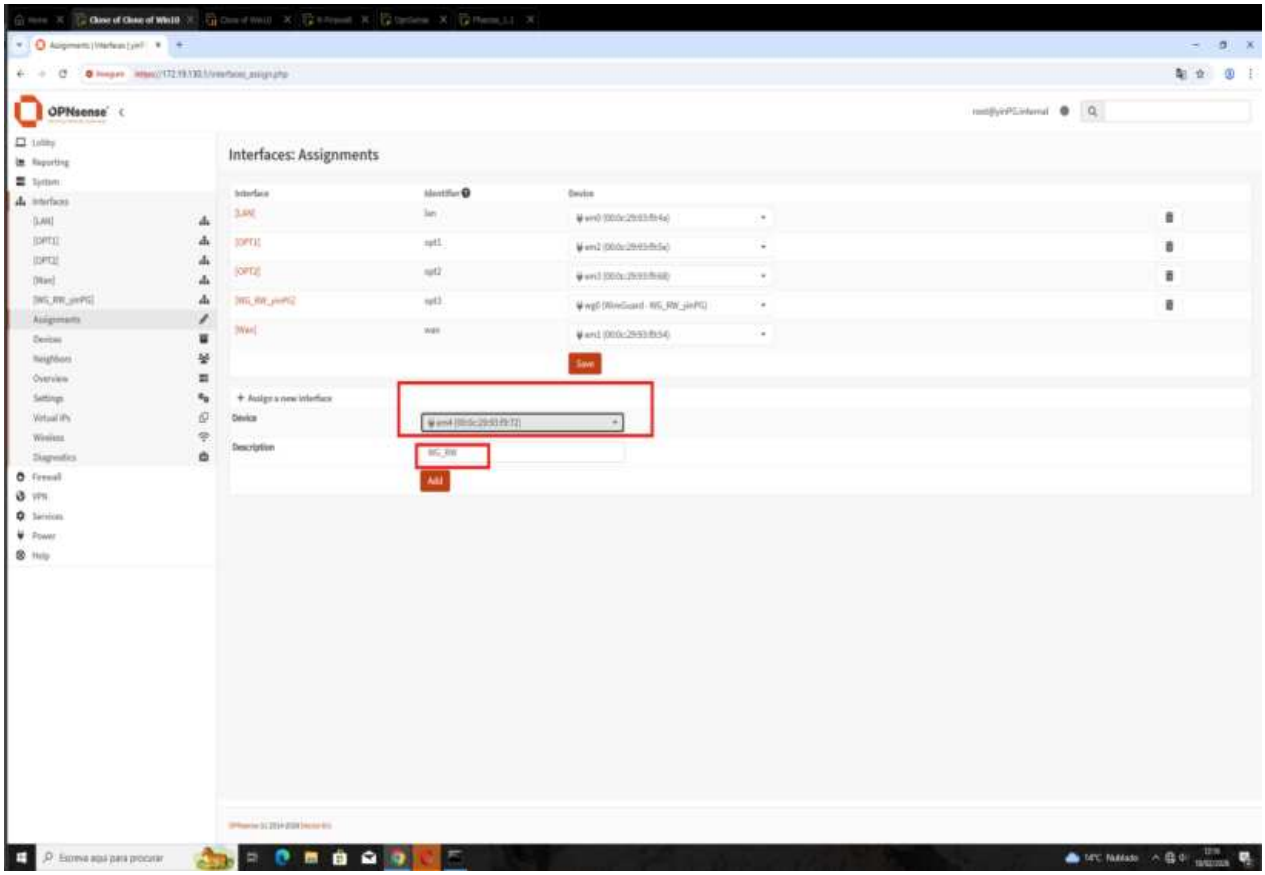


21.13 Editar interface [WG_RW_yingPG] -enable interface

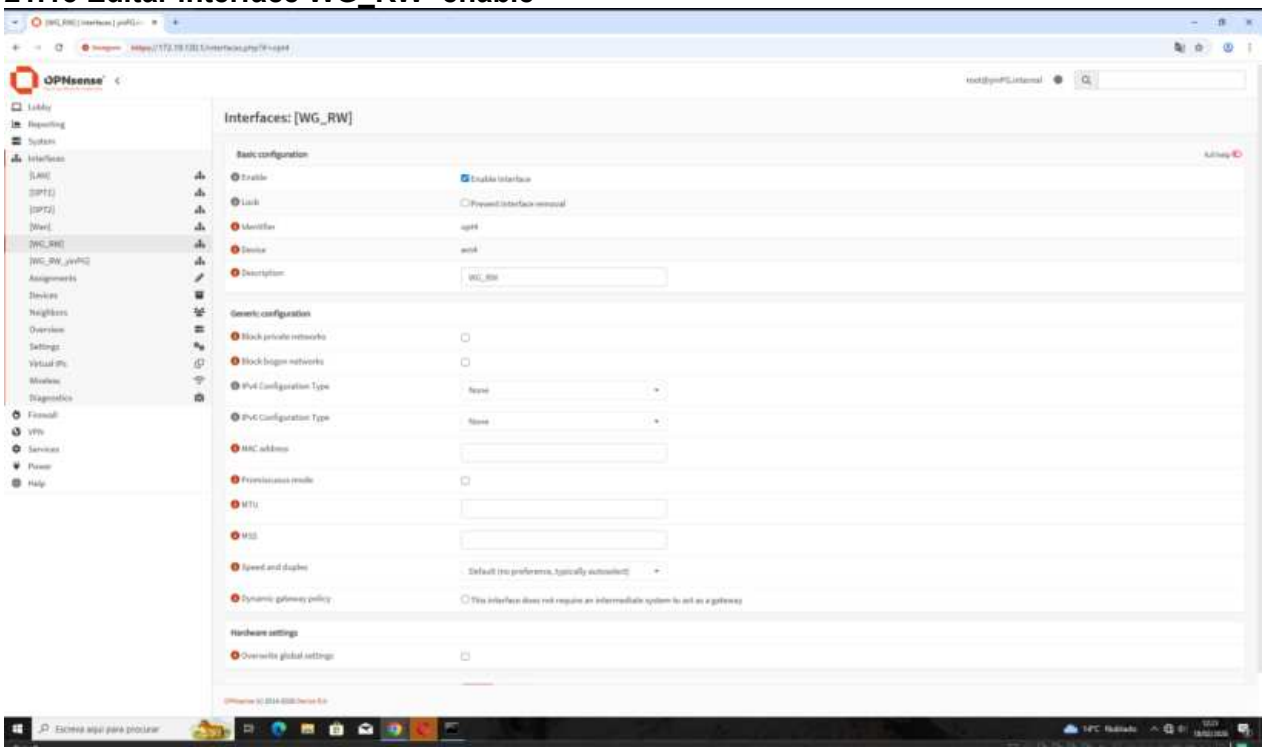


21.14 Entrar na nova interface

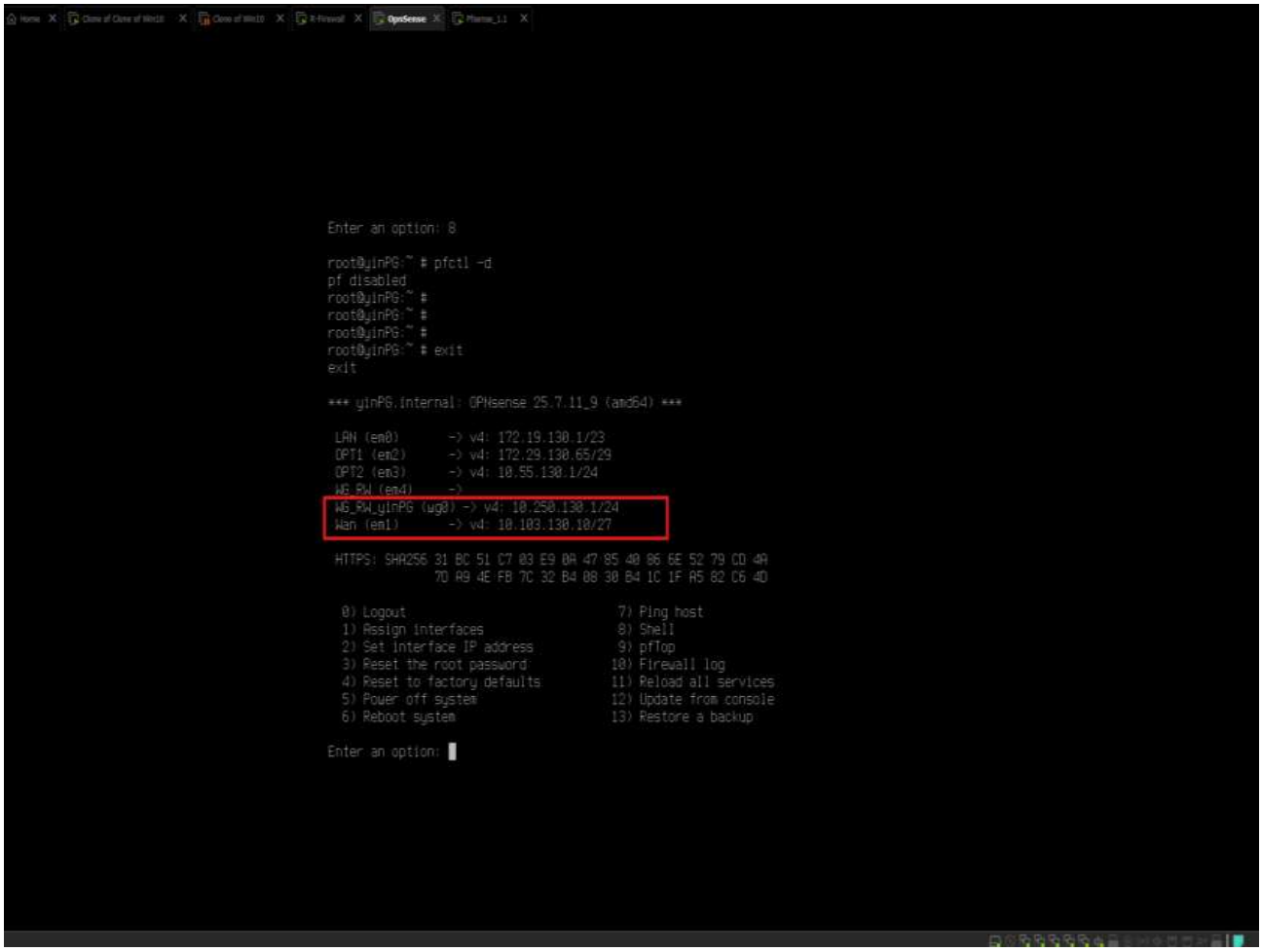
- Enable interface:
- Description: WG_RW



21.15 Editar interface WG_RW -enable



21.16.Verificação de interfaces em consola- cli

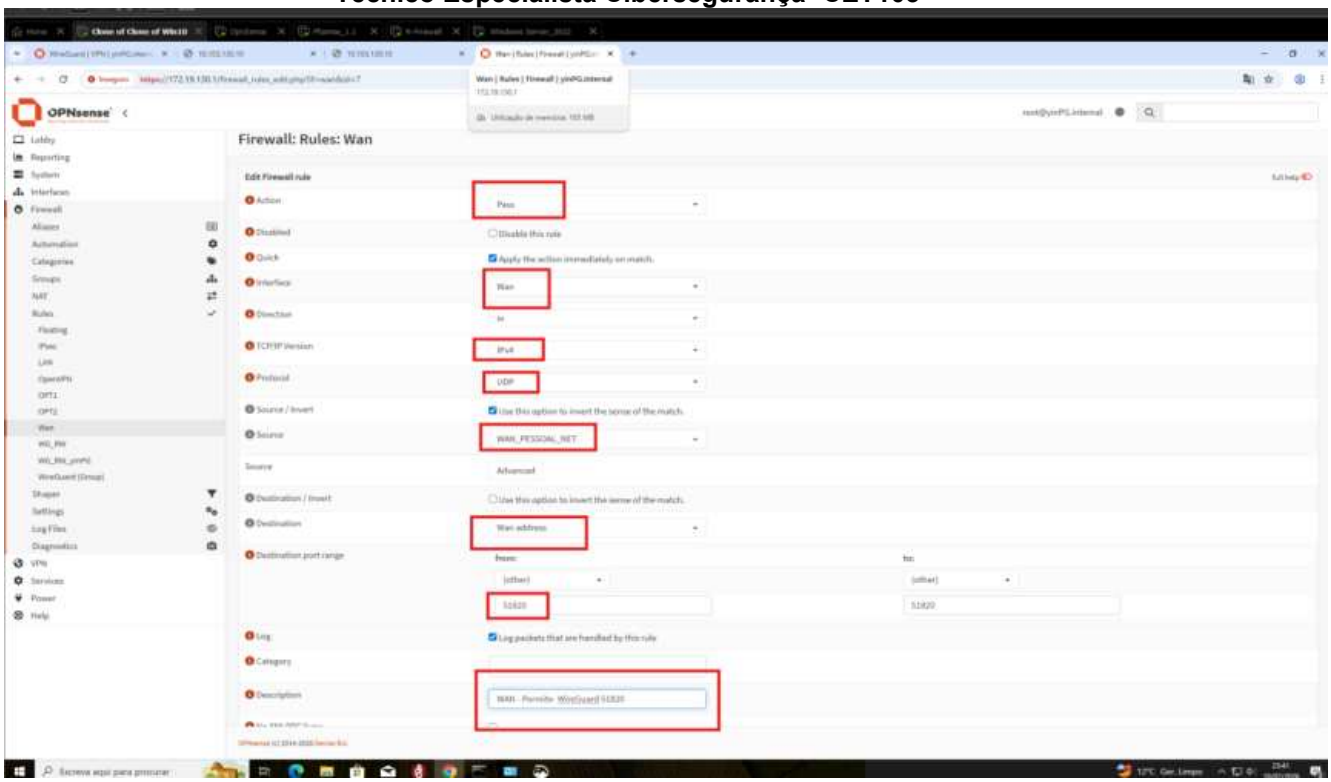


21.17- Regras Firewall no yinPG

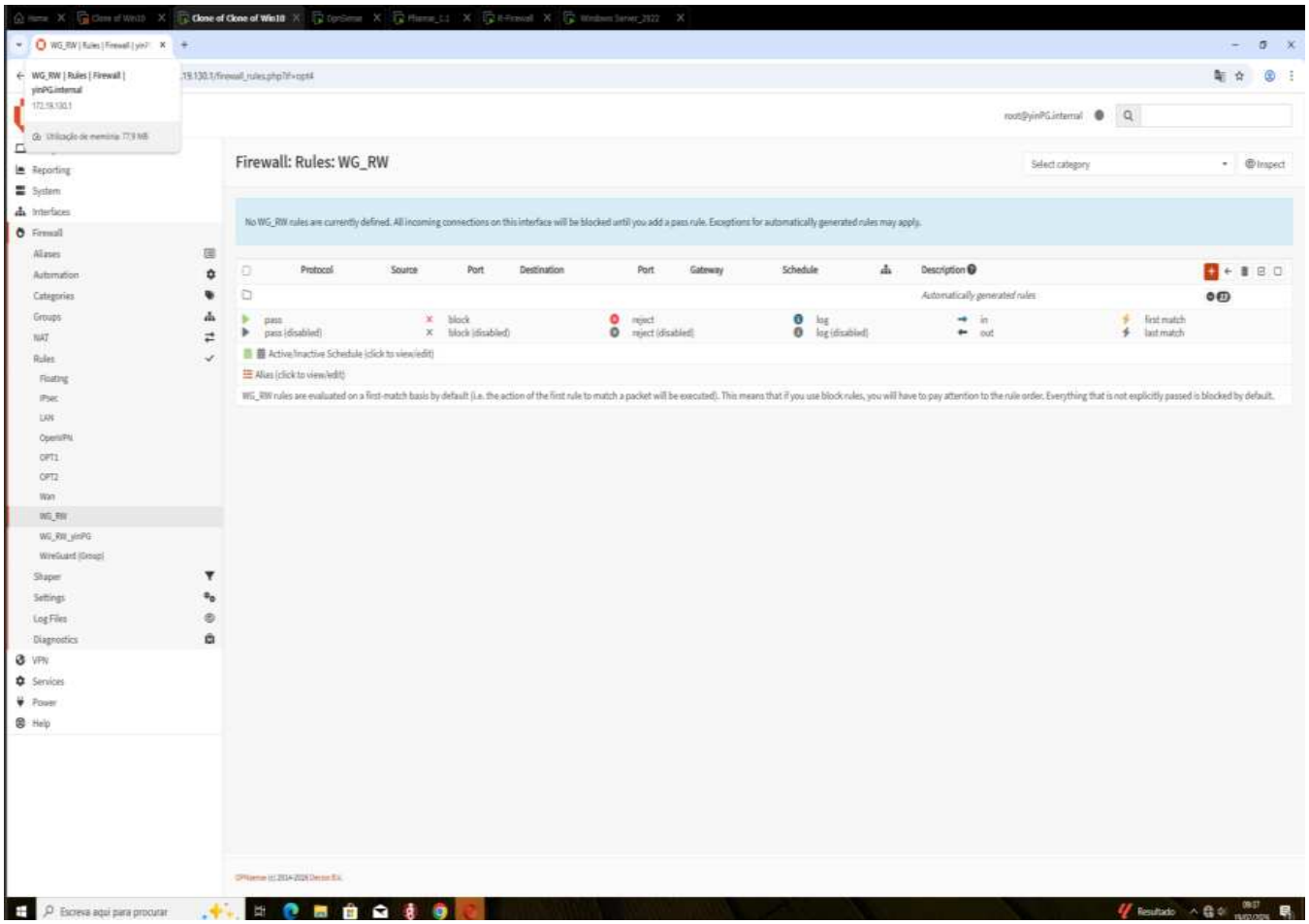
A) Regra na WAN para permitir "entrar" no WireGuard

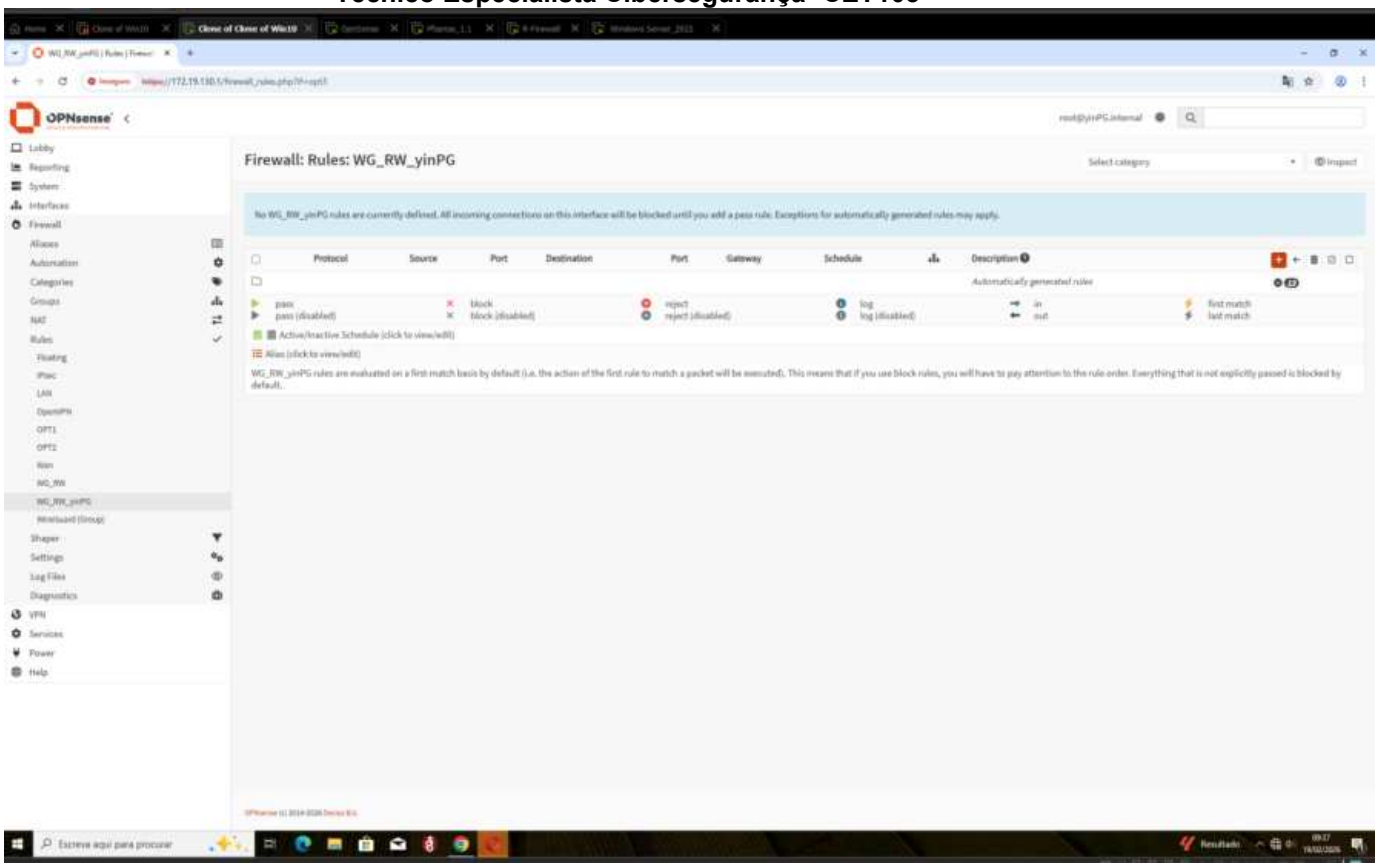
1. **Ir a Firewall → Rules → WAN**
2. **Add (Pass)**
3. **Campos:**

- Action: Pass
- Protocol: UDP
- Source: WAN_PESSOAL_NET (10.103.130.0/27) ou any
- Destination: WAN address (do yinPG)
- Destination port: 51820
- Description: WAN - PERMITE WireGuard-51820



ATENÇÃO: Se o WAN do yinPG for 10.x, confirma em Interfaces → WAN se NÃO tiver ativo "Block private networks", senão pode bloquear tráfego vindo da WAN pessoal.



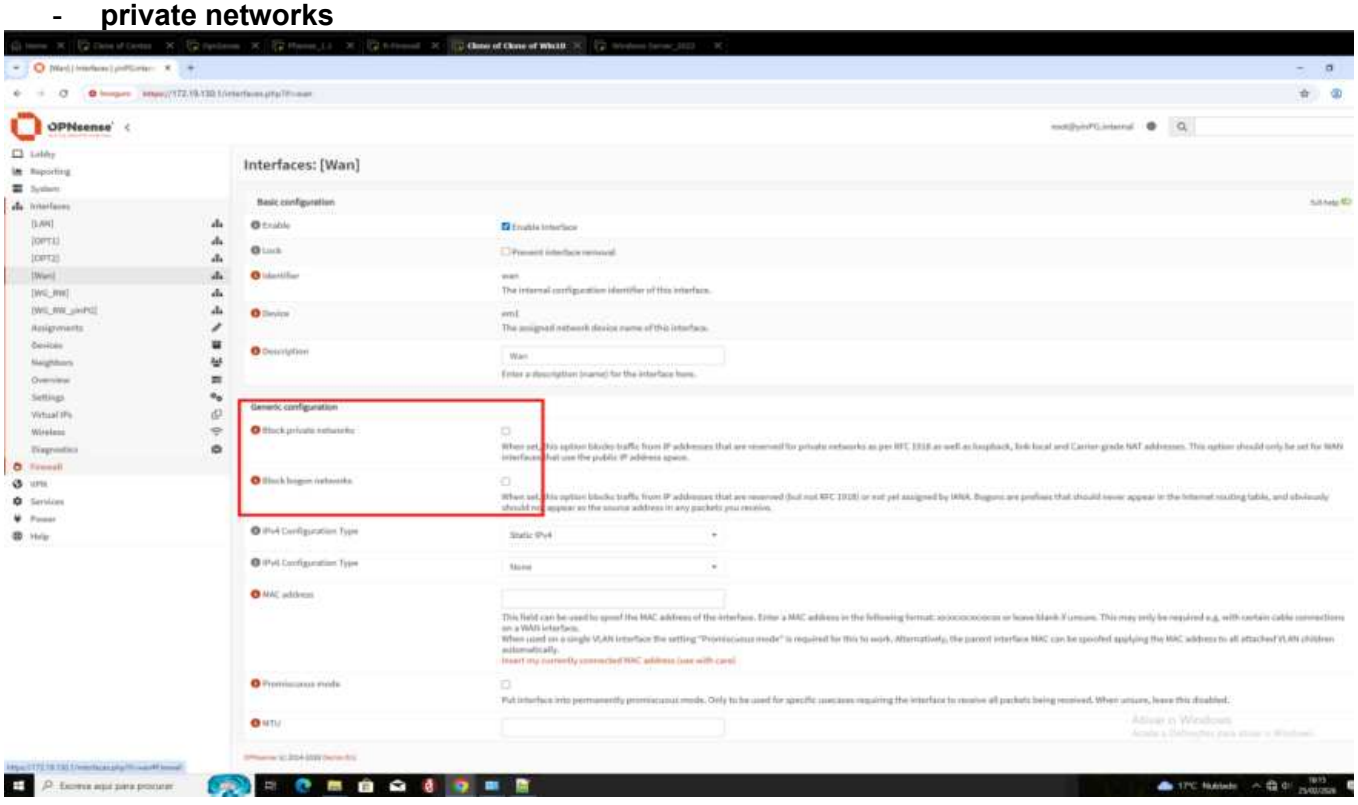


21.18- Regras na interface WG_RW (onde controlas RDP/SSH/DNS)

Verificações antes das regras

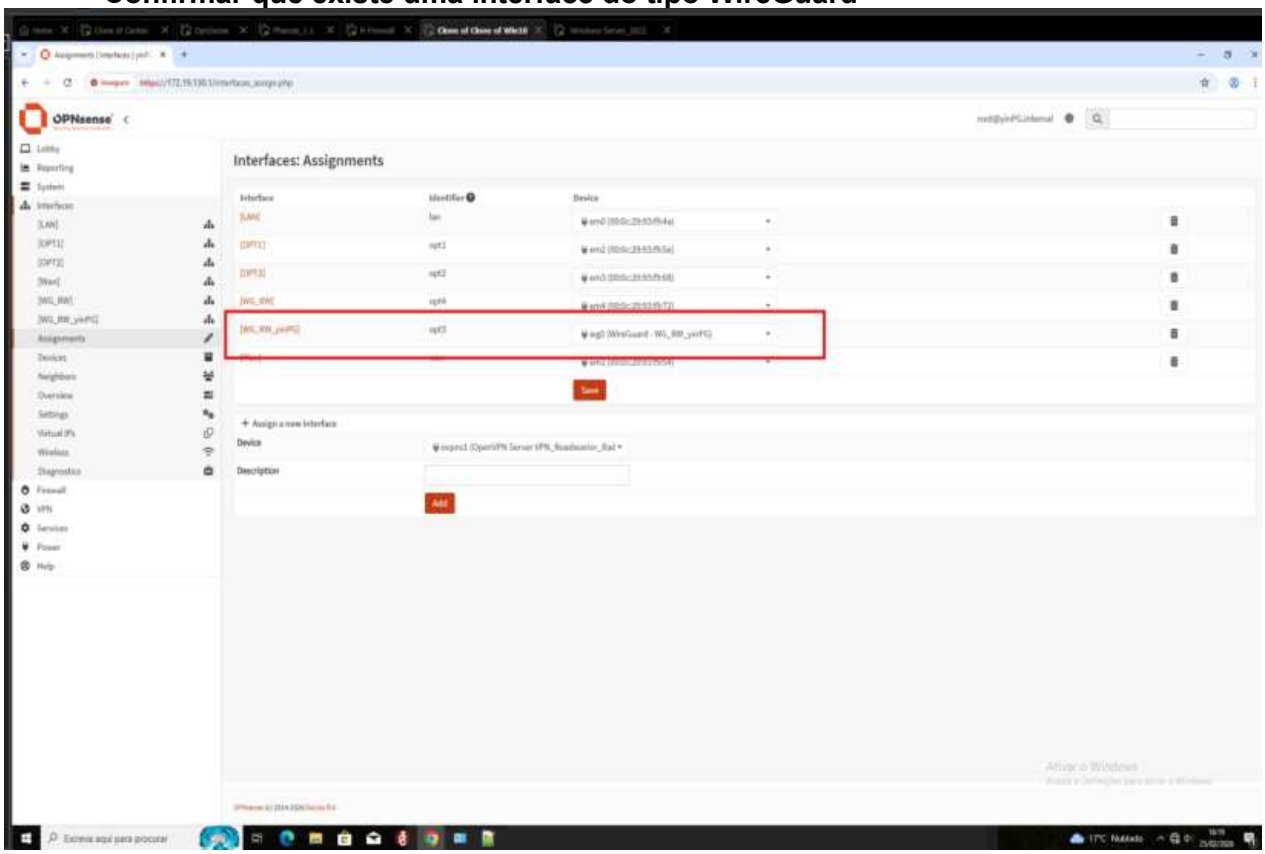
WAN do OPNsense (yinPG)

- **Ir a Interfaces → WAN**
- **Confirmar que está DESATIVO:**
- **private networks**

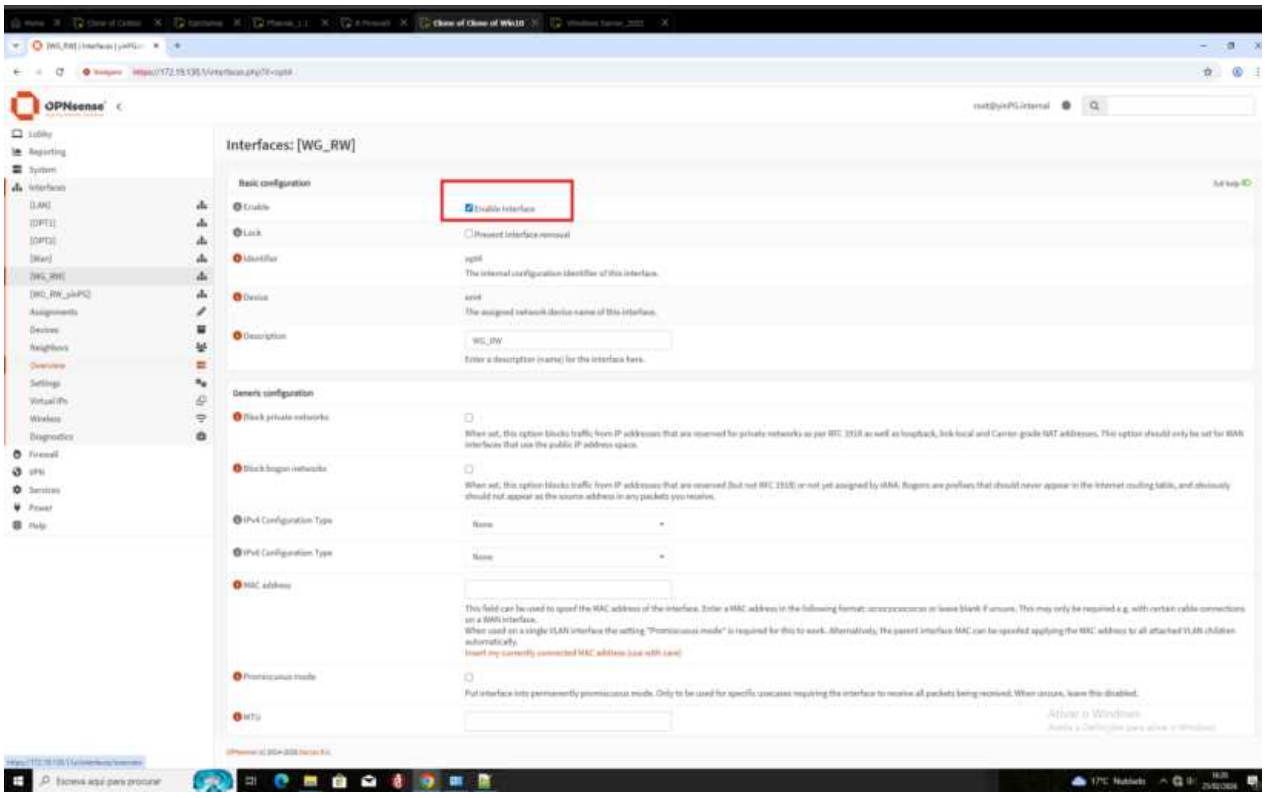


21.19 Interface WireGuard atribuída e ativa

- Ir a Interfaces → Assignments
- Confirmar que existe uma interface do tipo WireGuard



- Abrir a interface e confirmar:
  - Enable interface = sim



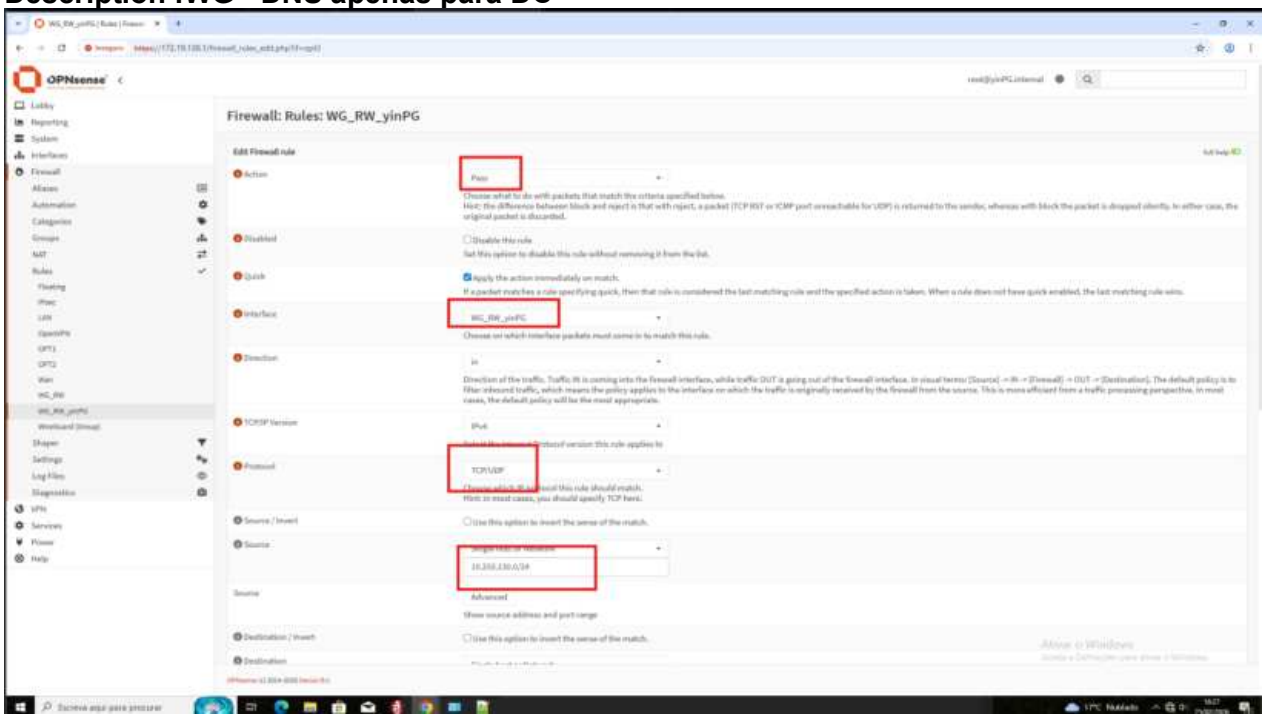
21.20 Firewall → Rules → WG_RW_yinPG

REGRA 1 — PASS DNS só para o DC

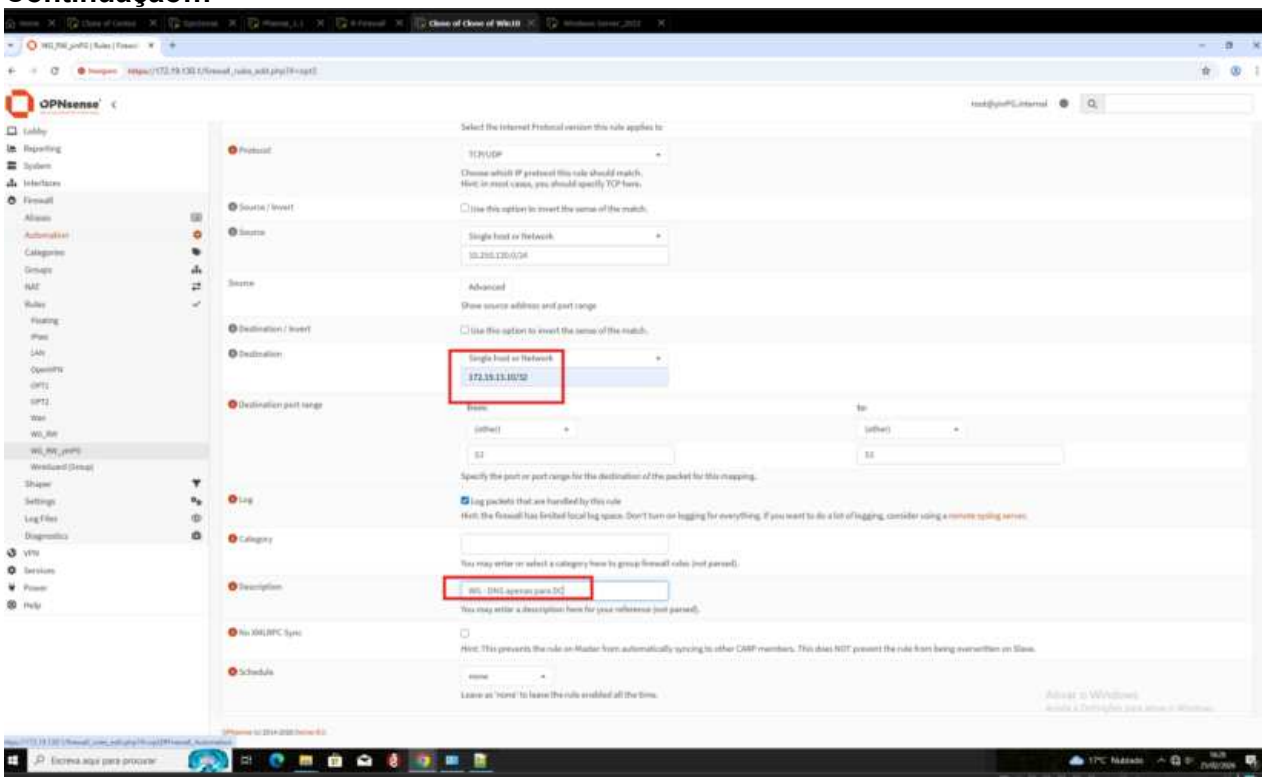
Add:

- **Action: Pass**
- **Protocol: TCP/UDP**
- **Source: 10.250.130.0/24**
- **Destination: Single host → 172.19.130.10**
- **Port: 53**
- **Description:**

Description :WG - DNS apenas para DC



Continuação…

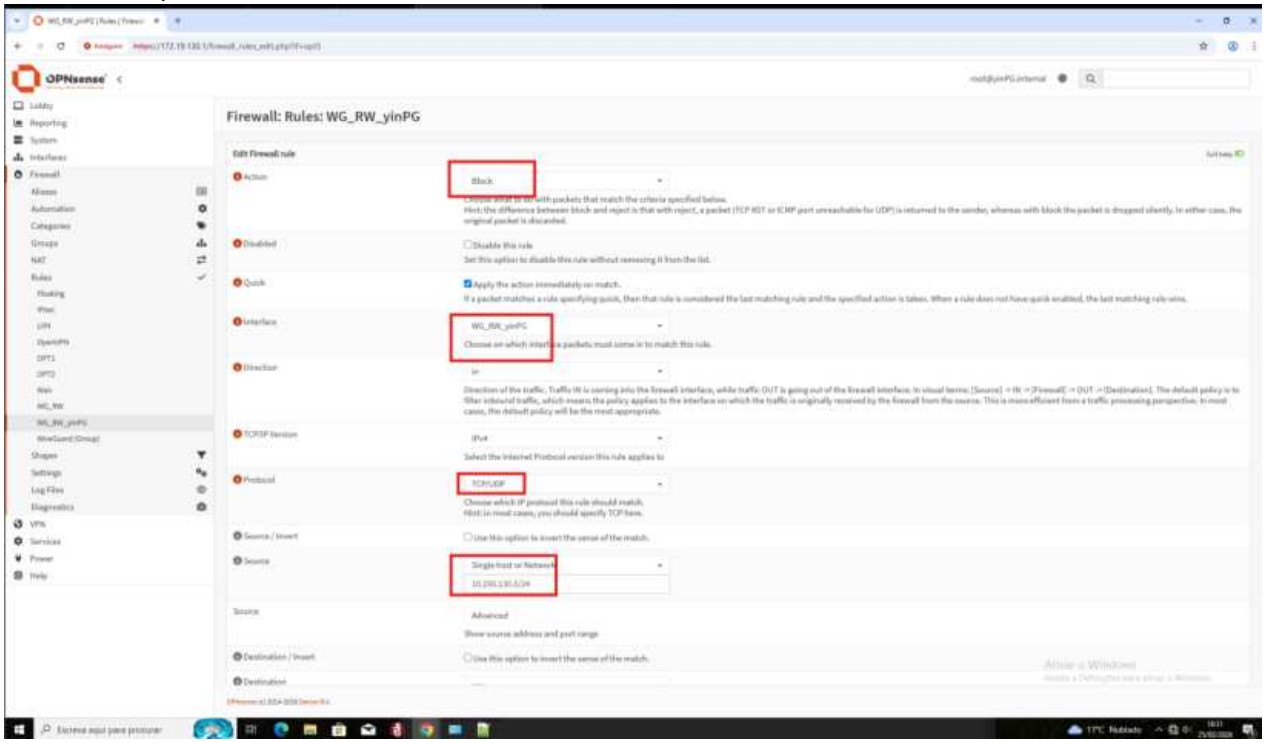


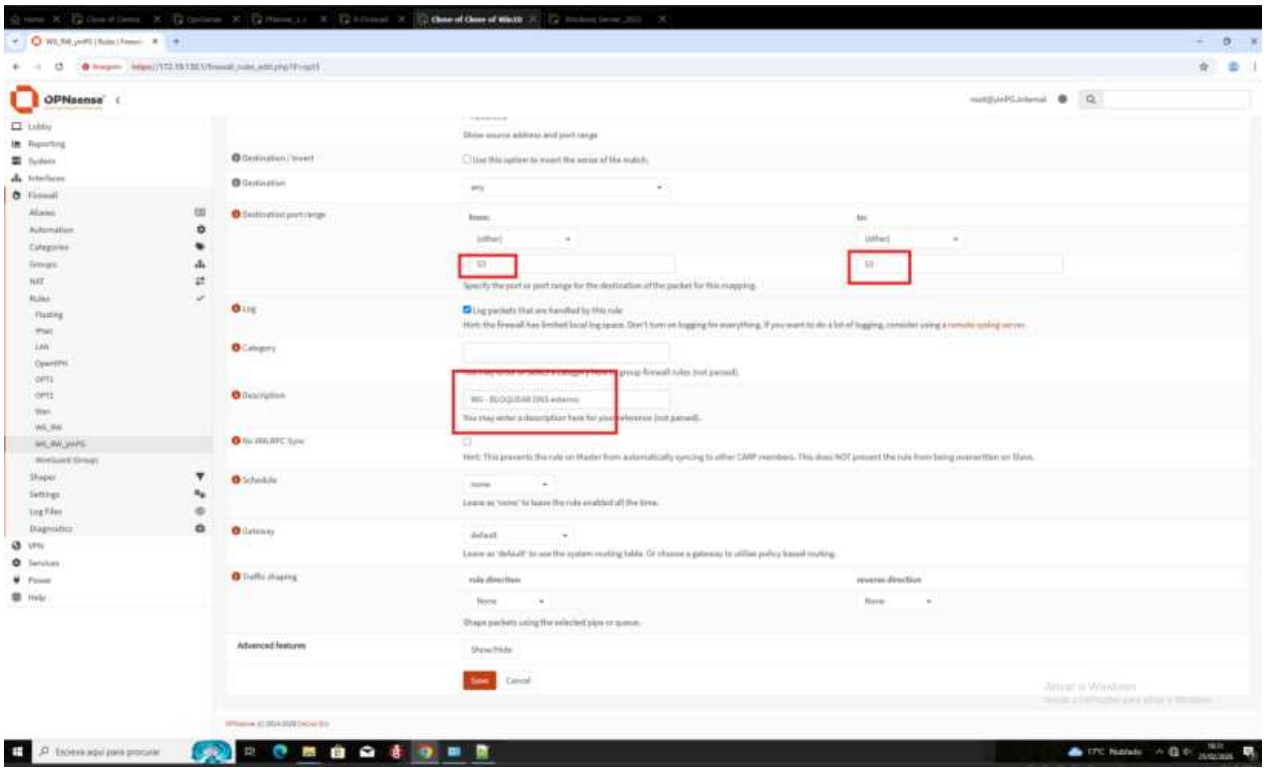
21.21 REGRA 2 — BLOCK DNS externo

Add:

- **Action: Block**
- **Protocol: TCP/UDP**
- **Source: 10.250.130.0/24**
- **Destination: any**
- **Port: 53**
- **Log:** on
- **Description:**

WG - BLOQUEAR DNS externo



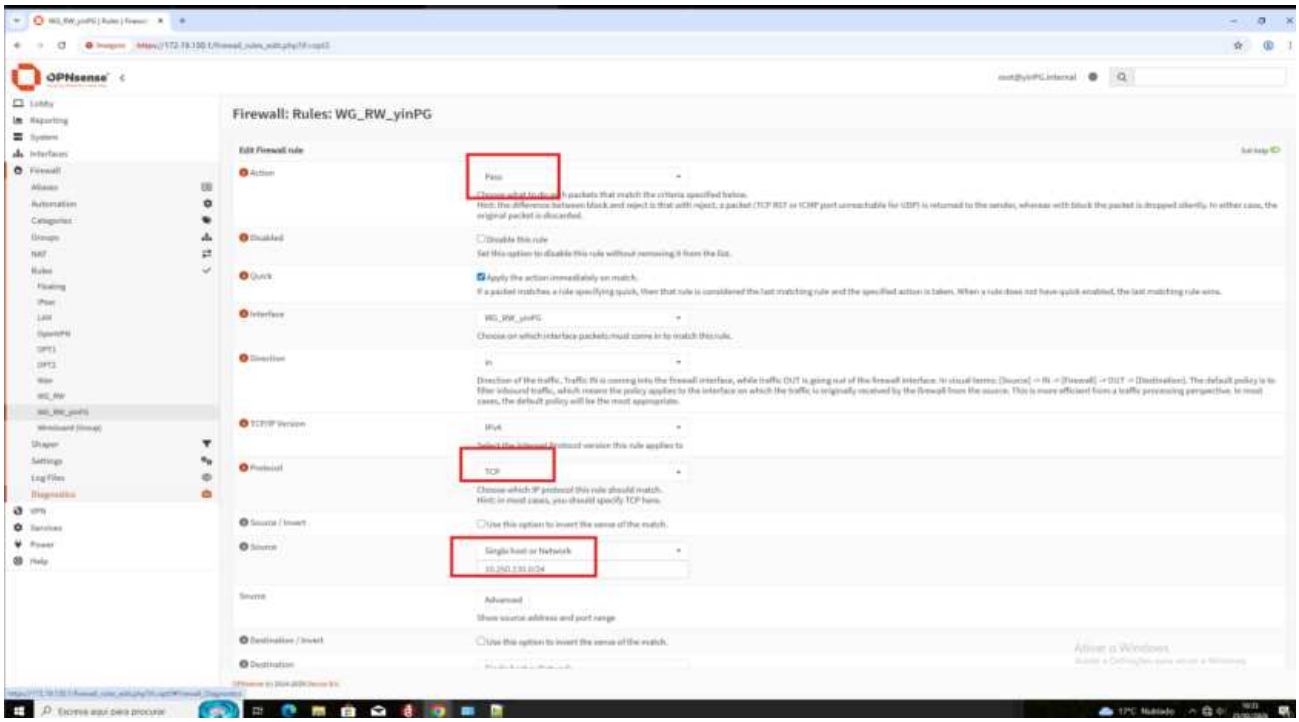


21.22 REGRA 3 — PASS RDP

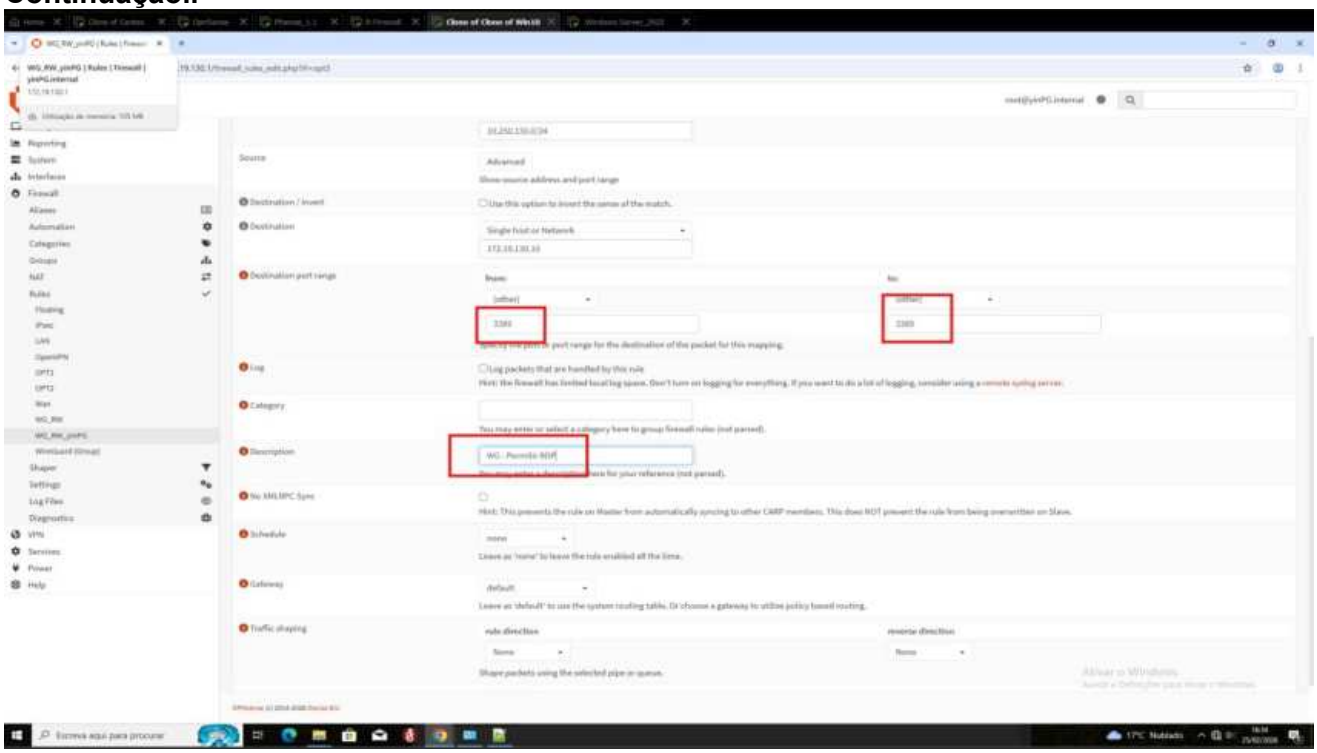
Add:

- **Action: Pass**
- **Protocol: TCP**
- **Source: 10.250.130.0/24**
- **Destination: 172.19.130.10**
- **Port: 3389**
- **Log: opcional**
- **Description:**

WG - Permitir RDP



Continuação..

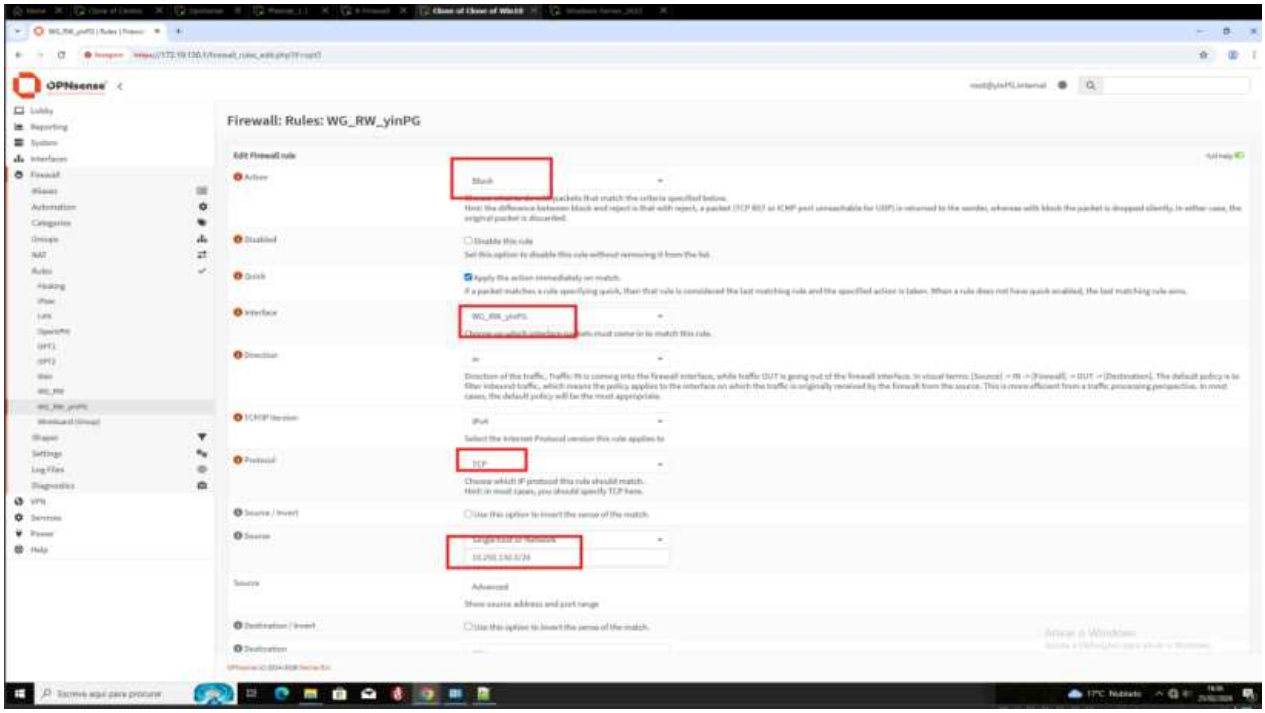


21.24 REGRA 4 — BLOCK SSH

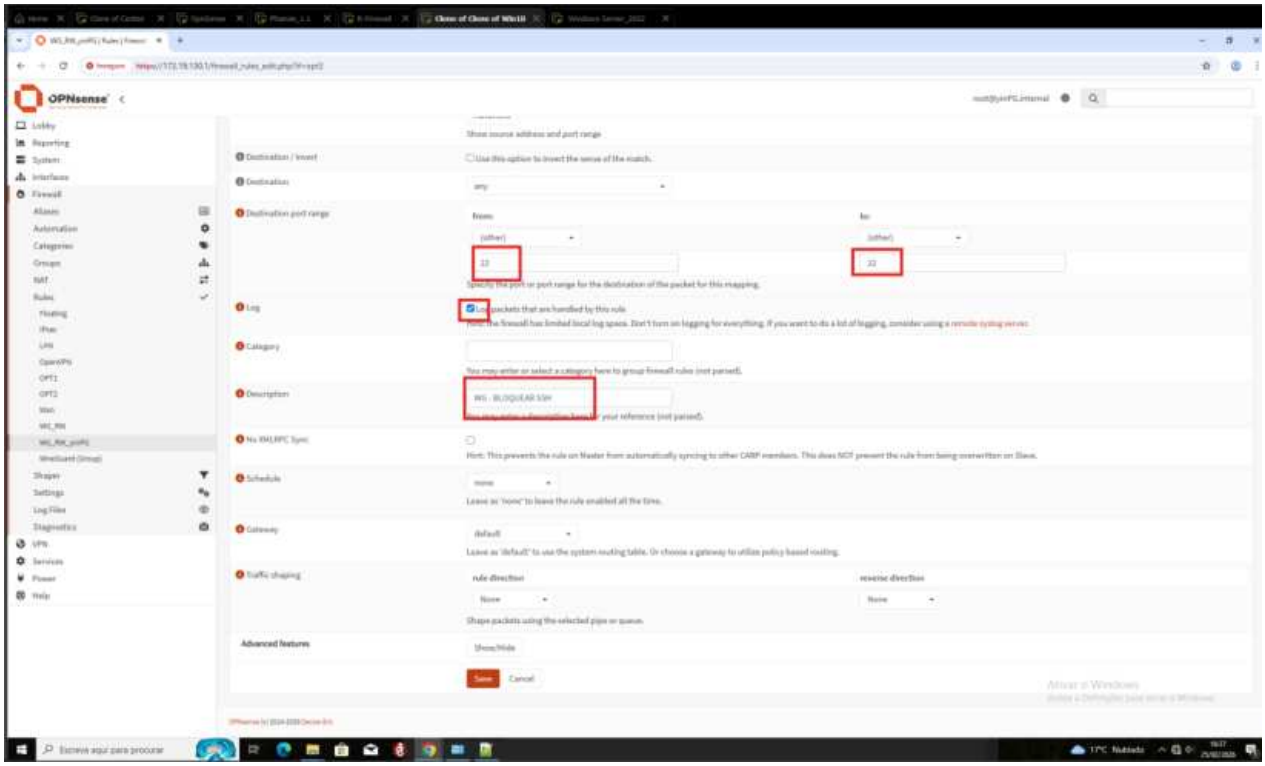
Add:

- **Action: Block**
- **Protocol: TCP**
- **Source: 10.250.130.0/24**
- **Destination: any**
- **Port: 22**
- **Log: ON**
- **Description:**

WG - BLOQUEAR SSH



Continuação…

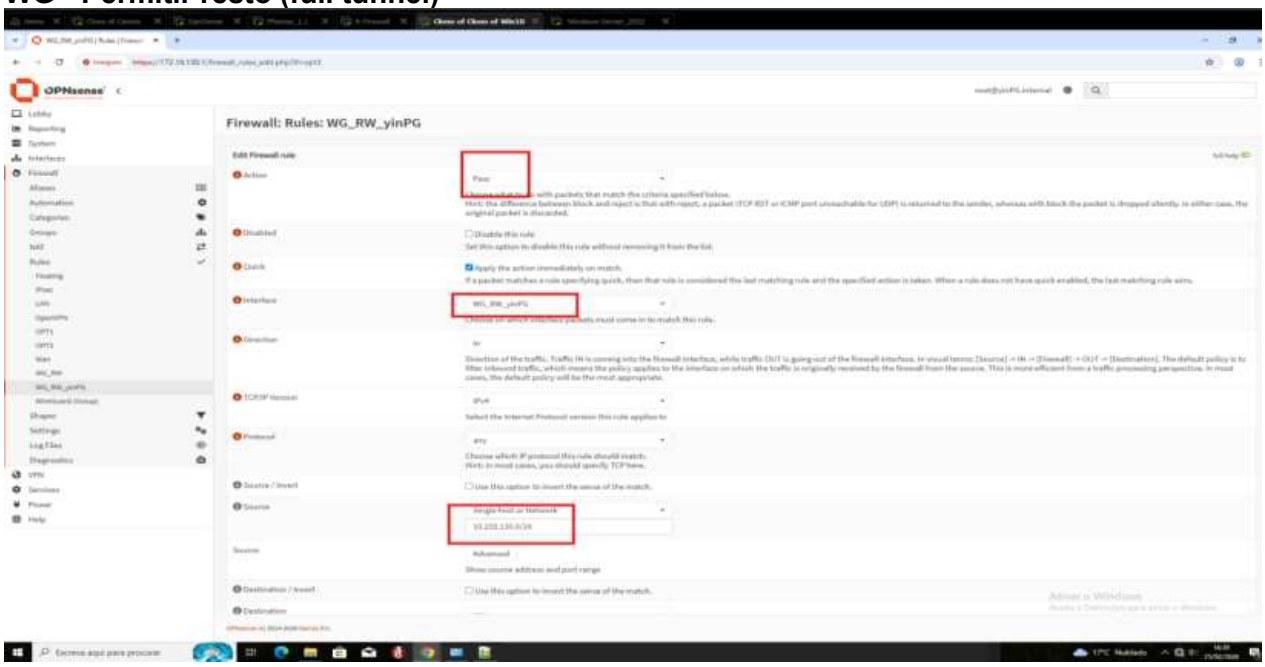


21.25 REGRA 5 — PASS resto (full tunnel)

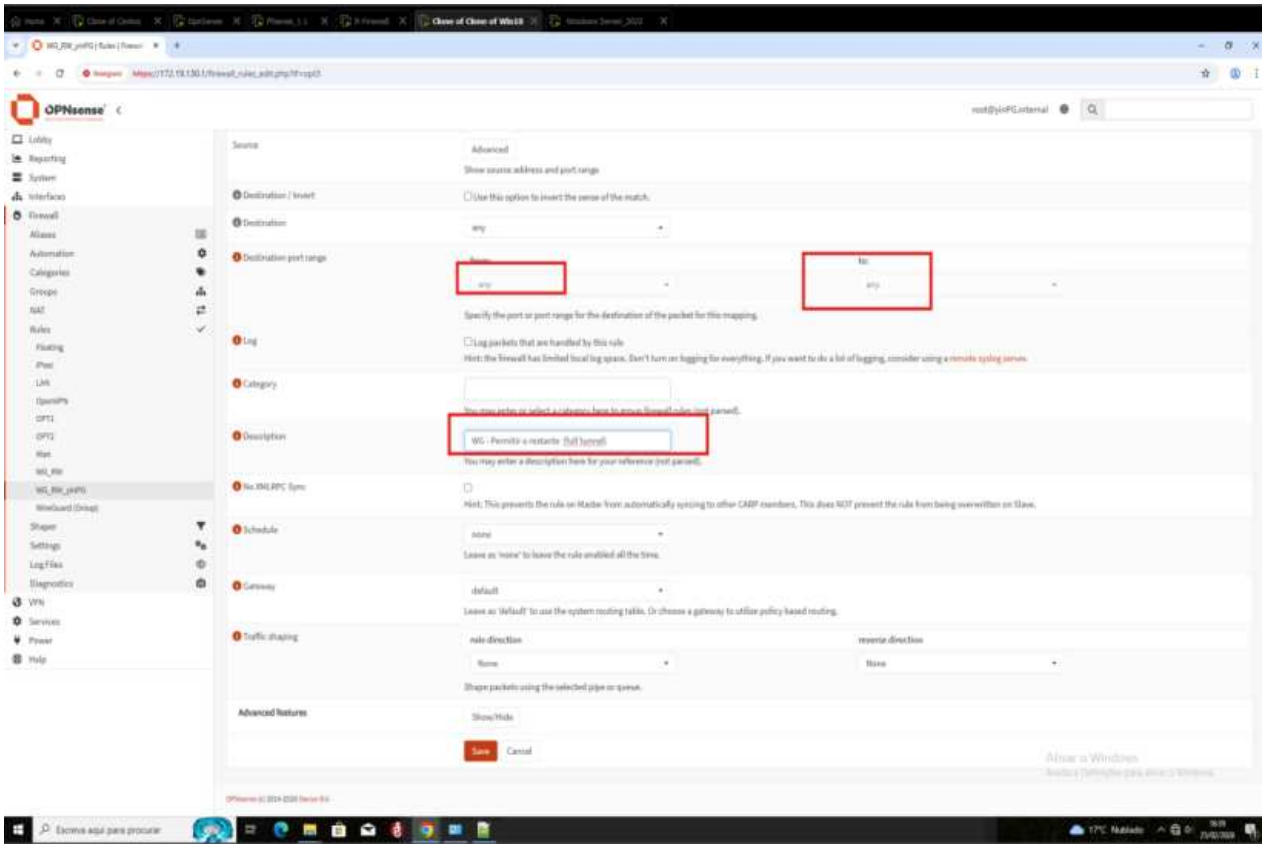
Add:

- **Action: Pass**
- **Protocol: any**
- **Source: 10.250.130.0/24**
- **Destination: any**
- **Description:**

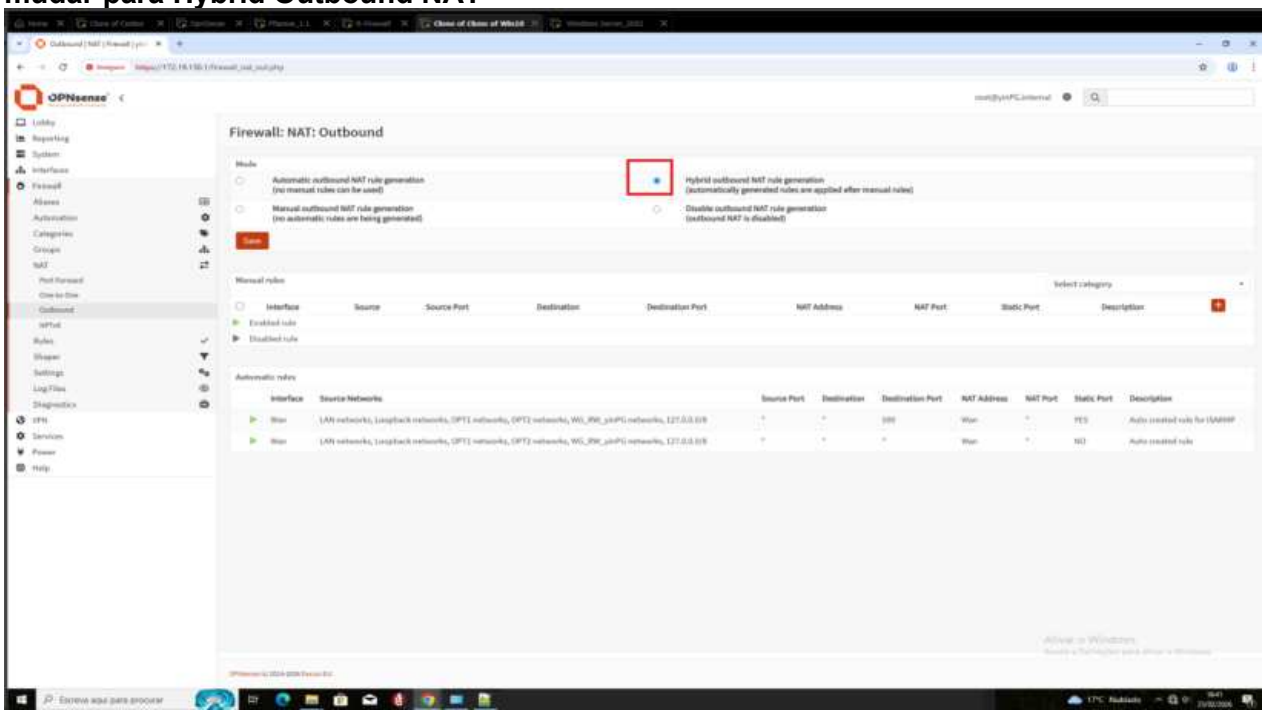
WG - Permitir resto (full tunnel)



Continuação..



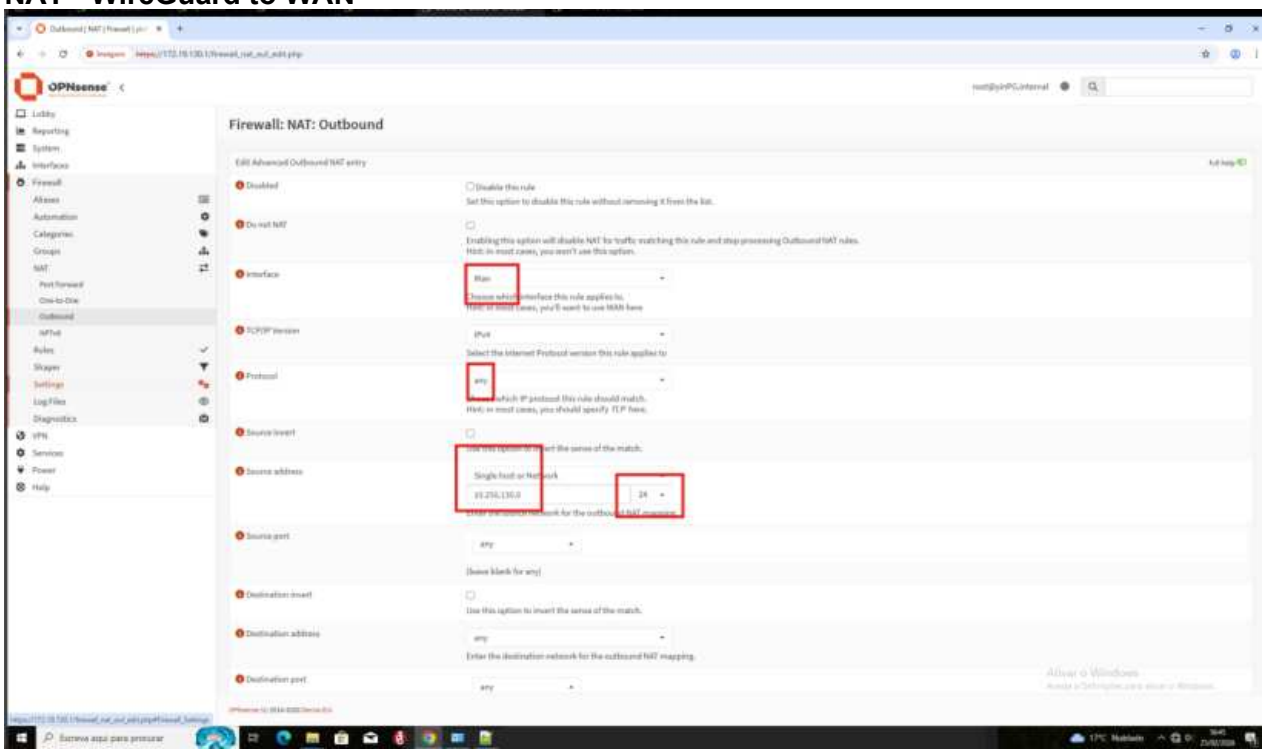
21.26 PASSO 4 — OUTBOUND NAT
Firewall → NAT → Outbound
Se estiver Automatic:
mudar para Hybrid Outbound NAT



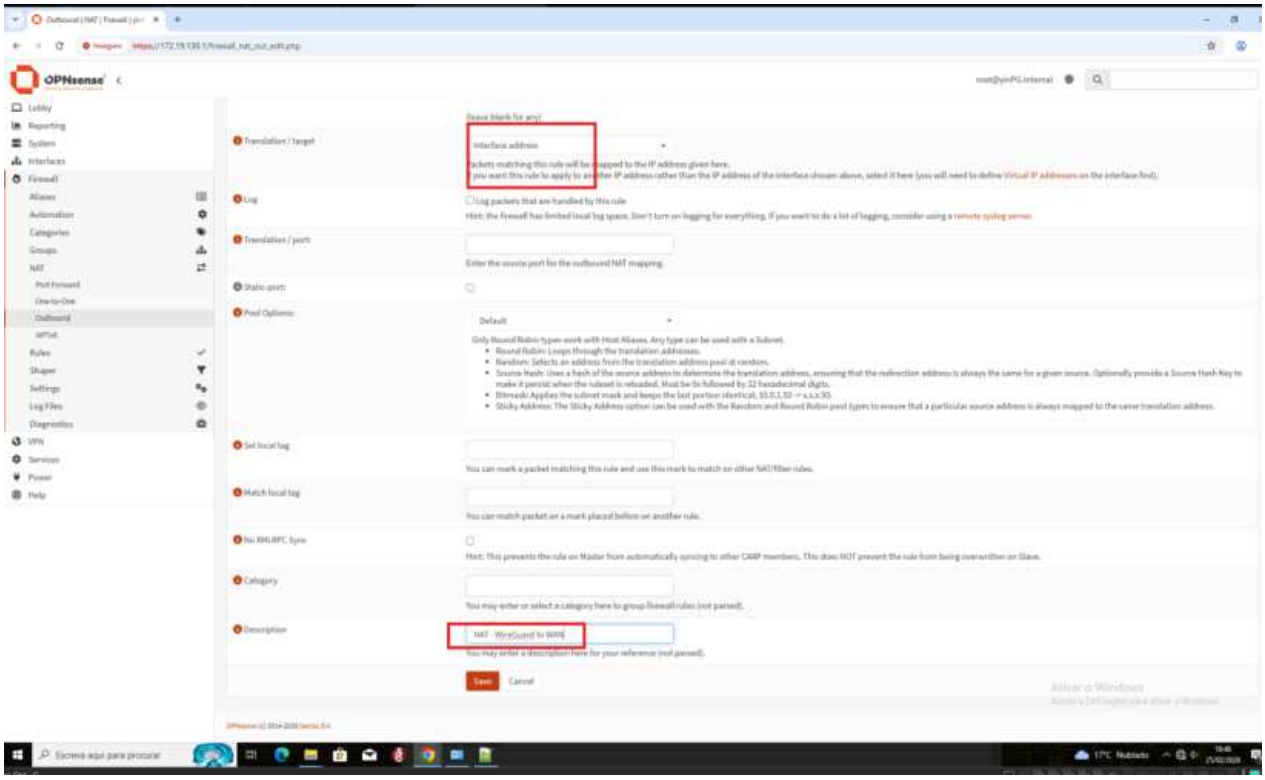
21.27 Criar regra nova
Add:

- **Interface: WAN**
- **Source: 10.250.130.0/24**
- **Translation: Interface address**
- **Description:**

NAT - WireGuard to WAN



Continuação..

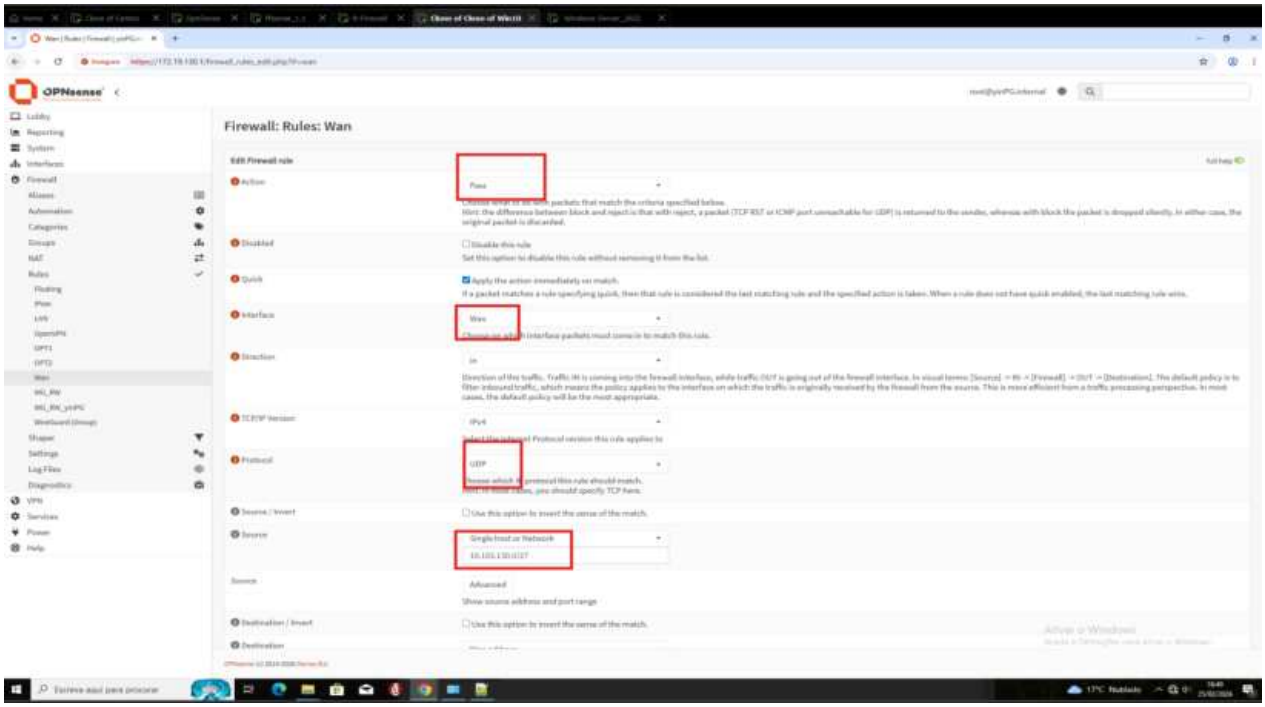


21.28 PASSO 5 — Regra na WAN para permitir WireGuard

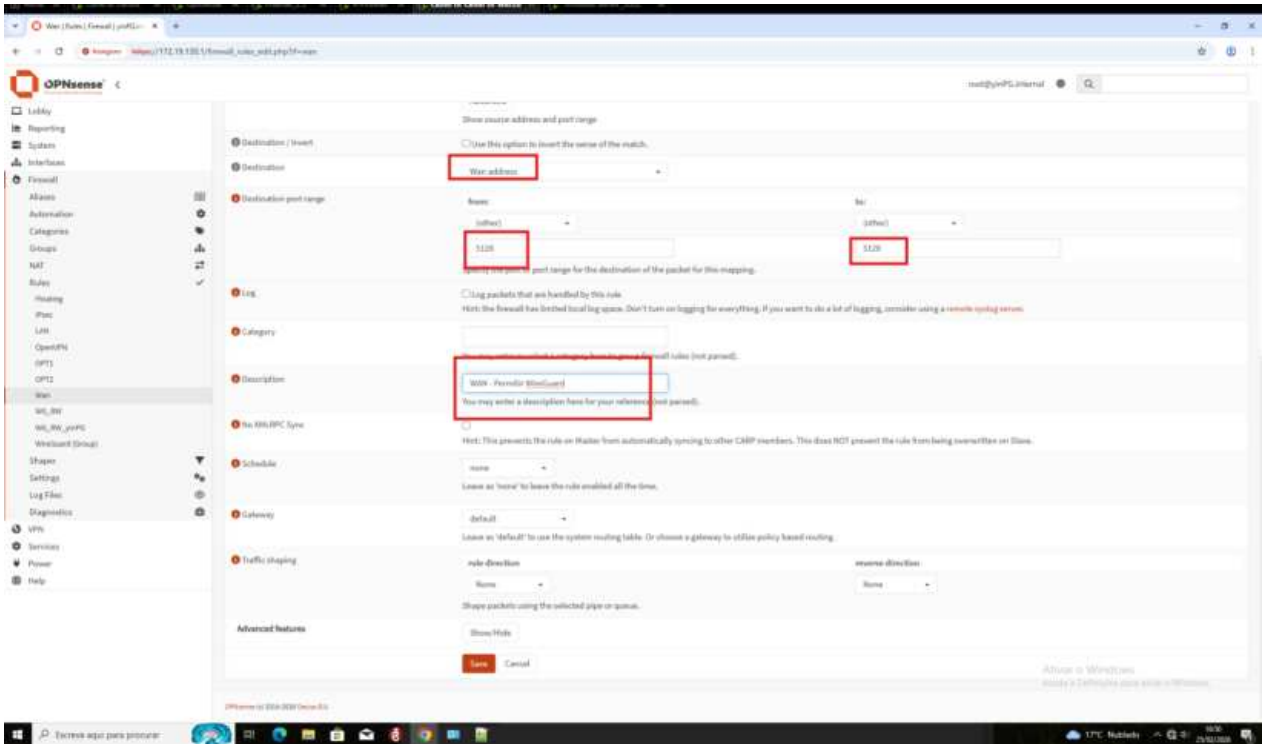
Firewall → Rules → WAN

- **Action: Pass**
- **Protocol: UDP**
- **Source: 10.103.130.0/27**
- **Destination: WAN address**
- **Port: 51820**
- **Description:**

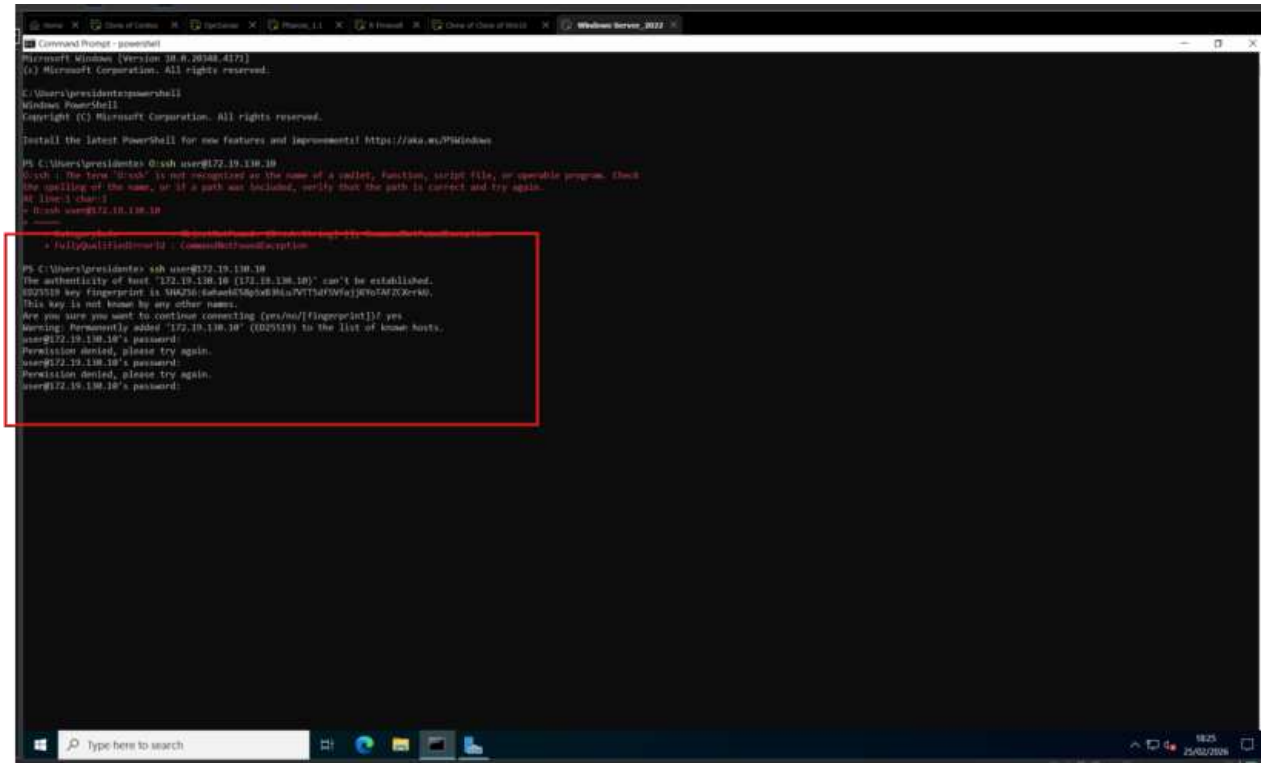
WAN - Permitir WireGuard



Continuação…

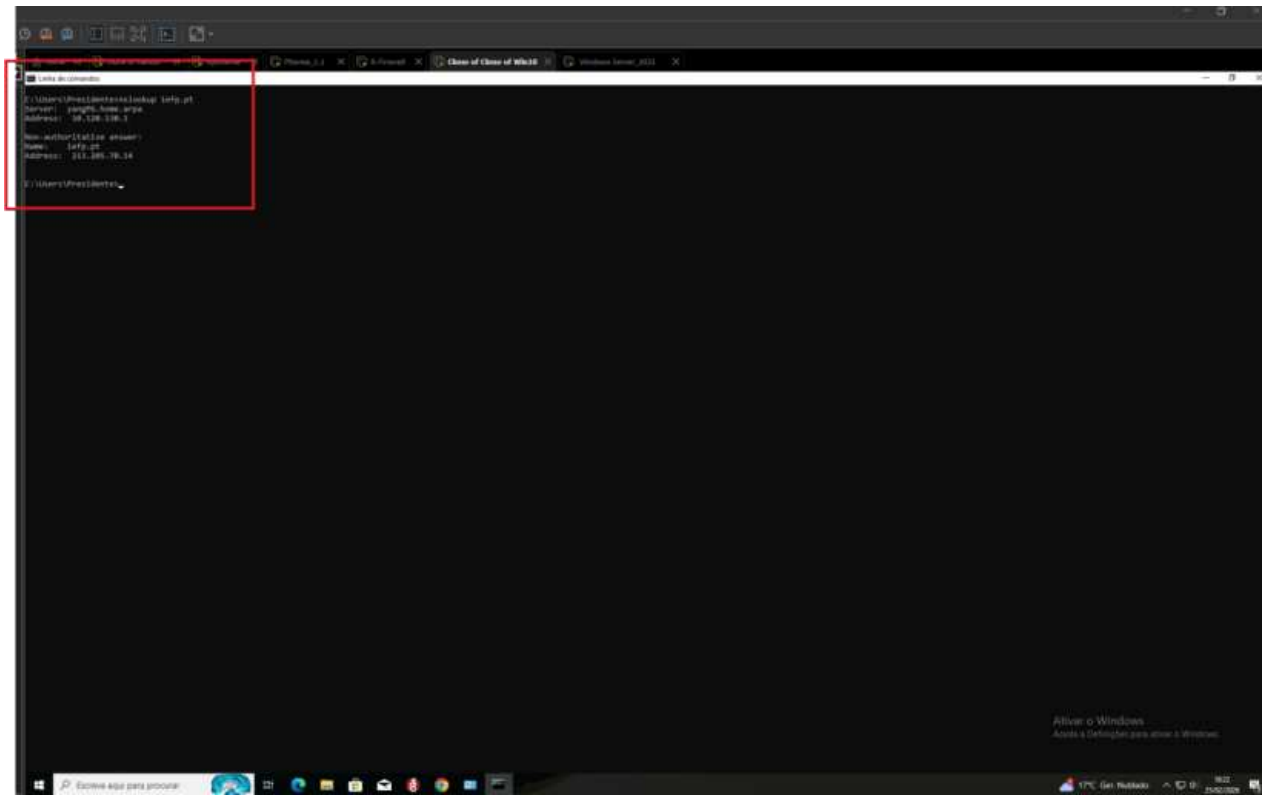


- Prova acesso ao Windows Server com RDP (mstsc → 172.19.130.10)
- Prova que SSH não dá:
    - ssh user@172.19.130.10 (ou ao CentOS)



- Prova DNS:

- nslookup iefp.pt e ver servidor DNS = 172.19.130.1



VPN j) Demostre que todo o tráfego desta VPN passa pelo túnel da VPN, quando o cliente acede à internet, mas o cliente não consegue pingar o IP 8.8.8.8 ou o 1.1.1.1. Pode utilizar o OpenDNS ou transfira o Windows Server para a zona green do yangNA para permitir a resolução de DNS e assim possibilitar navegar na internet
Crie uma VPN Wireguard entre as duas firewall yangNA e yinNA. Não pode passar trafego para acesso a SSH ou RDP nesta VPN. Mas qualquer outro tráfego é possível. Não se esqueça das imagens das regras! (Ping não é suficiente!!!!)

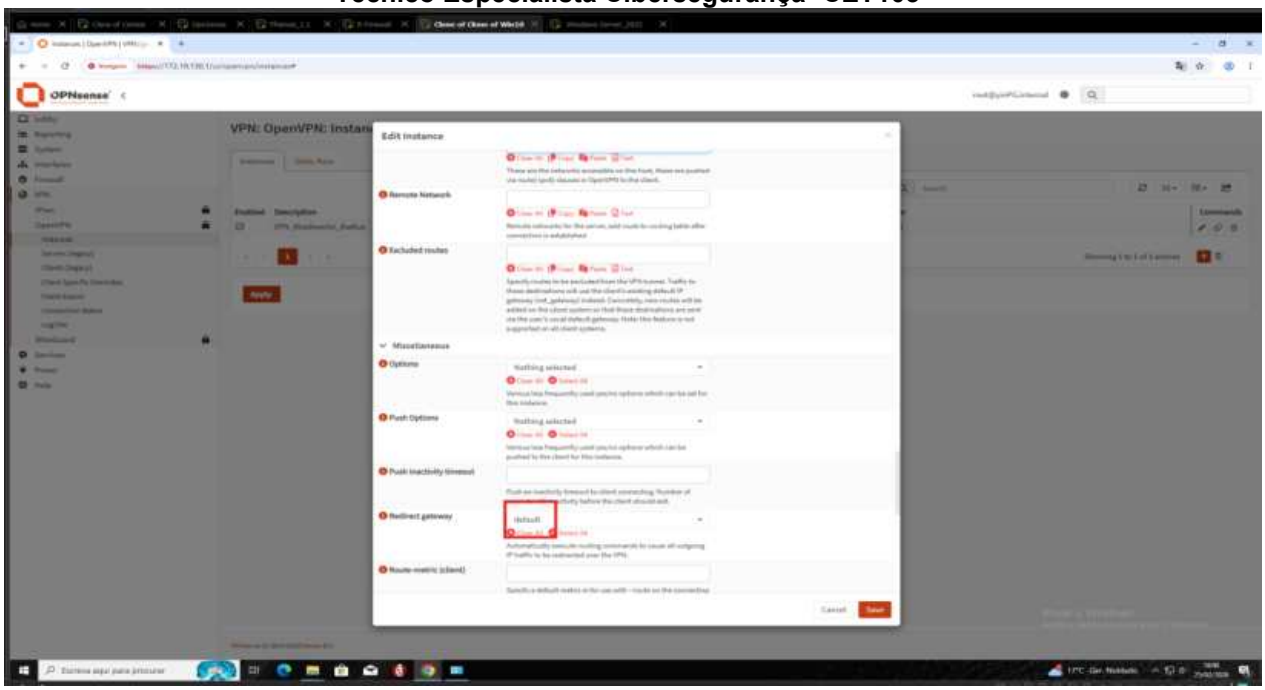
17) VPN j)

Demonstrar que TODO o tráfego da VPN passa pelo túnel

22- PASSO 1 — Forçar Full Tunnel (OpenVPN)

No OPNsense:

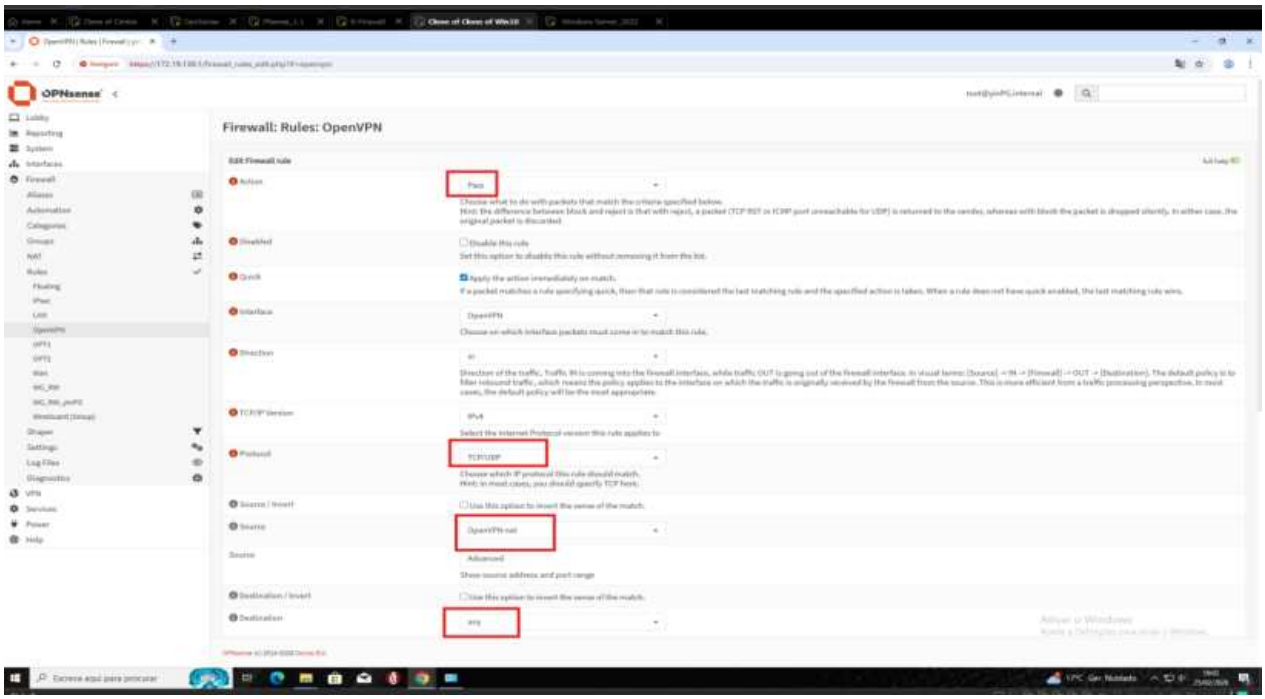
VPN → OpenVPN → Instances



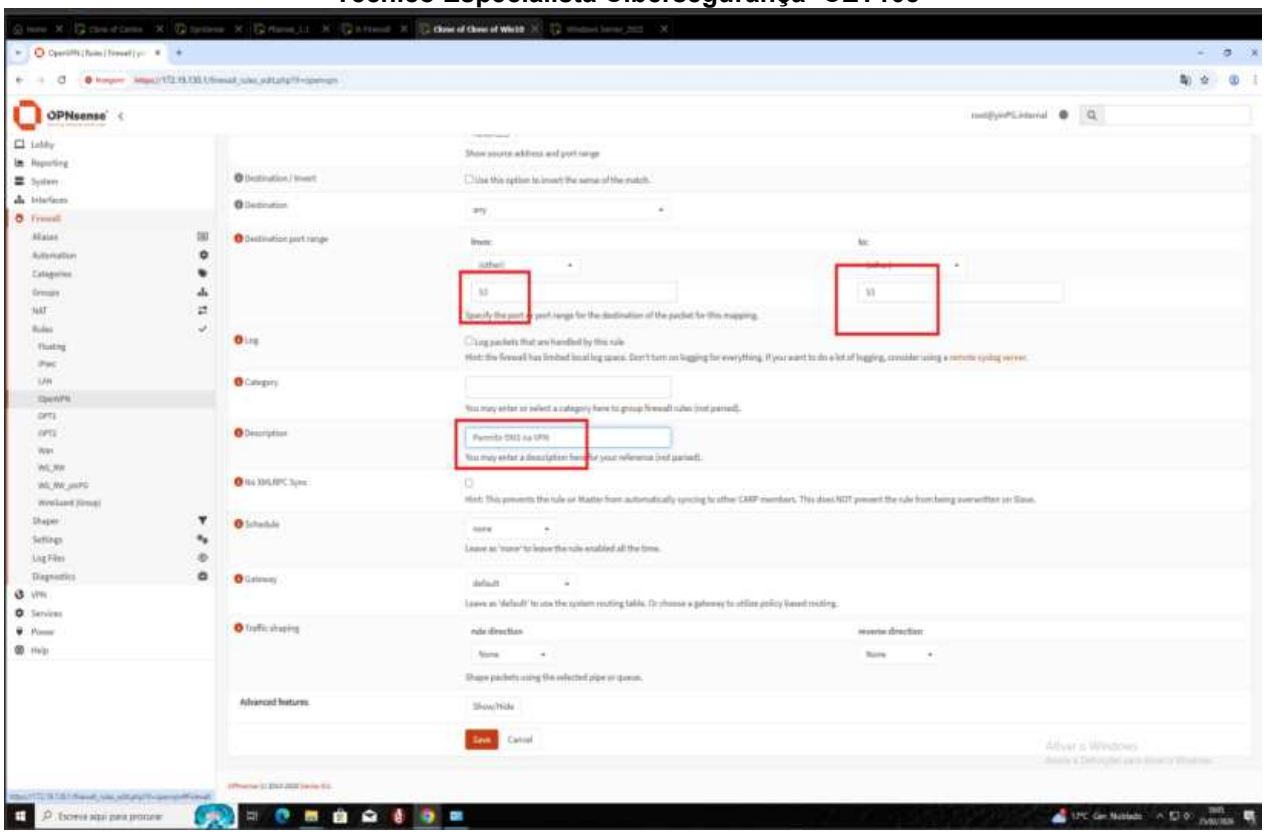
22.1 PASSO 2 — Permitir DNS na VPN

Firewall → Rules → OpenVPN

Pass
Protocol: TCP/UDP
Source: OpenVPN net
Destination: any
Port: 53

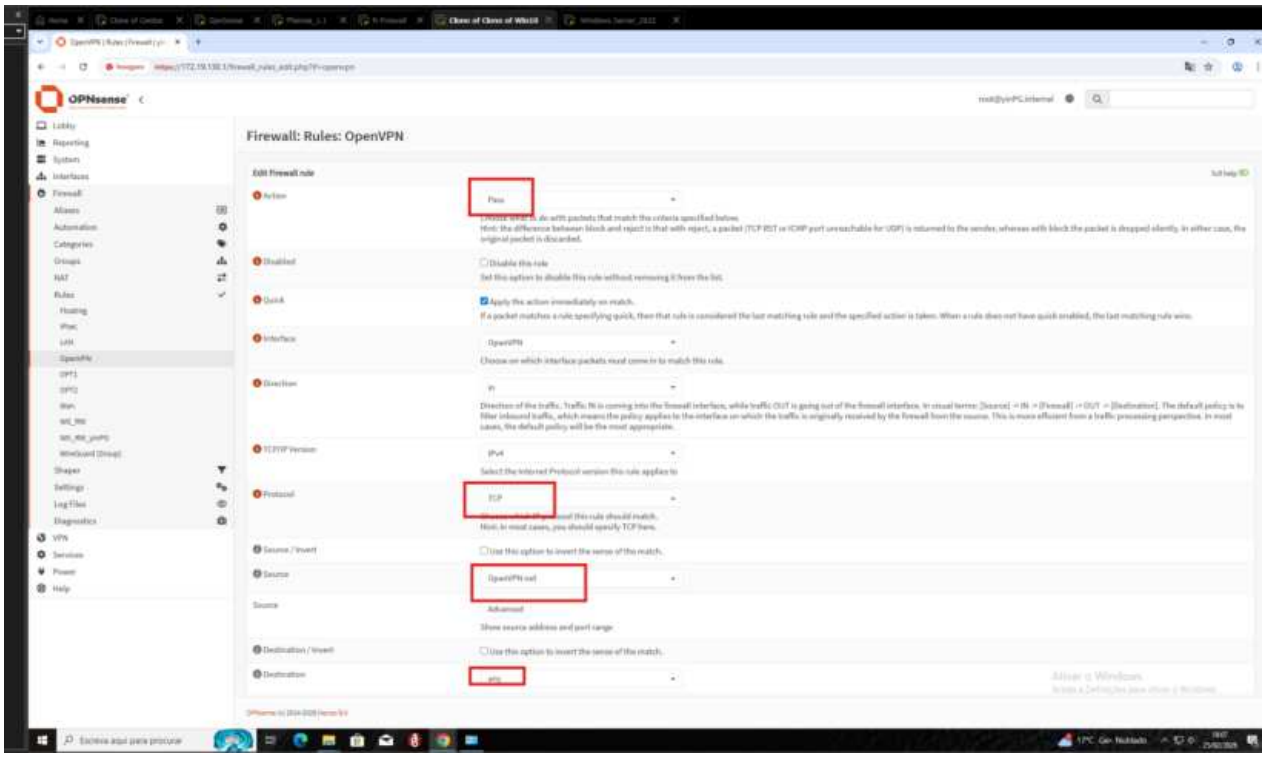


Continuação ..

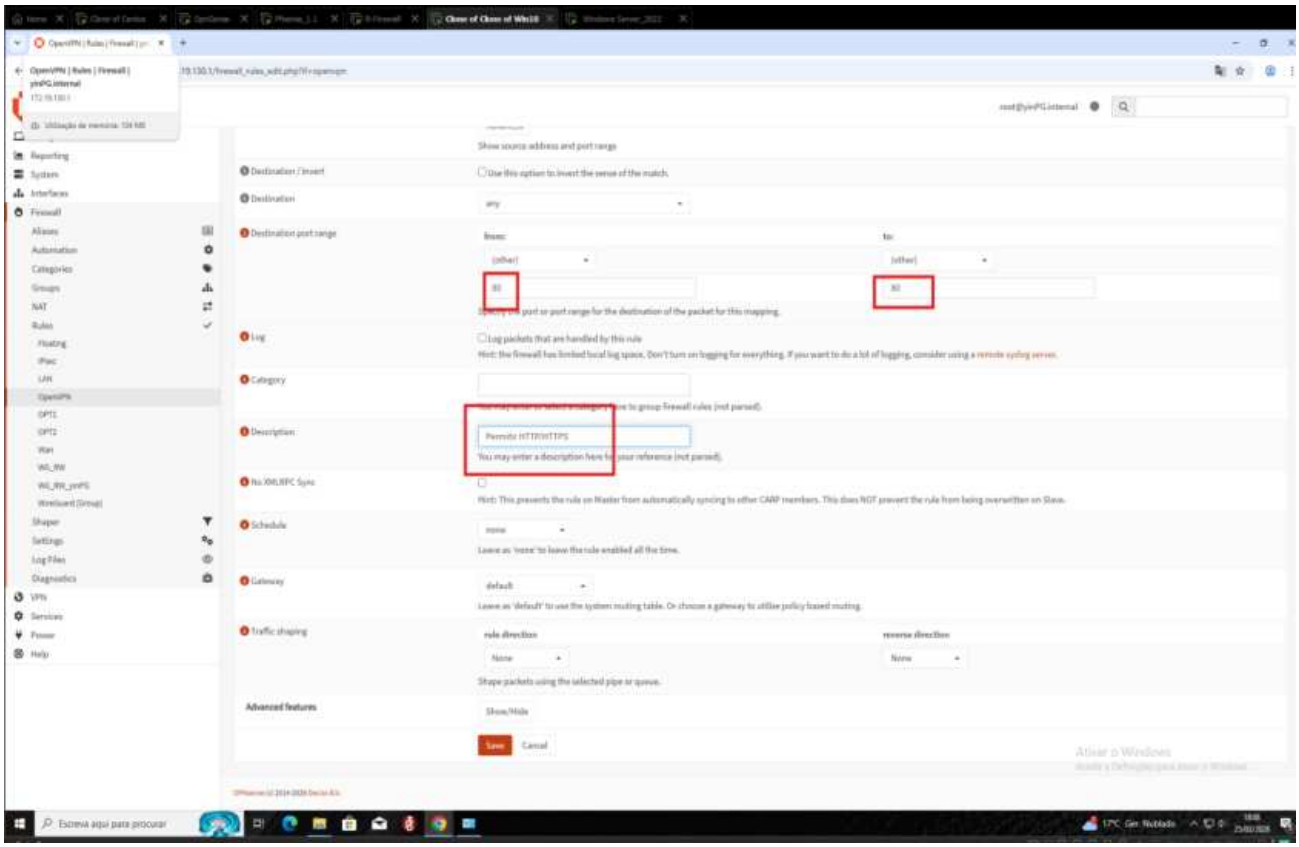


22.3 PASSO 3 — Permitir HTTP/HTTPS

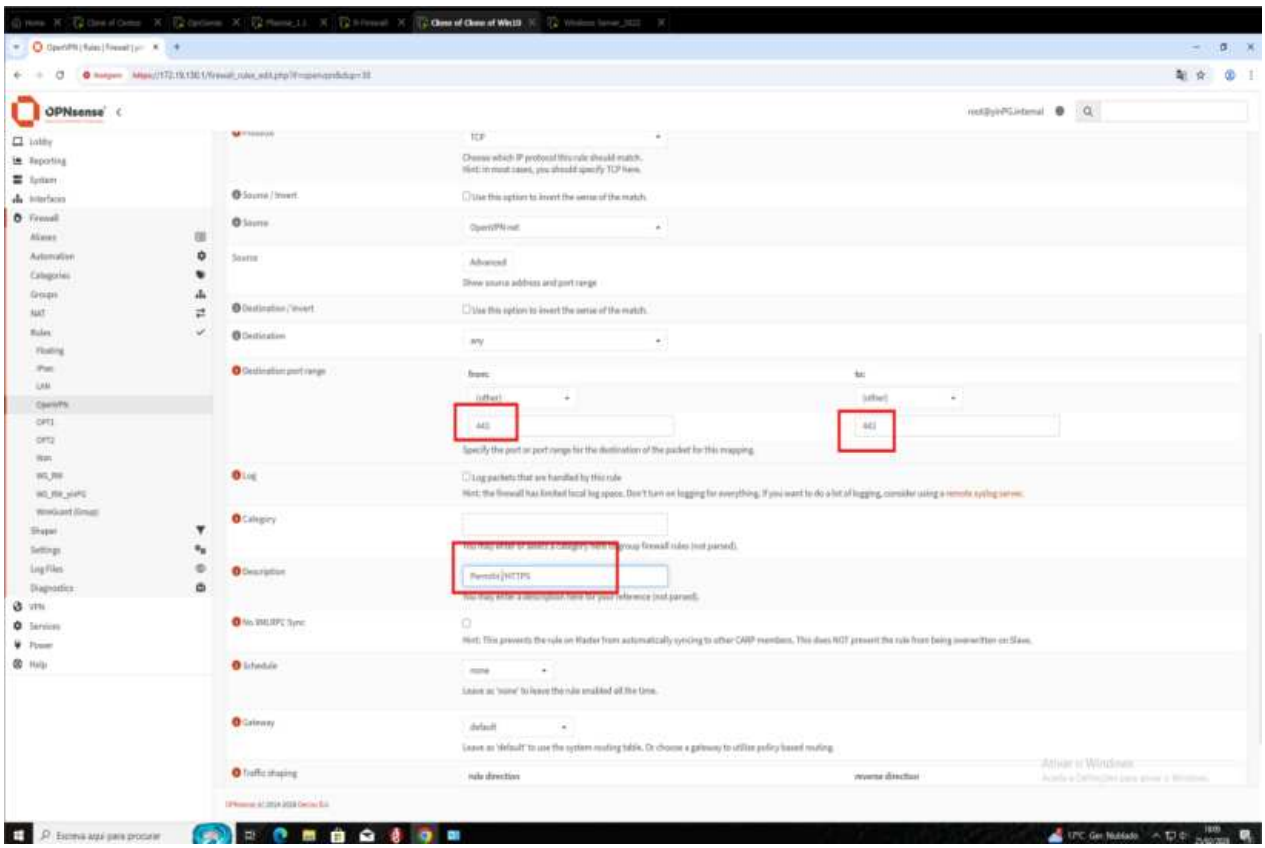
Pass
Protocol: TCP
Source: OpenVPN net
Destination: any
Ports: 80, 443



Continuação …



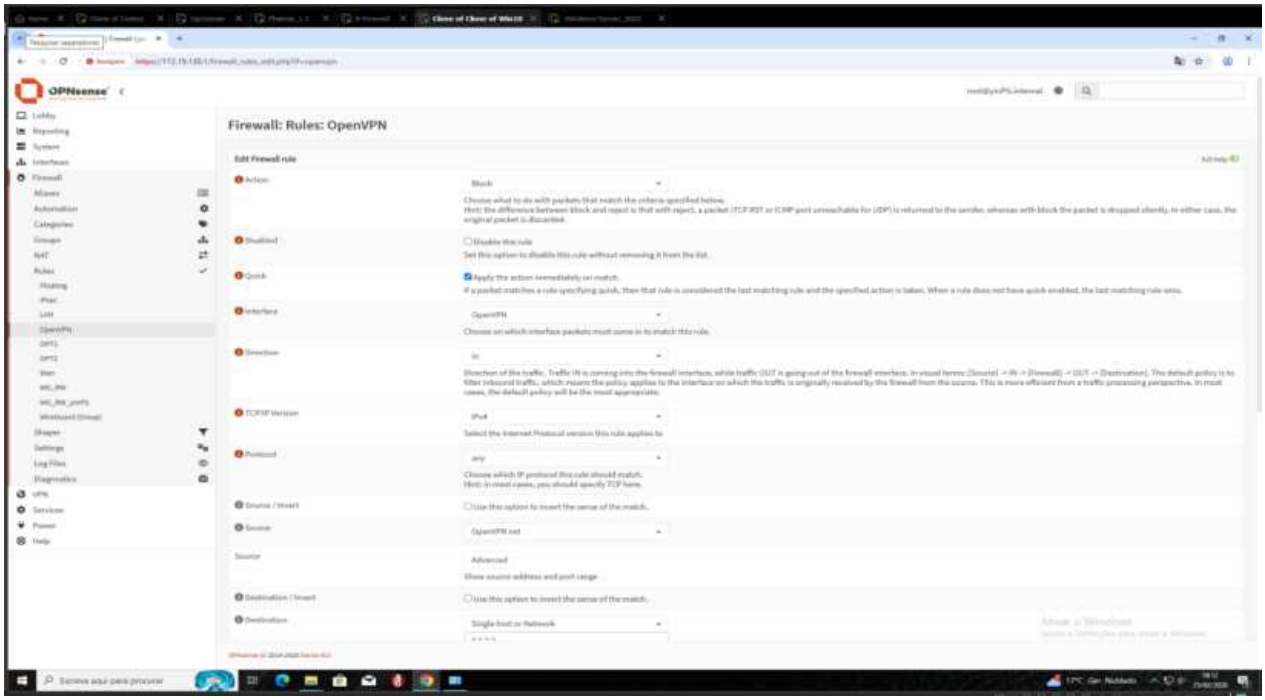
22.4 Porta 443



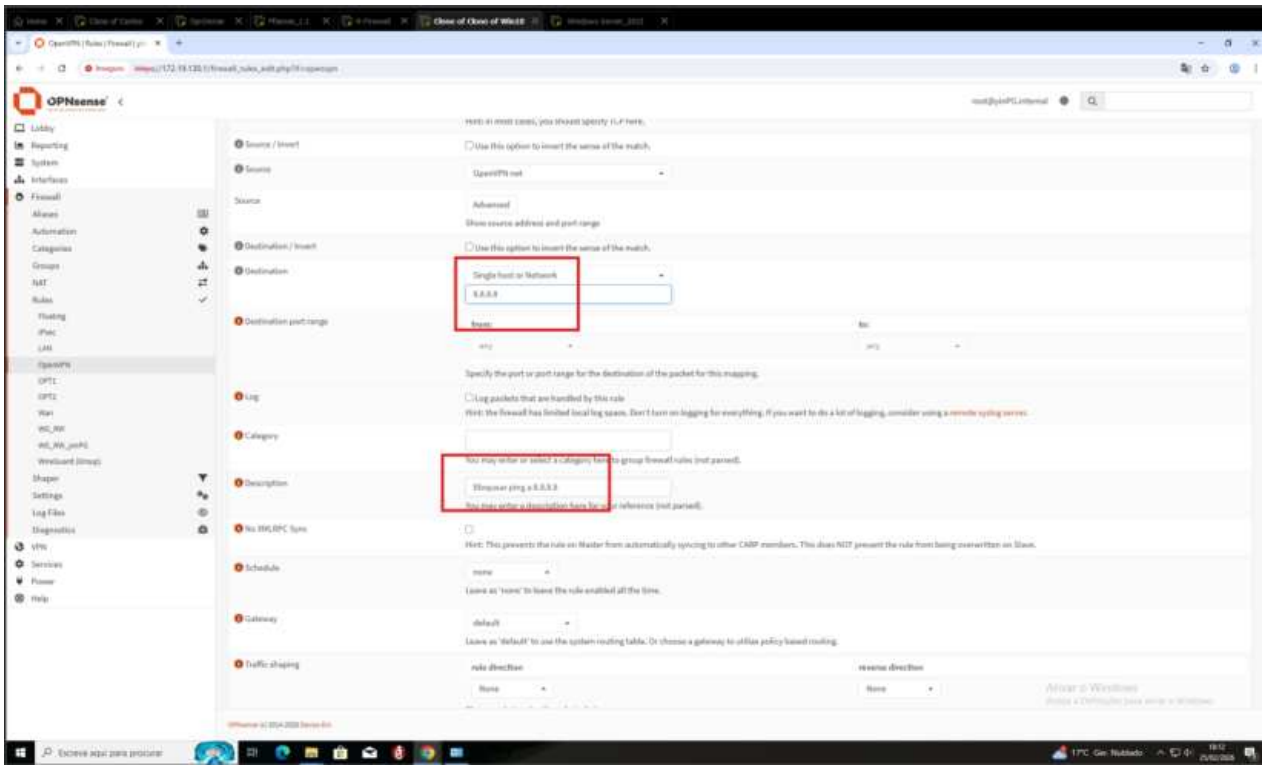
22.5 PASSO 4 — Bloquear ping a 8.8.8.8 e 1.1.1.1

Criar 2 regras acima das permissivas:

Block
Protocol: ICMP
Source: OpenVPN net
Destination: 8.8.8.8

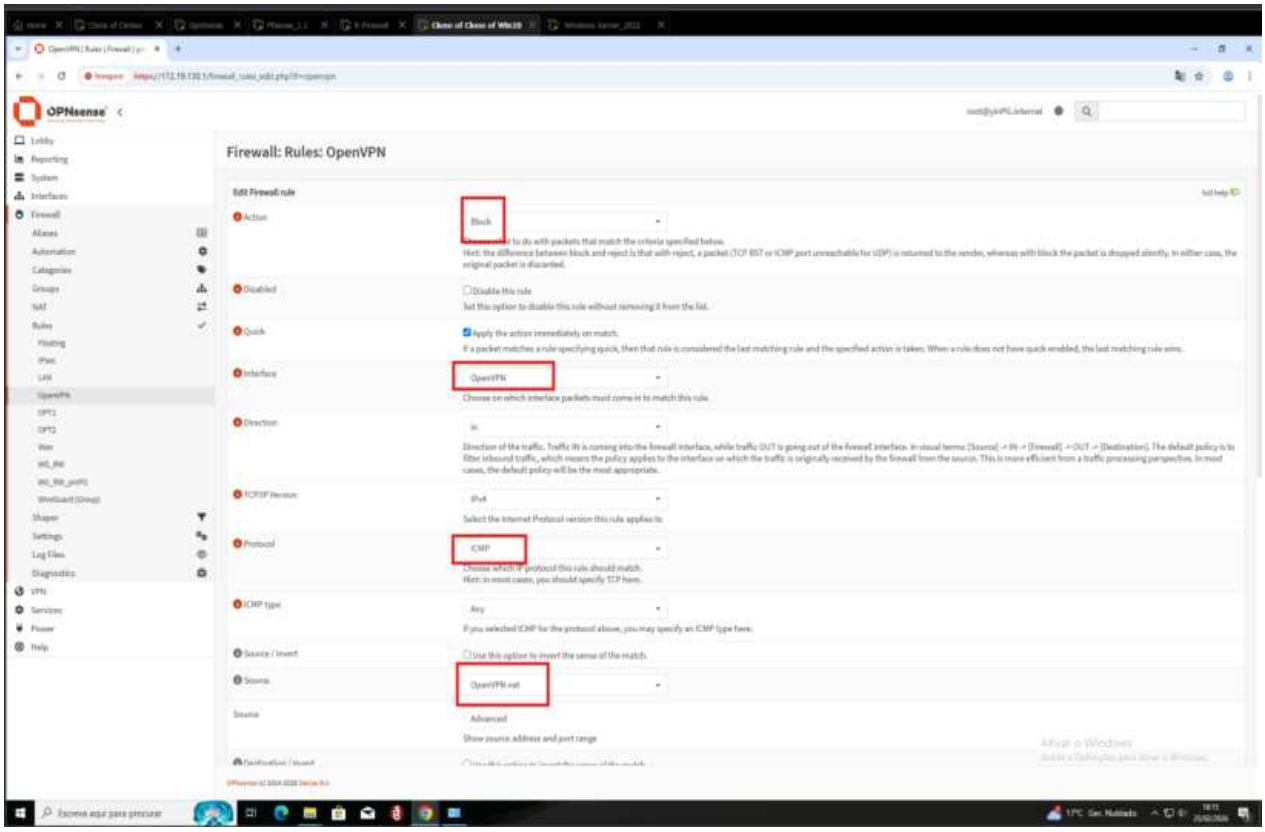


Continuação :

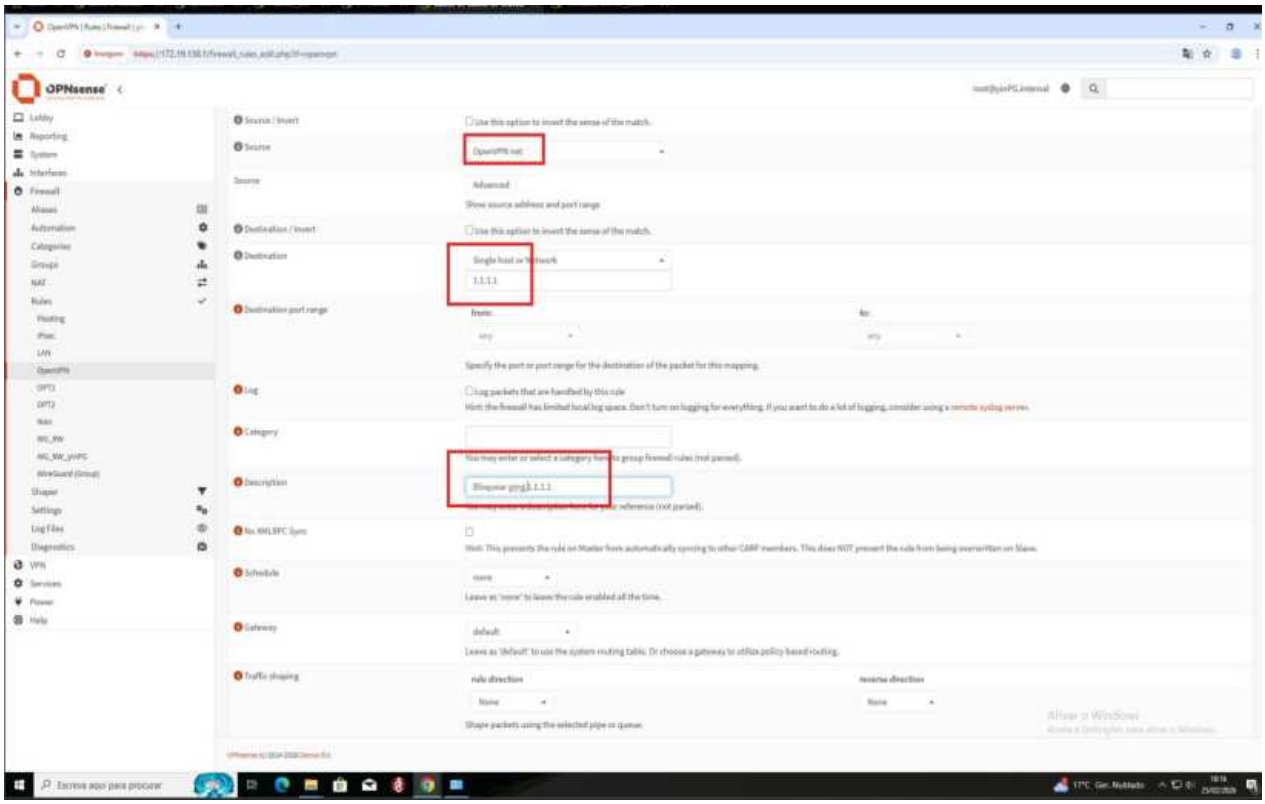


Block
Protocol: ICMP

22.6 Source: OpenVPN net
Destination: 1.1.1.1

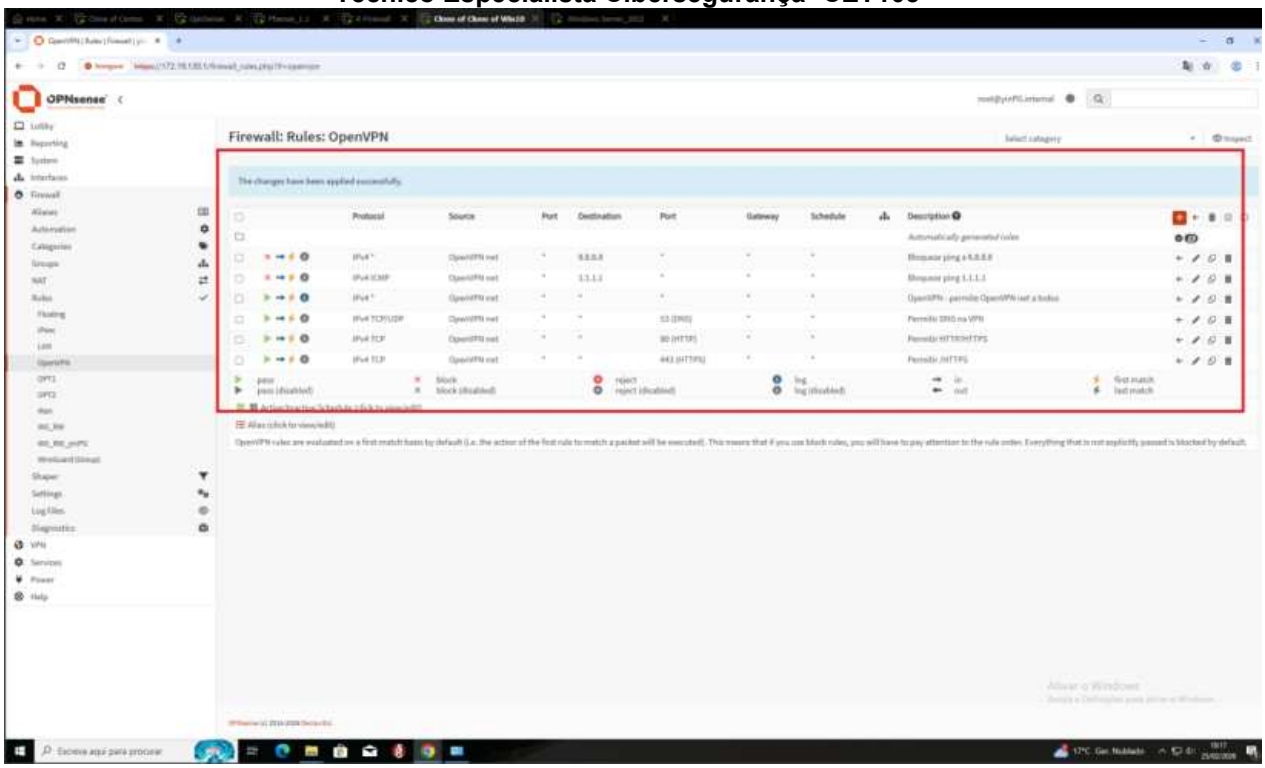


Continuação ..



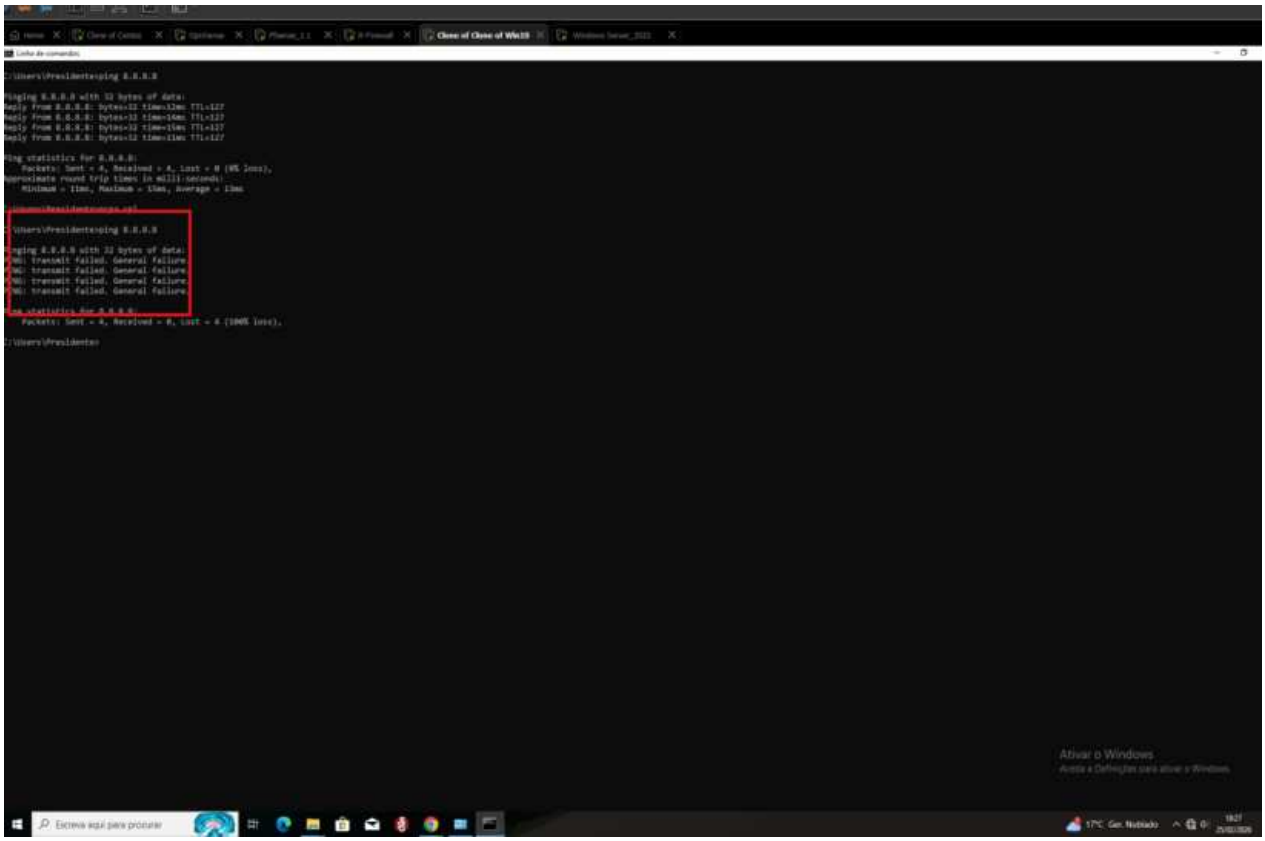
Muito importante:

As regras de block têm de estar acima das regras de pass.



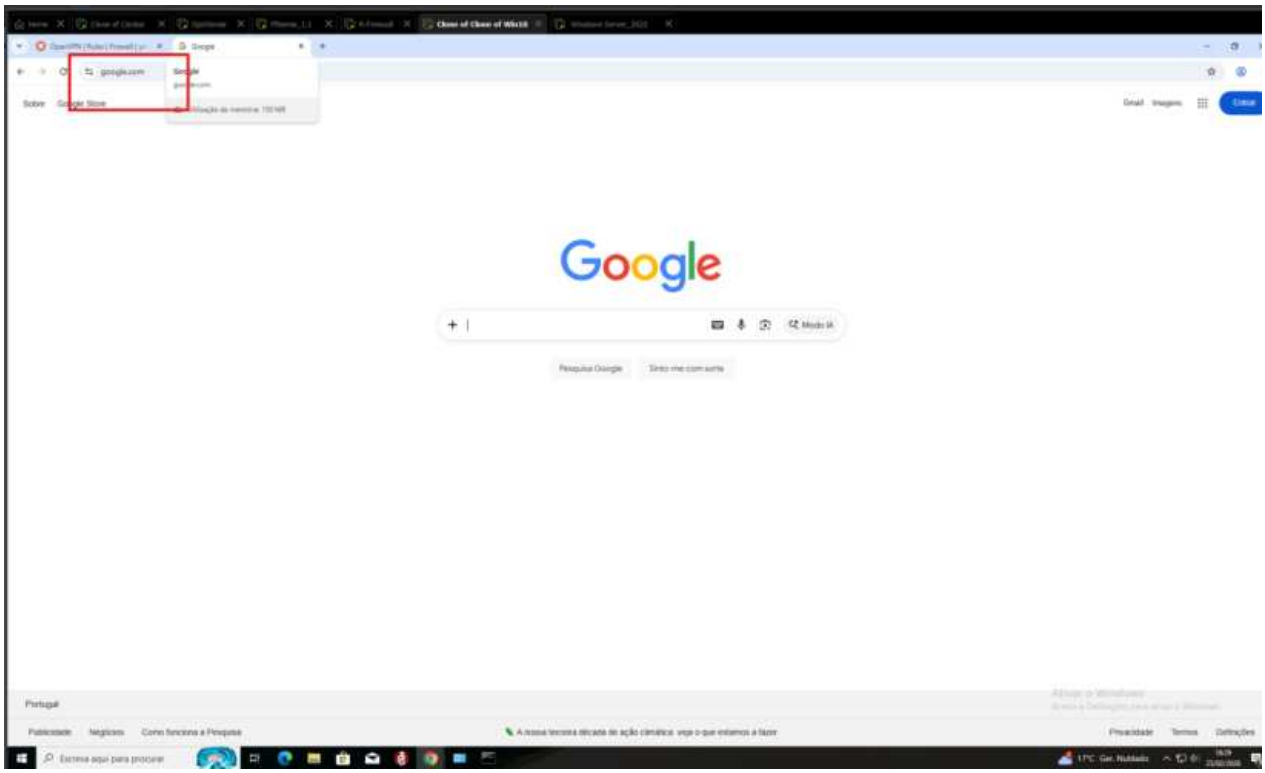
22.7 Testar ping

ping 8.8.8.8



22.8 Testar navegação

https://www.google.com



3) VPNs ----------

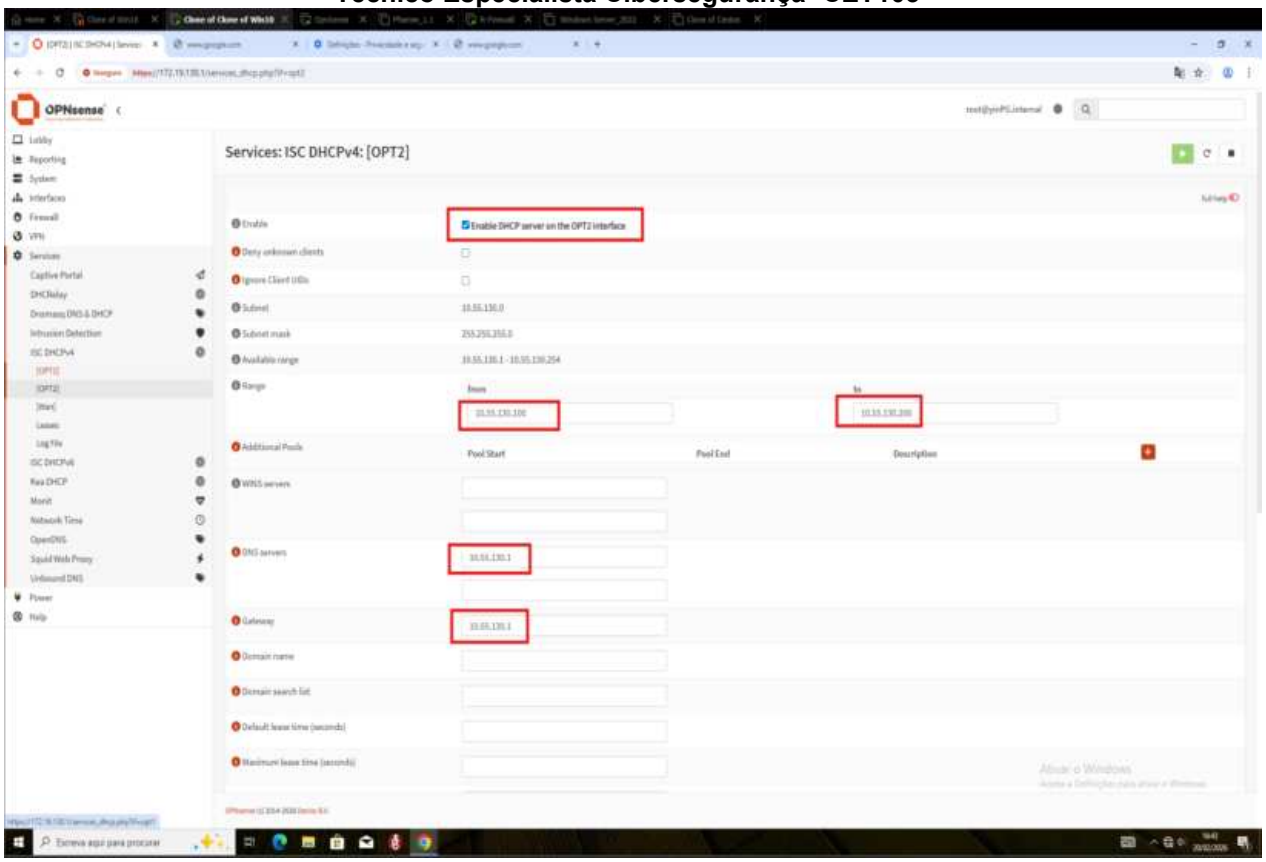
k) Na zona blue, do firewall yinNA considere a existência um Acess Point Wireless. Configure o Captive Portal para eventuais users wireless. O Captive Portal deve gerar Vouchers / Tickets para esses users (duração 2horas), e a largura de banda por utilizador tem um limite de upload de máximo de 5Mb e de download de 10MB . A atribuição de IPs nesta zona está a cargo do firewall (DHCP - configuração à sua escolha), no entanto o DNS só pode ser o DNS do próprio firewall

23) Captive Portal na BLUE do yinPG com vouchers 2h + bandwidth 10down/5up
Requisito: Captive Portal, vouchers 2h, shaper 10MB download e 5Mb upload, IPs por DHCP e DNS na BLUE só pode ser o do firewall.

23.1 DHCP na BLUE

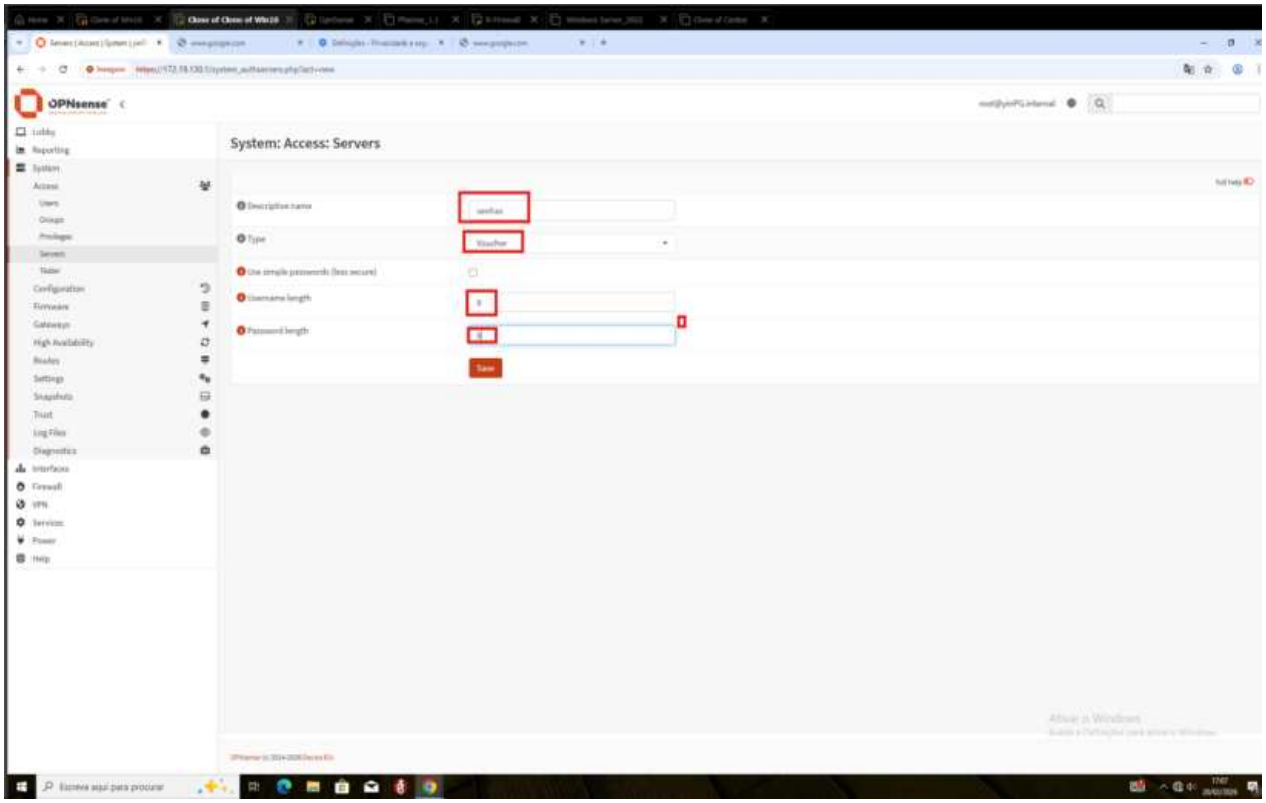
Services → DHCPv4 → [OPT2] --Blue

- Enable DHCP: yes
- Range: 10.55.130.100 – 10.55.130.200
- DNS servers: 10.55.130.1

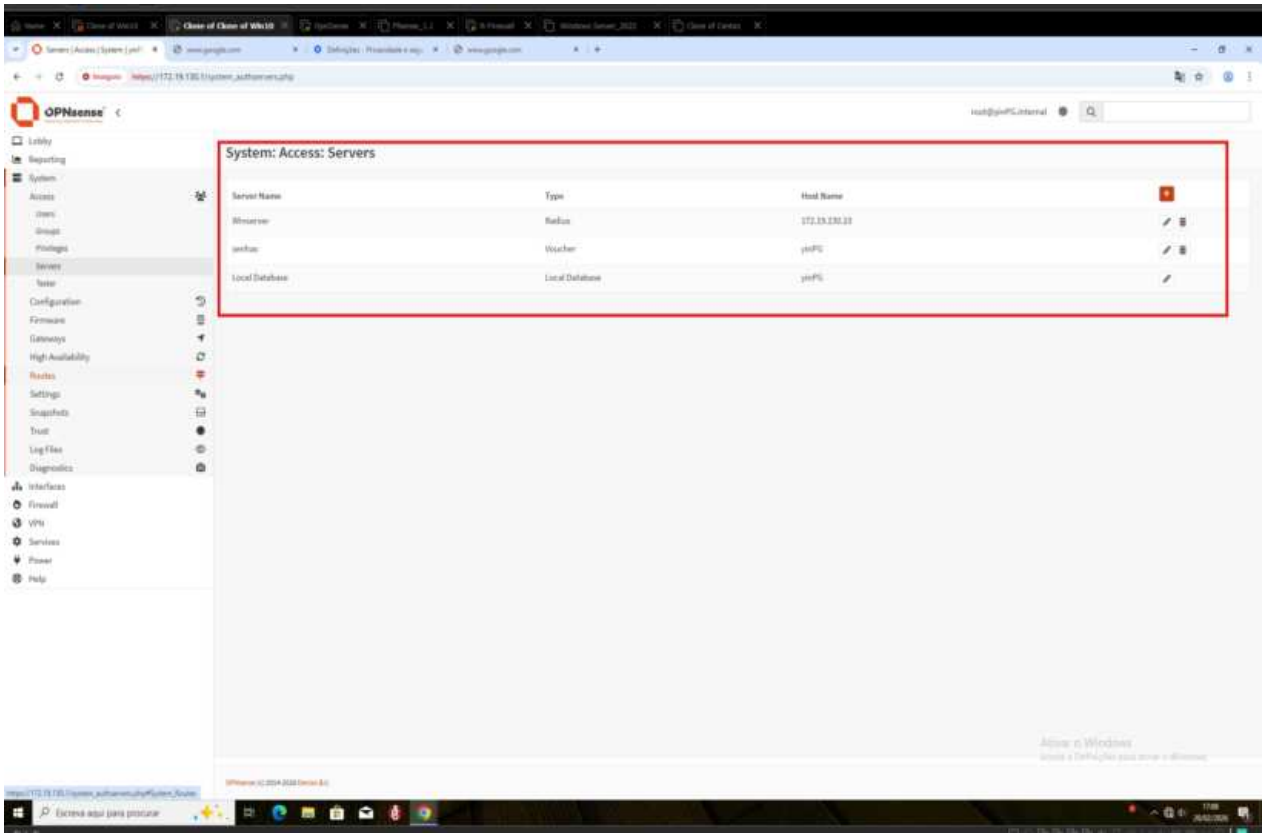


23.2 Criar Voucher Server

System → Access → Servers + add

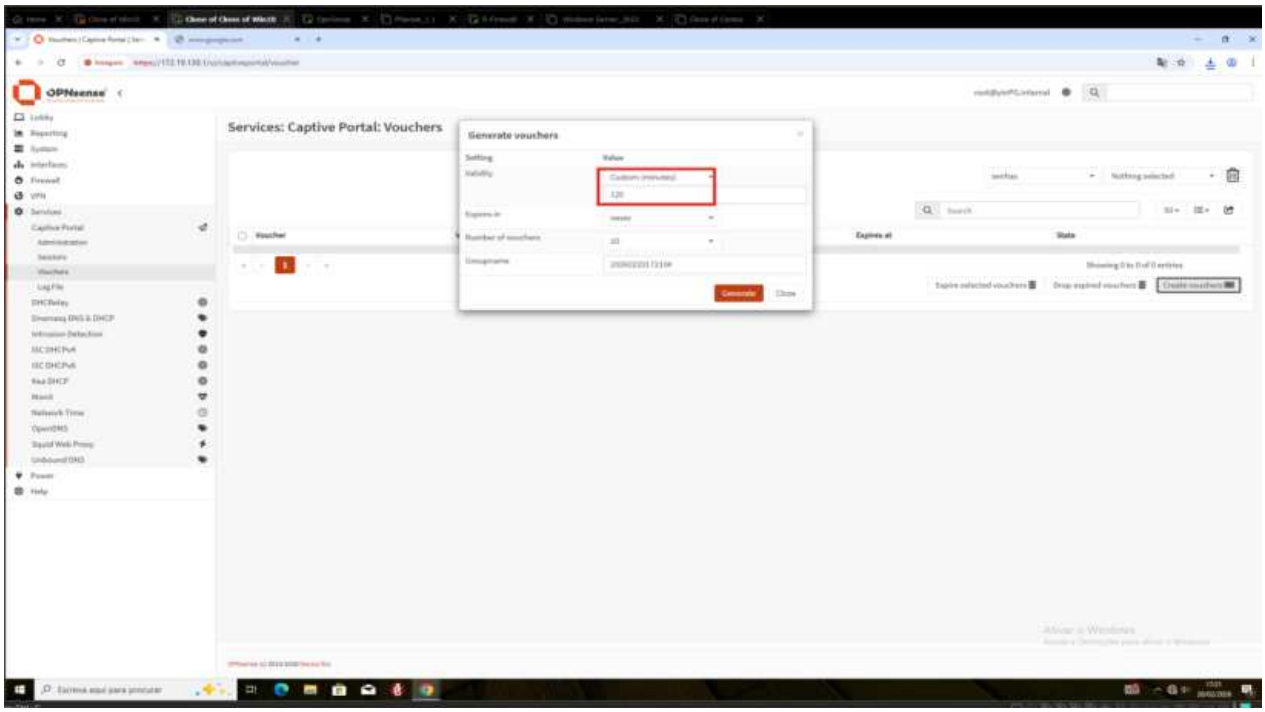


Resumo :

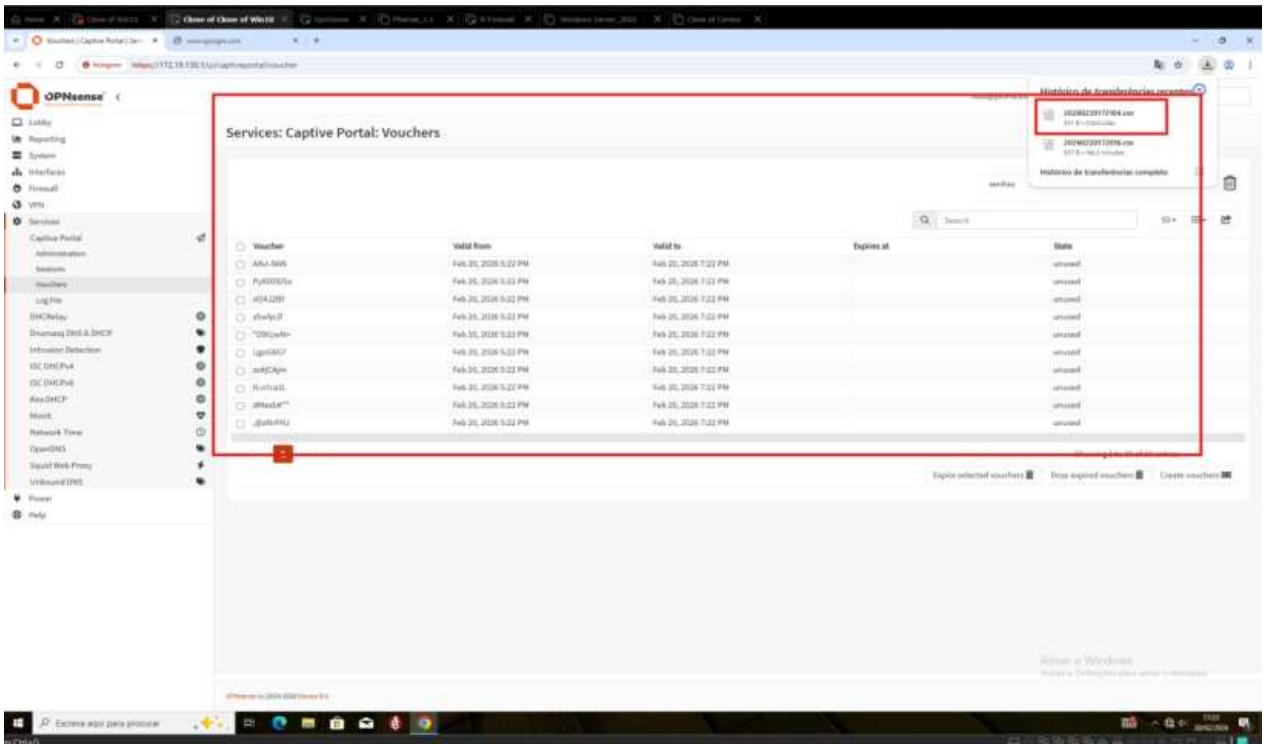


23.3 -Services → Captive Portal → Vouchers → (criar e gerar)
Config:

- Duração: **2 hours** (120m)

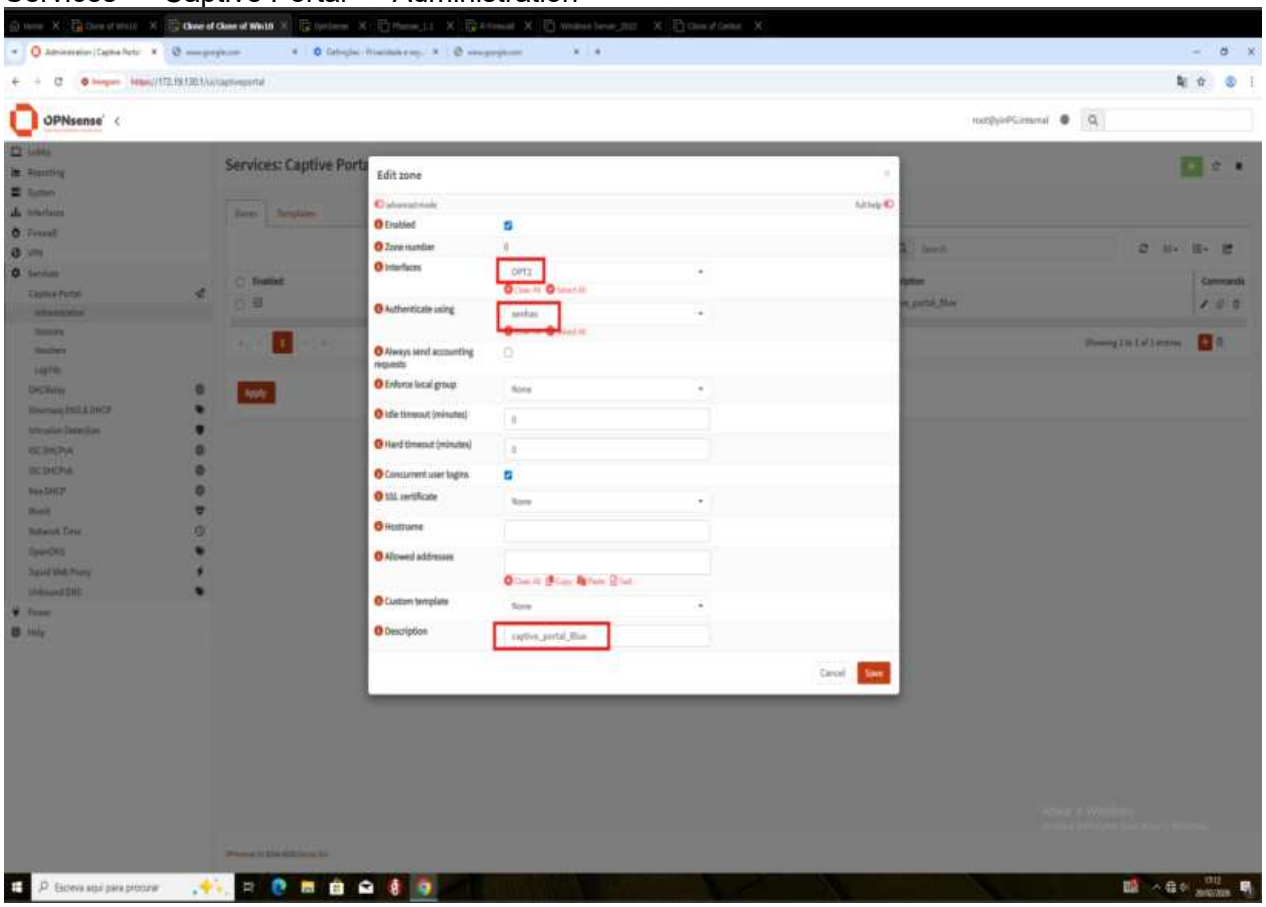


Resumo :



23.4 Criar zona Captive Portal

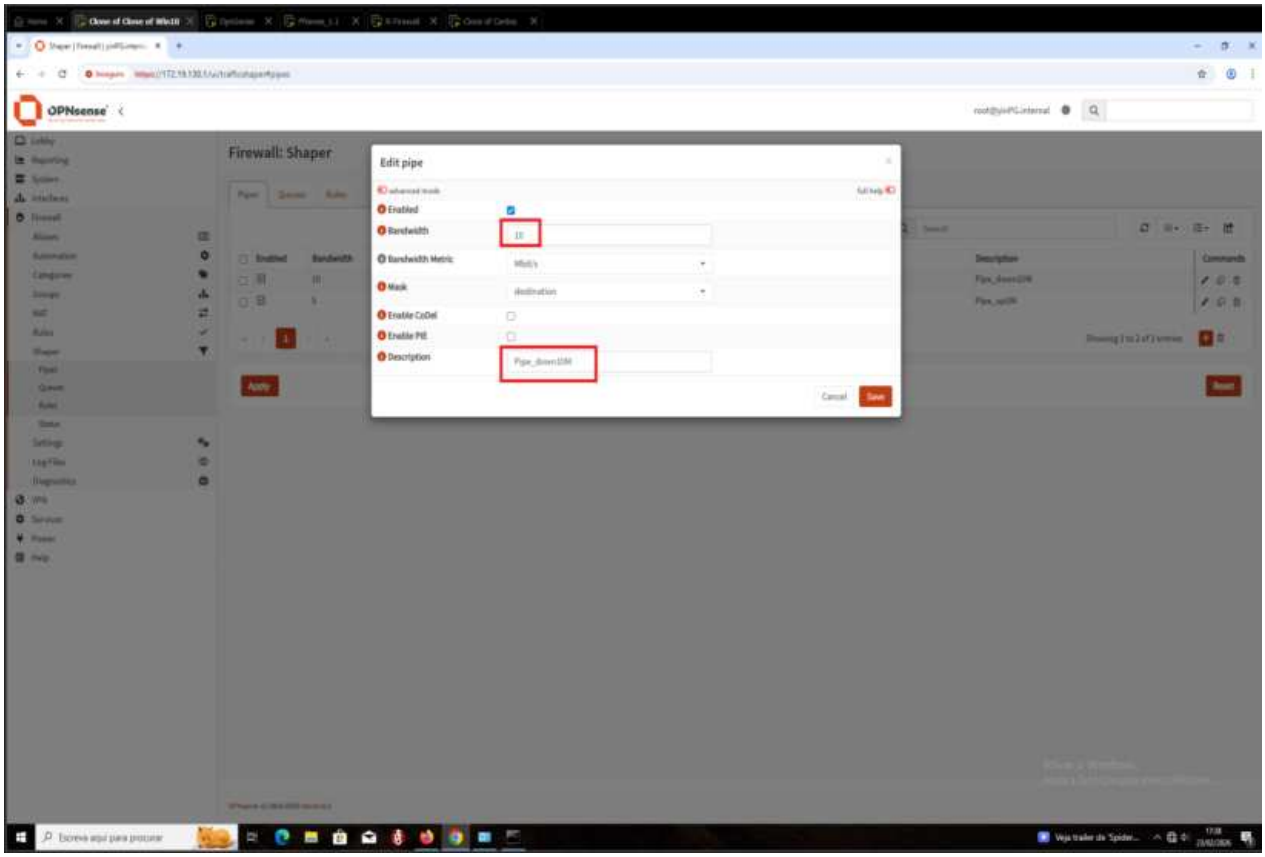
Services → Captive Portal → Administration



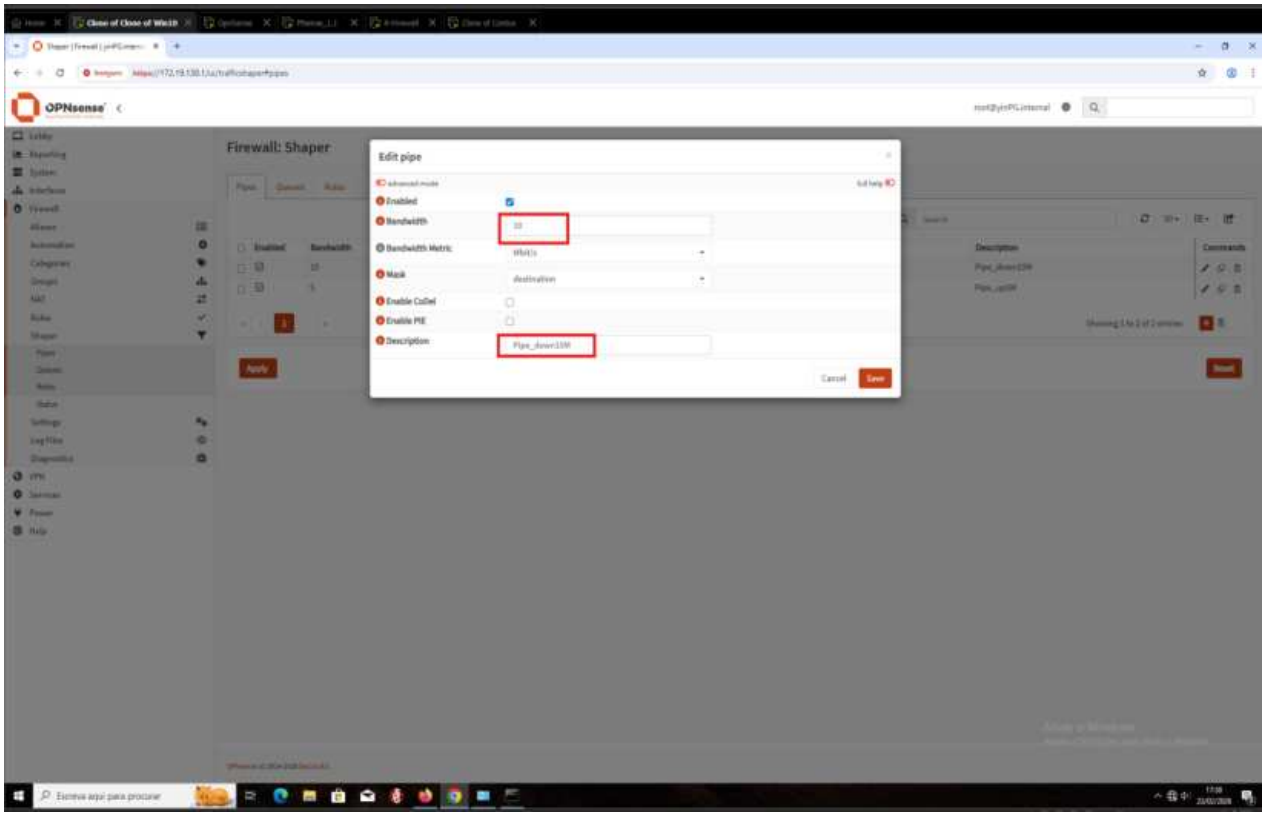
23.5 Criar shaper pipes (10MB down / 5MB up)

Firewall → Shaper → Pipes

- Add new pipe: 10Mbit (download)



- Add new pipe: 5Mbit (upload)



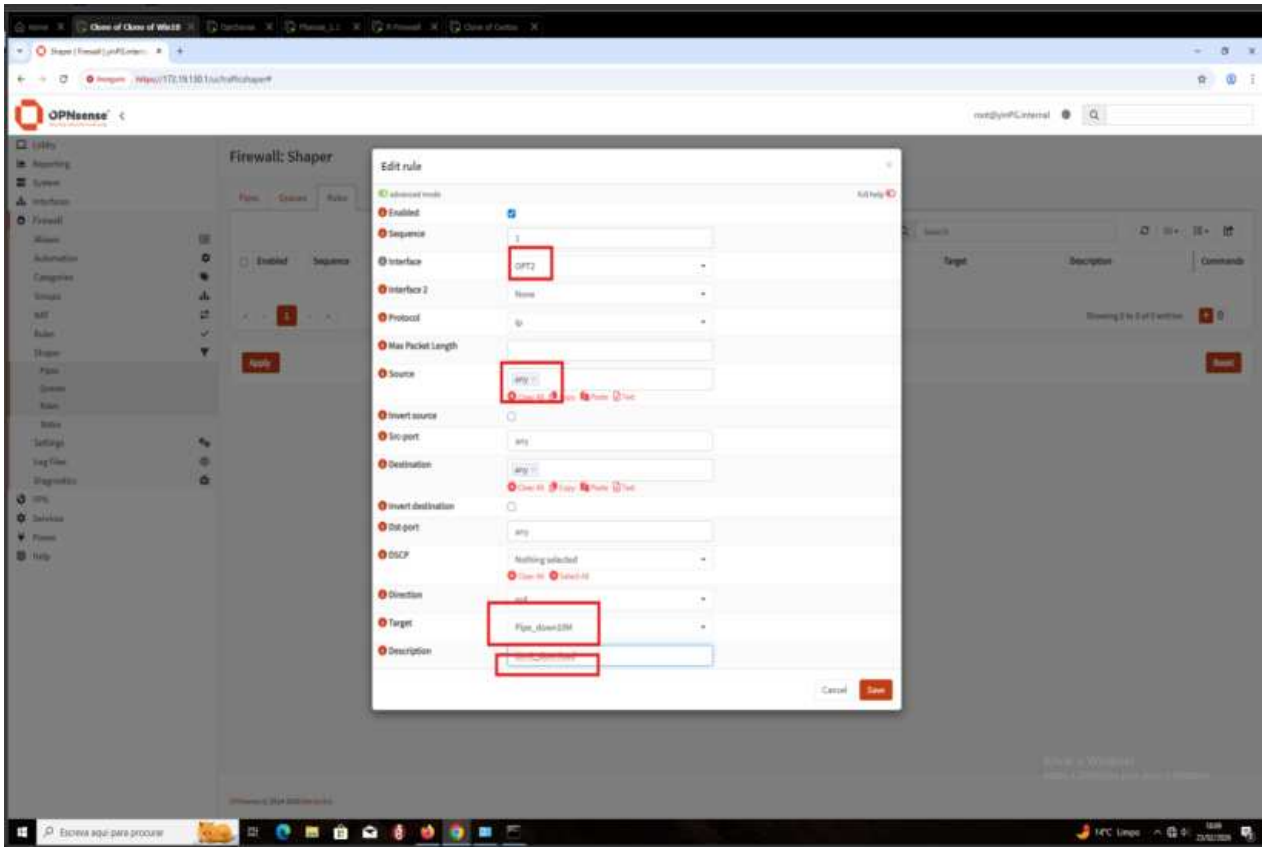
23.6 Shaper rules para aplicar aos utilizadores do Captive

Firewall → Shaper → Rules

- Criar regras associando:
    - Download → pipe 10M
    - Upload → pipe 5M

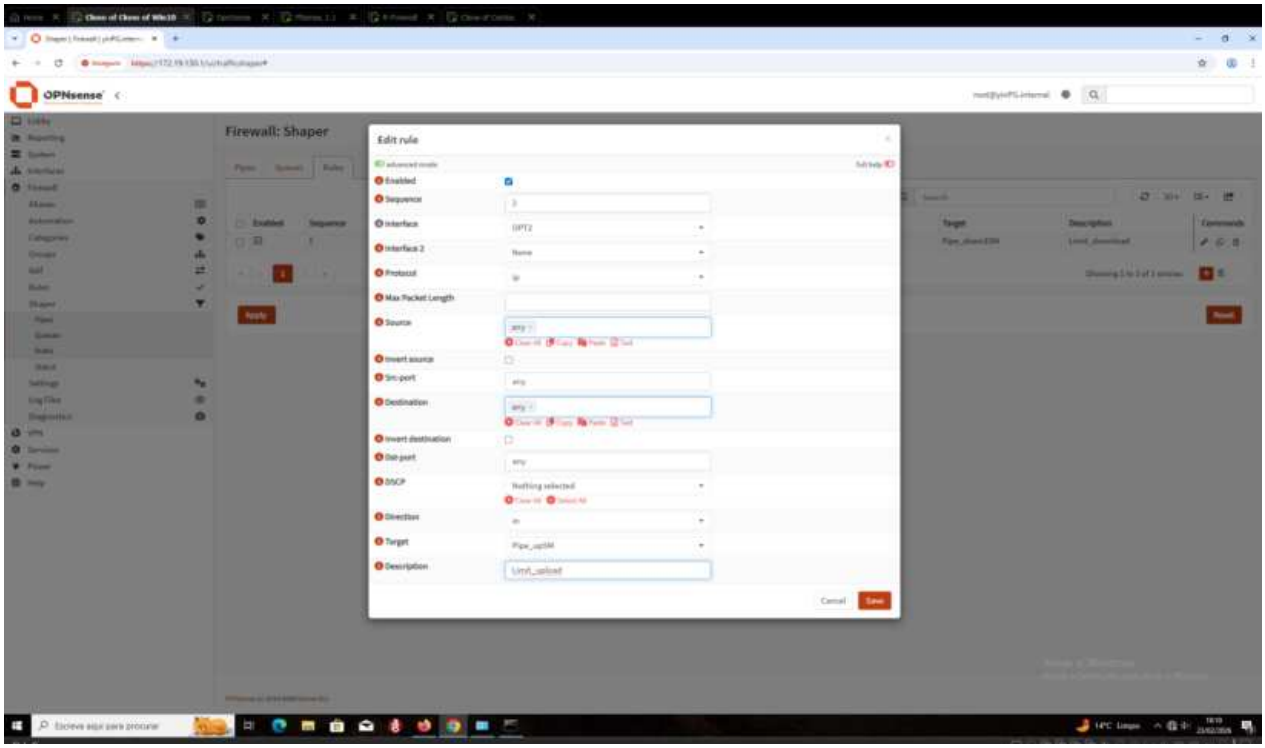
Regra Download

- Interface: BLUE
- Direction: out
- Source: any
- Destination: any
- Target: Limit_10MB
- Description: Limit_download



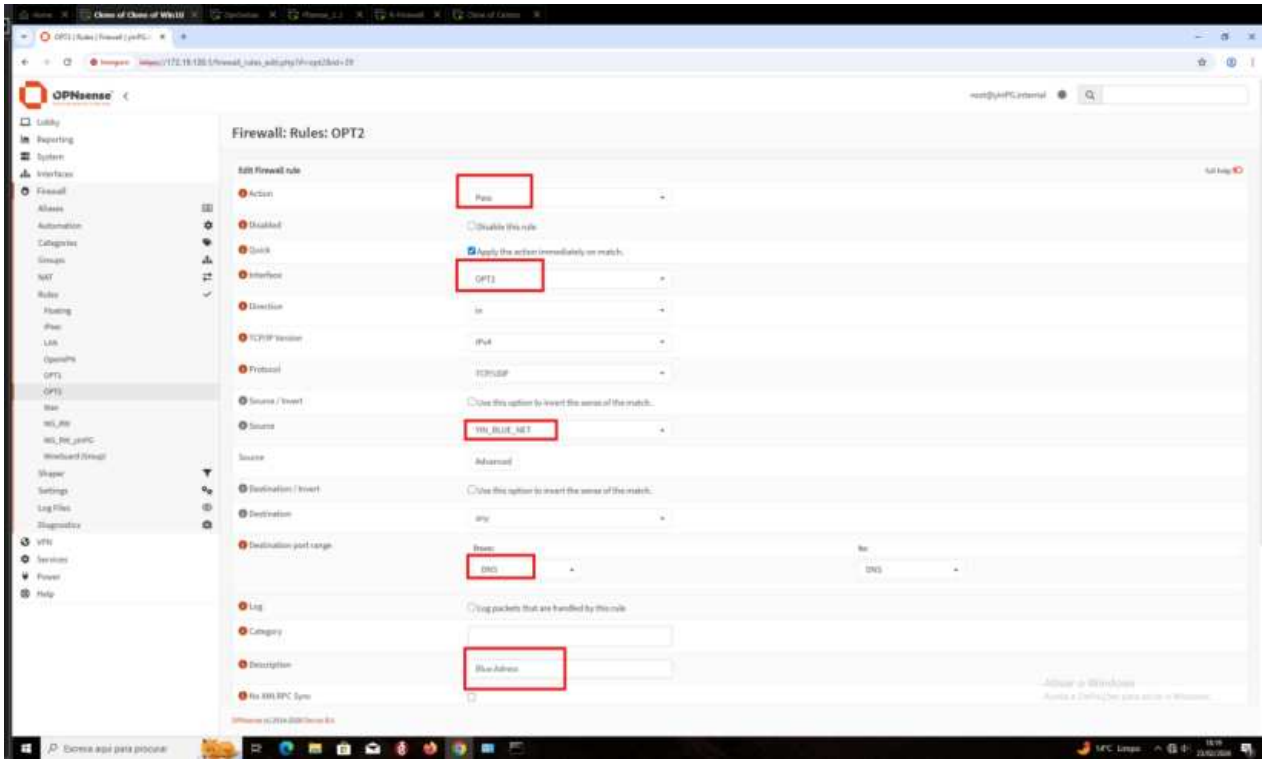
23.7 Regra Upload

- Interface: BLUE
- Direction: in
- Source: any-
- Destination: any-
- Target: Limit_5MB-
- Description: Limit_upload



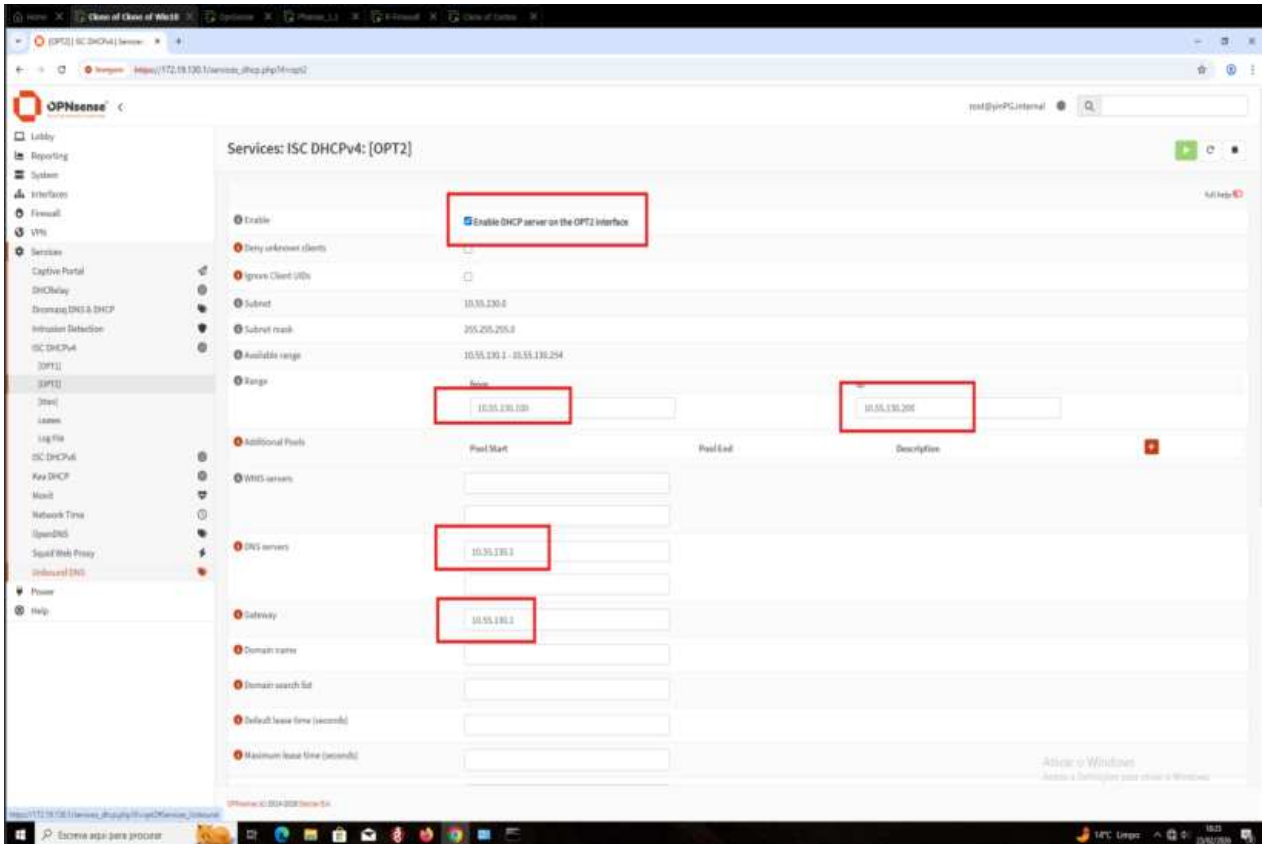
23.8 Regras firewall BLUE

Firewall → Rules → BLUE -OPT2



Services → DHCPv4 → OPT2 (Blue)

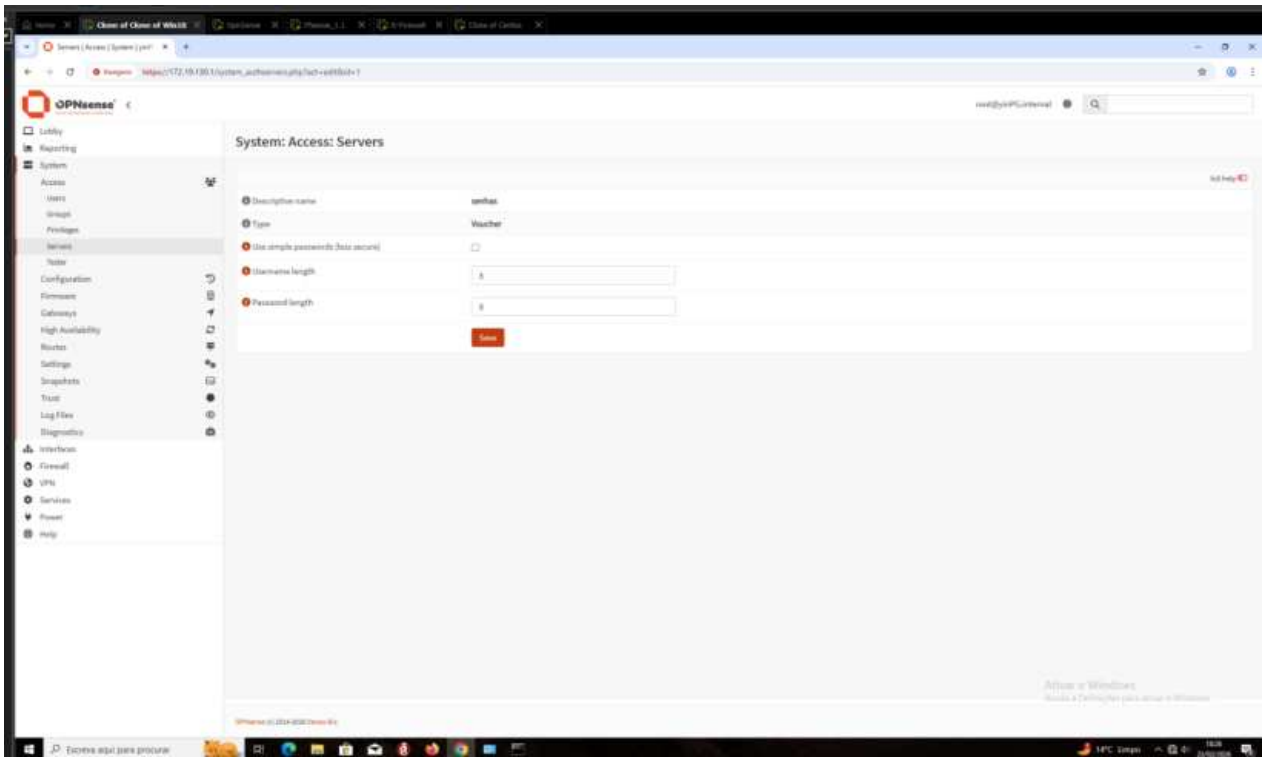
- Permitir DHCP e DNS para o firewall
- Enable DHCP: sim
- Range: 10.55.130.100 – 10.55.130.200
- DNS servers: 10.55.130.1 (IP do firewall na BLUE)
- Gateway: 10.55.130.1



23.8.1 Criar Voucher Server

System → Access → Servers → Add

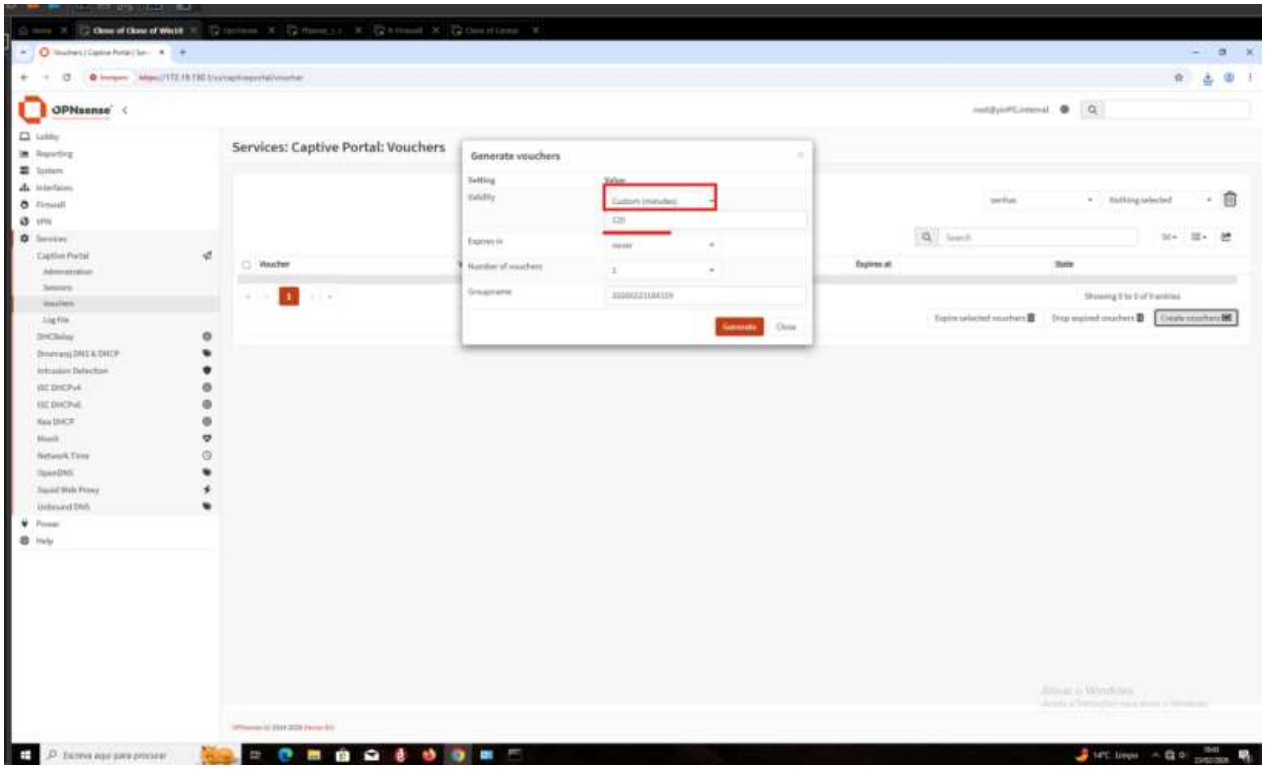
- Type: Voucher
- Nome: senhas
- Voucher length: 8
- Password length: 8

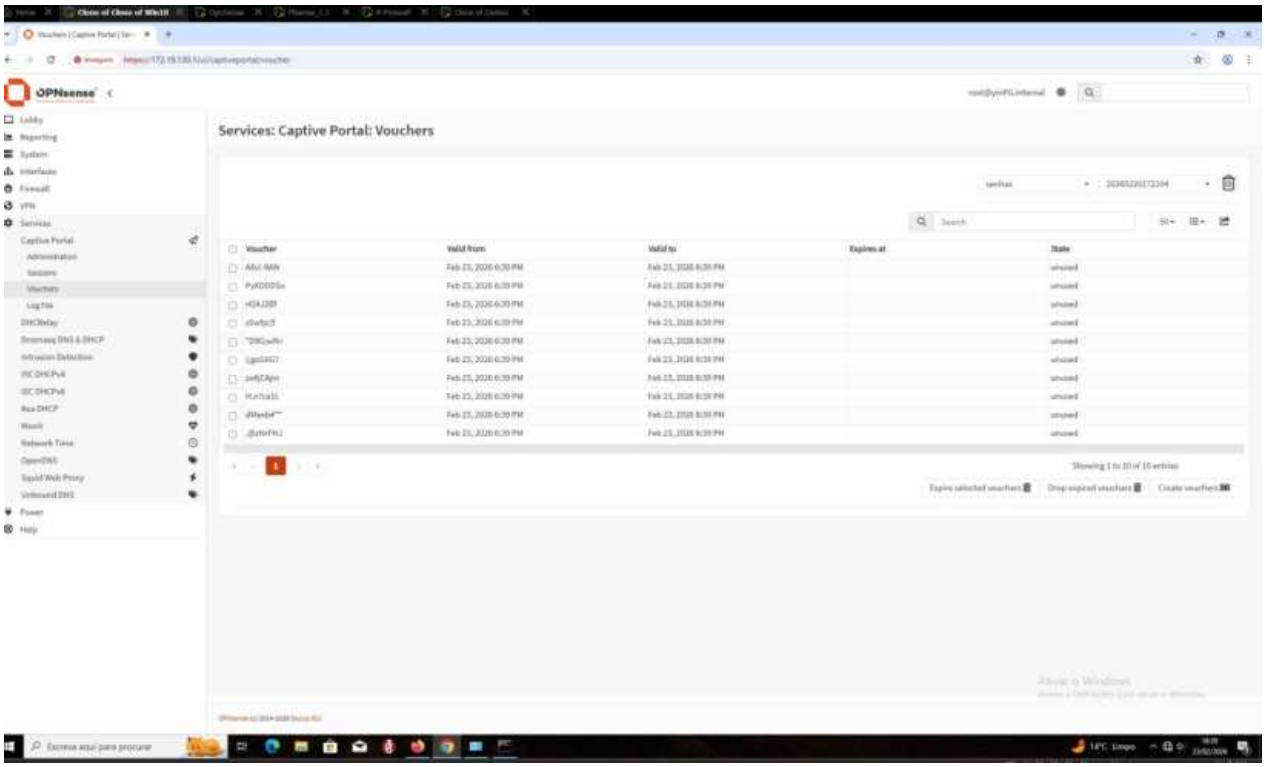


23.8.2 Gerar Vouchers 2h

Services → Captive Portal → Vouchers

- Server: Senhas
- Validity: 120 minutos
- Generate

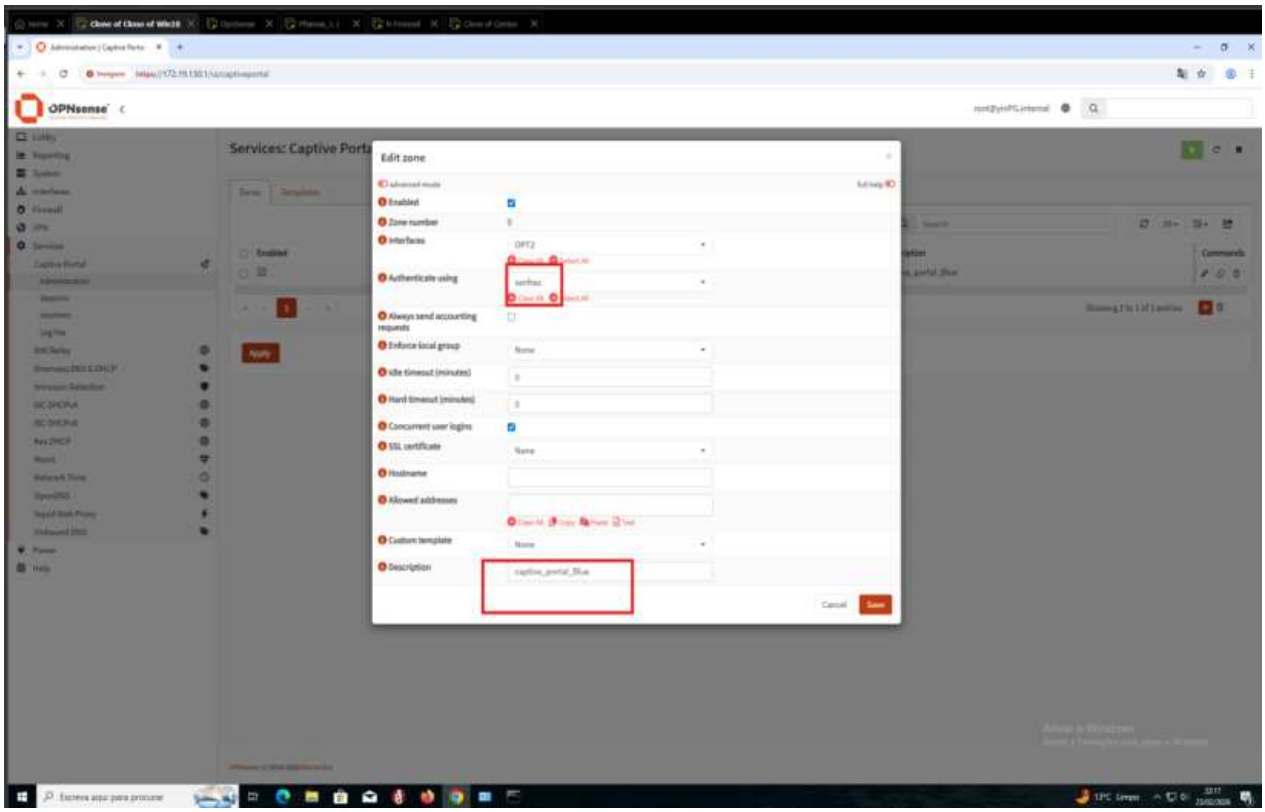




23.8.3 Criação de Zona Captive Portal

Services → Captive Portal → Administration → Add

- Interface: BLUE (OPT2)
- Authenticate using: Senhas

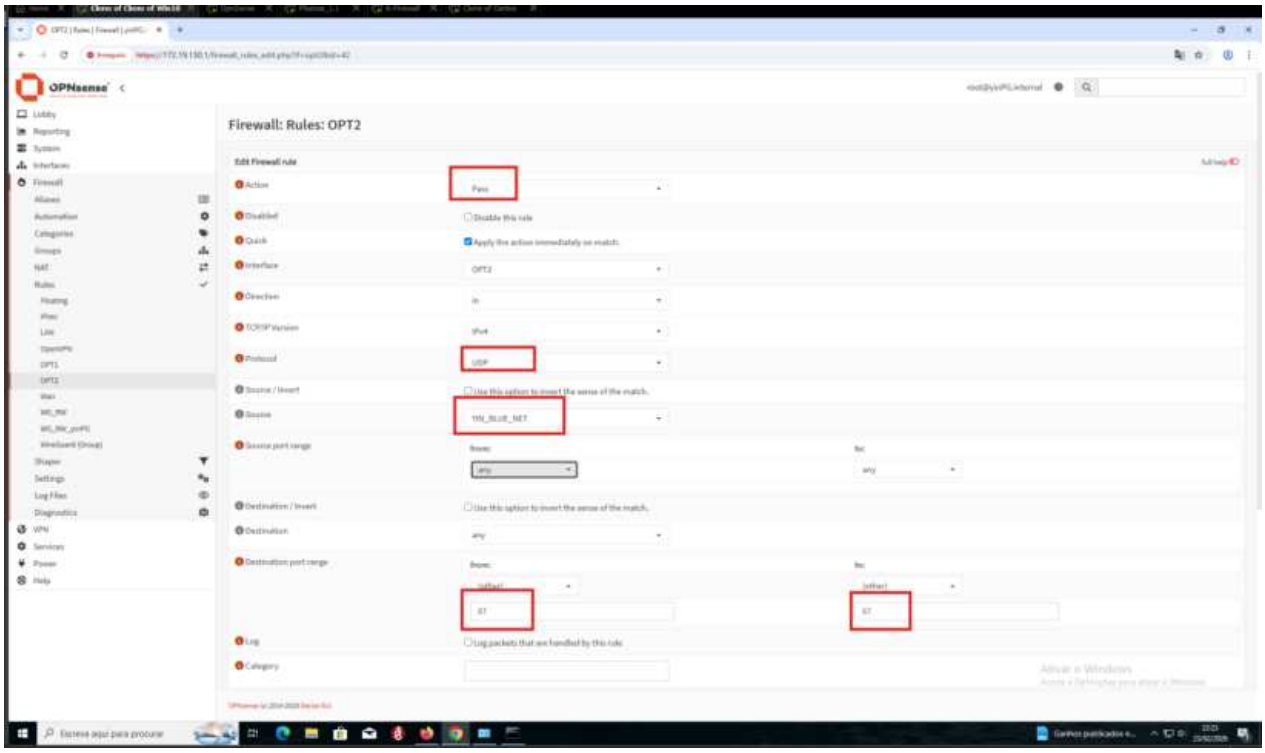


23.8.4 Firewall Rules BLUE

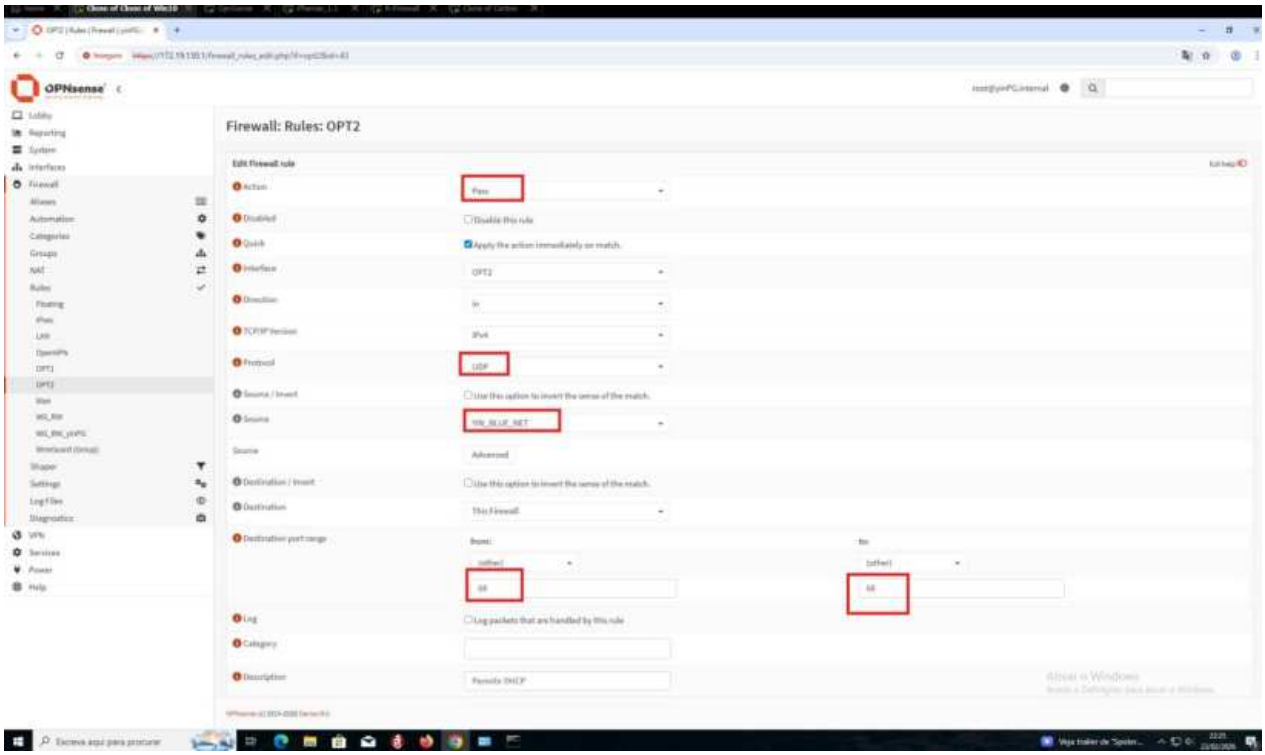
Firewall → Rules → OPT2

Permitir DHCP

- Protocol: UDP
- Source: BLUE net
- Destination: This Firewall
- Destination port: 67-68



Continuação…



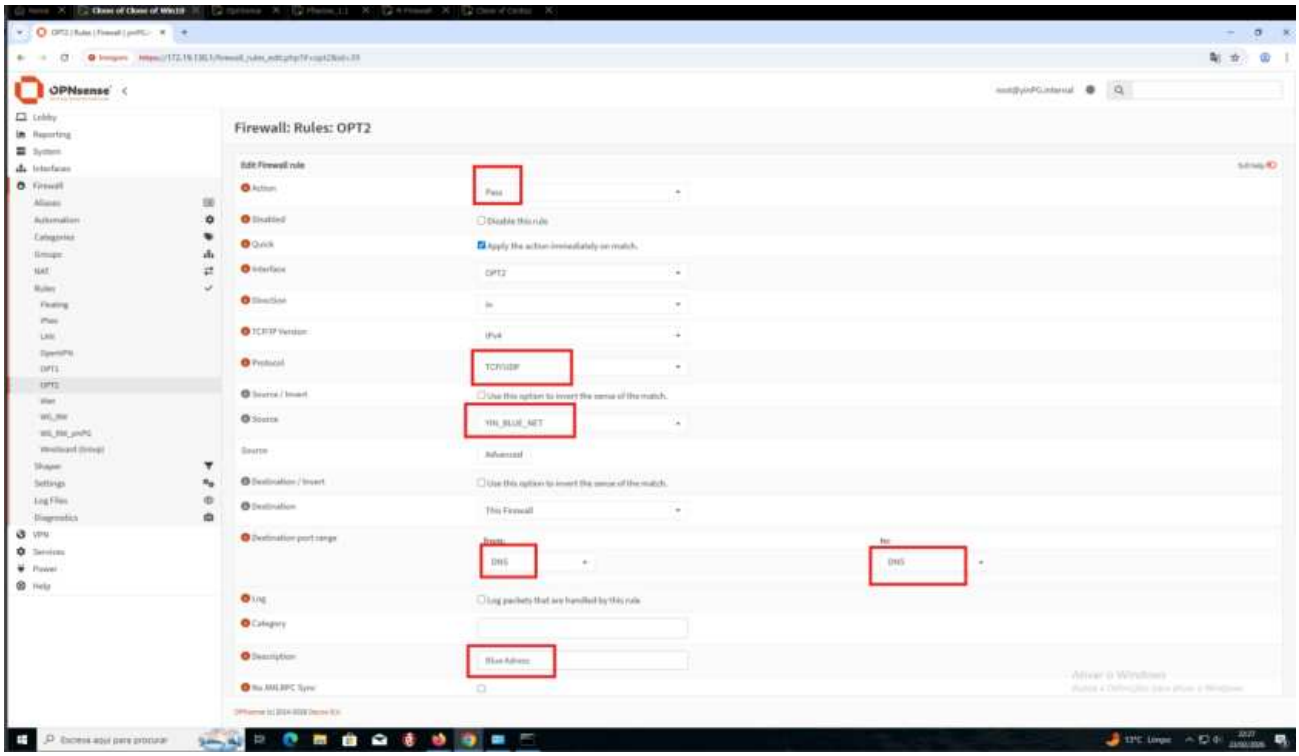
23.8.5 Permitir DNS apenas para firewall

- Protocol: TCP/UDP

- Source: BLUE net

- Destination: This Firewall

- Destination port: 53



Testes finais

No cliente (Windows):

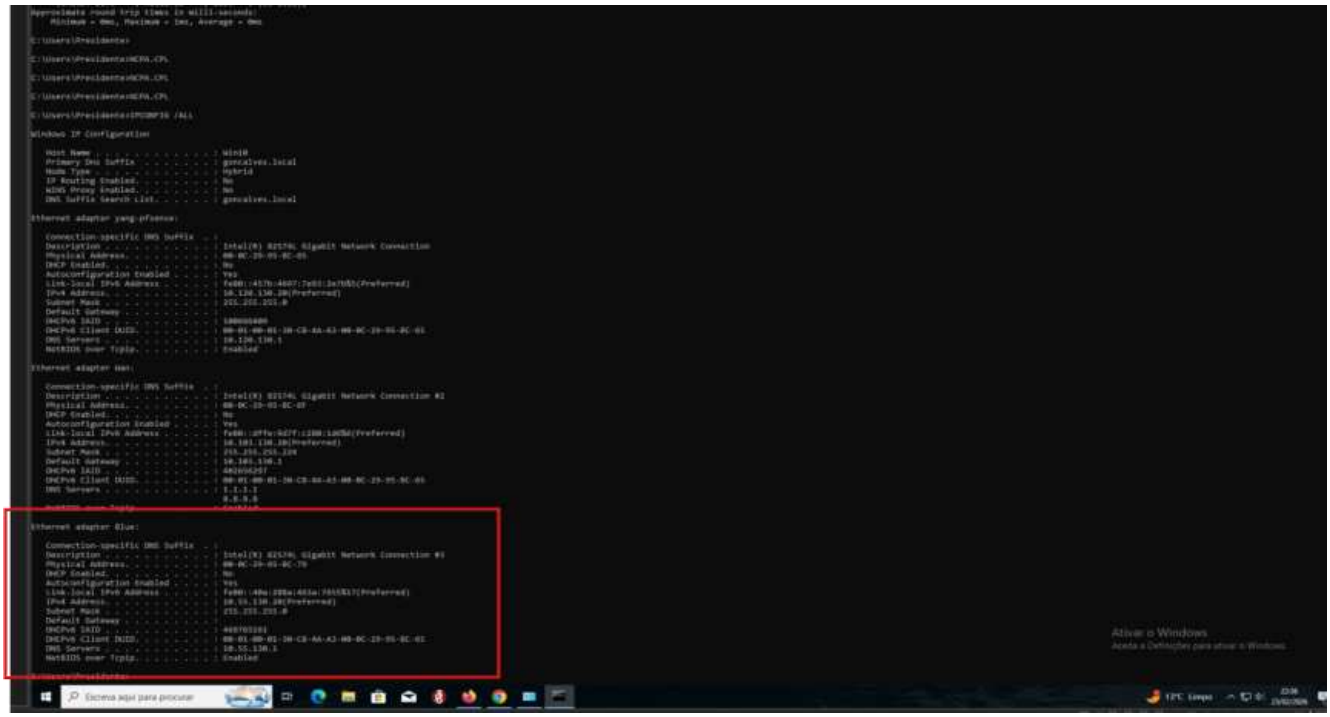
1. **Ligar ao SSID do Access Point da BLUE**
2. **Confirmar que a interface Wi-Fi fica conectada**

Se não ligar:

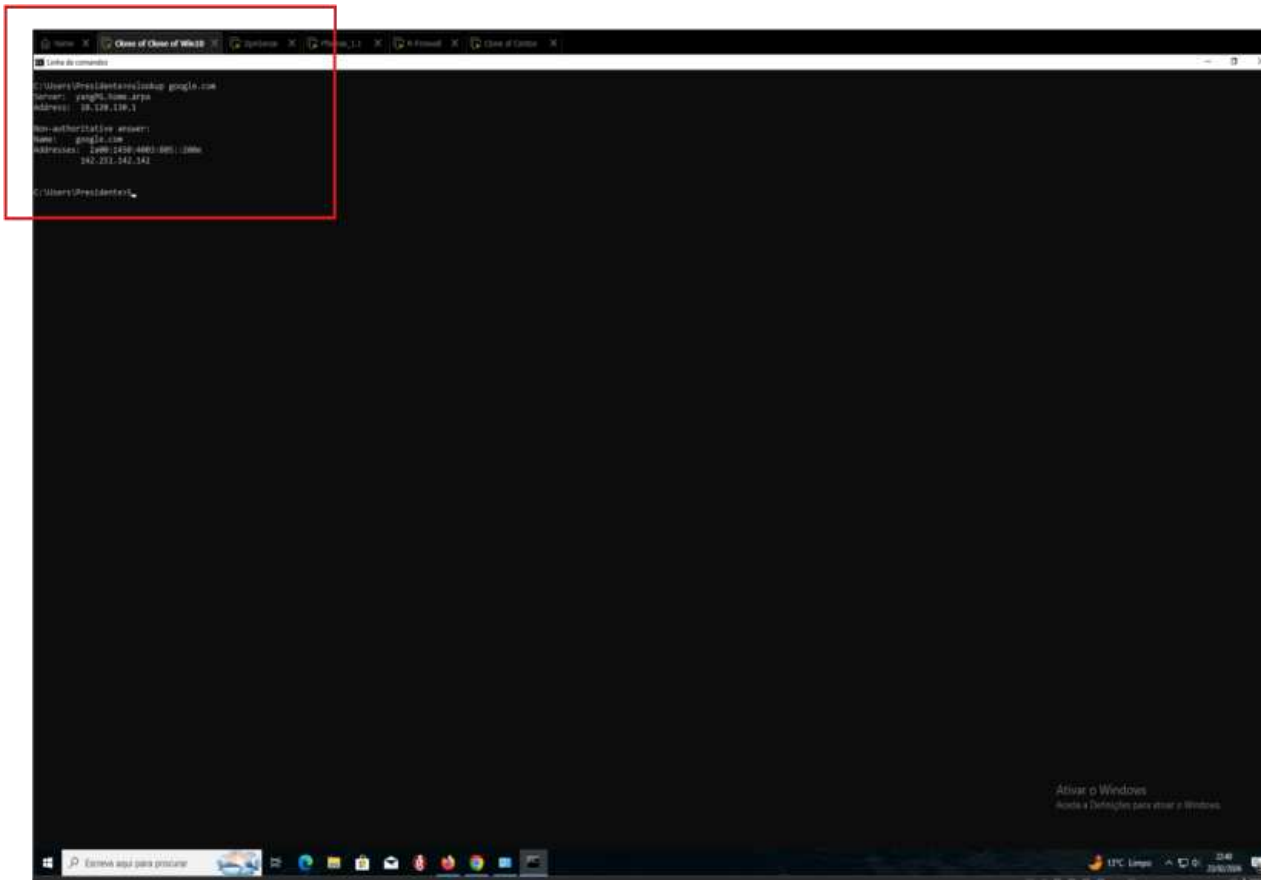
- **Verificar se AP está na rede 10.55.130.0/24**
- **Verificar se está ligado à interface BLUE (OPT2)**

23.8.6 -Verificação se recebe IP via DHCP

Comando : “cls”



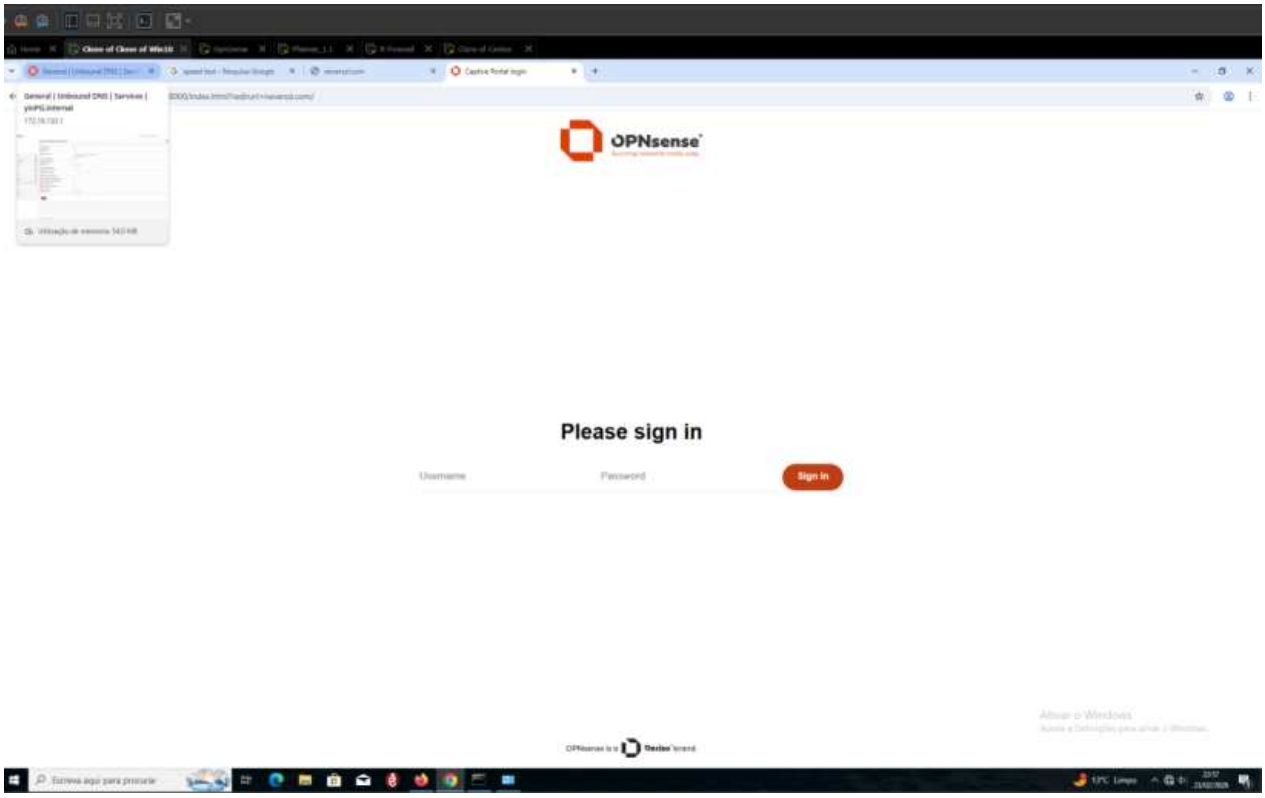
23.8.7 Confirmação DNS é o firewall



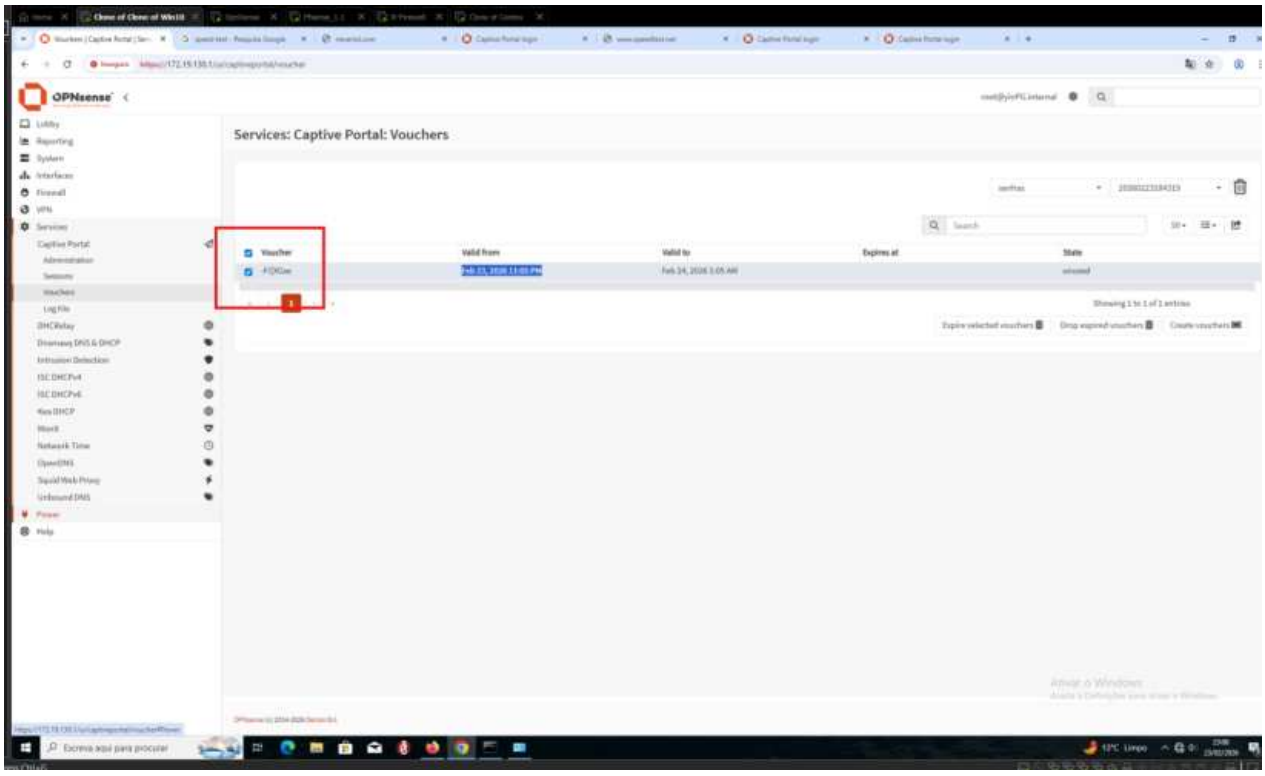
23.8.8 -Testar Captive

Abrir o browser, Escrever:

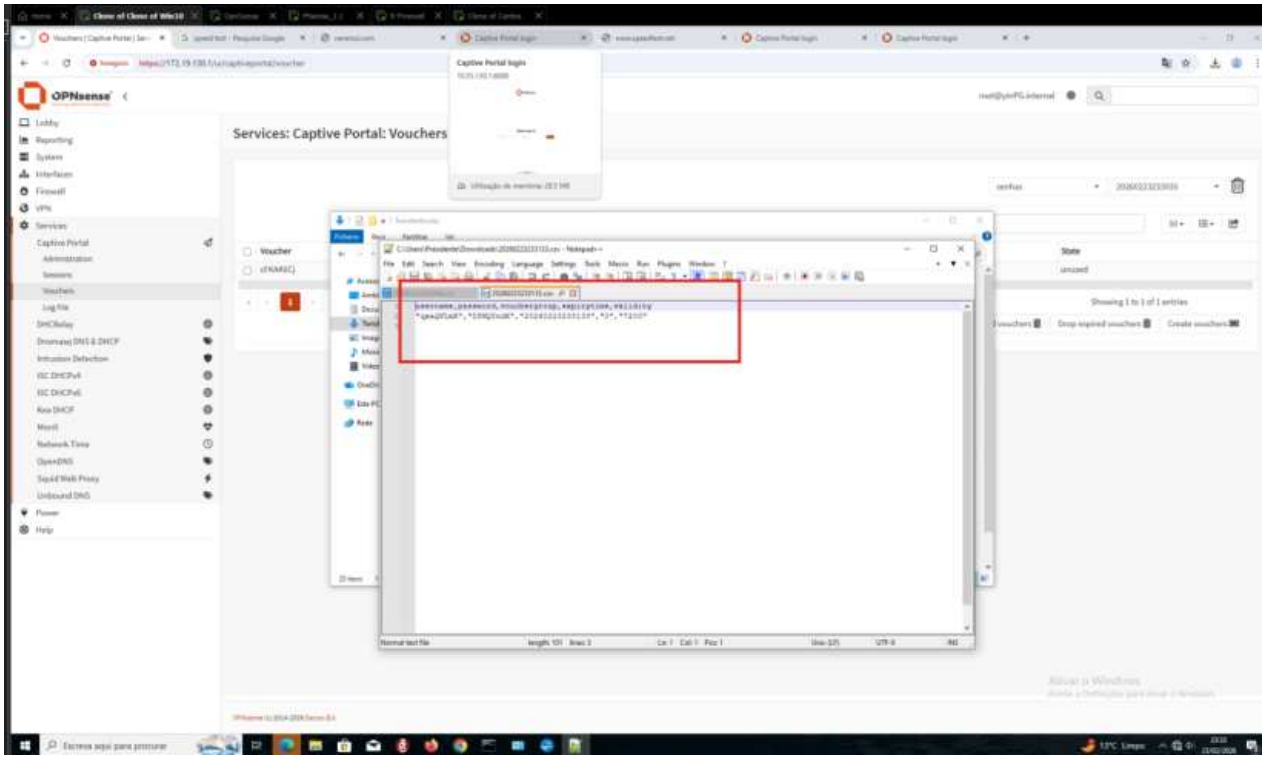
http://neverssl.com



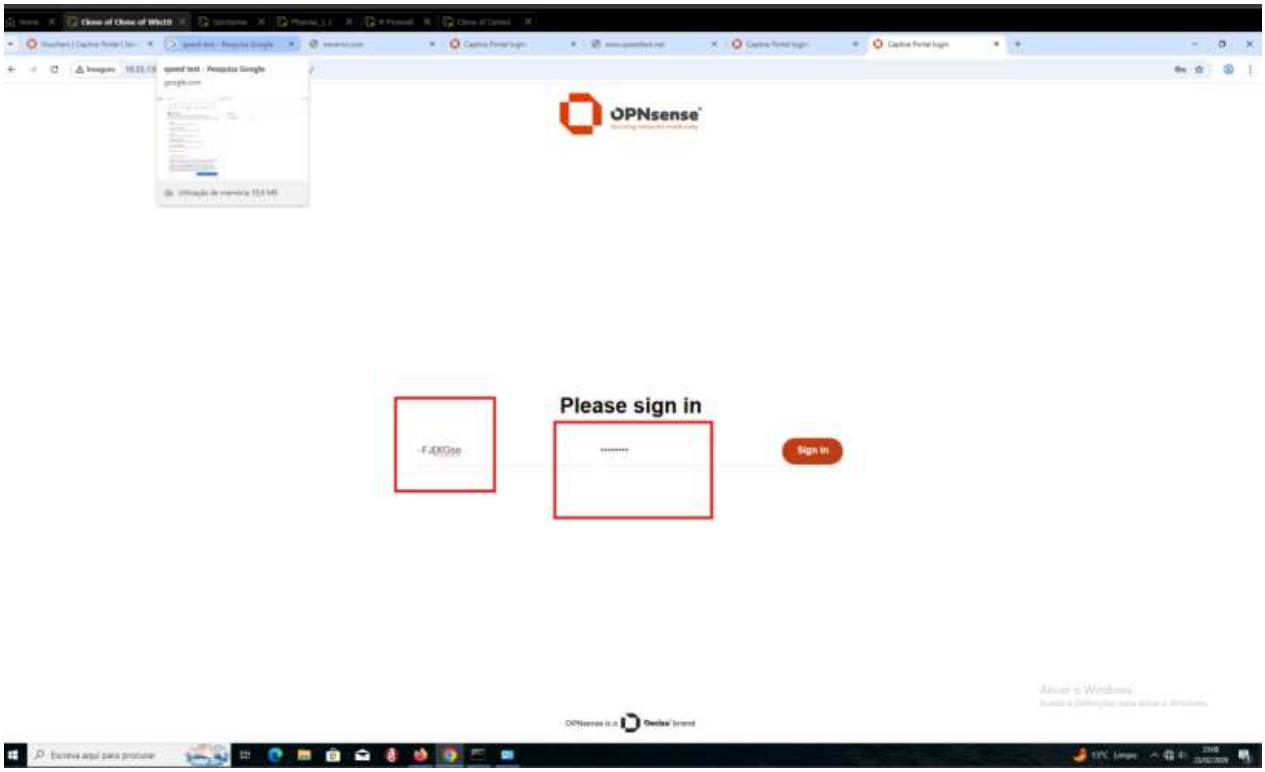
Continuação...



23.8.10 Username e password em ficheiros .csv



http://neverssl.com



Conclusão

O trabalho desenvolvido permitiu implementar uma arquitetura de segurança ativa completa, integrando múltiplas tecnologias de firewall, autenticação e redes privadas virtuais.

Através da segmentação adequada das zonas de rede e da aplicação rigorosa de regras de firewall, foi possível:

- Garantir controlo de resolução DNS
- Implementar autenticação centralizada
- Estabelecer túneis VPN seguros (IPsec, OpenVPN e WireGuard)
- Controlar navegação Web via proxy
- Aplicar políticas de bloqueio com blacklists
- Implementar controlo de largura de banda
- Simular cenários reais de segurança (DNS poisoning e MITM)

A comparação entre IPsec, OpenVPN e WireGuard permitiu compreender diferenças ao nível de:

- Complexidade de configuração
- Performance
- Gestão de chaves
- Escalabilidade

O trabalho demonstrou domínio prático das ferramentas utilizadas, bem como capacidade de diagnóstico e resolução de problemas ao longo do processo de implementação.

Em conclusão, os objetivos propostos foram atingidos com sucesso, tendo sido implementada uma infraestrutura segura, funcional e devidamente testada, simulando um ambiente empresarial real.